TATTVA SERIES

UPPSC PRELIMS | FAST REVISION POINTERS

Previous Year Questions | Topic-wise Analysis | Conceptual Linkages | Exam-Oriented Revision

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Every pointer in this book is drawn from original previous-year questions of UPPSC and other Uttar Pradesh state-level examinations held between 1990 and 2025, tagged with the years in which it was asked. Where an answer key is marked (*), is disputed, or has been overtaken by later changes in law or fact, the pointer says so and gives the correct position.

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TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Ancient History

PYQ-Based Revision Pointers

Chapters 1 to 15

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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How to Use This Book

Every pointer is a real question. Each pointer is the answer to a UPPSC question asked between 1990 and 2025, written as a fact you can revise in seconds, with the related fact you need for the next variation.

Frequency marks. ■ and a red count (e.g. 4×) mark a fact asked 3 or more times. • and a gold count mark facts asked once or twice. The years follow; "one-liner" means a short starred question in the source with no year given.

Asked Most box. Each chapter opens with the facts UPPSC repeats most. Revise these first.

Diagrams. Flowcharts and diagrams turn sequences (timelines, procedures, processes, river courses) into one picture.

Trap Alerts. The places where the obvious answer is wrong or two facts are commonly swapped.

Last-Minute 10. The ten lines to read on the morning of the exam.

Pairs with the Masterbook. Use this book for revision after the TATTVA PYQ Masterbook, which has the full questions, options and explanations.

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Chapter 1

Stone Age

Built from 26 UPPSC questions (1997–2023). Focus: Foote and the three-age system, Bhimbetka, the Mesolithic sites of the Vindhyas and Pratapgarh, Lahuradewa and Mehrgarh, Burzahom, Chalcolithic and OCP.

ASKED MOST · REVISE FIRST

- ■ **Mesolithic Ganga plain (Pratapgarh)**: **Mahadaha** bone tools, **Damdama** triple burial, **Lekhahia** most skeletons (Vindhyas) **6×** 2010, 2013, 2016, 2016, 2016, 2021
- ■ **Earliest agriculture**: **Lahuradewa** (UPPSC key, rice); **Mehrgarh** = earliest settled life **6×** 2004, 2007, 2008, 2008, 2008, 2010
- ■ **Domestication of animals** began in the **Mesolithic** (Bagor, Adamgarh); **cereals** in the **Neolithic** **3×** 2005, 2006, 2008
- ■ **Bhimbetka** = prehistoric rock paintings; found by **V.S. Wakankar** **3×** 2011, one-liner, one-liner
- • **Burzahom**: **dog buried with humans**, **pit dwellings** **2×** 2008, one-liner

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

1.1 Discovery of Indian Prehistory

- **Robert Bruce Foote** (first Palaeolithic tool, Pallavaram 1863): a **geologist** and archaeologist (both accepted) **1×** 2015
- Three-age system (stone, bronze, iron), Copenhagen Museum: **Thomsen** ("Thomson") **1×** 2010

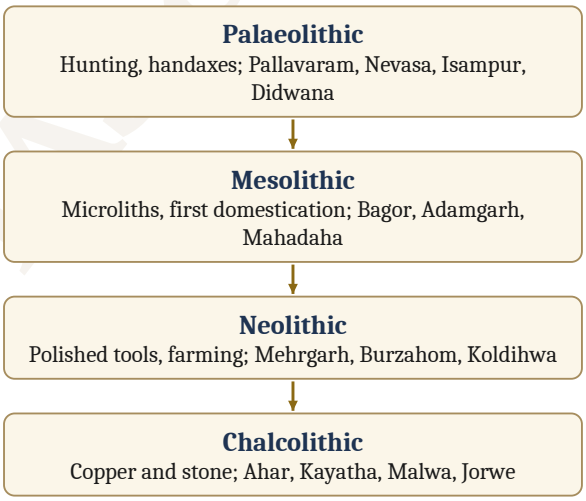
1.2 Palaeolithic Sites and Rock Art

- **Nevasa-Maharashtra, Isampur-Karnataka, Didwana-Rajasthan, Gudiyam cave-Tamil Nadu** **1×** 2023
- Prehistoric paintings: **Bhimbetka** **1×** 2011
- Most rock paintings: **Bhimbetka**; first noticed by **V.S. Wakankar** **2×** one-liner, one-liner

1.3 Mesolithic: Microliths and Domestication

- Domestication of animals began in the **Mesolithic** **1×** 2006
- Mesolithic animal husbandry: **Bagor and Adamgarh** **1×** 2008
- West to east: **Mahadaha, Lekhahia, Paisra, Birbhanpur** **1×** 2021
- Bone implements: **Mahadaha** (and Sarai Nahar Rai) **2×** 2010, 2013
- Three skeletons in one grave: **Damdama** **1×** 2016
- Most human skeletons in a Vindhyan rock shelter: **Lekhahia** **1×** 2016
- Vindhyan people moved to the Ganga valley in a **dry phase** at the end of the Pleistocene (A and R true, R explains A) **1×** 2016

STONE AGE TO METAL



1.4 Neolithic: Farming and Settled Life

- Cereals first cultivated in the **Neolithic Age** **1×** 2005
- First cereal used by man: **barley** **1×** 1997
- ■ Earliest agriculture (key): **Lahuradewa** (rice; date disputed) **4×** 2004, 2007, 2008, 2010
- Earliest settled life: **Mehrgarh** **1×** 2008
- Stone Age to Harappan remains at one site: **Mehrgarh** **1×** 2008
- Dog buried with human: **Burzahom** **1×** 2008
- Pit dwellings: **Burzahom** **1×** one-liner
- Ash mounds: **Sanganakallu** (Karnataka; Budihal also) **1×** 2009

1.5 Chalcolithic, OCP and Megaliths

- **Navdatoli** excavated by **H.D. Sankalia**; it is in **Madhya Pradesh** 2× 2009, one-liner
- Chalcolithic Maharashtra burials: **north-south** under house floors 1× 1997
- **Ochre-Coloured Pottery** named at **Hastinapur** 1× 2006
- **Megaliths**: **burial sites** 1× 2005

Site	Known for
Pallavaram (TN)	First Palaeolithic tool (Foote, 1863)
Bhimbetka (MP)	Rock paintings
Bagor, Adamgarh	Earliest animal domestication
Mahadaha, Damdama, Sarai Nahar Rai (UP)	Mesolithic burials, bone tools
Lahuradewa (UP)	Early rice (UPPSC key)
Mehrgarh (Pakistan)	Earliest settled farming village
Burzahom (Kashmir)	Pit dwellings, dog burials
Koldihwa (UP)	Neolithic rice
Hastinapur (UP)	OCP named

TRAP ALERT

- Animals were domesticated in the **Mesolithic**; cereals came in the **Neolithic**.
- Earliest rice in UPPSC keys = **Lahuradewa**; earliest settled village = **Mehrgarh**.
- **Mahadaha** = double burials; **Damdama** = triple burial.
- Megaliths are **burials**, not temples.

UP LINK

UP link: UP is rich in Mesolithic sites: **Sarai Nahar Rai, Mahadaha, Damdama** (Pratapgarh), **Lekhahia** and **Baghor** in the Vindhyas (Mirzapur-Sonbhadra), **Koldihwa** (Prayagraj, Neolithic rice), **Lahuradewa** (Sant Kabir Nagar) and **Hastinapur** (Meerut, OCP).

LAST-MINUTE 10

1. Foote = father of Indian prehistory (Pallavaram 1863)
2. Thomsen = three-age system
3. Bhimbetka paintings; Wakankar found them
4. Domestication: Mesolithic (Bagor, Adamgarh)
5. Mahadaha bone tools; Damdama triple burial; Lekhahia most skeletons
6. Barley = first cereal; cereals in Neolithic
7. Lahuradewa = earliest rice (key); Mehrgarh = earliest settled life
8. Burzahom = dog burial, pit houses
9. Navdatoli = Sankalia (MP); OCP = Hastinapur
10. Megaliths = burial sites

Chapter 2

Indus Valley Civilization

Built from 48 UPPSC questions (1990–2023). Focus: excavators and sites, site-river and site-state matching, UP's Harappan sites, Lothal, Kalibangan and Dholavira, seals and religion.

ASKED MOST · REVISE FIRST

- **Site-state matching**: **Kalibangan-Rajasthan, Rakhigarhi-Haryana, Manda-J&K, Daimabad-Maharashtra, Hulas and Alamgirpur-UP, Lothal and Dholavira-Gujarat** 7× 1996, 2006, 2010, 2012, 2018, 2020, 2021
- **Site-river matching**: **Harappa-Ravi, Kalibangan-Ghaggar, Lothal-Bhogava, Ropar-Sutlej** 3× 2008, 2010, one-liner
- **Lothal** = port and dockyard, ivory scale; **Kalibangan** = ploughed field; **Dholavira** = three-part city, water management 9× 1995, 2005, 2006, 2010, 2010, 2013, 2014, one-liner, one-liner
- **Great Bath** = **Mohenjo-daro** 2× 1992, 2010
- Harappan source: **archaeological excavations**; a **proto-historic** culture 4× 1993, 1994, 1996, 1996

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Nature and Sources

- Indus civilisation: **proto-historic** age (script not read) **1×** 1996
- Source of knowledge: **archaeological excavations** **3×** 1993, 1994, 1996
- Mohenjo-daro and Harappa are dead cities; being excavated is not the reason (A and R true, R does not explain A) **1×** 2009
- Human society depends mainly on **economy** (per key) **1×** 2004
- Non-Aryan because it was **urban** **1×** 2010
- Placed before the Aryans because of its **pottery** **1×** 1990
- Different from the Vedic culture: **all of the above** **1×** 2004

2.2 Discovery and Excavators

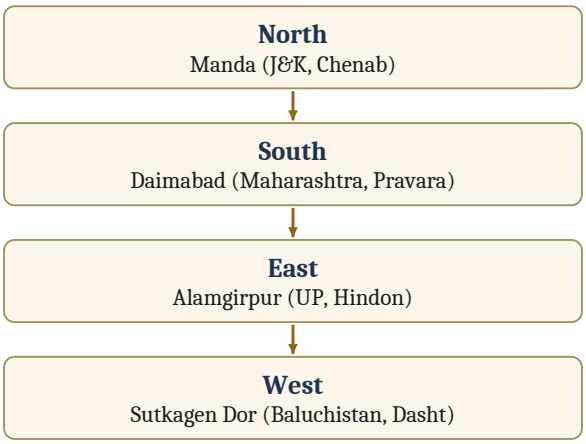
- First to dig Harappa without seeing its value: **Alexander Cunningham** **1×** 2006
- **Harappa-Daya Ram Sahni, Hastinapur-B.B. Lal, Takshashila-John Marshall, Kaushambi-N.G. Majumdar** (per key) **1×** 2017
- Wrong pair: **Dholavira-B.K. Thapar** (Dholavira = R.S. Bisht, found by J.P. Joshi) **1×** 2006
- **Chanhudaro** excavated by **J.H. Mackay** **1×** 2015

Site	Excavator
Harappa	Daya Ram Sahni (1921)
Mohenjo-daro	R.D. Banerji (1922)
Chanhudaro	N.G. Majumdar, J.H. Mackay
Lothal	S.R. Rao
Kalibangan	B.B. Lal, B.K. Thapar
Dholavira	R.S. Bisht
Rakhigarhi	Amarendra Nath, Vasant Shinde

2.3 Extent and Site Locations

- Eastern boundary: **Alamgirpur** (Meerut, on the Hindon) **1×** 2023
- **Manda-J&K, Daimabad-Maharashtra, Kalibangan-Rajasthan, Rakhigarhi-Haryana** **1×** 2012
- Rivers of the civilisation: **Sindhu, Chenab, Jhelum** (1, 2, 3); not Ganga **1×** 2009
- **Harappa-Ravi, Hastinapur-Ganga, Nagarjunakonda-Krishna, Paithan-Godavari** **1×** 2008
- **Harappa-Ravi, Kalibangan-Ghaggar, Lothal-Bhogava, Ropar-Sutlej** **1×** 2010
- Wrong pair: **Kalibangan-Haryana** (it is Rajasthan) **1×** 1996
- Wrong pair: **Rakhigarhi-Rajasthan** (it is Haryana) **1×** 2006
- **Balu-Haryana, Manda-J&K, Padri-Gujarat, Hulas-UP** **1×** 2020
- In Sindh: **Mohenjo-daro and Chanhudaro** (2 and 3) **1×** 2010
- Now in Pakistan: **Harappa** (Punjab province) **1×** 1994
- Largest Harappan site in India: **Rakhigarhi** **1×** 2004
- In UP: **Alamgirpur and Hulas** (III and IV) **1×** 2018
- **Mandi** (Muzaffarnagar): **UP** **1×** 2021
- **Dadheri**: late Harappan site in **Punjab** **1×** 2014
- **Sinauli** (Baghpat): **human burials** **1×** 2004

HARAPPAN FRONTIER SITES



2.4 Cities, Buildings and Crafts

- **Great Bath**: **Mohenjo-daro** **2×** 1992, 2010
- **Harappa-Cemetery R-37, Lothal-dockyard, Kalibangan-furrowed land, Mohenjo-daro-dancing girl** **1×** 2017
- House wells: several sites (*); best answer **Mohenjo-daro** **1×** 2004
- In India: **Lothal** **1×** 1995
- **Lothal**: on the **Bhogava**, in **Gujarat**; the **port town** **3×** one-liner, one-liner, one-liner

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- Ivory scale: **Lothal** 1× 2014
- Boat models: **Mohenjo-daro and Lothal** 1× 2022
- Ploughed field: **Kalibangan** 1× 2005
- Double burial: **Lothal** (and Kalibangan) 1× 2016
- Rock-cut (reservoir) architecture: **Dholavira** 1× 2006

■ **Dholavira**: **Gujarat**; city in **three parts**; advanced **water management** 3× 2010, 2010, 2013

Site	Famous for
Mohenjo-daro	Great Bath, granary, dancing girl, Pashupati seal
Harappa	Granaries, Cemetery R-37
Lothal	Dockyard, rice husk, ivory scale, double burial
Kalibangan	Ploughed field, fire altars, horned deity cake
Dholavira	Three-part city, reservoirs, signboard
Chanhudaro	Bead factory; no citadel
Banawali	Barley, toy plough

2.5 Economy, Religion and Contemporaries

- Not true: Harappans **knew iron** 1× 1992
- Cotton for cloth first grown in **India** 1× 2006
- Not on Harappan seals: **horse** 1× 2009
- Idol worship began in **pre-Aryan** times 1× 1992
- Horned deity on a terracotta cake: **Kalibangan** 1× 2008
- On the Nile: **Egyptian** civilisation 1× 2004
- North to south: **Aztec, Maya, Muisca, Inca** 1× 2016

TRAP ALERT

- Kalibangan = **Rajasthan** (Ghaggar); Rakhigarhi = **Haryana**; Harappa = Pakistan's **Punjab**, not Sindh.
- Lothal is on the **Bhogava**, not the Sabarmati.
- No iron, no horse on seals, no temples; but figurines show image worship.
- Alamgirpur is on the **Hindon**, not the Ganga.

UP LINK

UP link: UP's Harappan and late Harappan sites are **Alamgirpur** (Meerut, eastern limit), **Hulas** (Saharanpur), **Mandi** (Muzaffarnagar, gold hoard) and **Sinauli** (Baghpat, burials with copper objects and "chariots").

LAST-MINUTE 10

1. Proto-historic; known from excavations
2. Cunningham first; Daya Ram Sahni dug Harappa
3. Kalibangan = Rajasthan (Ghaggar); Rakhigarhi = Haryana (largest in India)
4. Harappa = Ravi; Ropar = Sutlej; Lothal = Bhogava
5. Alamgirpur, Hulas, Mandi, Sinauli = UP
6. Great Bath = Mohenjo-daro
7. Lothal = dockyard, ivory scale; Kalibangan = ploughed field
8. Dholavira = three parts, water system (Bisht)
9. No iron; no horse on seals; cotton first in India
10. Idol worship began before the Aryans

Chapter 3

Vedic Age

Built from 56 UPPSC questions (1991–2023). Focus: the term Arya, Boghazkoi, Rig Vedic rivers, the four Vedas and their layers, Upanishads, assemblies, cattle economy, Indra and Agni, the Later Vedic shift.

ASKED MOST · REVISE FIRST

- **Satyameva Jayate** from the **Mundaka Upanishad** 4× 1991, 2004, 2004, 2014
- **Upanishads** = **philosophy** and **moksha**; Nachiketa-Yama in the **Katha** Upanishad 6× 1998, 1999, 2002, 2002, 2003, 2004
- **Aghanya** = **cow**; cow = most valuable property; **yava** = barley 5× 2008, 2008, 2010, 2015, 2017
- **Indra** = most hymns (about 250), war god, thunder and rain; **Agni** next 5× 2002, 2010, 2011, 2017, one-liner
- **Rivers**: **Asikni-Chenab, Parushni-Ravi, Shutudri-Sutlej, Kubha-Kabul, Sadanira-Gandak**; Saraswati = "Naditama" 5× 2008, 2010, 2010, 2012, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 The Aryans and Their Home

- "Aryan" denotes a **speech group** (key: noble; never a race) 1× 1999
- Classical meaning of "Arya": a **noble man** 1× 1998
- Wrote on the Aryans' original home (Arctic): **Bal Gangadhar Tilak** 1× 1996
- **Boghazkoi** (Turkey, 14th century BCE): inscription naming **Vedic gods** 3× 1996, one-liner, one-liner
- Avesta resembles the Rig Veda; it is from **Iran** 1× 2004

3.2 Rig Vedic Geography and Tribes

- **Asikni** = **Chenab** 1× 2010
- Rivers linking the Aryans with Afghanistan: **Kubha, Kramu** 1× 2010
- **Kubha** is in **Afghanistan** (Kabul river) 1× one-liner
- **Kubha-Kabul, Parushni-Ravi, Sadanira-Gandak, Shutudri-Sutlej** 1× 2012
- "Matetama, Devitama, Naditama": **Saraswati** 1× 2008
- Not in the Panchajana: **Kikata** 1× 2009
- **Battle of Ten Kings**: on the **Parushni** (Ravi); Sudas won with Vasishtha 1× 2008

Vedic name	Modern river
Sindhu	Indus
Vitasta	Jhelum
Asikni	Chenab
Parushni	Ravi
Vipas	Beas
Shutudri	Sutlej
Kubha	Kabul
Gomati	Gomal
Sadanira	Gandak

3.3 The Four Vedas and Their Layers

- Oldest Veda: **Rig Veda** (a collection of **hymns**) 2× 1995, one-liner
- **Trayi**: the **three Vedas** (Rig, Sama, Yajur) 1× 2010
- **Rig-hymns and prayers, Yajur-hymns and rituals, Sama-musical hymns, Atharva-charms and spells** 2× 2002, 2003
- Compiled from the Rig Veda: **Samaveda** 1× 1997
- **Ayurveda** first appears in the **Atharvaveda** 1× 1994
- Rig Veda's "basic script" (key): **Brahmi** (it was oral) 1× 2004
- **Hota** priest: **Rig Veda** 1× one-liner
- **Gayatri Mantra**: **Rig Veda** (3rd Mandala, Vishvamitra) 1× 2013
- Order: **Samhitas, Brahmanas, Aranyakas, Upanishads** 1× 2014
- **Gopatha Brahmana**: **Atharvaveda** 1× 2014
- 800-600 BCE: the **Brahmana** era 1× 2002

Veda	Content	Priest	Upaveda
Rig	Hymns and prayers	Hota	Ayurveda (often)
Yajur	Rituals (prose)	Adhvaryu	Dhanurveda
Sama	Melodies	Udgata	Gandharvaveda
Atharva	Charms, medicine	Brahma	Shilpaveda (Ayurveda first appears here)

3.4 Upanishads

- Upanishads: **philosophy** 3× 1998, 2002, 2004

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- Talks of **moksha**: **Upanishads** 1× 2003
- **Nachiketa and Yama**: **Katha Upanishad** 2× 1999, 2002
- **Satyameva Jayate**: **Mundaka Upanishad** 4× 1991, 2004, 2004, 2014
- King **Ashvapati** ruled **Kekaya** 1× 1999

3.5 Polity, Economy and Society

- **Sabha and Samiti** as daughters of Prajapati: **Atharvaveda** 1× 2009
- **Bhagadugha**: **revenue collector** 1× 2023
- Aryans' chief livelihood (key): **agriculture** (Rig Vedic life was mainly pastoral) 1× 1993
- **Yava** = **barley** 1× 2008
- Most valuable early Rig Vedic property: **cow** 1× 2015
- **Aghanya** = **cows** (not to be killed) 3× 2008, 2010, 2017
- **Nishka**: a **neck** ornament 2× 2005, 2007
- **Indus-urban, Rig Vedic-pastoral, Later Vedic-agrarian, medieval-landlordism** 1× 2020
- **Gotra** first appears in the **Rig Veda** 1× 2005

3.6 Rig Vedic Religion

- People believed mainly in **sacrifice and rituals** 1× 1993
- Rig Vedic religion: **polytheistic** 1× 2014
- Most hymns: **Indra** (2010 key: Agni) 2× 2002, 2010
- War god and most popular deity: **Indra** 2× 2011, one-liner
- Indra: god of **thunder and rain** (1 and 4); Varuna guards moral order 1× 2017
- Priest of the gods: **Brihaspati** 1× 2013

EARLY TO LATER VEDIC

Rig Vedic (1500-1000 BCE)
Sapta Sindhu; pastoral; tribal; Indra, Agni

↓

Later Vedic (1000-600 BCE)
Ganga-Yamuna doab; Kuru-Panchala heartland

↓

Economy
Agriculture, iron (PGW), settled villages

↓

Society
Varna hardens; women's status declines

↓

Religion
Prajapati, Vishnu, Rudra rise; rituals grow

3.7 Later Vedic Age, Puranas and Epics

- Axis of Later Vedic culture: **Kuru-Panchala** 1× 2007
- Earliest iron (classic key): **Atranjikhera** 1× 1998
- Videgha Mathava's priest: **Gautama Rahugana** 1× 2015
- **Purushamedha**: in several texts (*) 1× 2008
- Sarga, pratisarga, vamsha, manvantara, vamshanucharita: **Puranas** 1× 2015
- Number of Puranas: **18** 2× 2009, 2010
- Mahabharata's original name: **Jaya Samhita**; also "**Shatasahasri Samhita**" 2× 2014, one-liner

TRAP ALERT

- **Asikni = Chenab**; **Parushni = Ravi** (students swap them).
- Gayatri = **Rig Veda**, 3rd Mandala; Satyameva Jayate = **Mundaka**, not Mandukya.
- Ayurveda first in the **Atharvaveda**; Samaveda is drawn from the Rig Veda.
- Sabha and Samiti = Prajapati's daughters in the **Atharvaveda**.

UP LINK

UP link: the Later Vedic heartland, **Kuru-Panchala**, lay largely in western UP (**Hastinapur, Ahichchhatra, Kampilya**); **Atranjikhera** (Etah) gave early iron with Painted Grey Ware, and **Naimisharanya** (Sitapur) is linked with the telling of the Puranas.

LAST-MINUTE 10

1. Arya = speech group (no race); Tilak = Arctic home
2. Boghazkoi = Vedic gods, 14th century BCE
3. Asikni = Chenab; Parushni = Ravi (Ten Kings); Kubha = Kabul
4. Saraswati = Naditama; Rig Veda oldest
5. Gayatri = Rig Veda; Ayurveda = Atharvaveda
6. Samhita, Brahmana, Aranyaka, Upanishad
7. Satyameva Jayate = Mundaka; Nachiketa = Katha
8. Aghanya = cow; yava = barley; nishka = neck ornament
9. Indra = most hymns; Brihaspati = gods' priest
10. Kuru-Panchala = Later Vedic axis; 18 Puranas; Jaya Samhita

Chapter 4

Buddhism

Built from 66 UPPSC questions (1991–2023). Focus: Buddha's life and the UP circuit, symbols, doctrine, Triratna, the Tripitaka, the four councils, sects, stupas, chaityas and art schools.

ASKED MOST · REVISE FIRST

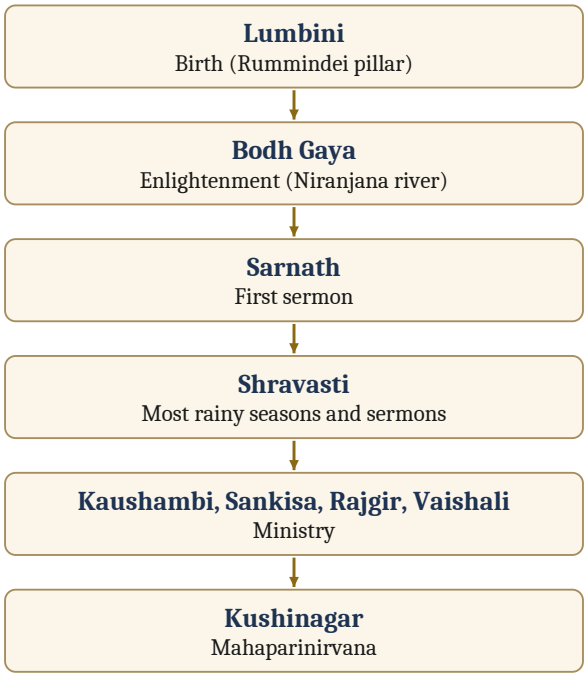
- **Kushinagar**: **Mahaparinirvana**, capital of the **Mallas** 5× 2004, 2005, 2011, 2015, one-liner
- **Sarnath**: first sermon (**Dharmachakrapravartana**); deer and wheel symbol 5× 2002, 2004, 2015, one-liner, one-liner
- **Lumbini** birthplace, proved by Ashoka's **Rummindei** pillar (Mauryan) 7× 2004, 2006, 2007, 2011, one-liner, one-liner, one-liner
- **Councils**: **Rajgriha** (Ajatashatru, Mahakashyapa), **Vaishali** (Kalashoka), **Pataliputra** (Ashoka), **Kundalavana** (Kanishka) 9× 2000, 2002, 2010, 2015, one-liner, one-liner, one-liner, one-liner, one-liner
- **Sanchi** = best stupa, not linked with any event of Buddha's life; built by **Ashoka** 4× 2008, 2008, 2011, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Life of the Buddha

- **Mahabhinishkramana**: his **departure from home** 1× 2014
- Mother's clan: **Koliya** 1× one-liner
- Born at **Lumbini** 1× one-liner
- Lumbini proven by **Mauryan** inscriptions: Ashoka's **Rummindei pillar**, marking the **birth** 6× 2004, 2006, 2007, 2011, one-liner, one-liner
- **Alara Kalama**: Buddha's **teacher** 1× 2010
- On the **Niranjana**: **Bodh Gaya** 1× 2012
- First sermon: **Sarnath** (Dharmachakrapravartana) 4× 2004, 2015, one-liner, one-liner
- Born in Kapilavastu (home town), enlightened at Bodh Gaya, rejected the Vedas, preached noble truths (all four per key) 1× 2010
- Wheel with deer: **first sermon** 1× 2002
- **Birth-lotus, first sermon-wheel, enlightenment-Bodhi tree, great departure-horse** 1× 2005
- Most sermons: **Shravasti** 1× 2011
- Visited Kaushambi under **Udayana** 1× 2010
- **Mahaparinirvana**: **Kushinagar** (Malla republic) 3× 2005, 2011, one-liner
- Kushinagar was the Malla capital; the parinirvana there does not explain that (A and R true, R does not explain A) 1× 2004
- **Mahaparinirvana Temple**: **Kushinagar** 1× 2015
- Last disciple: **Subhadda** 2× 2013, 2016
- Last rainy season: **Vaishali** 1× 2015

EIGHT GREAT PLACES



4.2 Doctrine and the Sangha

- Noble truths in order: **suffering, its cause, its cessation, the path** 1× 2006
- Buddha's teachings: **purity of conduct** 1× 1991
- "World is unstable and transient": **Buddhism** 1× 1992
- Buddha on karma, rebirth, nirvana and God: no option accepted (*) 1× 1992
- Buddhism **accepts rebirth**; the "no soul" idea is true (A false, R true) 1× 2006
- Jainism and Buddhism both accept **karma and transmigration** 1× 1996
- **Triratna**: **Buddha, Dhamma, Sangha** 1× 2017
- Wanted to lead the Sangha: **Devadatta** 1× 1999
- Women admitted as nuns at **Vaishali** 1× 2010

4.3 Texts

- **Tripitaka**: collection of Buddha's **teachings** 2× 2003, 2004
- Monastic rules: **Vinaya Pitaka** 1× one-liner
- **Vinaya-monastic rules, Sutta-teachings, Abhidhamma-philosophy, Mahavamsa-regional history** 1× 2023
- **Milindapanha**: Menander and **Nagasena** 1× 2023
- "Light of Asia": **Gautama Buddha**; Edwin Arnold's book draws on the **Lalitavistara** 2× 2010, 2014

4.4 The Four Councils

- First council presided by **Mahakashyapa**, under **Ajatashatru**, at **Rajgir** (Saptaparni cave) 5× 2000, 2010, one-liner, one-liner, one-liner
- Order: **Rajgriha, Vaishali, Pataliputra, Kundalavana** 1× 2002
- Second council: **Vaishali**; fourth under **Kanishka** 2× one-liner, one-liner
- Third council (Ashoka) in the **Dipavamsa and Mahavamsa** 1× 1999
- **Ashokarama** monastery: **Pataliputra** 1× 2015

Council	Place	Patron	President
First	Rajgriha (Saptaparni)	Ajatashatru	Mahakashyapa
Second	Vaishali	Kalashoka	Sabakami
Third	Pataliputra	Ashoka	Moggaliputta Tissa
Fourth	Kundalavana (Kashmir)	Kanishka	Vasumitra

4.5 Sects, Spread and Decline

- Hinayana vs Mahayana: **worship of god and goddess** (images, Bodhisattvas) 1× 1996
- **Shunyata** (void): **Nagarjuna** 1× 1998
- Karmapa Lama: **Kagyupa** sect 1× 2000
- Buddhism kept caste, challenged Brahmins and called some jobs low (1, 2, 3) 1× 1998
- Spread by simplicity, appeal to the lowly, missionary zeal, local language (1, 2, 3, 4) 1× 2009

4.6 Stupas, Caves and Art

- Not linked with Buddha's life: **Sanchi** 2× 2008, 2011
- Sarnath pillar: **Ashoka** 1× 2008
- **Kaushambi-Ghositarama, Kushinagar-Ramabhar stupa, Sarnath-Dhamekh, Shravasti-Sahet-Mahet** 1× 2010
- Stupa chronology: no option accepted (*) 1× 2014
- **Amaravati** stupa: **Satavahanas** 1× 2023
- Best stupa: **Sanchi**, built by **Ashoka** 2× 2008, one-liner

QUANTUM IAS & PCS

- Largest stupa: **Kesariya** (Bihar) 1× 2023
- **Chaitya** = worship hall; **vihara** = dwelling 1× 2001
- Largest Hinayana chaitya: **Karle** 1× 2014
- Eleven-headed Bodhisattva: **Kanheri** 1× 2017
- **Valabhi** university: **Gujarat** 1× 2014
- Laid the base of idol worship (key): **Buddhism** 1× 2008
- Standing Buddha statue: **Kushana** period 1× 1992
- Sarnath Bhumisparsha Buddha: **Gupta** period 3× 2009, 2010, 2010

TRAP ALERT

- Most sermons = **Shravasti**; last rainy season = **Vaishali**; death = **Kushinagar**.
- Buddhist Triratna = Buddha, Dhamma, Sangha; Jain Triratna = right faith, knowledge, conduct.
- Sanchi is **not** connected with any event of Buddha's life.
- Amaravati = **Satavahana**, not Mauryan; Kesariya is the largest stupa.

UP LINK

UP link: four of the eight great places are in UP: **Sarnath** (Dhamekh stupa), **Shravasti** (Sahet-Mahet, Jetavana), **Kushinagar** (Mahaparinirvana temple, Ramabhar stupa) and **Sankisa**; **Kaushambi** has the Ghositarama monastery, and **Piprahwa** (Siddharthnagar) is identified with Kapilavastu.

LAST-MINUTE 10

1. Mahabhinishkramana = leaving home; mother = Koliya
2. Lumbini = birth (Rummindei pillar)
3. Bodh Gaya on the Niranjana; Sarnath = first sermon
4. Shravasti = most sermons; Kushinagar = Malla, parinirvana
5. Subhadda = last disciple; Vaishali = last rainy season, nuns admitted
6. Triratna = Buddha, Dhamma, Sangha; Devadatta = rival
7. Councils: Rajgriha, Vaishali, Pataliputra, Kundalavana
8. Nagasena + Milinda; Nagarjuna = shunyata
9. Sanchi = Ashoka; Amaravati = Satavahana; Kesariya = largest
10. Karle = biggest chaitya; Sarnath Buddha = Gupta

Chapter 5

Jain Religion

Built from 25 UPPSC questions (1993–2023). Focus: Tirthankaras and their symbols, Parshvanatha, UP's Jain sites, Mahavira's life, Triratna, councils, sects and the Ajivikas.

QUANTUM

ASKED MOST · REVISE FIRST

- **Rishabhadeva** (Adinatha) founded Jainism; **Mahavira** is the 24th Tirthankara **3×** 1993, 2004, 2010
- **Parshvanatha** (23rd): **Varanasi**; nirvana at **Sammed Shikhar** **3×** 2002, 2002, 2016
- **Tirthankara-symbol and UP site matching**: **Adinatha-bull, Parshvanatha-snake, Mallinatha-water jar, Sambhavnath-horse; Shravasti-Sambhavnath, Ayodhya-Rishabhanath, Kakandi-Suvidhinath, Pabhosa-Padmaprabhu** **3×** 2017, 2021, 2021
- **Triratna** (right faith, knowledge, conduct) is the glory of **Jainism**; core point **non-violence** **2×** 1993, 2004
- **Ajivika** sect: **Makkhali Gosala**; "fate decides everything" **4×** 1996, 2006, 2017, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

5.1 Tirthankaras

- Originator of Jainism: **Rishabhadeva** **1×** 2010
- "Tirthankara": **Jainism** **1×** 1993
- Not a Tirthankara: **Nathamuni** (a Vaishnava acharya) **1×** 2004
- Order: **Abhinandan, Padmaprabhu, Vimalnath, Munisuvrata** **1×** 2016
- **Parshvanatha**: **Varanasi** **1×** 2016
- Parshvanatha's pilgrimage: **Sammed Shikhar** (Jharkhand) **2×** 2002, 2002
- **Adinatha-bull, Mallinatha-water jar, Parshvanatha-snake, Sambhavnath-horse** **1×** 2017
- Wrong pair: **Vasupujya-Sammed Shikhar** (Vasupujya attained nirvana at **Champa**) **1×** 2021

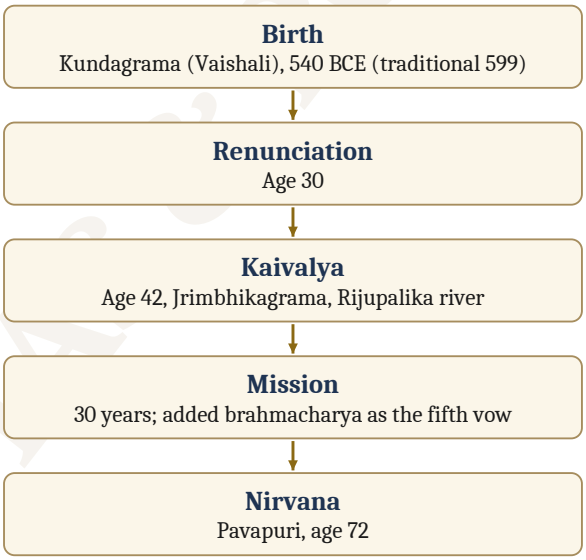
5.2 UP in Jain Geography

- **Shravasti-Sambhavnath, Kakandi-Suvidhinath, Ayodhya-Rishabhanath, Pabhosa-Padmaprabhu** **1×** 2021
- **Prabhasgiri** (Pabhosa): a **Jain** pilgrimage **1×** 2008
- Pilgrimage of both Buddhists and Jains in UP: **Kaushambi** **1×** 2010

5.3 Mahavira and Doctrine

- **Kundalpur**: birthplace of **Mahavira** **1×** 2004
- Enlightenment on the **Rijupalika** river **1×** 2017
- First follower (son-in-law): **Jamali** **1×** 2008
- **Triratna** (right faith, action, knowledge): **Jainism** **1×** 2004
- Basic point: **non-violence** **1×** 1993
- No "final annihilation of the world": **Jainism** **1×** 2014

MAHAVIRA'S LIFE



5.4 Literature, Councils and Sects

- Earliest Jain texts: **Ardha-Magadhi** **1×** 2006
- Shvetambara Agamas finally edited at **Valabhi** (key: Pataliputra) **1×** 2008
- **Yapaniya**: a **Jain** sect **1×** 2010
- **Ajivika** founder: **Makkhali Gosala** **1×** 1996
- "Fate assigns everything": **Ajivikas** **1×** 2006
- Ajivikas respected in the Maurya era (only II); they were not forest dwellers **1×** 2023
- **Lakulisha-Pashupata, Nagarjuna-Shunyavada, Bhadrabahu-Jainism, Gosala-Ajivika** **1×** 2017
- **Gommateshwara** (Shravanabelagola) built by **Chamundaraya** **1×** 2009

Council	Place	Head	Result
First	Pataliputra (c. 300 BCE)	Sthulabhadra	12 Angas compiled; split begins

QUANTUM IAS & PCS

Council	Place	Head	Result
Second	Valabhi (5th century CE)	Devardhi Kshamashramana	Agamas finally written

TRAP ALERT

- Mahavira is the **24th**, not the first; Rishabhadeva founded Jainism.
- Jain Triratna = right faith, knowledge, conduct; Buddhist = Buddha, Dhamma, Sangha.
- Parshvanatha gave **four** vows; Mahavira added **brahmacharya**.
- Ajivika = Gosala; Pashupata = Lakulisha; Yapaniya = Jain.

UP LINK

UP link: UP holds the birthplaces of many Tirthankaras: **Ayodhya** (Rishabhanatha and four others), **Shravasti** (Sambhavnath), **Kaushambi/Pabhosa** (Padmaprabhu), **Varanasi** (Parshvanatha, Suparshvanatha), **Kakandi** (Suvidhinath) and **Hastinapur**.

LAST-MINUTE 10

1. Rishabhadeva = founder; Mahavira = 24th
2. Parshvanatha = Varanasi; Sammed Shikhar
3. Adinatha = bull; Parshvanatha = snake; Mallinatha = water jar
4. Shravasti = Sambhavnath; Ayodhya = Rishabhanath
5. Kaushambi = Buddhist and Jain site; Pabhosa = Jain
6. Kundalpur birth; Rijupalika enlightenment; Jamali first follower
7. Triratna = Jainism; non-violence = core
8. Ardha-Magadhi texts; Valabhi = final edit
9. Yapaniya = Jain; Ajivika = Gosala
10. Gommateshwara = Chamundaraya

Chapter 6

Shaiva and Bhagavata Religion

Built from 16 UPPSC questions (1997–2022). Focus: iconography, Alvars and Nayanars, Heliodorus and the Bhagavatas, avatars of Vishnu, astika and nastika, the Gita and sacred geography.

ASKED MOST · REVISE FIRST

- **Nayanars** = **Shaiva** saints; **Alvars** = Vaishnava **3×** 2006, 2013, 2014
- **Heliodorus** pillar (**Besnagar**) = earliest evidence of Bhagavatism; Heliodorus came from **Taxila** **3×** 1997, 2008, 2010
- **Gita** gives equal weight to **jnana, karma, bhakti**; the Bhagavata Purana is **not** in the Prasthanatrayi **2×** 1997, 2005

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

6.1 Iconography and Shaivism

- **Shiva-trident, Vishnu-chakra, Ganesha-noose, Saraswati-veena** **1×** 2022
- **Ardhanarishvara**: union of god with his **Shakti** **1×** 1997

6.2 Tamil Bhakti

- **Nayanars**: **Shaiva** saints **2×** 2006, 2014
- Not an Alvar: **Tirugyan** (Tirujnana Sambandar is a Nayanar) **1×** 2013

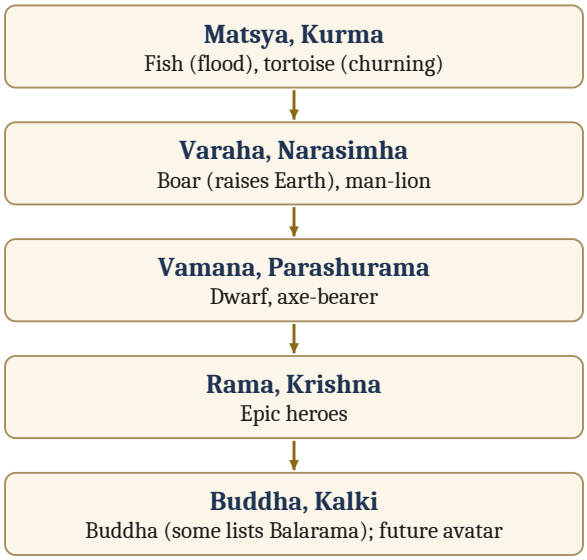
Group	God	Famous names
Alvars (12)	Vishnu	Andal (woman), Nammalvar, Periyalvar
Nayanars (63)	Shiva	Appar, Sambandar, Sundarar, Manikkavachakar

6.3 Bhagavatism and Vishnu

- First worshippers of Vasudeva Krishna: **Bhagavatas** **1×** 1997
- Earliest record of Bhagavatism: **Heliodorus' Besnagar inscription** **1×** 2008
- Heliodorus was from **Taxila** **1×** 2010

- Deity holding a plough: **Balarama** 1× 2007
- Avatar that saved the Earth from the sea: **Varaha** 1× 2010
- Forms of bhakti (Bhagavata): **9** 1× 2010

TEN AVATARS OF VISHNU



6.4 Philosophy, Epics and Sacred Places

- Astika vs nastika: belief in the **Vedas' authority** 1× 2005
- Equal place to jnana, karma, bhakti: **Bhagavad Gita** 1× 2005
- Not in the Prasthanatrayi: **Bhagavata** (it is Upanishads, Gita, Brahma Sutra) 1× 1997
- Mahabharata told to 60,000 hermits at **Naimisharanya** 1× 2006
- Rama meets Hanuman in the **Kishkindha Kanda** 1× 2004
- Nashik Kumbh: on the **Godavari** 1× 2003

TRAP ALERT

- Alvars = Vishnu; Nayanars = Shiva.
- Heliodorus was an Indo-Greek envoy from **Taxila** (of Antialkidas), not Megasthenes.
- Gita is in the Prasthanatrayi; the Bhagavata Purana is not.
- Varaha raises the Earth; Matsya saves Manu in the flood.

UP LINK

UP link: **Naimisharanya** (Sitapur) is where the Mahabharata and Puranas were recited; **Mathura** is the centre of Bhagavata worship of Vasudeva Krishna, and **Prayagraj** hosts the Kumbh at the Sangam.

LAST-MINUTE 10

1. Shiva = trident; Vishnu = chakra; Saraswati = veena
2. Ardhanarishvara = god + Shakti
3. Nayanars = Shaiva; Alvars = Vaishnava
4. Bhagavatas first worshipped Vasudeva Krishna
5. Heliodorus (Taxila) = Besnagar pillar
6. Balarama holds the plough; Varaha saves the Earth
7. Nine forms of bhakti
8. Astika = accepts the Vedas
9. Gita = jnana, karma, bhakti
10. Naimisharanya = Mahabharata recital; Nashik Kumbh = Godavari

Chapter 7

Sixth Century BCE: Political Condition

Built from 25 UPPSC questions (1992–2021). Focus: the sixteen Mahajanapadas and their capitals, UP's Mahajanapadas, kings of the Buddha's age, Haryanka, Shishunaga and Nanda Magadha, punch-marked coins.

ASKED MOST · REVISE FIRST

- **Mahajanapada-capital matching**: **Kuru-Indraprastha, Panchala-Ahichhatra, Kosala-Shravasti/Saketa, Vatsa-Kaushambi, Surasena-Mathura, Chedi-Suktimati, Anga-Champa, Matsya-Viratnagar, Asmaka-Potana** 7× 2008, 2011, 2014, 2016, 2017, 2020, one-liner
- Sixteen Mahajanapadas listed in the **Anguttara Nikaya**; **8** lay in today's UP 4× 2004, 2008, 2008, 2021
- **Ajatashatru** killed his father Bimbisara and was killed by his son 2× 2007, 2011
- **Mahapadma Nanda** = "Aparo Parashurama"; Nanda is named in **Kharavela's Hathigumpha** inscription 3× 1999, 2010, 2010
- Earliest Indian coins: **silver punch-marked** 1× 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 The Sixteen Mahajanapadas

- Listed in the **Anguttara Nikaya** 3× 2004, 2008, 2008
- Mahajanapadas in present UP: **8** 1× 2021
- **Kuru-Indraprastha, Panchala-Ahichhatra, Kosala-Saketa, Vatsa-Kaushambi** 1× 2008
- Wrong pair: **Kosala-Ahichhatra** 1× 2017
- Capital of North Panchala: **Ahichhatra** 1× 2006
- Wrong pair: **Surasena-Kaushambi** (Surasena = **Mathura**) 1× 2016
- **Suktimati**: capital of **Chedi** 1× 2011
- Ancient Shravasti: **crescent-shaped** town 1× 2010
- **Kalpi** stands on the **Yamuna** 1× 2015
- **Champa**: capital of **Anga** 1× 2014
- **Matsya**: **Rajasthan** 1× 2017
- **Matsya-Viratnagar, Kuru-Indraprastha, Surasena-Mathura, Asmaka-Potana** 1× 2020
- Ujjain's old name: **Avantika** 1× 2009

Mahajanapada	Capital	Today
Kashi	Varanasi	UP
Kosala	Shravasti, Saketa	UP
Vatsa	Kaushambi	UP
Panchala	Ahichhatra, Kampilya	UP
Surasena	Mathura	UP
Kuru	Indraprastha	Delhi-Haryana-west UP
Malla	Kushinara, Pava	UP
Chedi	Suktimati	Bundelkhand
Magadha	Rajagriha	Bihar
Anga	Champa	Bihar
Vajji	Vaishali	Bihar
Avanti	Ujjayini, Mahishmati	MP
Matsya	Viratnagar	Rajasthan
Asmaka	Potana	Godavari (only one in the south)
Gandhara, Kamboja	Taxila; Rajapura	North-west

7.2 Republics and Kings

- Not an early republic: **Yaudheya** (a later gana) 1× 1992
- **Pradyota-Avanti, Udayana-Vatsa, Prasenajit-Kosala, Ajatashatru-Magadha** 2× 2000, 2004
- Udayana-Vasavadatta story: **Ujjain** 1× 2008
- Physician **Jivaka**: **Bimbisara's** court 1× 2006

7.3 Rise of Magadha

- Killed his father and was killed by his son: **Ajatashatru** 2× 2007, 2011
- Magadha took Malwa (Avanti) under **Shishunaga** 1× 2008
- Varanasi as second capital: unclear (*); UPPSC accepted **Shishunaga** 1× 2007
- **Aparo Parashurama**: **Mahapadma Nanda** 1× 2010
- Nanda evidence: **Kharavela's Hathigumpha** inscription 1× 2010
- A Nanda king dug a canal in **Kalinga** 1× 1999
- Earliest extant coins: **silver** (punch-marked) 1× 2010

QUANTUM IAS & PCS

DYNASTIES OF MAGADHA

Haryanka
Bimbisara, Ajatashatru, Udayin (founded Pataliputra)

↓

Shishunaga
Shishunaga (took Avanti), Kalashoka (2nd Buddhist council)

↓

Nanda
Mahapadma Nanda (Ugrasena), Dhanananda

↓

Maurya
Chandragupta (322 BCE) with Chanakya

TRAP ALERT

- Kosala's capital is **Shravasti/Saketa**; Ahichhatra is Panchala.
- Surasena = **Mathura**; Vatsa = **Kaushambi**.
- **Asmaka** is the only southern Mahajanapada.
- Earliest coins = silver punch-marked; first king's portrait = Indo-Greek.

UP LINK

UP link: eight Mahajanapadas lay in present UP: **Kashi, Kosala, Vatsa, Panchala, Surasena, Malla, Chedi** (Bundelkhand part) and part of **Kuru**; capitals such as **Shravasti, Kaushambi, Ahichhatra** (Bareilly), **Mathura** and **Varanasi** are all in UP.

LAST-MINUTE 10

1. 16 Mahajanapadas = Anguttara Nikaya; 8 in UP
2. Kuru = Indraprastha; Panchala = Ahichhatra; Kosala = Shravasti, Saketa
3. Vatsa = Kaushambi; Surasena = Mathura; Chedi = Suktimati
4. Anga = Champa; Matsya = Viratnagar (Rajasthan); Asmaka = Potana
5. Ujjain = Avantika; Kalpi on the Yamuna
6. Pradyota = Avanti; Udayana = Vatsa; Prasenajit = Kosala
7. Jivaka = Bimbisara's physician
8. Ajatashatru: patricide and killed by his son
9. Shishunaga took Malwa; Mahapadma = Aparo Parashurama
10. Silver punch-marked = earliest coins

Chapter 8

Greek Invasion

Built from 5 UPPSC questions (1998–2015). Focus: Alexander's campaign, Porus, Massaga, the reasons for his success, and his companions versus later Greek envoys.

ASKED MOST · REVISE FIRST

- Alexander defeated **Porus** on the **Jhelum** (Hydaspes), 326 BCE **1×** 2013
- ■ Alexander succeeded through **no central power, a superior army, traitor allies** (1, 2, 3) **3×** 2000, 2002, 2003

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

8.1 Alexander in India

- Women took up arms against Alexander at **Massaga** (Assakenoi) **1×** 2013
- Defeated on the Jhelum: **Porus** **1×** 2013
- ■ Success: **no central power, superior army, help from traitors** (1, 2, 3) **3×** 2000, 2002, 2003
- Alexander left India in 325 BCE; he was **not** defeated by Chandragupta (A true, R false) **1×** 1998
- Did not come with Alexander: **Deimachus** (later envoy to Bindusara) **1×** 2015

ALEXANDER'S INDIAN CAMPAIGN

327 BCE
Crosses the Hindu Kush; Ambhi of Taxila submits

↓

Massaga
Fierce resistance; women fight

↓

326 BCE
Battle of the Hydaspes (Jhelum) with Porus

↓

Beas (Hyphasis)
Army refuses to go further

↓

325 BCE
Returns; dies at Babylon, 323 BCE

Greek	Role
Nearchus, Onesicritus	Alexander's companions (wrote accounts)
Megasthenes	Envoy of Seleucus to Chandragupta; Indica
Deimachus	Envoy to Bindusara
Dionysius	Sent by Ptolemy II to Ashoka's time
Heliodorus	Indo-Greek envoy; Besnagar pillar

TRAP ALERT

- Battle with Porus = **Jhelum**; Alexander turned back at the **Beas**.
- Ambhi **helped** Alexander; Porus fought him.
- Deimachus and Megasthenes came **after** Alexander.

UP LINK

UP link: Alexander never reached the Ganga valley; the Nanda power of Magadha, whose realm included eastern UP, is one reason his army refused to march beyond the Beas.

LAST-MINUTE 10

1. Massaga women fought Alexander
2. Porus defeated on the Jhelum (Hydaspes)
3. Ambhi of Taxila sided with Alexander
4. Army stopped at the Beas
5. Alexander left India in 325 BCE, undefeated
6. Success = disunity, strong army, traitors
7. Deimachus = envoy to Bindusara, not a companion
8. Megasthenes = envoy to Chandragupta
9. Nearchus = Alexander's admiral
10. Alexander died at Babylon, 323 BCE

Chapter 9

Maurya Empire

Built from 57 UPPSC questions (1990–2024). Focus: Chandragupta and Kautilya, Greek envoys, Megasthenes' Indica, the Junagadh inscription, Ashokan edicts and their decipherment, Dhamma, officials and revenue, Mauryan art.

ASKED MOST · REVISE FIRST

- **James Prinsep** first read Brahmi and Ashoka's edicts (1837) **4×** 2006, 2008, 2010, 2014
- **Junagadh** inscription of **Rudradaman**: names Chandragupta (Sudarshana lake) and Ashoka; Ashoka's edicts are on the same rock **7×** 1996, 2002, 2004, 2006, 2007, 2010, 2016
- **Megasthenes' Indica**: seven classes, no slavery, city administration; he came under **Chandragupta** **4×** 2008, 2009, 2013, 2017
- **Rock Edict XIII** = Kalinga war; **RE XII** = religious tolerance; **RE II** = southern kingdoms **5×** 2005, 2008, 2016, 2016, 2022
- Name "Ashoka" appears at **Gujarra** (Datia, MP) and Maski **2×** 2008, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

9.1 Chandragupta and Kautilya

- First Indian empire: **Chandragupta Maurya** **1×** 2002
- First to conquer Malwa, Gujarat, Maharashtra: **Chandragupta Maurya** **1×** 1991
- Defeated Seleucus: **305 BCE** **1×** 2014
- Mauryas: culture, provinces, rule up to the Hindu Kush (A, C, D); **no gold coins** **1×** 1990
- Kautilya: minister of **Chandragupta Maurya**; childhood name **Vishnugupta** **3×** 2002, 2002, 2006
- Arthashastra: a book on **governance**; compared with Machiavelli's **Prince** **2×** 1994, 2012
- Seventh limb of the Saptanga state: **Mitra** (ally) **1×** 2015

9.2 Greek Contacts and Megasthenes

- Sandrokottos met Alexander: **Justin** **1×** 2008

- **Megasthenes**: under **Chandragupta**; **Deimachus**: under **Bindusara** 2× 2015, 2017
- **Chandragupta-Sandrokottos, Bindusara-Amitraghata, Ashoka-Piyadasi, Chanakya-Vishnugupta** 1× 2001
- No slavery in India: **Megasthenes' Indica** 1× 2013
- Seven classes of society: **Indica** 1× 2008
- City administration: **Indica** 1× 2009
- **Kumrahar** and **Bulandibagh**: **Pataliputra** 2× 2008, 2011

9.3 Inscriptions

- Chandragupta's sway in the west: **Junagadh rock inscription of Rudradaman** 2× 1996, 2010
- Water management at Girnar: **Chandragupta and Ashoka** (2 and 3) 1× 2006
- First lake at Girnar (Sudarshana): **Chandragupta Maurya** 2× 2002, 2007
- Names both Chandragupta and Ashoka: **Junagadh** inscription 1× 2004
- Ashokan edicts also found at **Junagadh** (Girnar rock) 1× 2016
- Earliest stone inscriptions: **Prakrit** 1× 2009
- Brahmi deciphered from **stone** inscriptions 1× 2008
- First to read Ashoka's edicts and Brahmi: **James Prinsep** 4× 2006, 2008, 2010, 2014
- Pre-Ashokan Brahmi: **Anuradhapura** 1× 2008
- Ashoka named: **Gujarra** (Datia, MP) 2× 2008, 2015
- **Kalsi** (Uttarakhand): Ashoka's **rock edicts** 1× 2008

9.4 Kalinga and Dhamma

- Kalinga war known from **Rock Edict XIII** 1× 2016
- Ashoka annexed Kalinga as it held the routes to the south (A and R true, R explains A) 1× 2002
- Religious tolerance: **Rock Edict XII** 2× 2008, 2022
- Southern kingdom not named: **Satavahana** 1× 2005
- Southern kingdoms named in **Rock Edict II** 1× 2016
- **Ptolemy Philadelphus**: king of **Egypt**; the **Mauryas** dealt with Syria and Egypt 3× 2006, 2010, 2012
- Dhamma linked to Buddhist texts; **Dhamma Mahamattas** from the 14th year (both per key) 1× 2024
- Rajdharma of patience, liberty and compassion: **Ashoka** 1× 1993
- "All people should do good deeds...": **Ashoka** (initial key: Harsha) 1× 2023
- First hospitals and herbal gardens: **Ashoka** 1× 2015
- Pilgrimage route: **Lumbini, Kapilavastu, Gaya, Sarnath, Kushinagar, Shravasti** 1× 1999

Edict	Subject
Rock Edict I	Ban on animal sacrifice
Rock Edict II	Medical care; Cholas, Pandyas, Satiyaputras, Keralaputras
Rock Edict V	Dhamma Mahamattas
Rock Edict XII	Religious tolerance
Rock Edict XIII	Kalinga war; Greek kings
Minor RE (Maski, Gujarra)	Name "Ashoka"
Rummindei pillar	Lumbini, birth of Buddha

9.5 Administration, Economy and Art

- Revenue head: **Samaharta** 1× 2008
- Weights and measures: **Pautavadhyaksha** 1× 2012
- **Rajjukas**: Mauryan **officers** 1× 1996
- **Agronomoi**: **road building** (per key) 1× 2020
- **Bhaga and bali**: **revenue** 1× 1996
- **Sita**: income from **crown land** 1× 2013
- **Sarthavaha**: **merchants' caravan** 1× 2008
- Mauryas gave no religious land grants; no peasant revolt (A true, R false) 1× 2004
- Not true of Barabar caves: dated to the **6th century BCE** (they are Ashokan) 1× 2017
- Correct: **Lothal dockyard, Sarnath sermon, Nalanda learning** (1, 2, 4); Rajgir has no lion capital 1× 2004
- Remarriage prohibited: **Manusmriti** 2× 2002, 2003
- Manu called foreigners **fallen Kshatriyas** 1× 2014

QUANTUM IAS & PCS

TRAP ALERT

- Junagadh inscription is **Rudradaman's** (Shaka, 150 CE), not Mauryan.
- **Megasthenes** = Chandragupta; **Deimachus** = Bindusara.
- Prinsep read Brahmi (1837); the name Ashoka was found at **Maski** in 1915.
- Mauryas issued **silver** punch-marked coins, not gold.

UP LINK

UP link: Ashokan relics in UP include the **Sarnath lion capital** (national emblem), the **Allahabad (Kaushambi) pillar** with Samudragupta's later inscription, the **Meerut pillar** (now in Delhi) and Rummindei-area sites at **Piprahwa**; Kalsi was in UP until 2000.

LAST-MINUTE 10

1. Chandragupta = first empire; beat Seleucus 305 BCE
2. Kautilya = Vishnugupta; Arthashastra; Mitra = 7th limb
3. Megasthenes (Indica) under Chandragupta; Deimachus under Bindusara
4. Bindusara = Amitraghata; Ashoka = Piyadasi
5. Junagadh (Rudradaman) names Chandragupta and Ashoka
6. Prinsep read Brahmi; Gujarra and Maski name Ashoka
7. RE XIII = Kalinga; RE XII = tolerance; RE II = south kingdoms
8. Samaharta = revenue; Pautavadhyaksha = weights
9. Sita = crown land; sarthavaha = caravan
10. Kumrahar, Bulandibagh = Pataliputra

Chapter 10

Post-Mauryan Period

Built from 47 UPPSC questions (1990–2023). Focus: Shungas, Satavahanas, Kharavela, the order of foreign invaders, Indo-Greek and Kushan coins, Rudradaman, Kanishka's court, the Vikram and Saka eras, Gandhara art.

ASKED MOST · REVISE FIRST

- **Saka era = 78 CE**; **Vikram Samvat = 57 BCE**; gap **135 years** 6× 1990, 1992, 2002, 2006, 2007, 2013
- **Vima Kadphises** issued gold coins first and regularly; **Kushans** issued most copper coins; Buddha on **Kanishka's** coins 5× 2004, 2005, 2009, 2010, 2015
- **Kanishka's court**: **Ashvaghosha, Vasumitra** (not Patanjali, Kalidasa or Vishakhadatta) 4× 1994, 2005, 2008, 2008
- **Satavahanas**: strongest southern power after the Mauryas; founder **Simuka**; capital **Pratishthana** (Amaravati in options) 5× 1993, 2001, 2005, 2008, 2008
- **Kharavela** (Chedi, Kalinga): patron of **Jainism**, **Hathigumpha** inscription 4× 2008, 2014, 2015, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Shungas, Satavahanas and Kharavela

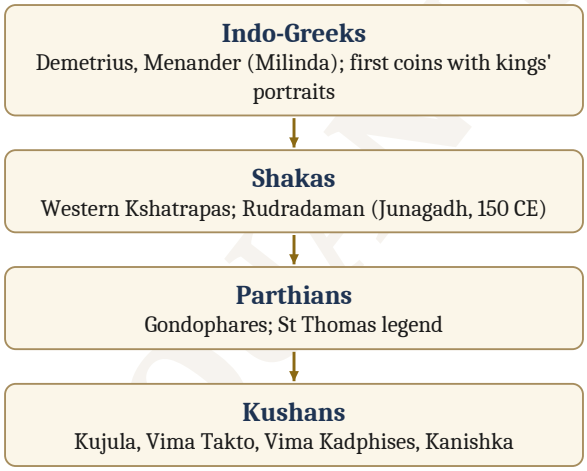
- Pushyamitra's two **Ashvamedhas**: **Ayodhya** inscription (Dhanadeva) 1× 2018
- **Patanjali** was a contemporary of **Pushyamitra Shunga** 1× 2011
- Strongest southern dynasty after the Mauryas: **Satavahana** 1× 1993
- **Simuka** founded the **Satavahana** dynasty 1× 2008
- Earliest dynasty among the options: **Satavahana** 1× 2001
- Satavahana capital: **Pratishthana** (Paithan); **Amaravati** in the 2005 options 2× 2005, 2008
- **Ekabrahmana**: **Gautamiputra Satakarni** 1× 2016
- Satavahanas promoted Sanskrit with Prakrit (A and R true, R explains A, per key) 1× 2017

- Satavahanas nurtured **Prakrit** and public art (both) 1× 2021
- "Shriparvatiya" in the Puranas: **Ikshvakus** 1× 2020
- Patron of Jainism: **Kharavela**, of the **Chedi** dynasty; **Hathigumpha** inscription 4× 2008, 2014, 2015, 2018
- Wrong pair: **Bhitari-Pulakeshin II** (Bhitari = Skandagupta) 1× 2017

10.2 Foreign Invaders

- Order: **Greeks, Sakas, Kushans** 1× 2023
- **Demetrius-Indo-Greek, Rudradaman-Scythian, Gondophares-Parthian, Vima-Kushan** 1× 2017
- Lead coins: **Strato II** 1× 2014
- Sankarshana and Vasudeva on coins of **Agathocles** 1× 2017
- Earliest **kavya**-style inscription: **Rudradaman** (Junagadh) 1× 1997
- Repaired Sudarshana lake without forced labour: **Rudradaman I** 1× 2014
- **Gandhara art-Kushans, Junagadh-Rudradaman I, Milindapanho-Menander, Taxila inscription-Patika** 1× 2017

ORDER OF INVADERS



10.3 Kanishka and the Kushans

- Sarnath image of Kanishka: **81 CE** (year 3, if he began in 78) 1× 2014
- Kanishka's coronation: **78 CE** 1× 1991
- Not in the Rabatak inscription: **Shravasti** 1× 2014
- Contemporaries: **Ashvaghosha, Vasumitra** (I and II) 1× 1994
- Ashvaghosha: contemporary of **Kanishka** 1× 2008
- Not at Kanishka's court: **Patanjali**, **Vishakhadatta** 2× 2005, 2008
- Taxila-educated physicians: **Jivaka** (and Charaka by tradition) 1× 2015
- **Takshashila**: between the **Indus and Jhelum** 1× 1996
- Most copper coins: **Kushans** 1× 2005
- First and regular gold coins: **Vima Kadphises** (Kushans) 3× 2004, 2009, 2015
- Buddha on coins of **Kanishka** 1× 2010

10.4 Coins, Eras, Trade and Art

- Wrong pair: **Satavahanas-gold** (they issued lead, potin, silver) 1× 2016
- Kardamaka Kshatrapas' rare coins: **copper** (per key) 1× 2019
- **Kartikeya** on **Yaudheya** coins 1× 2008
- **Vikram Samvat**: **57 BCE** 1× 1992
- **Saka Samvat**: **78 CE** 2× 1990, 2013
- Saka 556 = **634 CE** 1× 2002
- December 2009 = Saka **1931** 1× 2007
- Saka-Vikram gap: **135 years** 1× 2006
- Kushans traded via the Gulf and Red Sea; they had no great navy (A true, R false) 1× 1998
- **Gandhara** art: **Indian and Greek** blend 1× 1996
- **Bamiyan**: giant **Buddha statues** 1× 2008

School	Material	Patron	Feature
Gandhara	Grey schist	Kushans	Greek style, wavy hair
Mathura	Red sandstone	Kushans	Indian style; first Buddha images (also)
Amaravati	White marble	Satavahanas	Narrative reliefs

TRAP ALERT

- Vikram = CE **+ 57**; Saka = CE **- 78**; Saka date to CE: add 78.
- Satavahana coins are **lead**, not gold; gold = Kushans and Guptas.
- Order: Greeks, Shakas, Parthians, Kushans.
- Bhitari = **Skandagupta**; Hathigumpha = **Kharavela**; Ayodhya = **Pushyamitra**.

UP LINK

UP link: **Mathura** was the second Kushan capital and gave its name to the Mathura school of art; **Ayodhya** has Dhanadeva's inscription on Pushyamitra's two Ashvamedhas, and **Sarnath** has Kanishka's dated Buddha image.

LAST-MINUTE 10

1. Ayodhya inscription = Pushyamitra; Patanjali = his contemporary
2. Simuka founded the Satavahanas; Ekabrahmana = Gautamiputra
3. Kharavela = Chedi, Jain, Hathigumpha
4. Greeks, Shakas, Parthians, Kushans
5. Rudradaman = earliest kavya inscription; Sudarshana repair
6. Kanishka 78 CE; Ashvaghosha and Vasumitra at his court
7. Vima Kadphises = first regular gold coins
8. Vikram 57 BCE; Saka 78 CE; gap 135
9. Yaudheya coins = Kartikeya
10. Gandhara = Indo-Greek art; Bamiyan Buddhas

Chapter 11

Gupta and Post-Gupta Period

Built from 69 UPPSC questions (1990–2024). Focus: Gupta kings and titles, the Prayag Prashasti, Chandragupta II, coins and inscriptions, the Hunas, administration and trade, Harsha, Chinese travellers, court poets, the six schools of philosophy.

ASKED MOST · REVISE FIRST

- **Samudragupta**: **Prayag Prashasti** (Allahabad pillar, by Harishena), "**Napoleon of India**", Parakramanka **8×** 1990, 2004, 2004, 2009, 2009, 2015, 2015, 2016
- **Harsha**: capital **Kannauj**; assemblies at **Kannauj and Prayag**; defeated by **Pulakeshin II**; **Hiuen Tsang** came in his reign **7×** 1990, 1992, 1993, 2001, 2004, 2012, 2016
- **Skandagupta** beat the **Hunas** (**Bhitari** pillar); **Yashodharman** beat Mihirakula **5×** 1993, 2006, 2014, 2015, 2016
- **Six schools**: **Sankhya-Kapila, Yoga-Patanjali, Nyaya-Gautama** **9×** 1997, 1998, 2002, 2003, 2005, 2005, 2007, 2010, 2023
- **Coins**: gold **dinara**, silver **rupaka**; first coins **Chandragupta I**, first silver **Chandragupta II** **6×** 1992, 2002, 2010, 2010, 2011, 2017

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

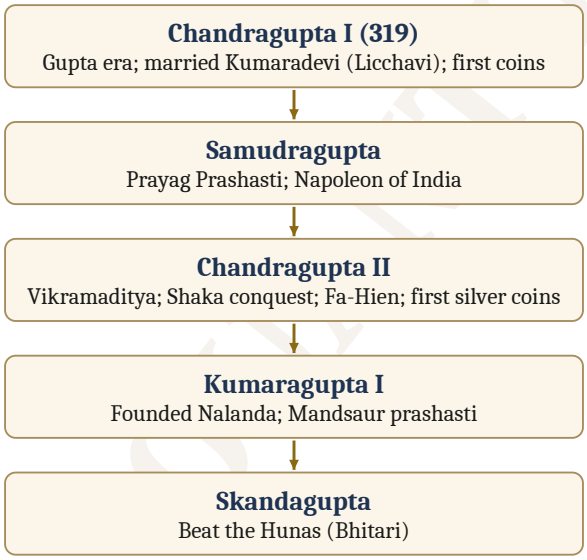
11.1 The Gupta Rulers

- **Chandragupta I-Kumaradevi, Samudragupta-Dattadevi, Chandragupta II-Kuberanaga, Kumaragupta I-Anantadevi** **1×** 2019
- **Ashoka-Priyadarshi, Samudragupta-Parakramanka, Chandragupta II-Vikramaditya, Skandagupta-Kramaditya** **1×** 2015
- **Devagupta**: **Chandragupta II** **1×** 2007
- Vikram 58 BCE, Saka 78 CE, Gupta 319 CE, Muslim rule 1192 (all four per key) **1×** 2011

11.2 Samudragupta and Chandragupta II

- Allahabad pillar: reign of **Samudragupta**; **Prayag Prashasti** = his campaigns **2×** 2004, 2004
- Also on the pillar: an inscription of **Jahangir** **1×** 2016
- **Dhananjaya-Kusthalapura, Nilaraja-Avamukta, Ugrasena-Palakka, Vishnugopa-Kanchi** **1×** 2018
- **Napoleon of India**: **Samudragupta** **2×** 1990, 2009
- First "Parama Bhagavata": **Samudragupta** (Commission accepted Chandragupta II) **2×** 2009, 2015
- "Prithivyah Pratham Veer": **Samudragupta** (question dropped) **1×** 2016
- Four Ashvamedhas: **Pravarasena I** (Vakataka) **2×** 2003, 2011
- **Shaka conqueror**: **Chandragupta II**; his silver coins (about 33 grains) prove it (both) **2×** 2010, 2022
- Delhi iron pillar: **Chandra** (Chandragupta II) **1×** 2002
- Navaratna linked with astrology: **Kshapanaka** **1×** 2008

IMPERIAL GUPTAS



11.3 Coins and Inscriptions

- Gold coin: **dinara**; silver coin: **rupaka** **3×** 1992, 2002, 2017
- First Gupta coins: **Chandragupta I**; first silver: **Chandragupta II** **3×** 2010, 2010, 2011
- **Mandsaur** prashasti: **Kumaragupta I** **1×** 2021
- Wrong pair: **Prabhavati Gupta-Udayagiri cave** **1×** 2017
- First record of sati: **Eran** (510 CE) **1×** 2010
- Most hurt by the Huna invasions: **Guptas** **1×** 1993
- Hunas came under **Skandagupta**, who defeated them (**Bhitari** pillar) **3×** 2006, 2014, 2015
- Defeated Mihirakula: **Yashodharman** **1×** 2016

11.4 Administration, Economy and Society

- Order: **bhukti, vishaya, vithi, grama** **1×** 2023
- Lawful tax: **one-sixth** of produce **1×** 1992
- Irrigation tax: **bidakabhagam** **1×** 2009
- North Indian port: **Tamralipti** **1×** 2014
- River ports and large entrepots existed (both) **1×** 2024
- **Warangal** (3rd century): **iron implements** **1×** 2001
- **Kausheya**: **silk** **1×** 2008
- "Golden Age": **Guptas** **1×** 2004
- Women equal to men: **none** of the ages **1×** 1994
- Post-Gupta feudalism: landlords and dependent peasants (A and R true, R explains A) **1×** 2004
- Gupta Ayurveda author: no option fits (*) **1×** 2002
- Chess began in **India** **1×** 2004

11.5 Harsha and the Post-Guptas

- Wrong pair: **Sarvavarman-Gaya copper plate** **1×** 2022
- Harsha's capital: **Kannauj** (moved from **Thanesar**) **2×** 1992, 1993
- Harsha's sources (key): **Kalhana** (main ones: Bana, Hiuen Tsang) **1×** 1995
- Assemblies at **Kannauj and Prayag**; Buddhist assembly in UP at **Prayag** **2×** 2001, 2004
- Harsha was defeated by **Pulakeshin II** **1×** 2016
- **Kushans-Devaputra, Pulakeshin II-Dakshinapatheshwara, Harsha-Sakalottarapathanatha** **1×** 2023
- Hiuen Tsang came under **Harsha**; his book **Si-Yu-Ki**; studied at **Nalanda** **4×** 1990, 1995, 2012, 2016

- Who sent Hiuen Tsang to Harsha: **none** (he came on his own) 1× 2007
- Chinese name for India: **Yin-tu** 1× 2013
- Travellers' dates: two pairs wrong (*) 1× 2021

Traveller	Came under	Note
Fa-Hien	Chandragupta II	399-414 CE; Fo-Kuo-Ki
Hiuen Tsang	Harsha	Si-Yu-Ki; studied at Nalanda
I-Tsing	Post-Harsha	671-695 CE; Nalanda

11.6 Poets, Religion and Philosophy

- Wrong pair: **Harishena-Harsha** (Harishena = Samudragupta) 1× 1997
- **Amir Khusrau-Alauddin, Kalidasa-Chandragupta II, Harishena-Samudragupta, Banabhatta-Harsha** 1× 2003
- Multan sun temple and Sambapur pilgrimage (both) 1× 2020
- Not true of Shankaracharya: he named **Prayag Tirtharaj** 1× 2005
- Shankaracharya: born in **Kerala**; maths at **Joshimath, Dwarka, Puri, Sringeri** 2× 1999, 2006
- Chief seat of the lunar dynasty: **Pratishthanpur** (Jhunsi, Prayagraj) 1× 2009
- **Sankhya**: **Kapila** 2× 1998, 2010
- **Yoga**: **Patanjali** 4× 1997, 2002, 2003, 2007
- **Nyaya**: **Gautama** 3× 2005, 2005, 2023
- **Nyaya-logic, Mimamsa-Vedic rites, Sankhya-real knowledge, Vedanta-self-knowledge** 1× 2019
- Theory of karma: **Mimamsa** 1× 1997
- Navya-Nyaya founder: **Gangesha** (Udayana also accepted) 1× 1995

School	Founder
Sankhya	Kapila
Yoga	Patanjali
Nyaya	Gautama (Akshapada)
Vaisheshika	Kanada
Mimamsa	Jaimini
Vedanta	Badarayana

TRAP ALERT

- Harishena wrote for **Samudragupta**, not Harsha.
- First coins = Chandragupta I; first **silver** = Chandragupta II.
- Skandagupta beat the first Hunas; **Yashodharman** beat Mihirakula.
- Four Ashvamedhas = **Pravarasena I**, not Samudragupta.

UP LINK

UP link: the **Allahabad pillar** (Prayag Prashasti, plus Ashoka and Jahangir), **Bhitari** (Ghazipur, Skandagupta's pillar), the **Deogarh Dashavatara temple** (Lalitpur), **Kannauj** as Harsha's capital, Harsha's **Prayag assembly**, and **Pratishthanpur** (Jhunsi) are all in UP.

LAST-MINUTE 10

1. Chandragupta I = Kumaradevi; first coins; Gupta era 319
2. Samudragupta = Prayag Prashasti, Napoleon, Parakramanka
3. Chandragupta II = Vikramaditya, Shaka conqueror, iron pillar, first silver
4. Kumaragupta = Mandsaur; Skandagupta = Bhitari, Hunas
5. Dinara gold; rupaka silver; Eran = first sati record
6. Bhukti, vishaya, vithi, grama
7. Harsha: Kannauj; Prayag assembly; beaten by Pulakeshin II
8. Hiuen Tsang = Si-Yu-Ki, Nalanda; Fa-Hien = Chandragupta II
9. Harishena = Samudragupta; Bana = Harsha; Kalidasa = Chandragupta II
10. Kapila Sankhya; Patanjali Yoga; Gautama Nyaya

Chapter 12

Architecture in Ancient India

Built from 54 UPPSC questions (1991–2023). Focus: Ellora and the Kailasa temple, Elephanta, Gupta temples of UP, Khajuraho, Konark and Lingaraja, Mamallapuram, Chola temples, Angkor Wat, Jain temples, site-state matching.

ASKED MOST · REVISE FIRST

- **Kailasa temple, Ellora** = rock-cut, built by the **Rashtrakutas**; Ellora has **Hindu, Buddhist and Jain** caves **14×** 1998, 1999, 2001, 2002, 2002, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2010, 2012
- **Khajuraho** = **Chandelas**; Hindu and Jain temples **7×** 1991, 1993, 1994, 1996, 2008, 2010, 2010
- **Konark Sun Temple** (Black Pagoda) = **Narasimhadeva I**; **Lingaraja** = Bhubaneswar, Nagara style **8×** 1993, 1995, 1997, 2001, 2004, 2006, 2009, 2010
- **Mamallapuram rathas** = **Pallavas** (Narasimhavarman I); smallest = **Draupadi ratha** **5×** 1994, 2007, 2011, 2014, 2015
- **Elephanta** = **Shaiva** caves, Trimurti (Maheshamurti) **5×** 2002, 2003, 2007, 2009, 2011

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

12.1 Rock-cut Caves

- Over 100 Buddhist caves: **Kanheri** **1×** 2002
- Ajanta and Ellora: **Maharashtra** **1×** 1993
- Ellora caves and Kailasa temple: **Rashtrakutas** (Chhatrapati Sambhajinagar) **8×** 1999, 2001, 2004, 2005, 2008, 2009, 2010, 2010
- Ellora: **Hindu, Buddhist and Jain** caves together **4×** 1998, 2002, 2002, 2006
- Rock-cut temple: **Kailasa, Ellora** **2×** 2007, 2012
- Rashtrakutas patronised **Jainism** (per key) **1×** 2010
- Elephanta: mainly **Shaiva** (with Buddhist remains); ascribed to the **Rashtrakutas** (per key) **4×** 2002, 2003, 2007, 2009
- **Trimurti**: **Elephanta** **1×** 2011

12.2 Stupas and Gupta Temples

- **Ayaka pillars**: **Amaravati** **1×** 2022
- **Stupa-Sanchi, painting-Bagh, headless Kanishka statue-Mathura, Dashavatara temple-Deogarh** **1×** 2017
- **Surkhi** introduced by the **Kushans** (per key) **1×** 2014
- **Brick temple-Bhitargaon, Dashavatara-Deogarh, Shiva temple-Bhumara, Vishnu temple-Eran** **1×** 2013
- **Bhitargaon** (Kanpur): 5th-6th century, one of the **oldest brick temples** (both) **1×** 2021

12.3 Temple Styles and Regional Temples

- **Gopuram**: ornate multi-storey **gateway** towers **1×** 1998
- Dravida-style pinnacle: **Teli ka Mandir, Gwalior** **1×** 2016
- Kandariya Mahadeva: **Chandelas** **1×** 1991
- Khajuraho temples: **Hindu and Jain** **1×** 1993
- **Matangeshvara**, Khajuraho: **Shiva** **1×** 2010
- Correct pair: **Khajuraho-Chandelas** **4×** 1994, 1996, 2008, 2010
- Konark: **Narasimhadeva**; "**Black Pagoda**"; the Sun temple **4×** 1993, 1995, 2006, 2009
- Largest and tallest Odisha temple: **Lingaraja**, at **Bhubaneswar** **3×** 2001, 2004, 2010
- Bhubaneswar and Puri temples: **Nagara** style **1×** 1997
- **Sun temple-Konark, Lingaraja-Bhubaneswar, Hawa Mahal-Jaipur, Gommateshwara-Karnataka** **1×** 1992

Style	Region	Mark
Nagara	North (Khajuraho, Odisha)	Curved shikhara
Dravida	South (Thanjavur, Madurai)	Pyramidal vimana, gopurams
Vesara	Deccan (Chalukyas, Hoysalas)	Mix of both

12.4 South India and Abroad

- Mamallapuram rathas: **Pallavas** (Narasimhavarman I) **4×** 1994, 2007, 2014, 2015

- Smallest ratha: **Draupadi** 1× 2011
- Order (key): **Sapta Pagoda, Shore Temple, Brihadishvara, Gangaikondacholapuram** 1× 2018
- Biggest Nandi (key): **Brihadishvara**, Thanjavur (Lepakshi also claimed) 1× 1999
- Brihadishvara built by **Rajaraja I** 1× 2008
- **Angkor Wat**: **Cambodia**; begun under **Suryavarman II** 4× 1992, 1993, 1995, 2006
- **Borobudur**: **Java** 1× 2013

12.5 Jain Temples, Sites and Pilgrimages

- Dilwara (Abu) Jain temples: **marble**; at **Mount Abu, Rajasthan** 3× 2010, one-liner, one-liner
- **Sonagiri** Jain temples: near **Datia, MP** 2× 1996, 2005
- **Elephanta-cave, Shravanabelagola-statue, Khajuraho-temple, Sanchi-stupa** 1× 2001
- **Hampi-Karnataka, Nagarjunakonda-Andhra, Shishupalgarh-Odisha, Arikamedu-Puducherry** 1× 2014
- **Bhimbetka-MP, Shore temple-TN, Hampi-Karnataka, Manas-Assam** 1× 2010
- **Baidyanath-Shiva, Sarnath-first sermon, Dilwara-Jain, Badrinath-Vishnu** 1× 2004
- Brahma temple of **Pushkar**: **Rajasthan** 1× 2023
- **Naimisharanya**: **Sitapur** 1× 2002

TRAP ALERT

- Kailasa (Ellora) = **Rashtrakutas** (Krishna I); not Chalukya or Pallava.
- Shore Temple is **structural**, not rock-cut; the rathas are rock-cut.
- Brihadishvara = **Rajaraja I**; Gangaikondacholapuram = Rajendra I.
- Konark = Narasimhadeva I (Eastern Ganga); Sonagiri = Datia (MP), not Jhansi.

UP LINK

UP link: the oldest surviving Gupta temples are in UP: the **Dashavatara temple, Deogarh** (Lalitpur) and the brick **Bhitargaon temple** (Kanpur); **Sarnath** and **Mathura** show early sculpture, and **Naimisharanya** (Sitapur) is a major pilgrimage.

LAST-MINUTE 10

1. Kanheri = 100+ caves; Ajanta, Ellora = Maharashtra
2. Ellora (three faiths), Kailasa = Rashtrakutas
3. Elephanta = Shaiva, Trimurti
4. Bhitargaon = brick (Kanpur); Deogarh = Dashavatara (Lalitpur)
5. Khajuraho = Chandelas; Kandariya = Shiva
6. Konark = Black Pagoda (Narasimhadeva); Lingaraja = Bhubaneswar
7. Mamallapuram rathas = Pallavas; Draupadi ratha smallest
8. Brihadishvara = Rajaraja I; Angkor Wat = Cambodia; Borobudur = Java
9. Dilwara = marble, Mount Abu; Sonagiri = Datia
10. Pushkar = Rajasthan; Naimisharanya = Sitapur

Chapter 13

South India: Chola, Chalukya, Pallava and Sangam Period

Built from 47 UPPSC questions (1991–2025). Focus: southern dynasties and capitals, Sangam literature and ports, Arikamedu, Pallavas, Pulakeshin II and Aihole, Rashtrakutas, Rajaraja I and Rajendra I, Chola village government, Nataraja bronzes.

ASKED MOST · REVISE FIRST

- **Rajendra I**: Bay of Bengal a "**Chola lake**", founded **Gangaikondacholapuram**, built the **Cholagangam** tank; **Rajaraja I** took northern Sri Lanka **5×** 2008, 2008, 2014, 2016, 2023
- **Arikamedu** (Poduke) = Roman trade and settlement; amphora jars **6×** 1997, 2001, 2002, 2003, 2009, 2014
- **Chola** special feature: **village autonomy**; famed for **naval power**; Nataraja bronze = **Chola art** **5×** 1991, 1992, 1995, 2004, 2006
- **Chalukyas of Badami**: capital **Vatapi**; greatest **Pulakeshin II**; **Aihole** inscription mentions **Kalidasa** **4×** 1991, 1991, 2004, 2013
- **Sangam**: third Sangam at **Madurai**; **Tolkappiyam** = grammar; Silappadikaram by **Ilango**; Pallavas were **not** a Sangam dynasty **5×** 1997, 2002, 2006, 2010, 2014

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

13.1 Dynasties and Capitals

- **Mahendravarman I-Pallava, Kadungon-Pandya, Amoghavarsha I-Rashtrakuta, Rajaraja I-Chola** **1×** 2025
- **Chalukyas-Badami, Pallavas-Kanchipuram, Harsha-Kannauj, Pandyas-Madurai** **1×** 2004

Dynasty	Capital
Chalukyas of Badami	Vatapi (Badami)
Eastern Chalukyas	Vengi
Pallavas	Kanchipuram
Rashtrakutas	Manyakheta
Cholas	Thanjavur, then Gangaikondacholapuram
Pandyas	Madurai
Kadambas	Vanavasi
Kakatiyas	Warangal

13.2 The Sangam Age

- Third Sangam: **Madurai** **1×** 2006
- Not a Sangam dynasty: **Pallava** **1×** 2010
- Pandya lifeline (key printed "Vengi"): the **Vaigai** **1×** 2010
- **Meenakshi** temple: **Madurai** **3×** 1991, 1992, 2004
- **Tolkappiyam**: **Tamil grammar** **2×** 1997, 2014
- **Silappadikaram**: **Ilango Adigal** **1×** 2002
- **Tirukkural-philosophy, Tolkappiyam-grammar, Silappadikaram-love story, Manimekalai-merchant's tale** **1×** 2014
- "Laghuveda": **Kural** **1×** 2013
- Tamil Ramayana (Ramavataram): **Kamban** **1×** 2010
- Roman trade: **Arikamedu** (Periplus' **Poduke**), a **Roman settlement** **5×** 1997, 2001, 2002, 2003, 2009
- **Amphora**: tall **two-handled jar** **1×** 2014
- West-coast Sangam ports: **Tondi and Muziris** (3 and 4) **1×** 2012

13.3 Pallavas, Chalukyas and Rashtrakutas

- Pallava order: **Mahendravarman I, Narasimhavarman I, Parameshvaravarman I, Nandivarman II** **1×** 2019
- Two naval expeditions to Ceylon (642): **Narasimhavarman I** **1×** 2016
- Chalukya capital: **Vatapi**; greatest king: **Pulakeshin II** **2×** 1991, 1991
- Kalidasa named in Pulakeshin II's **Aihole inscription** **1×** 2013
- Gave women high office: **Chalukyas** **1×** 2007
- China-India relations under the Chalukyas: **Ma Tuan-lin** **1×** 2014
- Sent ministers to stop cruelty to animals in Kashi: **Kumarapala** (Gujarat Chalukya) **1×** 2016
- **Kadamba** capital: **Vanavasi** **1×** 2005
- **Takkolam** (949): **Cholas vs Rashtrakutas** **1×** 2001

13.4 The Imperial Cholas

- Chola empire: **Coromandel coast and part of the Deccan** **1×** 1991
- Capital: **Thanjavur** **1×** 2009
- More known about Cholas: their rule left many records (A and R true, R explains A) **1×** 2020
- Naval power: **Cholas** **2×** 1992, 2004

- Northern Sri Lanka conquered by **Rajaraja I**; he also beat the Chera navy at Kandalur **2×** 2014, 2023
- "Chola lake", **Gangaikondacholapuram**, **Cholagangam**: **Rajendra I** **3×** 2008, 2008, 2016
- Freed Sri Lanka and married his daughter to a Sinhala prince: **Kulottunga I** **1×** 2012
- Heir named in the king's lifetime: unclear (*) **1×** 2001
- Who ended the Cholas: unclear (*); the Pandyas and Hoysalas **1×** 2004
- Special feature: **village self-government** **2×** 1991, 1995
- Garden committee: **Thotta Variyam** **1×** 2013
- Overseas guilds: **Nanadesis, Manigramam** **1×** 2018
- 72 merchants to China: **Kulottunga I** **1×** 1992
- **Nataraja** bronze: **Chola art** **1×** 2006
- Nataraja: four hands; a man's earring right, a woman's left (both) **1×** 2021
- **Dakshinamurti**: Shiva as **teacher** **1×** 2013
- **Rudramadevi** (Kakatiya): fourth ruler; beat Pandyas, Gangas and Yadavas (both) **1×** 2023

CHOLA GREATNESS



TRAP ALERT

- Pallavas are **post-Sangam**; the Sangam kingdoms are Chera, Chola, Pandya.
- Northern Sri Lanka = **Rajaraja I**; whole island and Ganga raid = **Rajendra I**.
- Kalidasa is named in the **Aihole** inscription (Ravikirti) of Pulakeshin II.
- Takkolam was a **Chola defeat** by the Rashtrakuta Krishna III.

UP LINK

UP link: Rajendra I's northern campaign reached the **Ganga** (in Bengal), and Kumarapala's ministers acted in **Kashi**; Varanasi also remained the great learning centre that southern scholars like Shankaracharya visited.

LAST-MINUTE 10

1. Mahendravarman = Pallava; Amoghavarsha = Rashtrakuta
2. Third Sangam = Madurai; Tolkappiyam = grammar
3. Silappadikaram = Ilango; Kural = Laghuveda; Kamban = Tamil Ramayana
4. Arikamedu = Poduke, Roman trade
5. Narasimhavarman I = Ceylon expeditions, Mamalla
6. Vatapi; Pulakeshin II; Aihole names Kalidasa
7. Takkolam = Chola vs Rashtrakuta
8. Rajaraja I = north Sri Lanka; Rajendra I = Chola lake
9. Chola = village autonomy; Thotta Variyam = gardens
10. Nataraja = Chola bronze; Dakshinamurti = teacher

Chapter 14

Pre-Medieval Period

Built from 32 UPPSC questions (1994–2024). Focus: the tripartite struggle for Kannauj, Palas and Vikramashila, Senas, Gurjara-Pratiharas, Rashtrakutas, Chauhans, Chandelas, Gahadavalas, Paramara Bhoja, Solankis, jurists and travellers.

ASKED MOST · REVISE FIRST

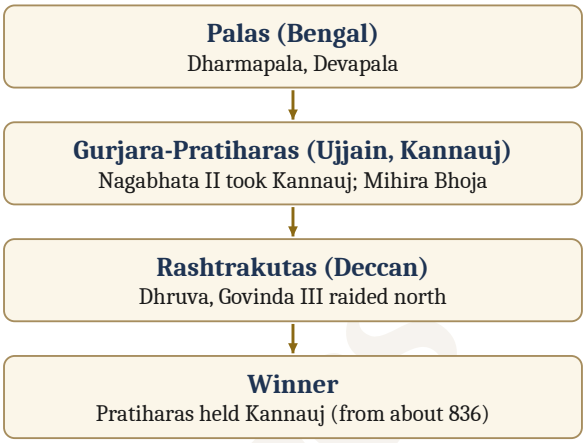
- **Vikramashila** (and Somapura) founded by the Pala **Dharmapala** 9× 1994, 1999, 2002, 2004, 2005, 2008, 2008, 2015, 2016
- **Tripartite struggle** for **Kannauj**: **Palas, Gurjara-Pratiharas, Rashtrakutas** 3× 2002, 2015, one-liner
- **Gurjara-Pratiharas**: founder **Nagabhata I**; greatest **Mihira Bhoja** ("Adivaraha") 5× 2002, 2003, 2010, 2017, 2017
- **Gopala** (Pala founder) was chosen by the people (with Nandivarman Pallavamalla) 2× 2015, 2021
- **Jejakabhukti** = **Bundelkhand** (Chandelas), not Kaushambi 3× 2008, 2017, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Kannauj, Palas and Senas

- Tripartite struggle: **Palas, Gurjaras, Rashtrakutas** (2, 3, 4) 2× 2002, 2015
- "Nagar Mahoday Shri": **Kannauj** 1× 2015
- **Gaudavaho**: **Vakpati** (court of Yashovarman of Kannauj) 1× 2014
- Pala founder: **Gopala** 1× 2015
- Chosen by the people: **Gopala and Nandivarman Pallavamalla** (2 and 4) 1× 2021
- **Vikramashila**: **Dharmapala** (Pala) 8× 1994, 2002, 2004, 2005, 2008, 2008, 2015, 2016
- **Somapura Mahavihara**: **Dharmapala** 1× 1999
- Sena order: **Hemanta, Vijaya, Ballala, Lakshmana** 1× 2024
- New Samvat (Lakshmana era): **Lakshmana Sena** (Senas) 2× 1999, 2005
- **Pundravardhana bhukti**: **North Bengal** 1× 2012

TRIPARTITE STRUGGLE FOR KANNAUJ



14.2 Pratiharas, Rashtrakutas and Rajputs

- Pratihara founder: **Nagabhata I** 2× 2002, 2003
- Greatest Pratihara: **Mihira Bhoja**, titled **Adivaraha** 2× 2010, 2017
- Order: **Vatsaraja, Nagabhata II, Mahendrapala, Mahipala** 1× 2017
- **Hiranyagarbha** ritual: **Dantidurga** (Rashtrakuta) 1× 2015
- Born in a military camp: **Amoghavarsha** 1× 2013
- "Last sun of Indian glory": **Prithviraj** 1× 2016
- Prithviraj Chauhan = **Prithviraja III** 1× 2010
- Wrong pair: **Abdur Rahman-Hammir Raso** 1× 2010

Raso	Poet
Prithviraj Raso	Chand Bardai
Alha Khand	Jagnik
Bisaldev Raso	Narpati Nalha
Hammir Raso	Sharangadhar (Jodhraj)

14.3 Chandelas, Gahadavalas, Paramaras, Solankis

- **Jejakabhukti**: **Bundelkhand** 1× 2008
- Wrong pair: **Jejakabhukti-Kaushambi** 1× 2021
- Chandela order: **Yashovarman, Dhanga, Vidyadhara, Kirtivarman** 1× 2017
- Gahadavala queen **Kumaradevi** built the Dharmachakra Jina Vihara at **Sarnath** 1× 2007
- Udaipur Prashasti praise: **Bhoja** (Paramara) 1× 2023

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- Book on mechanical devices: **Bhoja** (Samarangana Sutradhara) 1× 2013
- **Bhojshala** deity: **Saraswati** 1× 2001
- **Hemachandra** at **Kumarapala's** court 1× 2010
- Aid to rebuild a Khambhat mosque: **Jayasimha Siddharaja** 1× 1999

14.4 Jurists and Travellers

- Jurists: **Vijnaneshvara, Hemadri, Jimutavahana** (1, 2, 4) 1× 1995
- Order: **Fa-Hien, Hiuen Tsang, I-Tsing, Al-Biruni** 1× 2024

Jurist	Work
Vijnaneshvara	Mitakshara (Chalukya court)
Jimutavahana	Dayabhaga (Bengal)
Hemadri	Chaturvarga Chintamani (Yadavas)

TRAP ALERT

- Mihira Bhoja (Pratihara, Adivaraha) is **not** Bhoja Paramara (Dhar, Udaipur Prashasti).
- Vikramashila and Somapura = Dharmapala; Nalanda = **Kumaragupta I**.
- Jejakabhukti is a **region** (Bundelkhand), not a city.
- The Rashtrakutas raided Kannauj but never held it.

UP LINK

UP link: **Kannauj** was the prize of the tripartite struggle and the Pratihara capital; **Kalinjar** and **Mahoba** were Chandela strongholds in Bundelkhand; the Gahadavalas ruled from **Kannauj and Varanasi**, and Kumaradevi's vihara stands at **Sarnath**.

LAST-MINUTE 10

1. Kannauj = tripartite prize (Pala, Pratihara, Rashtrakuta)
2. Gopala elected; Dharmapala = Vikramashila, Somapura
3. Lakshmana Sena = Lakshmana era
4. Nagabhata I founded the Pratiharas; Mihira Bhoja = Adivaraha
5. Dantidurga = Hiranyagarbha; Amoghavarsha born in camp
6. Prithviraj III = last sun of Indian glory
7. Jejakabhukti = Bundelkhand (Chandelas)
8. Kumaradevi = Sarnath vihara (Gahadavala)
9. Bhoja Paramara = Bhojshala, Samarangana Sutradhara
10. Hemachandra = Kumarapala's court

Chapter 15

Ancient Literature and Litterateurs

Built from 39 UPPSC questions (1992–2025). Focus: author-book matching for grammar, drama, poetry, history, law, medicine, mathematics and music; Kalidasa, Harsha, Kalhana, Jayadeva, Aryabhata and Bhaskara.

ASKED MOST · REVISE FIRST

- **Author-book matching**: **Panini-Ashtadhyayi, Patanjali-Mahabhashya, Yaska-Nirukta, Katyayana-Varttika, Vatsyayana-Kamasutra, Chanakya-Arthashastra, Kalhana-Rajatarangini** 3× 1992, 2010, 2017
- **Harsha** wrote **Priyadarshika, Nagananda, Ratnavali**; Karpuramanjari is **Rajashekhara's** 3× 2006, 2012, 2014
- **Kalidasa**: Meghaduta, Raghuvamsha, **Malavikagnimitra** (hero **Agnimitra**); not Shringarashataka 2× 1998, 2025
- **Kalhana's Rajatarangini**: **eight** tarangas; continued by **Jonaraja and Srivara** 3× 2000, 2010, 2015
- **Bhaskara II**: algebra, **Lilavati**; **Aryabhata**: place value; zero by **Indians** 5× 2002, 2002, 2014, 2015, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Epics, Grammar and Classics

- Vishnu Purana on the Mauryas; Vayu Purana on the Guptas (both) 1× 2023
- "Thy right is to work only...": **Gita** 1× 1992
- "What is here is found elsewhere...": **Mahabharata** 1× 1992
- **Panini-Ashtadhyayi, Vatsyayana-Kamasutra, Chanakya-Arthashastra, Kalhana-Rajatarangini** 1× 1992
- **Ashtadhyayi**: **Panini** 1× 2010
- **Ashtadhyayi-Panini, Mahabhashya-Patanjali, Nirukta-Yaska, Varttika-Katyayana** 1× 2017
- Gupta-age works: **Amarakosha, Kamasutra, Meghaduta, Mudrarakshasa** (all, per key) 1× 2009
- Not by Kalidasa: **Shringarashataka** (Bhartrihari) 1× 2025
- Hero of Malavikagnimitra: **Agnimitra** 1× 1998

15.2 Dramatists and Prose Writers

- Harsha's works: **Priyadarshika, Nagananda, Ratnavali** (1, 2, 4) 1× 2014
- Wrong pair: **Karpuramanjari-Harsha** 1× 2012
- **Bharavi-Kiratarjuniya, Harsha-Nagananda, Kalidasa-Malavikagnimitra, Rajashekhara-Karpuramanjari** 1× 2006
- **Mudrarakshasa**: **Vishakhadatta** 1× 1992
- Wrong pair: **Subandhu-Kadambari** (Kadambari = Bana) 1× 2006
- **Dashakumaracharita**: **Dandin** 2× 2005, 2008
- **Mattavilasa Prahasana**: **Mahendravarman** (Pallava) 1× 2010
- **Amir Khusrau-Alauddin, Kalidasa-Chandragupta II, Harishena-Samudragupta, Banabhatta-Harsha** 1× 2002
- **Jayadeva** (Gita Govinda) at **Lakshmana Sena's** court 2× 1999, 2010

Work	Author
Mudrarakshasa	Vishakhadatta
Mrichchhakatika	Shudraka
Kadambari, Harshacharita	Banabhatta
Vasavadatta	Subandhu
Dashakumaracharita	Dandin
Kiratarjuniya	Bharavi
Karpuramanjari	Rajashekhara
Gita Govinda	Jayadeva

15.3 History, Stories and Law

- Kalhana's contemporary king: **Jayasimha** (Kashmir) 1× 2010
- Rajatarangini: **eight** tarangas 1× 2015
- Continued by **Jonaraja and Srivara** 1× 2000
- **Trivikrama Bhatta-Nala Champu, Somadeva-Kathasaritsagara, Jayadeva-Gita Govinda, Kshemendra-Brihatkathamanjari** 1× 2025
- **Manusmriti**: mainly the **social system** 1× 2007
- First law-giver: **Manu** 2× 2004, 2008
- **Sarvavarma-Katantra, Shudraka-Mrichchhakatika, Vijnaneshvara-Mitakshara, Kalhana-Rajatarangini** 2× 2002, 2003

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- **Devana Bhatta-Smriti Chandrika, Hemadri-Chaturvarga Chintamani, Madhavacharya-Parashara Madhaviya, Ballala Sena-Dana Sagara** 1× 2024

15.4 Medicine, Mathematics, Foreign Accounts and Music

- **Charaka Samhita**: **120 chapters, 8 sections** 1× 2023
- **Ashtanga Sangraha-medicine, Dasharupaka-drama, Lilavati-maths, Mahabhashya-grammar** 1× 2002
- **Zero**: invented by **Indians** 1× 2015
- Decimal place value: **Aryabhata** (key) 1× 2014
- Algebra: **Bhaskara** 2× 2002, 2002
- **Lilavati**: **Bhaskaracharya** 1× 2017
- **Panchasiddhantika** (Varahamihira): based on **Greek astronomy** (in part) 1× 2014
- **Milindapanha**: a **Pali** text 1× 1996
- Wrong pair: **Natural History-Ptolemy** (it is **Pliny**) 1× 2015
- "Bible" of North Indian music: **Natyashastra** 1× 2019
- Oldest instrument among options: **veena** 1× 2010
- **Ragamala-Pundarik Vitthal, Rasakaumudi-Srikantha, Ragavibodha-Somanatha, Chaturdandi Prakashika-Venkatamakhi** 1× 2022

Scientist	Work
Aryabhata	Aryabhatiya (499 CE); place value, earth's rotation
Varahamihira	Panchasiddhantika, Brihat Samhita
Brahmagupta	Brahmasphutasiddhanta; rules of zero
Bhaskara II	Lilavati, Bijaganita
Charaka	Charaka Samhita (medicine)
Sushruta	Sushruta Samhita (surgery)

TRAP ALERT

- Harsha wrote three plays; **Harshacharita** is Bana's and **Karpuramanjari** is Rajashekhara's.
- **Katyayana** wrote the Varttika; **Patanjali** the Mahabhashya.
- Kadambari = **Bana**; Vasavadatta = **Subandhu**.
- Natural History = **Pliny**; Geography = **Ptolemy**.

UP LINK

UP link: **Banabhatta** wrote the Harshacharita at Harsha's court in **Kannauj**; **Varanasi** remained the chief seat of Sanskrit learning, and **Ayodhya** and **Naimisharanya** are the settings of epic and Puranic narration.

LAST-MINUTE 10

1. Panini = Ashtadhyayi; Patanjali = Mahabhashya; Yaska = Nirukta
2. Vatsyayana = Kamasutra; Kalhana = Rajatarangini (8 tarangas)
3. Kalidasa = Malavikagnimitra (Agnimitra), Meghaduta
4. Harsha = Priyadarshika, Nagananda, Ratnavali
5. Vishakhadatta = Mudrarakshasa; Dandin = Dashakumaracharita
6. Mahendravarman = Mattavilasa Prahasana
7. Jayadeva = Gita Govinda (Lakshmana Sena)
8. Vijnaneshvara = Mitakshara; Manu = first law-giver
9. Bhaskara = Lilavati, algebra; Aryabhata = place value
10. Natyashastra = bible of music; Pliny = Natural History

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TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Medieval History

PYQ-Based Revision Pointers

Chapters 1 to 25

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
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Chapter 1

Muslim Invasion of India

Built from 22 UPPSC questions (1990–2024). Focus: the Prophet and the Arabs in Sindh, Mahmud of Ghazni and his court scholars, Muhammad Ghori's battles, Bakhtiyar Khalji, and why the Turks won.

ASKED MOST · REVISE FIRST

- **First Muslim invaders**: the **Arabs**; **Muhammad bin Qasim** (an Arab) took Sindh in **712 CE** 4× 1991, 1992, 2002, 2015
- **Muhammad Ghori**: beaten first by **Mularaja II** (Kasahrada, 1178); won **Second Tarain (1192)**; beat **Jayachandra at Chandawar (1194)** 5× 1990, 1995, 2004, 2008, 2016
- **Mahmud's court**: historian **Utbi**; **Al-Biruni** first Muslim to study the Puranas (came in the 11th century) 4× 1991, 2002, 2010, 2014
- **Bakhtiyar Khalji** conquered **Bihar and Bengal** for Ghori 2× 1991, 2018
- Turks won because North India had **no political unity** (A and R true, R explains A) 2× 2018, 2024

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

1.1 The Arabs in Sindh

- Prophet Muhammad born: **570 CE** (Hijra 622) 1× 1996
- First Muslim invaders: **Arabs** 1× 2015
- Muhammad bin Qasim conquered Sindh: **712 CE** 1× 1991
- First Muslim invader: **Muhammad bin Qasim**, an **Arab** 2× 1992, 2002

1.2 Mahmud of Ghazni

- Ghaznavid founder: **Alptigin** (Sabuktigin in many books) 1× 2015
- Mahmud raided **17 times**, but for **wealth**, not empire (A true, R false) 1× 2004
- Chandela king not defeated by Mahmud (key): **Vidyadhara** 1× 1991
- Silver coins with a Sanskrit legend: **Mahmud of Ghazni** 1× 2000
- Court historian: **Utbi** 1× 1991
- Shahnama author is **Firdausi** (Mahmud's court); the question named Firishta (*) 1× 2013
- First Muslim to study the Puranas: **Al-Biruni** 1× 2002
- Al-Biruni came in the **11th century** 1× 2010
- Not correct of Al-Biruni: a "**secular author**" 1× 2014

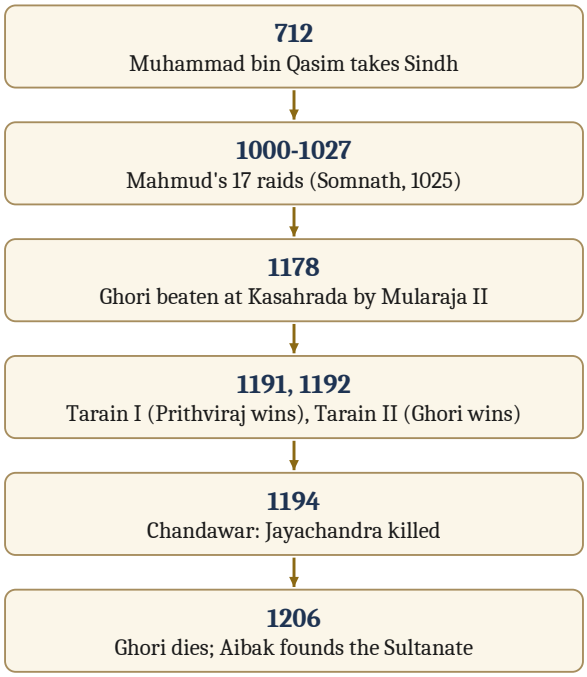
Scholar	Work
Al-Biruni	Kitab-ul-Hind (Tahqiq-i-Hind)
Utbi	Tarikh-i-Yamini
Firdausi	Shahnama

1.3 Muhammad Ghori

- Conquered North India in 1192: **Shihabuddin Muhammad Ghori** 1× 2004
- Battle that founded Muslim rule: **Second Battle of Tarain** 1× 1995
- First defeated Ghori: **Mularaja II** (Kasahrada, 1178) 1× 1990
- **Chandawar (1194)**: Ghori beat **Jayachandra** 2× 2008, 2016
- Bihar and Bengal conquered by **Bakhtiyar Khalji** 2× 1991, 2018
- Order: **Mahmud, Ghori, Genghis Khan, Timur** 1× 2008
- Turkish success: North India lacked **political unity** (A and R true, R explains A) 2× 2018, 2024

QUANTUM IAS & PCS

INVASIONS FROM THE NORTH-WEST



TRAP ALERT

- Hijra = **622**; Prophet's birth = **570**.
- Ghori's first defeat was by **Mularaja II**, not Prithviraj.
- Chandawar (1194) is not the Battle of Kannauj (1540, Sher Shah vs Humayun).
- Shahnama = **Firdausi**, not Firishta.

UP LINK

UP link: **Chandawar** (Firozabad, on the Yamuna) saw Ghori defeat **Jayachandra Gahadavala** of **Kannauj** in 1194; Mahmud raided **Mathura** and **Kannauj** (1018).

LAST-MINUTE 10

1. Prophet born 570; Hijra 622
2. Arabs first; Muhammad bin Qasim 712 (Sindh)
3. Alptigin founded the Ghaznavids
4. Mahmud: 17 raids for wealth
5. Utbi = court historian; Firdausi = Shahnama
6. Al-Biruni = Puranas, 11th century
7. Mularaja II beat Ghori first (1178)
8. Second Tarain 1192 = foundation of Muslim rule
9. Chandawar 1194 = Jayachandra
10. Bakhtiyar Khalji = Bihar and Bengal

Chapter 2

Delhi Sultanate: Slave Dynasty

Built from 14 UPPSC questions (1990–2024). Focus: Aibak and Lahore, Iltutmish and Delhi, Genghis Khan on the Indus, Razia and the Turkish nobles, Balban's kingship.

ASKED MOST · REVISE FIRST

- **Iltutmish** made **Delhi** the capital; "slave of a slave"; Genghis Khan came in his reign **10×** 1993, 2002, 2004, 2007, 2010, 2012, 2014, 2016, 2016, 2016
- **Balban**: "**blood and iron**"; centralised power **4×** 2009, 2013, 2014, 2024
- **Razia Sultan**: first woman ruler; removed by the **Turkish** nobles **3×** 2004, 2004, 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

2.1 Aibak and Iltutmish

- Founder of the Slave dynasty: **Qutbuddin Aibak** **1×** 1990
- Aibak's capital: **Lahore** **1×** 1990
- Made Delhi the capital: **Iltutmish** **5×** 2004, 2012, 2014, 2016, 2016
- First Muslim ruler of Delhi (key): **Iltutmish** (many books say Aibak) **1×** 2002
- "**Slave of a slave**": **Iltutmish** **1×** 2016
- Genghis Khan reached the Indus under **Iltutmish** **3×** 1993, 2007, 2010
- Genghis Khan's original name: **Temujin** **1×** 2015

2.2 Razia and Balban

- First woman ruler: **Razia Sultan** **2×** 2004, 2010
- Razia was overthrown by the **Turkish** nobles (the Forty) **1×** 2004
- "**Blood and iron**": **Balban** **1×** 2009
- Balban centralised power; the reason given does not explain it (A and R true, R does not explain A) **2×** 2013, 2024

- Called himself "helper of the Caliph" at **Garhmukteshwar**: **Balban** 1× 2014

SLAVE (MAMLUK) DYNASTY

Qutbuddin Aibak (1206-10)
Lahore capital; Lakh Bakhsh; Qutb Minar begun

↓

Iltutmish (1211-36)
Delhi capital; iqta; tanka and jital; Chahalgani

↓

Razia (1236-40)
First woman ruler

↓

Balban (1266-87)
Sijda and paibos; blood and iron; broke the Forty

TRAP ALERT

- Aibak's capital was **Lahore**; Delhi became the capital under **Iltutmish**.
- Genghis Khan came under **Iltutmish** (1221), not Alauddin.
- Razia was removed by **Turkish**, not Afghan, nobles.

UP LINK

UP link: **Garhmukteshwar** (Hapur, on the Ganga) has Balban's inscription calling himself helper of the Caliph; **Badaun** was Iltutmish's iqta before he became Sultan.

LAST-MINUTE 10

1. Aibak = founder; capital Lahore
2. Iltutmish = Delhi capital; slave of a slave
3. Genghis Khan (Temujin) came under Iltutmish
4. Razia = first woman ruler; ousted by Turks
5. Balban = blood and iron; helper of the Caliph
6. Chahalgani = the Forty (Iltutmish); Balban broke them
7. Iltutmish = silver tanka, copper jital
8. Aibak = Lakh Bakhsh
9. Balban = sijda, paibos
10. Slave dynasty 1206-1290

Chapter 3

Khalji Dynasty

Built from 15 UPPSC questions (1992–2022). Focus: Alauddin's titles, conquests and their order, Malik Kafur's Deccan raids, the Mongol frontier, land revenue by measurement, new taxes and market control.

ASKED MOST · REVISE FIRST

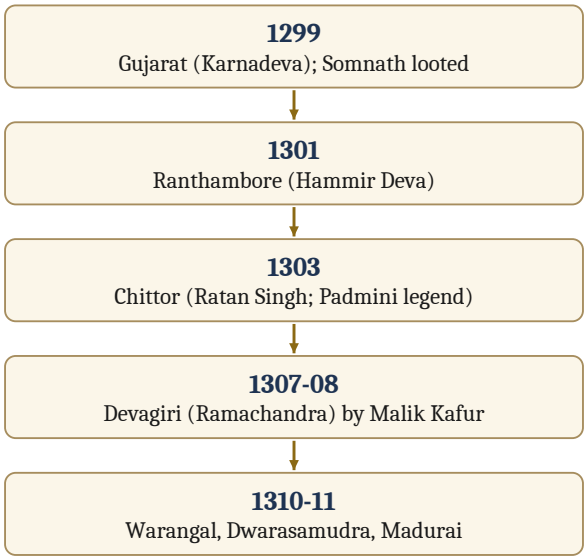
- ■ **Alauddin Khalji's market control** (price control, an early "public distribution system"), highly successful in Delhi 7× 1992, 1998, 2004, 2005, 2010, 2010, 2017
- **Sikandar-i-Sani** (second Alexander) = **Alauddin Khalji**, not Sikandar Lodi 2× 2008, one-liner
- Conquest order: **Gujarat, Ranthambore, Chittor, Warangal** 2× 2022, one-liner
- Alauddin first levied the **ghari** (house tax) and measured land for revenue 2× 1994, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Titles and Conquests

- Title "Alexander the Second": **Alauddin Khalji** 1× 2008
- "Yamin-ul-Khilafat", "Nasir-i-Amir-ul-Mominin": **Alauddin Khalji** 1× one-liner
- Declared himself **Khalifa**: **Mubarak Khalji** 1× 2016
- Order: **Gujarat, Ranthambore, Chittor, Warangal** 1× 2022
- Order (with Jaisalmer): **Jaisalmer, Ranthambore, Chittor, Warangal** 1× one-liner
- **Ranthambore-Hammir Deva, Chittor-Ratan Singh, Devagiri-Ramachandra, Gujarat-Karnadeva** 1× 2017
- Deccan raids were for **wealth**, not annexation (A true, R false) 1× 1997

ALAUDDIN'S CAMPAIGNS



3.2 Army, Revenue and Markets

- Frontier with the Mongols after 1306: the **Ravi** 1× 2014
- Special frontier army: **Alauddin Khalji** 1× 2021
- Revenue by **measuring land** and curbing governors (I and III) 1× 1994
- First **ghari** (house tax): **Alauddin Khalji** 1× 2011
- Price control in Delhi; it was not to cut artisans' wages (A true, R false) 1× 2005
- Strict price control: **Alauddin Khalji** 4× 1992, 1998, 2004, 2010
- Market control was **highly successful** 1× 2017
- Began a "public distribution system": **Alauddin Khalji** 1× 2010

Reform	Detail
Dagh and huliya	Branding horses, descriptive rolls of soldiers
Standing army	Paid in cash
Land revenue	Half the produce; based on measurement
Taxes	Ghari (house), charai (grazing)
Market control	Shahna-i-Mandi; fixed prices in Delhi

TRAP ALERT

- Sikandar-i-Sani = **Alauddin**, despite Sikandar Lodi's name.
- Ghari (house) and charai = Alauddin; haq-i-sharb (irrigation) = **Firuz**.
- Deccan campaigns sought **tribute**, not annexation.
- Order always ends Chittor (1303), then Warangal (1310).

UP LINK

UP link: Alauddin seized power by murdering his uncle **Jalaluddin Khalji at Kara** (Kara-Manikpur, Kaushambi-Pratapgarh) in 1296, funding his rise with the loot of Devagiri.

LAST-MINUTE 10

1. Alauddin = Sikandar-i-Sani
2. Mubarak Khalji called himself Khalifa
3. Gujarat, Ranthambore, Chittor, Warangal
4. Hammir Deva = Ranthambore; Ratan Singh = Chittor
5. Ramachandra = Devagiri; Karnadeva = Gujarat
6. Deccan raids = wealth, not annexation
7. Mongol frontier = Ravi
8. Ghari = house tax; land measured
9. Market control in Delhi; Shahna-i-Mandi
10. First "public distribution system"

Chapter 4

Tughluq Dynasty

Built from 20 UPPSC questions (1991–2017). Focus: Ghiyasuddin, Muhammad bin Tughluq's projects and Ibn Battuta, Firuz Shah's welfare state and taxes, the Ashokan pillars, Timur's invasion.

ASKED MOST · REVISE FIRST

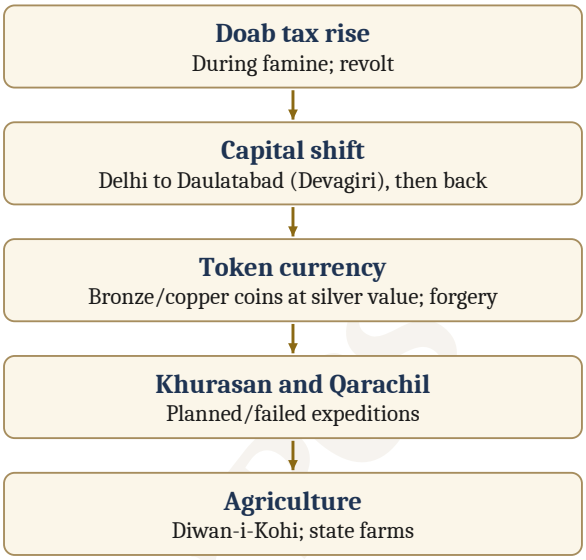
- **Muhammad bin Tughluq**: capital to **Daulatabad**, first **token currency**, the most learned Sultan; **Ibn Battuta** came in his reign **10×** 1991, 1994, 2002, 2004, 2004, 2004, 2006, 2011, 2012, 2017
- **Firuz Shah Tughluq**: employment bureau, public works, translation, Hajj, **Dar-ul-Shifa** **7×** 1998, 2000, 2004, 2006, 2010, 2010, one-liner
- **Haq-i-sharb** (irrigation tax) and **jizya on Brahmins**: **Firuz** **4×** 1996, 2008, 2010, 2011
- Firuz brought **Ashokan pillars** from **Topra and Meerut** to Delhi **2×** 1996, 2009

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 Ghiyasuddin and Muhammad bin Tughluq

- Founder: **Ghazi Malik** (Ghiyasuddin Tughluq) **1×** 1999
- First 14th-century canal builder: **Ghiyasuddin Tughluq** **1×** 2017
- Longest-ruling Sultanate dynasty: **Tughluqs** **1×** 2008
- Most learned Sultan: **Muhammad bin Tughluq** **1×** 2012
- Capital moved to **Daulatabad** **2×** 2002, 2004
- First **token currency**: **Muhammad bin Tughluq** **3×** 2004, 2004, 2017
- Token currency failed: he could not control minting (A and R true, R explains A) **1×** 2006
- **Ibn Battuta** came under **Muhammad bin Tughluq** **3×** 1991, 1994, 2011
- Postal system described by **Ibn Battuta** **1×** 2000
- "The king was freed from his people...": Badauni on **Muhammad bin Tughluq's** death **1×** 2008

MUHAMMAD BIN TUGHLUQ'S PROJECTS



4.2 Firuz Shah Tughluq

- Employment exchange: **Firuz Shah** **2×** 2010, 2010
- Public Works Department: **Firuz Shah** **2×** 2000, 2006
- Translation department: **Firuz Shah** **1×** 2004
- First state-funded Hajj (key): **Firuz Tughluq** **1×** 1998
- **Dar-ul-Shifa**: a **free hospital** **1×** 2013
- First **haq-i-sharb** (irrigation tax): **Firuz** **2×** 2008, 2010
- **Sharb**: a tax on **irrigation** **1×** 1996
- Jizya on Brahmins: **Firuz** **1×** 2011
- Ashokan pillars from **Topra and Meerut**: **Firuz** **2×** 1996, 2009
- Timur came in **1398** under **Nasiruddin Mahmud**; no option correct (*) **1×** 2002

Firuz's department	Work
Diwan-i-Khairat	Charity
Diwan-i-Bandagan	Slaves
Dar-ul-Shifa	Free hospital
Employment bureau	Jobs for the unemployed
Public works	Canals, towns (Jaunpur, Hisar, Firozabad)

QUANTUM IAS & PCS

TRAP ALERT

- Token currency = Muhammad bin Tughluq; it failed because of **forgery**, not a copper shortage.
- Irrigation tax = **Firuz**; house tax = **Alauddin**.
- Ashokan pillars moved by **Firuz**; the iron pillar is Gupta.
- Timur came in **1398**, after Firuz's death (1388).

UP LINK

UP link: **Jaunpur** was founded by **Firuz Shah Tughluq** (named after Muhammad bin Tughluq, Jauna Khan); the **Meerut** Ashokan pillar he moved now stands in Delhi.

LAST-MINUTE 10

1. Ghazi Malik = Ghiyasuddin; first canals
2. Tughluqs ruled longest
3. MBT: Daulatabad, token currency, most learned
4. Ibn Battuta under MBT; described the post
5. Firuz: employment, PWD, translation, Dar-ul-Shifa
6. Haq-i-sharb (irrigation) = Firuz
7. Jizya on Brahmins = Firuz
8. Topra and Meerut pillars to Delhi = Firuz
9. Firuz founded Jaunpur, Hisar, Firozabad
10. Timur 1398 (Nasiruddin Mahmud)

Chapter 5

Sayyid and Lodi Dynasties

Built from 6 UPPSC questions (1993–2023). Focus: the Sayyid order, the Lodis as the only Afghan dynasty, Sikandar Lodi and Agra, and the order of the five Sultanate dynasties.

ASKED MOST · REVISE FIRST

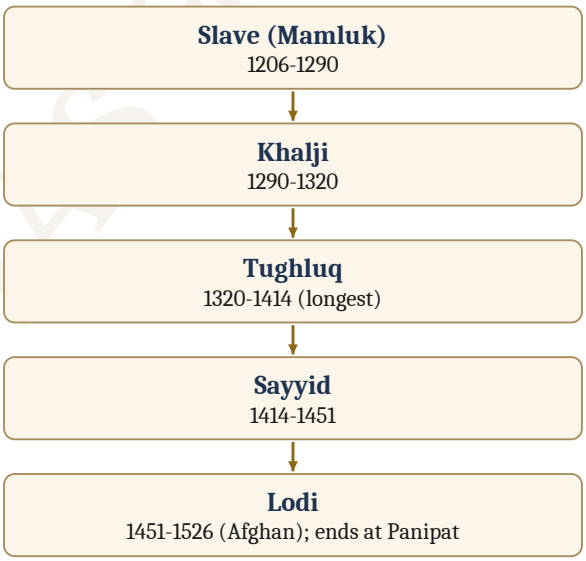
- **Agra** founded by **Sikandar Lodi** (1504) 7× 1993, 2004, 2005, 2008, 2010, 2011, 2011
- **Lodis** = the only **Afghan** Sultanate dynasty 4× 1993, 2008, 2011, 2016
- Order: **Slave, Khalji, Tughluq, Sayyid, Lodi** 1× 2012

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Sayyids and Lodis

- Sayyid order: **Khizr Khan, Mubarak Shah, Muhammad Shah, Alauddin Alam Shah** 1× 2023
- Afghan origin: **Lodis** 4× 1993, 2008, 2011, 2016
- Agra founded by **Sikandar Lodi** 7× 1993, 2004, 2005, 2008, 2010, 2011, 2011
- Pen-name "**Gulrukhi**": **Sikandar Lodi** 1× 2015
- Abolished the tax on grain: **Sikandar Lodi** 1× 2010
- Order: **Slave, Khalji, Sayyid, Lodi** (4, 1, 3, 2) 1× 2012

DELHI SULTANATE DYNASTIES



Lodi Sultan	Note
Bahlul Lodi	Founder; bahloli coin
Sikandar Lodi	Founded Agra (1504); gaz-i-Sikandari; Gulrukhi
Ibrahim Lodi	Killed by Babur at Panipat (1526)

TRAP ALERT

- The Khaljis were Turks; only the **Lodis** were Afghans.
- Agra = **Sikandar Lodi**, not Firuz Tughluq.
- Sayyid order: Mubarak Shah comes **before** Muhammad Shah.

QUANTUM IAS PCS

UP LINK

UP link: **Agra** was founded in 1504 by **Sikandar Lodi**, who made it his capital; the Lodis also fought the **Sharqi** sultans of **Jaunpur**, whom Bahlul Lodi ended.

LAST-MINUTE 10

1. Khizr Khan founded the Sayyids
2. Khizr, Mubarak, Muhammad, Alam Shah
3. Lodis = only Afghan dynasty
4. Bahlul Lodi = founder
5. Sikandar Lodi = Agra (1504)
6. Gulrukhi = Sikandar Lodi's pen-name
7. Sikandar abolished the grain tax
8. Gaz-i-Sikandari = his measuring yard
9. Ibrahim Lodi lost at Panipat, 1526
10. Slave, Khalji, Tughluq, Sayyid, Lodi

Chapter 6

Vijayanagara Empire

Built from 11 UPPSC questions (1993–2016). Focus: Sayana and the Sangamas, Ganga Devi, Krishnadeva Raya and the Ashtadiggajas, foreign travellers, Talikota and Hampi.

ASKED MOST · REVISE FIRST

- **Krishnadeva Raya**: golden age of **Telugu**; **Ashtadiggajas** = eight Telugu poets; founded **Nagalapura** 4× 2001, 2006, 2008, 2016
- **Talikota (1565)**: Vijayanagara vs **Bijapur, Ahmadnagar, Golconda** (and Bidar) 2× 1993, 2014
- **Hampi** = ruins of the capital 2× 2007, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Foundation and Literature

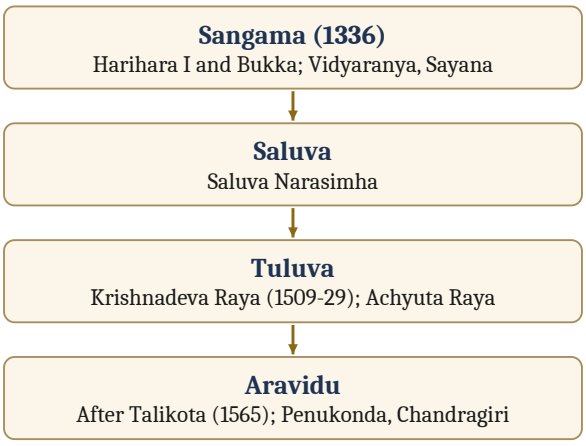
- **Sayana** (Vedic commentator): patronised by **Vijayanagara** rulers 1× 2008
- **Madhura Vijayam** (husband's victories): **Ganga Devi** 1× 2000
- **Ashtadiggajas**: **eight Telugu poets** of Krishnadeva Raya 1× 2001
- Krishnadeva Raya's age: golden age of **Telugu** 1× 2006
- Enthroned about 500 years before 2008: **Krishnadeva Raya** (1509) 1× 2008
- City founded by Krishnadeva Raya: **Nagalapura** 1× 2016
- Telugu translators: **Nannaya and Tikkana** (3 and 4) 1× 2012

6.2 Travellers, Talikota and Hampi

- **Abdur Razzaq** came under **Deva Raya II** 1× 2010
- **Talikota**: Vijayanagara vs **Bijapur, Ahmadnagar, Golconda** 1× 2014
- Talikota fought in **1565** 1× 1993
- Old capital ruins: **Hampi** 2× 2007, 2008

Traveller	Ruler	Origin
Nicolo Conti	Deva Raya I	Italy
Abdur Razzaq	Deva Raya II	Persia
Domingo Paes	Krishnadeva Raya	Portugal
Fernao Nuniz	Achyuta Raya	Portugal
Athanasius Nikitin	Bahmani kingdom	Russia

FOUR DYNASTIES OF VIJAYANAGARA



TRAP ALERT

- Ashtadiggajas = **poets**; Ashtapradhan = Shivaji's **ministers**.
- Abdur Razzaq = **Deva Raya II**; Paes = Krishnadeva Raya.
- Talikota was against the **Deccan sultanates**, not the Bahmanis (already split).

UP LINK

UP link: **Harihara and Bukka** had served Muhammad bin Tughluq of Delhi before founding Vijayanagara in 1336, during the same revolts that hit the Doab and Awadh.

LAST-MINUTE 10

1. Founded 1336 by Harihara and Bukka
2. Sayana = Vijayanagara patronage
3. Ganga Devi = Madhura Vijayam
4. Krishnadeva Raya = Telugu golden age
5. Ashtadiggajas = eight Telugu poets
6. Nagalapura founded by Krishnadeva Raya
7. Abdur Razzaq = Deva Raya II
8. Paes = Krishnadeva; Nuniz = Achyuta
9. Talikota 1565 vs Deccan sultanates
10. Hampi = capital ruins

Chapter 7

Delhi Sultanate: Administration

Built from 15 UPPSC questions (1991–2025). Focus: the Turkish nobility, the wazir and zawabit, khalisa and iqta, departments set up by individual Sultans, tanka and jital.

ASKED MOST · REVISE FIRST

- **Iqta** system introduced by **Iltutmish**; iqta revenue was **not** paid straight to the Sultan **3×** 2010, 2019, 2023
- **Diwan-i-Amir-Kohi** = **agriculture** department of **Muhammad bin Tughluq** **4×** 1997, 2004, 2017, 2017
- **Tanka** (silver) and **jital** (copper) = main Sultanate coins; started by **Iltutmish** **3×** 2001, 2008, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

7.1 Nobility, Ministries and Law

- Most Sultans and amirs were **Turkish** **1×** 1991
- **Wizarat** peaked under the **Tughluqs** **1×** 1997
- **Zawabit**: **state laws** **1×** 1997
- **Kharitadar**: **dispatcher of royal decrees** **1×** 2023

7.2 Land and Departments

- Land was **khalisa** and iqta (jagir); khalisa, not jagir, was state land (A true, R false) **1×** 2025
- Not an iqta feature: revenue paid **straight to the Sultan** **1×** 2019
- **Iqta** system: **Iltutmish** **1×** 2010
- **Iqta-land revenue, Mahrusa-protected town, Khalisa-state land, Shiq-a portion** **1×** 2023
- Wrong pair: **Diwan-i-Riyasat-Balban** (it was Alauddin's market department) **1×** 2008

■ **Diwan-i-Amir-Kohi**: **Muhammad bin Tughluq**; **agriculture** **4×** 1997, 2004, 2017, 2017

Department	Work	Founder
Diwan-i-Arz	Army	Balban
Diwan-i-Riyasat	Markets	Alauddin Khalji
Diwan-i-Mustakhraj	Arrears of revenue	Alauddin Khalji
Diwan-i-Amir-Kohi	Agriculture	Muhammad bin Tughluq
Diwan-i-Khairat	Charity	Firuz Tughluq
Diwan-i-Bandagan	Slaves	Firuz Tughluq

7.3 Coinage

- Main coins: **jital and tanka** (2 and 4) **1×** 2001
- Silver **tanka** started by **Iltutmish** **2×** 2008, 2013
- Last Baghdad Caliph's name first on coins of **Alauddin Masud Shah** **1×** 2012

TRAP ALERT

- Mansabdari is **Mughal**; the Sultanate used **iqta**.
- Diwan-i-Amir-Kohi = agriculture, not hills or forests.
- Diwan-i-Riyasat = **Alauddin**, not Balban.

UP LINK

UP link: **Awadh, Kara, Badaun** and **Sambhal** were major iqtas of the Sultanate in present UP; Iltutmish himself held **Badaun** before becoming Sultan.

LAST-MINUTE 10

1. Sultans and nobles mostly Turkish
2. Wizarat peaked under the Tughluqs
3. Zawabit = state laws; Kharitadar = royal dispatches
4. Khalisa = crown land; iqta = assigned revenue
5. Iltutmish = iqta system, tanka
6. Diwan-i-Arz = Balban; Diwan-i-Riyasat = Alauddin
7. Diwan-i-Amir-Kohi = MBT, agriculture
8. Tanka silver; jital copper
9. Mahrusa = protected town; shiq = portion
10. Masud Shah = last Caliph's name on coins

Chapter 8

Delhi Sultanate: Art and Architecture

Built from 5 UPPSC questions (2002–2022). Focus: the first Indo-Islamic tomb, the Qutb Minar's builders, early Sultanate monuments, Aibak's mosques at Delhi and Ajmer.

ASKED MOST · REVISE FIRST

- ■ **Qutb Minar**: begun by **Aibak**, raised by **Iltutmish**, fifth storey by **Firuz Tughluq**; **Alauddin** did not build it 4× 2002, 2003, 2004, 2013
- • **Sultan Ghari-Iltutmish, Red Palace-Balban, Jamaat Khana Masjid-Alauddin, Adhai Din ka Jhonpra-Aibak** 1× 2022

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Monuments

- First Indo-Islamic tomb (key): **Balban's tomb** (first true arch) 1× 2004
- ■ Did not build the Qutb Minar: **Alauddin Khalji** 3× 2002, 2003, 2013
- Fifth storey: **Firuz Shah Tughluq** 1× 2004
- **Sultan Ghari-Iltutmish, Red Palace-Balban, Jamaat Khana Masjid-Alauddin Khalji, Adhai Din ka Jhonpra-Aibak** 1× 2022
- Wrong pair: **Ajmer-Quwwat-ul-Islam** (that is Delhi; Ajmer has the Adhai Din ka Jhonpra) 1× 2011

Monument	Builder
Quwwat-ul-Islam mosque (Delhi)	Aibak
Adhai Din ka Jhonpra (Ajmer)	Aibak
Qutb Minar	Aibak, Iltutmish, Firuz (repairs)
Sultan Ghari	Iltutmish
Alai Darwaza, Siri	Alauddin Khalji
Tughlaqabad	Ghiyasuddin Tughluq
Atala Masjid (Jaunpur)	Sharqis

TRAP ALERT

- Alauddin built the **Alai Darwaza** and began the unfinished Alai Minar, not the Qutb Minar.
- Delhi = Quwwat-ul-Islam; Ajmer = Adhai Din ka Jhonpra.
- Balban's tomb has the first **true arch**.

UP LINK

UP link: the Sharqi sultans built the **Atala, Jhanjhri and Jama mosques** in **Jaunpur**, the finest regional Sultanate style in UP.

LAST-MINUTE 10

1. Balban's tomb = first true arch
2. Qutb Minar: Aibak, Iltutmish, Firuz
3. Alauddin not a Qutb Minar builder
4. Sultan Ghari = Iltutmish
5. Red Palace = Balban
6. Jamaat Khana Masjid = Alauddin
7. Adhai Din ka Jhonpra = Aibak (Ajmer)
8. Quwwat-ul-Islam = Aibak (Delhi)
9. Alai Darwaza = Alauddin
10. Atala Masjid = Jaunpur (Sharqi)

Chapter 9

Delhi Sultanate: Literature

Built from 24 UPPSC questions (1990–2023). Focus: Amir Khusrau, Amir Hasan, Persian translations, the chroniclers of the Sultanate and their books, regional histories and Hindavi poetry.

ASKED MOST · REVISE FIRST

- **Amir Khusrau**: born at **Patiali** (Etah then, Kasganj now); "**Parrot of India**"; court poet of **Alauddin**; father of **Khari Boli**; first Persian poet to describe India **10×** 1990, 1995, 2002, 2002, 2003, 2005, 2008, 2012, 2013, 2017
- **Chronicler-book matching**: **Barani-Tarikh-i-Firozshahi and Fatawa-i-Jahandari, Hasan Nizami-Taj-ul-Maasir, Minhaj-Tabaqat-i-Nasiri, Yahya-Tarikh-i-Mubarakshahi, Isami-Futuh-us-Salatin, Khusrau-Khazain-ul-Futuh** **4×** 1997, 2009, 2019, 2023
- **Amir Hasan Sijzi** = "Sadi of India"; compiled **Fawaid-ul-Fuad** (Nizamuddin Auliya's talks) **2×** 2013, 2021

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

9.1 Amir Khusrau and Amir Hasan

- Khusrau born at **Patiali** (Etah in 2002-03; now Kasganj) (*) **3×** 2002, 2003, one-liner
- "**Parrot of India**": **Amir Khusrau** **1×** 2013
- Court poet of **Alauddin Khalji** **1×** 1995
- Saw seven Sultans: **Amir Khusrau** and Nizamuddin both fit (*) **1×** 2005
- Father of Hindi **Khari Boli**; master of Hindi and Persian **4×** 1990, 2002, 2008, 2012
- First Persian poet to depict India: **Amir Khusrau** **1×** 2017
- **Tabla** introduced by **Amir Khusrau** (tradition) **1×** 2009
- "Sadi of India": **Amir Hasan** **1×** 2013
- **Fawaid-ul-Fuad**: compiled by **Amir Hasan Sijzi** **1×** 2021

9.2 Translations and Chronicles

- Shukasaptati into Persian as **Tutinama**: **Ziya-uddin Nakhshabi** **1×** 2022
- Translated Firuz's 300 Nagarkot volumes: **Azizuddin Khan** (per key) **1×** 2023
- **Futuh-us-Salatin-Isami, Futuhat-i-Firozshahi-Firuz, Tarikh-i-Firozshahi-Barani, Khazain-ul-Futuh-Khusrau** **1×** 2023
- Sultan who wrote his memoirs: **Firuz Tughluq** **1×** 2008
- **Barani-Tarikh-i-Firozshahi, Hasan Nizami-Taj-ul-Maasir, Minhaj-Tabaqat-i-Nasiri, Yahya-Tarikh-i-Mubarakshahi** **1×** 1997
- **Taj-ul-Maasir-Hasan Nizami, Khazain-ul-Futuh-Khusrau, Tarikh-i-Mubarakshahi-Yahya, Fatawa-i-Jahandari-Barani** **1×** 2019
- **Tarikh-i-Hind-Al-Biruni, Tarikh-i-Delhi-Khusrau, Rihla-Ibn Battuta, Tabaqat-i-Nasiri-Minhaj** **1×** 2009
- Order: **Kitab-ul-Hind, Prithviraj Raso, Tabaqat-i-Nasiri, Fatawa-i-Jahandari** **1×** 2020

Book	Author
Tabaqat-i-Nasiri	Minhaj-us-Siraj
Taj-ul-Maasir	Hasan Nizami
Tarikh-i-Firozshahi, Fatawa-i-Jahandari	Ziauddin Barani
Futuhat-i-Firozshahi	Firuz Shah Tughluq
Futuh-us-Salatin	Isami
Khazain-ul-Futuh, Tughlaqnama	Amir Khusrau
Tarikh-i-Mubarakshahi	Yahya bin Ahmad Sirhindi
Rihla	Ibn Battuta

9.3 Regional Histories and Hindavi

- **Mirat-i-Sikandari** (Gujarat), **Burhan-i-Maasir** (Bahmanis), **Riyaz-us-Salatin** (Bengal), **Riyaz-ul-Insha** (Mahmud Gawan's letters) **1×** 2023

QUANTUM IAS & PCS

- Mirat-i-Sikandari describes Mahmud Begada's prosperity and security (both) 1× 2023
- **Mulla Daud-Chandayan, Damodar-Padmavati Katha, Somnath-Raga Vibodh, Khusrau-Ashiqa** 1× 2019
- Jain scholar among options: **Hemachandra Suri** 1× 2009

TRAP ALERT

- Futuh-us-Salatin (**Isami**) vs Futuhat-i-Firozshahi (**Firuz**).
- Fatawa-i-Jahandari is **Barani's**, not Firuz's.
- Amir Hasan Sijzi (Sufi poet) is not Hasan Nizami (Aibak's historian).
- Patiali was in Etah until 2008; it is now **Kasganj**.

UP LINK

UP link: **Amir Khusrau** was born at **Patiali** (now Kasganj district, UP) in 1253; **Mulla Daud's Chandayan** (1379) was written at **Dalmau** (Rae Bareli), the first Sufi masnavi in Awadhi.

LAST-MINUTE 10

1. Khusrau = Patiali, Parrot of India, Alauddin's poet
2. Khusrau = Khari Boli; tabla by tradition
3. Amir Hasan = Sadi of India; Fawaid-ul-Fuad
4. Nakhshabi = Tutinama
5. Barani = Tarikh-i-Firozshahi, Fatawa-i-Jahandari
6. Hasan Nizami = Taj-ul-Maasir; Minhaj = Tabaqat-i-Nasiri
7. Isami = Futuh-us-Salatin; Firuz = Futuhat (memoirs)
8. Yahya = Tarikh-i-Mubarakshahi
9. Mirat-i-Sikandari = Gujarat; Burhan-i-Maasir = Bahmani
10. Mulla Daud = Chandayan

Chapter 10

Delhi Sultanate: Miscellaneous

Built from 12 UPPSC questions (1997–2021). Focus: Turkish court life, crops, ruler-measure matching, the order of travellers and rulers, battles and their dates.

ASKED MOST · REVISE FIRST

- **Ruler-measure matching**: **Firuz-canals, Balban-Nauroz, Alauddin-Diwan-i-Riyasat, Jahangir-Thomas Roe; Akbar-Ain-i-Dahsala, MBT-token currency, Iltutmish-Chahalgani, Sher Shah-Sarak-i-Azam** 2× 1997, 1997
- ■ **Battle-year matching**: **Chandawar 1194, Timur 1398, Ghaghra 1529, Talikota 1565; Kalinga 261 BCE, Tarain 1192, Haldighati 1576, Plassey 1757** 3× 2002, 2003, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Court, Society and Crops

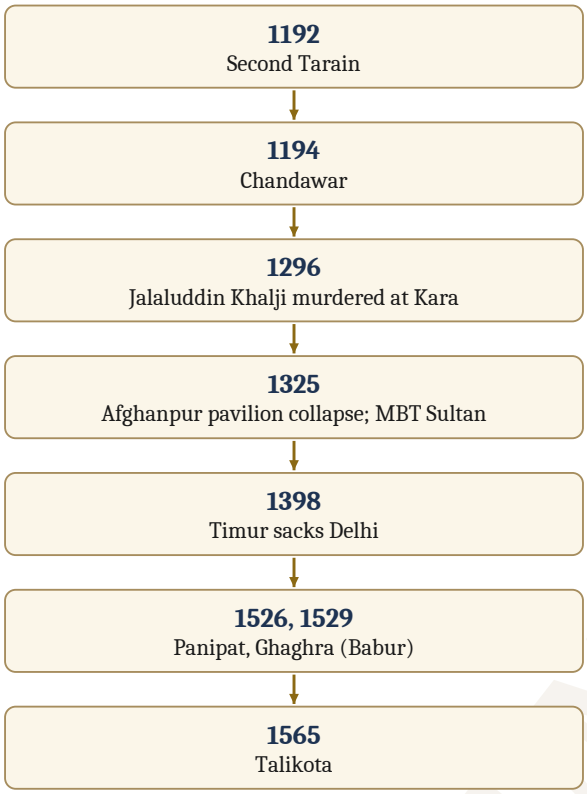
- **Polo** introduced by the **Turks** 1× 2001
- **Dastar-bandan**: the **ulema** 1× 2014
- Hindu gods worshipped in the palace: **Nasiruddin Khusrau Shah** 1× 2021
- Not grown in the 13th-14th centuries: **maize** (a New World crop) 2× 2008, 2011

10.2 Matching and Order

- **Firuz-canals, Balban-Nauroz, Alauddin-Diwan-i-Riyasat, Jahangir-Thomas Roe** 1× 1997
- **Akbar-Ain-i-Dahsala, MBT-token currency, Iltutmish-Chahalgani, Sher Shah-Sarak-i-Azam** 1× 1997
- Wrong pair: **Abdur Razzaq-Turkey** (he was **Persian**) 1× 2017
- Order: **Al-Biruni, Ibn Battuta, Tavernier, Manucci** 2× 1997, 2002
- Order: **Razia, Alauddin, Sher Shah, Akbar** 1× 2010

- Order: **Second Tarain, murder of Jalaluddin, Afghanpur conspiracy, capital transfer** 1× 2017
- **Chandawar-1194, Timur-1398, Talikota-1565, Ghaghra-1529** 1× 2017
- **Plassey-1757, Kalinga-261 BCE, Haldighati-1576, Tarain-1192** 2× 2002, 2003

KEY DATES OF THE PERIOD



TRAP ALERT

- Maize, tobacco, potato, chillies came from the Americas after 1500.
- Abdur Razzaq was Persian (from Herat), not Turkish.
- "Tarain" with no number means the **second** battle, 1192.

UP LINK

UP link: the **Battle of Ghaghra** (1529), where Babur beat the Afghans and the Bengal Sultan, was fought on the Ghaghra in eastern UP (Ballia region); **Chandawar** (1194) is near Firozabad.

LAST-MINUTE 10

1. Turks brought polo
2. Dastar-bandan = ulema
3. Khusrau Shah allowed Hindu worship in the palace
4. No maize before 1500
5. Balban = Nauroz; Firuz = canals
6. Iltutmish = Chahalgani; Sher Shah = Sarak-i-Azam
7. Al-Biruni, Ibn Battuta, Tavernier, Manucci
8. Chandawar 1194; Timur 1398
9. Ghaghra 1529; Talikota 1565
10. Haldighati 1576; Plassey 1757

Chapter 11

Provincial Dynasties of North India and the Deccan

Built from 24 UPPSC questions (1995–2023). Focus: Zain-ul-Abidin of Kashmir, the Sharqis of Jaunpur, Rana Kumbha, the Bahmanis and Deccan sultanates, provincial architecture, southern kingdoms and officer titles.

ASKED MOST · REVISE FIRST

- **Jaunpur**: founded by **Firuz Tughluq** in memory of **Muhammad bin Tughluq**; "**Shiraz of India**" (a centre of learning); last ruler **Hussain Shah** 9× 2001, 2002, 2003, 2004, 2004, 2005, 2006, 2017, 2023
- **Rana Kumbha**: built the victory tower (Kirti/Vijay Stambh) at **Chittor**; wrote on **music** 4× 2007, 2008, 2010, 2011
- **Bahmani kingdom**: **1347**, founded by **Hasan Gangu**; **Adil Shahi-Bijapur, Qutb Shahi-Golconda, Nizam Shahi-Ahmadnagar** 4× 1995, 2002, 2004, 2005
- **Southern capitals**: **Hoysala-Dwarasamudra (Halebid), Yadava-Devagiri, Kakatiya-Warangal, Pandya-Madurai** 6× 2003, 2003, 2008, 2014, 2018, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 Kashmir, Jaunpur and Mewar

- Abolished jizya and cow slaughter in Kashmir: **Zain-ul-Abidin** (real name **Shahi Khan**) 2× 2023, one-liner
- ■ Jaunpur founded in memory of **Muhammad bin Tughluq**, by **Firuz Shah Tughluq** 4× 2002, 2003, 2004, 2006
- ■ "**Shiraz of India**": **Jaunpur** 3× 2001, 2004, 2005
- Jaunpur was "Shiraz-i-Hind" as a great centre of learning (A and R true, R explains A) 1× 2023
- Last ruler of Jaunpur: **Hussain Shah** Sharqi 1× 2017
- Wrong pair: **Malik Sarwar Khwaja Jahan-Malwa** (he founded the Jaunpur Sharqi state) 1× 2021
- ■ Victory tower of Chittor: **Rana Kumbha** 3× 2008, 2010, 2011
- Rajput ruler who wrote on music: **Rana Kumbha** 1× 2007

11.2 The Deccan

- Bahmani kingdom: **1347**, founder **Hasan Gangu** (Alauddin Bahman Shah) 2× 1995, 2005
- **Adil Shahi-Bijapur, Qutb Shahi-Golconda, Nizam Shahi-Ahmadnagar, Sharqi-Jaunpur** 1× 2002
- **Kitab-i-Nauras**: **Ibrahim Adil Shah II** 1× 2020
- Wrong pair: **Yusuf Adil Shah-Ahmadnagar** (he founded Bijapur) 1× 2004

Deccan sultanate	Capital	Dynasty
Bijapur	Bijapur	Adil Shahi
Ahmadnagar	Ahmadnagar	Nizam Shahi
Golconda	Golconda	Qutb Shahi
Berar	Ellichpur	Imad Shahi
Bidar	Bidar	Barid Shahi

11.3 Architecture, the South and Officers

- Wrong pair: **Adina Masjid-Mandu** (it is at **Pandua**, Bengal) 1× 2018
- **Atala Masjid-Ibrahim Shah Sharqi, Chhota Sona Masjid-Wali Muhammad, Qadam Rasul-Nusrat Shah, Adina Masjid-Sikandar Shah** 1× 2021
- Correct: **Hoysala-Dwarasamudra**, Pandya-Madurai 2× 2003, 2008
- **Pallava-Kanchi, Pandya-Madurai, Yadava-Devagiri, Kakatiya-Warangal** 1× 2019
- Wrong pair: **Warangal-Ramachandra Dev** (Ramachandra ruled Devagiri) 1× 2018
- **Dwarasamudra** = modern **Halebid**; capital of the **Hoysalas** 2× 2003, 2014
- **Polygars**: territorial **administrative and military** chiefs 1× 2015
- **Talar-night guard, Pattakila-village tax collector, Sahasadhipati-dacoity officer, Baladhipa-octroi guard** 1× 2022

TRAP ALERT

- Jaunpur founder = **Firuz**; Sharqi state founder = **Malik Sarwar** (Khwaja Jahan).
- Zain-ul-Abidin **abolished** jizya; Sikandar Shah (Kashmir) imposed it.
- Yadava = **Devagiri**; Kakatiya = **Warangal** (often swapped).
- Adina Masjid = **Pandua** (Bengal), not Mandu.

UP LINK

UP link: the **Sharqi Sultanate of Jaunpur** (1394-1479) was UP's great medieval regional state; its **Atala Masjid** and **Jama Masjid**, and its fame as the "Shiraz of India", come from Ibrahim Shah Sharqi's patronage of learning.

LAST-MINUTE 10

1. Zain-ul-Abidin (Shahi Khan) = Budshah of Kashmir
2. Jaunpur = Firuz, for MBT; Shiraz of India
3. Malik Sarwar founded the Sharqis; Hussain Shah the last
4. Rana Kumbha = Chittor tower, music treatise
5. Bahmani 1347; Hasan Gangu
6. Bijapur = Adil Shahi; Golconda = Qutb Shahi; Ahmadnagar = Nizam Shahi
7. Ibrahim Adil Shah II = Kitab-i-Nauras
8. Atala Masjid = Ibrahim Sharqi; Adina = Sikandar Shah (Pandua)
9. Dwarasamudra = Halebid (Hoysala)
10. Polygars = military chiefs in the south

Chapter 12

Bhakti and Sufi Movement

Built from 49 UPPSC questions (1990–2025). Focus: Alvars and the acharyas, Ramananda and Kabir, the order of saints, Namdev, Chaitanya and Shankaradeva, Tulsidas, the Chishti chain and dargahs, other silsilas.

ASKED MOST · REVISE FIRST

- **Acharya-philosophy matching**: **Ramanuja-Vishishtadvaita, Madhva-Dvaita, Nimbarka-Dvaitadvaita, Vallabha-Shuddhadvaita** 5× 1991, 2002, 2005, 2017, 2022
- **Chishti chain**: **Usman Haruni, Muinuddin (Ajmer), Bakhtiyar Kaki, Farid, Nizamuddin (Delhi, Mahbub-i-Ilahi), Nasiruddin (Chirag-i-Delhi)** 10× 2002, 2006, 2007, 2010, 2014, 2014, 2014, 2014, 2017, 2020
- **Ramananda**: born at **Prayag**; Kabir's guru; "let no man ask a man's caste" 3× 2002, 2009, 2010
- **Shankaradeva** spread Vaishnavism in **Assam**; Chaitanya = Vaishnava 4× 1990, 2002, 2011, 2015
- **Naqshbandis** opposed music 3× 2002, 2003, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Origins and the Acharyas

- Bhakti started by the **Alvars** 1× 2001
- Revived in the North in the **15th-16th centuries** 1× 1993
- **Ramanuja**: **Vishishtadvaita**; followers are **Vaishnavas** 2× 1991, 1999
- **Shuddhadvaita**: **Vallabhacharya** 1× 2002
- Correct: **Dvaita-Madhvacharya** 1× 2005
- Acharya-philosophy match (A-2, B-3, C-1, D-4 per key) 1× 2017
- **Ramanuja-Vishishtadvaita, Madhva-Dvaita, Nimbarka-Dvaitadvaita, Vallabha-Shuddhadvaita** 1× 2022

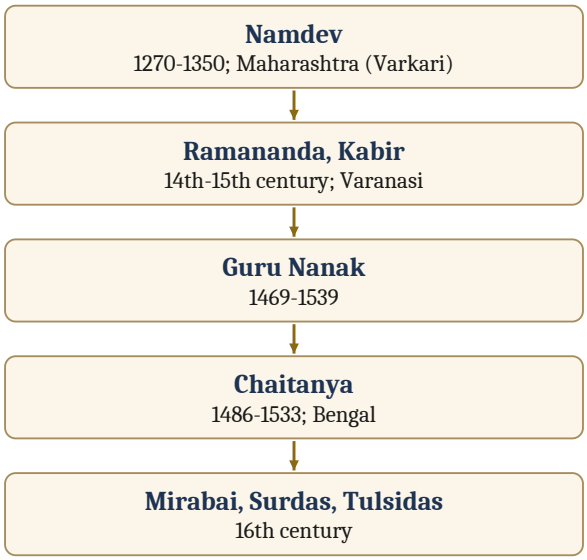
Acharya	Philosophy
Shankara	Advaita
Ramanuja	Vishishtadvaita
Madhva	Dvaita
Nimbarka	Dvaitadvaita
Vallabha	Shuddhadvaita

12.2 Northern Saints

- Born at **Prayag**: **Ramananda** 1× 2010
- "Let no man ask a man's sect or caste": **Ramananda** 1× 2009
- Kabir's guru: **Ramananda** 1× 2002
- Kabir-Dharamdas dialogues: **Amarmool** 2× 2002, 2003
- **Kabir-Ramananda, Khusrau-Nizamuddin, Surdas-Vallabhacharya, Mardana-Guru Nanak** 1× 2025
- Order: **Kabir, Nanak, Chaitanya, Mirabai** 1× 2014
- Order: **Namdev, Kabir, Nanak, Chaitanya** 1× 2025
- Order: **Namdev, Kabir, Nanak, Mirabai** 1× 2005
- Mirabai's contemporaries: several fit (*) 1× 1995
- Nanak's successor: **Guru Angad** 1× 2023
- Varkari saint: **Namdev** 2× 2002, 2004
- Influenced by Islam: **Namdev** 1× 2010
- **Chaitanya**: **Vaishnava** 1× 1990
- ■ Vaishnavism in Assam (Kamarupa, Cooch Bihar): **Shankaradeva** 3× 2002, 2011, 2015

- Tulsidas: born at Rajapur (now **Chitrakoot**); wife **Ratnavali** (neither statement) 1× 2023
- Tulsidas: contemporary of **Akbar and Jahangir** 1× 2004
- **Malukdas**: saint-poet of **Kara** 1× 2008
- Wrong pair: **Raskhan-Rasikpriya** (Rasikpriya = Keshavdas) 1× 2018

ORDER OF THE BHAKTI SAINTS



12.3 Sufism

- Not related to Sufism: **ulema** 1× 2014
- Sufis: **Nizamuddin Auliya, Muinuddin Chishti** (2 and 3) 1× 1998
- Sanskrit music works were translated into Persian; sama is true but not the reason (A and R true, R does not explain A) 1× 2018
- Chishti order in India: **Khwaja Muinuddin** (Ajmer) 2× 2002, 2014
- Chishti order's founder (abroad): **Khwaja Abu Abdal** 1× 2011
- Muinuddin's guru: **Usman Haruni** 2× 2014, 2014
- Baba **Farid** in the **Adi Granth** 1× 1994
- Nizamuddin's dargah: **Delhi**; his guru: **Baba Farid** 2× 2006, 2010
- Wrong pairs: **Nizamuddin's tomb-Ajmer**, **Nizamuddin-Multan** 2× 2007, 2020
- **Mahbub-i-Ilahi**: **Nizamuddin Auliya**; he saw seven Sultans 2× 2014, 2014
- **Chirag-i-Delhi**: **Nasiruddin** 1× 2017
- **Shaikh-ul-Hind**: **Salim Chishti** 2× 2004, 2008
- **Muinuddin-Chishti, Sirhindi-Naqshbandi, Dara Shikoh-Qadiri, Shihabuddin-Suhrawardi** 1× 2013
- Against music: **Naqshbandiya** 3× 2002, 2003, 2013
- Counted Krishna among the auliyas: **Shah Muhammad Ghaus** (Shattari) 2× 2002, 2003
- Hath Yoga: practised by Nathpanthis and adopted by Sufis (both) 1× 2019

Silsila	In India	Note
Chishti	Muinuddin (Ajmer)	Music (sama), simple life
Suhrawardi	Bahauddin Zakariya (Multan)	Close to the state
Qadiri	Shah Niamatullah; Dara Shikoh	Baghdad origin
Naqshbandi	Baqi Billah; Ahmad Sirhindi	Orthodox; no music
Shattari	Shah Muhammad Ghaus	Yoga links

TRAP ALERT

- **Dvaita** (Madhva) vs **Dvaitadvaita** (Nimbarka); Advaita vs Shuddhadvaita.
- Assam Vaishnavism = **Shankaradeva**, not Chaitanya.
- Shaikh-ul-Hind = **Salim Chishti**; Sultan-ul-Hind = Muinuddin.
- Rasikpriya = **Keshavdas**; Raskhan = Premvatika.

UP LINK

UP link: **Ramananda** (born Prayag) and **Kabir** taught at **Varanasi**; **Tulsidas** wrote the Ramcharitmanas at Varanasi (born Rajapur, Chitrakoot); **Malukdas** was of **Kara**, and **Salim Chishti's** dargah is at **Fatehpur Sikri**.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Alvars started Bhakti; revival 15th-16th century
2. Ramanuja Vishishtadvaita; Madhva Dvaita; Vallabha Shuddhadvaita
3. Ramananda = Prayag; Kabir's guru
4. Namdev, Kabir, Nanak, Chaitanya, Mirabai
5. Shankaradeva = Assam; Chaitanya = Bengal
6. Malukdas = Kara; Raskhan = Premvatika
7. Muinuddin = Ajmer (Usman Haruni's disciple)
8. Nizamuddin = Delhi, Mahbub-i-Ilahi; Farid's disciple
9. Nasiruddin = Chirag-i-Delhi; Salim Chishti = Shaikh-ul-Hind
10. Naqshbandi opposed music; Sirhindi was Naqshbandi

Chapter 13

Mughal Dynasty: Babur

Built from 15 UPPSC questions (1990–2025). Focus: Chaghtai Turk origins, Kabul and the Padshah title, Daulat Khan's invitation, Panipat, Khanwa, Chanderi and Ghaghra, the Baburnama, the Babri mosque.

ASKED MOST · REVISE FIRST

- ■ **First Panipat (1526)**: **Babur vs Ibrahim Lodi**; won by **skilled generalship**; first big use of **artillery** 5× 1990, 1996, 1996, 2012, 2016
- ■ **Khanwa (1527)**: Babur declared **jihad**, beat **Rana Sanga** 3× 2004, 2009, 2010
- Babur wrote the **Baburnama** in **Turki**; the court language was **Persian** (A true, R false) 1× 2025

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

13.1 Rise of Babur

- Mughals were **Chaghtai Turks** 1× 2010
- Babur lost at **Sar-i-Pul** to **Shaibani Khan** 1× 2015
- Took the title **Padshah** at **Kabul** 1× 2015
- Invited to India by **Daulat Khan** Lodi 1× 2021

13.2 Battles

- ■ **First Panipat**: **Babur vs Ibrahim Lodi, 1526** 3× 1996, 2012, 2016
- Babur's victory: **skilled generalship** (key; many guides say artillery) 1× 1990
- First use of artillery among options: **First Panipat** (Bajaur 1519 was earlier) 1× 1996
- **Jihad** declared at **Khanwa** 1× 2009
- **Rana Sanga** fought Babur at **Khanwa** (1527) 2× 2004, 2010
- Wrong pair: **Chanderi-1530** (it was **1528**) 1× 2015

BABUR'S BATTLES

1526
Panipat: Ibrahim Lodi (tulughma, artillery)

↓

1527
Khanwa: Rana Sanga (jihad; title Ghazi)

↓

1528
Chanderi: Medini Rai

↓

1529
Ghaghra: Afghans under Mahmud Lodi and Nusrat Shah of Bengal

↓

1530
Dies at Agra; buried at Kabul

13.3 Baburnama and Babri Mosque

- Babur wrote in **Turki**, but the court language was **Persian** (A true, R false) 1× 2025
- Hindu state named in the Baburnama: **Mewar** (Vijayanagara also named) 1× 1995
- **Babri mosque** (Ayodhya) built by **Mir Baqi** 1× 1992

TRAP ALERT

- Mughals = **Chaghtai Turks**, not Mongols or Persians.
- Jihad was declared at **Khanwa**, not Panipat.
- Chanderi = **1528**; Ghaghra = **1529**.
- Babur died at Agra but lies at **Kabul**.

UP LINK

UP link: **Mir Baqi** built the mosque at **Ayodhya** (1528) in Babur's name; Babur died at **Agra** (1530), and the **Battle of Ghaghra** (1529) was fought in eastern UP.

LAST-MINUTE 10

1. Mughals = Chaghtai Turks
2. Shaibani Khan beat Babur at Sar-i-Pul
3. Padshah title at Kabul
4. Daulat Khan invited Babur
5. Panipat 1526: Ibrahim Lodi
6. Khanwa 1527: Rana Sanga; jihad
7. Chanderi 1528; Ghaghra 1529
8. Baburnama in Turki
9. Mewar named in the Baburnama
10. Mir Baqi built the Babri mosque

Chapter 14

Humayun and Sher Shah

Built from 15 UPPSC questions (1993–2023). Focus: Humayun's family, Chausa and Kannauj, Sher Shah's rise, death at Kalinjar and tomb at Sasaram, the Purana Qila, revenue and the rupiya.

ASKED MOST · REVISE FIRST

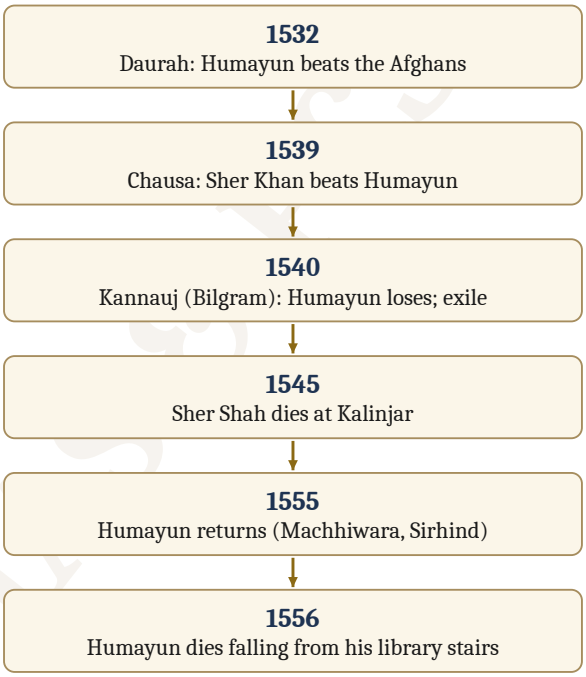
- ■ **Sher Shah**: died at **Kalinjar** (1545), buried at **Sasaram** 4× 1993, 2002, 2015, 2016
- ■ **Patta and qabuliyat**: **Sher Shah** 3× 2004, 2008, 2009
- • **Kannauj (Bilgram), 17 May 1540**: **Sher Shah beat Humayun** 1× 2019
- ■ **Purana Qila** in its present form and its **Qila-i-Kuhna** mosque: **Sher Shah** 3× 2009, 2016, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Humayun

- Wrong pair: **Hamida Banu-wife of Alauddin** (she was Humayun's wife) 1× 2023
- Not Humayun's brother: **Usman** (brothers: Kamran, Askari, Hindal) 1× 2008
- Kannauj, 17 May 1540: **Sher Shah vs Humayun** 1× 2019
- Order: **Daurah, Chausa, Kannauj, Samugarh** 1× 2017

HUMAYUN AND THE AFGHANS



14.2 Sher Shah

- Highly educated ruler: **Sher Shah** (studied at Jaunpur) 1× 2010
- **Jaita and Kumpa** of **Marwar** impressed Sher Shah 1× 2022
- "Lost the empire for a handful of millet": **Sher Shah** 1× 2007
- ■ Died at **Kalinjar**, his last campaign; buried at **Sasaram** 4× 1993, 2002, 2015, 2016
- ■ **Purana Qila** in present form: **Sher Shah**; **Qila-i-Kuhna** mosque too 3× 2009, 2016, 2021
- • Brilliant administrator (only 1) 1× 2021
- ■ **Patta and qabuliyat**: **Sher Shah** 3× 2004, 2008, 2009
- • Rupiya to dam: **1 : 64** (Sher Shah) 1× 2006

Sher Shah's reform	Detail
Land revenue	Measured land; one-third of produce
Patta, qabuliyat	Title deed and acceptance
Coinage	Silver rupiya, copper dam

Sher Shah's reform	Detail
Roads	Sarak-i-Azam (Grand Trunk Road), sarais
Dak	Horse post at sarais

TRAP ALERT

- Died at **Kalinjar** (UP); buried at **Sasaram** (Bihar).
- Chausa (1539) came **before** Kannauj (1540).
- Jaita and Kumpa were of **Marwar**, not Mewar.
- Rupiya:dam = 1:64 (Sher Shah) but 1:40 under Akbar.

UP LINK

UP link: **Chausa** lies on the UP-Bihar border near Buxar; **Kannauj (Bilgram)** in UP saw Humayun's final defeat (1540), and **Kalinjar fort** (Banda) is where Sher Shah died in 1545; he had studied at **Jaunpur**.

LAST-MINUTE 10

1. Hamida Banu = Humayun's wife (Akbar's mother)
2. Humayun's brothers: Kamran, Askari, Hindal
3. Daurah 1532; Chausa 1539; Kannauj 1540
4. Sher Shah educated at Jaunpur
5. Jaita and Kumpa = Marwar
6. Handful of millet = Sher Shah
7. Died at Kalinjar; buried at Sasaram
8. Purana Qila, Qila-i-Kuhna = Sher Shah
9. Patta, qabuliyat = Sher Shah
10. Rupiya:dam = 1:64

Chapter 15

Akbar

Built from 32 UPPSC questions (1990–2024). Focus: Bairam Khan, conquests, Rajput policy and Haldighati, religious policy, social reforms, mansabdari and revenue, court scholars and translations, Fatehpur Sikri.

ASKED MOST · REVISE FIRST

- **Todar Mal**: **land revenue**; the link between Sher Shah and Akbar 3× 1992, 2007, 2012
- **Religious order**: **jizya abolished (1564), Ibadat Khana (1575), Mahzar (1579), Din-i-Ilahi (1582)** 2× 2003, 2006
- First Rajput marriage alliance: **Kachhwahas** of Amber; **Mewar** never submitted 3× 2004, 2004, 2016
- **Haldighati (1576)**: Pratap's commander **Hakim Sur** 3× 2016, 2016, one-liner
- **Badauni** translated the **Mahabharata** (Razmnama) and the Ramayana 4× 2013, 2015, 2016, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Accession and Conquests

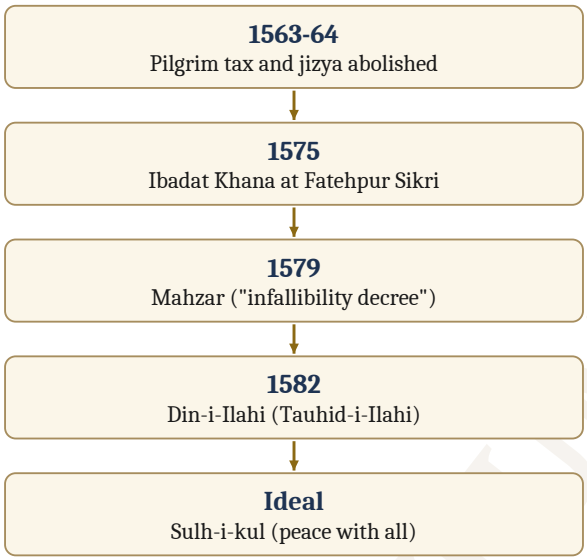
- Akbar gave Bairam Khan the title **Khan-i-Khanan** (only 1) 1× 2024
- Killed personally by Akbar: **Adham Khan** 1× 2010
- Order: **Malwa invasion, Amber marriage, Karrani conquest of Orissa, Tukaroi** 1× 2021
- First Rajput marriage: **Kachhwahas** (Amber) 2× 2004, 2016
- Did not accept Akbar willingly: **Mewar** 1× 2004
- Pratap's other commander at Haldighati: **Hakim Sur** 2× 2016, 2016
- **Haldighati**: **1576** 1× one-liner

15.2 Religion and Society

- Akbar visited **Muinuddin Chishti's** dargah 1× 2012
- Order: **jizya abolished, Ibadat Khana, Mahzar, Din-i-Ilahi** 2× 2003, 2006

- A fatwa against Akbar came from **Jaunpur** 1× 2011
- **Sulh-i-kul**: result of **all the above** influences 1× 1998
- Historian calling Din-i-Ilahi a religion: **none** of them 1× 2000
- Jain scholar honoured by Akbar: **Hiravijaya Suri** 2× 2002, 2012
- Akbar's popularity: **mansabdari, religious policy, revenue, social reform** (all) 1× 1990
- One wife unless the first was barren: **Akbar** 1× 2002
- Akbar tried to fix marriage age and let girls choose (both) 1× 2020

AKBAR'S RELIGIOUS PATH



15.3 Administration and Culture

- Military system: **mansabdari** 1× 1992
- First proper diwan: **Muzaffar Khan Turbati** 2× 2002, 2003
- **Todar Mal**: **land revenue** (dahsala); link between Sher Shah and Akbar 3× 1992, 2007, 2012
- Order: **Sheikh Mubarak's death, Faizi's death, Abul Fazl's murder, Daniyal's death** 1× 2024
- **Mahabharata** into Persian (**Razmnama**): **Badauni** 3× 2015, 2016, 2016
- **Ramayana** into Persian: **Badauni** 1× 2013
- "**Zarrin Qalam**": **Muhammad Husain** (calligrapher) 1× 1995
- Buland Darwaza was built for the **Gujarat** victory, not Jahangir's birth (A true, R false) 1× 1998
- Akbar built the **Buland Darwaza** and **Panch Mahal** 2× 1993, 2004
- Not Akbar's fort: **Red Fort, Delhi** (Shah Jahan) 1× 2015
- Contemporary of Elizabeth I: **Akbar**; an "**enlightened despot**" 2× 2002, 2009

Akbar's reform	Detail
Mansabdari	Zat and sawar ranks (from 1573-77)
Dahsala (Zabti)	Todar Mal; 10-year average (1580)
Ain-i-Dahsala	Revenue on average produce and prices
Translation bureau	Maktab Khana; Razmnama, Ramayana
Fatehpur Sikri	Buland Darwaza, Panch Mahal, Diwan-i-Khas

TRAP ALERT

- Jizya was abolished in **1564**, eleven years **before** the Ibadat Khana.
- First diwan = **Muzaffar Khan Turbati**, not Todar Mal.
- Zarrin Qalam = Muhammad Husain; Shirin Qalam = Abdus Samad.
- Buland Darwaza celebrates **Gujarat** (1573), not Salim's birth.

UP LINK

UP link: **Fatehpur Sikri** (Agra) has the Buland Darwaza, Panch Mahal, Ibadat Khana and Salim Chishti's tomb; Akbar's tomb is at **Sikandra**, and the **Agra fort** and **Allahabad fort** are his; a fatwa against him came from **Jaunpur**.

LAST-MINUTE 10

1. Bairam Khan = Khan-i-Khanan; Adham Khan killed by Akbar
2. Kachhwahas first Rajput alliance; Mewar resisted
3. Haldighati 1576; Hakim Sur
4. Jizya 1564; Ibadat Khana 1575; Mahzar 1579; Din-i-Ilahi 1582
5. Hiravijaya Suri = Jain scholar at court
6. Mansabdari; Muzaffar Khan Turbati first diwan
7. Todar Mal = land revenue (dahsala)
8. Badauni = Razmnama, Ramayana in Persian
9. Buland Darwaza = Gujarat victory; Panch Mahal
10. Akbar = Elizabeth I's contemporary

Chapter 16

Jahangir

Built from 14 UPPSC questions (1991–2023). Focus: the revolts of the reign, Nur Jahan's junta, the Mewar treaty, Hawkins and Thomas Roe, Pelsaert, the Tuzuk-i-Jahangiri, Mughal tombs.

ASKED MOST · REVISE FIRST

- ■ **Thomas Roe** (envoy of James I) met Jahangir at **Ajmer**; **Hawkins** came first ("English Khan") 6× 1993, 2007, 2008, 2013, 2015, 2023
- **Treaty with Mewar (1615)** with Amar Singh: under **Jahangir** 2× 2008, 2008
- **Pelsaert** (Dutch) wrote on Jahangir's reign 2× 2010, 2017
- Jahangir wrote his autobiography (**Tuzuk-i-Jahangiri**) in **Persian** 2× 2001, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Revolts and Nur Jahan

- Rewarded the killer of Abul Fazl (Bir Singh Bundela): **Jahangir** 1× 2003
- Rebels against Jahangir: **Khurram, Mahabat Khan, Khusrau** (2, 3, 4) 1× 2013
- Not in Nur Jahan's junta: **Jahangir** (junta: Itimad-ud-Daula, Asaf Khan, Khurram) 1× 2015
- Treaty with Mewar: **Jahangir** (1615) 2× 2008, 2008

16.2 Europeans at Court

- First EIC man at Jahangir's court: **Hawkins** 1× 1993
- "**English Khan**": **William Hawkins** 1× 2015
- Hawkins knew **Turkish** well; he came in 1608, not 1611 (only 2) 1× 2023
- **Thomas Roe** came under **Jahangir**, sent by **James I** 2× 2008, 2013
- Roe met Jahangir at **Ajmer** 1× 2007
- Dutch account of Jahangir's reign: **Francisco Pelsaert** 2× 2010, 2017

Visitor	Ruler	Note
William Hawkins	Jahangir (1608)	English Khan; first EIC envoy
Thomas Roe	Jahangir (1615-19)	James I's envoy; Surat factory rights
Francisco Pelsaert	Jahangir	Dutch; Remonstrantie
Peter Mundy, Manucci	Shah Jahan	Travel accounts
Bernier, Tavernier	Shah Jahan, Aurangzeb	French

16.3 Memoirs and Tombs

- Autobiography in Persian: **Jahangir** (Tuzuk-i-Jahangiri) 2× 2001, 2010
- No mausoleum in India: **Babur** (Kabul) and **Jahangir** (Shahdara, Lahore) 2× 1991, 2016

TRAP ALERT

- Hawkins = **first**; Roe = official royal envoy (James I).
- Roe's audience was at **Ajmer**, not Agra.
- "Baby Taj" = Itimad-ud-Daula (Agra); "Second Taj" = Bibi ka Maqbara.
- Asaf Khan never rebelled; Mahabat Khan did (1626).

UP LINK

UP link: Nur Jahan built the tomb of her father **Itimad-ud-Daula** ("Baby Taj") at **Agra**; Jahangir completed **Akbar's tomb at Sikandra** and set up his "chain of justice" at the **Agra fort**.

LAST-MINUTE 10

1. Bir Singh Bundela killed Abul Fazl; Jahangir rewarded him
2. Khurram, Mahabat Khan, Khusrau rebelled
3. Nur Jahan's junta: Itimad-ud-Daula, Asaf Khan, Khurram
4. Mewar treaty 1615 (Amar Singh)
5. Hawkins first; English Khan
6. Thomas Roe at Ajmer; James I's envoy
7. Pelsaert = Dutch account
8. Tuzuk-i-Jahangiri in Persian
9. Jahangir buried at Shahdara (Lahore)
10. Itimad-ud-Daula = Baby Taj (Agra)

Chapter 17

Shah Jahan

Built from 15 UPPSC questions (1992–2022). Focus: the move to Delhi and the end of sijda, the Red Fort and Jama Masjid, the Kohinoor, court poets and Sanskrit scholars, Dara Shikoh's Sirr-i-Akbar.

ASKED MOST · REVISE FIRST

- **Upanishads** into Persian (**Sirr-i-Akbar**) by **Dara Shikoh** under **Shah Jahan** 6× 1992, 2000, 2002, 2004, 2006, 2009
- **Shah Jahan** moved the capital to **Delhi**, built the **Red Fort** and **Jama Masjid** 4× 2002, 2004, 2008, 2010
- Abolished **sijda**: **Shah Jahan** (Balban began it) 2× 2010, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

17.1 Court and Buildings

- Capital moved from Agra to **Delhi**: **Shah Jahan** 1× 2008
- Abolished **sijda**: **Shah Jahan** 2× 2010, 2010
- **Lal Qila** (Delhi): **Shah Jahan** 2× 2002, 2010
- **Jama Masjid**, Delhi: **Shah Jahan** 1× 2004
- **Kohinoor** given by **Mir Jumla** 1× 2015
- Poet laureate: **Kalim** 1× 2015
- Patronised **Kavindracharya Saraswati** of Banaras: **Shah Jahan** 1× 2022
- Led Banaras pandits to end the pilgrim tax: **Kavindracharya** 1× 2000
- Deccan governor for most of the reign: **Aurangzeb** 1× 2009

17.2 Dara Shikoh

- Upanishads into Persian under **Shah Jahan** 3× 1992, 2006, 2009
- Title: **Sirr-i-Akbar**, by **Dara Shikoh** 3× 2000, 2002, 2004
- "**Shah Buland Iqbal**": **Dara Shikoh** 1× 2014
- First Muslim to study Hindu scriptures (key): **Dara Shikoh** 1× 2003
- Dara is buried in **Delhi** (Humayun's tomb) 1× 2017

Shah Jahan's buildings	Place
Taj Mahal	Agra
Moti Masjid (Agra fort), marble palaces	Agra
Red Fort, Jama Masjid, Shahjahanabad	Delhi
Peacock Throne	Delhi court

TRAP ALERT

- Sijda began under **Balban**, ended under **Shah Jahan**.
- Agra fort = Akbar; Delhi Red Fort = Shah Jahan.
- Moti Masjid: Agra = Shah Jahan; Delhi Red Fort = **Aurangzeb**.
- Sirr-i-Akbar (Dara) has nothing to do with Emperor Akbar.

UP LINK

UP link: the **Taj Mahal** at **Agra** is Shah Jahan's tomb for Mumtaz Mahal; **Kavindracharya** of **Banaras** won relief from the pilgrim tax on Banaras and Allahabad, and Shah Jahan spent his last years captive in the **Agra fort**.

TATTVA IAS & PCS

LAST-MINUTE 10

1. Capital to Delhi; Red Fort, Jama Masjid
2. Abolished sijda
3. Mir Jumla gave the Kohinoor
4. Kalim = poet laureate
5. Kavindracharya of Banaras
6. Aurangzeb = Deccan governor
7. Dara = Sirr-i-Akbar (Upanishads)
8. Dara = Shah Buland Iqbal
9. Dara buried in Delhi
10. Taj Mahal = Agra

Chapter 18

Aurangzeb

Built from 17 UPPSC questions (1992–2022). Focus: the war of succession, two coronations, Zinda Pir and jizya, Prince Akbar's revolt, Purandar, Bijapur and Golconda, family and buildings.

ASKED MOST · REVISE FIRST

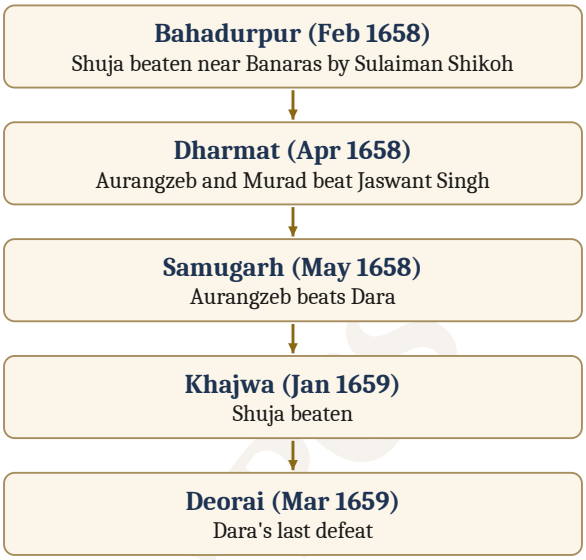
- ■ **Treaty of Purandar (1665)**: Shivaji with **Jai Singh** 5× 2008, 2009, 2010, 2010, 2011
- ■ Aurangzeb conquered **Bijapur (1686)** and **Golconda (1687, Abul Hasan Qutb Shah)** 4× 1992, 2004, 2004, 2020
- ■ Aurangzeb had the **most Hindu generals**; re-imposed **jizya**; called "**Zinda Pir**" 5× 1998, 2000, 2002, 2004, 2017
- • **Crowned twice**: **Aurangzeb** 2× 2006, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Succession

- Aurangzeb succeeded, but **not** by primogeniture (A true, R false) 1× 1994
- Crowned twice: **Aurangzeb** (1658, 1659) 2× 2006, 2009
- Order: **Shuja's defeat near Banaras, Dharmat, Samugarh, Deorai** 1× 2022
- Order: **Bilgram (1540), Sarnal (1573), Dharmat (1658), Jajau (1707)** 1× 2019

WAR OF SUCCESSION (1657-59)



18.2 Policy and Revolts

- "**Zinda Pir**": **Aurangzeb** 1× 2017
- Re-imposed **jizya**: **Aurangzeb** (1679) 1× 2002
- ■ Most Hindu generals: **Aurangzeb** 3× 1998, 2000, 2004
- Son who revolted: **Prince Akbar** (1681) 1× 2007
- ■ **Purandar**: Shivaji and **Jai Singh** 5× 2008, 2009, 2010, 2010, 2011
- Deccan conquests: **Bijapur and Golconda** 2× 2004, 2004
- Bijapur won in **1686** 1× 1992
- Golconda ruler in 1687: **Abul Hasan Qutb Shah** 1× 2020
- **Samarth Ramdas**: **Aurangzeb's** time 1× 2009

18.3 Family and Buildings

- Aurangzeb's daughter: **Mehrunnisa** 1× 2005
- "**Sahibat-uz-Zamani**": **Jahanara** 1× 2014
- "Second Taj": tomb of **Rabia-ud-Daurani** (Bibi ka Maqbara) 2× 2000, 2013
- **Moti Masjid**, Red Fort: **Aurangzeb** 2× 2001, 2002

TRAP ALERT

- Purandar = **Jai Singh**; Afzal Khan and Shaista Khan are other episodes.
- Bijapur 1686 = Sikandar Adil Shah; Golconda 1687 = Abul Hasan.
- Shuja's defeat near Banaras was by **Sulaiman Shikoh**, not Aurangzeb.
- Aurangzeb is buried at **Khuldabad**.

UP LINK

UP link: **Shuja** was defeated near **Banaras** (Bahadurpur, 1658) and at **Khajwa** (Fatehpur, 1659); Aurangzeb built the **Gyanvapi mosque** at Varanasi and mosques at **Mathura**, and Shivaji escaped from **Agra** in 1666.

LAST-MINUTE 10

1. Succession not by primogeniture
2. Crowned twice: 1658 and 1659
3. Dharmat, Samugarh, Deorai
4. Zinda Pir; jizya re-imposed 1679
5. Most Hindu nobles under Aurangzeb
6. Prince Akbar revolted (1681)
7. Purandar 1665 = Jai Singh
8. Bijapur 1686; Golconda 1687 (Abul Hasan)
9. Jahanara = Sahibat-uz-Zamani
10. Bibi ka Maqbara = Second Taj; Moti Masjid (Delhi)

Chapter 19

Mughal Administration

Built from 20 UPPSC questions (1992–2021). Focus: the central officers, subah-sarkar-pargana, mansabdari, land revenue terms, coins, Europeans in Mughal service and canals.

ASKED MOST · REVISE FIRST

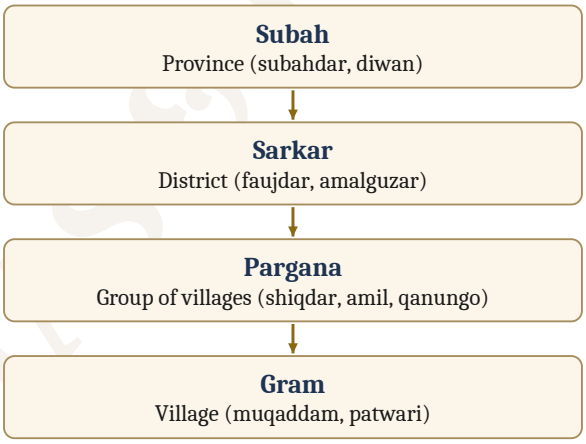
- ■ **Mansabdari**: started by **Akbar**; **zat and sawar** ranks; **not hereditary**; sawar could **not** exceed zat 4× 2009, 2010, 2010, 2019
- • **Mir Bakhshi** = head of the **army** (pay and recruitment) 2× 1992, 2004
- • **Land revenue** = main income; **mal** = land revenue 2× 1995, 2009
- ■ **Coins**: Babur's silver **shahrukhi**; copper **dam**; Akbar's **Ram-Siya** coins 3× 2008, 2011, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Officers and Units

- The Mughal state was first a **military state** (A and R true, R explains A) 1× 2021
- Army head: **Mir Bakhshi**; also oversaw officials 2× 1992, 2004
- Wazirs: **Akbar-Muzaffar Khan, Jahangir-Itimad-ud-Daula, Shah Jahan-Sadullah Khan, Aurangzeb-Asad Khan** 1× 2021
- **Diwan-i-Tan**: jagirs and salaries (with other officers' duties) 1× 2020
- District: **sarkar** 1× 2009
- **Chakla**: unit **between subah and pargana**, not the same as a sarkar 1× 2018

MUGHAL ADMINISTRATIVE UNITS



19.2 Mansabdari and Revenue

- Mansabdars were the official nobility started by **Akbar**; **not hereditary** (only 1) 1× 2019
- Not true: **sawar rank could exceed zat** 1× 2009
- Zat and mansab granted; **three classes** (1 and 3) 2× 2010, 2010
- Main income: **land revenue** 1× 1995
- **Mal**: **land revenue** 1× 2009
- Akbar's Deccan revenue: by the **number of ploughs** 1× 2017

Land class	Fallow period
Polaj	Cultivated every year
Parauti	Fallow 1-2 years
Chachar	Fallow 3-4 years
Banjar	Fallow 5 years or more

19.3 Coins, Europeans and Canals

- Silver **shahrukhi**: **Babur** 1× 2017

- Akbar regulated currency like Sher Shah; dam continued (A and R true, R explains A, per source) 1× 2019
- Copper coin: **dam** 1× 2008
- **Ram-Siya** coins: **Akbar** 1× 2011
- Physician in Mughal service: **Manucci** 1× 2008
- Banned tobacco: **Jahangir** 1× 2005
- Canal restoring Firuz's Rajabwah: **Shahab Nahar** 1× 2020

TRAP ALERT

- District = **sarkar**, not subah.
- Mansabs were **not** hereditary; sawar never exceeded zat.
- Mir Bakhshi = military paymaster; Diwan-i-Tan = jagirs and salaries.
- Rupiya:dam = 1:40 under Akbar (1:64 under Sher Shah).

UP LINK

UP link: the **subahs** of **Agra, Allahabad (Ilahabad)** and **Awadh** covered most of present UP; Akbar created the Allahabad subah and built its fort at the Sangam.

LAST-MINUTE 10

1. Mughal state = military state at first
2. Mir Bakhshi = army head
3. Subah, sarkar, pargana, village
4. Chakla = between subah and pargana
5. Mansabdari: Akbar; zat and sawar; not hereditary
6. Land revenue = main income; mal = revenue
7. Polaj, parauti, chachar, banjar
8. Shahrukhi = Babur; dam = copper; Ram-Siya = Akbar
9. Manucci = physician; Jahangir banned tobacco
10. Shahab Nahar restored Firuz's canal

Chapter 20

Music and Painting in the Mughal Period

Built from 15 UPPSC questions (1991–2021). Focus: Humayun and Akbar's painters, Jahangir's golden age, Abul Hasan and Mansur, regional schools, Swami Haridas, Tansen and Dhrupad, the emperors and music.

ASKED MOST · REVISE FIRST

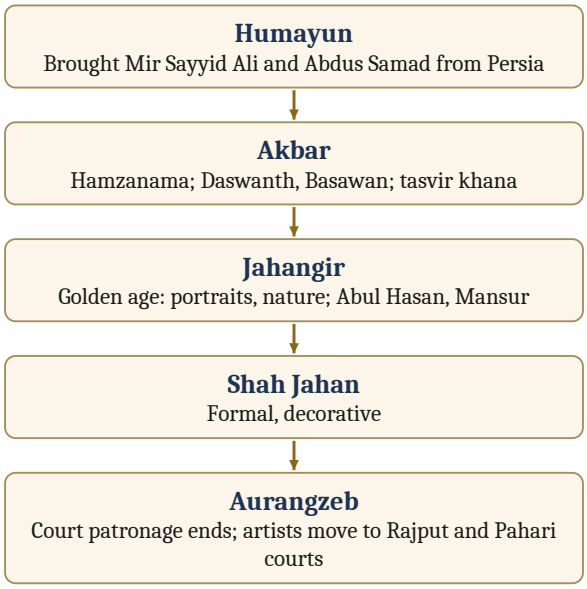
- **Mughal painting** founded by **Humayun** (Mir Sayyid Ali, Abdus Samad) 3× 2009, 2010, 2012
- **Jahangir** patronised **painting**: **Abul Hasan** (Nadir-uz-Zaman), **Mansur** (Nadir-ul-Asr, birds) 5× 1998, 2012, 2013, 2014, 2016
- **Tansen**: pupil of **Swami Haridas**; tomb at **Gwalior**; earlier patron Raja **Ramchandra of Rewa** 4× 1999, 2009, 2010, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Painting

- Foundation of Mughal painting: **Humayun** 3× 2009, 2010, 2012
- Statements on Mughal painting: **all** true 1× 1991
- Jahangir mainly patronised **painting** 1× 2016
- Jahangir's painters: **Abul Hasan and Aqa Riza** (2 and 3) 1× 2012
- "**Nadir-uz-Zaman**": **Abul Hasan** 1× 2013
- "**Nadir-ul-Asr**" and greatest bird painter: **Mansur** 2× 1998, 2014
- Pahari, Rajput, Mughal, Kangra: styles of **painting** 1× 1994

GROWTH OF MUGHAL PAINTING



20.2 Music

- Haridas' followers set up **5** music centres (per key) **1×** 2009
- Akbar's Dhrupad singers: **Tansen and Haridas** (1 and 2) **1×** 2010
- **Tansen's tomb**: **Gwalior** **1×** 1999
- Tansen's patron before Akbar: **Raja Ramchandra** (Baghela) **1×** 2019
- Not a great name in Hindustani music: **Mal Gurjari** (a raga) **1×** 2021
- Learnt Hindu music from Lala Kalawant: **Akbar** **1×** 2016
- Aurangzeb played the **veena** **2×** 2007, 2010

TRAP ALERT

- Nadir-uz-Zaman = **Abul Hasan**; Nadir-ul-Asr = **Mansur**.
- Mughal painting began under **Humayun**, not Akbar.
- Tansen's tomb is at **Gwalior**; he sang at Fatehpur Sikri.
- Aurangzeb ended court music but could play the veena.

UP LINK

UP link: **Swami Haridas** lived at **Vrindavan** (Mathura), where Tansen is said to have learnt from him; **Fatehpur Sikri** had Akbar's painting workshops and Tansen's performances.

LAST-MINUTE 10

1. Humayun founded Mughal painting
2. Mir Sayyid Ali and Abdus Samad = Persian masters
3. Jahangir = golden age of painting
4. Abul Hasan = Nadir-uz-Zaman
5. Mansur = Nadir-ul-Asr, birds
6. Kangra and Basohli = Pahari schools
7. Haridas = Tansen's guru (Vrindavan)
8. Tansen buried at Gwalior; patron Ramchandra of Rewa
9. Akbar learnt from Lala Kalawant
10. Aurangzeb played the veena

Chapter 21

Mughal Literature

Built from 20 UPPSC questions (1994–2023). Focus: the Baburnama, Gulbadan Begum and Maham Anaga, Persian at court and the Akbarnama, chronicler-book matching, Persian translations, Hindi Bhakti poets and Sufi premakhyans.

ASKED MOST · REVISE FIRST

- **Gulbadan Begum** (Babur's daughter) wrote the **Humayun-nama** **6×** 1994, 1998, 2002, 2004, 2004, 2004
- **Ain-i-Akbari** = third volume of the **Akbarnama**; court language = **Persian** **3×** 2012, 2014, 2023
- **Chronicler matching**: **Hasan Nizami-Taj-ul-Maasir, Khwandamir-Humayun-nama, Muhammad Kazim-Alamgirnama, Bhimsen-Nuskha-i-Dilkusha, Sujan Rai-Khulasat-ut-Tawarikh, Chandrabhan-Chahar Chaman** **2×** 2003, 2017
- **Maham Anaga** built the **Madrasa-i-Begum** and **Khair-ul-Manazil** mosque **2×** 2012, 2018

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

21.1 Memoirs and Women Writers

TATTVA IAS

- Baburnama into English: **Leyden and Erskine** 1× 2017
- Correct pair: **Babur-Tuzuk-i-Baburi** 1× 2009
- **Gulbadan Begum**: daughter of **Babur**; wrote the **Humayun-nama** 6× 1994, 1998, 2002, 2004, 2004, 2004
- **Madrasa-i-Begum** and **Khair-ul-Manazil** (opposite Purana Qila): **Maham Anaga** 2× 2012, 2018

21.2 Persian Histories and Translations

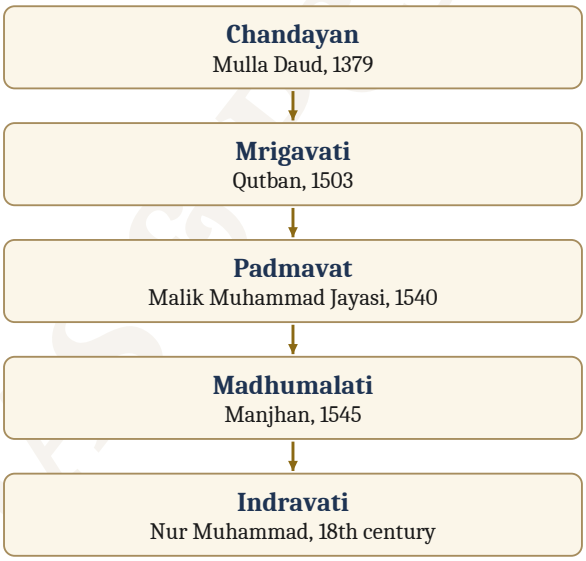
- Court language: **Persian** 1× 2012
- Akbarnama finished in **seven years** (per key) 1× 2014
- Third volume of the Akbarnama: **Ain-i-Akbari** 1× 2023
- Wrong pair: **Tughlaqnama-Ibn Battuta** (it is **Amir Khusrau's**) 1× 2019
- **Hasan Nizami-Taj-ul-Maasir, Khwandamir-Humayun-nama, Muhammad Kazim-Alamgirnama, Bhimsen-Nuskha-i-Dilkusha** 1× 2003
- **Bhimsen-Nuskha-i-Dilkusha, Chandrabhan-Chahar Chaman, Ishwardas Nagar-Futuhat-i-Alamgiri, Sujan Rai Bhandari-Khulasat-ut-Tawarikh** 1× 2017
- **Hitopadesha** into Persian: **Taj-ul-Mali** 1× 2013
- **Anwar-i-Suhaili**: translation of the **Panchatantra** 1× 1999

Persian work	Author	Subject
Akbarnama, Ain-i-Akbari	Abul Fazl	Akbar
Muntakhab-ut-Tawarikh	Badauni	Akbar (critical)
Tabaqat-i-Akbari	Nizamuddin Ahmad	Akbar
Padshahnama	Abdul Hamid Lahori	Shah Jahan
Alamgirnama	Muhammad Kazim	Aurangzeb
Muntakhab-ul-Lubab	Khafi Khan	Later Mughals

21.3 Hindi Poets and Premakhyans

- **Ramchandrika** and **Rasikpriya**: **Keshavdas** 1× 2008
- Krishna verses as "**Nagari Das**": **Raja Sawant Singh** (Kishangarh) 2× 2004, 2010
- **Ras Panchadhyayi-Nand Das, Chaitanya Bhagavata-Vrindavan Das, Chaitanya Mangal-Jayananda, Bhakti Ratnakar-Narahari Chakravarti** 1× 2021
- Wrong pair: **Gangabai-Ganesh Dev Leela** 1× 2023
- Written first: **Mrigavati** (Qutban, 1503) 1× 2014

SUFI PREMAKHYANS IN HINDI



TRAP ALERT

- Only **Babur** and **Jahangir** wrote their own memoirs.
- Two Humayun-namas: **Gulbadan** (memoir), **Khwandamir** (Qanun-i-Humayuni).
- Tughlaqnama = **Amir Khusrau**, not Ibn Battuta (Rihla).
- Mrigavati is older than Padmavat.

UP LINK

UP link: **Malik Muhammad Jayasi** wrote the **Padmavat** (1540) at **Jais** (Amethi); **Qutban's Mrigavati** came from the Jaunpur region, and **Tulsidas** and **Surdas** wrote in Awadhi and Braj in UP.

LAST-MINUTE 10

1. Leyden and Erskine = Baburnama in English
2. Gulbadan Begum = Humayun-nama
3. Maham Anaga = Madrasa-i-Begum, Khair-ul-Manazil
4. Persian = court language
5. Ain-i-Akbari = third volume of the Akbarnama
6. Kazim = Alamgirnama; Bhimsen = Nuskha-i-Dilkusha
7. Sujan Rai = Khulasat-ut-Tawarikh
8. Anwar-i-Suhaili = Panchatantra
9. Keshavdas = Rasikpriya, Ramchandrika
10. Mrigavati before Padmavat

Chapter 22

Mughal Period: Miscellaneous

Built from 26 UPPSC questions (1992–2025). Focus: English travellers, the order of rulers and battles, revolts against Aurangzeb, women rulers, burial places, monuments and their builders, court terms.

ASKED MOST · REVISE FIRST

- **Traveller-date matching: Ralph Fitch 1585-86, Hawkins 1608-11, Roe 1615-19, Manucci 1653-1708** 2× 2016, 2021
- **Burial places: Babur-Kabul, Humayun-Delhi, Akbar-Sikandra, Jahangir-Lahore, Shah Jahan-Agra, Aurangzeb-Khuldabad** 4× 1992, 2005, 2010, one-liner
- **Monument-builder: Babur-Sambhal Jama Masjid, Humayun-Dinpanah, Akbar-Jahangiri Mahal, Jahangir-Itimad-ud-Daula; Alai Darwaza-Alauddin, Buland Darwaza-Akbar, Moti Masjid Agra-Shah Jahan, Moti Masjid Delhi-Aurangzeb** 5× 2006, 2010, 2010, 2010, 2011
- **Battle order: Khanwa, Chanderi, Daurah, Chausa, Kannauj, Talikota, Samugarh** 3× 2016, 2016, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

22.1 Travellers and Rulers

- **Hawkins-1608-11, Roe-1615-19, Manucci-1653-1708, Ralph Fitch-1585-86** 1× 2016
- Order: **Ralph Fitch, Hawkins, Downton, Roe** 1× 2021
- Order: **Jahangir, Muhammad Shah, Ahmad Shah Abdali, Bahadur Shah II** 3× 2001, 2002, 2003
- True: **Abdali fought Third Panipat; Jahangir loved nature** (1 and 4) 1× 2010
- **Third Panipat-1761, Second Tarain-1192, Akbar's death-1605, Aurangzeb's death-1707** 1× 2005

22.2 Battles and Revolts

- **Haldighati-Akbar, Bilgram-Humayun, Khusrau's revolt-Jahangir, Khanwa-Babur** 1× 2012
- Wrong pair: **Jahangir-Battle of Balkh** (Balkh = Shah Jahan) 1× 2010
- Battles order (1998 item): Tarain, Haldighati, Asirgarh (1601), Third Panipat 1× 1998
- Order: **Daurah, Chausa, Kannauj, Samugarh** 1× 2025
- Order: **Chanderi, Chausa, Kannauj, Talikota** 1× 2016
- Order: **Khanwa, Chausa, Kannauj, Talikota** 1× 2016
- **Jat-1669, Satnami-1672, Sikh-1675**; Bundela dated 1554 per key (*) 1× 2023
- Women rulers in order: **Padmini, Durgavati, Tarabai, Ahilyabai** 2× 2001, 2002

22.3 Tombs and Monuments

- **Babur-Kabul, Humayun-Delhi, Jahangir-Lahore, Shah Jahan-Agra** 1× one-liner
- **Babur-Kabul, Akbar-Sikandra, Jahangir-Lahore, Shah Jahan-Agra** 1× 2005
- **Akbar's tomb: Sikandra** (Agra) 1× 1992
- Correct: **Delhi-tomb of Abdur Rahim Khan-i-Khanan** 1× 2010
- **Babur-Sambhal Jama Masjid, Humayun-Dinpanah, Akbar-Jahangiri Mahal, Jahangir-Itimad-ud-Daula's tomb** 2× 2010, 2011

TATTVA IAS & PCS

- **Alai Darwaza-Alauddin, Buland Darwaza-Akbar, Moti Masjid (Agra)-Shah Jahan, Moti Masjid (Delhi)-Aurangzeb** 2× 2006, 2010
- Order: **Atala Mosque, Sher Shah's tomb, Humayun's tomb, Rabia Daurani's tomb** 1× 2019
- **Govind Palace** (Hindu architecture): **Datia** 1× 2005

Emperor	Tomb
Babur	Kabul
Humayun	Delhi
Akbar	Sikandra (Agra)
Jahangir	Shahdara (Lahore)
Shah Jahan	Taj Mahal (Agra)
Aurangzeb	Khuldabad (Maharashtra)

22.4 Festivals, Learning and Terms

- **Nauroz** adopted from the **Zoroastrians** (Persians) 1× 2001
- Madrasa for Muslim jurisprudence: **Lucknow** (Firangi Mahal) 1× 2016
- Wrong pair: **jarib-a type of tax** (it is a measuring chain) 1× 2019
- **Muallim**: a ship's **navigator**; no option exact (*) 1× 2014

TRAP ALERT

- Jahangiri Mahal (Agra fort) was built by **Akbar**; Itimad-ud-Daula's tomb by **Nur Jahan**.
- Kannauj and Bilgram are the **same** battle (1540).
- Talikota (1565) had no Mughal side but often ends order questions.
- Jarib = measuring chain; muallim = navigator.

UP LINK

UP link: Babur's **Jama Masjid at Sambhal**, Akbar's **Sikandra** tomb, the **Taj Mahal** and **Itimad-ud-Daula** (Agra) are in UP, and **Firangi Mahal, Lucknow** was the great school of Islamic jurisprudence.

LAST-MINUTE 10

1. Fitch 1585; Hawkins 1608; Roe 1615; Manucci 1653
2. Jahangir, Muhammad Shah, Abdali, Bahadur Shah II
3. Third Panipat 1761; Akbar died 1605; Aurangzeb 1707
4. Jat 1669; Satnami 1672; Sikh 1675
5. Padmini, Durgavati, Tarabai, Ahilyabai
6. Babur Kabul; Humayun Delhi; Akbar Sikandra; Jahangir Lahore
7. Sambhal Jama Masjid = Babur; Dinpanah = Humayun
8. Moti Masjid Agra = Shah Jahan; Delhi = Aurangzeb
9. Nauroz from Zoroastrians; Firangi Mahal = Lucknow
10. Jarib = measuring chain

Chapter 23

Sikh Sect

Built from 10 UPPSC questions (1996–2021). Focus: Guru Nanak, succession and the ten Gurus, Gurmukhi and Amritsar, the executions of Arjan Dev and Tegh Bahadur, Gobind Singh and the Khalsa, Banda Bahadur.

ASKED MOST · REVISE FIRST

- Gurus executed by the Mughals: **Arjan Dev** (Jahangir) and **Tegh Bahadur** (Aurangzeb) 2× 2001, 2002
- **Guru Gobind Singh** founded the **Khalsa** (1699); his samadhi is at **Nanded** 2× 2002, 2006
- ■ **Guru Nanak**: born at **Nankana**; founded Sikhism under **Sikandar Lodi**; successor **Lehna (Angad)** 3× 2007, 2008, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

23.1 The Gurus

- Guru Nanak's birthplace: **Nankana** (Talwandi) 1× 2007
- Founded Sikhism under **Sikandar Lodi** 1× 2008

- Nanak's successor: **Lehna** (Guru Angad) 1× 2021
- **Gurmukhi** started by **Guru Angad** 1× 2017
- **Amritsar** founded by **Guru Ram Das** 1× 1996
- Put to death: **Arjan Dev and Tegh Bahadur** (2 and 4) 1× 2002
- Helped rebel prince **Khusrau**: **Guru Arjan Dev** 1× 2001
- **Khalsa** founded by **Guru Gobind Singh** 1× 2006
- **Nanded**: samadhi of **Guru Gobind Singh** 1× 2002
- Banda Bahadur's original name: **Lachhman Dev** 1× 2014

Guru	Known for
Nanak	Founder; Kartarpur
Angad	Gurmukhi script
Amar Das	Langar, manjis
Ram Das	Founded Amritsar
Arjan Dev	Adi Granth (1604); executed under Jahangir
Hargobind	Akal Takht; militarised the Sikhs
Tegh Bahadur	Executed under Aurangzeb (1675)
Gobind Singh	Khalsa (1699); died at Nanded

TRAP ALERT

- Amritsar = **Ram Das**; Harmandir and Adi Granth = **Arjan Dev**.
- Hargobind was imprisoned at Gwalior but **not** executed.
- Gobind Singh: born at **Patna**, Khalsa at **Anandpur**, died at **Nanded**.

UP LINK

UP link: Guru Nanak visited **Varanasi, Ayodhya** and **Prayag**; Guru Tegh Bahadur stayed at **Prayagraj** (Pakki Sangat) and Agra, where he was arrested before his execution in Delhi (1675).

LAST-MINUTE 10

1. Nanak born at Nankana
2. Sikhism under Sikandar Lodi
3. Lehna = Guru Angad; Gurmukhi
4. Amritsar = Ram Das
5. Arjan Dev executed (Jahangir); helped Khusrau
6. Tegh Bahadur executed (Aurangzeb, 1675)
7. Gobind Singh = Khalsa 1699
8. Nanded = Gobind Singh's samadhi
9. Banda Bahadur = Lachhman Dev
10. Hargobind = Akal Takht

Chapter 24

Maratha State and Confederacy

Built from 25 UPPSC questions (1990–2025). Focus: why the Marathas rose, Shivaji's campaigns and coronation, Ashtapradhan, chauth and coins, the successors and Tarabai, the Peshwas, Third Panipat.

ASKED MOST · REVISE FIRST

- **Ashtapradhan** = Shivaji's council of eight ministers 4× 1991, 1995, 1997, 2013
- **Third Panipat (1761)**: Marathas beaten by the **Afghans** under **Ahmad Shah Abdali**; Najib Khan Rohilla his ally 5× 1993, 1994, 2004, 2006, 2012
- **Peshwa order**: **Balaji Vishwanath, Baji Rao I, Balaji Baji Rao, Madhav Rao I, Narayan Rao, Raghunath Rao** 4× 2004, 2023, 2024, 2025
- Shivaji's capital: **Raigarh**, crowned there in **1674** 3× 1990, 2015, 2016
- **Tarabai** led the Marathas at Aurangzeb's death 3× 2008, 2012, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Shivaji

- Reasons for the Maratha rise: **all of the above** 1× 1992
- Big Deshmukhs opposed Shivaji as they wished to stay Bijapur's vassals (A and R true, R explains A) 1× 2024

- Born **1627** (key; 1630 is also used), crowned **1674** 1× 2015
- Bijapur sent **Afzal Khan** (1659) 1× 1999
- Order: **Chakan, Afzal Khan, clash with the Mughals, sack of Surat** 1× 2016
- Shivaji beat the Mughals at **Salher** 1× 2005
- Capital and coronation: **Raigarh** 2× 1990, 2016

24.2 Administration

- **Ashtapradhan**: **Maratha** council 3× 1991, 1995, 2013
- **Sar-i-Naubat**: **military** 1× 2014
- Ministers advised Shivaji but did **not** hold independent charge (A true, R false) 1× 1997
- Not a silver coin: **ruka** 1× 2016
- Protection levy: **chauth** 1× 2018

Ashtapradhan	Duty
Peshwa	Prime minister
Amatya (Majumdar)	Finance
Sachiv	Correspondence
Mantri (Waqianavis)	Records, intelligence
Sar-i-Naubat	Army
Sumant (Dabir)	Foreign affairs
Panditrao	Religion
Nyayadhish	Justice

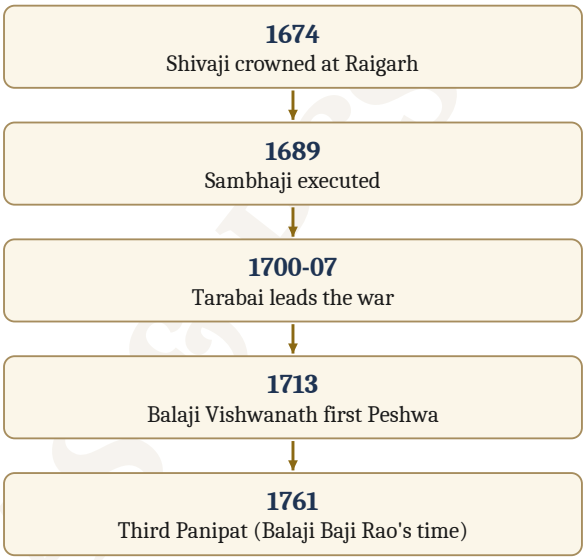
24.3 Successors and Peshwas

- Order: **Sambhaji, Rajaram, Shivaji II, Shahu** 1× 2005
- At Aurangzeb's death the leader was **Tarabai** 3× 2008, 2012, 2013
- Peshwa order: **Balaji Vishwanath, Baji Rao I, Balaji Baji Rao, Madhav Rao I** 2× 2004, 2025
- Order: **Balaji Vishwanath, Baji Rao I, Madhav Rao I, Narayan Rao** 1× 2023
- Order: **Balaji Vishwanath, Balaji Baji Rao, Narayan Rao, Raghunath Rao** 1× 2024
- By 1750 a confederacy under the Peshwa (A and R true, R explains A) 1× 1997
- Peshwa became the real ruler under **Balaji Baji Rao** 1× 2007
- Strongest native power after the Mughals, but the reason over-claims (A true, R false) 1× 2016

24.4 Third Panipat

- Marathas lost to the **Afghans** 3× 1993, 1994, 2012
- Fought between **Marathas and Ahmad Shah Abdali** 1× 2004
- Abdali's Rohilla confidant: **Najib Khan** 1× 2006

MARATHA TIMELINE



TRAP ALERT

- Chauth = **one-fourth** (protection); sardeshmukhi = **one-tenth**.
- Shahu ruled **after** Rajaram and Shivaji II (he was a Mughal prisoner).
- Balaji Vishwanath (first Peshwa) is not Balaji Baji Rao (Panipat).

UP LINK

UP link: the Marathas raided the **Doab** and **Rohilkhand**; **Najib Khan** of the Rohillas (Najibabad, Bijnor) and **Shuja-ud-Daula** of **Awadh** sided with Abdali at Third Panipat; Shivaji escaped from **Agra** in 1666.

LAST-MINUTE 10

1. Shivaji crowned 1674 at Raigarh
2. Afzal Khan (Bijapur) 1659; Salher victory
3. Ashtapradhan = eight ministers
4. Chauth = protection; sardeshmukhi = hereditary 10%
5. Sambhaji, Rajaram, Shivaji II, Shahu
6. Tarabai led after 1700
7. Balaji Vishwanath, Baji Rao I, Balaji Baji Rao, Madhav Rao I
8. Peshwa supreme under Balaji Baji Rao
9. Third Panipat 1761: Abdali (Afghans)
10. Najib Khan = Rohilla ally of Abdali

Chapter 25

Disintegration of the Mughal Empire

Built from 15 UPPSC questions (2004–2023). Focus: the later Mughals in order, Farrukhsiyar's farman and the Sayyid brothers, Muhammad Shah and Nadir Shah, the last emperors, Awadh and Hyderabad, Sawai Jai Singh.

ASKED MOST · REVISE FIRST

- **Later Mughals**: **Bahadur Shah I, Jahandar Shah, Farrukhsiyar, Muhammad Shah, Ahmad Shah, Alamgir II** 3× 2012, 2018, 2023
- **Farrukhsiyar's farman (1717)**: duty-free trade for the English in Bengal; **Sayyid brothers** = kingmakers 2× 2010, 2017
- **Nadir Shah** (1739) attacked under **Muhammad Shah**; tappa music grew at his court 3× 2004, 2004, 2023
- **Saadat Khan** = first Nawab of **Awadh**; **Chin Qilich Khan** founded **Hyderabad** 3× 2008, 2011, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

25.1 Later Mughals

- Aurangzeb's successor: **Bahadur Shah I** 1× 2012
- Order: **Bahadur Shah, Jahandar Shah, Farrukhsiyar, Muhammad Shah** 1× 2023
- Order: **Jahandar Shah, Muhammad Shah, Ahmad Shah, Alamgir II** 1× 2018
- Duty-free trade to the English in Bengal: **Farrukhsiyar** 1× 2010
- Kingmakers: **Hussain Ali and Abdullah** (Sayyid brothers) 1× 2017
- Nadir Shah's attack: under **Muhammad Shah** 1× 2004
- Eunuchs and ladies ruled under **Muhammad Shah** 1× 2004
- **Tappa** music refined at **Muhammad Shah's** court 1× 2023
- Bahadur Shah Zafar's father: **Akbar Shah II** 1× 2005
- Not correct of Zafar: pension **under one lakh** 1× 2006

DECLINE OF THE MUGHALS

1707
Aurangzeb dies; Bahadur Shah I

↓

1713-19
Farrukhsiyar; Sayyid brothers; 1717 farman

↓

1739
Nadir Shah sacks Delhi (Karnal)

↓

1761
Third Panipat

↓

1857
Bahadur Shah Zafar exiled to Rangoon

25.2 Successor States and Jai Singh

- First Nawab of Awadh: **Saadat Khan** (Burhan-ul-Mulk) 1× 2008
- Saadat Khan founded Awadh and joined the plot against the Sayyids (both) 1× 2023
- Founder of Hyderabad: **Chin Qilich Khan** (Nizam-ul-Mulk) 1× 2011
- **Jantar Mantar**, Delhi: **Jai Singh II** 1× 2005
- Jai Singh's observatories: **Delhi, Jaipur, Ujjain, Varanasi** (and Mathura) 1× 2007

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Successor state	Founder
Awadh	Saadat Khan (1722)
Hyderabad	Nizam-ul-Mulk Chin Qilich Khan (1724)
Bengal	Murshid Quli Khan (1717)
Rohilkhand	Ali Muhammad Khan

TRAP ALERT

- Order: Bahadur Shah I, **Jahandar Shah**, Farrukhsiyar, Muhammad Shah.
- Nadir Shah = **Muhammad Shah** (1739); Abdali = 1748-1761.
- First Nawab of Awadh = **Saadat Khan**, not Safdar Jang.
- No Mughal is called "Akbar Shah I"; Zafar's father was **Akbar II**.

UP LINK

UP link: **Awadh** grew from Saadat Khan's governorship (capital Faizabad, later **Lucknow** under Asaf-ud-Daula); Jai Singh's observatory at **Varanasi** (Man Mahal) still stands.

LAST-MINUTE 10

1. Bahadur Shah I succeeded Aurangzeb
2. Jahandar, Farrukhsiyar, Muhammad Shah, Ahmad Shah, Alamgir II
3. Farrukhsiyar = 1717 farman
4. Sayyid brothers = kingmakers
5. Nadir Shah 1739 under Muhammad Shah
6. Muhammad Shah = Rangeela; tappa
7. Akbar II = Zafar's father
8. Saadat Khan = first Nawab of Awadh
9. Chin Qilich Khan = Hyderabad
10. Jai Singh II = Jantar Mantar (Delhi, Varanasi too)

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Modern History

PYQ-Based Revision Pointers

Chapters 1 to 42

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Chapter 1

The Advent of European Companies

Built from 28 UPPSC questions (1990–2021). Focus: the order of arrival, Vasco da Gama and Albuquerque, European factories in Bengal, the English charters and Surat, the French and Danes, the Carnatic wars, why the English won.

ASKED MOST · REVISE FIRST

- **Portuguese** came first; **Vasco da Gama** reached **Calicut** in **1498** (welcomed by the **Zamorin**) **8×** 1990, 1992, 1993, 2000, 2004, 2004, 2007, 2013
- Order of entry into trade: **Portuguese, Dutch, English, French** **1×** 1997
- First English factory: **Surat** (1613), in **Akbar's** time the EIC was formed (1600) **5×** 1990, 2004, 2010, 2011, 2012
- **Carnatic wars**: **English vs French**; **Wandiwash (1760)** settled it **4×** 1995, 2015, 2016, 2017
- **Settlement matching**: **Pondicherry-French, Goa-Portuguese, Tranquebar-Danish, Sadras-Dutch** **1×** 2016

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

1.1 The Portuguese and the Dutch

- Order of entry: **Portuguese, Dutch, English, French** **1×** 1997
- First Europeans to trade: **Portuguese** **6×** 1990, 1993, 2000, 2004, 2004, 2007
- Vasco da Gama came in **1498**, welcomed by the **Zamorin** of Calicut **2×** 1992, 2013
- Real founder of Portuguese power: **Albuquerque** **1×** 2003
- First Portuguese fort: **Cochin** **1×** 2010
- Dutch trade centres: **all the above** **1×** 2017
- Portuguese Bengal factory: **Hooghly**; Dutch: **Chinsurah** **2×** 2004, 2010

Company	Founded	First factory	Main base
Portuguese	1498 (voyage)	Calicut, Cochin	Goa
Dutch	1602	Masulipatnam (1605)	Pulicat, Nagapattinam
English	1600	Surat (1613; Masulipatnam 1611)	Bombay, Madras, Calcutta
Danish	1616	Tranquebar (1620)	Serampore
French	1664	Surat (1668)	Pondicherry

1.2 The English, French and Danes

- First English charter: **Levant Company** **1×** 2014
- EIC formed in the reign of **Akbar** **2×** 2004, 2012
- First factory at Surat: **the British** (English) **1×** 2011
- First EIC factory: **Surat** **2×** 1990, 2010
- **Calcutta** founded by **Job Charnock** **1×** 2004
- First municipal corporation: **Madras** (1688) **1×** 2021
- British control of the Prayagraj fort: question erroneous (*); ceded in **1798** **1×** 2021
- French company under **Louis XIV**; founded by **Colbert** **3×** 2003, 2013, 2014
- Pondicherry: Portuguese did not occupy it first; only 1 per key **1×** 2006
- **Pondicherry-French, Goa-Portuguese, Tranquebar-Danish, Sadras-Dutch** **1×** 2016

1.3 Carnatic Wars and English Success

- Carnatic wars: **English vs French** **1×** 1995
- Wrong pair: **Second Carnatic War 1750-52** (it ran 1749-54) **1×** 2017
- **First Carnatic-Treaty of Aix-la-Chapelle, Second-inconclusive, Third-Treaty of Paris, First Mysore War-British lost** **1×** 2016
- Aix-la-Chapelle (1748): ended the first war; Madras returned to the English **1×** 2020
- **Wandiwash** (1760): **British defeated the French** **1×** 2015
- EIC's success: **all the above three** **1×** 1994
- French failed in the Deccan: the English had a **strong army** **1×** 1998

QUANTUM IAS & PCS

- English won through Bengal's wealth and **naval superiority** (both) 1× 2016

CARNATIC WARS

First (1746-48)
Dupleix took Madras; Aix-la-Chapelle returned it

↓

Second (1749-54)
Succession fights; Clive at Arcot (1751)

↓

Third (1756-63)
Wandiwash (1760): Eyre Coote beat Lally

↓

Result
Treaty of Paris (1763); French power ends

TRAP ALERT

- By **formation** English (1600) precede Dutch (1602); by **entry** Dutch precede English.
- First Portuguese fort and capital = **Cochin**; Goa from 1530.
- **Sadras** was Dutch, not English; Tranquebar and Serampore were Danish.
- Plassey fell during the Third Carnatic War but was not a Carnatic battle.

UP LINK

UP link: the **Allahabad fort** passed to the Company in **1798** when Nawab **Saadat Ali Khan II** of Awadh ceded it; Agra and Lucknow were early sites of English trade agents under Jahangir.

LAST-MINUTE 10

1. Portuguese first; Vasco da Gama 1498 at Calicut
2. Zamorin = ruler of Calicut
3. Albuquerque = real founder of Portuguese power
4. Cochin = first Portuguese fort
5. Hooghly = Portuguese; Chinsurah = Dutch
6. EIC 1600 under Akbar; Surat first factory
7. Job Charnock = Calcutta; Madras = first corporation
8. French company under Louis XIV; Colbert
9. Pondicherry French; Tranquebar Danish; Sadras Dutch
10. Wandiwash 1760: English beat French

Chapter 2

East India Company and Nawabs of Bengal

Built from 11 UPPSC questions (1990–2021). Focus: Murshid Quli Khan, Plassey and Clive, Mir Qasim and Buxar, the Diwani of 1765, Awadh's Hindu officials, and resistance to the Company.

ASKED MOST · REVISE FIRST

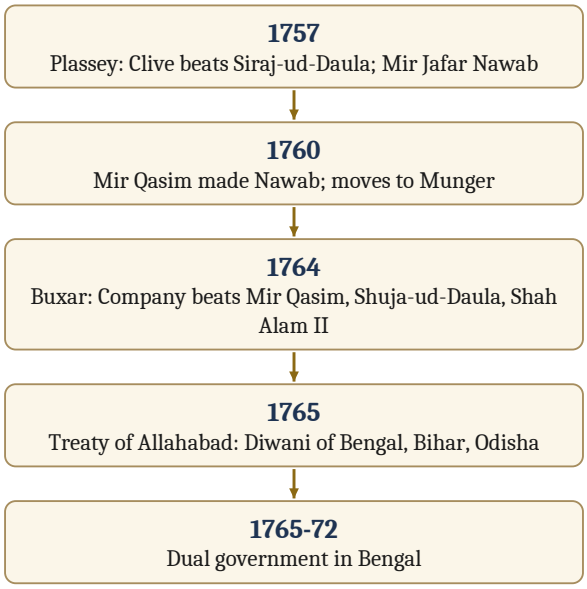
- **Buxar (1764)** = the most decisive battle for British supremacy 5× 1990, 2002, 2003, 2012, 2016
- **Diwani** of Bengal, Bihar, Odisha granted by **Shah Alam II** in **Clive's** time (1765) 4× 2003, 2004, 2006, 2010
- **Plassey (1757)**: **Clive** defeated **Siraj-ud-Daula**; Clive = "heaven-born general" 2× 2005, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Bengal

- Last Governor of Bengal appointed by the Mughal emperor: **Murshid Quli Khan** 2× 2013, 2021
- Siraj-ud-Daula lost at **Plassey** 1× 2005
- "**Heaven-born general**": **Robert Clive** 1× 2009
- Capital moved from Murshidabad to **Munger**: **Mir Qasim** 1× 2003
- Most decisive battle: **Buxar** 5× 1990, 2002, 2003, 2012, 2016
- Nawab of Bengal during Buxar: **Mir Jafar** (restored) 1× 2016
- Diwani granted by **Shah Alam II** (Mughal emperor), under **Clive** 4× 2003, 2004, 2006, 2010

FROM PLASSEY TO DIWANI



2.2 Awadh and Resistance

- Hindu ministers of Awadh: **Tikait Rai** was finance minister; Hindu castes held high posts (both) 1× 2016
- Opposed the British most vigorously (among options): **Marathas** 1× 1993

TRAP ALERT

- Buxar is in **Bihar**, opposite Ballia (UP).
- Diwani came from **Shah Alam II**; Clive was Governor at the time.
- Mir Jafar was Nawab during Buxar (Mir Qasim had been deposed).

UP LINK

UP link: **Shuja-ud-Daula** of Awadh fought at Buxar, and the **Treaty of Allahabad (1765)** was signed at Prayagraj, giving the Company the Diwani and returning Awadh to Shuja (minus Kora and Allahabad).

LAST-MINUTE 10

1. Murshid Quli Khan = last Mughal-appointed governor
2. Plassey 1757: Clive vs Siraj-ud-Daula
3. Clive = heaven-born general
4. Mir Qasim moved the capital to Munger
5. Buxar 1764 = most decisive battle
6. Mir Jafar was Nawab during Buxar
7. Treaty of Allahabad 1765
8. Diwani from Shah Alam II
9. Tikait Rai = Awadh finance minister
10. Marathas resisted most (among options)

Chapter 3

Regional States: Punjab and Mysore

Built from 17 UPPSC questions (1995–2023). Focus: Ranjit Singh's kingdom and court, the Anglo-Sikh wars, Duleep Singh, Haidar Ali and Tipu Sultan, the Anglo-Mysore wars and treaties.

ASKED MOST · REVISE FIRST

- ■ **Ranjit Singh**: **Sukerchakia** misl; capital **Lahore**; kingdom included **Srinagar**; Adalat-i-Ala at Lahore 4× 1995, 1997, 2004, 2021
- ■ **Duleep Singh** = last Sikh king; died in **Paris** (1893) 3× 2010, 2013, 2013
- ■ **Tipu Sultan**: beat the British at **Pollilur** (1780); set up modern embassies abroad; killed in the **Fourth** war (1799) 3× 2009, 2011, 2014
- • **Treaty-year matching**: **Allahabad 1765, Madras 1769, Salbai 1782, Mangalore 1784** 1× 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Ranjit Singh and Punjab

- Kingdom included **Srinagar** 1× 2004
- Misl: **Sukerchakia** 1× 1997
- Capital: **Lahore** 1× 1995
- **Adalat-i-Ala** set up at **Lahore** 1× 2021

- "God took away the light from my other eye...": **Ranjit Singh** 1× 2007
- Treaty of Lahore (1846): territories beyond the Sutlej and forts given up (I and II) 1× 2023
- Last Sikh king: **Duleep Singh** 2× 2010, 2013
- False of the merger: Duleep's pension was **Rs 2.5 lakh** 1× 2023
- Duleep Singh died in **Paris** (key: 23 October 1893) 1× 2013
- Not on the first Board of Three: **Elliot and Montgomery** 1× 2013

PUNJAB UNDER THE BRITISH

1809
Treaty of Amritsar: Sutlej the boundary

↓

1839
Ranjit Singh dies

↓

1845-46
First Anglo-Sikh war; Treaty of Lahore

↓

1848-49
Second war; Punjab annexed (Dalhousie)

↓

1849
Board of Three: Henry and John Lawrence, Mansel

3.2 Mysore

- Modern armoury at Dindigul (1755): **Haidar Ali** 1× 2020
- First Anglo-Mysore war won by **Haidar Ali** 1× 2015
- Tipu beat the British at **Pollilur** (1780) 1× 2009
- Correct: **Fourth Anglo-Mysore war-Tipu killed** (1799) 1× 2014
- Embassies abroad on modern lines: **Tipu Sultan** 1× 2011
- Order: **Shakarkheda, Bedara, Porto Novo, Mudki** 1× 2022
- **Allahabad-1765, Mangalore-1784, Salbai-1782, Madras-1769** 1× 2019

Anglo-Mysore war	Treaty	Governor-General
First (1767-69)	Madras (1769)	(Bengal: Verelst)
Second (1780-84)	Mangalore (1784)	Warren Hastings
Third (1790-92)	Seringapatam (1792)	Cornwallis
Fourth (1799)	None; Tipu killed	Wellesley

TRAP ALERT

- Treaty of Amritsar **1809** (Ranjit Singh) vs Treaty of Amritsar **1846** (Kashmir to Gulab Singh).
- Haidar **won** the first Anglo-Mysore war.
- Salbai (1782) ended the First Anglo-Maratha war; Mangalore (1784) the second Mysore war.

UP LINK

UP link: the **Treaty of Allahabad (1765)** was signed at Prayagraj; UP's Sikh heritage includes Guru Tegh Bahadur's stay at **Prayagraj**, but the Sikh kingdom itself lay in Punjab.

LAST-MINUTE 10

1. Ranjit Singh = Sukerchakia misl, Lahore
2. Kingdom included Srinagar (Kashmir)
3. Adalat-i-Ala at Lahore
4. Duleep Singh = last king; died in Paris
5. Treaty of Lahore 1846; annexation 1849
6. Haidar Ali = Dindigul armoury; won the first war
7. Tipu won at Pollilur 1780
8. Tipu = embassies abroad; died 1799
9. Madras 1769; Salbai 1782; Mangalore 1784
10. Shakarkheda, Bedara, Porto Novo, Mudki

Chapter 4

Governors-General and Viceroys

Built from 68 UPPSC questions (1990–2025). Focus: Warren Hastings to Mountbatten: their wars, reforms and laws; namesakes; who did what.

ASKED MOST · REVISE FIRST

- **Ripon** = father of **local self-government** (1882) **5×** 1994, 1996, 2010, 2015, 2017
- **Dalhousie**: **doctrine of lapse** (Satara first), **Awadh annexed for misrule (1856)**, first railway **Bombay-Thane** **11×** 1990, 1991, 1994, 2001, 2004, 2010, 2011, 2013, 2016, 2019, 2023
- **Wellesley** = **subsidiary alliance**; **Treaty of Bassein (1802)** with **Baji Rao II** **5×** 2010, 2011, 2012, 2014, 2016
- **Canning**: first Viceroy; read the **Queen's Proclamation at Allahabad** (1 Nov 1858); widow remarriage act **6×** 2001, 2002, 2003, 2004, 2009, 2010
- **Curzon**: longest tenure (among options), **Ancient Monuments Act**, ASI revived, **partition of Bengal** **7×** 2009, 2009, 2010, 2010, 2014, 2019, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 Warren Hastings to Wellesley

- Order: **Buxar, Clive's return, Treaty of Allahabad, Hastings as Governor** (*) **1×** 2016
- **Asiatic Society** (1784) under **Warren Hastings** **1×** 2014
- "**Ring fence**" policy: **Warren Hastings** **1×** 2006
- **Banaras rebellion**: **Raja Chait Singh** **1×** 2021
- Impeached by Parliament: **Warren Hastings** **2×** 2006, 2010
- Judicial organisation: **Cornwallis** **1×** 1991
- Civil service enforced under **Cornwallis** **1×** 2004
- Local face of "Company Bahadur": the **daroga** **1×** 2021
- **Cornwallis' tomb**: **Ghazipur** **1×** 2011
- Third Anglo-Mysore war: **Cornwallis** **1×** 2013
- Subsidiary alliance: **Wellesley** **1×** 2011
- Order of alliances: **Hyderabad, Mysore, Awadh, Scindia** **1×** 2014
- Who refused the alliance: all signed at some point (*) **1×** 2004
- **Treaty of Bassein** (1802): **British and Baji Rao II** **2×** 2012, 2016
- Wrong pair: **subsidiary alliance-1863** (it was 1798) **1×** 2010

4.2 Lord Hastings to Ellenborough

- Anglo-Nepal war: **Lord Hastings** **1×** 2010
- Treaty of Sugauli gave **Ranikhet and Nainital** areas (3 and 4) **1×** 2025
- Wrong pairs: **John Shore-Anglo-Nepal war**, **Cornwallis-Third Anglo-Maratha war**, **Lord Hastings-Second Anglo-Maratha war** **3×** 2001, 2010, 2014
- Peshwaship abolished: **1818**; Third Anglo-Maratha war: **Lord Hastings** **2×** 2013, 2015
- **Thomas Munro**, Madras: **1820-1827** **1×** 2016
- **Sati** abolished: **William Bentinck** (1829) **2×** 1990, 2012
- Took over Mysore for misrule: **Bentinck** **2×** 2003, 2004
- Not true: Bentinck "one of the greatest British rulers as conqueror" **1×** 2010
- "The Company is an anomaly...": **Macaulay** **1×** 2017
- Wrong pair: **Lytton-first Anglo-Afghan war** (that was Auckland) **1×** 2013
- Sindh conquered under **Ellenborough** (**1843**) **2×** 2012, 2015
- Slavery abolished: **Ellenborough** (1843) **3×** 2008, 2010, 2011

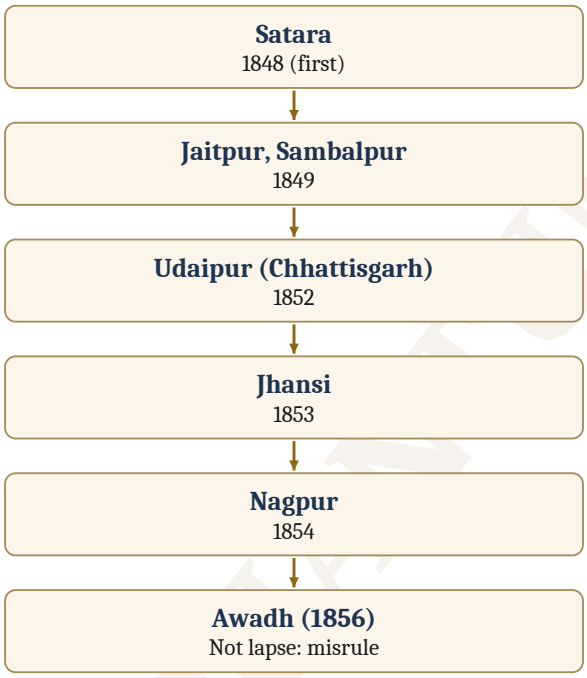
4.3 Dalhousie and Canning

- Awadh annexed for **maladministration** (1856) **5×** 1990, 1991, 1994, 2010, 2013
- Resident at Awadh then: **James Outram** **1×** 2011
- **Doctrine of lapse**: **Dalhousie**; first victim **Satara** **2×** 2011, 2016
- Lapse order: **Satara, Jaitpur-Sambalpur (Bhagat), Udaipur, Jhansi** **1×** 2019

QUANTUM IAS & PCS

- Governor-General match (A-2, B-1, C-4, D-3 per key) 1× 2017
- Not annexed by lapse: **Karnataka** (Carnatic) 1× 2023
- Correct: **Wellesley-Fourth Anglo-Mysore war** 1× 2013
- Order of mergers: **Sikkim (part, 1850), Punjab, Burma (Lower), Jhansi** per key 2× 2002, 2016
- Merged Sikkim (partly): **Dalhousie** 1× 2020
- First railway: **Bombay-Thane** (1853) 2× 2001, 2004
- Queen's Proclamation read at **Allahabad** by **Canning** 2× 2001, 2009
- First Viceroy: **Canning** 1× 2010
- Right of adoption restored by the **Queen's Proclamation (1858)** 1× 2004
- Widow Remarriage Act: under **Canning** 2× 2002, 2003

DOCTRINE OF LAPSE



4.4 Viceroys

- "**Masterly inactivity**": **John Lawrence** 1× 1997
- First census (1872): **Mayo** 2× 2000, 2004
- Viceroy killed in the Andamans: **Mayo** (1872) 3× 2004, 2013, 2023
- Local self-government (1882): **Ripon**, its father 4× 1994, 1996, 2010, 2015
- Wrong pair: **local self-government-Lytton** 1× 2017
- Order: **Awadh annexation, Indigo revolt, Second Afghan war, Ilbert Bill** 1× 2025
- Longest tenure among options: **Curzon** 1× 2009
- ASI under **Curzon** (key; founded 1861 under Canning) 2× 2009, 2010
- **Ancient Monuments Preservation Act** (1904): **Curzon** 2× 2005, 2010
- Curzon: "empire on granite rock"; Calcutta Municipal Act (both) 1× 2023
- 1905: provinces redrawn; new **East Bengal and Assam** (both) 1× 2019
- Compared Curzon with Aurangzeb: **Gokhale** 1× 2012
- "Divide and rule": **Curzon and Minto II** 1× 2014
- Capital to Delhi: **Hardinge II** 1× 2006
- Only Jewish Viceroy: **Reading** 1× 2014

4.5 Who Did What

- **Cornwallis-Permanent Settlement, Bentinck-sati ban, Curzon-partition of Bengal, Dalhousie-doctrine of lapse** 1× 2025
- **Sati ban, local self-government, partition, lapse**: Bentinck, Ripon, Curzon, Dalhousie 1× 1995
- **Metcalfe** removed press curbs (1835) 1× 1994
- Correct: **Cornwallis-Permanent Settlement** 3× 1998, 1998, 2008

Governor-General / Viceroy	Remember for
Warren Hastings	Asiatic Society, Regulating Act, impeachment
Cornwallis	Permanent Settlement, civil service, police
Wellesley	Subsidiary alliance, Fourth Mysore war
Lord Hastings	Nepal war, Third Maratha war
Bentinck	Sati ban, English education (1835)
Dalhousie	Lapse, railways, telegraph, Awadh
Canning	1857, first Viceroy, universities
Ripon	Local self-government, Ilbert Bill
Curzon	Partition of Bengal, Ancient Monuments Act

TRAP ALERT

- **Warren Hastings** (1772-85) is not **Lord Hastings** (1813-23).
- Awadh was annexed for **misrule**, not by lapse.
- First census = **Mayo** (1872); first regular census = **Ripon** (1881).
- Third Mysore war = Cornwallis; Third Maratha war = Lord Hastings.

UP LINK

UP link: **Cornwallis** is buried at **Ghazipur**; **Raja Chait Singh** of **Banaras** revolted against Warren Hastings (1781); **Awadh** was annexed in 1856 (Outram), and **Canning** read the Queen's Proclamation at **Allahabad** (Minto Park) on 1 November 1858.

LAST-MINUTE 10

1. Warren Hastings = Asiatic Society, ring fence, impeached
2. Chait Singh = Banaras rebellion
3. Cornwallis = Permanent Settlement; tomb at Ghazipur
4. Wellesley = subsidiary alliance; Bassein 1802
5. Lord Hastings = Nepal war, Peshwaship ended 1818
6. Bentinck = sati ban; Ellenborough = Sindh, slavery ban
7. Dalhousie = lapse (Satara first), Awadh 1856, railway 1853
8. Canning = Proclamation at Allahabad; first Viceroy
9. Mayo = first census, killed in the Andamans
10. Ripon = local self-government; Curzon = partition, monuments

Chapter 5

Economic Impact of British Rule on India

Built from 18 UPPSC questions (1991–2023). Focus: Permanent, Ryotwari and Mahalwari settlements, the village service systems, plantations, the drain of wealth, military spending, deindustrialisation.

ASKED MOST · REVISE FIRST

- **Permanent Settlement (1793)**: **Cornwallis** 4× 1991, 2007, 2010, 2010
- **Ryotwari**: **Thomas Munro**; **Madras and Bombay** (south and west India) 5× 2000, 2001, 2008, 2014, 2016
- **Drain of wealth**: **Dadabhai Naoroji**, "**Poverty and Un-British Rule in India**" (1901) 5× 1995, 2004, 2004, 2007, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Land Settlements

- Permanent Settlement: **Cornwallis, 1793** 4× 1991, 2007, 2010, 2010
- **Munro**: **Ryotwari** settlement (Madras) 2× 2000, 2008
- Ryotwari in **west and south India**; in **Madras and Bombay**; first in **Madras** 3× 2001, 2014, 2016
- Different systems in different regions (A and R true, R explains A) 1× 2020
- Wrong pair: **taluqdari-Bombay** (taluqdari was in **Awadh**) 1× 2023
- **Jajmani-North India, Bara Balute-Maharashtra, Mirasi-Tamil Nadu, Adade-Karnataka** 1× 2020

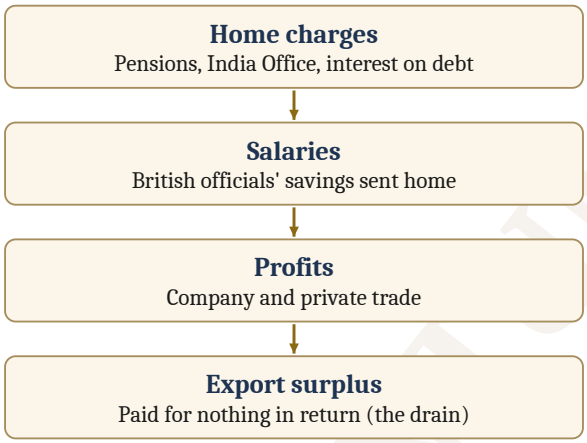
System	Year	Where	With whom
Permanent (Zamindari)	1793	Bengal, Bihar, Odisha, Banaras	Zamindars
Ryotwari	1792 (Read), 1820 (Munro)	Madras, Bombay, Assam	Cultivators

System	Year	Where	With whom
Mahalwari	1822 (Holt Mackenzie), 1833	North-West Provinces (UP), Punjab	Village community

5.2 Drain, Finance and Industry

- First tea company in Assam: **1839** 1× 2016
- **Drain theory**: **Dadabhai Naoroji** 2× 1995, 2004
- **"Poverty and Un-British Rule in India": 1901**, by **Naoroji** 3× 2004, 2007, 2021
- Favourable trade balance was the form the drain took (A and R true, R explains A) 1× 2017
- Central revenue on the army: about **40%** 1× 2021
- **Deindustrialisation** began around **1813**, hastened by ending the EIC monopoly (both) 1× 2017

HOW THE DRAIN WORKED



TRAP ALERT

- Mahalwari belongs to the **North-West Provinces (UP)** and Punjab, not Bengal.
- Taluqdari = **Awadh**, not Bombay.
- Ryotwari's pioneer was Read, but the standard answer is **Munro**.

UP LINK

UP link: the **Mahalwari** system was first applied in the **North-Western Provinces** (Agra region of UP) under Holt Mackenzie (1822) and R.M. Bird; **Awadh** had the taluqdari system, and the **Banaras** region came under the Permanent Settlement.

LAST-MINUTE 10

1. Permanent Settlement 1793 = Cornwallis
2. Ryotwari = Munro; Madras, Bombay
3. Mahalwari = North-West Provinces (UP), Punjab
4. Taluqdari = Awadh
5. Jajmani = North India; Bara Balute = Maharashtra
6. First Assam tea company 1839
7. Drain theory = Dadabhai Naoroji
8. Poverty and Un-British Rule 1901
9. About 40% of revenue on the army
10. Deindustrialisation from 1813

Chapter 6

Revolt of 1857

Built from 35 UPPSC questions (1990–2024). Focus: causes, Barrackpore and Meerut, leaders at each centre, the British side, why it failed, its nature, the Queen's Proclamation.

ASKED MOST · REVISE FIRST

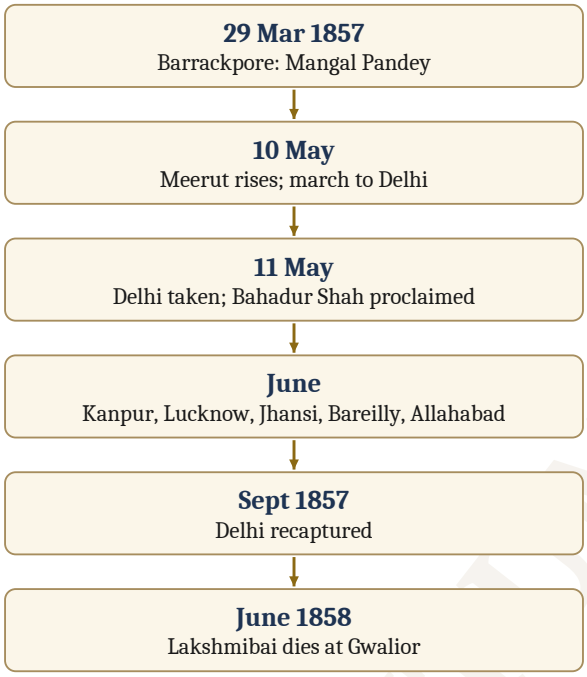
- **Centre-leader matching: Delhi-Bakht Khan/Bahadur Shah, Kanpur-Nana Saheb (Tatya Tope, Azimullah), Lucknow-Begum Hazrat Mahal, Jhansi-Lakshmibai, Faizabad-Maulvi Ahmadullah, Bareilly-Khan Bahadur, Allahabad-Liyaqat Ali, Assam-Maniram Dutta** 8× 1998, 2007, 2010, 2015, 2015, 2016, 2016, 2023
- Revolt began at **Meerut** (10 May 1857); **Mangal Pandey** at **Barrackpore** (29 March) 5× 1990, 1994, 2008, 2010, 2023
- **Canning** was Governor-General; made **Allahabad** his emergency HQ; **Palmerston** was British PM 5× 1990, 1991, 2005, 2012, 2013
- **Savarkar** called it the first war of independence; **R.C. Majumdar** said it was neither first, nor national, nor a war of independence 2× 2010, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Outbreak

- Main cause: **British imperial policy** 1× 1990
- **Mangal Pandey**: **Barrackpore** 1× 2010
- Mangal Pandey's regiment (34th BNI) not in the options (*) 1× 2014
- British officer at Barrackpore: **Hearsey** 1× 2014
- Began at **Meerut** 2× 1990, 1994
- First event: sepoys **marching to Delhi's Red Fort** 1× 2008
- Wrong pair: **Jhansi-11 May 1857** 1× 2023

SPREAD OF THE REVOLT



6.2 Leaders and Centres

- "Sahib-i-Alam Bahadur": **Bakht Khan** 1× 2016
- Recaptured by the English (key): **Delhi** 1× 2015
- Nana Saheb's commander-in-chief: **Tatya Tope** (real name **Ramchandra Pandurang**) 3× 2008, 2010, 2011
- Lucknow's leader: **Begum Hazrat Mahal** 1× 2015
- Most rebel soldiers came from **Awadh** 1× 2015
- Kunwar Singh of **Jagdishpur** led Bihar; Dalhousie did **not** recognise Lakshmibai's adopted son (only 1) 1× 2024
- Lakshmibai born at **Varanasi** 1× 2008
- Bareilly: **Khan Bahadur** 2× 1998, 2023
- Allahabad: **Maulvi Liyaqat Ali** 1× 2015
- Assam: **Maniram Dutta** 1× 2007
- Not linked to 1857: **Udham Singh** 1× 1996
- **Jhansi-Lakshmibai, Lucknow-Hazrat Mahal, Kanpur-Azimullah, Faizabad-Ahmadullah Shah** 1× 2010
- **Nana Saheb-Kanpur, Hamid Ali Khan-Delhi, Ahmadullah-Lucknow, Maniram-Assam** 1× 2016

6.3 British Side, Failure and Nature

- Governor-General: **Canning** 3× 1990, 2012, 2013
- Emergency HQ at **Allahabad**: **Canning** 1× 2005
- British PM: **Palmerston** 1× 1991
- Not linked with the revolt: **all of the above** 1× 2017
- Died at Lucknow: **Neill, Havelock, Henry Lawrence** (2, 3, 4) 1× 2008
- Failed for **all the above** reasons 1× 1998
- Educated middle class: **stayed neutral** 1× 1998
- **Mirza Ghalib**: originally of **Agra** 1× 2002
- "First war of independence": **Savarkar** 1× 2015
- Official historian of the freedom struggle: **S.N. Sen** (1857 volume) 1× 2010
- First Indian-language work on its causes: **Syed Ahmad Khan** 1× 2009
- "Neither first, nor national...": **R.C. Majumdar** 1× 2010
- It ended **Company rule** 1× 2010
- Proclamation promise kept: **end of annexation** 1× 1994

TRAP ALERT

- **Barrackpore** came first, but the revolt is said to start at **Meerut**.
- Lakshmibai died at **Gwalior**, not Jhansi; born at **Varanasi**.
- Udham Singh (1940) has nothing to do with 1857; Kunwar Singh does.
- Awadh was annexed for misrule; its sepoys formed the Bengal Army.

QUANTUM IAS & PCS

UP LINK

UP link: UP was the heart of 1857: **Meerut** (start), **Kanpur** (Nana Saheb, Tatya Tope, Azimullah), **Lucknow** (Begum Hazrat Mahal, Residency siege), **Jhansi** (Lakshmibai), **Bareilly** (Khan Bahadur), **Allahabad** (Liyaqat Ali), **Faizabad** (Ahmadullah Shah); Canning ran the war from **Allahabad**.

LAST-MINUTE 10

1. Mangal Pandey = Barrackpore, 29 March 1857
2. Meerut, 10 May = start
3. Bakht Khan = Delhi commander
4. Nana Saheb = Kanpur; Tatya Tope = Ramchandra Pandurang
5. Begum Hazrat Mahal = Lucknow; Ahmadullah = Faizabad
6. Lakshmibai = Jhansi (born Varanasi, died Gwalior)
7. Khan Bahadur = Bareilly; Liyaqat Ali = Allahabad; Maniram = Assam
8. Canning (Allahabad HQ); Palmerston PM
9. Savarkar = first war of independence; Majumdar disagreed
10. Ended Company rule; Queen's Proclamation 1858

Chapter 7

Other Civil Uprisings

Built from 28 UPPSC questions (1994–2025). Focus: the order and dates of revolts, Sanyasi and Anandamath, chiefs' risings, Wahabi, Faraizi and Kuka, Santhal and Munda, north-east risings, peasant movements, early mutinies.

ASKED MOST · REVISE FIRST

- **Revolt-year matching**: **Santhal 1855-56, Indigo 1859-60, Jaintia 1860-63, Pabna 1873-76 (to 1885), Rampa 1879-80, Moplah 1921, Eka 1922, Tana Bhagat 1914, Kol 1831-32, Ahom 1828** 6× 2016, 2016, 2017, 2018, 2019, 2019
- **Anandamath** (Bankim Chandra) is based on the **Sanyasi revolt** 3× 1998, 2015, 2017
- **Wahabi** centre **Patna**; **Faraizi** leader **Dadu Miyan**; **Kuka** in **Punjab** 3× 1994, 1999, 2017
- **Birsa Munda** led the Munda ulgulan; **Moplah** revolt (1921) in **Malabar** 3× 1998, 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 Chronology and Early Risings

- Order: **Sanyasi, Indigo, Kuka, Pabna** 1× 2025
- Year pairs: Sanyasi and Kol pairs wrong (*) 1× 2017
- **Anandamath**: on the **Sanyasi revolt**; made famous by **Bankim Chandra** 3× 1998, 2015, 2017
- **Paika-Jagabandhu, Bareilly-Mufti Muhammad Aiwaz, Malabar-Edachena Kungan, Sylhet-Radharam** 1× 2024
- **Velu Thampi**: **Kerala** (Travancore) 1× 2002
- **Gadkari** rebellion: **Kolhapur** 1× 1999
- **Kallar**: **Madurai** 1× 2017

7.2 Religious and Tribal Movements

- Wahabi centre: **Patna** 1× 1994
- **Faraizi** leader: **Dadu Miyan** 1× 1999
- **Kuka** movement: **Punjab** 1× 2017
- **Pagal Panthi**: rising of the **Garos** 1× 1999
- Santhal Hul: Major Burroughs lost near Bhagalpur; Gokko of Godda; Mahajan Deen Dayal Rai (1, 2, 3) 1× 2017
- Munda leader: **Birsa** 1× 1998
- Wrong pair: **Nayak-Tana Bhagat** 1× 2008
- In Maharashtra: **Bhil** uprising 1× 1998
- Wrong pair: **Chuar-** (and partly Koli) region 1× 2015
- **Kuki-Tripura, Kuka-Punjab, Pabna-Bengal, Birsa-Bihar** 1× 2010

QUANTUM IAS & PCS

- **Ahom-1828, Kol-1831-32, Santhal-1855-56, Moplah-1921** 1× 2017
- Wrong pair: **Ahom-1815** 1× 2018
- Revolt-year pairs: Kuka pair wrong (*) 1× 2019

Rising	Region	Leader
Sanyasi-Fakir	Bengal	Majnu Shah
Paika	Odisha	Bakshi Jagabandhu
Kol	Chotanagpur	Buddhu Bhagat
Santhal	Rajmahal hills	Sidhu, Kanhu
Munda (Ulgulan)	Ranchi	Birsa Munda
Kuka	Punjab	Ram Singh
Rampa	Andhra	Alluri Sitarama Raju (1922-24)

7.3 Peasant Revolts and Mutinies

- **Nil Darpan**: **Dinabandhu Mitra** 1× 2015
- **Rampa-1879-80, Pabna-1873-76, Indigo-1859-60, Jaintia-1860-63** 1× 2016
- Tebhaga-Moplah-Pabna-Indigo: Pabna date wrong (*) 1× 2016
- **Pabna-1873-85, Eka-1922, Santhal-1855-56, Tana Bhagat-1914** 1× 2019
- **Eka** movement: farmers of **Hardoi, Barabanki** (UP) 1× 2007
- **Moplah** revolt (1921): **Malabar** 2× 2010, 2016
- **Barrackpore-1824, Berhampur-February 1857, Santhal-1855-56, Vellore-July 1806** 1× 2006

TRAP ALERT

- **Kuka** = Punjab (Sikh reform); **Kuki** = north-east (Tripura, Manipur).
- Sanyasi revolt is **18th century**, not 1894.
- Nil Darpan = **Dinabandhu Mitra**; Anandamath = **Bankim Chandra**.
- Vellore mutiny = **1806**; Barrackpore mutiny = **1824** (and 1857).

UP LINK

UP link: the **Eka (unity) movement** (1921-22) of **Hardoi, Barabanki, Bahraich, Sitapur** was led by **Madari Pasi**; **Mufti Muhammad Aiwaz** led the **Bareilly** revolt of 1816 against a house tax.

LAST-MINUTE 10

1. Sanyasi, Indigo, Kuka, Pabna (order)
2. Anandamath = Sanyasi revolt (Bankim)
3. Paika = Jagabandhu; Bareilly 1816 = Mufti Aiwaz
4. Wahabi = Patna; Faraizi = Dadu Miyan; Kuka = Punjab
5. Pagal Panthi = Garos; Kallar = Madurai
6. Santhal 1855-56 (Sidhu, Kanhu); Birsa = Munda
7. Ahom 1828; Kol 1831-32
8. Indigo 1859-60 (Nil Darpan); Pabna 1873-76
9. Eka 1922 (Hardoi, Barabanki); Moplah 1921 (Malabar)
10. Vellore 1806; Barrackpore 1824

Chapter 8

Development of Education in Modern India

Built from 23 UPPSC questions (2002–2024). Focus: the Calcutta Madrasa and Fort William College, the Asiatic Society, the Charter Act of 1813 and Macaulay, education commissions, early colleges, Deccan Education Society, Karve, BHU.

ASKED MOST · REVISE FIRST

- **Asiatic Society of Bengal** founded by **William Jones** (1784, Warren Hastings' time); Jones translated **Shakuntala** 6× 2004, 2006, 2010, 2017, 2019, 2021
- **Charter Act 1813**: first **Rs 1 lakh** for education; **Macaulay** (1835, Bentinck) = English education, **filtration theory** 4× 2007, 2009, 2011, 2013
- **Women's university**: **D.K. Karve**; **Deccan Education Society**: **Tilak** 5× 2002, 2006, 2010, 2010, 2013
- **BHU**: grew from **Annie Besant's Central Hindu College**; foundation stone by **Hardinge**; first central university 3× 2003, 2011, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

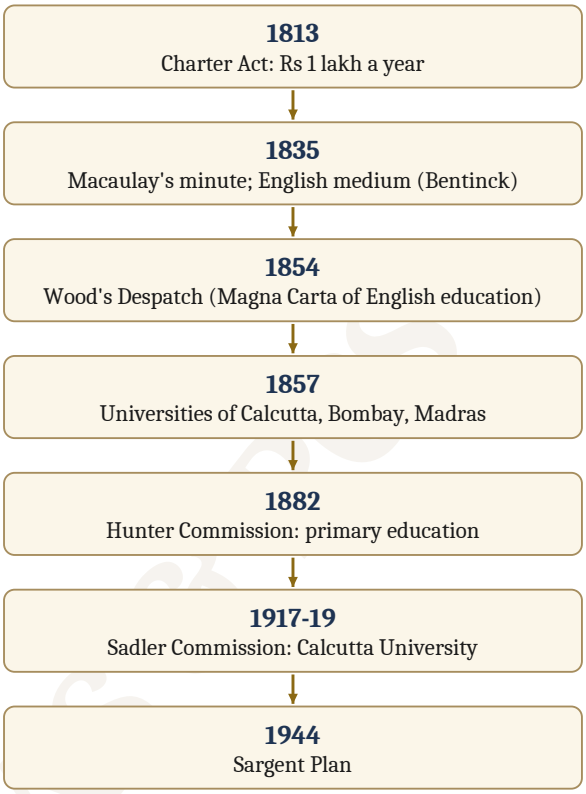
8.1 Early Institutions

- First British madrasa: **Calcutta** (1781) 1× 2006
- Fort William College: to teach Indian languages to British officers (only 2) 1× 2021
- Asiatic Society: **William Jones** 4× 2004, 2006, 2017, 2021
- Asiatic Society came under Warren Hastings, who backed Oriental learning (A and R true; source keys R as explanation) 1× 2019
- First English translation of **Shakuntala**: **William Jones** 1× 2010
- Offered membership of the Royal Asiatic Society of Paris: **Michael Madhusudan Dutt** 1× 2007

8.2 Policy

- First Rs 1 lakh for education: **Charter Act 1813** 1× 2009
- **Macaulay**: **English education** 1× 2007
- English education introduced under **William Bentinck** 1× 2011
- **Filtration theory**: **Macaulay** 1× 2013
- Order: **Wood's Despatch, Hunter, Sadler, Sargent** 1× 2018
- **Hunter Commission**: **primary education** 2× 2004, 2004
- **Sadler Commission**: **education** 1× 2010

EDUCATION POLICY MILESTONES



8.3 Colleges and Universities

- Founded first: **Hindu College, Calcutta** (1817, David Hare) 2× 2012, 2021
- **MAO College, Aligarh**: **1875** 1× 2016
- Not correct of Archbold: principal of **Lahore College** (he was at **MAO College**) 1× 2024
- **Deccan Education Society**: **Tilak** (with Agarkar, Chiplunkar) 2× 2010, 2013
- First women's university: **D.K. Karve** 3× 2002, 2006, 2010
- **Central Hindu College**, Banaras: **Annie Besant** 1× 2023
- BHU foundation stone: **Lord Hardinge** (1916) 1× 2003
- Advocated religious education in universities: **Madan Mohan Malaviya** 1× 2005
- First central university among options: **BHU** 1× 2011

QUANTUM IAS & PCS

TRAP ALERT

- Calcutta Madrasa (**Warren Hastings**, Muslim law) vs Fort William College (**Wellesley**, officers).
- Three universities came in **1857** (Canning), from Wood's Despatch (1854).
- Hunter Commission (education, 1882) is not the Hunter Committee on Jallianwala (1919).

UP LINK

UP link: **BHU** (1916, Malaviya and Besant's Central Hindu College) at Varanasi, **MAO College** (1875, Aligarh, Syed Ahmad Khan), **Muir Central College** (Allahabad) and **Thomason College, Roorkee** (then in the North-Western Provinces) were UP's landmark institutions.

LAST-MINUTE 10

1. Calcutta Madrasa 1781 (Warren Hastings)
2. Fort William College (Wellesley) for officers
3. William Jones = Asiatic Society, Shakuntala
4. Charter Act 1813 = Rs 1 lakh
5. Macaulay 1835 = English education, filtration theory
6. Wood, Hunter, Sadler, Sargent
7. Hindu College 1817; MAO College 1875
8. Tilak = Deccan Education Society
9. Karve = first women's university
10. BHU = Besant's college; Hardinge laid the stone

Chapter 9

Development of Press in Modern India

Built from 46 UPPSC questions (1990–2023). Focus: the first newspapers, Rammohan Roy's papers, press laws, the nationalist press, Tilak, the militant and revolutionary press, Besant and Gandhi, founders of papers, the press in UP.

ASKED MOST · REVISE FIRST

- **Paper-founder matching**: **Kesari-Tilak, Sudharak-Gokhale, Bengalee-Surendranath, Bombay Chronicle-Pherozeshah Mehta, Leader-Malaviya, Independent-Motilal Nehru, Al-Hilal-Azad, New India and Commonweal-Besant, Young India-Gandhi** **11×** 1990, 1992, 2002, 2003, 2004, 2008, 2008, 2012, 2014, 2017, 2017
- First newspaper: **Hicky's Bengal Gazette** (1780); first Indian-owned English paper: **Hindoo Patriot** **3×** 2004, 2006, 2010
- **Vernacular Press Act** (1878): **Lytton**; repealed by **Ripon** **2×** 2007, 2009
- **Annie Besant**: **Commonweal** and **New India** **5×** 2001, 2002, 2004, 2004, 2013

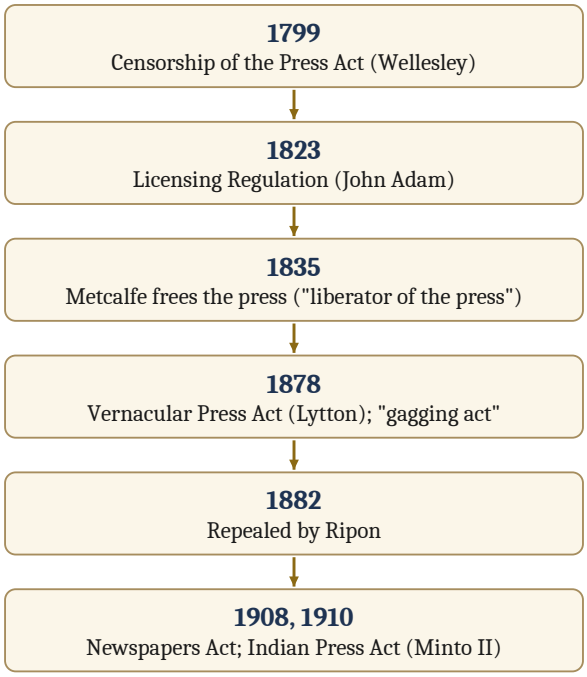
■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

9.1 Early Papers and Press Laws

- First newspaper: **(Hicky's) Bengal Gazette** **2×** 2004, 2010
- First Indian English paper: **Hindoo Patriot**; its editor **Harish Chandra Mukherjee** backed the indigo ryots **3×** 1998, 2006, 2010
- Rammohan Roy was not formally the editor of Sambad Kaumudi (A false, R true) **1×** 2023
- Wrong pair: **Tahzeeb-ul-Akhlaq-Jinnah** (it was **Syed Ahmad Khan's**) **1×** 2019
- **Mirat-ul-Akhbar** (Persian): **Rammohan Roy** **1×** 2000
- **Samachar Darpan-J.C. Marshman, Mirat-ul-Akhbar-Rammohan Roy, Kesari-Tilak, Young India-Gandhi** **1×** 2017

- **Udant Martand** (first Hindi paper, 1826): from **Kolkata** 1× 2016
- Press censorship first: **Wellesley** 1× 2001
- **Vernacular Press Act**: **Lytton**; repealed under **Ripon** 2× 2007, 2009

PRESS LAWS



9.2 The Nationalist Press

- **Indian Mirror** (1880s): **Calcutta** 1× 2007
- **Bengalee-Surendranath, Bangabasi-Jogendra Nath Bose, Vande Mataram-Aurobindo, Amrita Bazar Patrika-Motilal Ghosh** 1× 2018
- **Hindu-G. Subramania Iyer, Sudharak-Gokhale, Voice of India-Dadabhai Naoroji, Bengalee-Surendranath** 2× 2002, 2003
- **Kal-S.M. Paranjape, Bengalee-Surendranath, Sudharak-Gokhale, Somaprakash-Dwarkanath** 1× 2017
- Wrong pair: **Bombay Chronicle-Gokhale** (it was **Pherozeshah Mehta's**) 1× 2017
- Tilak's paper: **Kesari** 2× 1990, 1992
- First Indian journalist jailed (key): **Tilak** (Surendranath in 1883 by some accounts) 1× 1997
- Revolutionary magazines criticised the Congress: **all of the above** 1× 2011
- Advocated revolutionary terror: **Sandhya, Yugantar, Kal** (all) 1× 2008
- **Vande Mataram** (English): **Aurobindo Ghosh** 1× 1996
- **Swadeshabhimani**: **K. Ramakrishna Pillai** 1× 2005
- First **Ghadar** issue: **Urdu**; a **weekly** 2× 2007, 2013
- Paris Vande Mataram editor: none of the options (*) 1× 2015

9.3 Besant, Gandhi and Founders

- Besant's papers: **Commonweal and New India** 3× 2002, 2004, 2013
- Correct pair: **Annie Besant-Commonweal** 2× 2001, 2004
- First editor of **Indian Opinion**: **Mansukhlal Nazar** 1× 2014
- Indian Opinion was **not** published in **Urdu** 1× 2005
- **Young India** began as a weekly of the **Home Rule party** 1× 2003
- Gandhi had nothing to do with **Yugantar** 1× 2017
- **Harijan** first issue (1933): **Poona** 1× 2013
- **Bombay Chronicle-Mehta, Commonweal-Besant, Leader-Malaviya, Searchlight-Sachchidananda Sinha** 2× 2002, 2004
- **Leader-Malaviya, Bombay Chronicle-Mehta, Independent-Motilal Nehru, Justice-T.M. Nair** 1× 2014
- **Independent**: **Motilal Nehru** 1× 2012
- Wrong pair: **Ambedkar-Independent** 1× 2008
- **Azad-Al-Hilal, Mehta-Bombay Chronicle, Besant-New India, Gandhi-Young India** 1× 2008

9.4 Press in UP and Language Matching

- **Aaj-Shiv Prasad Gupta, Leader-Malaviya, National Herald-Nehru, Pioneer-George Allen** 1× 2010
- **Swadesh-Gorakhpur, Bharat Bandhu-Hathras, Satyawadi-Agra, Shakti-Almora** 1× 2022
- **Qaumi Awaz**: **Jawaharlal Nehru**; wrong pair: Qaumi Awaz-Azad 3× 2013, 2013, 2013
- **Bahishkrit Bharat** (Marathi): **Ambedkar** 2× 1991, 2004
- Urdu paper of Lajpat Rai (Lahore, 1920): **Vande Mataram** 1× 2008

QUANTUM IAS & PCS

- **Bharat Mitra-Hindi, Rashtramata-Marathi, Prajamitra-Gujarati, Nayak-Bengali** 2× 2012, 2013

TRAP ALERT

- Bombay Chronicle = **Mehta**; Sudharak = **Gokhale**.
- **Young India** (Gandhi) vs **New India** (Besant).
- Udant Martand and Bharat Mitra were Hindi papers from **Calcutta**, not UP.
- Three Bande Mataram papers: Calcutta (Aurobindo), Paris (Cama), Lahore (Lajpat Rai).

UP LINK

UP link: UP's major papers were **The Leader** (Allahabad, Malaviya), **The Pioneer** (Allahabad, then Lucknow), **Aaj** (Varanasi, Shiv Prasad Gupta), **National Herald** and **Qaumi Awaz** (Lucknow, Nehru), and **Swadesh** (Gorakhpur).

LAST-MINUTE 10

1. Hicky's Bengal Gazette 1780 = first paper
2. Hindoo Patriot = first Indian English paper (Harish Mukherjee)
3. Mirat-ul-Akhbar = Rammohan Roy; Udant Martand = Kolkata 1826
4. Wellesley = censorship; Lytton = VPA 1878; Ripon repealed it
5. Kesari = Tilak; Sudharak = Gokhale; Bengalee = Surendranath
6. Bombay Chronicle = Mehta; Leader = Malaviya; Independent = Motilal
7. Al-Hilal = Azad; Commonweal, New India = Besant
8. Young India, Harijan = Gandhi; Harijan from Poona
9. Aaj = Shiv Prasad Gupta; Pioneer = George Allen
10. Bahishkrit Bharat = Ambedkar

Chapter 10

Socio-Religious Movements

Built from 62 UPPSC questions (1990–2024). Focus: Rammohan Roy and the Brahmo Samaj, Vivekananda, Dayananda and the Arya Samaj, Prarthana Samaj, Phule, Dev Samaj and Radhasoami, Theosophy, Vidyasagar, social laws, Muslim reform.

ASKED MOST · REVISE FIRST

- **Rammohan Roy**: father of the Indian Renaissance; **Atmiya Sabha** first; Brahmo Samaj = **monotheism**; title "Raja" from **Akbar II** 7× 1994, 1999, 2005, 2005, 2009, 2009, 2012
- **Dayananda Saraswati**: **Arya Samaj (1875)**, **Satyarth Prakash**, "Martin Luther of India", **shuddhi** 6× 1995, 2005, 2009, 2010, 2013, 2017
- **Jyotiba Phule**: **Satyashodhak Samaj** (1873), **Gulamgiri**, anti-caste 4× 1997, 2000, 2022, 2022
- **Vivekananda**: Chicago **1893**, founded the **Ramakrishna Mission** 6× 1993, 2004, 2004, 2008, 2015, 2015
- **Servants of India Society** = **Gokhale** (1905) 6× 1990, 1993, 2004, 2015, 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Rammohan Roy and the Brahmos

- "Father of Indian Renaissance", "prophet of nationalism": **Rammohan Roy** 3× 1994, 2005, 2009
- **Gift to Monotheists** (1809): in **Persian** 1× 2021
- Title "Raja" from **Akbar II** 1× 2012
- He did **not** oppose **Western education** 1× 1992
- Opposed the **Jury Act of 1826**: **Rammohan Roy** 1× 2017
- First society: **Atmiya Sabha** 1× 2009
- Brahmo Samaj: **monotheism** 2× 1999, 2005
- **Bharatiya Brahmo Samaj** (Brahmo Samaj of India): **Keshab Chandra Sen** 3× 2008, 2010, 2010
- Wrong pair: **Rammohan-Adi Brahmo Samaj** (Debendranath's) 1× 1997
- Orthodox leader: **Radhakant Deb** 1× 2008

- Orthodox paper against Rammohan: **Samachar Chandrika** 1× 2017

10.2 Vivekananda, Dayananda and Maharashtra

- Chicago Parliament of Religions: **1893**, **Vivekananda** 4× 1993, 2004, 2008, 2015
- Jnana, Karma, Raja Yoga: **Vivekananda** 1× 2015
- Ramakrishna Mission founded by **Vivekananda** 1× 2004
- Arya Samaj: **1875** 1× 2009
- Revival of the Vedas: **Dayananda**; "Martin Luther of India" 2× 1995, 2005
- **Satyarth Prakash**: **Dayananda** 2× 2013, 2017
- **Shuddhi**: **Arya Samaj** 1× 2010
- Arya Samaj gave Hindus confidence; the reason does not explain it (A and R true, R does not explain A) 1× 2019
- **Prarthana Samaj** founder: **Atmaram Pandurang**; leading figure **M.G. Ranade** 2× 2004, 2010
- "**Lokahitvadi**": **Gopal Hari Deshmukh** 1× 2015
- **Satyashodhak Samaj** (1873), **Gulamgiri**, anti-caste: **Jyotiba Phule** 4× 1997, 2000, 2022, 2022
- **Bahujan Samaj**: **Mukund Rao Patil** 1× 2010
- "If God tolerated untouchability...": **Tilak** 1× 2004

Organisation	Founder	Year
Atmiya Sabha	Rammohan Roy	1815
Brahmo Samaj	Rammohan Roy	1828
Prarthana Samaj	Atmaram Pandurang	1867
Satyashodhak Samaj	Jyotiba Phule	1873
Arya Samaj	Dayananda Saraswati	1875 (Bombay)
Theosophical Society	Blavatsky, Olcott	1875 (New York); Adyar 1882
Ramakrishna Mission	Vivekananda	1897
Servants of India Society	Gokhale	1905

10.3 Other Movements

- **Dev Samaj**: **Shiv Narayan Agnihotri** 3× 2002, 2002, 2003
- **Radhasoami Satsang**: **Shiv Dayal Sahab** 1× 2002
- Correct: **Sanatan Dharma Rakshini Sabha-Calcutta** 1× 2022
- Theosophy succeeded mainly due to **Annie Besant** 1× 2010
- Widow remarriage legalised by **Vidyasagar's** campaign 1× 2012
- "No time to think about God...": **Vidyasagar** 1× 2016
- Sanskrit scholars: **all the above** 1× 2013
- Widow remarriage in Maharashtra: **Vishnu Parashuram Pandit** 1× 2013
- **D.K. Karve**: widow remarriage and **women's education** 1× 2016

10.4 Social Laws and Matching

- 1856 Act: only the **Widow Remarriage Act** (*) 1× 2001
- Act V of 1843 banned **slavery** 1× 2007
- **Native Marriage Act 1872**: **Keshab Chandra Sen**; girls' minimum age **14** 3× 2000, 2011, 2019
- Age of Consent Act 1891: urged by **Behramji Malabari** (only 1) 1× 2024
- Parsi editor of Indian Spectator: **B.M. Malabari** 1× 2016
- **Sharda Act** (1929): girls **14**, boys **18** 2× 2012, 2013
- **Bethune-girls' school, Karve-widow remarriage secretary, Vidyasagar-Sanskrit College principal, Malabari-child marriage crusade** 1× 2022
- Wrong pair: **Ranade-Theosophical Society** 2× 1990, 2011
- **Rammohan-Brahmo, Vivekananda-RK Mission, Dayananda-Arya, Ranade-Prarthana** 1× 1994
- Wrong pair: **Atmiya Sabha-Debendranath Tagore** 1× 2014
- **Servants of India Society**: **Gokhale**, **1905** 6× 1990, 1993, 2004, 2015, 2016, 2017
- Wrong pair: **Bahishkrit Hitkarini Sabha-Phule** (it was **Ambedkar's**) 1× 2016
- Places: Brahmo-Calcutta, Manav Dharma Sabha-Surat, Arya Samaj-Bombay 1× 2003

QUANTUM IAS & PCS

- Ideas: Rammohan-Upanishads, Dayananda-Vedas 1× 1998

10.5 Muslim Reform and Others

- **Deoband** movement: **1866** 1× 2016
- **Dar-ul-Ulum** (Nadwa, per key): **Shibli Nomani** 1× 2014
- **Tarakeswar** satyagraha (1924): **corruption in temples** 1× 2015
- **Hali** system: **bonded labour** 1× 2015

TRAP ALERT

- **Monotheism** (Brahmo), not monism.
- RK Mission was founded by **Vivekananda**, not Ramakrishna.
- Prarthana Samaj founder = **Atmaram Pandurang**; Ranade led it.
- Sharda Act: girls 14, boys 18; Native Marriage Act 1872: girls 14.

UP LINK

UP link: **Deoband** (Saharanpur, 1866) and **Nadwatul Ulama** (Lucknow) were UP's Muslim seminaries; **Aligarh** (Syed Ahmad Khan) led modern Muslim education; the **Radhasoami Satsang** is centred at **Dayalbagh, Agra**.

LAST-MINUTE 10

1. Rammohan = father of Renaissance; Atmiya Sabha; Akbar II's "Raja"
2. Keshab = Brahmo Samaj of India; Debendranath = Adi Brahmo Samaj
3. Vivekananda = Chicago 1893; RK Mission
4. Dayananda = Arya Samaj 1875, Satyarth Prakash, shuddhi
5. Atmaram Pandurang = Prarthana Samaj; Ranade led it
6. Phule = Satyashodhak 1873, Gulamgiri
7. Agnihotri = Dev Samaj; Shiv Dayal = Radhasoami
8. Vidyasagar = widow remarriage (1856)
9. Malabari = Age of Consent 1891; Sharda Act 1929
10. Gokhale = Servants of India 1905; Deoband 1866

Chapter 11

Political Organisations Established Before Congress

Built from 16 UPPSC questions (2006–2024). Focus: the first Bengal associations, London bodies, the Indian Association and Indian National Conference, the presidency associations, the United Indian Patriotic Association.

ASKED MOST · REVISE FIRST

- **Indian Association** (1876): **Surendranath Banerjee** (with Anand Mohan Bose); its **Indian National Conference** merged into the Congress (1886) 3× 2007, 2008, 2017
- **East India Association** (London, 1866): **Dadabhai Naoroji** 2× 2021, 2024
- **Bombay Presidency Association** (1885): **Pherozeshah Mehta** (with Telang, Tyabji) 1× 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 Bengal and London

- Order: **Bangabhasha Prakashika Sabha, Landholders' Society, Bengal British India Society, Indian League** 1× 2016
- Bengal British India Society (1843) 1× 2023
- Founding-year pairs: two wrong (*) 1× 2015
- East India Association: **Dadabhai Naoroji**, London, 1866 1× 2021
- "Poverty and Un-British Rule": **Dadabhai Naoroji** 1× 2016
- Order: **East India Association, National Indian Association, Indian Society, Indian Association** 1× 2024
- Wrong pairs: **East India Association-Calcutta** (it was London) 1× 2008
- Founder matching for the Landholders' Society, British India Society and others 1× 2014

PRE-CONGRESS BODIES



11.2 Indian Association and Others

- Indian Association: **S.N. Banerjee** 1× 2008
- Merged with the Congress in 1886: **Indian National Conference** 1× 2007
- Not pre-Congress (key): **The Indian Association** (*) 1× 2017
- **India League-Shishir Kumar Ghosh, Indian Association-Anand Mohan Bose, Liberal Federation-Surendranath, United Patriotic Association-Syed Ahmad Khan** 1× 2008
- Bombay Presidency Association founder: **Pherozeshah Mehta** 1× 2016
- Founder matching: **Ananda Charlu (Madras Mahajan Sabha), Surendranath, Ranade (Poona Sarvajanik Sabha), Mehta** 1× 2016
- United Indian Patriotic Association with Syed Ahmad: **Raja Shivaprasad** 1× 2016
- Pre-Congress master match (British Indian Association and others) 1× 2006

Body	Year	Founder
Landholders' Society	1838	Dwarkanath Tagore
British Indian Association	1851	Radhakant Deb, Debendranath
East India Association	1866	Dadabhai Naoroji
Poona Sarvajanik Sabha	1870	M.G. Ranade
Indian League	1875	Shishir Kumar Ghosh
Indian Association	1876	Surendranath, Anand Mohan Bose
Madras Mahajan Sabha	1884	Ananda Charlu, Raghavachari
Bombay Presidency Association	1885	Mehta, Telang, Tyabji

TRAP ALERT

- British India Society (London, 1839), Bengal British India Society (1843), British Indian Association (1851) are three bodies.
- The Indian Association continued; the **Conference** merged into the Congress.

UP LINK

UP link: **Raja Shivaprasad Sitara-e-Hind** of **Banaras** and **Syed Ahmad Khan** of **Aligarh** set up the **United Indian Patriotic Association** (1888) against the Congress.

LAST-MINUTE 10

1. Bangabhasha Prakashika Sabha 1836 = first
2. Landholders' Society 1838
3. East India Association 1866 = Naoroji (London)
4. Indian League 1875 = Shishir Ghosh
5. Indian Association 1876 = Surendranath, Anand Mohan Bose
6. Indian National Conference merged in 1886
7. Poona Sarvajanik Sabha = Ranade
8. Bombay Presidency Association 1885 = Mehta
9. United Patriotic Association = Syed Ahmad, Shivaprasad
10. Liberal Federation 1918 = Surendranath

Chapter 12

Indian National Congress

Built from 33 UPPSC questions (1995–2024). Focus: the 1885 session, viceroys' attitudes, first Muslim, European and woman presidents, Tilak, Amritsar and Nagpur, presidents of the 1920s-30s, Haripura and Tripuri, Azad.

ASKED MOST · REVISE FIRST

- **First session (1885)**: **Bombay**, president **W.C. Bonnerjee**, general secretary **A.O. Hume**; under **Dufferin** 12× 2004, 2006, 2010, 2010, 2011, 2012, 2014, 2015, 2015, 2017, 2023, 2023
- **First Muslim president**: **Badruddin Tyabji**; first woman: **Annie Besant (Calcutta 1917)**; first European: **George Yule** 9× 1995, 2003, 2004, 2005, 2010, 2012, 2013, 2015, 2023
- **Tilak** never presided; of Lal-Bal-Pal only **Lajpat Rai** did 4× 1997, 2006, 2008, 2021
- **Haripura (1938)**: **Subhas Bose**; **Gandhi** presided only at **Belgaum (1924)** 2× 2008, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Founding and British Attitude

- Congress founded **1885**, first session at **Bombay** 2× 2010, 2023
- First president: **W.C. Bonnerjee**; general secretary **A.O. Hume** 5× 2004, 2010, 2014, 2015, 2015
- Absent at the founding: **Surendranath Banerjee** 1× 1997
- Viceroy then: **Dufferin**; "microscopic minority" was his phrase 7× 2006, 2011, 2012, 2012, 2017, 2018, 2023
- "Assist it to a peaceful demise": **Curzon** 2× 2002, 2002
- Attended a Congress session: **Willingdon**, Bombay 1915 (as Governor) 1× 2008
- Early nationalists saw British rule as providential 1× 2024

12.2 Presidents and Firsts

- First Muslim president: **Badruddin Tyabji** (1887) 4× 1995, 2003, 2004, 2005
- Tyabji first Muslim; Sarojini Naidu first Indian woman president 1× 2023
- First woman president: **Annie Besant**, **Calcutta 1917** 3× 2010, 2012, 2015
- First European president: **George Yule** 1× 2013
- Never president: **Tilak** 3× 2006, 2008, 2021
- Of Lal-Bal-Pal, president: **Lala Lajpat Rai** (1920 special) 1× 1997
- "Swaraj is my birthright": **Tilak** (Lucknow 1916) 3× 2004, 2005, 2015
- Tilak's last session: **Amritsar 1919** 1× 2014
- Drafting the new constitution with Gandhi (Amritsar resolution) 1× 2014
- **C. Vijayaraghavachariar**: **Nagpur 1920** 1× 2016
- First policy on princely states: **Nagpur** session 1× 2015
- Gandhi presided only at **Belgaum (1924)** 1× 2011
- Presidents in order and matches (Azad, Naidu, Motilal and others) 4× 2002, 2004, 2012, 2014
- Nehru spoke of socialism at **Lucknow (1936)** 1× 2015
- **Haripura (1938)**: **Subhas Bose** 1× 2008
- **Jana Gana Mana** first sung at the **1911 Calcutta** session 1× 2010
- Six years in a row: **Abul Kalam Azad** (1940-46) 1× 2016
- Suggested winding up the Congress: **Gandhi** 1× 2004

Session	Year	President / event
Bombay	1885	W.C. Bonnerjee (first)
Madras	1887	Badruddin Tyabji (first Muslim)
Allahabad	1888	George Yule (first European)
Calcutta	1906	Dadabhai Naoroji (Swaraj)
Surat	1907	Split (Rash Behari Ghosh)
Lucknow	1916	Pact; Tilak's "Swaraj" slogan
Calcutta	1917	Annie Besant (first woman)
Belgaum	1924	Gandhi (only time)
Kanpur	1925	Sarojini Naidu (first Indian woman)
Lahore	1929	Jawaharlal Nehru; Purna Swaraj

QUANTUM IAS & PCS

Session	Year	President / event
Haripura, Tripuri	1938, 1939	Subhas Bose

TRAP ALERT

- Hume was **general secretary**, never president.
- First woman = **Besant (1917)**; first Indian woman = **Naidu (1925)**.
- "Microscopic minority" = **Dufferin**; "peaceful demise" = **Curzon**.
- Tilak never presided over a session.

UP LINK

UP link: UP hosted key sessions: **Allahabad 1888** (George Yule) and 1892, **Lucknow 1916** (Lucknow Pact) and **1936** (Nehru's socialism), **Kanpur 1925** (Sarojini Naidu), and **Faizpur's** predecessor **Meerut 1946** (Kripalani).

LAST-MINUTE 10

1. 1885 Bombay; W.C. Bonnerjee; Hume secretary; Dufferin Viceroy
2. Dufferin = microscopic minority; Curzon = peaceful demise
3. Tyabji = first Muslim (1887); George Yule = first European (1888)
4. Besant = first woman (1917); Naidu = first Indian woman (1925)
5. Tilak never president; Lajpat Rai did (1920)
6. Tilak's last session = Amritsar 1919
7. Nagpur 1920 = Vijayaraghavachariar
8. Gandhi presided only at Belgaum 1924
9. Bose = Haripura 1938; Jana Gana Mana first sung 1911
10. Azad = president 1940-46

Chapter 13

Moderates and Extremists in Congress

Built from 17 UPPSC questions (1995–2016). Focus: telling the two camps apart, Naoroji and swaraj, the Extremist critique, Tilak's methods, trials and titles.

ASKED MOST · REVISE FIRST

- **Tilak** called the Moderates' politics "**prayer, petition and protest**"; Chirol called Tilak "**father of Indian unrest**" **8×** 2001, 2002, 2002, 2002, 2004, 2005, 2013, 2013
- Not a Moderate: **Tilak, Lajpat Rai**; not an Extremist: **Gokhale** **4×** 1995, 2011, 2014, 2016
- **Tilak** made the **Ganapati festival** political and founded lathi clubs **2×** 2007, one-liner

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

13.1 The Two Camps

- Not a Moderate: **Tilak** **1×** 1995
- Not with the Moderates: **Lala Lajpat Rai** **1×** 2011
- Not an Extremist: **Gokhale** **2×** 2014, 2016
- An Extremist: **Bipin Chandra Pal** **1×** 2013
- Self-rule demanded from 1904: **Dadabhai Naoroji** **1×** 2016
- "Prayer, petition and protest": **Tilak** **4×** 2002, 2002, 2002, 2005
- Congress a "begging institute": **B.C. Pal** **1×** 2008
- Supporter of swadeshi: **Aurobindo Ghosh** **1×** 2009
- Lajpat Rai's political guru: **Mazzini** **1×** 2013

Moderates	Extremists
Dadabhai Naoroji, Pherozeshah Mehta	Bal Gangadhar Tilak
G.K. Gokhale, Surendranath Banerjee	Lala Lajpat Rai, Bipin Chandra Pal
Methods: petitions, constitutional reform	Methods: swadeshi, boycott, mass action

13.2 Tilak

- Tilak was **not** communal, though he used religious festivals (A false, R true) 1× 2001
- **Lathi club**: **Tilak** 1× one-liner
- Ganapati festival made national: **Tilak** 1× 2007
- "Father of Indian unrest": **Tilak**, so called by **Valentine Chirol** 4× 2001, 2004, 2013, 2013
- Title "Lokmanya": linked (per key) with the **revolutionary movement** period (disputed) 1× 2012
- Pleaded as "a Sanskrit scholar": **Max Muller** 1× 2014
- Carried Tilak's bier with Gandhi: **Shaukat Ali** 1× 2014

TRAP ALERT

- Swaraj was first demanded by a **Moderate**, Naoroji (Calcutta 1906).
- Lal-Bal-Pal = Lajpat Rai (Punjab), Tilak (Maharashtra), Pal (Bengal).
- "Begging institute" = **Pal**; "prayer, petition, protest" = **Tilak**.

UP LINK

UP link: **Madan Mohan Malaviya** of Allahabad stood with the Moderates; the 1916 **Lucknow** session reunited Moderates and Extremists after the Surat split.

LAST-MINUTE 10

1. Tilak, Lajpat Rai, Pal, Aurobindo = Extremists
2. Gokhale, Mehta, Naoroji, Surendranath = Moderates
3. Naoroji = swaraj from 1904-06
4. Tilak = prayer, petition, protest
5. Pal = begging institute
6. Mazzini = Lajpat Rai's guru
7. Tilak = Ganapati festival, lathi club
8. Chirol = father of Indian unrest
9. Max Muller pleaded for Tilak
10. Shaukat Ali carried Tilak's bier

Chapter 14

Revolutionary Movements in India

Built from 45 UPPSC questions (1994–2023). Focus: Abhinav Bharat and Anushilan, the Alipore case, HRA and Kakori, HSRA and Bhagat Singh, Azad, Bengal revolutionaries, matching founders.

ASKED MOST · REVISE FIRST

- **Kakori** (1925): **Azad** escaped; Bismil hanged at **Gorakhpur**; Ashfaqullah the first Muslim hanged 8× 2004, 2006, 2006, 2006, 2008, 2011, 2015, 2021
- **Assembly bomb** (8 April 1929): **Bhagat Singh and B.K. Dutt**; "Inquilab Zindabad" = **Bhagat Singh** 5× 2004, 2010, 2010, 2013, 2013
- **HSRA** = **Chandra Shekhar Azad**, Delhi 1928; **HRA** founded at **Kanpur** 4× 1996, 2006, 2008, 2010
- **Abhinav Bharat** = Savarkar, **Maharashtra**, 1905 4× 2000, 2002, 2010, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Maharashtra and Bengal

- Abhinav Bharat organised by **V.D. Savarkar** 2× 2002, 2010
- Abhinav Bharat (1905) set up in **Maharashtra** 2× 2000, 2015
- Mitra Mela started by **V.D. Savarkar** 1× 2011
- Anushilan Samiti founded by **P. Mitra** 1× 2008
- **Barindra Ghosh**: Anushilan Samiti 2× 2008, 2016
- Yugantar led by **Jatindranath Mukherjee** 1× 2009
- Revolutionary bodies: **Abhinav Bharat, Anushilan Samiti** (not the Indian Patriotic Association) 1× 2004
- Aurobindo defended in the **Alipore case** by **C.R. Das** 2× 2004, 2012
- Bomb on Lord Hardinge (Delhi, 1912): **Anushilan Samiti** (per key) 1× 2023

14.2 HRA and Kakori

- HRA founded at **Kanpur** (1924) 1× 2010
- Not an HRA member: **Shiv Verma** 1× 2006

- Kakori case: **1925** 1× 2008
- Escaped from the Kakori case: **Chandra Shekhar Azad** 3× 2006, 2006, 2011
- Not linked with Kakori: **Bhagat Singh** 1× 2004
- Government advocate in Kakori: **Jagat Narain Mulla** 1× 2015
- Defence committee chaired by **G.B. Pant** (revised key) 1× 2014
- Refused milk before hanging, "only my mother's milk": **Bismil** (question later deleted) 1× 2014
- "Daro-deewar pe hasrat ki nazar karte hain...": keyed to **Wajid Ali Shah** 2× 2002, 2003
- Hanged at **Gorakhpur** jail: **Ram Prasad Bismil** 3× 2006, 2015, 2021
- First Muslim hanged for freedom: **Ashfaqullah Khan** 1× 2004

14.3 Conspiracy Cases

- Not linked with the Kanpur case (1924): **M.A. Ansari** 1× 2021
- Order (per key): **Lahore (1915), Kanpur, Kakori, Meerut** 1× 2010
- Wrong pair: **Kakori case, 1924** (it was 1925) 1× 2015
- Wrong pair: **Nasik case, Ras Bihari Bose** 1× 2017

Case	Year	Accused
Lahore (I)	1915-16	Ghadar revolutionaries
Kanpur Bolshevik	1924	Dange, Muzaffar Ahmad
Kakori	1925	Bismil, Ashfaqullah, Lahiri, Roshan Singh
Meerut	1929	Communists, trade unionists
Lahore (II)	1929-30	Bhagat Singh, Sukhdev, Rajguru

14.4 HSRA and Bhagat Singh

- HSRA formed by **Chandra Shekhar Azad** 2× 1996, 2006
- HSRA set up at **Delhi**, 1928 1× 2008
- "Hindustan Socialist Democratic Union": no such body; question marked (*) 1× 2022
- Not an early Naujawan Sabha member: **Ambika Chakravarti** 1× 2021
- Assembly bomb with Bhagat Singh: **Batukeshwar Dutt** 2× 2013, 2013
- "Inquilab Zindabad": **Bhagat Singh** 3× 2004, 2010, 2010
- "Criticism and independent thinking...": **Bhagat Singh** 1× 2014
- Hanged in the Lahore case: **Bhagat Singh, Sukhdev, Rajguru** 2× 2002, 2010
- Bhagat Singh's memorial: **Ferozepur** 1× 2013
- Jatin Das arrested in the **Lahore Conspiracy** 1× 1994
- Died on hunger strike in jail: **Jatin Das** 1× 2004
- Azad killed **by bullets in an encounter** (Alfred Park, Allahabad, 1931) 1× 1994
- Tried to kill Tegart: **Gopinath Saha** 1× 2016
- Fired at the Governor at convocation: **Bina Das** 1× 2010

14.5 Matching Founders

- Per key: Abhinav Bharat **G.D. Savarkar**, Mitra Mela **V.D. Savarkar**, Indian Republican Army **Surya Sen**, HRA **Sanyal** 1× 2014
- Revolt group **Surya Sen**, HRA **Bismil**, HSRA **Azad**, Naujawan Bharat Sabha **Bhagat Singh** 1× 2002
- Order: **Ghadar (1913), Berlin Committee (1915), Assembly bomb (1929), Chittagong raid (1930)** 1× 2022

TRAP ALERT

- HRA = Kanpur 1924 (Sanyal); HSRA = Delhi 1928 (Azad); Naujawan Bharat Sabha = Lahore 1926 (Bhagat Singh).
- **Saunders**, not Scott, was shot. **B.K. Dutt** threw the bomb but was not hanged.
- Abhinav Bharat: V.D. Savarkar in 2002 and 2010, but G.D. Savarkar in the 2014 match. Read the options.

UP LINK

UP link: **Kakori** (Lucknow) 1925, Bismil hanged at **Gorakhpur**, Roshan Singh at **Allahabad (Naini)**, Lahiri at **Gonda**; **Azad** fell at Alfred Park, **Allahabad** (now Chandrashekhar Azad Park).

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Abhinav Bharat = Savarkar, Maharashtra 1905
2. Anushilan = P. Mitra, Barindra Ghosh
3. Yugantar = Jatindranath Mukherjee
4. C.R. Das defended Aurobindo
5. HRA = Kanpur; HSRA = Azad, Delhi 1928
6. Kakori = 1925; Azad escaped
7. Mulla for government; Pant for defence
8. Bismil = Gorakhpur; Ashfaqullah = first Muslim
9. Bhagat Singh + B.K. Dutt = Assembly bomb
10. Jatin Das = hunger strike death

Chapter 15

Revolutionary Activities Outside India

Built from 18 UPPSC questions (2002–2019). Focus: Shyamji Krishna Varma and India House, Dhingra and Udham Singh, Madame Cama, the Ghadar Party, Komagata Maru, the Kabul government, Rash Behari Bose.

ASKED MOST · REVISE FIRST

- **Ghadar Party**: **Sohan Singh Bhakna** (first chairman), Lala Hardayal; HQ **San Francisco**, USA **6×** 2004, 2008, 2008, 2013, 2014, 2015
- **Shyamji Krishna Varma**: Indian Home Rule Society and India House, **London 1905** **5×** 2002, 2004, 2007, 2014, 2014
- **Bhikaji Cama**: "Mother of Indian Revolution"; first to unfurl the tricolour abroad **3×** 2002, 2003, 2010

■ asked 3 or more times · asked once or twice **4×** times asked, followed by the exam years

15.1 London

- Indian Home Rule Society started by **Shyamji Krishna Varma** **2×** 2002, 2007
- Indian Home Rule Society set up in **London** **1×** 2004
- Year of the Indian Home Rule Society: **1905** **1×** 2014
- India House, London: **Shyamji Krishna Varma** **1×** 2014
- Hanged for killing British officers in England: **Madan Lal Dhingra and Udham Singh** **1×** 2004

Revolutionary	Act in England	Year
Madan Lal Dhingra	Shot Curzon Wyllie	1909
Udham Singh	Shot Michael O'Dwyer	1940

15.2 Madame Cama

- "Mother of Indian Revolution": **Bhikaji Rustam Cama** **2×** 2002, 2003
- First woman to unfurl the Indian tricolour (Stuttgart, 1907): **Bhikaji Cama** **1×** 2010

15.3 The Ghadar Party (1913)

- Founded the Ghadar Party: **Lala Hardayal** **1×** 2008
- Ghadar movement founded by **Sohan Singh Bhakna** **1×** 2004
- First chairman: **Sohan Singh Bhakna** **1×** 2015
- Not linked with Ghadar: **Khudiram Bose** **1×** 2019
- Headquarters: **San Francisco** **1×** 2014
- Founded in the **USA** **2×** 2008, 2013
- Komagata Maru episode: **Baba Gurdit Singh** **1×** 2014
- Komagata Maru was **a ship** **1×** 2015

15.4 Governments and Leagues Abroad

- Republican government abroad: **Raja Mahendra Pratap** **1×** 2008
- Provisional Government (WWI) in **Afghanistan** (Kabul, 1915) **2×** 2010, 2012
- ■ Indian Independence League: **Rash Behari Bose** **3×** 2008, 2008, 2010

TRAP ALERT

- Home Rule **Society** (Shyamji, London 1905) is not the Home Rule **League** (Tilak and Besant, 1916).
- Udham Singh shot **O'Dwyer** (Punjab Lt. Governor), not General Dyer.
- Kabul 1915 = Mahendra Pratap (President), Barkatullah (PM); Singapore 1943 = Subhas Bose.

UP LINK

UP link: **Raja Mahendra Pratap** came from **Mursan (Hathras)** and founded Prem Mahavidyalaya at **Vrindavan**; the state university at **Aligarh** is named after him.

LAST-MINUTE 10

1. Shyamji = India House, Home Rule Society, London 1905
2. Dhingra 1909, Udham Singh 1940
3. Cama = Mother of Indian Revolution, Stuttgart 1907
4. Ghadar = 1913, San Francisco, USA
5. Sohan Singh Bhakna = first chairman
6. Hardayal = Ghadar founder
7. Komagata Maru = ship, Gurdit Singh
8. Mahendra Pratap = Kabul 1915
9. Rash Behari Bose = Indian Independence League
10. Khudiram Bose not in Ghadar

Chapter 16

Partition of Bengal and Swadeshi Movement

Built from 17 UPPSC questions (1991–2023). Focus: Curzon's partition and its motive, 7 August 1905, boycott, leaders and critics, Vande Mataram, swadeshi outside Bengal, the 1911 annulment.

ASKED MOST · REVISE FIRST

- Anti-partition movement began **7 August 1905** (Calcutta Town Hall); boycott adopted the same year **3×** 1994, 2014, 2023
- **Vande Mataram** became the theme song during the **Swadeshi Movement** **3×** 2002, 2005, 2010
- Partition by **Curzon**, cancelled in **1911** **3×** 1991, 1993, 2011
- First to suggest boycott: **Krishna Kumar Mitra** **2×** 2004, 2011

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

16.1 The Partition (1905)

- Partitioned Bengal: **Lord Curzon** **1×** 2011
- Occurred last among the options: **Partition of Bengal** **1×** 2000
- Main aim: **to weaken nationalism in Bengal** **1×** 1998
- Lt. Governor of Bengal at the time: **Sir Andrew Fraser** **1×** 2014
- Anti-partition movement began on **7 August 1905** **1×** 1994
- Mass demonstration at Calcutta Town Hall: **7 August 1905** **1×** 2023
- First to suggest boycott: **Krishna Kumar Mitra** (in Sanjivani) **2×** 2004, 2011
- Boycott adopted as national policy: **1905** **1×** 2014
- Partition cancelled: **1911** (Delhi Durbar) **2×** 1991, 1993

BENGAL, 1905 TO 1911

Curzon announces partition
July 1905

↓

Town Hall meeting, boycott resolution
7 August 1905

↓

Partition takes effect, day of mourning
16 October 1905

↓

Calcutta Congress adopts swadeshi and swaraj
1906

↓

Annulled at the Delhi Durbar
1911

16.2 Leaders, Critics and Spread

- Led the anti-partition agitation: **Surendranath Banerjee** **1×** 2011
- Critic of swadeshi, pleaded for East-West understanding: **R.N. Tagore** **1×** 2009
- Theme song: **Vande Mataram**, during Swadeshi **3×** 2002, 2005, 2010
- Not correct about Swadeshi: "it was **limited to Bengal**" **1×** 2002

QUANTUM IAS & PCS

- Swadeshi leader in Madras: **V.O. Chidambaram Pillai** 1× 2008
- Swadeshi leader in Delhi: **Syed Haider Raza** 1× 2011
- British journalist **H.W. Nevinson**: Swadeshi Movement 1× 2014
- Indian Society of Oriental Art: **Abanindranath Tagore** 1× 2007

Region	Swadeshi leader
Bengal	Surendranath Banerjee, Aurobindo, B.C. Pal
Maharashtra	Tilak
Punjab	Lajpat Rai, Ajit Singh
Madras	V.O. Chidambaram Pillai
Delhi	Syed Haider Raza

TRAP ALERT

- **7 August** = movement began; **16 October** = partition took effect (Raksha Bandhan day).
- Curzon partitioned (1905); **Hardinge** annulled (1911). Minto faced the agitation.
- Tagore led Raksha Bandhan but later **criticised** the movement's turn to violence.

UP LINK

UP link: The **Benaras Congress** (1905) under **Gokhale** backed swadeshi and boycott for Bengal.

LAST-MINUTE 10

1. Curzon = partition 1905
2. Aim = weaken Bengal nationalism
3. Andrew Fraser = Lt. Governor
4. 7 August 1905 = movement began
5. K.K. Mitra = boycott idea
6. Surendranath = led agitation
7. Vande Mataram = theme song
8. V.O.C. Pillai = Madras
9. Abanindranath = Oriental Art
10. 1911 = partition annulled

Chapter 17

Congress: Benaras, Calcutta and Surat Sessions

Built from 13 UPPSC questions (1990–2014). Focus: Gokhale at Benaras, swaraj at Calcutta, Naoroji's titles and British Parliament, the Surat split of 1907.

ASKED MOST · REVISE FIRST

- Congress split into **Naram Dal** and **Garam Dal** at **Surat (1907)**, president **Rash Behari Ghosh** 5× 1990, 1991, 2007, 2010, 2012
- **Dadabhai Naoroji** = "Grand Old Man of India"; first Indian elected to the British Parliament 4× 1991, 1992, 2007, 2014
- **Gokhale** presided over the **1905 Benaras** session 3× 1999, 2003, 2004

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

17.1 Benaras and Calcutta

- Gokhale presided over Congress in **1905** 2× 2003, 2004
- Benaras session (1905) president: **G.K. Gokhale** 1× 1999
- "Swaraj" first uttered at **Calcutta, 1906** (question later deleted) 1× 2014
- First leader to use "Swaraj" (per key): **Tilak** 1× 2013

17.2 Dadabhai Naoroji

- Known as **Grand Old Man of India** 3× 1991, 2007, 2014
- First Indian elected to the British Parliament: **Naoroji** 1× 1992
- First Indian to contest for the House of Commons (per key): **Naoroji** (strictly, Lal Mohan Ghose in 1885) 1× 2004
- Not correct: elected on a **Conservative** ticket (he was a **Liberal**, Finsbury Central 1892) 1× 2008
- Not correct: presided over Congress **four** times (he presided **three** times: 1886, 1893, 1906) 1× 2014

17.3 Surat 1907

- Split into Naram Dal and Garam Dal at **Surat** 2× 1990, 2012
- Surat session president: **Rash Behari Ghosh** 2× 2007, 2010
- First split of the Congress: **1907** 1× 1991

Session	Year	President	Remembered for
Benaras	1905	G.K. Gokhale	Support for swadeshi
Calcutta	1906	Dadabhai Naoroji	Swaraj as goal
Surat	1907	Rash Behari Ghosh	Split
Lucknow	1916	A.C. Mazumdar	Reunion, Lucknow Pact

TRAP ALERT

- Gokhale presided **once** (1905); Naoroji **three** times.
- Session that first used "swaraj" = Calcutta 1906 (Naoroji); leader keyed = Tilak.
- The 1908 **Allahabad** convention came **after** the split; the split was at Surat.

UP LINK

UP link: The **Benaras** session of 1905 was held in UP; the Moderates drew up the Congress constitution at the **Allahabad convention** (1908) after Surat.

LAST-MINUTE 10

1. Gokhale = Benaras 1905
2. Naoroji = Calcutta 1906, swaraj
3. Naoroji = Grand Old Man
4. Naoroji = first Indian MP, Liberal, Finsbury
5. Naoroji = 3 presidencies
6. Surat 1907 = split
7. Rash Behari Ghosh = Surat president
8. Naram Dal = Moderates; Garam Dal = Extremists
9. Lal Mohan Ghose = first to contest (1885)
10. Tilak = "Swaraj" (per key)

Chapter 18

Muslim League and Communal Representation

Built from 8 UPPSC questions (1994–2024). Focus: the founding at Dacca, early sessions, the 1909 Act and separate electorates, the Justice Party.

ASKED MOST · REVISE FIRST

- **Morley-Minto Act (1909)** gave **communal representation** (separate electorates) 2× 1994, 1996
- Muslim League founder: **Nawab Salimullah**; first president: **Aga Khan** 3× 1997, 2006, 2007
- Seed of the division of India: **separate electorates for Muslims** 2× 2001, 2004

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 The League

- Founder of the All India Muslim League (Dacca, 1906): **Nawab Salimullah Khan** 2× 2006, 2007
- First president: **Aga Khan** 1× 1997
- 1907 annual session: **Karachi** 1× 2008
- League rejected Congress claims (A true); "only the League had the right" (R false) 2× 2002, 2003

18.2 Separate Electorates

- Morley-Minto Reform passed in **1909** 1× 1994
- 1909 Act provided **communal representation** 1× 1996
- Seeds of discord: **reservation of seats and separate electorates for Muslims** 2× 2001, 2004
- Justice Party: called the Congress Brahmin-dominated, and claimed communal representation for non-Brahmins; **both correct** 1× 2024

Act	Keyword
1909 (Morley-Minto)	Separate electorates for Muslims

Act	Keyword
1919 (Montagu-Chelmsford)	Dyarchy in provinces
1935	Federation, provincial autonomy

TRAP ALERT

- Three firsts: founder **Salimullah**; chaired the Dacca meeting **Viqar-ul-Mulk**; first president **Aga Khan**.
- The Congress **accepted** separate electorates in the Lucknow Pact (1916).
- Simla Deputation (1906) was led by the **Aga Khan** to Minto.

UP LINK

UP link: The **Aligarh** group (Mohsin-ul-Mulk, Viqar-ul-Mulk) organised the Simla Deputation; the League's 1916 session met at **Lucknow** alongside the Congress.

LAST-MINUTE 10

1. League = Dacca 1906
2. Salimullah = founder
3. Aga Khan = first president
4. 1907 session = Karachi
5. 1909 = communal representation
6. Separate electorates = seed of partition
7. Lucknow Pact accepted separate electorates
8. Justice Party = non-Brahmin, Madras
9. Simla Deputation = Aga Khan to Minto
10. Viqar-ul-Mulk chaired Dacca meet

Chapter 19

Delhi Durbar and Change of Capital

Built from 4 UPPSC questions (1990–2025). Focus: the 1911 Durbar announcements, Hardinge and the transfer of the capital, the 1912 bomb.

ASKED MOST · REVISE FIRST

- ■ Capital moved from Calcutta to Delhi under **Lord Hardinge**; announced 1911, transferred **1912** (per key) 3× 1990, 2008, 2008
- 1911: **partition revoked** and **capital transfer** announced (not the Lucknow Pact) 1× 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 The Durbar and the Capital

- Events of 1911: **partition of Bengal revoked, capital shifted to Delhi** 1× 2025
- Capital transfer under **Lord Hardinge** 2× 1990, 2008
- Year of transfer: **1912** 1× 2008
- Bomb at his state entry into Delhi: **Lord Hardinge** (1912) 1× 2008

CAPITAL TIMELINE

Delhi Durbar of George V, announcement
12 December 1911
↓
Delhi becomes the capital
1912
↓
Bomb on Hardinge, Chandni Chowk
23 December 1912
↓
New Delhi inaugurated
1931

TRAP ALERT

- "Announced" = 1911; "transferred" = 1912; New Delhi inaugurated = 1931.
- The **1903** Durbar was Curzon's; the **1911** Durbar was George V's under Hardinge.
- **Mayo** was killed at **Port Blair** (1872), not Delhi.

UP LINK

UP link: The Lucknow Pact (1916) is **not** a 1911 event; it is the usual distractor in 1911 questions.

LAST-MINUTE 10

1. 1911 Durbar = George V
2. Partition annulled 1911
3. Capital to Delhi = Hardinge
4. Transfer year = 1912 (key)
5. Bomb on Hardinge = 1912
6. New Delhi inaugurated = 1931
7. Curzon = 1903 Durbar
8. Mayo = Port Blair 1872
9. Lucknow Pact = 1916
10. Calcutta capital till 1911

Chapter 20

Lucknow Session of Congress (The Lucknow Pact)

Built from 8 UPPSC questions (1992–2013). Focus: the 1916 sessions at Lucknow, A.C. Mazumdar, reunion of the two camps, Champaran, the Pact and its makers.

ASKED MOST · REVISE FIRST

- ■ Lucknow session (1916) president: **Ambika Charan Mazumdar** 5× 2006, 2009, 2010, 2010, 2013
- ■ **Lucknow Pact** (1916) between Congress and League; made by **Tilak** (with Jinnah) 3× 2004, 2006, 2010

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

20.1 Lucknow 1916

- Congress and League both met in December 1916 at **Lucknow** 1× 2004
- ■ President: **A.C. Mazumdar** 4× 2006, 2009, 2010, 2013
- Chief architect of the Moderate-Extremist reunion: **Annie Besant** 1× 2004
- Not correct: "**Mazumdar was not the president**" 1× 2010
- Gandhi heard of Champaran's peasants at **Lucknow, 1916** (from Raj Kumar Shukla) 1× 2013

20.2 The Pact

- Lucknow Pact signed in **1916** 2× 2006, 2010
- Agreement made by **Tilak** (for the Congress) 1× 2004
- Period of Congress-League unity: **1916 to 1922** 1× 1992

TRAP ALERT

- Lucknow 1916 = Mazumdar; Lucknow **1936** = Nehru.
- Two reunions: Moderates and Extremists (**Besant, Tilak**); Congress and League (**Tilak, Jinnah**).
- The Pact **accepted** separate electorates.

UP LINK

UP link: **Lucknow** hosted both sessions in 1916; **Raj Kumar Shukla** met Gandhi there and took him to Champaran.

LAST-MINUTE 10

1. Lucknow 1916 = A.C. Mazumdar
2. Lucknow Pact = 1916
3. Tilak + Jinnah = Pact
4. Besant = reunion architect
5. Champaran told to Gandhi at Lucknow
6. Unity period = 1916-1922
7. Pact accepted separate electorates
8. Lucknow 1936 = Nehru
9. Raj Kumar Shukla = Champaran
10. Extremists rejoined in 1916

Chapter 21

Home Rule League Movement

Built from 6 UPPSC questions (1991–2010). Focus: Tilak's and Besant's leagues, the movement's timing, who was not in it.

ASKED MOST · REVISE FIRST

- First Home Rule League movement launched by **Annie Besant** (per key) **2×** 1991, 1993
- Two leagues (1915-16): **Tilak and Annie Besant**; popular during **World War I** **2×** 2001, 2008

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

21.1 The Leagues

- Launched the first Home Rule movement: **Annie Besant** **2×** 1991, 1993
- Annie Besant mostly linked with the **Home Rule Movement** **1×** 2010
- Two leagues led by **Tilak and Annie Besant** **1×** 2001
- Popular during World War I: **Home Rule Movement** **1×** 2008
- Not associated: **C.R. Das** **1×** 2010
- "Not correct" statement on the leagues: question marked (*) **1×** 2004

League	Founder	Founded	Area
Indian Home Rule League	Tilak	April 1916, Belgaum	Maharashtra (not Bombay city), Karnataka, CP, Berar
All India Home Rule League	Annie Besant	September 1916, Madras	Rest of India

TRAP ALERT

- Besant **planned** first (1915); Tilak **founded** first (April 1916).
- Besant's papers: **New India** and **Commonweal**.
- The **August Declaration** (1917) promised responsible government by stages, not swaraj.

UP LINK

UP link: **Motilal Nehru, Jawaharlal Nehru and Tej Bahadur Sapru** joined Besant's league at **Allahabad**.

LAST-MINUTE 10

1. Tilak = Belgaum, April 1916
2. Besant = Madras, September 1916
3. Besant = launched first (key)
4. Home Rule = World War I period
5. C.R. Das not in it
6. New India, Commonweal = Besant
7. Kesari, Maratha = Tilak
8. August Declaration = 1917
9. Montagu = Secretary of State
10. 1920: Besant's league renamed Swarajya Sabha

Chapter 22

Gandhi and his Early Movements

Built from 58 UPPSC questions (1991–2023). Focus: South Africa, Gokhale, Champaran, Ahmedabad and Kheda, titles, satyagraha, truth and non-violence, Gandhian politics and economics, Vaikom, Yerwada.

ASKED MOST · REVISE FIRST

- First satyagraha in India: **Champaran (1917)**, for **indigo** workers under **tinkathia** (3/20 of land) **6×** 2004, 2009, 2011, 2012, 2013, 2015
- First hunger strike: **Ahmedabad mill strike (1918)**, for cotton mill workers **4×** 1998, 2002, 2010, 2015
- Political guru: **Gokhale**, who told Gandhi to keep "ears open, mouth shut" for a year **3×** 1991, 2014, 2015
- "Mahatma" first used by **Tagore** (Champaran period); "Father of the Nation" first by **Subhas Bose** **5×** 2001, 2002, 2008, 2010, 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

22.1 Early Life and South Africa

- Karamchand Gandhi Diwan of "all of these" (per key; really Porbandar, Rajkot, Wankaner) **1×** 2001
- True: schooled at **Rajkot**, married at 13, law at **Inner Temple**, influenced by **Unto This Last**: **all four** **1×** 2010
- Thrown off the train at **Pietermaritzburg** **1×** 2015
- Adopted "Seva Dharma" in **South Africa** **1×** 2015
- First Congress session attended: **Calcutta, 1901** **2×** 2003, 2004
- South Africa prepared him for leadership (A true); "Indians would be ready for militant struggle" (R false) **1×** 2023
- Paper in South Africa: **Indian Opinion** **1×** 2002
- Influences: **Ruskin, Thoreau, Tolstoy** (all) **1×** 2013

Thinker	Gave Gandhi
Ruskin (Unto This Last)	Dignity of labour, good of all
Thoreau (Civil Disobedience)	Civil disobedience, refusing taxes
Tolstoy (The Kingdom of God Is Within You)	Non-violence; Tolstoy Farm

22.2 Return to India (1915)

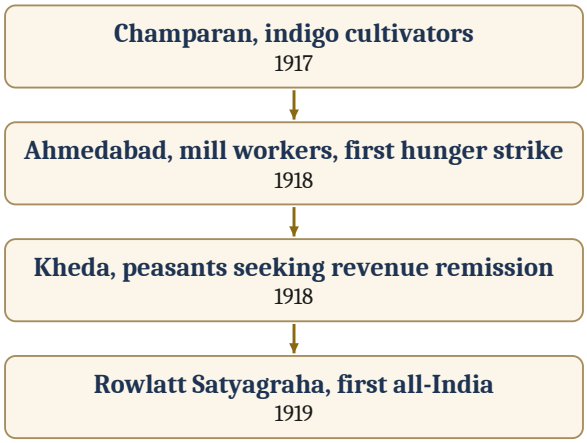
- Political guru: **G.K. Gokhale** **1×** 1991
- "Ears open, mouth shut" for the first year: **Gokhale** **1×** 2014
- Advised a year as observer and student: **Gokhale** **1×** 2015
- First public speech in India: **Varanasi** (BHU, 1916) **1×** 2015

22.3 Champaran, Ahmedabad, Kheda

- First successful satyagraha after return: **Champaran** **1×** 2011
- First satyagraha in India: **Champaran** **2×** 2004, 2009
- Mission to free bonded labour: question marked (*); intended **Champaran** **1×** 2015
- Tinkathia: **indigo on 3/20 of the land** **1×** 2013
- Champaran movement: problems of **indigo workers** **1×** 2012
- Tinkathia law: **Champaran, indigo** **1×** 2015
- Not correct: Champaran was "**the first all-India movement**" **1×** 2011
- Joined Gandhi at Champaran: **Rajendra Prasad and Anugrah Narayan Sinha** **1×** 1999
- Invited Gandhi to Champaran: **Raj Kumar Shukla** **1×** 2016
- Linked with Champaran Satyagraha: **Raj Kumar Shukla** **1×** one-liner
- Not on the Champaran Agrarian Enquiry Committee: **Anugrah Narayan Sinha** **1×** 2017
- Opposed Champaran (per key, by elimination): **N.G. Ranga** **1×** 2007
- First satyagraha (per key) against **low wages of workers** (Ahmedabad) **1×** 1998
- First hunger strike: **Ahmedabad strike** **1×** 2010
- Struggle of industrial workers: **Ahmedabad** **1×** 2002
- Ahmedabad satyagraha for **cotton mill workers** **1×** 2015
- Not led by Gandhi: **Bardoli** (1928, Vallabhbhai Patel) **1×** 2005

QUANTUM IAS & PCS

FIRST THREE SATYAGRAHAS



22.4 Titles

- "Mahatma" added during **Champaran** 2× 2001, 2002
- First called him "Mahatma": **Tagore** 1× 2010
- First called him "Father of the Nation": **Subhas Chandra Bose** 2× 2008, 2010
- "One-man boundary force": **Mountbatten** 1× 2015

22.5 Satyagraha and Ideas

- Coined "Satyagraha": **Gandhi** 1× 2015
- Best known for **passive resistance** for independence 1× 2013
- Not Gandhian: "the aim of a satyagrahi is **to defeat the enemy**" 1× 2013
- Last step in satyagraha (per key): **strike** 1× 2013
- Twin principles of Ram Rajya: **truth and non-violence** 1× 1993
- Non-violence is **a way to attain truth** 1× 1994
- Cruelest violence: **persistence of poverty** 1× 2015
- "Wrong means never lead to right ends": **Gandhi** 1× 2015
- "Morally wrong can never be politically right": **Gandhi** 1× 2015
- Supporter of **philosophical anarchism** 1× 2009
- Not a feature of Gandhian politics: **authority** 1× 2013
- Politics meant **activity for public welfare** 1× 2013
- "An individualist among socialists..." (per key) 1× 2013

22.6 Gandhian Economics and Swadeshi

- Trusteeship propounded in **South Africa, 1903** (per key) 1× 2013
- Not correct (per key): "he did not favour mechanisation in the USA" 1× 2013
- Uplift of untouchables through **cottage industry** 1× 2013
- "Gandhian innovation": **more from less for more people** 1× 2015
- Opposed planned development: **Gandhi** 1× 2019
- Favoured by Nehru, not Gandhi: **heavy industrialisation** 1× 2006
- "Destruction is the best way to deal with foreign cloth": **Gandhi** 1× 2010
- Boycott worked because India was a **market for manufactured goods** 1× 2013
- Not related to Gandhi: **Swadeshi Movement (1905)** 1× 1998

22.7 Other Points

- Correct: worked for untouchables, launched Non-Cooperation, began Civil Disobedience: **all the above** 1× 2013
- Vaikom: against untouchability, and Gandhi took part: **both correct** 1× 2021
- Jail Gandhi called a "Mandir": **Yerwada** 1× 2015

TRAP ALERT

- First in India = **Champaran**; first all-India = **Rowlatt**; first hunger strike = **Ahmedabad**.
- Bardoli **1928** was Patel's; Gandhi's planned Bardoli campaign (1922) was dropped after Chauri Chaura.
- Tagore called Gandhi **Mahatma**; Gandhi called Tagore **Gurudev**.

UP LINK

UP link: Gandhi's first public speech in India was at the opening of **BHU, Varanasi** (February 1916); he first heard of Champaran at the **Lucknow** Congress (1916).

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Indian Opinion = South Africa
2. Pietermaritzburg = thrown off train
3. Gokhale = political guru
4. BHU, Varanasi = first speech
5. Champaran 1917 = indigo, tinkathia
6. Raj Kumar Shukla = invited Gandhi
7. Ahmedabad 1918 = first hunger strike
8. Bardoli 1928 = Patel
9. Tagore = Mahatma; Bose = Father of the Nation
10. Yerwada = Mandir

Chapter 23

Peasant Movement and Kisan Sabha

Built from 11 UPPSC questions (1998–2024). Focus: the UP and Awadh Kisan Sabhas, regional peasant leaders, the All India Kisan Sabha at Lucknow, Bardoli, the Workers and Peasants Party, Bhoodan.

ASKED MOST · REVISE FIRST

- ■ First **All India Kisan Sabha**: **Lucknow** (1936); president **Sahajanand Saraswati**, general secretary **N.G. Ranga** 4× 2008, 2010, 2015, 2016
- • **Baba Ramchandra** organised the peasants of **Awadh** 2× 2008, 2010
- ■ **Bardoli Satyagraha** (1928): **Sardar Patel** 3× 2002, 2006, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

23.1 UP and the Kisan Sabhas

- UP Kisan Sabha (1918) formed by **Indra Narayan Dwivedi** (with Gauri Shankar Mishra) 1× 2023
- Baba Ramchandra organised farmers in **Awadh** 2× 2008, 2010
- General secretary of the All India Kisan Congress at Lucknow: **N.G. Ranga** 1× 2016
- First All India Kisan Sabha formed at **Lucknow** 2× 2008, 2010
- Founder president: **Swami Sahajanand Saraswati** 1× 2015
- Congress leader at the first meeting: **Jawaharlal Nehru** 1× 2017

23.2 Leaders and Regions

- Sahajanand **Bihar**, Khudai Khidmatgars **NWFP**, Swami Ramanand Tirtha **Hyderabad**, Abdul Hamid Khan **Southern Assam** 1× 1998
- Bardoli **Patel**, Bhartiya Kisan Vidyalaya **N.G. Ranga**, Bengal Praja Party **Fazlul Haq**, Bakasht **Shraddhanand** (per key; really Sahajanand) 1× 2006
- Patel's key role: **Bardoli Satyagraha** 2× 2002, 2010
- Workers and Peasants Party: "formed in 1927 as an all-India body" wrong; "work inside the Congress" right: **only 2** 1× 2024
- Bhoodan first started in **Andhra Pradesh** (Pochampalli, now Telangana, 1951) 2× 2013, 2014

Movement	Year	Region	Leader
UP Kisan Sabha	1918	UP	Indra Narayan Dwivedi, Gauri Shankar Mishra
Awadh Kisan Sabha	1920	Awadh	Baba Ramchandra
Eka Movement	1921-22	Hardoi, Bahraich	Madari Pasi
Bardoli	1928	Gujarat	Vallabhbhai Patel
All India Kisan Sabha	1936	Lucknow	Sahajanand, N.G. Ranga
Tebhaga	1946	Bengal	Kisan Sabha
Bhoodan	1951	Pochampalli	Vinoba Bhave

TRAP ALERT

- Lucknow 1936: Sahajanand **president**, Ranga **general secretary**.
- Bardoli **1922** = called off after Chauri Chaura; Bardoli **1928** = Patel's revenue satyagraha.
- Bhoodan began in Pochampalli: pick **Telangana** if offered.

UP LINK

UP link: The **UP Kisan Sabha** (1918), **Baba Ramchandra's** Awadh Kisan Sabha (1920), the **Eka** movement of Madari Pasi, and the first **All India Kisan Sabha** at **Lucknow** (1936) are all UP stories.

LAST-MINUTE 10

1. UP Kisan Sabha 1918 = Indra Narayan Dwivedi
2. Baba Ramchandra = Awadh
3. Eka = Madari Pasi
4. AIKS = Lucknow 1936
5. Sahajanand = president
6. N.G. Ranga = general secretary
7. Bardoli 1928 = Patel
8. Fazlul Haq = Bengal Praja Party
9. Abdul Hamid Khan = Southern Assam
10. Bhoodan = Pochampalli 1951

Chapter 24

Trade Union and Communist Party

Built from 10 UPPSC questions (1997–2023). Focus: the first trade union, AITUC and its presidents, the Kanpur and Meerut cases, M.N. Roy and Soumyendranath Tagore.

ASKED MOST · REVISE FIRST

- ■ First president of **AITUC** (Bombay, 1920): **Lala Lajpat Rai** 5× 1997, 2002, 2006, 2011, 2023
- AITUC at **Nagpur (1929)** presided by **Jawaharlal Nehru** 2× 2013, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Trade Unions

- First trade union (1918, Madras Labour Union): **B.P. Wadia** 1× 2007
- First AITUC president: **Lala Lajpat Rai** 2× 1997, 2002
- Presided over the first AITUC session in Bombay: **Lala Lajpat Rai** 1× 2023
- First AITUC session (Bombay, 1920): **Lajpat Rai** 2× 2006, 2011
- Presided over AITUC at Nagpur, 1929: **Nehru** 1× 2013
- AITUC president in 1929: **Nehru** 1× 2021

24.2 Communists and the Left

- Kanpur conspiracy case: against **communist** leaders 2× 2001, 2010
- World-wide sympathy (Einstein, H.G. Wells, Laski): **Meerut Conspiracy Case** 1× 2017
- Radical Democratic Party (1940): **M.N. Roy** 1× 2007
- Party of Soumyendranath Tagore: **Revolutionary Communist Party** 1× 2015

Event	Year	Key name
Madras Labour Union	1918	B.P. Wadia
AITUC, Bombay	1920	Lajpat Rai (president), N.M. Joshi
CPI at Tashkent	1920	M.N. Roy
Kanpur Bolshevik case	1924	Dange, Muzaffar Ahmad, Shaukat Usmani
CPI conference, Kanpur	1925	Satyabhakta
Meerut case	1929-33	31 trade union and communist leaders
AITUC, Nagpur	1929	Nehru; split

TRAP ALERT

- AITUC founder **N.M. Joshi**; first president **Lajpat Rai**.
- Ahmedabad 1918 = Gandhi's strike; first union 1918 = **Wadia** (Madras).
- Meerut case drew world sympathy; the Lahore case (1929-30) was Bhagat Singh's.

UP LINK

UP link: **Kanpur** gave its name to the 1924 Bolshevik case and hosted the 1925 communist conference; **Meerut** gave its name to the 1929 trial.

LAST-MINUTE 10

1. Wadia = first union, Madras 1918
2. AITUC = Bombay 1920
3. Lajpat Rai = first AITUC president
4. N.M. Joshi = founder
5. Nehru = AITUC Nagpur 1929
6. Bose = AITUC 1931
7. Kanpur case = communists
8. Meerut case = world sympathy
9. M.N. Roy = Radical Democratic Party 1940
10. Soumyendranath Tagore = Revolutionary Communist Party

Chapter 25

Rowlatt Act and Jallianwala Bagh Massacre (1919)

Built from 15 UPPSC questions (1990–2025). Focus: the Rowlatt Act under Chelmsford, the satyagraha, 13 April 1919, Tagore and Sankaran Nair, the Congress inquiry, Udham Singh.

ASKED MOST · REVISE FIRST

- **Tagore** returned his **knighthood** over Jallianwala Bagh **5×** 2001, 2004, 2006, 2016, 2022
- **Sankaran Nair** resigned from the Viceroy's Executive Council **4×** 2007, 2013, 2013, 2025
- Rowlatt Act (1919) and the massacre: under **Lord Chelmsford** **3×** 1990, 1993, 2014

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

25.1 The Rowlatt Act

- Passed in **1919** **1×** 1993
- Viceroy: **Lord Chelmsford** **1×** 2014
- Viceroy at Jallianwala Bagh: **Chelmsford** **1×** 1990
- Suggested a no-tax campaign against it: **Swami Shraddhanand** **1×** 2008
- Event just before Jallianwala: **Rowlatt Act** **1×** 2012

25.2 Jallianwala Bagh

- Crowd protested the arrest of **Dr. Saifuddin Kitchlew and Dr. Satyapal** **1×** 2002
- Order: **Satyapal's arrest, the massacre, Amritsar Congress (1919)** **2×** 2002, 2003
- Returned knighthood: **Tagore** **4×** 2001, 2004, 2016, 2022
- Tagore gave up knighthood over the **Jallianwala Bagh massacre** **1×** 2006
- Tagore renounced knighthood (A true); to join Non-Cooperation (R false) **1×** 2021
- Resigned from the Executive Council: **only Sankaran Nair** **1×** 2025
- Resigned from the Council: **Sankaran Nair** **3×** 2007, 2013, 2013
- Drafting the Congress inquiry report: **Gandhi** **1×** 2014
- Called "preventive murder" by Montagu: **Jallianwala Bagh** **1×** 1998
- Killed Michael O'Dwyer: **Udham Singh** (London, 1940) **1×** 1994

1919 IN PUNJAB

Rowlatt Act passed
March 1919
↓
Rowlatt Satyagraha, first all-India protest
April 1919
↓
Kitchlew and Satyapal arrested
10 April 1919
↓
Massacre under General Dyer
13 April 1919
↓
Hunter Committee and Congress inquiry
1919-20
↓
Amritsar Congress session
December 1919

TRAP ALERT

- **General Dyer** fired; **O'Dwyer** was Lt. Governor and was shot by Udham Singh.
- Tagore gave up a **title**; Sankaran Nair gave up an **office**; Jinnah resigned over the **Rowlatt Act**.
- Rowlatt Act passed under **Chelmsford**, repealed under **Reading** (1922).

UP LINK

UP link: **Motilal Nehru** of Allahabad presided over the **Amritsar Congress** (1919) held after the massacre.

LAST-MINUTE 10

1. Rowlatt Act = 1919, Chelmsford
2. No-tax idea = Shraddhanand
3. Massacre = 13 April 1919
4. Kitchlew + Satyapal arrested
5. Dyer fired; O'Dwyer = Lt. Governor
6. Tagore = returned knighthood
7. Sankaran Nair = resigned from Council
8. Gandhi = drafted Congress report
9. Montagu = "preventive murder"
10. Udham Singh = O'Dwyer, 1940

Chapter 26

Khilafat Movement

Built from 10 UPPSC questions (1993–2016). Focus: the Ali brothers, Hakim Ajmal Khan, Gandhi as Khilafat president, Congress motives, Jinnah and Malaviya, Hindu-Muslim unity.

ASKED MOST · REVISE FIRST

- Khilafat started and led by the **Ali brothers** (Mohammad Ali, Shaukat Ali) **2×** 2008, 2016
- Warned Gandhi against it and denounced his part: **Jinnah**; did not support it: **Malaviya** **5×** 2002, 2008, 2010, 2010, 2011

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

26.1 Leaders

- Started the movement: **Shaukat Ali and Mohammad Ali** **1×** 2008
- Prominent leaders: **Maulana Mohammad Ali and Shaukat Ali** **1×** 2016
- Renounced "Haziq-ul-Mulk": **Hakim Ajmal Khan** **1×** 2014
- President of the All India Khilafat Conference (1919): **Gandhi** **1×** 1993
- Congress backed it for **reinstating the Caliph** and **Muslim sympathy** **1×** 2000

26.2 Support and Opposition

- Warned Gandhi against fanaticism: **Jinnah** **2×** 2002, 2010
- Denounced Gandhi's role: **M.A. Jinnah** **1×** 2011
- Did not support Khilafat: **Madan Mohan Malaviya** **2×** 2008, 2010
- Speech on unity from the Jama Masjid pulpit (4 April 1919): **Swami Shraddhanand** **1×** 2004
- Result: **Hindu-Muslim differences narrowed** **1×** 1996

Person	Title given up
Hakim Ajmal Khan	Haziq-ul-Mulk
Gandhi	Kaiser-i-Hind
Tagore	Knighthood

TRAP ALERT

- **Shariatullah** was a Faraizi reformer (1818), not a Khilafat leader.
- The aim was to **restore** the Caliph; Turkey **abolished** the Caliphate in 1924 and the movement ended.
- Jinnah opposed it; Azad backed it.

UP LINK

UP link: The Ali brothers were from **Rampur**; the Khilafat drive found strong support in UP towns like **Lucknow** and **Aligarh**.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Ali brothers = Khilafat
2. Ajmal Khan = Haziq-ul-Mulk
3. Gandhi = Khilafat Conference president 1919
4. Aim = restore Caliph
5. Jinnah = warned Gandhi
6. Malaviya = did not support
7. Shraddhanand = Jama Masjid speech
8. Result = Hindu-Muslim unity
9. Caliphate abolished 1924
10. Khilafat merged into Non-Cooperation

Chapter 27

Non-Cooperation Movement

Built from 27 UPPSC questions (1990–2023). Focus: causes and launch, swaraj in a year, Nagpur 1920, titles and boycotts, Tagore's critique, national institutions, Chauri Chaura and the withdrawal.

ASKED MOST · REVISE FIRST

- ■ **Chauri Chaura** (Gorakhpur), **4 February 1922**; Gandhi withdrew the movement because of the violence **6×** 1990, 2002, 2006, 2006, 2008, 2013
- ■ Burning foreign cloth "a ruthless waste": **Tagore** **5×** 2002, 2003, 2004, 2010, 2013
- ■ "Swaraj in a year": aim of **Non-Cooperation** (1920), Gandhi's first mass movement **4×** 2007, 2008, 2010, 2012

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

27.1 Launch and Programme

- Not a cause of Non-Cooperation: **Salt Law** **1×** 2017
- Gandhi's first mass movement: **Non-Cooperation** **1×** 2007
- Started in **1920** **1×** 2008
- "Swaraj in a year": during **Non-Cooperation** **1×** 2012
- "Swaraj in a year" was its aim **1×** 2010
- Congress sought swaraj after the 1919 reforms under **Gandhi** **1×** 2017
- Moved the resolution at Nagpur (1920): **C.R. Das** **1×** 2011
- Supported it but did not live to see it: **Tilak** (died 1 August 1920) **1×** 2010
- Title Gandhi gave up: **Kaiser-i-Hind** **1×** 2004
- Gave up legal practice: **C.R. Das** **1×** 1999
- Not correct: "**Jinnah supported it**" **1×** 2013
- ■ "A ruthless waste": **Tagore** **4×** 2002, 2003, 2004, 2010
- Opposed burning foreign cloth: **Tagore** **1×** 2013
- Founded during the movement: **Kashi Vidyapeeth, Gujarat Vidyapeeth, Jamia Millia** (not BHU) **1×** 2005

27.2 Chauri Chaura and After

- Chauri Chaura: **4 February 1922** **2×** 2002, 2006
- District: **Gorakhpur** **2×** 2008, 2013
- Suspended because of **violence at Chauri Chaura** **2×** 1990, 2006
- Gandhi was at **Bardoli** at the time **1×** 2011
- Order: **Chauri Chaura, Bardoli resolutions, suspension** **1×** 2001
- Ended in **1922** **1×** 1990
- Censure vote against Gandhi (AICC, Delhi, 24 February 1922): **Dr. Moonje** **1×** 2002
- Gandhi postponed it (A true); C.R. Das and Motilal opposed (R true, not the reason) **1×** 2010
- Main outcome: **Hindu-Muslim unity** **1×** 2005
- E.V. Ramasamy joined Non-Cooperation and left the Congress in 1925: **both correct** **1×** 2023
- Wrong pair: **1930, Non-Cooperation** **1×** 1996
- Right pair: **1931, execution of Rajguru** **1×** 2004

NON-COOPERATION, 1920 TO 1922

Special session at Calcutta approves the programme
September 1920

↓

Nagpur session confirms; C.R. Das moves it
December 1920

↓

Boycott of titles, courts, schools, councils, cloth
1921

↓

Chauri Chaura, 22 policemen killed
4 February 1922

↓

Bardoli resolutions; movement withdrawn
12 February 1922

↓

Gandhi arrested, six years' sentence
March 1922

TRAP ALERT

- **Salt Law** belongs to Civil Disobedience (1930), not Non-Cooperation.
- Jamia Millia began at **Aligarh** (1920) and moved to Delhi later; **BHU** (1916) predates the movement.
- Bardoli 1922 = withdrawal; Bardoli 1928 = Patel's satyagraha.

UP LINK

UP link: **Chauri Chaura** is in **Gorakhpur** district; **Kashi Vidyapeeth** (Varanasi) and **Jamia Millia** (Aligarh) were born in the movement.

LAST-MINUTE 10

1. Non-Cooperation = 1920, first mass movement
2. C.R. Das = moved Nagpur resolution
3. Tilak died 1 August 1920
4. Gandhi returned Kaiser-i-Hind
5. Tagore = "ruthless waste"
6. Kashi, Gujarat Vidyapeeth, Jamia = new institutions
7. Chauri Chaura = 4 February 1922, Gorakhpur
8. Gandhi at Bardoli then
9. Moonje = censure vote
10. Outcome = Hindu-Muslim unity

Chapter 28

Formation of Swaraj Party (1923)

Built from 10 UPPSC questions (1991–2017). Focus: why the Swaraj Party was formed, C.R. Das and Motilal Nehru, Swarajists and no-changers, Vithalbhai Patel, allies in the Assembly.

ASKED MOST · REVISE FIRST

- ■ Swaraj Party (1923) formed by **C.R. Das and Motilal Nehru** 3× 1998, 1998, 2016
- • **C.R. Das** ("Deshbandhu") resigned the Congress presidency to form it 2× 2004, 2006

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

28.1 Formation

- Reasons: **withdrawal of Non-Cooperation** and **entering councils to wreck the 1919 Act** 1× 1998
- Resigned the Congress presidency to form it: **C.R. Das** 1× 2004
- Formed by **C.R. Das and Motilal Nehru** 2× 1998, 1998
- Party of Motilal and Das (1923): **Swaraj Party** 1× 2016
- "Deshbandhu": **Chittaranjan Das** 1× 2006

28.2 Members and Allies

- Not in the Swaraj Party: **C. Rajagopalachari** 2× 1991, 1993
- Not associated: **Rajendra Prasad** 1× 2014
- Elected President of the Central Legislative Assembly (1925): **Vithalbhai Patel** 1× 2012
- Supported the Swarajists in the Assembly: **Jinnah** 1× 2017
- Independent Party (16 December 1922), per key: **Malaviya and Motilal Nehru** 1× 2006

Pro-changers (Swarajists)	No-changers
C.R. Das, Motilal Nehru	Rajagopalachari, Rajendra Prasad
Vithalbhai Patel, N.C. Kelkar	Vallabhbhai Patel, M.A. Ansari
Hakim Ajmal Khan	Kasturi Ranga Iyengar

TRAP ALERT

- **Gaya** 1922 (Das presiding) rejected council entry; the **Delhi** special session (1923, Azad) allowed it.
- **Vithalbhai** was a Swarajist; his brother **Vallabhbhai** was a no-changer.
- Deshbandhu = C.R. Das; Deshpriya = J.M. Sengupta.

UP LINK

UP link: **Motilal Nehru** of Allahabad (Anand Bhawan) led the Swarajists in the Central Assembly.

LAST-MINUTE 10

1. Swaraj Party = 1923
2. C.R. Das + Motilal Nehru = founders
3. Das resigned Congress presidency
4. Aim = wreck 1919 Act from inside
5. Deshbandhu = C.R. Das
6. Rajagopalachari, Rajendra Prasad = no-changers
7. Vithalbhai Patel = Assembly President 1925
8. Jinnah = Assembly ally
9. Gaya 1922 = rejected council entry
10. Delhi 1923 = allowed council entry

Chapter 29

Simon Commission

Built from 12 UPPSC questions (1990–2023). Focus: dates and visits, an all-white commission, the boycott, Lajpat Rai, dominion status, the Independence for India League, Jinnah's Fourteen Points.

ASKED MOST · REVISE FIRST

- ■ Boycotted because it had **no Indian member** 5× 2003, 2004, 2010, 2010, 2010
- **Lala Lajpat Rai** injured in the lathi charge against it (Lahore, 1928) 2× 1993, 2004
- Visited India in **1928** 2× 1990, 1996

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

29.1 The Commission

- Visited India in **1928** 1× 1996
- Purpose: **administrative reform** (review of the 1919 Act) 1× 1990
- Made two visits (true); "second visit October 1929 to April 1930" (false): **only 1** 1× 2023
- Liberal Party member: **Sir John Simon** 1× 2013
- Indians kept out on the advice of **Lord Irwin** (per key; Birkenhead is also cited) 1× 2013

29.2 Boycott and Response

- ■ Congress boycotted it (A), no Indian member (R): **both true, R explains A** 3× 2010, 2010, 2010
- Boycotted because there was **no Indian member** 2× 2003, 2004
- Reviewed the 1919 Act, headed by Simon, recommended federation, opposed by Indians: **all true** 1× 2005
- Lajpat Rai injured in the **lathi charge** against the Commission 2× 1993, 2004
- First demanded dominion status (per key): **Tej Bahadur Sapru and Jayakar** 1× 2013
- Independence for India League (1928): **Nehru and Subhas Bose** 1× 2008
- Jinnah's Fourteen Points: **1929** 1× 2015

SIMON COMMISSION TIMELINE



TRAP ALERT

- Appointed **1927**, arrived **1928**, reported **1930**.
- Lajpat Rai was hurt at **Lahore** and died; Nehru and Pant were beaten at **Lucknow** and survived.
- Nehru Report (1928) asked for **dominion status**; Jinnah answered with the **Fourteen Points** (1929).

UP LINK

UP link: **Jawaharlal Nehru and G.B. Pant** were lathi-charged at **Lucknow** (November 1928) while leading the Simon boycott.

LAST-MINUTE 10

1. Simon Commission = 1927 appointed, 1928 arrived
2. All seven members British
3. Simon = Liberal
4. Boycott = no Indian member
5. Lajpat Rai = lathi charge, Lahore
6. Nehru, Pant = Lucknow lathi charge
7. Nehru Report = 1928, dominion status
8. Independence League = Nehru, Bose
9. Fourteen Points = Jinnah 1929
10. Report = 1930

Chapter 30

Lahore Session of Congress: Proposal of Complete Independence

Built from 10 UPPSC questions (1993–2023). Focus: Purna Swaraj at Lahore 1929, Nehru's first presidency, the resolutions and who moved them, the flag on the Ravi.

ASKED MOST · REVISE FIRST

- **Purna Swaraj** resolution passed at **Lahore, 1929** 7× 1993, 1999, 2004, 2008, 2010, 2011, 2023
- President: **Jawaharlal Nehru** (his first presidency) 5× 2004, 2006, 2009, 2010, 2013
- Tricolour hoisted on the Ravi at midnight, **31 December 1929**, by **Nehru** 2× 2004, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

30.1 The Session

- Purna Swaraj passed in **1929** 3× 1993, 1999, 2004
- Passed at the **Lahore session** 1× 2023
- Declared the goal at **Lahore, 1929** 3× 2008, 2010, 2011
- Nehru's first presidency: **Lahore, 1929** 1× 2013
- Lahore president: **Jawaharlal Nehru** 2× 2004, 2010
- Congress president when Purna Swaraj passed: **Nehru** 2× 2006, 2009
- Not in the Lahore resolutions: **abolition of untouchability** 1× 2010
- Resolution moved by **Gandhi** (some keys give Nehru) 1× 2009
- Flag hoisted at midnight, 31 December 1929: **Nehru** 1× 2004
- New tricolour first hoisted: **31 December 1929** 1× 2014

ROAD TO PURNA SWARAJ

Madras session names independence as goal
1927

↓

Nehru Report asks for dominion status
1928

↓

Calcutta session gives a one-year ultimatum
1928

↓

Lahore session declares Purna Swaraj
December 1929

↓

Independence Day observed
26 January 1930

TRAP ALERT

- **Presided** = Nehru; **moved** the resolution = Gandhi.
- Motilal at **Calcutta 1928**; Jawaharlal at **Lahore 1929**.
- Lahore 1929 = Purna Swaraj; Lahore **1940** = Muslim League's Pakistan resolution.

UP LINK

UP link: **Jawaharlal Nehru** of Allahabad became Congress president at 40; 26 January was later chosen for the Constitution to come into force (1950).

LAST-MINUTE 10

1. Lahore 1929 = Purna Swaraj
2. Nehru = president, first time
3. Gandhi = moved resolution
4. Flag on the Ravi = 31 December 1929
5. 26 January 1930 = Independence Day
6. Calcutta 1928 = one-year ultimatum
7. Motilal = Calcutta 1928
8. Untouchability not in Lahore resolutions
9. Civil Disobedience authorised at Lahore
10. Republic Day = 26 January 1950

Chapter 31

Civil Disobedience Movement

Built from 23 UPPSC questions (1993–2021). Focus: the salt choice, the Dandi March, arrests and Abbas Tyabji, Dharasana and Webb Miller, the Red Shirts and Garhwal Rifles, women's role.

ASKED MOST · REVISE FIRST

- **Dandi March** (12 March to 6 April 1930) from **Sabarmati** to break the **salt law**; began Civil Disobedience **7×** 1993, 1999, 2000, 2002, 2004, 2010, 2012
- After Gandhi's arrest the leader was **Abbas Tyabji** (then Sarojini Naidu) **3×** 2002, 2006, 2012
- Foreign journalist at Dharasana and with Gandhi: **Webb Miller** **4×** 2002, 2003, 2015, 2021
- **Lal Kurti** (Red Shirts): **Khan Abdul Ghaffar Khan** **4×** 1993, 2009, 2016, 2016

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

31.1 Salt and Dandi

- Salt agitation in 1930 (A true); "to make salt free for the poor" (R false) **2×** 2002, 2003
- Began with the Dandi March: **Civil Disobedience** **1×** 2000
- March started from **Sabarmati** **1×** 1999
- Civil Disobedience started with the **Dandi March** **1×** 2010
- Aim: **break the salt law** **1×** 1993
- Dandi March = violation of the **Salt Law** **1×** 2004
- 6 April 1930: **Dandi March** (salt law broken) **2×** 2002, 2012

■ Not correct (per key): "**Gandhi prepared salt at the seashore**" (he picked up natural salt) **3×** 2004, 2008, 2010

- Most marchers from **Gujarat** **1×** 2013
- "World sympathy in this battle of Right against Might": **Dandi March** **1×** 2013

31.2 Arrests and Dharasana

- "Gandhi not punished" false; "Malaviya, Devdas Gandhi, K.M. Munshi punished" true: **2 only** 1× 2016
- Vinoba Bhave first arrested in **Civil Disobedience** (per key) 1× 2001
- During the Dharasana raid Gandhi was in **Yerwada Jail** 1× 2010
- Led after Gandhi's arrest: **Abbas Tyabji** 3× 2002, 2006, 2012
- Foreign journalist at Sabarmati during Dandi: **Webb Miller** 2× 2002, 2003
- Reported Dharasana: **Webb Miller** 1× 2021
- American with Gandhi during Dandi: **Webb Miller** 1× 2015

31.3 Across India

- Ghaffar Khan's movement: **Lal Kurti (Red Shirts)** 1× 2016
- Leader of Lal Kurti: **Khan Abdul Ghaffar Khan** 2× 2009, 2016
- Red Shirts founded to **throw out the British** 1× 1993
- Garhwal soldiers refused to fire (Peshawar, 1930): **Civil Disobedience** 1× 2004
- Zeliangrong (Jiatrang) movement: **Manipur** (Rani Gaidinliu) 1× 2008
- Highest women's participation: **Salt Satyagraha** 1× 2016

Region	Leader of Civil Disobedience
Dandi, Gujarat	Gandhi
Dharasana, Gujarat	Abbas Tyabji, Sarojini Naidu
Vedaranyam, Tamil Nadu	C. Rajagopalachari
Malabar	K. Kelappan
NWFP	Khan Abdul Ghaffar Khan
Peshawar	Chandra Singh Garhwali (Garhwal Rifles)
Manipur, Nagaland	Rani Gaidinliu

TRAP ALERT

- Gandhi wanted the salt **tax and monopoly** abolished, not free salt.
- March began **12 March**; salt law broken **6 April 1930**.
- Webb Miller (1930) is not **Louis Fischer** (1942).

UP LINK

UP link: **Chandra Singh Garhwali** of the Garhwal Rifles (Peshawar, 1930) came from the Garhwal hills, then part of the **United Provinces**.

LAST-MINUTE 10

1. Dandi = 12 March to 6 April 1930
2. Start = Sabarmati
3. Salt law broken = 6 April
4. Abbas Tyabji = after Gandhi's arrest
5. Sarojini Naidu = Dharasana
6. Webb Miller = Dharasana reporter
7. Gandhi = Yerwada Jail
8. Red Shirts = Ghaffar Khan
9. Garhwal Rifles = Peshawar 1930
10. Rani Gaidinliu = Manipur

Chapter 32

Gandhi-Irwin Pact and Round Table Conference

Built from 20 UPPSC questions (1993–2021). Focus: the Delhi Pact, Karachi 1931, ordering the events of 1931, the three Round Table Conferences, Gandhi in London, Ambedkar.

ASKED MOST · REVISE FIRST

- Congress attended only the **Second Round Table Conference** (1931), with **Gandhi** 3× 2004, 2008, 2015
- Order: **Gandhi-Irwin Pact, Bhagat Singh's execution, Karachi session, Second RTC** 4× 2006, 2009, 2009, 2013
- Attended all three RTCs: **B.R. Ambedkar** (Sapru too) 3× 2006, 2011, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

32.1 The Pact and Karachi

- Main purpose: **ease Congress participation** in the Round Table 1× 1998
- Pact suspended **Civil Disobedience** 1× 1993
- "The Two Mahatmas": **Sarojini Naidu** 1× 2001
- Gandhi's gains as "consolation prizes": **Alan Campbell-Johnson** 1× 2014
- Karachi session president: **Vallabhbhai Patel** 1× 2005
- Karachi, the "pinnacle of Gandhi's popularity": **S.C. Bose** 1× 2014
- ■ Order: **Gandhi-Irwin Pact, execution of Bhagat Singh, Karachi, Second RTC** 4× 2006, 2009, 2009, 2013
- Order: **Gandhi-Irwin Pact, execution of Bhagat Singh, Karachi, Poona Pact** 1× 2004

32.2 Round Table Conferences

- Not true of the First RTC: "**attended by the Congress**" 1× 2010
- First RTC summoned in **1930** 1× 2014
- Indian Christians at the First RTC: **K.T. Paul** 1× 2000
- Congress represented at the **Second RTC** 1× 2004
- Congress first participated at the **Second RTC** 1× 2008
- Gandhi present at the **Second** RTC 1× 2015
- "Nehru represented the Congress at the Second RTC" (A false); Pact implied Congress participation (R true) 1× 2016
- At the Second RTC: **Gandhi, Sarojini Naidu, Malaviya** (not Azad) 1× 1998
- Not at the Second RTC: **Rajendra Prasad** 1× 2020
- Gandhi stayed at **Kingsley Hall** in London 1× 2003
- Sailed on **S.S. Rajputana** 1× 2000
- ■ At all three RTCs: **Ambedkar** 3× 2006, 2011, 2021

Conference	Dates	Congress
First RTC	November 1930 to January 1931	Boycotted
Second RTC	September to December 1931	Gandhi (sole representative)
Third RTC	November to December 1932	Absent

TRAP ALERT

- The Pact **suspended** Civil Disobedience; it did not abolish the salt tax.
- Karachi 1931 = **Patel**; Lahore 1929 = **Nehru**.
- Ambedkar **and** Sapru attended all three; pick Ambedkar if both appear.

UP LINK

UP link: **Madan Mohan Malaviya** of Allahabad attended the Second RTC with Gandhi and Sarojini Naidu.

LAST-MINUTE 10

1. Gandhi-Irwin Pact = 5 March 1931
2. Bhagat Singh hanged = 23 March 1931
3. Karachi = Patel, end March 1931
4. Sarojini = "Two Mahatmas"
5. First RTC = 1930, no Congress
6. Second RTC = Gandhi, S.S. Rajputana, Kingsley Hall
7. Naidu, Malaviya also at Second RTC
8. K.T. Paul = Indian Christians
9. Ambedkar = all three RTCs
10. Third RTC = 1932

Chapter 33

Communal Award and Poona Pact (1932)

Built from 14 UPPSC questions (1994–2017). Focus: MacDonald's Award, the Poona Pact and who signed it, its effect, the Harijan Sevak Sangh, Dalit organisations, Karachi 1931.

ASKED MOST · REVISE FIRST

- ■ **Poona Pact** (1932) concerned the **depressed classes** 4× 1996, 1997, 2007, 2008
- ■ **Harijan Sevak Sangh** (1932): founded by **Gandhi**, president **G.D. Birla** 4× 1994, 2006, 2011, 2017
- • **Communal Award** issued by **Ramsay MacDonald** (August 1932) 2× 1999, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

33.1 Award and Pact

- Communal Award: **Ramsay MacDonald** 1× 1999
- Not given separate electorates: **Buddhists** 1× 2001
- Award announced in August 1932 (A); separate electorates for minorities (R): **both true, R explains A** 1× 2017
- ■ Poona Pact concerned **depressed classes** 4× 1996, 1997, 2007, 2008
- "Signed between": question marked (*); intended **Gandhi and Ambedkar** 1× 2004
- Pact that followed the Award: **Poona Pact** 1× 2010
- Did not sign the Poona Pact: **Gandhi** (Ambedkar and Malaviya signed) 1× 2001
- Poona Pact undid the Award (A); led to reserved seats for SC and ST (R): **both true, R explains A** 1× 2005

33.2 Harijan Work and Dalit Bodies

- Founded the Harijan Sevak Sangh: **Gandhi** 1× 2017
- ■ Its president: **G.D. Birla** 3× 1994, 2006, 2011
- Former name: **All India Anti-Untouchability League** 1× 2014
- "Gandhi like a fleeting phantom raises dust but not the level": **Ambedkar** 1× 2004
- All India Depressed Classes League: **Jagjivan Ram** 1× 2000
- "Gandhi may die, but Gandhism will remain": **Karachi, 1931** 2× 2008, 2013

FROM AWARD TO PACT, 1932

Communal Award, separate electorates for depressed classes
16 August 1932
↓
Gandhi's fast unto death in Yerwada
20 September 1932
↓
Poona Pact: reserved seats in joint electorates
24 September 1932
↓
Harijan Sevak Sangh founded
1932
↓
Harijan weekly started
1933

TRAP ALERT

- Award by the **Prime Minister** (MacDonald), not the Viceroy (Willingdon).
- Poona Pact signed by **Ambedkar and Malaviya**; Gandhi did not sign.
- Depressed Classes **League** = Jagjivan Ram; **Association** = M.C. Rajah; SC **Federation** = Ambedkar.

UP LINK

UP link: **Madan Mohan Malaviya** signed the Poona Pact for the caste Hindus.

LAST-MINUTE 10

1. Communal Award = MacDonald, 1932
2. Buddhists = no separate electorate
3. Poona Pact = depressed classes
4. Gandhi fasted at Yerwada
5. Signed by Ambedkar, Malaviya
6. Reserved seats replace separate electorates
7. Harijan Sevak Sangh = Gandhi; Birla president
8. Former name = Anti-Untouchability League
9. Jagjivan Ram = Depressed Classes League
10. "Gandhism will remain" = Karachi 1931

Chapter 34

Congress Socialist Party, 1937 Elections and World War II

Built from 19 UPPSC questions (1992–2024). Focus: the Congress Socialist Party, the 1937 elections and ministries, the UP ministry, League reports, Haripura and Tripuri, the war and the resignations.

ASKED MOST · REVISE FIRST

- ■ **Tripuri (1939)**: Bose re-elected by defeating **Pattabhi Sitaramayya**; after he resigned, **Rajendra Prasad** became president **5×** 1997, 2004, 2007, 2010, 2011
- • **Haripura (1938, Gujarat)**: **S.C. Bose** presided **2×** 2000, 2006
- ■ 1937: no full Congress majority in **Punjab**; no ministry in **Bengal and Punjab** **3×** 1992, 2004, 2016
- ■ Congress ministries lasted **28 months** (1937-39) **3×** 2010, 2013, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

34.1 Congress Socialist Party (1934)

- "Formed January 1934" and "Nehru formally joined": **neither correct** **1×** 2024
- Prominent CSP leader: **Acharya Narendra Dev** **1×** 1996
- Organised by **Jayaprakash Narayan and Narendra Dev** **1×** 2008
- Convenor at Patna (1934): **Jayaprakash Narayan** **1×** 2008

34.2 Elections of 1937

- No full majority: **Punjab** **2×** 1992, 2004
- No Congress ministry: **Bengal and Punjab** **1×** 2016
- ■ No absolute majority: question marked (*); both **Bombay and Assam** fit **4×** 2003, 2004, 2004, 2008
- Number of Congress ministries: question marked (*); 6 in July 1937, 8 later **1×** 2015
- Finance in the UP ministry: **G.B. Pant** himself (some keys give Rafi Ahmad Kidwai) **1×** 2012
- League reports in order: **Pirpur, Shareef, Muslim Sufferings under Congress Rule** **1×** 2013

34.3 Haripura, Tripuri and the War

- Haripura president: **S.C. Bose** **1×** 2006
- Haripura is in **Gujarat** **1×** 2000
- Bose's second presidency: **Tripuri** **1×** 2011
- Bose defeated **Pattabhi Sitaramayya** **1×** one-liner
- Defeated Sitaramayya at **Tripuri, 1939** **2×** 2004, 2010
- After Bose resigned: **Rajendra Prasad** **2×** 1997, 2007
- Congress policy in WWII: **cooperation if complete independence was assured** **1×** 1993
- "Congress helped Britain in the war, expecting independence": **both false** **1×** 1992
- ■ Tenure of the 1937 ministries: **28 months** **3×** 2010, 2013, 2013

Session	Year	President
Lucknow	1936	Jawaharlal Nehru
Faizpur (first village session)	1936	Jawaharlal Nehru
Haripura	1938	Subhas Chandra Bose
Tripuri	1939	Bose, then Rajendra Prasad
Ramgarh	1940	Maulana Azad

TRAP ALERT

- Haripura 1938 = Bose **unopposed**; Tripuri 1939 = Bose **by contest**, then resigned.
- CSP worked **inside** the Congress; it is not the CPI or the Forward Bloc.
- Ministries resigned in **October-November 1939** over India being dragged into the war.

UP LINK

UP link: **G.B. Pant** led the UP Congress ministry (July 1937) with **Rafi Ahmad Kidwai** in charge of revenue; Narendra Dev, a CSP founder, came from **Faizabad**.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. CSP = 1934, Patna, JP and Narendra Dev
2. Nehru never joined CSP
3. 1937 = no ministry in Bengal, Punjab
4. Pant = UP Premier 1937
5. Pirpur Report = first League report
6. Haripura 1938 = Bose, Gujarat
7. Tripuri 1939 = Bose beat Sitaramayya
8. Rajendra Prasad = after Bose
9. Ministries = 28 months
10. Ramgarh 1940 = Azad

Chapter 35

Demand for Pakistan

Built from 10 UPPSC questions (1991–2014). Focus: Iqbal at Allahabad, Jinnah's changing role, the Lahore Resolution of March 1940, Pakistan Day.

ASKED MOST · REVISE FIRST

- Idea of a separate Muslim nation: **Muhammad Iqbal**, Allahabad session of the League (1930) 3× 2003, 2013, 2014
- **Pakistan Resolution** adopted at **Lahore**, **March 1940**, Jinnah presiding 4× 2010, 2012, 2012, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

35.1 Iqbal and Jinnah

- Separate nation idea: **Sir Muhammad Iqbal** 2× 2003, 2014
- First definite expression: Iqbal's speech at the **Allahabad session (1930)** 1× 2013
- "Nehru is a patriot, Jinnah a politician": **Iqbal** 2× 1991, 2010
- "Ambassador of Hindu-Muslim Unity": **Sarojini Naidu** 2× 2000, 2004
- Not true about Jinnah (per key): "**he always kept a diary**" 1× 2008
- Chaired the Lahore session (1940): **Jinnah** 1× 2013

35.2 Lahore Resolution

- Pakistan Resolution: **March 1940** 2× 2010, 2012
- Session that gave effect to the two-nation theory: **Lahore** 1× 2012
- Moved the 1940 resolution (per key): **Khaliquzzaman** (Fazlul Haq moved it; Khaliquzzaman seconded) 1× 2000
- League's Pakistan Day: **23 March 1943** (per key) 1× 2008

Step	Year	Person
Separate Muslim state idea	1930	Iqbal, Allahabad
Name "Pakistan"	1933	Choudhry Rahmat Ali
Lahore Resolution	1940	Jinnah presided; Fazlul Haq moved
Direct Action Day	16 August 1946	Muslim League

TRAP ALERT

- Iqbal gave the **idea**; Rahmat Ali gave the **name**; Jinnah led the **demand**.
- Lahore 1929 = Purna Swaraj (Congress); Lahore 1940 = Pakistan (League).
- If **Fazlul Haq** is an option, he is the mover; never mark Jinnah, who presided.

UP LINK

UP link: Iqbal's 1930 address was given at **Allahabad**; **Choudhry Khaliquzzaman** was a League leader from **Lucknow**.

LAST-MINUTE 10

1. Iqbal = Allahabad 1930
2. Rahmat Ali = name, 1933
3. Sarojini = "Ambassador of Unity"
4. Iqbal: "Nehru patriot, Jinnah politician"
5. Lahore = March 1940
6. Jinnah presided Lahore
7. Fazlul Haq moved; Khaliquzzaman seconded
8. Two-nation theory = Lahore session
9. Pakistan Day = 23 March
10. Direct Action = 16 August 1946

Chapter 36

Individual Satyagraha, Cripps Mission and Quit India Movement

Built from 33 UPPSC questions (1990–2024). Focus: 1939-42 order, Vinoba Bhave, Cripps proposals, Bombay 8 August 1942, "Do or Die", Nehru and Azad, the underground, Ballia, Louis Fischer, Linlithgow and Wavell.

ASKED MOST · REVISE FIRST

- First individual satyagrahi (1940): **Vinoba Bhave** 3× 1995, 2002, 2007
- **Quit India** (1942) launched at **Bombay**; "**Do or Die**" by Gandhi 6× 1990, 1991, 1993, 2009, 2012, 2015
- Parallel government in UP: **Ballia** (also Satara, Midnapur) 4× 2002, 2013, 2013, 2016
- Not with Quit India: **Shanti Ghosh**; underground: **Aruna Asaf Ali, Usha Mehta, JP** 7× 1997, 2009, 2010, 2010, 2011, 2013, 2023
- American with Gandhi in 1942: **Louis Fischer** 4× 2001, 2002, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

36.1 1939 to 1942

- Order: **ministers resign (1939), Ramgarh (March 1940), August Offer (August 1940), Cripps (1942)** 1× 2024
- Limited satyagraha in October 1940 (A true); "to embrace Britain's war effort" (R false) 1× 2023
- First individual satyagrahi: **Vinoba Bhave** 2× 1995, 2002
- Started the individual satyagraha: **Vinoba Bhave** 1× 2007
- Cripps: dominion status, Indian Union, right of provinces to stay out, Indians to frame the constitution: **all of the above** 2× 1998, 2005

36.2 Quit India

- Launched at **Bombay** 1× 1990
- Year: **1942** 1× 1993
- Gandhi arrested at **Bombay** 1× 2012
- "Free India or die in the attempt": **Quit India** 1× 2009
- "Do or Die": **Gandhi** 1× 1991
- "Do or Die" during **Quit India** 1× 2015
- Supported the resolution: **Vallabhbhai Patel** 1× 2009
- Moved the Quit India resolution: **Jawaharlal Nehru** 1× 2013
- Congress president then: **Maulana Azad** 1× 1996
- President for six consecutive years: **Azad** (1940-46) 2× 2008, 2012
- Not true: "**led by Gandhi**" (he was arrested at once) 1× 2016
- Did not support Quit India: **all of the above** (Muslim League, Hindu Mahasabha, CPI) 1× 2004
- British and League drew closer (A); Jinnah kept Muslims away (R): **both true, R explains A** 1× 2003

36.3 Underground and Parallel Governments

- Ran Congress Radio: **Usha Mehta** 1× 1997
- Broadcast on Congress Radio: **Ram Manohar Lohia** 1× 2008
- Escaped jail and led the underground: **Jayaprakash Narayan** (Hazaribagh) 1× 2011
- Aruna Asaf Ali: **Quit India** underground 1× 2009
- ■ Woman not linked with Quit India: **Shanti Ghosh** 4× 2010, 2010, 2013, 2023
- Parallel governments: **Ballia and Satara** 1× 2002
- Jatiya Sarkar in Midnapur: **1942** 1× 2016
- UP district with a parallel government: **Ballia** (Chittu Pandey) 2× 2013, 2013

36.4 Observers and the British

- American with Gandhi during Quit India: **Louis Fischer** 2× 2001, 2002
- Louis Fischer was **an American journalist** 1× 2008
- Fischer linked with **Quit India** 1× 2010
- "Political blackmail": **Lord Linlithgow** 1× 2007
- Commander-in-Chief during Quit India: **Lord Wavell** 1× 2013

QUANTUM IAS & PCS

- "Operation Rubicon": plan for **Gandhi's fast unto death in jail** 1× 2017
- Quit India was the culmination (A); transfer of power needed a mechanism (R): **both true, R explains A** 1× 2005

Place	Parallel government	Leader
Ballia (UP)	1942	Chittu Pandey
Satara (Maharashtra)	1943-46	Y.B. Chavan, Nana Patil
Tamluk, Midnapur (Bengal)	Jatiya Sarkar, 1942-44	Satish Samanta

TRAP ALERT

- Gandhi was **arrested** in Bombay but **held** at the Aga Khan Palace, Poona.
- Hazaribagh = JP's jail, not a parallel government; Meerut is a distractor.
- Wavell: **C-in-C** in 1942, **Viceroy** from 1943.

UP LINK

UP link: **Ballia** under **Chittu Pandey** set up a parallel government in August 1942, earning the name "Baghi Ballia".

LAST-MINUTE 10

1. Vinoba = first individual satyagrahi
2. Cripps = March 1942
3. Quit India = Bombay, 8 August 1942
4. Do or Die = Gandhi
5. Nehru moved; Azad presided
6. Usha Mehta = Congress Radio
7. JP = Hazaribagh escape
8. Ballia = Chittu Pandey
9. Louis Fischer = 1942
10. Linlithgow = "political blackmail"

Chapter 37

Subhas Chandra Bose and Azad Hind Fauj

Built from 22 UPPSC questions (1990–2023). Focus: Bose's early life and titles, the Forward Bloc, the first INA, Rash Behari Bose, the Provisional Government, the Rani Jhansi Regiment, the Red Fort trials.

ASKED MOST · REVISE FIRST

- ■ INA officers defended at the **Red Fort (1945)**: **Bhulabhai Desai** led; **T.B. Sapru** pleaded; **Patel** did not 7× 2004, 2008, 2010, 2010, 2010, 2013, 2019
- ■ First INA (1942): idea and first commander **Mohan Singh** 3× 1997, 2003, 2005
- Provisional Government of Free India: **21 October 1943** 1× 2013
- Women's unit: **Rani Jhansi Regiment**, raised by **Bose** 2× 2015, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

37.1 Bose before the INA

- Born at **Cuttack** 1× 2006
- Political guru: **C.R. Das** 1× 2017
- "Desh Nayak": **Tagore** 2× 2006, 2009
- Forward Bloc founded in **1939** 2× 2008, 2016

37.2 The INA and Azad Hind

- INA brainchild (per key): **Gyani Pritam Singh and Major Fujiwara** 1× 2001
- INA founded in **1942** 1× 2003
- Suggested the INA: **Mohan Singh** 1× 1997
- First commander: **Mohan Singh** 1× 2005
- Helped Subhas Bose set up the Azad Hind Fauj: **Rash Behari Bose** 1× 2002
- Provisional Government announced on **21 October 1943** 1× 2013
- Did not recognise Azad Hind: **Indonesia** 1× 2023
- Women's regiment: **Rani Jhansi Regiment** 1× 2023
- Founded the Rani Laxmibai Regiment: **Subhas Bose** 1× 2015

- "Jai Hind" slogan: **S.C. Bose** 1× 1990

37.3 Red Fort Trials

- Not linked with the INA: **B.C. Dutt** (RIN mutiny, 1946) 1× 2016
- Sentenced to seven years: **Rashid Ali** 1× 2008
- Led the defence lawyers: **Bhulabhai Desai** 1× 2010
- Pleaded for the INA officers: **Sir T.B. Sapru** 3× 2008, 2010, 2013
- Not a defence lawyer: **Vallabhbhai Patel** 1× 2004
- Trials held in **1945** 1× 2010
- Held at the **Red Fort, Delhi** 1× 2019
- INA Day: **12 November 1945** 1× 2013

INA TIMELINE

Mohan Singh raises the first INA
1942

↓

Rash Behari Bose hands over the League and INA
July 1943

↓

Provisional Government at Singapore
21 October 1943

↓

Imphal-Kohima campaign; "Give me blood"
1944

↓

Red Fort trials of Shah Nawaz, Sahgal, Dhillon
November 1945

TRAP ALERT

- **Rash Behari Bose** (Indian Independence League) helped **Subhas Chandra Bose**.
- **B.C. Dutt** = navy revolt 1946; **B.K. Dutt** = Assembly bomb 1929. Neither was INA.
- "Delhi Chalo" = Singapore 1943; "Give me blood" = Burma 1944.

UP LINK

UP link: The **Red Fort** defence team included **Jawaharlal Nehru**, **T.B. Sapru** and **Kailash Nath Katju**, all lawyers from **Allahabad**.

LAST-MINUTE 10

1. Bose born at Cuttack
2. C.R. Das = guru
3. Tagore = "Desh Nayak"
4. Forward Bloc = 1939
5. Mohan Singh = first INA, 1942
6. Rash Behari Bose = handed over INA
7. Azad Hind = 21 October 1943
8. Rani Jhansi Regiment = women
9. Bhulabhai Desai = led defence
10. INA Day = 12 November 1945

Chapter 38

Cabinet Mission Plan (1946)

Built from 10 UPPSC questions (1994–2024). Focus: the C.R. Formula, the order of events in 1946, the three ministers, the three-tier plan, the Congress and League responses.

ASKED MOST · REVISE FIRST

- **Three-tier polity** proposed by the **Cabinet Mission** 4× 1994, 2002, 2010, 2015
- Led by **Pethick-Lawrence** (with Cripps and A.V. Alexander) 3× 2015, 2015, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

38.1 Before and During the Mission

- C.R. Formula: **C. Rajagopalachari** 1× 2019
- Order: **UK Parliamentary Delegation, RIN revolt, Cabinet Mission announced, Interim Government** 1× 2024
- Arrived in India in 1946: **Cabinet Mission** 1× 2004
- Led by **Sir Pethick-Lawrence** 2× 2015, 2015
- Not a member: **John Simon** 1× 2016
- Three-tier polity: **Cabinet Mission** 4× 1994, 2002, 2010, 2015
- Congress president during the visit: **Maulana Azad** 1× 2008

- Fully in favour of the Plan (per key): **Sardar Patel** 1× 2001
- League withdrew its acceptance (A true); League joined the interim government (R true, not the reason) 1× 2018

Tier	Content
Union	Defence, foreign affairs, communications
Groups of provinces	A (Hindu majority), B (north-west), C (Bengal, Assam)
Provinces and states	All other subjects

TRAP ALERT

- **Cripps** led the 1942 mission; in 1946 he was only a **member**.
- C.R. Formula = **Rajagopalachari**, not C.R. Das.
- Mission **announced** 19 February, **arrived** 24 March 1946.

UP LINK

UP link: **Acharya Kripalani** became Congress president at the **Meerut** session (November 1946), after Azad and Nehru in the same year.

LAST-MINUTE 10

1. C.R. Formula = Rajaji, 1944
2. Cabinet Mission = 1946
3. Pethick-Lawrence = leader
4. Cripps, A.V. Alexander = members
5. John Simon not a member
6. Three-tier plan = Cabinet Mission
7. Azad = Congress president then
8. Patel = favoured Plan (key)
9. League withdrew acceptance
10. Plan date = 16 May 1946

Chapter 39

Formation of Interim Government (1946)

Built from 7 UPPSC questions (2004–2015). Focus: Direct Action Day, formation date, members and portfolios, the League's entry.

ASKED MOST · REVISE FIRST

- ■ **Direct Action Day: 16 August 1946** 3× 2007, 2008, 2010
- Interim Government under Nehru formed in **September 1946** 2× 2006, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

39.1 The Government

- ■ Direct Action Day: **16 August 1946** 3× 2007, 2008, 2010
- Formed under Nehru: **September 1946** (2 September) 2× 2006, 2015
- Not ministers on 2 September 1946: **Liaquat Ali Khan and Maulana Azad** 1× 2013
- Not in the interim cabinet: **B.R. Ambedkar** 1× 2004
- Railways: **Asaf Ali** 1× 2008
- Finance Minister: **Liaquat Ali Khan** (League joined on 26 October 1946) 1× 2014

Member	Portfolio
Jawaharlal Nehru	External Affairs, Commonwealth Relations
Vallabhbhai Patel	Home, Information and Broadcasting
Baldev Singh	Defence
Rajendra Prasad	Food and Agriculture
Asaf Ali	Railways
Liaquat Ali Khan (League)	Finance

TRAP ALERT

- Announced **24 August**, formed **2 September 1946**.
- Liaquat and the League joined only in **October 1946**.
- Ambedkar was **not** in the interim government; he became Law Minister in 1947.

UP LINK

UP link: **Jawaharlal Nehru** of Allahabad headed the interim government as Vice-President of the Executive Council.

LAST-MINUTE 10

1. Direct Action Day = 16 August 1946
2. Interim Government = 2 September 1946
3. Nehru = Vice-President of Council
4. Patel = Home
5. Baldev Singh = Defence
6. Asaf Ali = Railways
7. Liaquat = Finance
8. League joined October 1946
9. Ambedkar not a member
10. Azad joined later (Education)

Chapter 40

Partition of India and Independence

Built from 26 UPPSC questions (1990–2016). Focus: why Britain left, the Mountbatten Plan, the AICC vote on partition, the Partition Council and Radcliffe, the Independence Act, 15 August, first office-holders, princely states.

ASKED MOST · REVISE FIRST

- ■ Congress president at independence: **J.B. Kripalani** (Meerut 1946; Delhi AICC, June 1947) 4× 1994, 2007, 2010, 2015
- ■ Britain left in 1947 (A) because it was weakened by WWII (R): **R explains A**; Labour in power 4× 2001, 2003, 2010, 2012
- ■ **Indian Independence Act**: passed **July 1947**, Royal Assent **18 July 1947** 3× 2012, 2014, 2014
- First Governor-General of free India: **Mountbatten** 2× 2006, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

40.1 The Plan and the Partition Vote

- Party in power in the UK: **Labour** 1× 2012
- ■ Britain freed India (A); weakened by WWII (R): **both true, R explains A** 3× 2001, 2003, 2010
- Platform of independence: **Mountbatten Plan** 1× 2006
- Mountbatten Plan was the basis for **transfer of power and partition** 1× 2013
- Congress accepted it (A true); "believed in the two-nation theory" (R false) 1× 1998
- Presided over the AICC that passed partition (15 June 1947): **J.B. Kripalani** 2× 1994, 2007
- Congress president on 15 August 1947: **J.B. Kripalani** 1× 2015
- Meerut session (1946): **J.B. Kripalani** 1× 2010
- Seconded the partition resolution: **Maulana Azad** (Pant moved it) 1× 2014
- "Surrender of nationalism to communalism": **Dr. Kitchlew** 1× 2014
- Voted against partition: **Khan Abdul Ghaffar Khan** 1× 2002
- Congress on the Partition Council: **Patel and Rajendra Prasad** (some keys: Nehru and Patel) 1× 2009
- Boundary Commission head: **Radcliffe** 1× 2015

40.2 Independence

- Independence Act passed: **July 1947** 1× 2012
- Came into force (per key): **18 July 1947** 1× 2014

QUANTUM IAS & PCS

- Royal Assent: **18 July 1947** 1× 2014
- Sang "Hindustan Hamara" and "Jana Gana Mana" at midnight: **Sucheta Kripalani** 1× 2010
- Took no part in Independence Day celebrations: **Gandhi** (in Calcutta, fasting) 1× 2016
- First PM appointed by the **Governor-General** 1× 2001
- First Governor-General of free India: **Mountbatten** 2× 2006, 2014
- First Law Minister: **B.R. Ambedkar** 1× 2012
- Most princely states integrated in **1947** 1× 1990
- Delayed accession: **Junagadh, Hyderabad, Jammu and Kashmir** 1× 1998
- Standstill Agreement with India: **Hyderabad** 1× 1997
- "Indian nationalism the child of British rule": **R. Coupland** 1× 2010
- "Unification of India the notable British achievement": **K.M. Panikkar** 1× 2004

1947



TRAP ALERT

- Mountbatten headed the **Partition Council**; **Radcliffe** drew the boundaries.
- First Governor-General of free India = **Mountbatten**; first Indian = **Rajagopalachari** (also the last).
- At the AICC: **Pant** moved, **Azad** seconded, **Ghaffar Khan** opposed.

UP LINK

UP link: **J.B. Kripalani** became Congress president at **Meerut** (1946); **Sucheta Kripalani**, who sang at midnight, later became UP's (and India's) first woman Chief Minister (1963).

LAST-MINUTE 10

1. Labour (Attlee) in power
2. Mountbatten Plan = 3 June 1947
3. Kripalani = Congress president 1947
4. Pant moved, Azad seconded
5. Radcliffe = boundaries
6. Act = 18 July 1947
7. Sucheta Kripalani sang at midnight
8. Mountbatten = first Governor-General
9. Ambedkar = first Law Minister
10. Hyderabad = Standstill Agreement

Chapter 41

Constitutional Development of India

Built from 28 UPPSC questions (1990–2023). Focus: the Regulating and Pitt's Acts, Charter Acts, the Agra Presidency, 1858 to 1909, dyarchy in 1919, simultaneous ICS examination, the 1935 Act and Indian verdicts.

ASKED MOST · REVISE FIRST

- **Regulating Act (1773)**: first Governor-General of Bengal and the first Supreme Court (Calcutta) 2× 1998, 2013
- **Indian Councils Act (1861)**: ordinance power, portfolio system; High Courts in **1861** 5× 1997, 2001, 2010, 2013, 2021
- **Dyarchy** in the provinces: **Act of 1919** 3× 1998, 2004, 2005
- **Simon Commission** report was the base of the **1935 Act**; Nehru called it a "**charter of slavery**" and "a car with brakes but no engine" 4× 2007, 2010, 2010, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

41.1 Company Rule

- First Governor-General of Bengal: **Regulating Act, 1773** 1× 2013
- First Supreme Court: **Regulating Act, 1773** 1× 1998
- Board of Control: **Pitt's India Act, 1784** 1× 2015
- Governor-General may overrule his council: **Act of 1786** 1× 1990
- Charter Act 1813 important: **money set aside for Indian education** 1× 2016
- Company lost its tea monopoly: **Charter Act, 1833** 1× 2015
- First functioning Legislative Council: **Charter Act, 1853** 1× 2016
- Most of present UP put under the **Agra Presidency**: **1834** 1× 2021

41.2 Crown Rule to 1919

- Rule passed from the Company to the Crown: **Government of India Act, 1858** 2× 2007, 2010
- Ordinance power to the Governor-General: **Indian Councils Act, 1861** 2× 1997, 2001
- Portfolio system replacing collective working: **Indian Councils Act, 1861** 1× 2021
- High Courts at Bombay, Madras, Calcutta: **1861** (Act; opened 1862) 2× 2010, 2013
- First indirect elections: **1892** 1× 2016
- Dyarchy divided administration (A); aimed at responsible government in provinces (R): **both true, R explains A** 1× 1998
- Dyarchy introduced in the provinces by the **1919** Act 2× 2004, 2005
- 1919 kept the structure unitary (A); much authority went to provinces (R): both true, R not the explanation (keys differ) 1× 1998
- Chamber of Princes (120 members): **Act of 1919** 1× 2023
- Simultaneous ICS examination recommended by the **Montagu-Chelmsford Report** 1× 2010
- Simultaneous ICS examination finally agreed in **1922** 1× 2014

41.3 The 1935 Act and After

- Main elements: **federation and provincial autonomy** (not provincial dyarchy) 1× 1998
- Council of India abolished by the **1935 Act** 1× 2016
- Basis of the 1935 Act: **Simon Commission** 2× 2010, 2015
- 1909 **Morley-Minto**, 1919 **Montagu-Chelmsford**, 1935 **Simon Report and JSC**, 1947 **Mountbatten Plan** 1× 2009
- Congress rejected the 1935 Act at **Lucknow, 1936** 1× 2013
- "Charter of slavery": **Nehru** 1× 2010
- "A car with brakes but no engine": **Nehru** 1× 2007
- "No need to beg pardon" for borrowing from the 1935 Act: **Ambedkar** 1× 2015
- "One free country with separate electorates?": **Sardar Patel** 1× 2021

Act	Key change
Regulating Act 1773	Governor-General of Bengal, Supreme Court at Calcutta
Pitt's India Act 1784	Board of Control
Charter Act 1813	End of trade monopoly (not tea, China); Rs 1 lakh for education
Charter Act 1833	Governor-General of India; tea monopoly ends
Charter Act 1853	Separate legislative council; open ICS competition
Act of 1858	Crown rule; Secretary of State
Councils Act 1861	Portfolios, ordinances
Councils Act 1892	Indirect election
Councils Act 1909	Separate electorates
Act of 1919	Provincial dyarchy, bicameral centre
Act of 1935	Federation, provincial autonomy, Federal Court

TRAP ALERT

- First **indirect** election 1892; first **separate** electorate 1909; first **direct** election 1919.
- 1935 brought dyarchy to the **centre** (never worked) and **ended** it in the provinces.
- Aitchison Commission **opposed** simultaneous ICS examinations.

QUANTUM IAS & PCS

UP LINK

UP link: The **Agra Presidency** (1834) and the North-Western Provinces became the base of today's UP; the **Queen's Proclamation** was read at **Allahabad** (1 November 1858).

LAST-MINUTE 10

1. 1773 = Governor-General of Bengal, Supreme Court
2. 1784 = Board of Control
3. 1813 = education money
4. 1833 = tea monopoly lost
5. 1853 = Legislative Council
6. 1858 = Crown rule
7. 1861 = ordinances, portfolios, High Courts
8. 1892 = indirect election
9. 1919 = dyarchy, Chamber of Princes
10. 1935 = federation, provincial autonomy

Chapter 42

Modern History: Miscellaneous

Built from 136 UPPSC questions (1990–2025). Focus: wars and treaties, famines and commissions, Acts in sequence, chronology drills from 1905 to 1947, famous quotations, Tagore, Nehru, Ambedkar, leaders and organisations, spot-the-error items.

ASKED MOST · REVISE FIRST

- Order of mass movements: **Khilafat, Non-Cooperation, Civil Disobedience, Quit India** 3× 2000, 2008, 2012
- **Communal Award (16 August 1932)** came before the **Poona Pact (24 September 1932)** 4× 2010, 2016, 2017, 2019
- Wartime plans: **August Offer (1940), Cripps (1942), C.R. Formula (1944), Wavell Plan and Shimla (1945), Cabinet Mission (1946), Mountbatten Plan (1947)** 6× 2000, 2002, 2011, 2016, 2023, 2025
- 1919-22 order: **Champaran, Rowlatt Satyagraha, Jallianwala, Moplah, Chauri Chaura** 4× 2004, 2007, 2008, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

42.1 Wars, Treaties and Company Rule

- Anglo-Afghan policy driven by fear of **Russia** 1× 2004
- Order: **Second Anglo-French (Carnatic) War, First Anglo-Mysore, First Anglo-Afghan, First Anglo-Sikh** 1× one-liner
- Second Anglo-Mysore **1780-84**; Second Anglo-French **1750-54**; Second Anglo-Maratha **1803-05**; Second Anglo-Afghan **1878-80** 1× 2021
- Treaty of Allahabad **1765**; Aix-la-Chapelle **1748**; La Bourdonnais takes Madras **1746**; Dupleix recalled **1754** 1× 2016
- Order: **death of Nana Fadnavis (1800), Wellesley recalled (1805), Vellore Mutiny (1806), Anglo-Nepal War (1814)** 1× 2024
- Occurred first among the options: **First Anglo-Sikh War** 1× 2017
- Wrong pair: **end of slavery, Malcolm** 1× 2013
- "Paternalism in governance": **Thomas Munro** 1× 2016
- Child marriage **Ranade**; Thuggee **Sleeman**; widow remarriage **Vidyasagar**; Pindaris **Lord Hastings** 1× 1990

42.2 Famines and Commissions

- "Sea of calamity": **Orissa famine, 1866-67** 1× 2013
- Famine Code (1883) from the **Strachey Commission** 1× 2015
- Wrong pair (per key): **famine in western UP, 1871-72** 1× 2017
- Bengal famine with millions dead: **1943** 1× 2013
- Andrew Fraser **Police**; MacDonnell **Famine**; Scott-Moncrieff **Irrigation**; Robertson **Railway** commission 1× 2019

42.3 Acts and Constitutional Steps

- Order: **Dramatic Performances Act (1876), Vernacular Press Act (1878), Bengal Tenancy Act (1885), NWP and Oudh Act (1890)** 1× 2016
- Order: **Lex Loci (1850), Bengal Tenancy (1885), Age of Consent (1891), Indian Universities Act (1904)** 1× 2023

QUANTUM IAS & PCS

- Order: **Sati ban (1829), Wood's Despatch (1854), Vernacular Press Act (1878), Rowlatt Act (1919)** 1× 2017
- Indirect election began in **1892** 1× 2009
- 1909 Act **Minto**; 1919 Act **dyarchy**; Lucknow Pact **1916**; 1935 Act **provincial autonomy** 1× 2010
- Separate electorates **1909**; Congress-League pact **1916**; Communal Award **1932**; Deliverance Day **1939** 1× 1999
- MacDonald **Communal Award**; Linlithgow **August Offer**; Dalhousie **Doctrine of Lapse**; Chelmsford **dyarchy** 1× 2002
- Simon Commission **1927**; Quit India **1942**; Congress formed **1885**; Morley-Minto **1909** 1× 2008
- August Declaration **Montagu**; August Offer **Linlithgow**; August Resolution **Gandhi**; Direct Action Day **Jinnah** 1× 1997

42.4 Press and Scholarship

- Order: **Hicky's Bengal Gazette (1780), Asiatic Society (1784), Servants of India Society (1905), Ghadar Party (1913)** 1× 2021
- Order: **Kesari (1881), Khudiram executed (1908), Curzon Wyllie shot (1909), Al-Hilal (1912)** 1× 2021
- Imperial Gazetteer compiled by **W.W. Hunter** 1× 2016

42.5 Chronology, 1905 to 1922

- Order: **partition decided, Swadeshi declared, Swaraj made the goal, Surat split** 1× 2010
- Order: **Regulating Act, Partition of Bengal, Muslim League, Surat split** 1× 1992
- Order: **Surat split, Simon Commission, Civil Disobedience** 1× 1992
- Order: **Partition of Bengal, Lucknow Pact, Rowlatt Act, dyarchy** 1× 1998
- Order: **Swadeshi, Home Rule, Non-Cooperation, Civil Disobedience** 1× 1998
- Order: **Home Rule, Non-Cooperation, Civil Disobedience, Quit India** 1× 1998
- Order: **Swadeshi, Jallianwala, Dandi, Quit India** 1× 2002
- Third in order: **Dandi March** 1× 1996
- Order: **Champaran, Non-Cooperation, Dandi, Quit India** 1× 1994
- Order: **Champaran, Jallianwala, Non-Cooperation, Civil Disobedience** 1× 2010
- Order: **Khilafat, Non-Cooperation, Civil Disobedience, Quit India** 3× 2000, 2008, 2012
- Order: **Non-Cooperation, Simon Commission, Nehru Report, Quit India** 1× 1990
- Non-Cooperation **1920**; Civil Disobedience **1930**; Congress ministries **1937**; Quit India **1942** 1× 2009
- Took some women out of seclusion: **Swadeshi, Home Rule, Non-Cooperation, Civil Disobedience** (all four) 1× 2012
- Order: **Home Rule, Champaran, Rowlatt Satyagraha, Jallianwala** 1× 2004
- Order: **Home Rule, Champaran, Jallianwala, Amritsar session** 1× 2003
- Order: **Lucknow Pact, Champaran, Jallianwala, Khilafat** 1× 2013
- Last among the options: **Moplah Revolt** 1× 2023
- Order: **Champaran, Rowlatt Satyagraha, Jallianwala, Chauri Chaura** 1× 2004
- Order: **Champaran, Jallianwala, Moplah, Chauri Chaura** 1× 2008
- Order: **Champaran, Amritsar case, Moplah, Chauri Chaura** 1× 2007

42.6 Chronology, 1919 to 1934

- Order: **Congress report on Jallianwala, peace terms to Turkey, Tilak's death, Calcutta special session** (all 1920) 1× 2014
- Order: **Rowlatt Act, Jallianwala, Amritsar session, Tilak's death** 1× 2014
- Order: **Rowlatt Act, Rowlatt Satyagraha, Jallianwala, Amritsar session** 1× 2010
- Rowlatt Satyagraha, Jallianwala, Khilafat, Amritsar session: question marked (*) 1× 2013
- Order: **Simon Commission set up, Nehru Report, Sharda Act, Dandi** 1× 2021
- Order: **Pherozeshah Mehta's death (1915), Jallianwala, Simon Commission, Dandi** 1× 2019
- Order: **Home Rule, Rowlatt Act, Simon Commission, Gandhi-Irwin Pact** 1× 1994

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- Deepavali Declaration (1929) was about **dominion status** 1× 2015
- Order: **Purna Swaraj, Civil Disobedience, Gandhi-Irwin Pact, Poona Pact** 1× 2016
- Order: **Nehru Report, Civil Disobedience, Gandhi-Irwin Pact, Poona Pact** 1× 2002
- Order: **Azad's death, Gandhi-Irwin Pact, Bhagat Singh's execution, Karachi session** 1× 2007
- Order: **Lahore session, Gandhi-Irwin Pact, Rajguru's execution, Karachi session** 1× 2011
- Order: **Gandhi-Irwin Pact, Second RTC, Poona Pact, end of Civil Disobedience** 1× 2025
- Last among the options: **Communal Award** 1× 2017
- Order: **Simon Commission, Dandi, Gandhi-Irwin Pact, Poona Pact** 1× 2015
- Order: **Dandi, Gandhi-Irwin Pact, Communal Award, Poona Pact** 2× 2010, 2016
- Order: **Nehru Report, Gandhi-Irwin Pact, Second RTC, Communal Award** 1× 2017
- Order: **Communal Award, Poona Pact, Third RTC, August Offer** 1× 2017
- Order: **Lucknow Pact, Gandhi-Irwin Pact, Poona Pact, Civil Disobedience withdrawn** 1× 2008
- Wrong pair: **Communal Award, 1933** (it was 1932) 1× 2019

42.7 Chronology, 1930 to 1947

- Order: **Civil Disobedience, Gandhi-Irwin Pact, Poona Pact, Cripps** 1× 2019
- Order: **Simon Commission, Gandhi-Irwin Pact, Cripps, Partition** 1× 2009
- Order: **Simon Commission, Poona Pact, Cripps, Cabinet Mission** 1× 2009
- Order: **Poona Pact, Quit India, Simla Conference, Cabinet Mission** 1× 2005
- Order: **August Offer, Cripps, C.R. Formula, Wavell Plan** 1× 2016
- Order: **August Offer, Cripps, Quit India, Desai-Liaquat Pact** 1× 2017
- Order: **Nehru Report, August Offer, Cripps, Wavell Plan** 2× 2017, 2017
- Order: **Civil Disobedience, Individual Satyagraha, Cripps, Quit India** 1× 2004
- Order: **Cripps, Wavell Plan, Shimla Conference, Cabinet Mission** 1× 2025
- Order: **Cripps, Wavell Plan, Cabinet Mission, Mountbatten Plan** 1× 2000
- Order: **Cripps, Quit India, Wavell Offer, Cabinet Mission** 1× 2010
- Order: **August Offer, Cripps, Wavell Plan, Cabinet Mission** 1× 2011
- Order: **August Offer, Cripps, Shimla Conference, Cabinet Mission** 1× 2023
- Order: **C.R. Formula, Wavell Plan, Cabinet Mission, Mountbatten Plan** 1× 2002
- Order: **Cripps, Quit India, Cabinet Mission, Mountbatten Plan** 1× 2023
- Order: **Shimla Conference wrecked, Cabinet Mission arrives, Direct Action, Interim Government** 1× 2010

Event	Date
August Offer	August 1940
Individual Satyagraha	October 1940
Cripps Mission	March 1942
Quit India	August 1942
C.R. Formula	1944
Desai-Liaquat Pact	January 1945
Wavell Plan, Shimla Conference	June 1945
Cabinet Mission	March to June 1946
Direct Action Day	16 August 1946
Interim Government	2 September 1946
Mountbatten Plan	3 June 1947

42.8 Famous Statements

- "The British Government is not responsible for partition": **Attlee** 1× 2009
- Imperialism dead like the dodo: **Attlee** 1× 2008
- "A revolution is going to occur here; we should leave at once": **Cripps** 1× 2007
- "Kings are made for the public": **Dadabhai Naoroji** 1× 2015
- "Political freedom is the life-breath of a nation": **Aurobindo Ghosh** 1× 2004

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- "I am an Indian tom-tom": **Annie Besant** 1× 2019
- "Freedom is almost within reach, we have to seize it": **Gandhi** 2× 2002, 2004
- "India's salvation will not come under Gandhi's leadership": **Subhas Bose** 1× 2014

42.9 Tagore, Nehru, Ambedkar

- Tagore's Nobel Prize: **1913** 1× 2017
- Called Tagore "the Great Sentinel": **Gandhi** 1× 2010
- Tagore became a painter at about **seventy** 1× 2001
- Not correct of Tagore: "**he favoured the economic side of Swadeshi**" 1× 2013
- "I am a socialist and a republican...": **Nehru** 1× 2011
- "Everything else can wait, but not agriculture": **Nehru** 1× 2006
- Nehru had no faith in the Upanishads (A false); scientific outlook (R true) 1× 2006
- "High priest of communism in India" (per key): **Nehru** 1× 2013
- Chaired the first All India Socialist Youth Congress: **Nehru** 1× 2013
- Nehru's biographer: **Frank Moraes** 1× 2014
- Ambedkar statements: question marked (*); **Siddharth College** and **Mook Nayak (1920)** are correct 1× 2005
- Said Ambedkar's leadership was "thrust upon him" by the British: **B.S. Moonje** 1× 2000
- Ambedkar died on **6 December 1956** 1× 1990

42.10 Leaders and Organisations

- Lajpat Rai first opposed then joined Non-Cooperation, and helped form the Swaraj Party: **both correct** 1× 2019
- Served social regeneration (per key): **Gandhi, Tilak** 1× 1992
- "Bombay Trimurti": **Pherozeshah Mehta, K.T. Telang, Badruddin Tyabji** 1× 2017
- Workers' delegate at the ILO, Washington 1919: **N.M. Joshi** (official key gave B.P. Wadia, his adviser) 1× 2020
- Samadhis of Kasturba and Mahadev Desai: **Aga Khan Palace, Poona** 2× 2008, 2011
- Azad Muslim Conference **Allah Baksh**; Khaksar **Allama Mashriqi**; Khudai Khidmatgar **Ghaffar Khan**; Krishak Praja **Fazlul Haq** 1× 2002
- All Parties Muslim Conference (Delhi, 31 December 1928): **Aga Khan** 1× 2015
- Home Rule League **Tilak**; Nationalist Party **Malaviya**; National Liberal Federation **Sapru**; Swaraj Party **C.R. Das** 1× 1998
- Organisation-founder pairs: question marked (*); all four pairs held correct 1× 2014
- Event-year pairs: question marked (*); several pairs wrong 1× 2016
- Wrong pair: **People's Plan, Vinoba Bhave** (it was M.N. Roy's) 1× 2017
- Vinoba **Individual Satyagraha**; Tilak **Home Rule**; Aruna Asaf Ali **Quit India**; Sarojini **Dharasana** 1× 2014
- Bardoli **Patel**; Champaran **Gandhi**; Kuka **Ram Singh**; Lal Kurti **Ghaffar Khan** 1× 1992
- Home Rule **Besant**; Bardoli **Patel**; Non-Cooperation **Gandhi**; Swaraj Party **C.R. Das** 1× 2010
- Besant **Home Rule**; Rajendra Prasad **Champaran**; Nehru **Lahore session**; A.C. Mazumdar **Lucknow session** 1× 2007
- Bose **Haripura**; Patel **Operation Polo**; Iqbal **Allahabad League session**; B.K. Dutt **Assembly bomb** 1× 2013
- Khilafat **Ali brothers**; Home Rule **Tilak**; Civil Disobedience **Khan brothers**; Quit India with **Ambedkar** wrong: question marked (*) 1× 2002
- Wrong pair: **Jinnah, Khilafat Movement** 1× 2015

42.11 Spot the Wrong Fact

- True: Asaf Ali had Railways; Ancient Monuments Act under Curzon; Shraddhanand's no-tax idea. False: "**Haripura presided by C.R. Das**" 1× 2013
- Kunwar Singh, Khan Bahadur Khan, Deliverance Day: only **1 and 2** correct; question marked (*) 1× 2002
- True (per key): Arya Samaj 1875, Al-Hilal by Azad, Hindu College by Ram Mohan Roy. False: "**Lajpat Rai not linked with The People**" 1× 2013
- Wrong pair: Governor-General at Pitt's Act was "**Lord Minto**" (it was Warren Hastings) 1× 2013
- Not correct: "**Lajpat Rai was never Congress president**" (he presided at Calcutta, 1920) 1× 2013

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42.12 After 1947 and General

- Operation Polo: **military action in Hyderabad** (1948) 1× 2013
- Order: **Khrushchev and Bulganin visit (1955), Dalai Lama flees (1959), Zhou Enlai visits (1960), Goa liberated (1961)** 1× 2007
- "Writers' Building": **West Bengal** secretariat 1× 2013
- Kisan Diwas: **23 December** (Charan Singh's birthday) 2× 2005, 2010
- Good Friday: **crucifixion of Jesus** 1× 1990
- Not true of Mother Teresa: "**not an Indian citizen at her death**" 1× 2004
- Pope Benedict XVI: German, first Pope to resign in 600 years: **both correct** 1× 2023

TRAP ALERT

- **Communal Award (Aug 1932)** comes before the **Poona Pact (Sep 1932)**; 1933 is the White Paper.
- **Moplah (1921)** comes before **Chauri Chaura (1922)**.
- **Shimla Conference (1945)** comes before the **Cabinet Mission (1946)**; Direct Action (16 Aug) before the Interim Government (2 Sep 1946).
- **Vellore (1806)** came after Wellesley's recall (1805).

UP LINK

UP link: **Kisan Diwas** (23 December) marks the birthday of **Chaudhary Charan Singh** of Meerut; the **NWP and Oudh Act** (1890) and the **Agra Presidency** shaped today's UP.

LAST-MINUTE 10

1. Khilafat, NCM, CDM, QIM
2. Moplah 1921 before Chauri Chaura 1922
3. Communal Award before Poona Pact
4. August Offer, Cripps, C.R. Formula, Wavell, Cabinet Mission
5. Shimla 1945 before Cabinet Mission 1946
6. Strachey = Famine Code 1883
7. Attlee = "imperialism dead like dodo"
8. Besant = "Indian tom-tom"
9. Bombay Trimurti = Mehta, Telang, Tyabji
10. Operation Polo = Hyderabad 1948

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TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Indian Polity

PYQ-Based Revision Pointers

Chapters 1 to 40

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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Chapter 1

Constitutional Development in British India

Built from 21 UPPSC questions (2002–2023). Focus: which Act did what first; the 1935 Act alone gives 12 questions.

ASKED MOST · REVISE FIRST

- ■ **GoI Act, 1935**: **dyarchy at the Centre**, **All-India Federation**, residuary powers with the **Governor-General**, Federal Court, ordinance power, Burma separated **8×** 2008, 2008, 2008, 2010, 2011, 2012, 2013, one-liner
- • **1935 Act** = main source of the Constitution (about 250 Articles drawn from it) **2×** 2010, 2015
- ■ **Regulating Act, 1773** = Supreme Court at Calcutta (1774); first Chief Justice **Elijah Impey** **4×** 2002, 2003, 2010, 2016
- ■ **1919 Act** = first **bicameral** central legislature; Chamber of Princes **3×** 2008, 2021, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

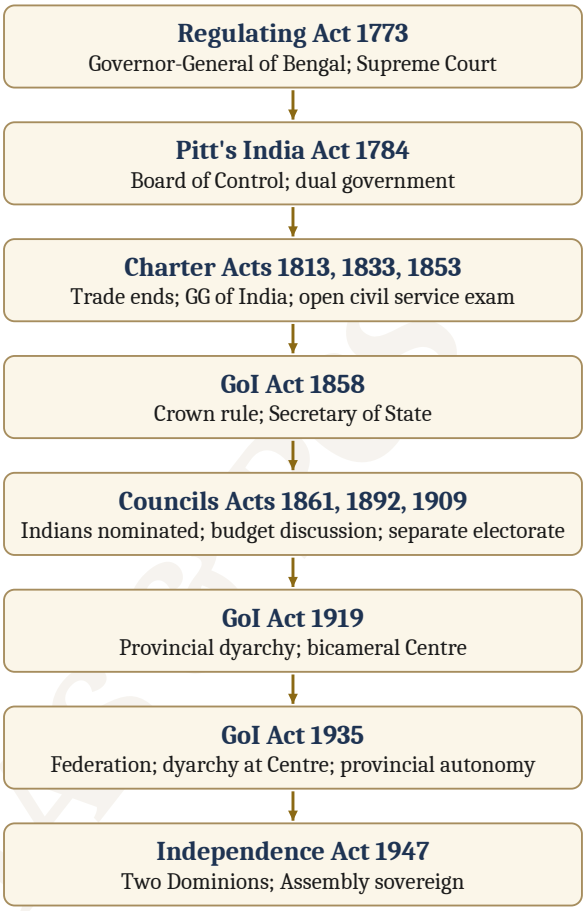
1.1 Regulating Act and Pitt's India Act

- Supreme Court at Calcutta: **Regulating Act, 1773** **1×** 2010
- First Chief Justice: **Sir Elijah Impey**; Le Maistre was only a puisne judge **1×** 2016
- Match: **Board of Control-Pitt's India Act 1784, Supreme Court-1773, missionaries-Charter Act 1813, Law Member-Charter Act 1833** **2×** 2002, 2003

1.2 The Charter Acts

- Trade monopoly ended **except tea and China trade**: **Charter Act, 1813** **1×** 2017
- First legal opening of offices to Indians (Section 87): **Charter Act, 1833** **1×** 2012

CONSTITUTIONAL MILESTONES



1.3 Councils Acts, 1858 to 1909

- Power to **discuss the budget** (no vote): **Indian Councils Act, 1892** **1×** 2003
- 1909 (Morley-Minto): separate electorate for Muslims; resolutions on budget; Minto = "Father of Communal Electorate"

1.4 Government of India Act, 1919

- Legislature made **bicameral** first: **1919 Act** **1×** 2008
- 1919 Act provided an ICS exam in India; by 1941 Indians outnumbered Europeans in the ICS (both true) **1×** 2021
- **Chamber of Princes** (120 members): **1919 Act** **1×** 2023

1.5 Government of India Act, 1935

- **Dyarchy at the Centre**: **1935 Act** **2×** 2008, 2013
- **Federal system** and dyarchy at Centre: **1935 Act** **1×** one-liner
- ■ **All-India Federation**: **1935 Act**; federal government provided by the 1935 Act **3×** 2010, 2012, 2008

QUANTUM IAS & PCS

- **Residuary powers** under the 1935 federation: **Governor-General** 1× one-liner
- Federal Court: Act 1935, began **1937** (key: "none of the above" when asked for the Act) 2× 2014, one-liner
- **Ordinance** power of the President: relic of the **1935 Act** 1× 2008
- **Burma** separated from India: **1935 Act** 1× 2011
- Main source of the Constitution: **1935 Act** 1× 2015
- **Not** a feature of the 1935 Act: a **written Constitution** for the country 1× 2010

Act	First / key provision
1773	Supreme Court at Calcutta; GG of Bengal (Warren Hastings)
1784	Board of Control
1813	Missionaries; trade monopoly ends (except tea, China)
1833	GG of India (Bentinck); Law Member (Macaulay); Section 87
1853	Open competition for civil services
1861	Indians nominated; portfolio system; ordinances
1892	Budget discussion; questions
1909	Separate electorate for Muslims
1919	Provincial dyarchy; bicameral Centre; direct election
1935	Federation; Central dyarchy; Federal Court (1937); RBI

TRAP ALERT

- **1919** = dyarchy in the **provinces**; **1935** = dyarchy at the **Centre**.
- 1892 = discuss budget; 1909 = resolutions on budget; 1919 = vote on part of budget.
- Residuary powers: 1935 = **Governor-General**; today = **Parliament**.

UP LINK

The **Federal Court** idea, provincial autonomy and dyarchy shaped the United Provinces' first elected ministry (Govind Ballabh Pant, 1937) under the 1935 Act. The Allahabad High Court began at Agra in 1866 (moved to Allahabad in 1869) under the Indian High Courts Act, 1861.

LAST-MINUTE 10

1. 1773 = Supreme Court Calcutta; Impey first CJ
2. 1784 = Board of Control
3. 1813 = trade ends except tea and China
4. 1833 = Indians may hold office (Section 87)
5. 1892 = discuss budget
6. 1909 = Muslim separate electorate
7. 1919 = bicameral Centre; Chamber of Princes
8. 1935 = Central dyarchy, federation, residuary with GG
9. Federal Court Act 1935, working **1937**
10. Ordinance power and Burma separation = **1935**

Chapter 2

Constituent Assembly and Making of the Constitution

Built from 33 UPPSC questions (1990–2023). Focus: who proposed and set up the Assembly, key dates, B.N. Rau, Objectives Resolution, critics.

ASKED MOST · REVISE FIRST

- **9 December 1946** = Assembly's first meeting, chaired by **Sachchidananda Sinha** 6× 1990, 1990, 1990, 1995, 2011, 2013
- **26 November 1949** = adopted, enacted, completed; **26 January 1950** = in force 7× 1990, 2002, 2006, 2010, 2012, 2013, 2015
- **B.N. Rau** = Constitutional Adviser and first draft 3× 1998, 2009, 2014
- **Cabinet Mission 1946** set up the Assembly; one seat per **10 lakh** people 4× 2001, 2002, 2003, 2009
- Idea of an Assembly: **M.N. Roy**; elected Assembly proposed by **Cripps Mission** 4× 2001, 2008, 2010, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Idea and First Acceptance

- Idea of a Constituent Assembly: **M.N. Roy** (1934) **2×** 2001, 2017
- Proposal for an elected Assembly: **Cripps Mission** (1942) **2×** 2008, 2010

2.2 Cabinet Mission and Composition

- Assembly set up under the **Cabinet Mission, 1946** **1×** 2009
- Not a Cabinet Mission member: **William Wood** (members: Pethick-Lawrence, Cripps, A.V. Alexander) **1×** 2015
- One member per **10 lakh** population **3×** 2001, 2002, 2003
- Members elected **indirectly by Provincial Assemblies** **1×** 1993
- Women members: **15** **1×** 2014
- Strength: **389** originally; **299** after Partition

2.3 Interim Government, 1946

- Announced: **24 August 1946**; took office 2 September 1946 **1×** 2022
- Not a member: **Dr. S. Radhakrishnan** **1×** 2010

2.4 First Meeting and President

- First meeting chaired by **Dr. Sachchidananda Sinha** (temporary President) **2×** 1990, 2013
- Permanent President: **Dr. Rajendra Prasad** (11 December 1946) **1×** one-liner
- First meeting: **9 December 1946** **4×** 1990, 1995, 2011, 1990

2.5 Objectives Resolution

- Moved by **Nehru** on 13 December 1946; adopted **22 January 1947** **2×** 2007, 1998
- True: the Objectives Resolution shaped the Constitution; Preamble states objectives; people gave the Constitution. False: Head of State directly elected **1×** 2010

2.6 B.N. Rau and Readings

- Constitutional Adviser: **B.N. Rau**; first draft by **B.N. Rau** **3×** 1998, 2014, 2009
- Third reading began **17 November 1949** (some texts 14 November) (*) **1×** 2014

FROM ASSEMBLY TO REPUBLIC



2.7 Adoption and Commencement

- Completed and adopted: **26 November 1949** **3×** 2002, 2006, 2010
- Came into force: **26 January 1950** **1×** 1990
- Enacted by the **Constituent Assembly**; adopted and enacted 26 November 1949 **3×** 2012, 2013, 2015
- Order: **first meeting, Drafting Committee, adoption, commencement** **1×** 2023

2.8 Sessions and Time

- Sessions: **11** for framing; key says **12** (counting 24 Jan 1950) **1×** 2005
- Last meeting: **24 January 1950** **1×** 2018
- Time taken: **2 years, 11 months, 18 days** **1×** 2007

2.9 Nature and Critics

- Constitution provides for all needs; called a "borrowed" Constitution (both true, R not the explanation) **1×** 2008
- Founding Fathers minimised minority interests: **Ivor Jennings** **1×** 2004
- "Assembly was Congress and Congress was India": **Granville Austin** **1×** 2013

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- Congress should be disbanded after independence: **Mahatma Gandhi** 1× 2011

TRAP ALERT

- **Proposed** an elected Assembly = Cripps (1942); **set up** = Cabinet Mission (1946).
- First meeting chair = **Sinha**; President = **Rajendra Prasad**; Drafting Committee = **Ambedkar**; first draft = **B.N. Rau**.
- 26 Nov 1949 = adopted; 26 Jan 1950 = in force; 24 Jan 1950 = signed, last meeting.

UP LINK

UP (United Provinces) sent the largest provincial group to the Assembly; **Govind Ballabh Pant** moved the resolution on minorities' advisory committee. Allahabad's **Purushottam Das Tandon** argued for Hindi.

LAST-MINUTE 10

1. Idea = **M.N. Roy**; proposal = **Cripps**; set up = **Cabinet Mission**
2. 1 seat per **10 lakh**; indirectly elected; 15 women
3. Interim Govt announced **24 Aug 1946**
4. First meeting **9 Dec 1946** under **Sinha**
5. President **Rajendra Prasad**; adviser **B.N. Rau**
6. Objectives Resolution: moved 13 Dec 1946, adopted **22 Jan 1947**
7. Adopted **26 Nov 1949**; in force **26 Jan 1950**
8. Last meeting **24 Jan 1950**
9. Time: **2 y 11 m 18 d**; 11 sessions (key 12)
10. Austin = "Assembly was Congress"; Jennings = minorities minimised

Chapter 3

Important Committees of the Constituent Assembly

Built from 8 UPPSC questions (1991–2025). Focus: committee chairmen, Drafting Committee members and date.

ASKED MOST · REVISE FIRST

- **Drafting Committee**: chairman **Ambedkar**, **6 other members**, set up **29 August 1947** 5× 2008, 2008, 2012, 2014, one-liner
- Chairmen: **Union Constitution-Nehru, Provincial Constitution-Patel, Rules of Procedure-Rajendra Prasad, FR sub-committee-Kripalani** 2× 2008, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Committee Chairmen

- Match: **Union Constitution-Nehru, Rules of Procedure-Rajendra Prasad, Drafting-Ambedkar, FR sub-committee-J.B. Kripalani** 1× 2025
- Provincial Constitution Committee: **Sardar Patel** 1× 2008
- Drafting Committee chairman: **Dr. B.R. Ambedkar** 1× one-liner

Committee	Chairman
Union Powers; Union Constitution; States	Nehru
Provincial Constitution; Advisory (FR, minorities)	Patel
Rules of Procedure; Steering; Flag (ad hoc); Finance	Rajendra Prasad
Drafting	B.R. Ambedkar
FR sub-committee	J.B. Kripalani
Minorities sub-committee	H.C. Mukherjee
Order of Business	K.M. Munshi

3.2 Drafting Committee

- Members besides Ambedkar: **6** 1× 2008
- Members included **N. Gopalaswami Ayyangar and Alladi Krishnaswami Ayyar** 1× 2014
- Not a member: **Jawaharlal Nehru** 1× 2012
- Set up: **29 August 1947** 1× 2008

DRAFTING COMMITTEE (7 MEMBERS)



3.3 National Flag Committee

- Chairman of the (ad hoc) National Flag Committee: **Dr. Rajendra Prasad** 1× 1991

TRAP ALERT

- **Union** Constitution = Nehru; **Provincial** Constitution = Patel.
- Nehru, Patel, Prasad were **not** on the Drafting Committee; **Munshi** was.
- Congress Flag Committee 1931 = Patel; Constituent Assembly Flag Committee = Prasad.

UP LINK

No direct UP item. **Govind Ballabh Pant** of UP moved the motion creating the Advisory Committee on Fundamental Rights and Minorities (chaired by Patel).

LAST-MINUTE 10

1. Drafting Committee: **Ambedkar + 6**, set up **29 Aug 1947**
2. Members: Gopalaswami Ayyangar, Alladi Krishnaswami Ayyar, Munshi
3. Nehru = Union Powers, Union Constitution, States
4. Patel = Provincial Constitution, Advisory
5. Prasad = Rules, Steering, Flag
6. Kripalani = FR sub-committee
7. Nehru not on Drafting Committee
8. Madhava Rao and T.T. Krishnamachari = replacements

Chapter 4

Sources of the Constitution

Built from 13 UPPSC questions (1998–2021). Focus: country-feature pairs, judicial review, federation, Concurrent List, DPSP.

ASKED MOST · REVISE FIRST

- **USA**: judicial review, Preamble idea, Fundamental Rights **8×** 1998, 2006, 2008, 2010, 2012, 2014, 2015, 2017
- **Australia**: **Concurrent List** **4×** 2011, 2012, 2014, 2015
- **Canada**: federation with a strong Centre, "Union of States" (British North America Act) **3×** 2006, 2017, 2017
- **Ireland**: DPSP and nomination to Rajya Sabha **3×** 1998, 1998, 2004

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 USA

- **Judicial review**: **USA**; exists in **India and USA** **3×** 1998, 2008, 2012
- Idea of a **Preamble**: **USA** 1× 2015
- **Fundamental Rights**: **USA** **4×** 2006, 2010, 2014, 2017
- India's Constitution is the longest; FRs from the USA (both true, R not the explanation) 1× 2015

4.2 Canada

- **Federal system**: **Canada** 2× 2006, 2017
- "**Union of States**": the **British North America Act** (Canada) 1× 2017

4.3 Australia

- **Concurrent List**: **Australia** 4× 2011, 2012, 2014, 2015

4.4 Ireland

- **DPSP**: **Ireland** 2× 1998, 2004
- Nomination of talent to the **Rajya Sabha**: **Ireland** 1× 1998

4.5 USSR

- **Fundamental Duties**: **USSR** 1× 2004

4.6 Mixed Pairs

- Wrong pair: **residuary powers-Australia** (it is **Canada**) 1× 2021
- Match: **Rule of Law-England, procedure established by law-Japan, Governor reserving bills-Canada, Concurrent List-Australia** 1× 2010

Country	Borrowed feature
Britain	Parliamentary system, rule of law, single citizenship, writs, bicameralism
USA	FRs, judicial review, Preamble, impeachment, VP as RS chair
Canada	Strong-Centre federation, residuary with Centre, Governor appointed
Australia	Concurrent List, joint sitting, trade and commerce
Ireland	DPSP, RS nomination, President's election
USSR	Fundamental Duties, five-year plans
France	Republic; Liberty, Equality, Fraternity
Germany (Weimar)	Suspension of FRs in Emergency
South Africa	Amendment procedure; RS election
Japan	Procedure established by law

TRAP ALERT

- Preamble **idea** = USA; Liberty-Equality-Fraternity **words** = France.
- Federation = **Canada**; Concurrent List = **Australia**; residuary = **Canada**.
- Rights = USA; **Duties** = USSR. "Procedure established by law" = **Japan**, not USA.

UP LINK

No UP-specific item. Note the UP link in Concurrent List subjects (education, forests) that UP legislates on with the Centre.

LAST-MINUTE 10

1. Judicial review, FRs, Preamble idea = **USA**
2. Federation, Union of States, residuary = **Canada**
3. Concurrent List, joint sitting = **Australia**
4. DPSP, RS nomination, President's election = **Ireland**
5. Fundamental Duties = **USSR**
6. Procedure established by law = **Japan**
7. Emergency suspension of FRs = **Germany**
8. Amendment, RS election = **South Africa**
9. Rule of law, parliamentary system = **Britain**
10. Republic, Liberty-Equality-Fraternity = **France**

Chapter 5

Parts and Articles of the Constitution

Built from 37 UPPSC questions (2002–2025). Focus: Part-to-subject pairs and Article-number matching.

ASKED MOST · REVISE FIRST

- **Part matching**: III FRs, IV DPSP, **IV-A Duties**, V Union, VI States, **VIII UTs**, **IX Panchayats**, **IX-A Municipalities**, **XIV-A Tribunals**, XV Elections, XVIII Emergency **10×** 2004, 2007, 2009, 2013, 2016, 2016, 2016, 2020, 2023, 2025
- **Article matching**: 14 equality, 36 DPSP, 54 President's election, 74 CoM, 75 PM, 76 AG, 148 CAG, 155 Governor, 280 FC, 312 AIS, 323A tribunals, 324 EC, 352 Emergency, 360 financial emergency, 368 amendment **11×** 2002, 2005, 2010, 2011, 2015, 2015, 2016, 2016, 2020, 2023, 2024
- **Article 312** = All India Services **3×** 2012, 2014, 2016
- Constitution "divided into" **22 Parts** (numbering; 25 by count now) **2×** 2008, 2012

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

5.1 Number of Parts

- Divided into **22 Parts** (original numbering; now 25 by count) **2×** 2008, 2012

5.2 Parts I to V

- **Fundamental Duties-Part VI-A** is wrong; it is **Part IV-A**; IV-A = Duties is correct **2×** 2004, 2007
- Order: **Union and territory (I), Citizenship (II), DPSP (IV), Duties (IV-A)** **1×** 2022
- Match: **FRs-Part III, Duties-Part IV-A, Union-Part V, Emergency-Part XVIII** **1×** 2013
- **Attorney General-Part XIII** is wrong (AG is in **Part V**, Art 76) **1×** 2020

5.3 Parts VI to IX-B

- **UTs-Part VII** is wrong; UTs = **Part VIII** (Part VII repealed in 1956) **1×** 2016
- Match: **IX-Panchayats, VIII-UTs, IV-A-Duties, IX-A-Municipalities** **1×** 2009
- Three-tier Panchayats and Panchayati Raj: **Part IX** **3×** 2013, 2016, 2020
- Wrong: **Part XI-Municipalities** (it is IX-A) **1×** 2025
- **Tribunals-Part X** is wrong (Tribunals = **XIV-A**) **2×** 2016, 2016

5.4 Parts X to XIV-A

- Union-State **legislative** relations: **Part XI, Chapter I** **1×** 2012
- **Tribunals-Part XI** is wrong **1×** 2023

5.5 Parts XV to XVIII

- Elections: **Part XV** **1×** 2023
- Match: **XV-Elections, XVI-special provisions for certain classes, XVII-Official Language, XVIII-Emergency** **1×** 2013
- Part XVI: **special provisions for certain classes** (SC, ST, Anglo-Indians) **1×** 2017

Part	Subject
I / II / III	Union and territory / Citizenship / FRs
IV / IV-A	DPSP / Fundamental Duties
V / VI	Union / States
VIII	Union Territories
IX / IX-A / IX-B	Panchayats / Municipalities / Co-operatives
X	Scheduled and Tribal Areas
XI / XII	Centre-State relations / Finance
XIV / XIV-A	Services / Tribunals
XV / XVI / XVII / XVIII	Elections / Special classes / Language / Emergency
XX / XXI	Amendment / Temporary provisions

5.6 Articles I: Territory, Rights, DPSP

- **New states-Art 3, Citizenship-Part II, FRs-Part III, Admin Tribunals-Art 323A** **1×** 2015
- **Admit new state-Art 2, equality-Art 14, arrest and detention-Art 22, assent to bills-Art 111** **1×** 2016
- **Art 352-National Emergency** is correct **2×** 2002, 2002
- **Art 14-equality, Art 36-DPSP, Art 74-CoM, Art 368-amendment** **1×** 2010
- **Art 48-separation of judiciary** is wrong (that is **Art 50**; 48 = agriculture, animal husbandry) **1×** 2013
- **Equality-14, right to work-41, humane conditions of work-42, education-45** (key *; since 2002 education 6-14 = **Art 21A**, Art 45 = under-6 care) **1×** 2015

QUANTUM IAS & PCS

5.7 Articles II: Executive and Parliament

- **President's election-54, PM-75, Governor-155, CM-164** 3× 2002, 2005, 2011
- **CAG-148, SC jurisdiction-131, AG-76, Union Ministers-75(1)** 1× 2015
- **Impeachment of President-61, removal of VP-67(b), Deputy Chairman-90, Speaker-94** 1× 2020
- Financial Bills: **Art 117** 1× 2010
- **Religious affairs-26, village panchayats-40, Parliament secretariat-98, UT administration-239** 1× 2023
- High Courts for UTs: **Art 241** 1× 2016

5.8 Articles III: Bodies and Finance

- **Consolidated Fund-266, Finance Commission-280, Financial Emergency-360, CAG-148** 1× 2024
- **Amendment-368, FC-280, Financial Emergency-360, AIS-312** 1× 2016
- **All India Services: Art 312** 2× 2012, 2014
- **CAG-148, FC-280, Admin Tribunal-323A, UPSC-315** 1× 2016
- All four pairs correct (official key wrong, *) 1× 2020
- **Tribunals-323A, Election-324, SC/ST seats in LS-330, PSC functions-320** 1× 2015
- **EC-324, PSC-315, FC-280, NCSC-338** 1× 2023
- None correct (*): NCBC is **Art 338B**; Art 340 = backward-classes commission 1× 2024

Article	Subject
14 / 17 / 21A	Equality / Untouchability / Education (6-14)
32 / 36-51 / 51A	Writs / DPSP / Duties
52, 54, 61	President, election, impeachment
63, 67(b)	Vice-President, removal
74, 75, 76	CoM, PM and ministers, AG
110, 112, 117	Money Bill, Budget, Financial Bills
123 / 213	Ordinance (President / Governor)
148 / 280 / 312 / 315	CAG / FC / AIS / PSC
324 / 330 / 338 / 338B	EC / SC-ST seats / NCSC / NCBC
352 / 356 / 360 / 368	Emergency / President's Rule / Financial Emergency / Amendment

TRAP ALERT

- Duties = **IV-A** (not VI-A); UTs = **VIII** (VII repealed); Municipalities = **IX-A**; Tribunals = **XIV-A**.
- **Art 50** = separation of judiciary; **Art 48** = agriculture and cows.
- RS Chairman (VP) removed under **67(b)**; Deputy Chairman under **90**.
- Education: **21A** now; Art 45 = under-6 care.

UP LINK

Art 243-243ZT (Parts IX, IX-A) are the base of UP's 3-tier Panchayats and urban bodies. UP's Advocate General (Art 165) sits at Prayagraj; Art 213 lets the UP Governor issue ordinances.

LAST-MINUTE 10

1. 22 Parts by number, 25 by count
2. Duties **IV-A**; UTs **VIII**; Panchayats **IX**; Municipalities **IX-A**
3. Tribunals **XIV-A** (Art 323A); Elections **XV**; Emergency **XVIII**
4. Art **312** = AIS; **315** = PSC; **324** = EC
5. Art **148** = CAG; **280** = FC; **76** = AG
6. Art **54** President's election; **61** impeachment
7. Art **75** PM; **155** Governor; **164** CM
8. Art **50** separation of judiciary
9. Art **117** Financial Bills; **266** Consolidated Fund
10. NCBC = **338B**

Chapter 6

Schedules of the Constitution

Built from 30 UPPSC questions (1997–2025). Focus: Schedule-subject matching, Seventh Schedule lists, Sixth Schedule states, Ninth and Tenth Schedules.

ASKED MOST · REVISE FIRST

- **7-8-9-10 match**: **7th-legislative lists, 8th-languages, 9th-validated Acts, 10th-defection** 3× 2016, 2016, 2025
- **Seventh Schedule** questions dominate: lists, railway police (State), criminal procedure (Concurrent) 9× 1997, 1997, 2001, 2004, 2006, 2009, 2010, 2011, 2021
- **Sixth Schedule** = **Assam, Meghalaya, Tripura, Mizoram** 4× 2016, 2023, one-liner, one-liner
- **Education** moved to Concurrent by the **42nd Amendment** 2× 2008, 2017
- **Fourth Schedule** = Rajya Sabha seats; not Second or Ninth 4× 2015, 2019, 2022, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Master Map

- States and territories: **First Schedule**; new state needs amending the **First** 2× 2014, 2008
- **Fourth Schedule**: seats in the **Council of States** 1× 2015

Schedule	Subject
1st	States and UTs
2nd	Salaries and allowances
3rd	Oaths and affirmations
4th	Rajya Sabha seats
5th	Scheduled Areas
6th	Tribal areas of Assam, Meghalaya, Tripura, Mizoram
7th	Union, State, Concurrent Lists
8th	22 languages
9th	Protected laws (1st Amendment, 1951)
10th	Anti-defection (52nd, 1985)
11th	Panchayats, 29 subjects (73rd)
12th	Municipalities, 18 subjects (74th)

6.2 Schedule Matching

- **7th-legislative distribution, 8th-languages, 9th-validation of Acts, 10th-disqualification for defection** 3× 2016, 2016, 2025
- **3rd-oaths, 4th-RS seats, 7th-lists of subjects, 8th-languages** 1× 2022
- **4th-Council of States, 6th-tribal areas, 8th-language, 9th-land reforms** 1× 1999
- **Ninth-RS seats** is wrong; **RS seats-Second** is wrong 2× 2023, 2019

6.3 Third Schedule

- In the Third Schedule: oaths of a **Union Minister, SC Chief Justice, MP**; **not** the President (Art 60) 1× 2017

6.4 Fifth and Sixth Schedules

- Sixth Schedule: **Assam, Meghalaya, Tripura, Mizoram** 3× 2016, 2023, one-liner
- Of Sikkim, Tripura, Nagaland: only **Tripura** 1× one-liner

6.5 Seventh Schedule

- Three lists: **VII Schedule** 1× 2021
- **Union-census, State-police and public order, Concurrent-population control, residuary-space research** 1× 2004
- **Stock exchange-Concurrent** is wrong (it is **Union**) 2× 2009, 2010

6.6 State List

- **Railway police**: State List 2× 1997, 2001
- **Gas** (and gas-works) is not in the Union List (it is State) 1× 1997
- **Criminal procedure**: Concurrent 2× 2006, 2011

6.7 Concurrent List and 42nd Amendment

- **Criminal matters**: Concurrent 1× 2005

- **Economic planning**: Concurrent 1× 2006
- **Marriage, divorce, adoption**: Concurrent 1× 2010
- Concurrent includes **adulteration and succession** (trade and commerce taken as not) 1× 2024
- **Education** to Concurrent: **42nd Amendment** 2× 2008, 2017

42ND AMENDMENT (1976): STATE TO CONCURRENT



6.8 Ninth Schedule

- State land-reform laws protected in the **9th Schedule** 1× 2003
- **Coelho case** (2007): **Ninth** Schedule; laws after 24 April 1973 open to basic-structure review 1× 2017

6.9 Tenth Schedule

- Anti-defection: **Tenth Schedule** 2× 2013, one-liner

6.10 Eleventh and Twelfth

- **Eleventh**: **Panchayati Raj**; it has **29** items 3× 2014, 2015, 2019

TRAP ALERT

- **2nd** = salaries; **4th** = RS seats. **5th** = Scheduled Areas; **6th** = AMTM tribal areas.
- President, VP, Governor oaths are in **Arts 60, 69, 159**, not the 3rd Schedule.
- Railways = Union; **railway police = State**. Stock exchange = Union.

UP LINK

UP has no Fifth or Sixth Schedule areas. UP has **31 Rajya Sabha** and **80 Lok Sabha** seats (Fourth Schedule for RS). UP Panchayats take the 29 Eleventh Schedule subjects.

LAST-MINUTE 10

1. 1st = States; 4th = **RS seats**; 3rd = oaths (not President)
2. 6th = **Assam, Meghalaya, Tripura, Mizoram**
3. 7th = three lists; 8th = languages
4. 9th = land reforms, **Coelho** (2007); 10th = **anti-defection**
5. 11th = Panchayats (**29**); 12th = Municipalities (18)
6. Railway police = **State**; criminal procedure = Concurrent
7. Marriage, planning, adulteration = Concurrent
8. Education to Concurrent by **42nd** (1976)
9. Space research = residuary (Union)
10. Stock exchange = **Union**

Chapter 7

The Preamble

Built from 27 UPPSC questions (1991–2025). Focus: 42nd Amendment words, order of words, types of justice and liberty, legal nature, cases, "soul".

ASKED MOST · REVISE FIRST

- **42nd Amendment (1976)** added **Socialist, Secular, Integrity**; Sovereign and Republic were original 9× 1991, 2006, 2008, 2009, 2010, 2010, 2013, 2017, 2025
- Preamble = "**soul**" of the Constitution 5× 2006, 2008, 2008, 2013, 2015
- **Justice**: social, economic, political (3); **no** religious justice; **no** economic freedom under Liberty 5× 2008, 2016, 2018, 2021, one-liner
- **Kesavananda Bharati** (1973): Preamble **is part** of the Constitution; Berubari (1960) said not 2× 2014, 2017
- **Non-justiciable**, can be amended, cannot override specific provisions 3× 2004, 2009, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 Text and Purpose

- Preamble ideals elaborated in the **DPSP and FRs** 2× 2002, 2004
- Constitution's objectives reflected in the **Preamble** 1× 2013

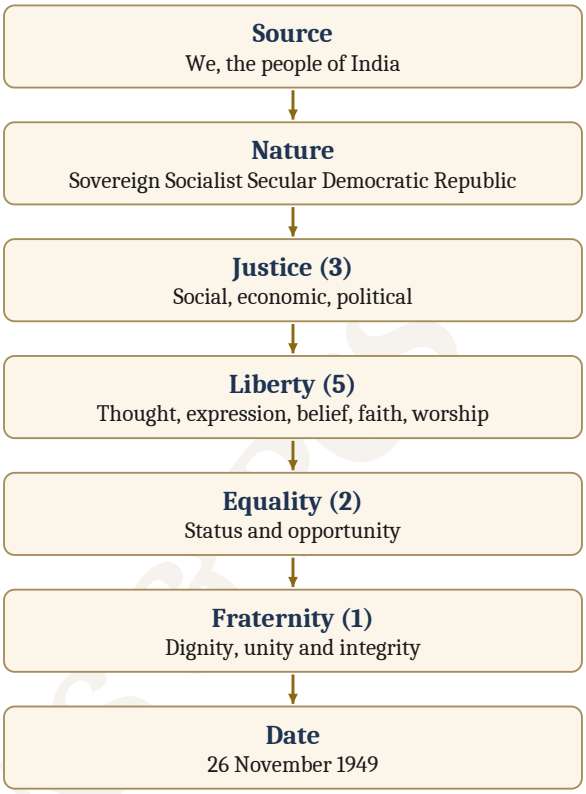
7.2 Original Text and 42nd Amendment

- Status on 26 January 1950: **Sovereign Democratic Republic** 2× 2008, 2009
- Absent in 1949: **Socialist, Integrity, Secular** (Republic was present) 1× 2009
- Not in the Preamble in 1975: **Integrity** 1× 2006
- ■ **Secular and Socialist** added by the **42nd Amendment**; Secular by the 42nd 5× 2008, 2017, 1991, 2010, 2013
- Inserted by the 42nd: **Socialist and Secular** (not Sovereign, Republic) 1× 2025
- Added: **Socialist and Secular**; "Gram Swarajya" is not in the Preamble 1× 2010
- Balram Singh case (Nov 2024): SC upheld the insertion of Socialist and Secular

7.3 Order and Absent Words

- Order: **Sovereign, Socialist, Secular, Democratic (Republic)** 1× 2023
- Not in the Preamble: **Federal** 1× 2021

PREAMBLE AT A GLANCE



7.4 Secular

- "Secular" in India: **no state religion** 1× 2005
- India described as Secular in the **Preamble** 1× 2005

7.5 Four Objectives

- Justice, liberty, equality, fraternity: **3, 5, 2, 1** 1× 2018
- Types of justice: **three**: **social, economic, political** 2× 2016, 2021
- Not mentioned: **religious justice** 1× one-liner
- Liberty not mentioned: **economic freedom** 1× 2008
- Goals: justice, liberty of thought, equality of opportunity, dignity: **all** 1× 1998
- **Social equality** is guaranteed 1× 2008

7.6 Legal Nature

- True: came from the **Objectives Resolution**, **non-justiciable**, **cannot override** provisions; false: cannot be amended 1× 2009
- Legal nature: **not enforceable** 1× 2019
- Not enforceable, useful, states aims, helps interpretation: all true (*) 1× 2004

7.7 Cases

QUANTUM IAS & PCS

- Preamble is **part** of the Constitution: **Kesavananda Bharati** 2× 2014, 2017
- Preamble (secularism) part of basic structure: **S.R. Bommai** (1994) 1× 2012

Case	Holding
Berubari (1960)	Preamble **not** part of Constitution
Kesavananda Bharati (1973)	Preamble **is** part; can be amended, not basic structure
S.R. Bommai (1994)	Secularism part of basic structure
LIC v. Consumer Education (1995)	Preamble integral part
Balram Singh (2024)	42nd Amendment words upheld

7.8 Descriptions

■ "**Soul**" of the Constitution: the **Preamble** 5× 2006, 2008, 2008, 2013, 2015

TRAP ALERT

- **Economic justice** is in; **economic freedom** is not. **Religious justice** is not; faith and worship are under Liberty.
- **Berubari** = not part; **Kesavananda** = part.
- Ambedkar's "heart and soul" = **Article 32**, not the Preamble.

UP LINK

No UP-specific item. Note: the original handwritten Preamble page was illuminated by **Beohar Rammanohar Sinha** of Jabalpur at Santiniketan.

LAST-MINUTE 10

1. 42nd (1976) added **Socialist, Secular, Integrity**
2. 1950 status = **Sovereign Democratic Republic**
3. Order: Sovereign, Socialist, Secular, Democratic, Republic
4. Justice 3, Liberty 5, Equality 2, Fraternity 1
5. No **religious justice**, no **economic freedom**, no **Federal**
6. Secular = no state religion
7. Non-justiciable; amendable; cannot override
8. Berubari not part; **Kesavananda** part
9. **Bommai** = secularism basic structure
10. Preamble = **soul**; Art 32 = Ambedkar's heart and soul

Chapter 8

The Governance System

Built from 40 UPPSC questions (1990–2025). Focus: parliamentary vs presidential, features of the polity, republic, quasi-federalism and the scholars behind each label.

ASKED MOST · REVISE FIRST

■ India = **parliamentary** system; executive responsible to legislature; not presidential 6× 1990, 2005, 2008, 2010, 2015, 2025

■ **Quasi-federal**: **K.C. Wheare** 4× 2010, 2019, 2023, 2024

- **Single citizenship** is **not** a federal feature 1× 2015
- Constitution = **partly rigid, partly flexible** 2× 2005, 2012
- **Republic** = elected head, no hereditary ruler 2× 1997, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Sovereignty

- Most prominent element of a state: **sovereignty** 1× 2008

- India is in the **Commonwealth**; its sovereignty is **not** curtailed 1× 2017

8.2 Parliamentary System

- India parliamentary; executive responsible to legislature (R explains A; option d in 2025) 1× 2025
- Constitution provides **parliamentary** governance 1× 1990
- In a parliamentary system the **legislature controls the executive** 1× 2010
- First in **Great Britain** 1× 2018
- Incorrect about India: **presidential system**; not a feature: **presidential government** 2× 2008, 2015
- Correct: India is a **parliamentary democracy** 1× 2005

8.3 Presidential System

- All executive power in the **President** 1× 2005
- Basic feature: **single executive** 1× 2006
- Main features: all of the listed options (key) 1× 2014

Parliamentary (India, UK)	Presidential (USA)
Dual executive (nominal + real)	Single executive
CoM from legislature	Secretaries from outside
Collective responsibility to LS	Fixed term, no confidence vote
Lower house can be dissolved	No dissolution

8.4 Cardinal Features

- True: **democratic republic, parliamentary form, sovereignty in the people**; false: unified authority 2× 2008, 2009
- India: democratic polity, sovereign state, democratic society, welfare state (all true) 1× 2012
- India democratic; has its own Constitution (both true, R not the explanation) 1× 2017
- Not a feature: **authoritarian government** 1× 2011

8.5 Supremacy and Republic

- Supreme: the **Constitution**; Centre and States get authority from it 2× 2013, 2017
- Republic: **no hereditary rulers**; head of State **elected** 2× 1997, 2017
- Not democratic: supreme power in one person like a king 1× 2010
- Constitution: **partly rigid, partly flexible** 1× 2012
- "Good balance between rigidity and flexibility": **K.C. Wheare** (per key) 1× 2005
- Bulkiest Constitution: all reasons valid (*) 1× 1997
- Not its philosophy: **communist state** 1× 2015

8.6 India vs UK and USA

- India differs from UK by **judicial review** 1× 2010
- Common to India and USA: **pocket veto** (only 3) 1× 2017

8.7 Federal and Quasi-Federal

- Not federal: **single citizenship** 1× 2015
- Federal system; strong Centre (both true, R not the explanation) 1× 1995
- "Federalism not practical" true per key; "India is not federal" false 1× 1998
- True: federal structure with **unitary bias**; false: both legislatures sovereign 1× 2004
- **Cooperative-Granville Austin, bargaining-Morris-Jones, quasi-federal-K.C. Wheare, centralising tendency-Ivor Jennings** 1× 2024
- "Quasi-federal": **K.C. Wheare** 3× 2010, 2019, 2023
- "Not set in a tight mould of federalism": **Ambedkar** 1× 2008
- Quasi-federal; independent judiciary (both true, R not the explanation) 1× 2020
- Quasi-federal because neither federal nor unitary (R explains A) 1× 2007
- Shift from centralism to federalism: **1960s** 1× 2008

Scholar	Label for Indian federalism
K.C. Wheare	Quasi-federal
Granville Austin	Cooperative federalism
Morris-Jones	Bargaining federalism; three styles of politics
Ivor Jennings	Federation with a strong centralising tendency
Ambedkar	Not in a tight mould of federalism

QUANTUM IAS & PCS

8.8 Parties and Styles

- Parties are the life-blood of democracy (key: R explains A) 1× 2009
- Continuity and change: modern, traditional, saintly styles (Morris-Jones) (R explains A) 1× 2020

TRAP ALERT

- **Single citizenship**, integrated judiciary, All India Services = **unitary** features.
- "Unified authority" and "both legislatures sovereign" are wrong statements.
- Common to India and USA: pocket veto; residuary is with Centre in India, States in USA.

UP LINK

UP's own government follows the same parliamentary model under Articles 163-164: Governor nominal, **Chief Minister and Council of Ministers** responsible to the **Vidhan Sabha**.

LAST-MINUTE 10

1. India = **parliamentary**; first in **Britain**
2. Presidential = **single executive**
3. Republic = **elected head**, no hereditary ruler
4. Constitution supreme; partly rigid, partly flexible
5. Commonwealth does not curtail sovereignty
6. India vs UK = **judicial review**
7. **Single citizenship** = unitary feature
8. Quasi-federal = **Wheare**; cooperative = **Austin**
9. Bargaining = **Morris-Jones**; centralising = **Jennings**
10. India-USA common: **pocket veto**

Chapter 9

National Symbols

Built from 4 UPPSC questions (1990–2020). Focus: National Flag facts and the Saka-based National Calendar.

ASKED MOST · REVISE FIRST

- ■ National Flag adopted by the Constituent Assembly on **22 July 1947** 3× 1990, 2004, 2020
- National Calendar = **Saka era**, adopted **22 March 1957**; first month **Chaitra** 2× 2016, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 National Flag

- True: adopted **22 July 1947**; false: 21 spokes (it is **24**), ratio 3:4 (it is **2:3**) 1× 2004
- Adopted by the Constituent Assembly: **22 July 1947** 2× 1990, 2020

9.2 National Calendar

- Based on **Saka** era (not Vikram), adopted **22 March 1957** (not 26 January 1950): neither statement true 1× 2019
- First month: **Chaitra** 1× 2016

Symbol	Fact
National Flag	Adopted 22 July 1947; ratio 2:3; Ashoka Chakra 24 spokes
National Emblem	Sarnath Lion Capital; adopted 26 Jan 1950; "Satyameva Jayate" (Mundaka Upanishad)
National Anthem	Jana Gana Mana; adopted 24 Jan 1950; 52 seconds
National Song	Vande Mataram; Anandamath (Bankim Chandra)
National Calendar	Saka era; from 22 March 1957; Chaitra first month
National Animal / Bird	Tiger / Peacock
National Aquatic Animal	Ganges river dolphin

TRAP ALERT

- Spokes **24**, not 21; ratio **2:3**, not 3:4.
- Calendar: **Saka**, not Vikram; **1957**, not 1950.
- Saka year = Gregorian minus **78** (from 22 March).

UP LINK

The National Emblem comes from the **Sarnath** Lion Capital (Varanasi, UP). The **Ganges river dolphin**, the national aquatic animal, is UP's state aquatic animal too.

LAST-MINUTE 10

1. Flag adopted **22 July 1947**
2. Ratio **2:3**; chakra **24** spokes
3. Emblem = **Sarnath** Lion Capital, 26 Jan 1950
4. Satyameva Jayate = **Mundaka Upanishad**
5. Anthem adopted **24 Jan 1950**; 52 seconds
6. Calendar = **Saka**; from **22 March 1957**
7. First month = **Chaitra**
8. Saka = AD minus 78
9. Song = Vande Mataram (Anandamath)
10. Aquatic animal = Ganges dolphin

Chapter 10

States and the Union Territories

Built from 32 UPPSC questions (1990–2025). Focus: Articles 1-4, Parliament's power to redraw states, state-formation order, Sikkim, Delhi, linguistic reorganisation.

ASKED MOST · REVISE FIRST

- ■ **Parliament** alone can create states and alter boundaries (Art 2-4), by **simple majority** 6× 1990, 2008, 2013, 2014, 2015, 2016
- ■ **State-formation order** questions every few years 9× 2006, 2008, 2009, 2010, 2013, 2013, 2014, 2021, 2025
- ■ India = "**Union of States**" (Art 1); no right to secede 3× 2008, 2008, 2019
- • **Sikkim**: 35th Amendment (associate) and **36th (full State, 1975)** 2× 2005, 2013
- • First linguistic state: **Andhra (1953)** 2× 2009, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Article 1

- India is a **Union of States** 1× 2008
- States **cannot secede**; advocacy of secession is not protected speech (only 1) 1× 2019

10.2 Articles 2 to 4

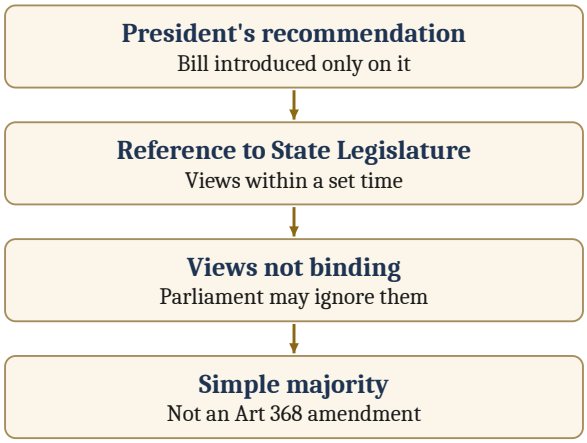
- ■ Power to carve a new state: **Parliament**; alter boundaries: **Parliament** 5× 1990, 2008, 2013, 2014, 2015
- Not under **Art 3**: **admission of new states** (that is Art 2) 1× 2020
- Wrong: such a law is **not** deemed an amendment of the Constitution (Art 4) 1× 2011
- New-state Bill: **simple majority** 1× 2016
- Not a provision: "a State may include a UT" (*; Parliament can merge UT areas under Art 3) 1× 2013
- Wrong pair: **altering a State's name-State Legislature** (it is Parliament) 2× 2006, 2024

10.3 Article 3 Procedure

- President refers the Bill to the State for views in a set time; views **not binding** (both true, R not the explanation) 1× 2023
- "India is not a Union" false; Parliament can alter states without consent true 1× 1999

QUANTUM IAS & PCS

ARTICLE 3 PROCEDURE



10.4 Counting States and UTs

- 2002: 28 States and 7 UTs; now **28 and 8** (key "none") 1× 2002
- After Telangana: 29 (2014); key 28 = after J&K became a UT in 2019 1× 2013
- Not a UT: **Goa** 1× 2013

10.5 Delhi and Sikkim

- Delhi made **National Capital Territory**: **69th Amendment** (1991) 1× 2014
- Sikkim: **35th** (associate state, 1974) per key; full state by **36th** (1975) 2× 2005, 2013

10.6 Formation Order

- **Haryana, Goa, Jharkhand, Telangana** 1× 2021
- **Sikkim, Arunachal, Chhattisgarh, Jharkhand** 2× 2006, 2009
- Uttarakhand created **2000** 1× 2014
- **Nagaland, Haryana, Meghalaya, Sikkim** 2× 2008, 2010
- **Andhra Pradesh, Haryana, Himachal Pradesh, Sikkim** 1× 2014
- **Nagaland, Meghalaya, Sikkim, Arunachal Pradesh** 1× 2013
- Descending: **Haryana, Maharashtra, Rajasthan** 1× 2013
- **Nagaland, Punjab (1966 form), Sikkim, Arunachal** (key) 1× 2025

State	Year
Andhra (first linguistic)	1953
Maharashtra, Gujarat	1960
Nagaland	1963
Haryana, Punjab (present)	1966
Himachal Pradesh	1971
Meghalaya, Manipur, Tripura	1972
Sikkim	1975
Arunachal, Mizoram, Goa	1987
Chhattisgarh, Uttarakhand, Jharkhand	2000 (Nov 1, 9, 15)
Telangana	2014
J&K to UT	2019

10.7 Act Year vs Statehood

- **Jharkhand-2000, Telangana-2014, Sikkim-1975**; Nagaland 1963 (Act 1962) (*) 1× 2017
- Formed in 1948: none as State (*; Himachal became a province in 1948) 1× 2013

10.8 Linguistic Reorganisation

- SRC (Fazal Ali) set up **1953**, reported 1955 (*) 1× 1990
- Fazal Ali members: **K.M. Panikkar, H.N. Kunzru** 1× 2025
- Union of States (no secession), Dhar favoured administrative convenience, JVP against language basis: all true 1× 2008
- First linguistic state: **Andhra Pradesh**; created **1953** 2× 2018, 2009

TRAP ALERT

- **Art 2** = admit new states; **Art 3** = form and alter existing states.
- Art 3 laws are **not** Art 368 amendments; simple majority.
- 1953 = Andhra State (linguistic); 1956 = Andhra Pradesh, SR Act.
- 2000 order: **Chhattisgarh, Uttarakhand, Jharkhand**.

QUANTUM IAS & PCS

UP LINK

Uttarakhand was carved out of UP on **9 November 2000** under Art 3 (UP Reorganisation Act, 2000). Parts of UP (Noida, Ghaziabad, Meerut) are in the **NCR**, a planning region, not the constitutional NCT.

LAST-MINUTE 10

1. Art 1 = **Union of States**; no secession
2. Art 2 admit; Art 3 form/alter; Art 4 not an amendment
3. **Parliament**, simple majority; State's views not binding
4. 69th = Delhi **NCT**
5. Sikkim: 35th associate, **36th** full state (1975)
6. Andhra **1953** first linguistic state
7. SRC 1953: Fazal Ali, **Panikkar, Kunzru**
8. 2000: Chhattisgarh, Uttarakhand, Jharkhand
9. Telangana **2014**; now 28 States, 8 UTs
10. Goa is a State (1987)

Chapter 11

Citizenship

Built from 8 UPPSC questions (1994–2020). Focus: Part II, single citizenship, loss of citizenship, OCI, CAA 2019.

ASKED MOST · REVISE FIRST

- Citizenship = **Part II, Articles 5-11**; **Parliament** lays down conditions (Art 11) **4×** 2010, 2013, 2018, 2020
- India has **single citizenship** **2×** 1994, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

11.1 Constitutional Provisions

- Citizenship provisions: **Part II**; **Articles 5 to 11** **3×** 2010, 2018, 2020
- Competent to prescribe conditions: **Parliament** **1×** 2013

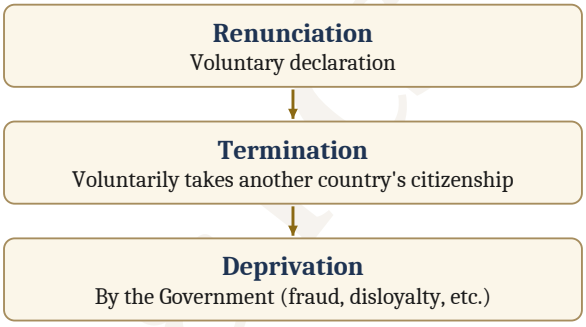
11.2 Single Citizenship

- Citizenship under the Constitution: **single**; single citizenship for all India **2×** 1994, 2015

11.3 Loss of Citizenship

- Lost by **renunciation** and **voluntarily acquiring another citizenship**; not by marrying a foreigner or criticising the government **1×** 2017

CITIZENSHIP ACT, 1955: LOSING CITIZENSHIP



11.4 OCI

- Not eligible for OCI: an Indian who **migrated to Pakistan** (or Bangladesh) **1×** 2016

11.5 CAA 2019

- Passed by Parliament: **11 December 2019** (RS); assent 12 December **1×** 2016

Item	CAA 2019
Countries	Afghanistan, Bangladesh, Pakistan
Communities	Hindu, Sikh, Buddhist, Jain, Parsi, Christian
Cut-off	Entered on or before 31 Dec 2014
Residence	5 years (instead of 11) for naturalisation
Rules	Notified March 2024

TRAP ALERT

- **OCI is not dual citizenship**; OCI holders are foreign nationals.
- Marriage to a foreigner or criticism of the Government does **not** end citizenship.
- CAA passed 11 Dec 2019; **assent 12 Dec**.

UP LINK

No UP-specific question. UP districts with Pakistani Hindu migrant settlements were among those authorised to receive CAA applications.

LAST-MINUTE 10

1. Citizenship = **Part II, Art 5-11**
2. **Parliament** decides conditions (Art 11)
3. Act = Citizenship Act, **1955**
4. **Single** citizenship (British model)
5. Loss: renunciation, termination, deprivation
6. Not lost by marriage or criticism
7. OCI not for migrants to Pakistan/Bangladesh
8. CAA passed **11 Dec 2019**
9. CAA: 3 countries, 6 communities, **31 Dec 2014**
10. OCI ≠ dual citizenship

Chapter 12

Fundamental Rights

Built from 76 UPPSC questions (1992–2024). Focus: Article groups, Articles 14-32, writs, Right to Property, citizens-only rights, basic structure.

ASKED MOST · REVISE FIRST

- **Right to Property** removed from FRs by the **44th Amendment** (1978); now **Art 300A** 10× 1992, 1996, 2004, 2004, 2007, 2010, 2013, 2013, 2014, 2015
- **Art 17** abolishes untouchability 5× 1994, 2003, 2013, 2015, 2017
- **Art 24**: no child labour in factories and mines 5× 2005, 2009, 2011, 2017, one-liner
- **Guardian of FRs**: **Supreme Court / judiciary** (not the President) 7× 1992, 2004, 2010, 2012, 2013, 2015, 2015
- **Art 32** = "heart and soul" (Ambedkar) 4× 2004, 2010, 2010, 2016
- **Basic structure**: **Kesavananda Bharati** (1973) 3× 2007, 2013, 2015

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

12.1 Origin and Nature

- **Nehru Report (1928)** advocated FRs 1× 2009
- "A pledge to our people and a pact with the civilised world": **Dr. S. Radhakrishnan** 1× 2021
- Rights are **claims of citizens against the State** 1× 2017
- FRs guard against State action, are in Part III, aim at justice (1, 2, 3 correct) 1× 1998
- Incorrect: right to die is an FR, Constitution wholly federal, FRs can be abrogated (B, C, D) 1× 2023

12.2 Scheme of Part III

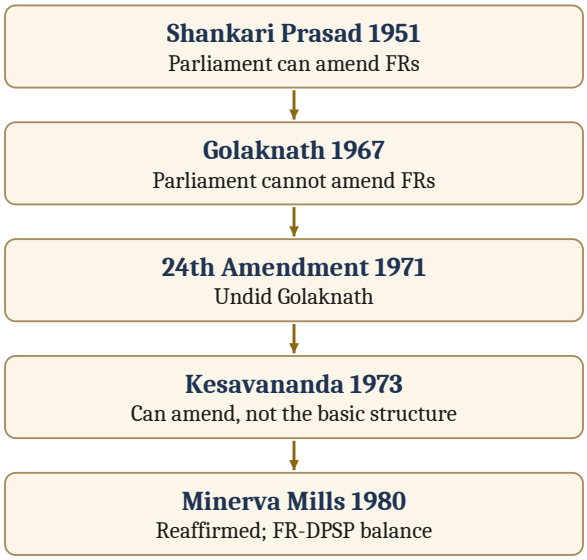
- FRs in **Articles 12 to 35** 1× 2015
- Not among the six FRs: **right to protest** 1× 2015
- **Freedom of Religion-Arts 23-24** is wrong (it is **25-28**) 1× 2023
- **Arrest and detention-Art 23** is wrong (it is **Art 22**) 1× 2016
- **Titles-18, religious affairs-26, minority language-29, education-21A** 1× 2019

Right	Articles
Equality	14-18
Freedom	19-22
Against exploitation	23-24
Freedom of religion	25-28
Cultural and educational	29-30
Constitutional remedies	32

12.3 Article 13 and Basic Structure

- Art 13 secures the Constitution's paramountcy for **FRs** 1× 1999
- Basic structure doctrine: **Kesavananda Bharati** 2× 2013, 2015
- **Duties-42nd Amendment, Parliament can amend FRs-Kesavananda, basic structure-Minerva Mills, trafficking-Art 23** 1× 2007

AMENDING FRS: THE CASE TRAIL



12.4 Emergency

- FRs **can be suspended** during Emergency 1× 2002
- Not suspendable: **right to life and personal liberty** (Arts 20, 21; 44th Amendment) 1× 2008

12.5 Right to Equality

- Five Articles: **14 to 18** 2× 2014, 2015
- Equality from **Arts 14, 15, 16** (not Art 13) 1× 2009
- Not included: **economic equality** 1× 2008
- Odd one out: **Art 19** (freedom group) or Art 14 (all persons) (*) 1× 2000
- State may treat unequals differently; equal law among equals (R explains A) 1× 2016

12.6 Article 14 and Equal Pay

- Absolute limit on legislative power: **Art 14** 1× 1999
- Equality before law: **Art 14** 1× 2015
- Equal pay for equal work deduced from **Arts 14, 16 + Preamble ("socialist")** 3× 1999, 2001, 2002

12.7 Articles 15 and 16

- **Champakam Dorairajan** (1951) led to amending the **right against discrimination** (Art 15(4), 1st Amendment) 1× 1997
- Equal opportunity in public employment: **Art 16(1) and 16(2)** 1× 2009
- "Descent, residence" in non-discrimination in State jobs: **Art 16(2)** 1× 2023

12.8 Articles 17 and 18

- Untouchability abolished: **Art 17** 5× 1994, 2003, 2013, 2015, 2017
- Achieved by **laws, education, awareness** (1, 2, 3) 1× 1992
- Untouchability is structural violence rooted in religious sanction (R explains A) 1× 2019

12.9 Article 19

- Right to Freedom: **Arts 19-22** 1× 2016
- Not under Art 19 now: **right to property** (19(1)(f) deleted) 1× 2024
- Freedom of the **press**: **Art 19(1)(a)** 2× 2001, 2004

12.10 Articles 20 and 22

- No self-incrimination: **Art 20(3)** 1× 2004
- Double jeopardy: **Art 20** 1× 2010
- Not in Art 20: **prohibition against torture** (from Art 21) 1× 2023
- Rights of arrested persons: **Art 22** 1× 2013
- Preventive detention without Advisory Board: **3 months** 2× 2009, 2013

12.11 Articles 21 and 21A

- Due process (as read now): **Art 21** 1× 2014
- Travel abroad: **Art 21** 1× 1997
- Recognised FRs: **shelter, travel abroad, equal pay** 1× 1998
- Under 19(1)(d) with 21: **right to privacy** 2× 2003, 2004
- Not under Art 21: **capital punishment** 1× 2017
- Education for 6-14: a **Fundamental Right** (Art 21A, 86th Amendment, 2002) 1× 2006
- Free education 6-14; education key to development (R explains A) 1× 2016

12.12 Against Exploitation

- Covers **trafficking and forced labour, child labour in factories** (2, 3) 1× 2024
- Child exploitation: **Art 24**; children in factories: **Art 24** 5× 2005, 2009, 2011, 2017, one-liner
- **Bonded Labour** abolished by an Act of **1976** 2× 2001, 2010

QUANTUM IAS & PCS

- Child labour in hazardous work banned by all listed (Constitution and UN) 2× 2002, 2003

12.13 Freedom of Religion

- Includes **propagation, Sikhs' kirpans, social-reform laws**; not conversion by bodies 1× 2001
- **Art 25**: freedom of religion 1× 1996
- "Hindu" in Art 25 excludes **Parsis** (includes Sikh, Jain, Buddhist) 1× 2014

12.14 Cultural and Educational

- **Art 29-minority educational institutions** is wrong (that is **Art 30**) 2× 2009, 2015
- Minorities' right to run educational institutions: **Art 30** 2× 1997, 2017
- "Minorities" not defined; only religious and linguistic recognised (both true, R not the explanation) 1× 2019

12.15 Article 32

- Guardian of FRs: **judiciary**; **Supreme Court** 6× 2010, 2012, 2015, 1992, 2004, 2015
- Not correct: **President** is guardian of FRs 1× 2013
- Art 32 enforces **FRs** only 1× 1997
- "Heart and soul": **right to constitutional remedies**; Art 32 gives a remedy (R explains A) 4× 2004, 2010, 2010, 2016

12.16 Writs

- Greatest safeguard of liberty: **habeas corpus**; issued by High Court for liberty 2× 2015, 2015
- **Prohibition-"to be certified"** is wrong (that is certiorari) 1× 2019
- Order to perform duty: **mandamus** 2× 2001, 2007

Writ	Meaning
Habeas corpus	"Have the body"; against unlawful detention
Mandamus	"We command"; perform public duty
Prohibition	"To forbid"; stops a lower court midway
Certiorari	"To be certified"; quashes an order made
Quo warranto	"By what authority"; illegal holding of office

12.17 Right to Property

- Omitted by the **44th Amendment**; removed from FRs 8× 2004, 2004, 2010, 2013, 2013, 2015, 2007, 2014
- Now a **constitutional legal right** (Art 300A) 2× 1992, 1996

12.18 Citizens vs All Persons

- Not for foreigners: **freedom of expression** 2× 2007, 2015
- For all persons: **equal protection of laws** 1× 2010
- Not for all persons: **non-discrimination, free movement, contesting elections** (2, 3, 4) 1× 2002

TRAP ALERT

- Religion = **25-28**; exploitation = **23-24**; Art 29 = culture, **Art 30** = minority institutions.
- Art 20 has 3 clauses; torture protection is from **Art 21**.
- **Kesavananda** propounded basic structure; **Minerva** reaffirmed; **Golaknath** barred amending FRs.
- Citizens only: **15, 16, 19, 29, 30**.

UP LINK

UP links: the **Kesavananda** era 24th-25th Amendments followed **Indira Gandhi v. Raj Narain** (Allahabad HC, 1975); Allahabad HC hears many **Art 226** writs. UP's Anti-Conversion Act 2021 is tested against **Art 25**.

LAST-MINUTE 10

1. FRs: Arts **12-35**; six groups
2. Equality **14-18**; Freedom **19-22**; Religion **25-28**
3. **17** untouchability; **24** child labour; **23** trafficking
4. **21A** education 6-14 (86th, 2002)
5. **20(3)** self-incrimination; **22** arrest; 3 months detention
6. Property out by **44th**; now **Art 300A**
7. **Art 32** = heart and soul; SC = guardian
8. Habeas corpus = liberty; mandamus = duty; certiorari = "to be certified"
9. Kesavananda = basic structure; Parsis not "Hindu"
10. Arts 20, 21 never suspended

Chapter 13

Directive Principles of State Policy

Built from 35 UPPSC questions (1994–2025). Focus: welfare state, Irish source, non-justiciability, Article-subject pairs, equal pay, added DPSPs, spot-the-DPSP.

ASKED MOST · REVISE FIRST

- DPSP borrowed from **Ireland** 4× 1994, 2005, 2008, 2015
- **Welfare state** = DPSP (and Preamble) 4× 1997, 2004, 2004, 2013
- **Equal pay for equal work** = DPSP, **Art 39(d)**, not an FR 5× 1998, 2004, 2010, 2012, 2014
- **Art 40** = village panchayats 4× 2001, 2002, 2014, 2017
- **Art 43A** = workers in management (42nd) 2× 2009, 2019
- **Uniform Civil Code** (Art 44) is a DPSP, original 3× 1998, 2010, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Objectives

- DPSPs aim at economic and social democracy (R explains A) 1× 2025
- Not an objective: **a religious state** 2× 2013, 2017
- Welfare state: **DPSP**; in both **Preamble and DPSP** 4× 2004, 2004, 2013, 1997

13.2 Source and Types

- Source: **Ireland** 4× 1994, 2005, 2008, 2015
- Classified as **socialist, Gandhian, liberal-intellectual** (all) 1× 2018

13.3 Nature (Art 37)

- **Not enforceable** by courts, yet **fundamental in governance** (both true) 1× 2023
- DPSPs are **non-justiciable** 1× 2004
- "A cheque payable at the bank's convenience": **K.T. Shah** 1× 2007

13.4 Article Map

- Village panchayats: **Art 40** 2× 2014, 2017
- **Art 40-village panchayats, 41-right to work, 44-UCC, 48-agriculture** 2× 2001, 2002
- **Art 48-separation of judiciary** is wrong (it is **Art 50**) 1× 2020
- **Co-operatives-Art 43A** is wrong (it is **43B**) 1× 2021

Article	Directive
38	Social order, reduce inequality
39	Livelihood, equal pay (d), wealth not concentrated (b, c)
39A	Free legal aid (42nd)
40	Village panchayats
41 / 42 / 43	Work, education, aid / Maternity relief / Living wage, cottage industry
43A / 43B	Workers in management (42nd) / Co-operatives (97th)
44	Uniform Civil Code
45	Care under 6 (86th)
46 / 47	SC, ST, weaker sections / Nutrition, prohibition
48 / 48A	Agriculture, cattle / Environment (42nd)
49 / 50 / 51	Monuments / Separate judiciary / International peace

13.5 Article 39

- Equal pay for equal work: a **DPSP**; included in DPSP 4× 1998, 2004, 2010, 2012
- Not an FR: **equal pay for equal work** 1× 2014
- Protection from exploitation for **children, women, tribals** (1, 2, 3) 1× 2009

13.6 Right to Work

- MGNREGA gives 100 days; right to work is **not** in Part III (A true, R false) 1× 2013
- MGNREGA implements **Art 43** (with 41) 2× 2009, 2010
- Update: MGNREGA replaced by **VB-G RAM G Act 2025** (125 days), from 1 July 2026

13.7 Education

- Not a DPSP now: free education up to 14 (moved to **Art 21A**) 1× 2010

QUANTUM IAS & PCS

- Not a DPSP: free education up to **16** (never in Constitution) 1× 2013
- DPSPs silent on **adult education** 1× 2008

13.8 Environment

- Environment, forests, wildlife: in **both** DPSP (48A) and Duties (51A(g)) 2× 2019, 2023
- "State shall endeavour to protect the environment": a **DPSP** 1× 2009

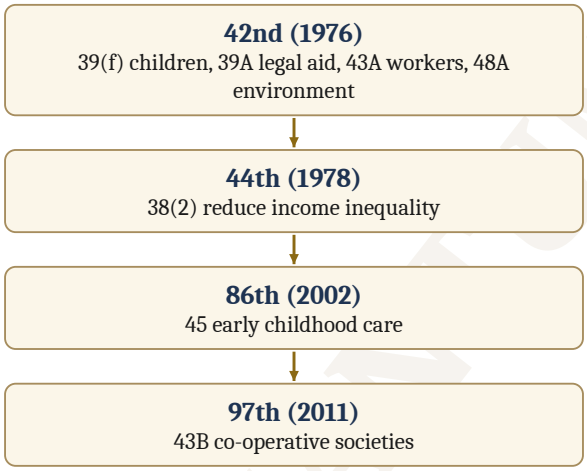
13.9 Article 51

- International peace and security: **DPSP**; foreign policy: **Art 51** 2× 2017, 2016

13.10 Added Later

- Workers in management: **Art 43A** (42nd) 2× 2009, 2019
- Not added by 42nd: **Uniform Civil Code** (original) 1× 1998
- Added later: **free legal aid** (39A) 1× 2014

DPSPS ADDED BY AMENDMENTS



13.11 Spot the DPSP

- A DPSP: **Uniform Civil Code** 2× 2010, 2010
- DPSP of the list: only **prohibition of liquor** (others are FRs) 1× 2002
- Not a DPSP: **abolition of untouchability** (Art 17) 1× 2008
- Not a DPSP: **right to information** 2× 2006, 2010

TRAP ALERT

- **43** living wage, **43A** workers in management, **43B** co-operatives.
- **48** agriculture and cattle; **50** separation of judiciary; **48A** environment (State) vs **51A(g)** (citizen).
- **Art 51** foreign policy (DPSP) vs **51A** Duties.

UP LINK

Art 40 is the base of UP's 3-tier Panchayati Raj. **Uttarakhand** (carved from UP) became the first State to enact a **UCC** (2024, in force January 2025). UP's prohibition debates link to **Art 47**.

LAST-MINUTE 10

1. DPSP = **Ireland**; Part IV, Arts **36-51**
2. Non-justiciable (Art 37); "cheque" = **K.T. Shah**
3. Welfare state = DPSP + Preamble
4. Equal pay = **39(d)**, not FR
5. **40** panchayats; **44** UCC; **50** separation of judiciary
6. **43A** workers (42nd); **43B** co-operatives (97th)
7. **39A** legal aid; **48A** environment (42nd)
8. MGNREGA = **Art 43** (now VB-G RAM G)
9. Silent on adult education; no "16 years"
10. **Art 51** = international peace

Chapter 14

Fundamental Duties

Built from 18 UPPSC questions (1993–2022). Focus: Swaran Singh Committee, 42nd Amendment, Part IV-A/Art 51A, 11 duties, spot-the-non-duty.

ASKED MOST · REVISE FIRST

- **Swaran Singh Committee** recommended; **42nd Amendment** (1976) added 10 duties **4×** 1993, 2002, 2008, 2012
- **Part IV-A, Article 51A** **5×** 2006, 2010, 2011, 2012, 2015
- **11** duties now (11th by the 86th Amendment, 2002) **2×** 2014, 2014
- **Not** duties: voting, paying taxes, respecting parents, abolishing untouchability, protecting monuments, organising panchayats **7×** 2007, 2008, 2014, 2015, 2019, 2021, 2022

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

14.1 Origin

- Recommended by the **Swaran Singh Committee** **2×** 2008, 2012
- Added as **Art 51A** by the 42nd Amendment **1×** 2015
- In **Part IV-A** **4×** 2006, 2010, 2011, 2012
- Ten duties added by the **42nd** **2×** 1993, 2002
- **Part V-Fundamental Duties** is wrong **1×** 2009
- Incorrect: Duties are part of FRs **1×** 2009

14.2 Eleven Duties

- Number: **eleven** **2×** 2014, 2014

DUTIES TIMELINE

Swaran Singh Committee
1976; suggested 8 duties

↓

42nd Amendment
1976; 10 duties, Part IV-A

↓

86th Amendment
2002; 11th duty: parents educate children 6-14

↓

Verma Committee
1999; ways to make duties work

14.3 Nature

- Not true: duties **can be enforced by mandamus** **2×** 2000, 2003

14.4 Spot the Non-Duty

- Not a duty: **voting** **1×** 2021
- Not a duty: **abolition of untouchability** (Art 17) **1×** 2008
- Not a duty: **organising village panchayats** (Art 40) **1×** 2015
- Duties: **defend the country, composite culture, public property**; not paying income tax **1×** 2019
- Not in Part IV-A: **respect parents and teachers** **1×** 2014

14.5 Monuments vs Culture

- Not a duty: **protect monuments** (that is Art 49, DPSP) **2×** 2007, 2022

14.6 Environment

- Protecting **wildlife** is a duty **1×** 2008
- "Duty of every citizen to protect the environment": **Art 51A** **1×** 1998

14.7 Cross-Part Matching

- **Respect international law-Art 51(c), compassion for living creatures-51A(g), scientific temper-51A(h), separate judiciary-Art 50** **1×** 2019

51A clause	Duty
(a)	Abide by Constitution; respect Flag and Anthem
(b), (c), (d)	Freedom struggle ideals; sovereignty; defend country
(e)	Brotherhood; renounce practices against women
(f)	Composite culture heritage
(g)	Environment, compassion for living creatures
(h)	Scientific temper, inquiry, reform
(i)	Public property; abjure violence
(j)	Strive for excellence
(k)	Parents educate children 6-14 (86th)

TRAP ALERT

- "Duty of every citizen" = **51A(g)**; "State shall endeavour" = **48A**.
- **Composite culture** = duty (51A(f)); **monuments** = DPSP (Art 49).
- Duties are **non-justiciable**; no writ.

UP LINK

No UP-specific question. UP schools observe **Samvidhan Divas** (26 Nov) with duty pledges, a likely current-affairs link.

LAST-MINUTE 10

1. **Swaran Singh** Committee, 1976
2. **42nd** Amendment: 10 duties
3. **Part IV-A, Art 51A**; from **USSR**
4. **11** duties; 11th by **86th** (2002)
5. Not enforceable by writ
6. Not duties: vote, tax, respect parents
7. Not duties: untouchability (17), panchayats (40), monuments (49)
8. Environment and wildlife = **51A(g)**
9. Scientific temper = **51A(h)**
10. Parents educate = **51A(k)**

Chapter 15

The President

Built from 57 UPPSC questions (1992–2025). Focus: Art 53, electoral college, impeachment, vacancy and acting President, veto and ordinance, the sequence of Presidents.

ASKED MOST · REVISE FIRST

- No President and no VP: **Chief Justice of India** acts **6×** 1992, 2005, 2014, 2014, 2015, 2021
- Executive power of the Union in the President: **Article 53** **4×** 2008, 2010, 2015, 2021
- Vacancy filled **within six months** **4×** 1998, 2005, 2013, 2016
- Impeachment: **Art 61**, by **either House**, **14 days'** notice **6×** 1994, 2007, 2008, 2014, 2015, 2017
- **Neelam Sanjiva Reddy**: only unanimous election; ex-Speaker; not VP before **5×** 1998, 2002, 2004, 2021, 2022
- Joint sitting (Art 108) summoned by the **President** **3×** 2012, 2015, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

15.1 Nominal Head

- Head of the Republic: the **President** **1×** 2013
- Indian President differs from the British King; **not** like the US President (A true, R false) **1×** 1998
- "Represents the nation but does not lead it": **President** **1×** 2007
- Authority: **titular, de jure, constitutional, nominal** (2 and 4) **1×** 2006

15.2 Executive Power

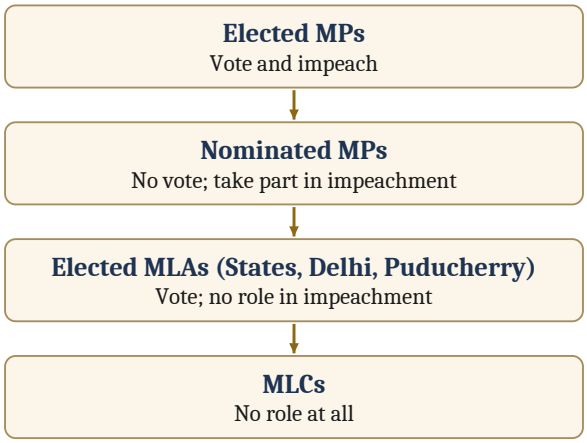
- Vested by **Art 53**; in the **President** only **4×** 2008, 2010, 2015, 2021
- Heads Union Executive; power is **not** unlimited (A true, R false) **1×** 2012
- Executive power in the President, the constitutional head (R explains A per key) **1×** 2023
- Executive power is carrying on government (both true, R not the explanation) **1×** 2010
- Supreme command is his, but **regulated by law** (A true, R false) **2×** 2005, 2010

15.3 Election (Arts 54-55)

- **Indirect** election by an electoral college (R explains A) **1×** 2015
- Electoral college: **elected MPs, elected MLAs of all States, elected MLAs of Delhi and Puducherry** **1×** 2001
- Delhi and Puducherry added by the **70th Amendment** **1×** 2017
- Not included: **elected MLCs** **1×** 2015
- Wrong: election can be postponed if the electoral college is incomplete **1×** 2016
- Nomination needs **50 proposers and 50 seconders** (no option correct, *) **1×** 2004
- Disputes decided by the **Supreme Court** (Art 71)

TATTVA IAS & PCS

WHO VOTES, WHO REMOVES



15.4 Qualifications

- Only true: **35 years**; not "citizen by birth"; qualified for **Lok Sabha** (not "Parliament") 1× 1998
- Not a qualification: **education** 1× 1992
- An MP or MLA elected President **vacates the seat** on taking office 1× 2008

15.5 Impeachment (Art 61)

- Procedure in **Art 61** 2× 2015, 2017
- Removed before term only by **impeachment** 1× 1994
- Initiated by **either House** 2× 2007, 2008
- Notice: **14 days** 1× 2014
- Ground: **violation of the Constitution**; two-thirds of total membership of each House

15.6 Vacancy and Acting President

- Vacancy filled **within six months** 4× 1998, 2005, 2013, 2016
- No VP: **Chief Justice of India** acts 6× 1992, 2005, 2014, 2014, 2015, 2021
- CJI who acted as President: **M. Hidayatullah** (1969) 2× 2003, 2004

15.7 Legislative Powers

- President is **part of Parliament**; no Bill becomes law without assent (R explains A) 2× 2007, 2023
- Wrong: President **takes part in debates** 1× 2010
- Joint sitting summoned by the **President** 3× 2012, 2015, 2015
- Withhold assent: **Art 111** 3× 2004, 2008, 2011
- Not his power: **returning a Money (financial) Bill** 1× 2007

Veto	Use
Absolute	Withhold assent (Private Member's Bills, after cabinet resigns)
Suspensive	Return once (not Money Bill); bound if passed again
Pocket	No time limit; Postal Bill 1986 (Zail Singh)
Money Bill	Cannot be returned; prior recommendation
Amendment Bill	Must assent (24th Amendment)

15.8 Ordinance (Art 123)

- Ordinance power: **Art 123** 2× 2013, 2016
- 2002 electoral-reform ordinance signed under **Art 123** 1× 2015

15.9 Appointments

- Appoints all listed officials 1× 2002
- Not appointed by him: the **Vice-President** (elected) 1× 1997
- Appoints **FC chairman and UT administrators/CMs** (1 and 3) 1× 2005
- Governors' duties in emergencies: **Art 160** 1× 2007
- Declares **Scheduled Areas**: President 1× 2006

15.10 Art 143 and Art 72

- Consult the Supreme Court: **Art 143** 1× 2012
- Incorrect: the President **must accept** SC advice 1× 2008
- Pardon is an **executive power** (Art 72) 1× 2010

15.11 Presidents of India

- Order: **Rajendra Prasad, Radhakrishnan, Zakir Husain, V.V. Giri** 1× 2009
- Fourth President: **V.V. Giri** 1× 1994
- **Fakhruddin Ali Ahmed 1974-77, Sanjiva Reddy 1977-82, Zakir Husain 1967-69, V.V. Giri 1969-74** 1× 2013
- Order: **Sanjiva Reddy, Venkataraman, K.R. Narayanan, Kalam** 1× 2025
- **Pratibha Patil**: **12th** President 1× 2015

QUANTUM IAS & PCS

- Elected **unanimously**: **N. Sanjiva Reddy** 2× 1998, 2004
- Also Lok Sabha **Speaker**: **Sanjiva Reddy** 1× 2021
- Not VP before: **Sanjiva Reddy** 2× 2002, 2022
- Was Governor of Bihar: **Zakir Husain** 1× 2004
- "Philosopher King": **Radhakrishnan** 1× 2014
- Trade union movement: **V.V. Giri** 1× 2012
- Rashtrapati Bhavan designed by **Edwin Lutyens** 1× 2014

No.	President	Term
1	Rajendra Prasad	1950-62
2	S. Radhakrishnan	1962-67
3	Zakir Husain	1967-69 (died)
4	V.V. Giri	1969-74
5	Fakhruddin Ali Ahmed	1974-77 (died)
6	N. Sanjiva Reddy	1977-82
7, 8	Zail Singh; R. Venkataraman	1982-87; 1987-92
9, 10	S.D. Sharma; K.R. Narayanan	1992-97; 1997-2002
11, 12	A.P.J. Kalam; Pratibha Patil	2002-07; 2007-12
13, 14, 15	Pranab Mukherjee; Ram Nath Kovind; Droupadi Murmu	2012-17; 2017-22; 2022-

TRAP ALERT

- Qualified for **Lok Sabha** (President) vs **Rajya Sabha** (VP).
- **MLCs** never vote; nominated MPs impeach but do not elect.
- Acting Presidents (Giri, Hidayatullah, Jatti) are **not counted** in the sequence.

UP LINK

UP has the largest share of the presidential vote: its MLA vote value is **208** (highest), and 80 LS + 31 RS MPs. **Ram Nath Kovind** (14th President) is from **Kanpur Dehat**, UP; **Zakir Husain** studied and taught at Aligarh.

LAST-MINUTE 10

1. Art **53** executive power; **54-55** election; **58** qualifications
2. Electoral college: elected MPs, MLAs, **Delhi, Puducherry** (70th)
3. Art **61** impeachment; either House; **14 days**
4. Vacancy filled in **6 months**; CJI acts if no VP
5. **Hidayatullah** = CJI who acted as President
6. Art **111** assent; **123** ordinance; **143** SC advice; **72** pardon
7. Joint sitting summoned by President (Art 108)
8. **Sanjiva Reddy**: unanimous, ex-Speaker, not VP
9. **Pratibha Patil** = 12th; Giri = 4th, trade unions
10. Radhakrishnan = Philosopher King; Lutyens = Rashtrapati Bhavan

Chapter 16

The Vice-President

Built from 12 UPPSC questions (2003–2025). Focus: electoral college of all MPs, Rajya Sabha qualification, removal from the Rajya Sabha, Art 71.

ASKED MOST · REVISE FIRST

- VP elected by **all members of both Houses** (elected and nominated); no MLAs 3× 2012, 2012, 2016
- Removal resolution only in the **Rajya Sabha** 4× 2003, 2004, 2007, 2013
- VP is ex-officio **Chairman of the Rajya Sabha** but **not its member** 3× 2004, 2010, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Election (Art 66)

- Electoral college: **all members of both Houses** 1× 2012
- Votes: **Lok Sabha and Rajya Sabha members** only (1 and 2) 1× 2012
- Election indirect like the President; State legislators have **no role** in VP election (only 1) 1× 2016

16.2 Qualification and RS Chair

- Must be qualified for the **Upper House** because he chairs it (R explains A) 1× 2010
- Not correct: qualified for **Lok Sabha**; removed by **impeachment** (1 and 4) 1× 2005
- Ex-officio Chairman of RS; **not** a member (only 1) 1× 2025
- Presided by a non-member: **Rajya Sabha** 1× 2004

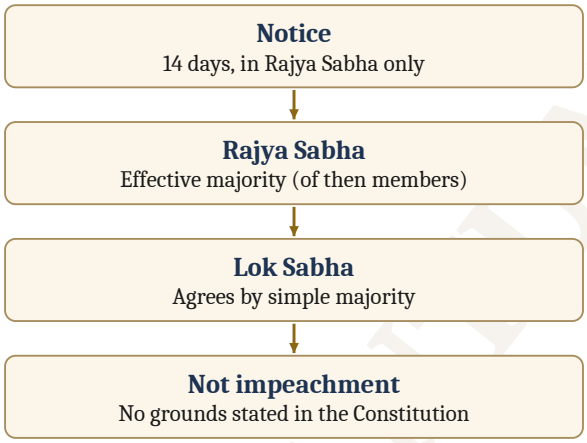
16.3 Acting President (Art 65)

- Second highest dignitary; no formal VP functions; acts in absence and in vacancy (all four) 1× 2013

16.4 Removal (Art 67)

- Removed by a resolution of the **Council of States** (effective majority), agreed by the Lok Sabha (simple majority) 1× 2013
- Proposed only in the **Rajya Sabha** 3× 2003, 2004, 2007

REMOVING THE VICE-PRESIDENT



16.5 Disputes (Art 71)

- Election voided by SC: acts done **on or before** the decision stay valid (only 1) 1× 2023

16.6 Vice-Presidents

- Wrong: **Zakir Husain** was the first VP (it was **S. Radhakrishnan**, 1952) 1× 2013

Item	President	Vice-President
Electors	Elected MPs + elected MLAs	All MPs (incl. nominated)
Qualification	Eligible for Lok Sabha	Eligible for Rajya Sabha
Removal	Impeachment, either House	RS resolution + LS agreement
Disputes	Supreme Court (Art 71)	Supreme Court (Art 71)

TRAP ALERT

- Nominated MPs **vote for VP**, not for President.
- VP chairs RS but casts only a **casting vote**.
- VP removal = **not impeachment**; starts only in RS.

UP LINK

Vice-Presidents linked to UP: **Mohammad Hamid Ansari** (born in Kolkata, family from Ghazipur, UP; ex-AMU VC). UP's large MP group shapes VP elections since only MPs vote.

LAST-MINUTE 10

1. Electors: **all MPs**, no MLAs
2. Qualified for **Rajya Sabha**; 35 years
3. Ex-officio **RS Chairman**, not a member
4. Acts as President under **Art 65**
5. Removal: RS effective majority + LS simple majority
6. Removal starts only in **Rajya Sabha**
7. Disputes: **Supreme Court**, Art 71
8. Acts before SC decision stay valid
9. First VP = **S. Radhakrishnan**
10. Six VPs became President

Chapter 17

The Council of Ministers

Built from 29 UPPSC questions (1992–2021). Focus: PM and CoM, six-month rule, collective responsibility to the Lok Sabha, no-confidence, Deputy PM, PM facts.

ASKED MOST · REVISE FIRST

- ■ CoM **collectively responsible to the Lok Sabha** only (Art 75(3)) **5×** 2012, 2012, 2016, 2019, 2021
- ■ Non-member minister: **six months** **3×** 1993, 1995, 2015
- ■ No-confidence motion needs **50** LS members **3×** 2009, 2013, 2015
- • Government can move a **confidence motion** **2×** 2006, 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

17.1 PM and the Union Executive

- True: PM **appointed by President**, **heads the CoM**, need not be an MP at appointment; executive power is **not** in the PM **1×** 2011
- PM is the **real head** of the CoM **1×** 2013
- **Art 78**: duties of the **PM** (inform the President) **1×** 2014
- Legal name for suits (Art 300): **Union of India** **1×** 2021

17.2 Cabinet vs Council

- Cabinet = **ministers of Cabinet rank** **1×** 1992
- CoM capped at **15%** of Lok Sabha (91st Amendment, 2003)

17.3 Appointment (Art 75)

- Appointment and tenure of ministers: **Art 75** **1×** 2001
- ■ Non-member minister: **6 months** **3×** 1993, 1995, 2015
- Not true: Cabinet membership limited to the **Lower House** **1×** 2011

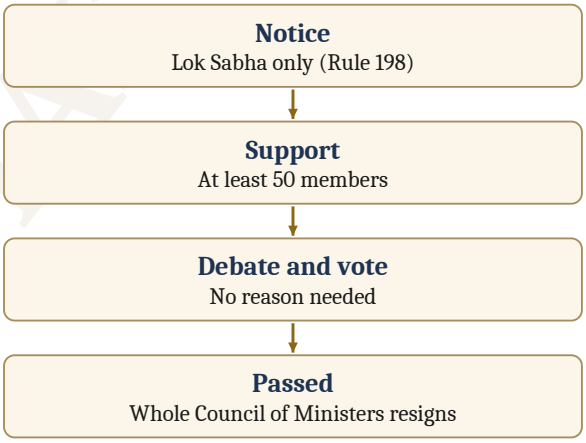
17.4 Responsibility

- Collectively responsible: **to Lok Sabha, a constitutional duty, Art 75(3)** (1, 2, 3) **1×** 2012
- Responsible to the **Lok Sabha** **1×** 2012
- Ministers hold office at the **President's** pleasure, not the PM's (only 1) **1×** 2016
- Responsible to both Houses: **false**; both Houses can supply ministers: true **2×** 2019, 2021
- Ministers make policy; collective responsibility (both true, R not the explanation) **1×** 1993

17.5 Confidence and No-Confidence

- CoM can move a **confidence motion** **2×** 2006, 2010
- Resigns on a no-confidence vote in the **Lok Sabha** **1×** 2012
- ■ No-confidence needs **50** members **3×** 2009, 2013, 2015
- Wrong: **President's Rule** follows CoM resignation **1×** 2010

NO-CONFIDENCE MOTION



17.6 Deputy PM

- Calling someone Deputy PM is political; gives no PM power (R explains A) **1×** 2004
- Deputy PM: **not in the Constitution** (K.M. Sharma v. Devi Lal, 1990, upheld it) **1×** 2009

17.7 Prime Ministers

- 12th PM (counting Nanda): **Deve Gowda** **1×** 2005
- PM more than once with gaps: **Nanda, Indira Gandhi, Vajpayee** **1×** 2010
- Never a CM before PM: **Chandra Shekhar** **1×** 2011
- Was a Lok Sabha member as PM: **Chandra Shekhar** **1×** 2015
- LPG policy: **P.V. Narasimha Rao** (1991) **1×** 2013
- Not true of Manmohan Singh: chaired the **Finance Commission** **1×** 2004

QUANTUM IAS & PCS

- **Rajiv Gandhi-Karam Bhumi** is wrong (his samadhi is **Veer Bhumi**) 1× 1996

17.8 Machinery

- Cabinet reorganisation report: **Gopalaswami Ayyangar** 1× 2010
- Official document: **White Paper** 1× 2008

TRAP ALERT

- Ministers serve at the **President's** pleasure, responsible to **Lok Sabha** only.
- Ministers may come from **either House**.
- **50 members** is a Lok Sabha rule, not an Article.
- Deputy PM is **extra-constitutional**, not unconstitutional.

UP LINK

Nine PMs sat for UP seats: Nehru (Phulpur), Shastri (Allahabad), Indira Gandhi (Rae Bareli), Charan Singh (Baghpat), Rajiv Gandhi (Amethi), V.P. Singh (Fatehpur), Chandra Shekhar (Ballia), Vajpayee (Lucknow), Modi (Varanasi). **Charan Singh** and **V.P. Singh** were UP Chief Ministers.

LAST-MINUTE 10

1. Art **74** CoM aids President; **75** appointment; **78** PM's duties
2. **Art 300** = Union of India (legal name)
3. Collective responsibility to **Lok Sabha** (Art 75(3))
4. Pleasure of the **President**
5. Non-member minister: **6 months**
6. CoM cap **15%** (91st)
7. No-confidence: **50** members, LS only
8. Deputy PM: extra-constitutional (Devi Lal case)
9. Gaps in PM-ship: Nanda, Indira, Vajpayee
10. **Narasimha Rao** = LPG 1991

Chapter 18

The Attorney-General and the C.A.G.

Built from 34 UPPSC questions (1991–2023). Focus: AG's pleasure tenure and rights in Parliament, Advocate General, CAG's tenure, audit role and the PAC.

ASKED MOST · REVISE FIRST

- **Attorney General** may **speak in Parliament** but **cannot vote** (Art 88) 5× 2003, 2005, 2011, 2017, 2023
- AG: appointed by President, **SC-judge qualification**, holds office **at President's pleasure** 5× 2014, 2016, 2020, 2023, 2023
- AG = first law officer and chief legal adviser of the Government of India 3× 1995, 2008, 2010
- **CAG** (Art 148): **6 years**, removed like an SC judge, "guardian of the public purse" 6× 1993, 2001, 2006, 2008, 2014, 2015
- CAG is the **friend and guide of the PAC** 3× 2005, 2006, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Attorney General (Art 76)

- Holds office at the **President's pleasure** 1× 2016
- First law officer: **Attorney-General**; chief legal adviser; adviser to the **Government of India** 3× 2010, 1995, 2008
- Right of audience in all courts: **AG** 1× 2015
- Qualified as an SC judge (true); removed like SC judges (false) (only 1) 1× 2023
- SC-judge qualification (true); pay fixed by **Parliament** (false, President fixes it) (only 1) 1× 2020

18.2 AG and Parliament

- Appointed by President; SC-judge qualification (1, 2); not an MP, not impeached 1× 2014
- Right to address Parliament: **AG** 1× 2011
- Speaks and joins committees, **no vote**: AG 1× 2003
- Takes part in **both Houses**: AG 1× 2005

- Pleasure of the PM (false); no MP privileges (false): neither 1× 2023
- Not correct: AG **can vote** in Parliament 1× 2017

18.3 SG and Advocate General

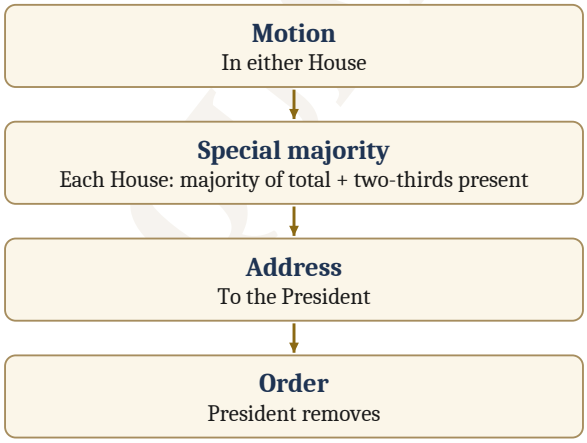
- Solicitor General: a **legal adviser** 2× 1991, 1994
- State legal adviser: **Advocate General** (Art 165) 2× 2014, 2015

Office	Article	Key fact
Attorney General	76	SC-judge qualification; pleasure of President; Art 88 speaks
Solicitor General	Statutory	Assists AG; not constitutional
Advocate General	165	HC-judge qualification; pleasure of Governor; Art 177
CAG	148-151	6 years / 65; removed like SC judge

18.4 CAG: Appointment and Tenure

- Appointed by the **President**; under **Art 148** 2× 2001, 2015
- Union audit reports go first to the **President** (Art 151) 1× 2013
- Office created **by the Constitution** 1× 2012
- Tenure: **6 years** (or 65) 1× 1993
- Not true: removable by the President at will 1× 2014
- Removal: **President's order after address by both Houses** (*) 1× 2006

CAG REMOVAL (LIKE AN SC JUDGE)



18.5 Audit, Not Control

- Not his duty: **controlling the issue of public money** 1× 2004
- "**Guardian of the public purse**": CAG 1× 2008
- Accounts separated from audit: **1976** 2× 2004, 2019
- State accounts are **not** controlled by any listed body (one CAG audits all) 1× 2003

18.6 CAG and the PAC

- Action on CAG remarks: **Parliament** 1× 2014
- Attends PAC meetings: **CAG**; CAG = friend and guide of the **PAC** 2× 2011, 2006
- **VP-Rajya Sabha, CAG-PAC, Solicitor General-Supreme Court, AG-Lok Sabha meetings** 1× 2005

18.7 Public Finance

- Not part of public finance: **functioning of commercial banks** 1× 2014

TRAP ALERT

- AG pay fixed by the **President**; CAG pay by **Parliament**.
- AG serves at **pleasure**; CAG has **security of tenure**.
- AG speaks but **never votes**; not an MP.
- There is no "Advocate General of India" and no State CAG.

UP LINK

UP's **Advocate General** sits at the **Allahabad High Court** (Prayagraj) and can speak in the UP Legislature (Art 177). CAG's UP reports go to the **Governor** of UP (Art 151(2)), then to the Vidhan Sabha PAC.

LAST-MINUTE 10

1. AG: **Art 76**; SC-judge qualification; President's pleasure
2. AG speaks in Parliament (**Art 88**), no vote; MP privileges (105(4))
3. AG = first law officer; audience in all courts
4. Advocate General = **Art 165**, State adviser
5. CAG: **Art 148**; 6 years or 65
6. Removed like an SC judge
7. CAG = guardian of the **public purse**
8. Accounts separated from audit **1976**
9. CAG = friend and guide of the **PAC**
10. Union reports to **President**; State to **Governor**

Chapter 19

The Parliament I: Lok Sabha

Built from 40 UPPSC questions (1995–2025). Focus: Lok Sabha strength, 1971 Census freeze, state seats, reserved seats, quorum, Speaker and pro tem Speaker.

ASKED MOST · REVISE FIRST

- ■ Maximum Lok Sabha strength **550** (530 States + 20 UTs; Anglo-Indian seats lapsed 2020) 4× 2006, 2007, 2008, 2018
- • **UP** has the most seats in both Houses (80 LS, 31 RS) 2× 1995, 2012
- ■ Quorum: **one-tenth** of total members 4× 2004, 2006, 2008, 2017
- ■ Speaker resigns to the **Deputy Speaker**; removed by LS majority of members 4× 2007, 2008, 2008, 2014
- ■ First Speaker: **G.V. Mavalankar**; first no-confidence against a Speaker too 4× 2000, 2004, 2011, 2023
- • First woman Speaker: **Meira Kumar** 2× 2007, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Elections

- Not correct: rolls prepared under the **State** Government's control (it is the EC) 1× 2011

19.2 Strength (Art 81)

- 525 to 545: **31st Amendment** (1973) 1× 2004
- ■ Maximum: **550**; UT maximum: **20** 3× 2006, 2008, 2018
- 2009: **530** constituencies from States 1× 2007

19.3 Census Freeze

- State seats based on the **1971 Census** 1× 2003
- **84th Amendment** froze seats till the first census after **2026** 1× 2005

SEAT FREEZE TIMELINE

42nd Amendment 1976
Froze seats on 1971 Census till 2000

↓

84th Amendment 2001
Extended freeze till census after 2026

↓

87th Amendment 2003
Boundaries redrawn on 2001 Census

↓

106th Amendment 2023
One-third women's seats after next delimitation

19.4 State Seats

- Most seats in **both Houses**: **UP** 2× 1995, 2012
- After UP: **Maharashtra** (48) 1× 2008
- Bihar then = Assam + undivided MP (54) (*) 1× 2000
- Equal seats then: AP and WB (42) (*) 1× 2005
- Wrong then: Assam seats; AP 42 is wrong now (25) 1× 1997
- Two Lok Sabha seats: **Tripura** 1× 2016
- Largest UP seat by electors: **Ghaziabad** (*) 1× 2007

State	LS seats
Uttar Pradesh	80
Maharashtra	48
West Bengal	42
Bihar	40
Tamil Nadu	39
Madhya Pradesh	29

State	LS seats
Karnataka	28
Gujarat / Rajasthan, AP	26 / 25 each
Assam, Jharkhand	14 each

19.5 Reserved Seats

- No ST seats in LS: **Kerala and Tamil Nadu** 1× 1998
- No SC/ST LS seats: **Arunachal Pradesh and J&K** (then) 1× 2000
- SC seats are **proportional** to SC population (WB 42 seats: 10 SC) (*) 1× 2014
- Most ST seats: **Madhya Pradesh** (6) 2× 1997, 2005
- Anglo-Indian nomination: 2 then; **lapsed in 2020** (104th) (*) 1× 2014

19.6 Sessions and Quorum

- Sitting ended by adjournment, prorogation or dissolution (all) 1× 2000
- Prorogues the House: **President** 1× 2015
- Quorum: **one-tenth** 4× 2004, 2006, 2008, 2017

19.7 Speaker: Removal

- Resignation to the **Deputy Speaker** 2× 2008, 2014
- Removed by an LS resolution passed by a **majority of all members** (effective majority) 2× 2007, 2008

19.8 Pro tem, Casting Vote, Deputy

- Pro tem Speaker **administers oath**; usually a **senior-most** member 2× 2010, 2017
- Casting vote only in a **tie**; under **Art 100** 2× 2008, 2015
- Deputy Speaker presiding: may **vote in a tie** 1× 2016

19.9 Speakers

- First Speaker: **G.V. Mavalankar**; first no-confidence against a Speaker 4× 2011, 2023, 2000, 2004
- First woman: **Meira Kumar** 2× 2007, 2008
- First tribal: **P.A. Sangma** 1× 2015
- **11th-Sangma, 12th-Balayogi, 14th-Somnath Chatterjee, 17th-Om Birla** 1× 2025

19.10 Leader of Opposition

- Recognition needs **one-tenth** of House (55) (*) 1× 2006

19.11 Secretariat (Art 98)

- LS Secretariat under the **Speaker** 1× 1997
- Common posts allowed; Parliament regulates staff service (both) 1× 2023

TRAP ALERT

- Seat **allocation** = 1971 Census; **boundaries** = 2001 Census.
- Only the **Lok Sabha** is dissolved; RS is permanent.
- SC/ST seats are **proportional**; one-third is for **women** (Art 330A).
- Speaker resigns to the **Deputy Speaker**, not the President.

UP LINK

UP: 80 Lok Sabha seats, the most of any State, with **17 reserved for SCs**; it is also the largest State in the Rajya Sabha (**31**). Ghaziabad is UP's largest constituency by electors. The UP Vidhan Sabha has **403** elected seats.

LAST-MINUTE 10

1. Max LS **550**; States 530, UTs **20**
2. Anglo-Indian seats lapsed **2020**
3. Seats frozen on **1971 Census** till after **2026** (84th)
4. UP **80**; Maharashtra 48; WB 42
5. Most ST seats: **MP**; no ST seats: Kerala, TN
6. Quorum **1/10**; President prorogues
7. Speaker resigns to **Deputy Speaker**; casting vote **Art 100**
8. Pro tem: senior member, oaths
9. First Speaker **Mavalankar**; woman **Meira Kumar**; tribal **Sangma**
10. LoP needs **1/10** seats

Chapter 20

The Parliament II: Rajya Sabha

Built from 28 UPPSC questions (1990–2025). Focus: permanent House, 6-year term, state seat numbers, Art 249 and Art 312 powers, money-bill limits, Chairman.

ASKED MOST · REVISE FIRST

- Rajya Sabha is **permanent**, never dissolved; **one-third retire every 2 years** 4× 2011, 2012, 2014, 2016
- **Exclusive** RS power: create new **All India Services** (Art 312) 3× 1997, 2018, 2025
- **Art 249**: Parliament legislates on a State subject after an RS resolution 3× 2011, 2012, 2017
- State seat matching: Maharashtra 19, TN 18, Bihar 16, Karnataka 12, Gujarat 11, Rajasthan, Odisha 10, Kerala 9, Punjab 7 3× 2017, 2021, 2025
- Presiding officer who is **not a member**: VP as RS Chairman 4× 1992, 2010, 2011, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Composition (Art 80)

- Elected by **elected MLAs** of States 1× 1990
- President nominates **12** 1× 2006

20.2 Term and Age

- True: **not dissolved, one-third retire every 2 years**; not 5 years, not 25 years (1 and 3) 1× 2014
- Not dissolved because one-third retire every two years (R explains A) 1× 2016
- Member's term: **6 years** (RPA 1951) 1× 1992
- RS is **not subject to dissolution**; dissolved by **none** 2× 2011, 2012
- Minimum age: **30** 1× 2006
- First constituted: **1952** (3 April) 1× 2017

20.3 State Seats (Fourth Schedule)

- Equal then: undivided AP and TN (18) (*) 1× 1999
- UP's 34 then = AP 18 + MP 16 (*); UP now **31** 2× 2000, 2002
- **Maharashtra 19, Karnataka 12, Bihar 16, Tamil Nadu 18** 1× 2025
- **Rajasthan 10, Gujarat 11, Karnataka 12, Punjab 7** 1× 2021
- **Gujarat 11, Karnataka 12, Kerala 9, Odisha 10** 1× 2017
- Fewest elected members among options: **Himachal Pradesh** (3) 2× 2006, 2010
- UTs represented in **both Houses** (as a class) 1× 2015

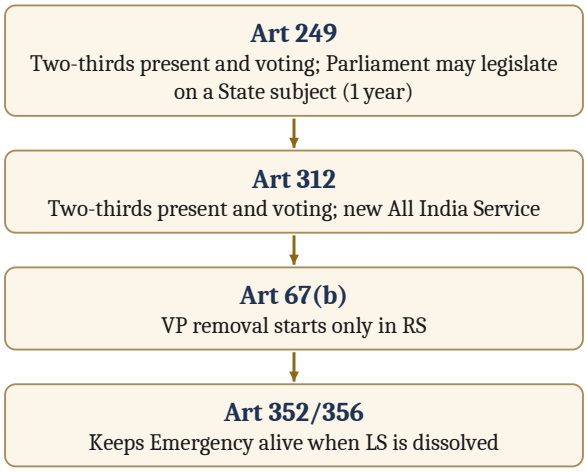
State	RS seats
Uttar Pradesh	31
Maharashtra	19
Tamil Nadu	18
West Bengal, Bihar	16 each
Karnataka	12
Gujarat, AP, MP	11 each
Rajasthan, Odisha	10 each
Kerala	9
Punjab	7
Delhi / J&K / Puducherry	3 / 4 / 1

20.4 Special Powers

- Wrong: **only the RS** makes laws on State List matters (Parliament does) 1× 2011
- Legislate on a State subject: **Art 249** 2× 2012, 2017
- Exclusive: **recommend a new All India Service** 2× 1997, 2018
- RS can resolve for a new AIS (true); needs two-thirds of **total** (false, it is of present and voting) (only 1) 1× 2025

QUANTUM IAS & PCS

RAJYA SABHA-ONLY POWERS



20.5 Limits

- LS need not accept RS recommendations on Money Bills; **RS** has 14 days (A false); Money Bill not introduced in RS (R true) **1×** 2017
- Cannot start in RS: **Money Bill** **1×** 2021
- LS-only: **Money Bill introduction, rejecting or amending it, CoM accountability**; not AIS (I, II, III) **1×** 2004

20.6 Chairman

- Chairman: **Vice-President**, ex officio **2×** 2011, 2013
- Presiding officer not a member: **Rajya Sabha** **2×** 1992, 2010

TRAP ALERT

- RS term **6 years** comes from the **RPA 1951**, not the Constitution.
- Art 312 and 249: two-thirds of members **present and voting**.
- Many UTs (Ladakh, Chandigarh, A&N) have **no RS seat**.

UP LINK

UP sends **31** members, the most of any State; they are elected by UP's **403** MLAs by single transferable vote. UP's 34 seats fell to 31 when Uttarakhand (3 seats) was formed in 2000.

LAST-MINUTE 10

1. Max **250** (238 + 12 nominated); now 245
2. Elected by **elected MLAs**; STV
3. Permanent; **1/3** retire every 2 years; term **6 years**
4. Minimum age **30**; first constituted **1952**
5. UP **31**; Maharashtra 19; TN 18
6. **Art 249** State subject; **Art 312** new AIS
7. Money Bill never starts in RS; 14 days for RS
8. Chairman = **VP**, not a member
9. Deputy Chairman presides when VP acts as President
10. Only some UTs have RS seats

Chapter 21

The Parliament III: Work Management and Legislative Process

Built from 59 UPPSC questions (1991–2024). Focus: what Parliament consists of, joint sitting, Money Bills, Consolidated and Contingency Funds, Budget stages, zero hour.

ASKED MOST · REVISE FIRST

- Parliament = **President + Lok Sabha + Rajya Sabha**; the VP is **not** a part **9×** 2004, 2011, 2012, 2013, 2013, 2014, 2015, 2017, 2024
- **Joint sitting** (Art 108) presided by the **Speaker**; only for ordinary Bills **12×** 1997, 2005, 2006, 2009, 2010, 2013, 2013, 2013, 2014, 2020, 2021, 2023
- **Money Bill**: Art 110; **Speaker** certifies; only in LS; RS has **14 days** **12×** 1992, 1992, 2000, 2006, 2006, 2007, 2011, 2012, 2014, 2015, 2016, 2019
- **Zero hour**: Indian innovation, from **12 noon** (key: 11-12), up to one hour **5×** 2004, 2007, 2015, 2015, 2021
- **Cut motion** is linked to the Budget **3×** 2002, 2002, 2003

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

21.1 Composition and Address

- President's special address: at the first session after each election and each year; tells the causes of summons (both) 1× 2021
- Constitution says: President **not** a member of either House; Parliament = President + both Houses (both) 1× 2024
- Parliament = **President, Rajya Sabha, Lok Sabha** 7× 2004, 2011, 2012, 2013, 2014, 2015, 2017
- Not part of Parliament: **Vice-President** 1× 2013

21.2 Sessions and Officers

- Parliament must meet at least **twice** a year (gap under 6 months) 1× 1995
- Officers: **Speaker, Deputy Speaker, Secretary-General, RS Chairman** (all four) 2× 2010, 2013

21.3 Membership

- Seat vacated after **60 days'** absence without leave 1× 2018
- First disqualified for absence: a **Rajya Sabha** member 1× 2004
- Office of profit decided by the **President** (MPs) and **Governor** (MLAs) on EC opinion 2× 2000, 2006

21.4 Privileges

- Not a collective privilege: **freedom from attending as witness** (individual) 1× 2017
- Courts cannot inquire into proceedings: **Art 122** 1× 2010

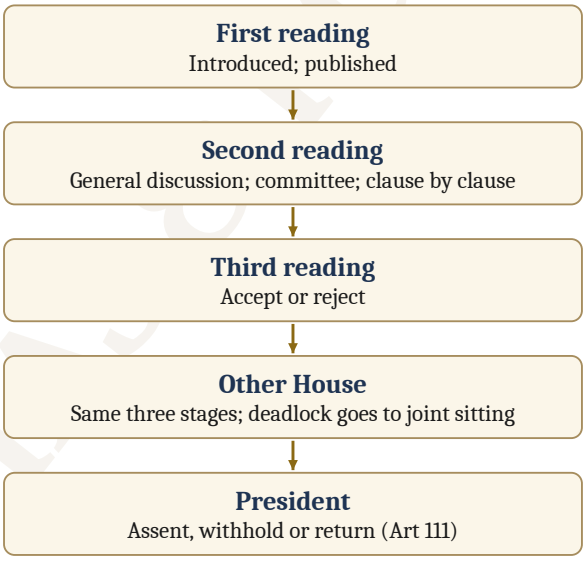
21.5 State List and Residuary

- Parliament legislates on State List when the **States pass a resolution** (Art 252) 1× 2012
- Also for treaties, by consent, in Emergency, after RS resolution (all) 1× 2005
- Treaties: **Art 253**; **without** any State's consent 3× 2016, 2016, 2006
- Only: **RS resolution** and **national emergency** (2 and 4) 1× 2005
- Residuary subjects: **Parliament** 1× 2015

21.6 Question Hour, Zero Hour, Motions

- Zero hour time (key): **11 am-12 noon** (it actually starts at 12 noon) 3× 2015, 2015, 2021
- Zero hour: India's contribution; maximum **one hour** 2× 2004, 2007
- Control over CoM: **adjournment motion, question hour, supplementaries** 1× 2017
- Budget-related: **cut motion** 3× 2002, 2002, 2003
- Calling attention restricted the **adjournment motion** 1× 2010
- Rule 184 debates end in a **vote** (R explains A) 2× 2001, 2002

HOW AN ORDINARY BILL BECOMES LAW



21.7 Passing Bills and Joint Sitting

- General discussion: **second reading** 1× 2006
- Joint sitting is **Art 108** (not 109); **Speaker** presides (only 2) 1× 2023
- Joint sitting: **Art 108** 1× 2021
- Neither: Art 109 is not joint sitting; no joint sitting for amendment Bills 1× one-liner
- Presides: **Speaker of Lok Sabha** 8× 2005, 2006, 2009, 2010, 2013, 2013, 2013, 2014
- First joint sitting: **Dowry Prohibition Bill** (1961) 1× 2004
- Joint sitting resolves **ordinary legislation** 2× 1997, 2020

21.8 Money Bills

- Defined in **Art 110** 2× 2000, 2019
- Not a Money Bill matter: **fines or penalties** 1× 2010
- Decides a Money Bill: **Speaker** 3× 1992, 2007, 2014

- Introduced in **Lok Sabha** 4× 2006, 2011, 2012, 2015
- Deemed passed by RS after **14 days** 3× 1992, 2006, 2016
- RS amendments: deemed passed if LS passes it again (accept or reject) 1× 1994
- Bill only involving expenditure: either House 1× 2016
- President's prior sanction: **new State, State-interest taxes, boundaries, Money Bill** (all) 1× 2005
- True: property not an FR; RS can stop a (non-money) Financial Bill (1 and 2) 1× 2007

21.9 Funds

- No tax without authority of law: **Art 265** 1× 2007
- Issue from Consolidated Fund: none listed (needs **appropriation law**) 1× 2000
- Not charged: **Election Commissioners' pay** (per key) 1× 2004
- Not charged: **Vice-President's salary** (paid as RS Chairman) 1× 2006
- Not charged: **PM's salary** 1× 2011
- Contingency Fund: **Art 267**; advance before Parliament's approval 2× 2021, 1991

Charged on Consolidated Fund (not voted)	Voted
President's emoluments; RS Chairman, Speaker, Deputy Speaker	PM and ministers' salaries
SC judges' salary; HC judges' pension	Most ministries' grants
CAG, UPSC members; debt charges	MPs' salaries

21.10 Budget

- Not in the Constitution: **"Budget"** (it says annual financial statement, Art 112) 1× 2011
- Economic Survey: **before** the Budget 1× 2004
- Normal stages: presentation, discussion, **Appropriation Bill, Finance Bill** (not vote on account) 2× 2005, 2010
- Vote on account: withdraw money for part of the year; when the Budget will be late 2× 2004, 2008
- **Vote on account-sanction of all budget** is wrong 1× 2017

21.11 Limits and History

- Sovereignty limited by **judicial review** 1× 2004
- Golden Jubilee: **13 May 2002** 1× 2008
- First Indian Speaker of the Central Legislative Assembly (1925): **Vithalbhai Patel** 1× 2007

TRAP ALERT

- Art **108** joint sitting; **109** Money Bill procedure; **110** definition; **118(4)** Speaker presides.
- No joint sitting for **Money** or **Amendment** Bills.
- Every Money Bill is a Financial Bill, not the reverse.
- Demands for grants are voted only in the **Lok Sabha**.

UP LINK

The UP Legislature mirrors this: Money Bills only in the **Vidhan Sabha** (Art 198), its Speaker certifies, and the Vidhan Parishad has **14 days**. UP's budget is the largest of any State.

LAST-MINUTE 10

1. Parliament = **President + LS + RS**; VP not part
2. Meets **twice** a year; 60 days' absence vacates seat
3. Art **122** courts barred; **253** treaties; **249** RS resolution
4. Zero hour = India; up to **1 hour**
5. Cut motion = Budget; calling attention limits adjournment
6. Joint sitting **Art 108**; **Speaker** presides; first: Dowry Bill 1961
7. Money Bill **Art 110**; Speaker certifies; RS **14 days**
8. Art **265** tax by law; **267** Contingency Fund
9. "Budget" not in Constitution; Art 112
10. **Vithalbhai Patel** = first Indian Speaker (1925)

QUANTUM

Chapter 22

Parliamentary Committees

Built from 18 UPPSC questions (2002–2025). Focus: financial committees, Estimates Committee (LS only, largest), PAC and the CAG, JPC ratio.

ASKED MOST · REVISE FIRST

- ■ **Estimates Committee**: largest financial committee, **Lok Sabha only**, 1-year term **6×** 2008, 2010, 2010, 2016, 2017, 2020
- ■ **PAC**: 15 LS + 7 RS; chairman named by the **Speaker**; examines **CAG reports** **8×** 2004, 2006, 2013, 2014, 2014, 2021, 2022, 2025
- • Financial committees: **PAC, Estimates, Public Undertakings** **1×** 2007

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

22.1 Classification

- Financial committees: **PAC, Estimates, Public Undertakings** (all three) **1×** 2007
- Not a standing committee: none (all four were standing, *) **1×** 2015
- **PAC-financial, Petitions-functional, JPC on stock scam-ad hoc, DRSCs-standing** **1×** 2005
- No parliamentary committee on **welfare of minorities** **1×** 2012

22.2 Estimates Committee

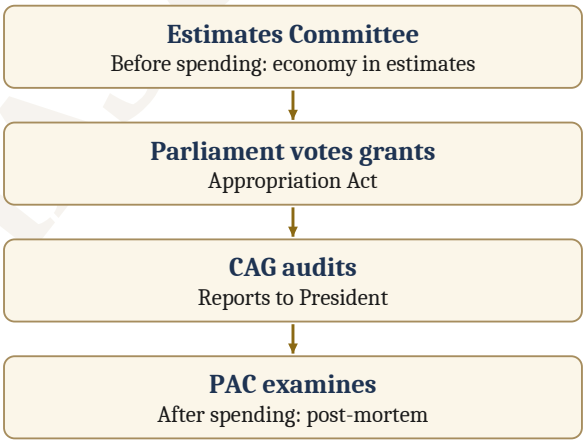
- Largest (among options): **Estimates Committee** (30 members) **2×** 2017, 2020
- Term: **one year** **1×** 2016
- ■ No RS member: **Estimates Committee**; drawn from **Lok Sabha** **3×** 2008, 2010, 2010

Committee	Members	Note
Estimates	30 (LS only)	"Continuous economy committee"; 1950
Public Accounts	22 (15 LS + 7 RS)	Oldest (1921); CAG = friend and guide
Public Undertakings	22 (15 LS + 7 RS)	1964 (Krishna Menon Committee)
DRSCs	31 each (21 LS + 10 RS)	24 committees; 1993

22.3 PAC

- PAC reports to the **Lok Sabha** (true); 15 RS members (false) (only 1) **1×** 2025
- PAC reports to **Parliament** **2×** 2002, 2015
- Members: **15 LS + 7 RS = 22** **1×** 2014
- ■ Chairman named by the **Speaker** **4×** 2004, 2006, 2013, 2014
- CAG audit report examined by the **PAC** (and COPU for PSUs) **1×** 2022
- PAC scrutinises the **CAG's report** **1×** 2021

MONEY TRAIL AND ITS WATCHDOGS



22.4 Joint Committees

- Standing committees: LS and RS members in a **2:1** ratio **1×** 2013
- 2G JPC: **20 LS + 10 RS**; headed by **P.C. Chacko** **2×** 2010, 2010

TRAP ALERT

- Estimates = **LS only**; PAC and COPU have **7 RS** members.
- PAC chairman is **appointed by the Speaker**; Opposition chair is a convention.
- A JPC is **ad hoc**; it ends after reporting.

UP LINK

The UP Legislature has its own **PAC, Estimates Committee and Public Undertakings Committee**; CAG's UP reports go to the UP PAC via the Governor.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Financial committees: **PAC, Estimates, COPU**
2. Estimates: **30**, LS only, 1 year, largest
3. PAC: **22** (15 + 7); oldest (1921)
4. PAC chairman: named by **Speaker**
5. PAC examines **CAG** reports; reports to LS
6. COPU: 22; public sector audit
7. DRSCs: 2:1 ratio (21 LS + 10 RS)
8. 2G JPC: 30 members, **P.C. Chacko**
9. No committee on minorities' welfare
10. JPC = ad hoc

Chapter 23

Parliamentary Acts

Built from 25 UPPSC questions (1991–2021). Focus: years of colonial Acts, RTI 2005, Code on Wages, Disaster Management Act and lockdown powers, environment Acts.

ASKED MOST · REVISE FIRST

- **RTI Act, 2005**: assent **15 June 2005**; a **legal right** (also read into Art 19(1)(a)); MKSS movement **5×** 2010, 2013, 2013, 2015, 2019
- **Disaster Management Act, 2005**: NDMA chaired by the **PM**; with Epidemic Diseases Act 1897 backed the lockdown **3×** 2020, 2021, 2021
- Act-year matching: Arms Act 1878, Royal Titles 1876, High Courts 1861, Divorce 1869, Criminal Tribes 1871 **3×** 2016, 2016, 2020

■ asked 3 or more times · asked once or twice **4×** times asked, followed by the exam years

23.1 Article 245

- Parliament may legislate for the **Union, territory and Indians and their property abroad** (all) **1×** 2021

23.2 Colonial Acts

- **Arms Act-1878, Royal Titles Act-1876, Indian High Courts Act-1861, Divorce Act-1869** **1×** 2020
- Criminal Tribes Act: **1871** **1×** 2016
- Wrong pair: Prevention of Seditious Meetings Act-**1908** (it was 1907) **1×** 2016

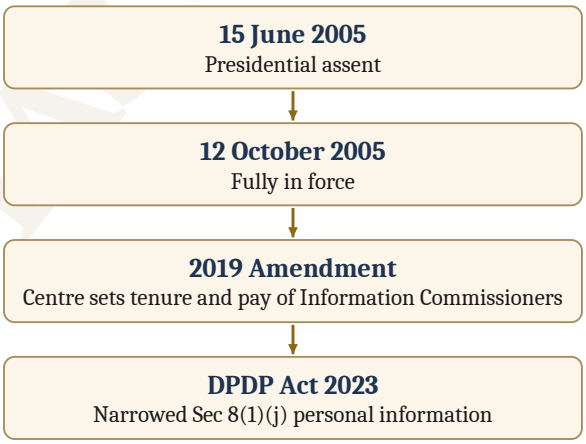
23.3 Social Laws

- Not a social Act: **MISA** (security law) **1×** 1991
- 73rd = rural panchayats; 74th = urban bodies (swap is the trap) **1×** 2009
- Domestic Violence Act in force: **26 October 2006** **1×** 2015

23.4 RTI: Nature

- Assent: **15 June 2005**; passed **2005** **2×** 2013, 2013
- RTI is a **legal right**; both legal and (implied) fundamental **2×** 2015, 2010
- **MKSS** (Aruna Roy): **RTI** movement **1×** 2019

RTI ACT TIMELINE



23.5 RTI: Provisions and Cases

- Not provided: applicant need **not give reasons** (Sec 6(2)) (*) **1×** 2006
- Applicant must give reasons (overturned view): **Madras High Court** **1×** 2014
- RTI case: **Namit Sharma v. Union of India** **1×** 2015
- RTI raised accountability; still far to go (both true, R not the explanation) **1×** 2010

23.6 Commercial Laws

- Wrong: **Indian Contract Act, 1870** (it is **1872**) **1×** 2021

Act	Year
Indian Contract Act	1872
Sale of Goods Act	1930
Partnership Act	1932

Act	Year
Companies Act	2013
Prevention of Food Adulteration	1954
Forest Conservation	1980
PPV and Farmers' Rights	2001
National Green Tribunal	2010

23.7 Code on Wages

- Merged in Code on Wages 2019: **Minimum Wages, Payment of Bonus, Equal Remuneration** (and Payment of Wages); not Contract Labour 1× 2021

23.8 Disaster Management

- Disaster Management Act: **2005** 1× 2021
- NDMA chaired by the **Prime Minister** 1× 2021
- Lockdown legal basis: **Disaster Management Act 2005 and Epidemic Diseases Act 1897** 1× 2020

23.9 Environment and Food

- **NGT**: **2010** 1× 2016
- PFA Act: **1954** 1× 2006
- Forest Conservation Act: **1980** 1× 2017
- PPV and FR Act: **2001** 1× 2021

TRAP ALERT

- RTI: assent **15 June**; in force **12 October** 2005.
- NDMA = **PM**, though MHA is the nodal ministry; SDMA = **CM**.
- Contract labour goes to the **OSH Code**, not Code on Wages.

UP LINK

UP has a **State Disaster Management Authority** chaired by the **Chief Minister**. The **UP Information Commission** (Lucknow) hears RTI appeals in the State.

LAST-MINUTE 10

1. Art **245**: laws for India, can act abroad
2. Arms 1878; Royal Titles 1876; HC Act 1861; Divorce 1869
3. Criminal Tribes **1871**; Contract Act **1872**
4. RTI: assent **15 June 2005**; MKSS; legal right
5. Domestic Violence Act in force **26 Oct 2006**
6. Code on Wages 2019: wages, bonus, equal pay
7. DM Act **2005**; NDMA = **PM**
8. Lockdown: DM Act + Epidemic Diseases Act **1897**
9. NGT 2010; Forest Conservation 1980
10. PFA 1954; PPV&FR 2001

Chapter 24

The Supreme Court

Built from 56 UPPSC questions (1990–2025). Focus: strength, collegium, removal, Art 129 Court of Record, jurisdictions, Arts 32/137/142/143, judicial review, basic structure, PIL.

ASKED MOST · REVISE FIRST

- **Basic structure**: **Kesavananda Bharati** (1973), a product of **judicial interpretation** 4× 2010, 2012, 2013, 2014
- **Guardian of the Constitution**: **Supreme Court**; judicial review with HCs too 4× 2015, 2015, 2017, 2019
- **Advisory opinion** (Art 143): sought only by the **President** 4× 1998, 2010, 2012, 2016
- Fundamental Rights enforced by **both SC (Art 32) and HCs (Art 226)** 5× 2004, 2004, 2006, 2010, 2023
- **Court of Record** (Art 129): power to punish for contempt; SC and HCs both 3× 2008, 2008, 2023
- **PIL**: judicial initiative; idea from the **USA**; core of judicial activism 5× 2004, 2008, 2011, 2014, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Strength (Art 124)

- Sanctioned strength changed: 26 (2004), 31 (2015), 34 (2019), 38 per the 2026 law cited in the masterbook (*) 2× 2004, 2015
- Strength fixed by **Parliament by law**, not by amendment

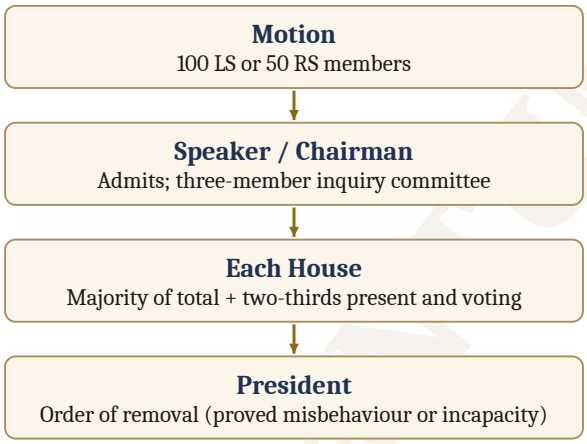
24.2 Appointment and Oath

- Collegium: CJI + **4** senior-most judges 1× 2010
- President's reference on appointments: **Art 143(1)** (Third Judges case, 1998) 1× 1998
- Executive, legislature and CJI in appointments: **2nd ARC** 1× 2014
- Oath quoted ("true faith and allegiance… uphold the Constitution"): **Chief Justice of India** 1× 2004

24.3 Tenure and Removal

- Resignation to the **President** 2× 2014, 2023
- Retirement age: **65** 1× 1990
- Salaries determined by **Parliament** 1× 2008
- Retired SC judges may practise in **no court** 1× 1997

REMOVING AN SC JUDGE (ART 124(4))



24.4 Acting CJI, Ad hoc, Seat, Benches

- Acting CJI appointed by the **President** 2× 2000, 2001
- **Ad hoc judges**: only for the **Supreme Court** (Art 127) 2× 2004, 2010
- Can sit elsewhere with the **President's approval** (Art 130) 1× 2008
- Constitutional cases: at least **five** judges 2× 2012, 2012
- English in SC proceedings: **Art 348** 1× 2013

24.5 Court of Record (Art 129)

- Court of Record: can **punish for its contempt** 1× 2008
- Wrong pair: **Art 124-Court of Record** (it is **Art 129**) 1× 2023
- Courts of Record: **High Courts and Supreme Court** 1× 2008

24.6 Jurisdiction

- Has **original, appellate and advisory** jurisdiction 2× 2010, 2013
- Not true: **exclusive** power to issue writs for FRs 1× 2017
- Appeal in constitutional cases: **Art 132 with 134A** 2× 2002, 2004
- Interpretation cases come under **appellate** jurisdiction 1× 2013
- TDSAT judgments challenged in **High Court and SC** 1× 2003

24.7 Writs (Art 32)

- FRs safeguarded under **Art 32** 1× 2010
- FR enforcement: **High Court and Supreme Court** 3× 2004, 2004, 2006
- SC issues five writs; Parliament may empower other courts (Art 32(3)) (both) 1× 2023

24.8 Advisory (Art 143)

- Advisory opinion sought by the **President** 2× 2010, 2012
- Advice to the President on law or fact of public importance 1× 2016

24.9 Review and Curative

- Curative petition rests on **Art 142** (per key) 1× 2014
- Review: **Art 137**; SC **can review** and **change** its judgment 3× 2009, 2011, 2005

24.10 Judicial Review

- Declares laws **unconstitutional** 1× 1994
- Enjoyed by **SC and High Courts** 1× 2017
- SC has **no exclusive** power over Central laws (Art 131A removed 1977); guardian (A false, R true) 1× 2019
- Judicial review in **India and USA** 1× 2013

- Review limited; "bag of borrowings" (both true, R not the explanation) 1× 2017
- Guardian of the Constitution: **Supreme Court** 2× 2015, 2015

24.11 Basic Structure and Benches

- Basic structure: **Kesavananda Bharati** 3× 2012, 2013, 2014
- Source: **judicial interpretation** 1× 2010
- Inviolable: **Arts 32, 226, 227** (judicial review) 1× 1999
- "FRs let a man chalk out his life": **Golaknath** 1× 2012
- Second largest bench (11): Golaknath, R.C. Cooper, T.M.A. Pai; Minerva was 5 (*) 1× 2004

Case	Year	Point
Golaknath	1967	11 judges; FRs not amendable
Kesavananda Bharati	1973	13 judges; basic structure
Minerva Mills	1980	Limited amending power
S.R. Bommai	1994	Secularism, Art 356 limits
Navtej Johar	2018	Section 377 read down

24.12 PIL

- PIL era CJI: Y.V. Chandrachud; architect **P.N. Bhagwati** (*) 1× 2018
- PIL introduced by **judicial initiative** 1× 2008
- PIL lets public-minded citizens act for those who cannot (R explains A) 1× 2004
- PIL "essential" for rule of law (false per source); access to justice (true) 1× 2025
- PIL originated in the **USA** 1× 2011
- Judicial activism relates to **PIL** 1× 2014

24.13 Landmark Judgments

- IMDT Act struck down under **Art 355** (Sarbananda Sonowal, 2005) 1× 2002
- Glorified women's status (Sept 2003): **Supreme Court** 1× 2009
- CBI a "**caged parrot**": **Coalgate** case 1× 2015
- Homosexuality ruling: **Section 377 IPC** 1× 2018

24.14 Judges in the News

- **Fathima Beevi**: first woman SC judge; first Muslim woman in higher judiciary (1 and 3); she was Governor of **Tamil Nadu** 1× 2024
- Fourth woman SC judge (April 2010): **Gyan Sudha Misra** 1× 2008
- Not earlier an HC judge (direct from Bar): **L. Nageswara Rao** 1× 2016

TRAP ALERT

- **Ad hoc** judges = SC (Art 127); **additional/acting** = HC (Art 224).
- Writs are **not exclusive** to the SC; Art 131 federal disputes are.
- Art 141 binds all courts **except** the SC.
- Retired HC judges can practise in SC and other HCs; retired SC judges nowhere.

UP LINK

UP's link: **Indira Gandhi v. Raj Narain** began in the **Allahabad High Court** (Justice Jagmohanlal Sinha, 1975) and reached the SC, which struck down part of the 39th Amendment, applying basic structure.

LAST-MINUTE 10

1. Basic structure = **Kesavananda** (13 judges), judicial interpretation
2. Ad hoc judges = **SC only** (Art 127)
3. Constitution Bench = **5** judges
4. Retirement **65**; resign to **President**; no practice after
5. Art **129** Court of Record; **137** review; **142** complete justice
6. Art **143** advisory: President only
7. Art **32** with HC's **226** for FRs
8. Collegium = CJI + **4**
9. PIL = **USA** idea; Bhagwati; judicial activism
10. Caged parrot = **Coalgate**; Fathima Beevi = first woman judge

Chapter 25

The Governor

Built from 26 UPPSC questions (1992–2024). Focus: appointment and pleasure tenure, qualifications, oath, discretion, Arts 200 and 213, immunity, first woman Governor.

ASKED MOST · REVISE FIRST

- ■ Governor holds office at the **pleasure of the President**; no impeachment **7×** 1992, 1997, 2004, 2004, 2004, 2009, 2010
- • Appointed by the **President** (Art 155) **2×** 2012, 2015
- • Oath given by the **Chief Justice of the High Court** **2×** 2013, 2014
- ■ Ordinance: **Art 213**; reserve Bill for President: **Art 200** **3×** 2004, 2016, 2017
- ■ One person can be Governor of **two or more States** (7th Amendment) **3×** 1996, 2016, 2024

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

25.1 Appointed, Not Elected

- Elected Governor dropped: **another election, rivalry with the CM, parliamentary system** (1, 3, 4) **2×** 2002, 2003

25.2 Article Map

- **Art 155-removal** is wrong (155 = appointment; tenure is Art 156) **1×** 2024
- Same person can be Governor of two States (A false); Art 153 says a Governor for each State (R true) **1×** 2024
- Two-State Governor's pay shared as the **President** decides **1×** 2016

25.3 Appointment and Qualifications

- Appointed by the **President**; under **Art 155** **2×** 2012, 2015
- True: **35 years**; can govern two or more States (2 and 4); no "by birth", no LS-eligibility test **1×** 1996
- Oath by the **High Court Chief Justice** **2×** 2013, 2014

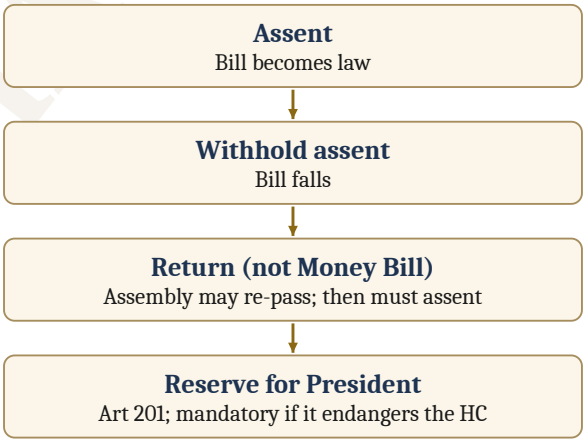
25.4 Term and Removal

- Appointed by President, at his pleasure, head of State executive, normally 5 years (all four) **2×** 2004, 2010
- Not true: removed by the **State Legislature** **2×** 1997, 2004
- Responsible to the **President** **1×** 1992
- In office as long as the President wishes: **Governor** **1×** 2004
- No impeachment provision: **Governor** **1×** 2009
- Rajasthan Governor dismissed (1981): **Raghukul Tilak** **1×** 2003

25.5 Discretion (Art 163)

- Acts on his own: **testing majority, dismissing CM, reserving a Bill, returning a Bill** (not HC consultation) **1×** 2006

GOVERNOR AND A STATE BILL (ART 200)



25.6 Bills and Ordinances

- Reserve a Bill for the President: **Art 200** **1×** 2004
- ■ Ordinance: **Art 213**; approved by the **State Legislature** **3×** 2016, 2017, 2008

25.7 Legislative and Judicial Powers

- True: does **not appoint HC judges**; **nominates MLCs** (1 and 3); he is part of the Legislature and has pardon powers **1×** 2017
- Not true: Governor can **pardon a death sentence** (only commute/suspend) **1×** 2005

25.8 Immunity (Art 361)

- "Cannot be sued for any act" false; President impeached, Governors dismissed (A false, R true)

QUANTUM IAS & PCS

1× 2004

- Not true: courts can inquire into **ministers' advice** to the Governor (barred by Art 163(3)) 1× 2016

25.9 J&K and UTs

- Sadar-e-Riyasat renamed Governor (1965): **6th Amendment of the J&K Constitution** 1× 2007
- UTs administered by the **President** 1× 1995

25.10 Firsts

- First woman Governor: **Sarojini Naidu** (UP, 1947) 1× 2013
- 13 February (her birthday) is **National Women's Day** 1× 2002

Governor	Article
Office	153
Executive power	154
Appointment / tenure	155 / 156
Qualifications / oath	157 / 159
Council of Ministers aid	163
Pardon	161
Assent / reservation	200 / 201
Ordinance	213
Immunity	361

TRAP ALERT

- President's oath from **CJI**; Governor's from the **HC Chief Justice**.
- Governor can't pardon a **death** sentence; President can (Art 72).
- **Art 200** = Governor reserves; **Art 201** = President decides.

UP LINK

Sarojini Naidu, India's first woman Governor, governed the **United Provinces** (1947-49) and died in office at Lucknow. The UP Governor is chancellor of State universities and nominates **10** UP MLCs.

LAST-MINUTE 10

1. Appointed by **President** (155); tenure at pleasure (156)
2. Qualifications: citizen, **35** years (157)
3. Oath by **HC Chief Justice** (159)
4. No impeachment; responsible to President
5. One Governor for two or more States (7th)
6. Pardon **161** (not death); ordinance **213**
7. Reserve Bill **200**; President decides **201**
8. Immunity **361**; advice not examined (163(3))
9. **Raghukul Tilak** dismissed (Rajasthan)
10. First woman Governor **Sarojini Naidu** (UP)

Chapter 26

The State Legislature

Built from 32 UPPSC questions (2003–2023). Focus: Legislative Councils (Art 169, 171), Assembly size (Art 170), Speaker, Money and ordinary Bills, Contingency Fund, Chief Minister.

ASKED MOST · REVISE FIRST

- ■ Council created or abolished by **Parliament** after the **Assembly's resolution** (Art 169) 4× 2008, 2007, 2018, one-liner
- ■ **Madhya Pradesh** and **Rajasthan** have **no** Legislative Council 4× 2006, 2008, 2008, 2013
- ■ Assembly strength **60 to 500** (Art 170); **Sikkim** an exception; **UP** largest 4× 2006, 2014, 2015, 2017
- • Council detains ordinary Bills: **3 months** first time, **4 months** in all 2× 2005, 2017
- • State CoM: minimum **12** 1× 2020

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

26.1 Creating a Council (Art 169)

- Created or abolished by **Parliament on the Assembly's resolution** 1× 2008
- Article: **169** 1× one-liner
- Abolished but not dissolved: **State Legislative Councils** 2× 2007, 2018

26.2 Bicameral States

- No Council: **Rajasthan** 1× 2013
- Not bicameral: **MP** 3× 2006, 2008, 2008
- 1956 bicameral States: 7 (per source; none of options) (*) 1× 2004
- Bicameral now (6): **UP, Bihar, Maharashtra, Karnataka, AP, Telangana**

26.3 Council Composition (Art 171)

- Not more than **one-third** of Assembly; at least **40** (both) 1× 2023
- Nominated share: "none of the above" per key (about 1/6 by Art 171) 1× 2017

LEGISLATIVE COUNCIL (ART 171)



26.4 Assembly (Art 170)

- Minimum and maximum: **60 and 500**; maximum **500** 2× 2017, 2014
- Exception to 60: **Sikkim** (32) 1× 2015
- Largest Assembly: **UP** (403) 1× 2006
- **Manipur-62** is wrong (it has **60**) 1× 2022

26.5 Elections, Speaker, Quorum

- Assembly elections conducted by the **Election Commission of India** 1× 2008
- Final authority on MLA disqualification (Art 192): **Governor** (on EC opinion) 1× 2016
- Speaker stays until **just before the new Assembly's first sitting** 1× 2004
- Quorum: **10 members or one-tenth**, whichever is more 1× 2021

26.6 Money and Ordinary Bills

- State Money Bill only on the **Governor's** recommendation 1× 2012
- Council may hold an Assembly Bill **3 months** first time 1× 2017
- Total detention of ordinary Bills: **4 months** (3 + 1) 1× 2005

26.7 State Contingency Fund

- Established by the **State Legislature** 1× 2014
- Operated by: none listed (at the Governor's disposal) (*) 1× 2004

26.8 State Executive and CM

- Art 162 heading: **extent of executive power of State** 1× 2019
- Not correct: Governor acts on CM's advice in **all** functions 1× 2010
- Not correct: CM is "normally selected" by the Governor 1× 2010
- CM's duties: **Art 167** 1× 2013
- First woman CM: **Sucheta Kripalani** (UP, 1963) 1× 2004

26.9 State Ministers

- Minimum CoM: **12** 1× 2020
- Ministers' pay fixed by the **State Legislature** 1× 2008
- J&K CM tenure: no fixed term (*) 1× 2008

26.10 Reservation and Civil Code

- Muslim quota proposed: **Andhra Pradesh** 1× 2003
- Only common civil code (2009): **Goa**; Uttarakhand UCC since 2025 1× 2009

Item	Assembly	Council
Size	60-500	Max 1/3 of Assembly, min 40
Term	5 years, dissolvable	Permanent; 1/3 retire every 2 years
Age	25	30
Money Bill	Starts here	14 days only
Ordinary Bill	Final say	Can delay 4 months

QUANTUM IAS & PCS

TRAP ALERT

- **MP** never got its Council; **J&K** Council ended in 2019.
- Council: **3 months** first time, **4 months** total; Money Bill **14 days**.
- MLA disqualification (non-defection) = **Governor** on EC's opinion; defection = **Speaker**.

UP LINK

UP is **bicameral**: Vidhan Sabha **403**, Vidhan Parishad **100** (10 nominated). **Sucheta Kripalani** became India's first woman CM in UP (1963). UP has the largest Assembly in India.

LAST-MINUTE 10

1. Council by **Parliament** after Assembly resolution (169)
2. No Council: **MP, Rajasthan**; six bicameral States
3. Council max **1/3** of Assembly, min **40**
4. Assembly **60-500**; **Sikkim 32**; UP **403**
5. ECI runs Assembly polls
6. Speaker until first sitting of the new House
7. Quorum: 10 or 1/10
8. Council delays **3 + 1 months**
9. CoM minimum **12**; Art **167** CM's duties
10. First woman CM **Sucheta Kripalani**

Chapter 27

The High Court and Subordinate Courts

Built from 13 UPPSC questions (1997–2016). Focus: HC judges' appointment and pension, number of High Courts, March 2013 courts, writs, Art 226 vs Art 32.

ASKED MOST · REVISE FIRST

- **Acting and additional judges** exist only in **High Courts** (Art 224) 1× 2016
- Retired HC judge's **pension**: **Consolidated Fund of India** (salary from the State fund) 1× 2013
- New High Courts in **March 2013**: **Meghalaya, Manipur, Tripura** 2× 2005, 2013
- **Certiorari** = transfer/quash a lower court's record; **Prohibition** = case pending 3× 2008, 2008, 2010
- **Art 226** is wider than **Art 32** (covers legal rights too) 2× 1997, 2004

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

27.1 Appointment and Tenure

- Not correct: the **Governor** appoints HC judges (the **President** does) 1× 2016
- Acting judges: **High Court** 1× 2016

27.2 Salary and Pension

- HC judge's pension: **Consolidated Fund of India** 1× 2013

27.3 Number and Benches

- High Courts: 21 in 2008; **25** now (*) 1× 2008
- No own HC in 2005: **Manipur** (got one in 2013) (*) 1× 2005
- Established March 2013: **Meghalaya and Tripura** (and Manipur) 1× 2013
- Most benches: **Bombay** now; Gauhati had six in 2007 1× 2007

High Court	Covers
Gauhati	Assam, Nagaland, Mizoram, Arunachal
Bombay	Maharashtra, Goa, Dadra-Nagar Haveli and Daman-Diu
Calcutta	West Bengal, Andaman and Nicobar
Kerala	Kerala, Lakshadweep
Madras	Tamil Nadu, Puducherry
Punjab and Haryana	Punjab, Haryana, Chandigarh

High Court	Covers
J&K and Ladakh	Both UTs

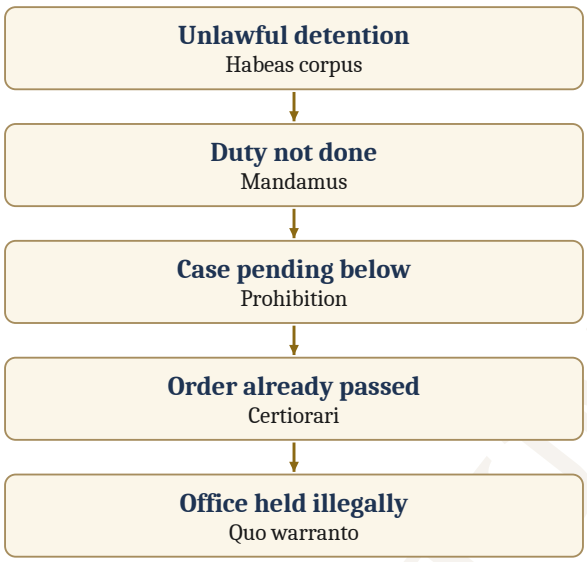
27.4 Jurisdiction

- Not a main HC jurisdiction: **advisory** 1× 2007

27.5 Writs

- Certiorari: superior court calls up the **record** from an inferior court 1× 2008
- Examines subordinate courts' work: **certiorari** 1× 2008
- While a case is **pending**: **prohibition** 1× 2010

WHICH WRIT, WHEN



27.6 Art 226 vs Art 32

- HC writ power covers FRs, legal rights and more (all) 1× 1997
- HCs better placed: SC's Art 32 covers only FRs (R explains A) 1× 2004

27.7 Subordinate Courts

- District judges appointed by the **Governor** in consultation with the HC (Art 233); HC controls them (Art 235)

TRAP ALERT

- **Governor** gives the HC judge's **oath** (Art 219), but the **President** appoints.
- Salary: **State** fund; pension: **Union** fund.
- Gauhati HC still serves **four** States.

UP LINK

The **Allahabad High Court** (est. Agra 1866, Allahabad 1869) is India's largest HC by sanctioned strength (160), with a **Lucknow Bench** (1948). UP district judges are appointed by the **UP Governor** with the HC.

LAST-MINUTE 10

1. President appoints HC judges; Governor gives oath
2. Acting and additional judges = **HC only** (Art 224)
3. Retirement **62**; resign to President
4. Pension from **Consolidated Fund of India**
5. 25 High Courts; **Manipur, Meghalaya, Tripura** in 2013
6. Gauhati HC = 4 States
7. Certiorari = record; Prohibition = pending
8. **Art 226** wider than Art 32
9. No advisory jurisdiction for HCs
10. District judges: **Governor** + HC (233)

Chapter 28

Centre-State Relations

Built from 33 UPPSC questions (1991–2025). Focus: Part XI and the Seventh Schedule, residuary power, Art 249, Inter-State Council, Sarkaria, Part XII taxes.

ASKED MOST · REVISE FIRST

- **Sarkaria Commission** = Centre-State relations; led to the **Inter-State Council** (1990) 9× 2003, 2004, 2006, 2008, 2014, 2016, 2017, 2018, 2021
- **Inter-State Council**: **Art 263**, set up by the **President** 3× 2000, 2009, 2017
- Legislative relations: **Part XI, Arts 245-246, Seventh Schedule** 5× 1993, 2001, 2004, 2008, 2014
- **Residuary powers**: **Parliament** 2× 1991, 1995
- Art 249 resolution lasts **one year** 3× 2009, 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

28.1 Basis

- Rests on **provisions, conventions, judicial rulings, dialogue** (all four) 1× 2009
- Federal polity: **Union-State, inter-State, cooperation, dispute settlement** (all) 1× 2010

28.2 Legislative Relations

- **Part XI** 2× 2004, 2014
- Three lists regulated by **Arts 245 and 246** 1× 2001
- **Seventh Schedule** 2× 1993, 2008
- Both can legislate: **Concurrent List** 1× 2006
- Economic and social planning, infants and minors: **both** Centre and States 1× 2019

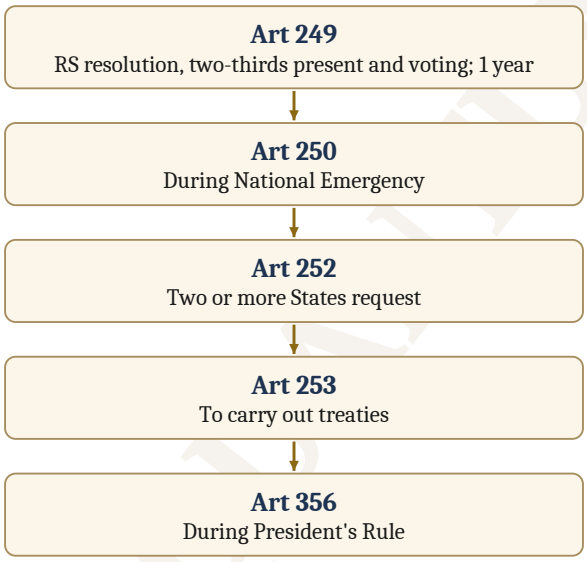
28.3 Residuary

- Residuary powers: **Parliament**; with the **Union** 2× 1991, 1995

28.4 Art 249

- Art 249: Parliament legislates on a **State List** subject in national interest 1× 2009
- Resolution in force up to **one year** (renewable) 2× 2016, 2017

PARLIAMENT ON THE STATE LIST



28.5 Administrative Relations

- States execute Union laws; Centre may give directions (Art 256) (R explains A) 1× 2016

28.6 Inter-State Council

- Inter-State relations: **Art 263** 1× 2009
- Source: the **Constitution** 1× 2000
- Set up by the **President** 1× 2017
- On recommendation of **Sarkaria Commission** 1× 2021

28.7 Zonal Councils

- Zonal councils formed by **law of Parliament** (SR Act 1956) 1× 2000
- Not a feature: being a **constitutional** body 1× 2017
- National integration has many dimensions (true); NIC set up in 1961 (key: false, 1962) 1× 2025

28.8 River Waters (Art 262)

- Wrong: **River Boards Act, 1970** (it is **1956**, like the Inter-State River Water Disputes Act) 1× 2004

28.9 Commissions

- Sarkaria reviewed **Union-State relations**; recommendations on Centre-State relations 7× 2006, 2008, 2016, 2003, 2004, 2014, 2018
- Member: **Dr. S.R. Sen** (with B. Sivaraman) 1× 2017
- Not on Centre-State: **Indrajit Gupta Committee** (state funding of elections) 1× 2015
- Demand for review; States lack resources (R explains A) 1× 2007

Body	Year	Note
Rajamannar Committee	1969	Tamil Nadu's own
Sarkaria Commission	1983-87	Led to Inter-State Council (1990)
Punchhi Commission	2007-10	Second Union commission
Zonal Councils	1956	Statutory (SR Act)
Inter-State Council	1990	Art 263; Presidential order

28.10 Financial Relations

- **Part XII**; **Arts 268-281** 2× 2019, 2014
- Union tax shared with States: all now (80th Amendment) (*) 1× 2006
- Stock-exchange transaction taxes: **levied by the Union** (only 1) 1× 2006

28.11 Borrowing and Disputes

- States **cannot** take foreign loans directly 1× 2012

- Fiscal disputes resolved by the **Supreme Court** (Art 131) 1× 1995
- Constitutional cap: **profession tax** (Rs 2,500) 2× 2002, 2004

28.12 Inter-State Trade

- Freedom of trade: **Art 301** 1× 1999

TRAP ALERT

- Inter-State Council: **in** the Constitution (Art 263) but **set up** by Presidential order; Zonal Councils are **statutory**.
- Art 249 needs only the **Rajya Sabha**.
- Profession tax ceiling **Rs 2,500** (60th Amendment).

UP LINK

UP sits in the **Central Zonal Council** (with Uttarakhand, MP, Chhattisgarh). Under the 16th Finance Commission UP gets the largest share of devolved taxes (17.62% per the masterbook).

LAST-MINUTE 10

1. Part **XI** legislative; Part **XII** financial
2. Arts **245-246**; Seventh Schedule
3. Residuary = **Parliament**
4. Art **249** = 1 year; RS only
5. Art **263** Inter-State Council (1990); President sets up
6. Zonal Councils = **SR Act 1956**
7. **Sarkaria** = Centre-State; member S.R. Sen
8. Indrajit Gupta = election funding
9. Art **301** free trade; States no foreign loans
10. Profession tax cap **Rs 2,500**

Chapter 29

Emergency Provisions

Built from 16 UPPSC questions (1992–2024). Focus: three emergencies, Art 352 grounds and approval, Arts 358-359, President's Rule, Art 360 never used.

ASKED MOST · REVISE FIRST

- **Three** types of emergency: Arts **352, 356, 360** 5× 2007, 2009, 2010, 2010, 2014
- Art 352 grounds: **war, external aggression, armed rebellion** (not internal disturbance since 44th) 3× 2016, 2016, 2017
- Parliament approves a National Emergency within **one month** 3× 2003, 2004, 2006
- **Art 360** (Financial Emergency) **never used** 3× 2006, 2015, 2016
- **Arts 20 and 21** cannot be suspended (44th Amendment) 2× 2009, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

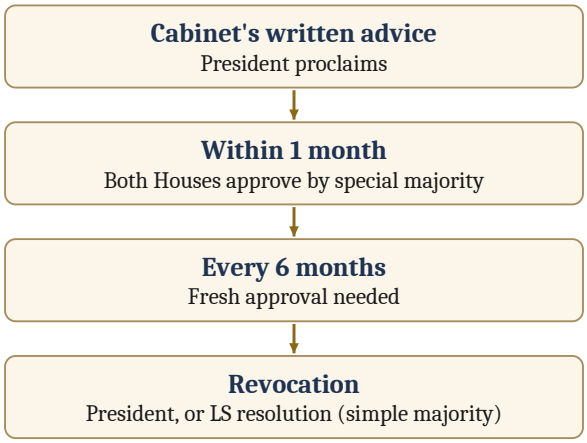
29.1 Three Types

- Types: **three** 2× 2009, 2014
- Duty to protect States from aggression and disturbance: **Art 355** 1× 2001
- Grounds for an emergency: **external aggression, breakdown in a State, financial crisis** (1, 3, 4) 3× 2007, 2010, 2010

29.2 National Emergency (Art 352)

- Art 352 deals with emergency; it can come any time (both true, R not the explanation) 1× 2024
- Grounds: **war, external aggression, armed rebellion** 3× 2016, 2016, 2017
- Parliament's approval: **one month** 3× 2003, 2004, 2006

NATIONAL EMERGENCY AFTER THE 44TH AMENDMENT



29.3 FRs in Emergency

- Suspend enforcement of FRs (except 20, 21): **Art 359** 1× 2008
- What the President can suspend: disputed (*); Art 21 can never be 1× 2015
- Cannot be suspended: **Arts 20 and 21** 2× 2009, 2015

29.4 Lok Sabha Term

- Can be extended **one year at a time** during Emergency (6 months after it ends) 1× 2010

29.5 President's Rule (Art 356)

- Maximum: **3 years** 1× 1992
- First use (among options): **PEPSU** (1953); Punjab 1951 was the very first 1× 2010
- Governor's Rule (old J&K Section 92): historical only (*) 1× 2015

Feature	Art 352	Art 356	Art 360
Approval	1 month	2 months	2 months
Majority	Special	Simple	Simple
Maximum	Indefinite (6-monthly)	3 years	Indefinite
Used	1962, 1971, 1975	100+ times	Never

29.6 Financial Emergency (Art 360)

- Never used: **Art 360**; declared **never** 3× 2015, 2016, 2006

29.7 Constituent Assembly

- "Emergency power is a fraud on the Constitution": **K.M. Nambiar** (per key) 1× 2004

TRAP ALERT

- "Internal disturbance" is in **Art 355**, not a ground under Art 352.
- **Art 358** suspends Art 19 only in **external** emergency; **Art 359** suspends enforcement.
- Art 352 approval **1 month**; Arts 356 and 360 **2 months**.

UP LINK

UP has had President's Rule **10 times** (first in 1968; last in 2002). The 1975 Emergency followed the **Allahabad High Court** verdict in *Indira Gandhi v. Raj Narain*.

LAST-MINUTE 10

1. Three emergencies: **352, 356, 360**
2. Art **355**: Union protects States
3. Grounds: war, external aggression, **armed rebellion**
4. Approval **1 month**; special majority
5. Art **358** suspends 19; Art **359** enforcement
6. **Arts 20, 21** never suspended
7. LS term extended **1 year** at a time
8. President's Rule max **3 years**; first **Punjab 1951** (PEPSU 1953)
9. Art **360** never used
10. Emergencies: 1962, 1971, 1975

Chapter 30

The Finance Commission

Built from 15 UPPSC questions (1993–2023). Focus: tax sharing, Art 280, five-year cycle, chairman plus four members, chairmen of the 13th and 14th, State Finance Commissions.

ASKED MOST · REVISE FIRST

- **Finance Commission** shares taxes between Centre and States 7× 1993, 2002, 2006, 2008, 2010, 2013, 2023
- Constituted **every 5 years** under **Art 280** by the **President** 4× 2008, 2010, 2011, 2021
- **Chairman + 4 members**; reports to the **President** 2× 2008, 2023
- **State Finance Commission** is a **constitutional** body 3× 2015, 2015, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

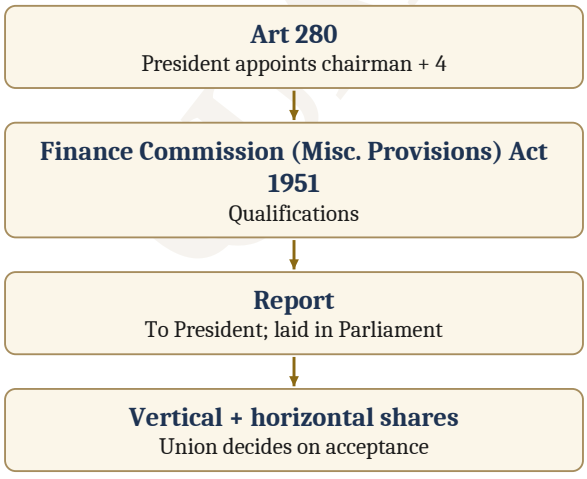
30.1 Functions

- Distributes taxes between Centre and States; finances shared on its advice 4× 2023, 2008, 2010, 2013
- Determines the States' share of taxes and grants 2× 1993, 2002
- Advises the **President** on dividing net tax proceeds 1× 2006

30.2 Constitution (Art 280)

- Every **5 years** 3× 2008, 2010, 2011
- Appointed under **Art 280** 1× 2021
- Chairman and six members; reports to NITI Aayog: **neither** true 1× 2023
- Chairman and **four** other members 1× 2008
- Qualifications set by **Parliament** (1951 Act), not the President (A false, R true) 1× 2021

HOW A FINANCE COMMISSION AWARD WORKS



30.3 Classification

- "Statutory" body (per key; strictly **constitutional**) 1× 2023

30.4 Chairmen

- 13th FC: **Vijay Kelkar** 1× 2012
- 14th FC: **Y.V. Reddy** (42% share) 1× 2016

FC	Chairman	States' share
1st (1951)	K.C. Neogy	-
13th	Vijay Kelkar	32%
14th	Y.V. Reddy	42%
15th	N.K. Singh	41%
16th (2026-31)	Arvind Panagariya	41%

30.5 State Finance Commissions

- SFC is constitutional (true); Union FC **can** recommend aid to Panchayats (R false) 1× 2015
- SFC is a **constitutional** body (Arts 243I, 243Y) 2× 2015, 2016

TRAP ALERT

- Report goes to the **President**, not NITI Aayog or Parliament directly.
- President appoints; **Parliament** fixes qualifications.
- FC is **constitutional**; Planning Commission and NITI are **executive** bodies.

UP LINK

UP is the largest gainer from FC awards: about **17.6%** of the States' pool under the 15th and 16th FCs. The **UP State Finance Commission** (Art 243I) shares State taxes with Panchayats and urban bodies.

LAST-MINUTE 10

1. FC: **Art 280**; every **5 years**
2. Chairman + **4** members; President appoints
3. Qualifications: **1951 Act** of Parliament
4. Report to **President**
5. Shares taxes and grants
6. 13th **Kelkar**; 14th **Y.V. Reddy** (42%)
7. 15th N.K. Singh; 16th **Panagariya** (41%)
8. First FC: **K.C. Neogy**
9. SFC: **Arts 243I, 243Y**; constitutional
10. UP: largest share

Chapter 31

The Planning Commission / NITI Aayog

Built from 17 UPPSC questions (1991–2023). Focus: extra-constitutional status, PM as chairman, NDC, NITI Aayog structure.

ASKED MOST · REVISE FIRST

- **Planning Commission** is **not** a constitutional body; set up by a **Cabinet resolution** (1950) **8×** 1999, 2003, 2004, 2008, 2010, 2013, 2013, 2014
- **NITI Aayog** = extra-constitutional; set up by Cabinet resolution (1 Jan 2015) **3×** 2015, 2016, 2018
- **NDC** (1952) approved Five-Year Plans; linked Commission and States **5×** 2002, 2005, 2010, 2012, 2016

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

31.1 Status

- Planning Commission set up by a **Cabinet resolution** **1×** 1999
- Not constitutional: **Planning Commission** **7×** 2003, 2004, 2008, 2010, 2013, 2013, 2014
- Extra-constitutional: **NITI Aayog** **2×** 2015, 2018

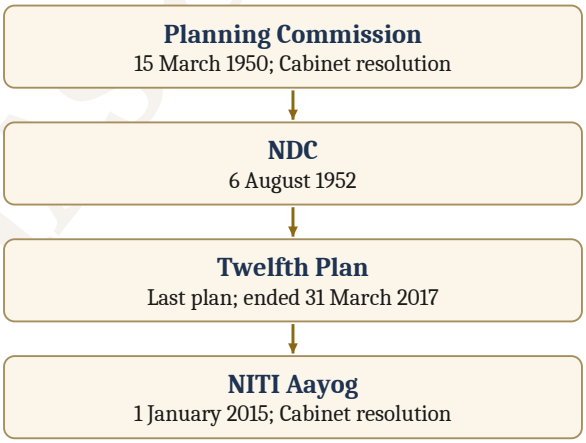
31.2 Planning Commission

- Chairman: **Prime Minister** **1×** 2006
- Deputy Chairman: rank of **Cabinet Minister** **2×** 1991, 1994
- Never Deputy Chairman: **P. Chidambaram** **1×** 2015
- "Economic cabinet" for Union and States; makes Plans (both true, R not the explanation) **1×** 2010

31.3 NDC

- Formed **6 August 1952** **1×** 2016
- Approves **Five-Year Plans** **1×** 2002
- Coordinator between the Commission and States: **NDC** **1×** 2012
- Not true: Planning Commission was the **highest** body for plans (NDC was) **1×** 2004
- NDC discusses progress, suggests ways, gives guidelines; **not** money allocation (1, 2, 3) **2×** 2005, 2010

FROM PLANNING TO NITI



31.4 NITI Aayog

- Planning Commission abolished by PM **Narendra Modi** **1×** 2015
- NITI set up by a **Union Cabinet resolution** **1×** 2016
- Not correct: NITI has a **full-time chairman** (PM is chairperson) **1×** 2015
- Not in Governing Council: **Governors** (CMs and LGs are) **1×** 2023

Body	Basis	Head
Planning Commission	Cabinet resolution 1950	PM
NDC	Cabinet resolution 1952	PM
NITI Aayog	Cabinet resolution 2015	PM; full-time Vice-Chairperson
Finance Commission	Art 280	Chairman appointed by President

TRAP ALERT

- "Economic and social planning" is a **Concurrent List** subject, but the bodies are **executive**.
- NITI has **full-time members** but no full-time chairman.
- Governing Council: **CMs and LGs**, not Governors.

UP LINK

UP's own **State Planning Institute** and the UP CM sits on NITI Aayog's **Governing Council**. UP launched its State-level think tank on NITI lines (State Transformation Commission).

LAST-MINUTE 10

1. Planning Commission: **15 March 1950**, Cabinet resolution
2. Chairman **PM**; Deputy Chairman = Cabinet rank
3. Not constitutional; extra-constitutional
4. NDC **6 Aug 1952**; approved plans
5. Allocation of plan money: Commission, not NDC
6. NITI Aayog **1 Jan 2015**; Cabinet resolution
7. PM chairs NITI; no full-time chairman
8. Governing Council: **CMs + LGs**
9. Last Plan: **Twelfth** (2012-17)
10. Chidambaram never Deputy Chairman

Chapter 32

Lokpal and Important Commissions

Built from 37 UPPSC questions (1991–2024). Focus: Lokpal and UP Lokayukta, NHRC, minorities bodies, CVC, Mandal, Venkatachaliah, UPSC and State PSCs.

ASKED MOST · REVISE FIRST

- ■ **Lokpal**: Indian ombudsman; first recommended by the **ARC (1966-67, Morarji Desai)**; first Bill **1968** 3× 1998, 2014, 2017
- ■ **UPSC/PSC**: constitutional; annual report to the **President**; State PSC member removed only by the **President** after SC inquiry 6× 2010, 2013, 2014, 2020, 2021, 2022
- ■ **NHRC** is **statutory** (1993), not constitutional 4× 2004, 2007, 2007, 2015
- • **Mandal Commission**: set up by **Morarji Desai** (1979), report **1980** 2× 1991, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

32.1 Lokpal

- Indian ombudsman: **Lokpal** 1× 2014
- ARC that first proposed Lokpal and Lokayukta: headed by **Morarji Desai** 1× 2017
- First Lokpal Bill: **1968** 1× 1998
- 2011 Bill: introduced 4 Aug 2011, sent to Standing Committee, called "weak" (1, 3, 4) 1× 2010
- Civil society drafters: **Anna Hazare, Santosh Hegde, Shanti Bhushan** 1× 2010
- Selection committee: **PM, CJI (or nominee), Speaker, LoP**, eminent jurist; not RS Chairman 1× 2024

32.2 Lokayukta

- First Lokayukta: **Maharashtra** (1971) 1× 2015
- First to pass a Lokayukta Bill in 2011: **Uttarakhand** 1× 2012
- UP Lokayukta reports to the **Governor** 1× 2007
- UP Lokayukta tenure: **5 years** (or 70) per current law; 8 years in 2017 1× 2017

32.3 Minorities

- "Minority" undefined; Minority Commission not constitutional (both true, R not the explanation) 1× 1991
- Special Officer for Linguistic Minorities: one for India (not each State), **Art 350B** by the **7th** Amendment (neither) 1× 2024

32.4 Human Rights

- Human rights stress the **dignity of man** 1× 2014
- Second-generation rights: **right to work** and education 1× 2010
- NHRC head need not be ex-CJI; term 3 years or 70 since 2019 (both false now) 1× 2015
- NHRC Chairperson: former **CJI or SC judge** (2019 law) 1× 2007
- Not constitutional: **NHRC**; status different from the others: **NHRC** (statutory) 2× 2007, 2004

Body	Basis
UPSC, State PSCs, FC, EC, CAG	Constitutional
NCSC (338), NCST (338A), NCBC (338B)	Constitutional
Linguistic minorities officer	Constitutional (350B)
NHRC, NCM, NCW, CVC, Lokpal	Statutory
CBI, NITI Aayog	Executive

32.5 Vigilance and Years

- CVC in no Article (**statutory**, 2003 Act) 1× 2020
- Wrong pairs: Prevention of Corruption Act (**1988**), Enforcement Directorate (**1956**) 1× 2017
- Politician-criminal nexus: **Vohra Committee** (1993) 1× 1995
- Wrong: **NDC-1954** (it is **1952**) 1× 2016

32.6 Review and Mandal

- Constitution review commission: recommendatory, chaired by **M.N. Venkatachaliah** (1 and 3) 2× 2002, 2003
- Chairman (2000): **M.N. Venkatachaliah** 1× 2002
- Mandal set up by **Morarji Desai**; report **1980** 2× 2008, 1991

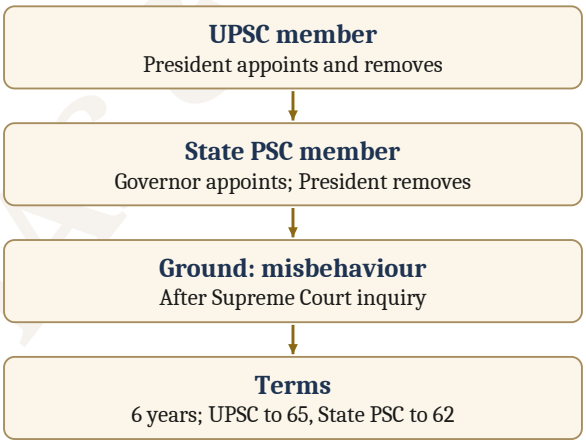
32.7 UPSC

- UPSC: **constitutional** body 1× 2014
- First PSC provided by the **GoI Act, 1919** (set up 1926) 1× 2008
- UPSC functions come from the **Constitution, laws, rules and conventions** (all) 1× 2024
- Annual report to the **President** 2× 2010, 2013
- P.C. Hota Committee on civil services: **2004** 1× 2021

32.8 PSCs (Arts 315-323)

- State PSC member removed by the **President after SC inquiry** 1× 2021
- Parliament plays **no** role in removing the **UPSC Chairman** (only 3) 1× 2020
- Joint PSC: by **Parliament** on States' request 1× 2022
- UPSC can serve a State on the **President's** approval 1× 2014
- UPPSC functions extended by the **UP Legislature** (Art 321) 1× 2021
- UPPSC expenses charged on the **State Consolidated Fund** 1× 2010

WHO APPOINTS, WHO REMOVES (PSCS)



TRAP ALERT

- RS Chairman is **not** on the Lokpal panel.
- Linguistic minorities officer: **Art 350B** (7th Amendment), one for India.
- Mandal: **set up** by Morarji Desai, **implemented** by V.P. Singh (1990).
- State PSC member: Governor appoints, **President removes**.

UP LINK

UPPSC (Prayagraj, 1937) is a State PSC: its members are appointed by the UP Governor, removed by the President; its expenses are charged on UP's Consolidated Fund. The **UP Lokayukta** (1975 Act) reports to the Governor.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Lokpal: ARC (**Morarji Desai**); first Bill **1968**; Act 2013
2. Lokpal panel: PM, CJI, Speaker, LoP, jurist
3. First Lokayukta **Maharashtra** (1971); UP Lokayukta → Governor
4. NHRC, CVC = **statutory**; UPSC = constitutional
5. NHRC head: ex-CJI or SC judge (2019)
6. **Vohra** = crime-politics nexus
7. Mandal: Morarji set up; report **1980**
8. **Venkatachaliah** = Constitution review (2000)
9. First PSC: **1919 Act**, set up 1926
10. State PSC: President removes after SC inquiry

Chapter 33

Temporary Special Provisions

Built from 9 UPPSC questions (1997–2020). Focus: Art 370, Arts 371-371J state map, J&K constitutional history.

ASKED MOST · REVISE FIRST

- **Art 371**: special provisions for **Maharashtra and Gujarat**; series 371A-371J for other States **2×** 1997, 2011
- ■ **Art 370**: **Jammu and Kashmir**; made inoperative in August 2019 **3×** 2010, 2014, 2016
- **371B** = Assam; **371J** = Hyderabad-Karnataka region **2×** 2010, 2020

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

33.1 Article Map

- Special provisions for States: **Art 371** **1×** 1997
- **Art 370**: **Jammu and Kashmir** **1×** 2010

33.2 Arts 371-371J

- Art 371: **Maharashtra and Gujarat** **1×** 2011
- Art 371-B: **Assam** **1×** 2010
- Art 371-J: **Hyderabad-Karnataka** region **1×** 2020

Article	State
371	Maharashtra, Gujarat
371A	Nagaland
371B	Assam
371C	Manipur
371D, 371E	Andhra Pradesh (Telangana)
371F	Sikkim
371G	Mizoram
371H	Arunachal Pradesh
371I	Goa
371J	Karnataka (Hyderabad-Karnataka)

33.3 Art 370 and Its End

- State with its own Constitution in 2014: **J&K** (none now) (*) **1×** 2014
- Articles applying to J&K on their own: **Arts 1 and 370** (then) (*) **1×** 2016

J&K TIMELINE

26 Oct 1947
Instrument of Accession (Hari Singh)

↓

26 Jan 1950
Constitution of India in force

↓

1954
Accession ratified; Presidential Order; Art 35A

↓

26 Jan 1957
J&K Constitution in force

↓

1965
Sadr-e-Riyasat renamed Governor

↓

5 Aug 2019
Art 370 inoperative (C.O. 272, 273)

↓

31 Oct 2019
UTs of J&K and Ladakh

↓

Dec 2023
SC upholds abrogation

33.4 J&K History

- Order: **Instrument of Accession, Constitution of India, ratification by J&K Assembly, J&K Constitution** 1× 2019
- Sadr-e-Riyasat became Governor in **1965** (none of options) (*) 1× 1997

TRAP ALERT

- **371A** = Nagaland; **371B** = Assam (do not swap).
- Art 370 is **inoperative**, not deleted; Art 35A was never in the main text.
- Sadr-e-Riyasat change = **1965**, not 1952.

UP LINK

No UP item. Related: Article 371 provisions never covered UP; but **Uttarakhand** (carved from UP) has no 371 clause either; its hill areas are protected by State law.

LAST-MINUTE 10

1. Art **370** = J&K; inoperative Aug **2019**
2. Art **371** = Maharashtra, Gujarat
3. **371A** Nagaland; **371B** Assam; **371C** Manipur
4. **371D** Andhra/Telangana; **371F** Sikkim
5. **371G** Mizoram; **371H** Arunachal; **371I** Goa
6. **371J** Hyderabad-Karnataka
7. Accession **26 Oct 1947**
8. J&K Constitution **1957**
9. Sadr-e-Riyasat to Governor **1965**
10. UTs from **31 Oct 2019**

Chapter 34

Election Commission

Built from 29 UPPSC questions (1991–2025). Focus: Art 324, three-member Commission, CEC tenure and removal, functions, voting age, MCC, electoral reforms.

ASKED MOST · REVISE FIRST

- Election Commission: **Art 324** 3× 2010, 2016, 2017
- Voting age cut from 21 to 18 by the **61st Amendment (1988)**; first used in **1989** 6× 2000, 2004, 2007, 2008, 2010, 2011
- Right to vote is a **legal right**; adult suffrage in **Art 326** 4× 2010, 2013, 2015, 2017
- CEC removed **like an SC judge**; ECs on the CEC's recommendation 4× 1991, 2002, 2006, 2014
- MP disqualification (non-defection): **President** on **EC's opinion** 2× 2009, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

34.1 Art 324

- Election Commission under **Art 324** 3× 2010, 2016, 2017

34.2 Composition and Removal

- Made three-member in **1989** (permanent from 1993) 1× 2006
- CEC tenure: **6 years or 65** 1× 2012
- CEC's status: like an SC judge (no option right, *) 1× 2016
- Independent EC (true); executive removes ECs (false) 1× 2006
- CEC removed like an **SC judge** 2× 1991, 2002
- EC removed by the **President on CEC's recommendation** 1× 2014

APPOINTING THE CEC AND ECS (2023 ACT)



34.3 Functions

- Functions: **conduct Parliament, Assembly, President, VP polls; electoral rolls** (I, II, IV); not recommending President's Rule 1× 2001
- Not EC's: **making laws on elections** (Parliament does) 1× 2009
- President's election conducted by the **Election Commission** 1× one-liner
- EC does **none** of: Speaker's election, municipal polls, their disputes 1× 2024
- Not EC's: **settling election validity** (courts do) 1× 1992
- MP disqualification: **President in consultation with EC**; EC's opinion binding 2× 2009, 2015

34.4 Model Code of Conduct

- MCC applies from announcement (true); **not** enacted by Parliament (false) 1× 2017

34.5 Voting

- Right to vote and be elected: **legal right** 3× 2010, 2013, 2015
- Adult suffrage: **Art 326** 1× 2017
- Assembly elections: adult suffrage, one roll per seat, no religion bar, parties free (all four) 1× 2011
- First general election with 18-year voters: **1989** 1× 2011
- 21 to 18: **61st Amendment, 1988** 4× 2004, 2007, 2008, 2010
- Voting age **18** (1989 law) 1× 2000

34.6 Electors and Years

- Electors in 2009: **714 million** 1× 2004
- Wrong: **Seventh Lok Sabha-1982** (it was **1980**) 1× 2022

34.7 Proportional Representation

- PR helps minorities; represents all groups (R explains A) 1× 2013
- PR ensures **minority representation** 2× 2013, 2013

34.8 Reforms

- Order: **EVM (1982), EPIC (1993), VVPAT (2013), NOTA (2013)** 1× 2025

Reform	Year
EVM first used	1982 (Paravur, Kerala)
EPIC (voter ID)	1993
EVM in all LS seats	2004
VVPAT first used	2013 (Noksen, Nagaland)
NOTA	2013 (PUCL case)
National Voters' Day	25 January (from 2011)

34.9 Firsts

- **Sukumar Sen** first CEC; **V.S. Ramadevi** first woman CEC (both) 1× 2021
- National Voters' Day: **25 January** 1× 2015

TRAP ALERT

- **MCC** has no statutory backing.
- Defection = Speaker/Chairman; other disqualification = **President on EC's opinion**.
- Local body polls = **State Election Commission** (243K, 243ZA).

UP LINK

UP's Panchayat and municipal polls are run by the **UP State Election Commission**, not the ECI. In 2025 the ECI extended the **Special Intensive Revision** of rolls to UP and other States.

LAST-MINUTE 10

1. EC: **Art 324**; multi-member since **1989**
2. CEC: 6 years or **65**; removed like SC judge
3. 2023 Act: PM, LoP, Cabinet Minister select
4. Adult suffrage **Art 326**; vote = legal right
5. **61st** (1988): voting age 18; first in **1989**
6. MCC: not a law
7. MP disqualification: President + EC opinion
8. EVM 1982; EPIC 1993; VVPAT, NOTA 2013
9. First CEC **Sukumar Sen**; woman CEC **Ramadevi**
10. National Voters' Day **25 Jan**

Chapter 35

Political Parties

Built from 21 UPPSC questions (1997–2023). Focus: national and state party recognition, founding years, trade-union links, anti-defection law.

ASKED MOST · REVISE FIRST

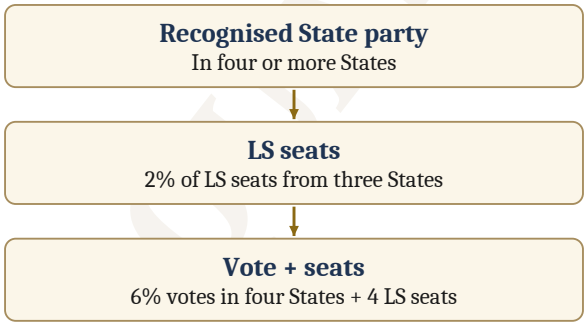
- ■ **Anti-defection** (52nd Amendment, Tenth Schedule): passed **15 February 1985**; parties first got constitutional recognition in **1985** 3× 2008, 2010, 2010
- **National party**: recognised as a State party in **four** States 1× 2000
- State party: **8%** votes alone, or 6% with seats 1× 2014
- **CPI** lost national status in **2023** (now a State party) 2× 2008, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

35.1 Recognition

- National party: State party in at least **four** States 1× 2000
- State party: **8%** vote alone (key's 6% needs seats) 1× 2014
- Regional party: **Akali Dal** (CPI also now) 1× 2008
- Not a national party: **TDP** (and CPI now) 1× 2009

BECOMING A NATIONAL PARTY (ANY ONE)



35.2 Founding

- "National" in INC reflected **reaction against British rule** 1× 2009
- **Independent Labour Party**: **B.R. Ambedkar** (1936) 1× 2009
- Order: **TDP (1982), BSP (1984), SP (1992), TMC (1998)** 1× 2022
- **CPI-1920 (Tashkent), CPM-1964, AIADMK-1972, TDP-1982** 1× 2011

35.3 Trade Unions

- **BMS-BJP, INTUC-Congress, AITUC-CPI**; UTUC is RSP's (CITU is CPI(M)'s) (*) 1× 2022

Union	Party
INTUC	Congress
AITUC	CPI
CITU	CPI(M)
BMS	BJP (Sangh Parivar)
HMS	Socialists
UTUC	RSP

35.4 Elections

- 1999 seats: **TDP-30** wrong (29) 1× 2000
- Lok Sabha 2024: **7 phases** 1× 2023

35.5 Party History

- First communist government: **Kerala, 1957** (E.M.S. Namboodiripad) 2× 2014, one-liner
- **Kamaraj Plan** (1963): **revitalise the Congress** 1× 1997

35.6 Party System

- Coalition agenda gives guidance; but not in detail (both true, R not the explanation) 1× 1998
- ■ No fixed party system; many parties (both true, R not the explanation) 3× 2000, 2004, 2004
- Written Constitution; regional parties show aspirations (both true, R not the explanation) 1× 2015
- Inner-party democracy: **periodic internal elections** 1× 2004

35.7 Anti-Defection

- Bill passed: **15 February 1985** 1× 2008
- Not covered: **large defections in spells** 1× 2010
- Parties got constitutional recognition: **1985** 1× 2010

TRAP ALERT

- **CITU** = CPI(M); **UTUC** = RSP.
- 6% vote alone never gives State party status.
- **Kihoto Hollohan** (1992) struck down only Para 7; Speaker's decision is reviewable.

UP LINK

UP-born parties: **Samajwadi Party** (Mulayam Singh Yadav, 1992) and **BSP** (Kanshi Ram, 1984; national party). CPI was founded at **Kanpur** in 1925.

LAST-MINUTE 10

1. National party: State party in **4** States
2. State party: **8%** votes, or 6% + seats
3. CPI lost national status **2023**
4. Independent Labour Party: **Ambedkar**
5. CPI 1920 (Tashkent)/1925 (Kanpur); CPM **1964**
6. TDP 1982, BSP 1984, SP 1992, TMC 1998
7. First communist govt: **Kerala 1957**
8. Kamaraj Plan **1963**: revive Congress
9. Anti-defection: **15 Feb 1985**, 52nd
10. CITU = CPI(M); AITUC = CPI

Chapter 36

Constitutional Amendment

Built from 37 UPPSC questions (1990–2024). Focus: Art 368 procedure, types of amendment, basic-structure cases, landmark amendments and their numbers.

ASKED MOST · REVISE FIRST

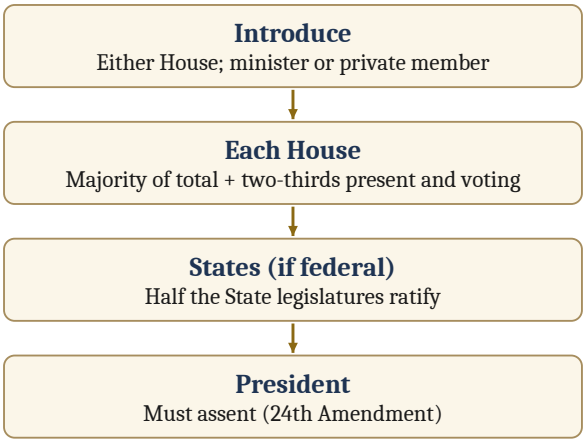
- Amendment under **Art 368**; introduced in **either House**, never by State legislatures **5×** 2008, 2013, 2013, 2015, 2024
- **Basic structure**: **Kesavananda Bharati (1973)**; **Golaknath** first limited the amending power **5×** 2002, 2005, 2016, 2020, 2022
- **52nd** = anti-defection; **61st** = voting age 18; **69th** = Delhi NCT **9×** 1992, 1996, 2006, 2008, 2009, 2014, 2015, 2021, 2022
- **86th** = Art 21A education; **91st** = 15% ministry cap; **97th** = co-operatives **6×** 2004, 2006, 2015, 2016, 2017, 2022

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

36.1 Art 368 Procedure

- Parliament can amend any part under Art 368 (key: R explains A) **1×** 2015
- Special majority in each House separately: **Constitution Amendment Bill** **1×** 2007
- Provision: **Art 368** **1×** 2008
- Initiated by **Lok Sabha or Rajya Sabha** (1 and 2); in **either House** **3×** 2024, 2013, 2013
- RS rejects an amendment Bill: the Bill **is dropped** (no joint sitting) **1×** 2004

PASSING A CONSTITUTION AMENDMENT BILL



36.2 Types and Ratification

- Half the States must ratify: **federal provisions, Supreme Court, Art 368 itself** (1, 2, 3); not new States **1×** 2004

QUANTUM IAS & PCS

Method	Examples
Simple majority (outside Art 368)	New States (Arts 2-4), citizenship, Rajya Sabha seats, quorum
Special majority	FRs, DPSP, most provisions
Special majority + half the States	President's election, SC and HCs, Seventh Schedule, Art 368, representation in Parliament

36.3 FRs and Basic Structure

- Order: **Shankari Prasad, Sajjan Singh, Golaknath, Kesavananda** 1× 2024
- Order: **Golaknath (1967), 24th (1971), Kesavananda (1973), 42nd (1976)** 1× 2021
- First limited the amending power: **Golaknath** 1× 2002
- Kesavananda: SC set out the **basic features** 1× 2005
- Basic structure: **Kesavananda Bharati** 3× 2016, 2020, 2022
- Competent to amend FRs: **Parliament** 1× 2016
- Laws for Art 39(b), (c) saved from Arts 14, 19: **25th Amendment** (Art 31C) 1× 2009

36.4 Master Table

- **42nd-1976, 52nd-1985, 86th-2002, 96th-2011** 1× 2024

36.5 First and Sikkim

- 1st Amendment (1951): **protected agrarian reforms** (Ninth Schedule) 1× 2009
- Sikkim a State: **36th Amendment** 1× 2006
- Sikkim: joined 1975; botanist's paradise (I and II); Lepchas are a minority 1× 2004

36.6 1970s

- **38th-curtailed courts' power (Emergency), 44th-property right abolished, 61st-voting age, 42nd-"socialist" added** 1× 2017
- Fundamental Duties: **42nd** 1× 2003

36.7 1985-1999

- Banned floor-crossing: **52nd**; 52nd = **defection** 4× 1992, 1996, 2008, 2006
- Hindi authoritative text: **58th** (1987) 1× 2016
- Voting age 18: **61st** 2× 2014, 2021
- Delhi NCT: **69th** 3× 2009, 2015, 2022
- SC/ST seats extended: **79th** 2× 2008, 2013

36.8 From 2002

- Art 21A: **86th**; education an FR by the 86th (2002) 2× 2006, 2022
- 15% ministry cap: **91st** 3× 2004, 2015, 2017
- "Co-operative societies" in Art 19(1)(c): **97th** (2011) 1× 2016
- Co-operative board: up to **21** directors (Art 243ZJ) 1× 2016
- **91st-Art 164, 92nd-Eighth Schedule, 95th-Art 334, 103rd-Art 15** 1× 2016

Amendment	Change
1st (1951)	Ninth Schedule; Art 15(4)
7th (1956)	States reorganised; Art 350A, 350B
24th (1971)	Parliament can amend FRs; assent compulsory
42nd (1976)	Socialist, Secular, Integrity; Duties
44th (1978)	Property out; "armed rebellion"
52nd (1985)	Anti-defection
61st (1988)	Voting age 18
73rd, 74th (1992)	Panchayats, Municipalities
86th (2002)	Art 21A, 45, 51A(k)
101st (2016)	GST
103rd (2019)	10% EWS quota
106th (2023)	One-third seats for women

36.9 Count

- Amendment Acts implemented up to 2013: **98** 1× 2014

TRAP ALERT

- No **joint sitting** and no **veto** for Amendment Bills.
- Ratification (half the States) is different from Art 3 consultation.
- **25th** created Art 31C for 39(b), (c); **42nd** widened it.
- **35th** = associate state; **36th** = Sikkim full State.

QUANTUM IAS & PCS

UP LINK

The **73rd and 74th** Amendments shaped UP's 3-tier Panchayats and 17 Municipal Corporations. The **Indira Gandhi v. Raj Narain** case (from Allahabad HC) led to the 39th Amendment, partly struck down.

LAST-MINUTE 10

1. Art **368**; either House; no State initiative
2. Special majority; half the States for federal parts
3. Golaknath 1967 → 24th 1971 → **Kesavananda 1973** → 42nd 1976
4. **25th** = Art 31C; **1st** = land reforms
5. **36th** Sikkim; **42nd** Duties; **44th** property out
6. **52nd** defection; **58th** Hindi text; **61st** age 18
7. **69th** Delhi; **79th** SC/ST seats extended
8. **86th** Art 21A; **91st** 15% cap; **97th** co-operatives
9. Co-op board max **21** directors
10. 98 Acts by 2013; **106th** latest (women)

Chapter 37

Official Language

Built from 13 UPPSC questions (1990–2025). Focus: English for 15 years, Kher Commission, Art 350A, Eighth Schedule languages, State official languages.

ASKED MOST · REVISE FIRST

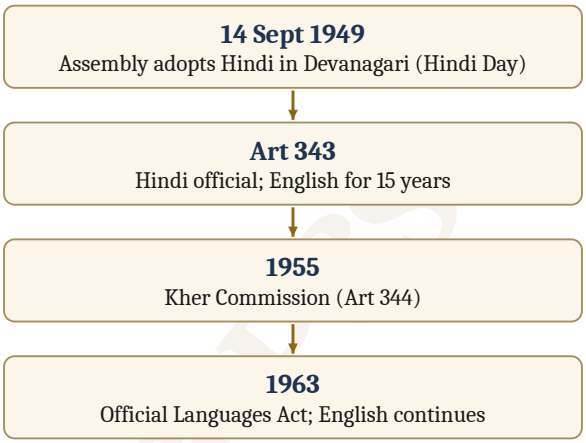
- ■ **22** languages in the **Eighth Schedule**; Bhojpuri and Rajasthani are **not** 3× 1993, 2010, 2025
- English allowed for **15 years** after 1950 (Art 343) 1× 2023
- **Art 350A**: primary education in the **mother tongue** 2× 2002, 2003
- First Official Language Commission: **1955**, **B.G. Kher** 1× 1998

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

37.1 Union Language

- English permitted for all official purposes for **15 years** 1× 2023
- Amending language provisions: special majority (Art 368) for Articles; simple majority for laws under Art 343(3) (*) 2× 2005, 2018

HINDI AS OFFICIAL LANGUAGE



37.2 Commission (Art 344)

- First Commission: **1955**, chaired by **B.G. Kher** 1× 1998

37.3 No National Language

- Correct: **Bodo** is in the Eighth Schedule 1× 2016
- Mother-tongue instruction at primary level: **Art 350A** 2× 2002, 2003

37.4 Eighth Schedule

- Not included: **Bhojpuri** (Nepali, Dogri, Bodo are in) 1× 2025
- In 1993 none of Bodo, Dogri, Maithili, Santhali was in (all added 2003) (*) 1× 2021
- Added by amendment: **Sindhi** (1967) and **Konkani** (1992) 1× 1990
- Not included: **Rajasthani** 1× 1993
- Number: **22** 1× 2010

Amendment	Languages added
21st (1967)	Sindhi
71st (1992)	Konkani, Manipuri, Nepali
92nd (2003)	Bodo, Dogri, Maithili, Santhali
Total now	22 (original 14)

37.5 State Languages (Art 345)

- Sanskrit as second official language: **Uttarakhand** (2010) 1× 2011

37.6 Speakers

- Most spoken Scheduled language (after Hindi): **Bengali** 1× 1998
- Order (2011 Census): **Bengali, Marathi, Telugu, Tamil** 1× 2001

TRAP ALERT

- India has **no national language**; Hindi is the official language of the **Union**.
- Union numerals are **international** form (1, 2, 3), not Devanagari.
- **Pali and Prakrit** are classical languages but **not** Scheduled.

UP LINK

UP's official language is **Hindi** with **Urdu** as the second official language (1989). Bhojpuri and Awadhi, widely spoken in UP, are **not** in the Eighth Schedule.

LAST-MINUTE 10

1. Union language: **Hindi in Devanagari** (Art 343)
2. English for **15 years**; continued by 1963 Act
3. **Kher** Commission 1955 (Art 344)
4. Hindi Day **14 September**
5. **Art 350A** mother tongue; **350B** linguistic minorities officer
6. **22** Scheduled languages
7. Not Scheduled: **Bhojpuri, Rajasthani**
8. 92nd (2003): Bodo, Dogri, Maithili, Santhali
9. **Uttarakhand**: Sanskrit second language
10. Bengali second most spoken

Chapter 38

Panchayati Raj and Community Development

Built from 83 UPPSC questions (1991–2025). Focus: Balwant Rai Mehta and Nagaur, later committees, 73rd and 74th Amendments, Gram Sabha, reservation, State Election and Finance Commissions, UP bodies.

ASKED MOST · REVISE FIRST

- **Balwant Rai Mehta** (1957) = architect; **three-tier** system 7× 2006, 2010, 2012, 2013, 2015, 2016, 2017
- First State: **Rajasthan**; launched at **Nagaur**, **2 October 1959** by Nehru 6× 1991, 1994, 2003, 2010, 2012, 2015
- **73rd Amendment** = constitutional status; in force **1993** (24 April) 7× 2004, 2012, 2015, 2019, 2020, 2022, 2024
- **State Election Commission** conducts Panchayat polls; appointed by the **Governor** 6× 2008, 2013, 2015, 2020, 2023, 2023
- Women: at least **one-third** seats (Art 243D) 5× 2010, 2012, 2016, 2017, 2025
- Committees order: **B.R. Mehta, Ashok Mehta, G.V.K. Rao, L.M. Singhvi** 3× 2015, 2017, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

38.1 Art 40 and Community Development

- Art 40: **organise village panchayats** 1× 2015
- First community programme (among options): **National Extension Service** (1953); CDP itself began 1952 1× 2013

38.2 The Idea

- Grassroots democracy: **Panchayati Raj** 1× 2013
- PRI: local self-government, three tiers, decentralisation (1, 2, 3) 1× 1998
- Adopted to **decentralise** power 1× 2016
- Purpose: people's **participation in development** 1× 2004

- Not an attribute of local government: **central control** 1× 2008
- Name "Panchayati Raj" suggested by **Nehru** 1× 2017

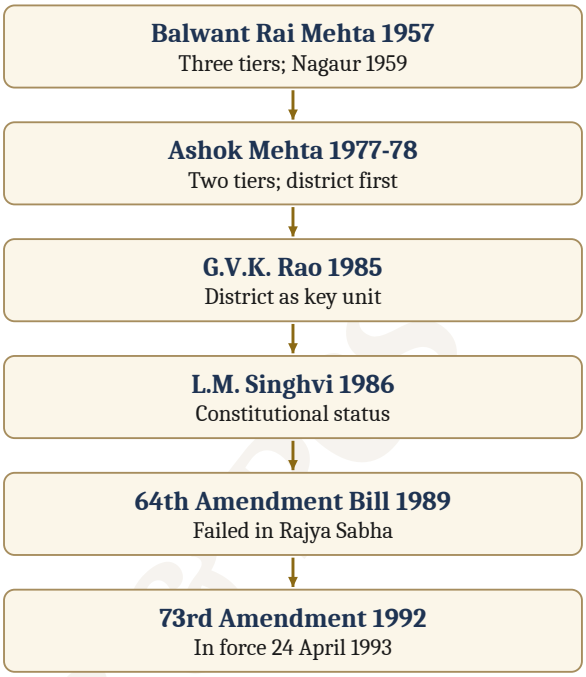
38.3 Balwant Rai Mehta and Nagaur

- Architect: **B.R. Mehta**; recommended three tiers 3× 2015, 2016, 2017
- Chaired the committee on **democratic decentralisation** 2× 2012, 2013
- Recommended Panchayati Raj in its **1957** report 2× 2006, 2010
- Not a B.R. Mehta tier: **Nyaya Panchayat** 1× 2024
- First State: **Rajasthan** 4× 2003, 2010, 2012, 2015
- Inaugurated **2 October 1959** at **Nagaur** 2× 1991, 1994

38.4 Later Committees

- Ashok Mehta: **two-tier**; replace three tiers with two (true); block as first point (false; it said **district**) 2× 2009, 2023
- Constitutional status suggested by **L.M. Singhvi** 1× 2008
- **B.R. Mehta-three-tier, Ashok Mehta-two-tier, Singhvi-constitutional local self-government, G.V.K. Rao-improving rural administration** 1× 2013
- Order: **B.R. Mehta, Ashok Mehta, G.V.K. Rao, Singhvi** 3× 2015, 2017, 2021
- Committee match: code A-4, B-3, C-1, D-2 (key) 2× 2017, 2022
- Not on Panchayati Raj: **P.V.N. Rao Committee**; **V.K.R.V. Rao** 2× 2014, 2008
- Wrong: **G.V.K. Rao-block-level planning** (that was **Dantwala**) 1× 2024

PANCHAYATI RAJ COMMITTEES



38.5 73rd and 74th Amendments

- Constitutional status: **73rd**; 73rd = Panchayati Raj 2× 2015, 2004
- Local government: **73rd and 74th** 1× 2024
- Third tier added: **1993** 2× 2020, one-liner
- 73rd a watershed because it gave constitutional status (R explains A) 1× 2022
- Less political interference; 73rd revived PRIs (R explains A) 1× 1998
- 73rd recognised **Panchayats only** (only 1) 1× 2017
- Part IX-A: **urban** areas only 1× 2021
- Wrong: 73rd in force **15 August 1993** (it was **24 April 1993**) 1× 2012
- National Panchayati Raj Day: **24 April** 1× 2019

38.6 Part IX Architecture

- Three tiers: **Part IX** 1× 2013
- Composition fixed by the **State Legislature** 1× 2016
- Not correct: every State must have all three tiers 2× 2013, 2019
- Intermediate tier optional below **20 lakh** population 1× 2017
- 73rd covers **Zila, Kshetra, Gram Panchayat** (not Nagar Panchayat) 2× 2010, 2013
- Functions in the **Eleventh Schedule**; **29** subjects 2× 2006, 2009

38.7 Gram Sabha

- Gram Sabha: persons on the **electoral roll** of the village 1× 2015
- Not true: its powers set by the **Central Government** (State law does) 1× 2023

38.8 Duration

- Five years from its **first meeting** 1× 2013
- Election after early dissolution: within **6 months** 1× one-liner
- Not on Panchayats: the **74th Amendment** 1× 2010

38.9 Reservation

- Women's one-third: **Art 243D** 1× 2010
- 243D(3) is true; not tied to Art 334 (only 1) 1× 2017
- "30%" in no amendment; 73rd gives one-third (*) 1× 2016
- UP women's seats: **one-third** 1× 2012
- 5-year tenure (true); 1/4 for SC/ST women (false) (only 1) 1× 2025
- Reservation brought change (true); long demand (false) 2× 2002, 2003
- No SC reservation in Panchayats: **Arunachal Pradesh** 1× 2016

38.10 State Election Commission

- Control of Panchayat polls: **State Election Commission**; conducts them 2× 2020, 2023
- SEC appointed by the **Governor** 1× 2013
- Decides to hold polls: **SEC** 1× 2008
- Panchayati Raj is in the **State List** 2× 2015, 2016
- SEC constitutional (true); ECI runs rural polls (false) 1× 2015
- Correct statement: Panchayat polls under State law and SEC 1× 2023

38.11 Finances

- State Finance Commission constituted by the **Governor** 1× 2015
- Revenue sharing with local bodies: **State Finance Commission** 1× 2013
- Union FC has **no** role (false); local government a State subject (true) 1× 2011
- Constitutional: **SEC, SFC, District Panchayat** (1, 2, 3) 2× 2012, 2012

38.12 Exempted Areas

- No PRIs: **Nagaland** 2× 2005, 2011
- **PESA**: **1996** 1× 2022
- Nyaya Panchayats imprisonment: Bihar intended; none now (*) 1× 2005

38.13 UP Bodies

- Panchayat Samiti members **directly elected** 1× 2013
- Panchayat Samiti at **block** level 1× 2015
- UP Zila Panchayat chairman elected **by members from among themselves** 1× 2010

UP tier	Level	Head
Gram Panchayat	Village	Pradhan (direct)
Kshetra Panchayat	Block	Pramukh (indirect)
Zila Panchayat	District	Adhyaksh (indirect)
Nagar Nigam	Big city	Mayor (direct, 5 years)
Nagar Palika Parishad	Smaller city	Chairman (direct)
Nagar Panchayat	Transitional area	Chairman (direct)

38.14 74th Amendment

- Municipalities: **Part IX-A** 1× 2015
- Constitutional status: **74th Amendment, 1992** 1× 2013
- 74th: new Schedule, restructures municipalities, women's reservation (1, 2, 3) 1× 2019
- Municipal area notified by the **Governor** 1× 2015
- Member age 21 (25 false); one-third seats for women (only 2) 1× 2025

38.15 UP Urban Bodies

- UP municipal chairman: elected by **voters on the list**, from **ward electors** (2 and 4) 1× 2010
- Mayor's term: **5 years** 1× 1992
- Wrong: **Ghaziabad-Nagar Palika Parishad** (it is a **Nagar Nigam**) 1× 2010

QUANTUM IAS & PCS

- Census urban test: **75%** male main workers non-farm, not 60% 1× 2010

38.16 District Planning

- 73rd/74th created **SEC, DPC, SFC** (1, 2, 3) 1× 2010
- UP district plan includes **Nagar, Gram, Kshetra Panchayats** 1× 2016

TRAP ALERT

- Gandhi = idea; **B.R. Mehta** = architect; **Nehru** = name and Nagaur launch.
- 2 Oct **1952** = CDP; 2 Oct **1959** = Nagaur.
- Acts are of **1992**; in force **1993** (73rd: 24 April; 74th: 1 June).
- **State Election Commission** (243K) runs local polls, not the ECI.

UP LINK

UP follows a 3-tier system under the **UP Panchayat Raj Act 1947** and **Kshettra and Zila Panchayat Act 1961**: Gram Panchayat (Pradhan), Kshetra Panchayat (Pramukh), Zila Panchayat (Adhyaksh). **Ghaziabad** is a Nagar Nigam; UP Mayors are directly elected for 5 years.

LAST-MINUTE 10

1. Art **40**; CDP 1952; NES 1953
2. **B.R. Mehta** 1957: three tiers; Nagaur **2 Oct 1959**
3. Ashok Mehta: two tiers; **Singhvi**: constitutional status
4. **73rd**: Part IX, 11th Schedule (**29**); 24 April 1993
5. **74th**: Part IX-A, 12th Schedule (18); 1 June 1993
6. Women **1/3** (243D); intermediate tier optional below **20 lakh**
7. **SEC** (243K) by Governor; **SFC** (243I)
8. PESA **1996**; Nagaland no PRIs
9. Panchayati Raj Day **24 April**
10. Gram Sabha = village electors

Chapter 39

Special Provisions Relating to Certain Classes

Built from 8 UPPSC questions (1991–2023). Focus: Part XVI articles, SC/ST seats, Anglo-Indian lapse, National Commissions, who notifies SCs and Scheduled Areas.

ASKED MOST · REVISE FIRST

- **Anglo-Indian** nomination to Lok Sabha: **Art 331**; lapsed **25 January 2020** 5× 2010, 2013, 2015, 2015, 2017
- SC/ST seats in Lok Sabha: **Art 330**; no reservation in **Rajya Sabha** or Councils 2× 1998, 2010
- **President** notifies SCs (Art 341) and Scheduled Areas 3× 2006, 2016, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

39.1 Part XVI Map

- Part XVI (Arts 330-342): **reservation for SCs, STs and Anglo-Indians** 1× 2004

Article	Provision
330 / 332	SC-ST seats in Lok Sabha / Assemblies
330A, 332A, 334A	Women's one-third (106th, 2023)
331 / 333	Anglo-Indians in LS / Assemblies (lapsed 2020)
334	Time limit (SC-ST seats till 2030)
335	SC-ST claims to services
338 / 338A / 338B	NCSC / NCST (89th) / NCBC (102nd)
340	Backward classes commission (Kalelkar, Mandal)
341 / 342 / 342A	President notifies SCs / STs / SEBCs

39.2 Seats

- ST seats in the Lok Sabha: **Art 330** 1× 2010

- No caste reservation: **Rajya Sabha** and **Legislative Councils** 1× 1998

39.3 Anglo-Indians

- Anglo-Indian LS nomination: **Art 331** (lapsed 2020) (*) 5× 2010, 2013, 2015, 2015, 2017

39.4 Commissions

- National Commissions for SCs and STs: **Arts 338 and 338A** 2× 2010, 2013

COMMISSIONS FOR CLASSES

NCSC (Art 338)
Joint commission split in 2003

↓

NCST (Art 338A)
89th Amendment, 2003

↓

NCBC (Art 338B)
Statutory 1993; constitutional 2018 (102nd)

39.5 Who Decides

- SC list specified by the **President** (Art 341) 2× 2006, 2016
- Most STs by number: **Madhya Pradesh** 1× 1991

39.6 Scheduled Areas

- President notifies Scheduled Areas; States with them need a **Tribes Advisory Council** (both) 1× 2023

TRAP ALERT

- **Rajya Sabha** has no SC/ST or women's reservation.
- Art 331 is not deleted; it **lapsed** via Art 334 (104th Amendment).
- TAC = **Fifth** Schedule; District Councils = **Sixth** Schedule.

UP LINK

UP has **17 SC-reserved Lok Sabha seats** and the largest SC population of any State. UP's STs are few (Tharu, Buksa, Gond and others); Sonbhadra has the State's highest ST share.

LAST-MINUTE 10

1. Part XVI: Arts **330-342A**
2. **330** LS seats; **332** Assembly seats
3. **331** Anglo-Indians; lapsed **25 Jan 2020**
4. **334**: SC/ST seats till 2030
5. **338** NCSC; **338A** NCST; **338B** NCBC (2018)
6. **341** President notifies SCs; **342** STs
7. No reservation in RS or Councils
8. Most STs by number: **MP**
9. TAC with Fifth Schedule
10. Women's quota: **330A, 332A** (106th)

Chapter 40

Miscellaneous

Built from 46 UPPSC questions (1990–2025). Focus: UN facts, world leaders, India's foreign policy, mixed constitutional statements, press, security bodies.

ASKED MOST · REVISE FIRST

- **UN**: ICJ at **The Hague**; Hindi is **not** a UN official language; Amnesty is **not** a UN body 4× 1994, 2002, 2003, 2004
- India's foreign policy: **Non-Alignment** by **Nehru**; Commonwealth "free will" link 2× 2014, 2017
- **Press Council** = statutory quasi-judicial body; press = **Fourth Estate** 2× 1995, 2004
- Security bodies: **BHARATPOL (CBI)**, **Five Eyes**, **Djibouti Code** (piracy) 3× 2021, 2023, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

40.1 United Nations

- Not a UN body: **Amnesty International** 1× 2004
- ICJ: **The Hague** 1× 1994
- Not a UN language: **Hindi** 2× 2002, 2003
- Non-member in 1996: **Switzerland** (joined 2002) (*) 1× 1996
- Newest country: **South Sudan** (2011) 1× 2013
- Not elected to UNSC for 2021-22: **Canada** 1× 2020

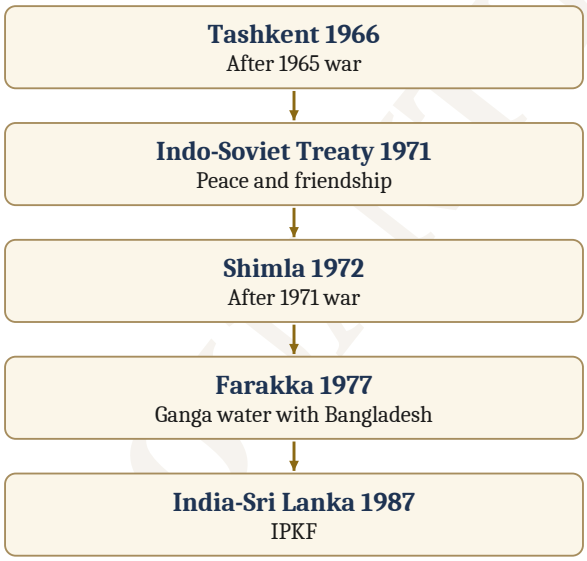
40.2 World Politics

- Wrong pair: **Bahasa-Thailand** (it is Indonesia/Malaysia; Thai in Thailand) 1× 2021
- Pakistan more ethnically diverse (true); Punjabi official language (false; Urdu and English) 1× 2016
- "**Pivot to Asia**": **USA** 1× 2015
- First woman PM: **Sirimavo Bandaranaike** (Sri Lanka, 1960) 1× 2013
- Countries with compulsory voting: no fixed count (*) 1× 2016

40.3 India's Foreign Policy

- Non-Alignment: **Nehru** 1× 2014
- "Agreement by free will, ended by free will": **Commonwealth** 1× 2017
- First PM to visit Afghanistan: **Nehru** 1× 2005
- Six-point formula for South Asia: **Vajpayee** 1× 2021
- Order: **Tashkent (1966), Indo-Soviet Treaty (1971), Shimla (1972), Farakka (1977)** 1× 2002
- Not in the India-Sri Lanka Accord: **complete freedom** 1× 1990
- First Festival of India abroad: **Britain, 1982** 1× 1991
- Pravasi Bharatiya Divas began **2003** (9 January) 1× 2014

INDIA'S KEY ACCORDS



40.4 Mixed Constitutional Statements

- HC judges to 62, national party in four States, Speaker certifies Money Bill, cognizable arrest without warrant: **all true** 1× 2002
- Art 21 not suspended, VP removal by RS + LS (only 1 and 2 now; Jaitley was RS LoP in 2010) 1× 2010
- Wrong: **Madam Cama** first woman INC president (it was **Annie Besant**) 1× 2013
- Two wrong: Eighth Schedule has 22; 1858 Proclamation = Magna Carta (*) 1× 2013
- Extra-constitutional: power **not in accordance with** the Constitution 2× 2002, 2002
- **21-Panchayat member, 25-MLA, 30-MLC, 35-President** 1× 2021
- Wrong: Constitution Day 27 Jan (it is **26 Nov**); Human Rights Day 29 Jan (it is **10 Dec**) 1× 2025

Minimum age	Office
18	Voter
21	Panchayat, municipality member
25	MP (LS), MLA
30	MP (RS), MLC
35	President, VP, Governor

40.5 Press

- Press Council: **statutory quasi-judicial** body 1× 2004
- Chairman in 2010: **G.N. Ray** (none of options) (*) 1× 2010
- Fourth Estate: **Press** 1× 1995
- "Bahujan Hitaya Bahujan Sukhaya": **All India Radio** 1× 2002

40.6 Political History

- Most coalition governments 1967-71: **Bihar** 1× 2013
- Causes of regionalism: **all** listed 1× 2008
- OBC status to Gurkhas (2003): **Uttarakhand** 1× 2010
- Wrong: **Silent Valley-Baba Amte** (Amte = Narmada; Silent Valley = Kerala movement) 1× 2009
- Anti-war organisation chairman (1988): **Narayan Desai** 1× 1994

40.7 Security

- **ULFA**: **Assam** 1× 1993
- Lashkar-e-Taiba founder: **Hafiz Saeed** 1× 2004

- LTTE banned by **India, Sri Lanka** (USA, later EU) 1× 1998
- BHARATPOL: built by the **CBI** 1× 2025
- Five Eyes alliance (true); Stone Ghost network (true, R not the explanation) 1× 2023
- **Djibouti Code**: anti-**piracy** maritime grouping 1× 2021
- Internal Security Academy: **Mount Abu** 1× 2016

40.8 Ministries and Agencies

- **Adi Karmayogi** (June 2025): **Ministry of Tribal Affairs** 1× 2025
- Not an HRD department (2006): **Women and Child Development** 1× 2006
- National Water Development Agency: **1982** 1× 1999
- First national law university: **NLSIU Bengaluru, 1986** (*) 1× 1990

TRAP ALERT

- **ICJ** (UN organ) vs **ICC** (Rome Statute), both at The Hague.
- 26 Nov = Constitution Day; 25 Jan = Voters' Day; 26 Jan = Republic Day.
- "At present" statements about office-holders change with time.

UP LINK

UP links: the **Sarnath** Lion Capital (emblem), **Pravasi Bharatiya Divas** 2019 held in **Varanasi**, and **Annie Besant** (first woman INC president, 1917) worked at Varanasi's Central Hindu College.

LAST-MINUTE 10

1. ICJ = **The Hague**; Hindi not a UN language
2. Switzerland joined UN **2002**; South Sudan **2011**
3. First woman PM = **Sirimavo Bandaranaike**
4. NAM = **Nehru**; six-point SAARC formula = **Vajpayee**
5. Tashkent, Indo-Soviet, Shimla, Farakka
6. PBD from **2003**, 9 January
7. Ages: 21 Panchayat, 25 MLA, 30 MLC, 35 President
8. Press Council: statutory, quasi-judicial
9. BHARATPOL = **CBI**; Djibouti = piracy
10. Internal Security Academy = **Mount Abu**

QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Indian Geography

PYQ-Based Revision Pointers

Chapters 1 to 26

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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QUANTUM IAS & PCS

Chapter 1

India: Location, Extent and Standard Time

Built from 26 UPPSC questions (1990–2022). Focus: Tropic of Cancer, IST and time sums, border states, boundary lines.

ASKED MOST · REVISE FIRST

- **Tropic of Cancer** crosses **8 states**: Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram; **not UP, not Manipur** 3× 1990, 2008, 2017
- Town **nearest the Tropic**: **Gandhinagar** (Jabalpur when Gandhinagar is absent) 4× 1999, 2002, 2010, 2017
- **IST = 82°30' E**; passes through **Mirzapur** near Prayagraj; 5 states (UP, MP, Chhattisgarh, Odisha, AP); not Maharashtra 3× 2006, 2010, 2013
- **Time sums**: 1° = 4 minutes, 15° = 1 hour; **east is ahead** 4× 1992, 2001, 2002, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

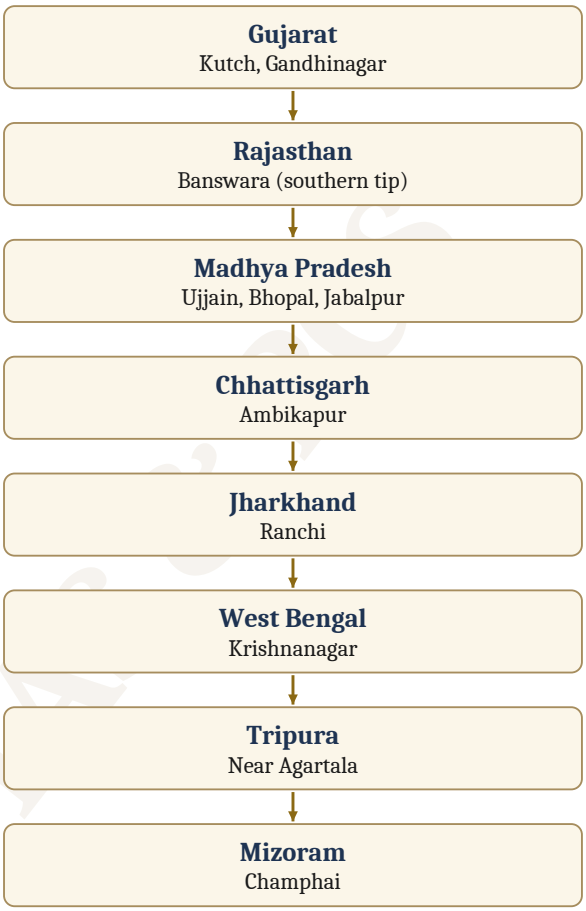
1.1 Size and Extent

- India is the **7th** largest country (not 5th or 6th) and has about **2.4%** of world land 2× 2006, 2022
- **Tropic of Cancer** divides India into two nearly equal halves; India is **not wholly** in the torrid zone 2× 2006, 2022
- India is now the **most populous** country (passed China, April 2023) 1× 2016
- Mainland: **8°4' N to 37°6' N, 68°7' E to 97°25' E**; area **32.87 lakh sq km**

1.2 The Tropic of Cancer

- States crossed: **Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram** (west to east) 2× 1990, 2017
- **Manipur** lies wholly **north** of the Tropic 1× 2008
- Of Tripura, Mizoram, WB and Manipur, the Tropic does **not** cross **Manipur** 1× one-liner
- Nearest to the Tropic (23°26' N): **Gandhinagar**; next option set: **Jabalpur** 4× 1999, 2002, 2010, 2017
- **Mahi** is the river that crosses the Tropic **twice** 1× 2016

TROPIC OF CANCER, WEST TO EAST (8 STATES)



TRAP ALERT

- **UP never touches the Tropic** (its southern tip is about 23°52' N). Strike out UP first.
- "Nearest town" answers change with the option set: Gandhinagar first, then Jabalpur.

1.3 Longitude and Indian Standard Time

- IST meridian is **82°30' E** (GMT + 5:30) 1× 2013
- It does **not** pass through **Maharashtra** 1× 2010
- IST is taken near **Prayagraj (Mirzapur)** 1× one-liner
- Gujarat to Arunachal (Walong): about 29° of longitude = about **2 hours** 1× 1992
- Noon at IST, local time at 120° E: 37°30' × 4 = 150 min ahead = **14:30** 1× 2001
- Midnight at IST, 6 a.m. at a place: 6 h = 90° east, so **172°30' E** 1× 2002

QUANTUM IAS & PCS

- Sunrise 5 a.m. at **Tirap**: Kandla sees it about **7 a.m.** 1× 2010

HOW TO SOLVE ANY TIME QUESTION

Find the gap
Subtract the two longitudes

↓

Convert
1° = 4 minutes; 15° = 1 hour

↓

Decide direction
East = ahead (add); West = behind (subtract)

↓

Check IST
82°30' ÷ 15 = 5.5 h, so GMT + 5:30

1.4 Extreme Points

- Southernmost point of **India**: **Indira Point**, **Great Nicobar** 2× 2002, one-liner
- Southernmost point of the **mainland**: **Kanyakumari**

Direction	Point	State / UT
North	Indira Col (Siachen)	Ladakh
South	Indira Point	Andaman and Nicobar
South (mainland)	Kanyakumari	Tamil Nadu
East	Kibithu / Dong	Arunachal Pradesh
West	Guhar Moti (Sir Creek)	Gujarat

1.5 Neighbours and Border States

- **Manipur** does **not** border Bangladesh 2× 2002, 2004
- **Nagaland** does **not** border Bangladesh 1× one-liner
- **Assam** does **not** border Myanmar 1× 2012
- Border with Pakistan: **Gujarat, Rajasthan, Punjab, J&K** (now UTs J&K and Ladakh) 2× 1990, 2001
- **Tripura** has an international border on **three sides** (Bangladesh) 1× 2008

Neighbour	Border (km)	Indian states / UTs
Bangladesh	4,097 (longest)	WB, Assam, Meghalaya, Tripura, Mizoram
China	3,488	Ladakh, HP, Uttarakhand, Sikkim, Arunachal
Pakistan	3,323	Gujarat, Rajasthan, Punjab, J&K, Ladakh
Nepal	1,751	Uttarakhand, **UP**, Bihar, WB, Sikkim
Myanmar	1,643	Arunachal, Nagaland, Manipur, Mizoram
Bhutan	699	Sikkim, WB, Assam, Arunachal
Afghanistan	106 (shortest)	Ladakh (PoK area)

UP LINK

UP borders only **Nepal**, through 7 districts: Pilibhit, Lakhimpur Kheri, Bahraich, Shravasti, Balrampur, Siddharthnagar, Maharajganj. The IST meridian passes through **Mirzapur**.

1.6 Boundary Lines

- India-Pakistan boundary: **Radcliffe Line** (1947) 1× 1996
- **Durand Line**: with **Afghanistan** (1893) 1× 2006
- **McMahon Line**: **India-China** (1914, Simla) 1× 2018

Line	Year	Between
Radcliffe	1947	India-Pakistan, India-Bangladesh
Durand	1893	British India-Afghanistan
McMahon	1914	India-China (Arunachal)
LoC	1972	India-Pakistan (J&K)
Maginot	1930s	France-Germany (never Indian)

LAST-MINUTE 10

1. India: **7th** largest, **2.4%** of land, **most populous** since 2023
2. Tropic of Cancer: **8 states**; not UP, not Manipur
3. **Gandhinagar** nearest the Tropic; **Mahi** crosses it twice
4. IST **82°30' E, Mirzapur**; 5 states; not Maharashtra
5. **1° = 4 min**; east is ahead
6. Southernmost: **Indira Point**; mainland: **Kanyakumari**
7. Longest border **Bangladesh**; shortest **Afghanistan**
8. **Tripura**: Bangladesh on three sides
9. Assam does not touch Myanmar; Manipur does not touch Bangladesh
10. **Radcliffe**-Pakistan, **Durand**-Afghanistan, **McMahon**-China

ASKED MOST · REVISE FIRST

- **Passes and states** are asked almost every year: **Nathu La, Jelep La = Sikkim**; **Bomdi La, Se La, Bum La = Arunachal**; **Lipulekh, Niti, Mana = Uttarakhand**; **Shipki La, Rohtang = HP**; **Zoji La = Ladakh** 13× 1998, 2001, 2004, 2006, 2008, 2010, 2011, 2013, 2014, 2015, 2016, 2017, 2023
- **Highest peak of each state** (match): Anaimudi-Kerala, Doddabetta-TN, Saramati-Nagaland, Guru Shikhar-Rajasthan, Dhupgarh-MP, Nanda Devi-Uttarakhand 3× 2018, 2021, 2025
- **Anaimudi** (2,695 m) = highest peak of **South India / the Peninsula** 4× 2005, 2012, 2013, one-liner
- **Aravalli** = **oldest** and a **residual** mountain 4× 1991, 2005, 2010, 2017
- Largest glacier: **Siachen** (Gangotri if Siachen is not offered) 3× 1995, 2013, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

Chapter 2

Physical Divisions: Structure, Mountains, Peaks, Passes and Glaciers

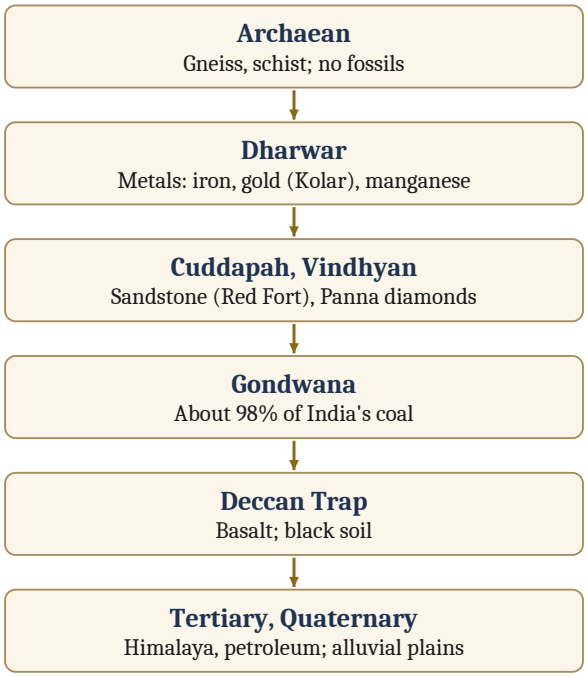
Built from 77 UPPSC questions (1990–2025), the biggest geography chapter. Focus: passes and their states, peaks, hills, Himalayan ranges, glaciers.

2.1 Geology and Rock Systems

- Palaeomagnetism shows the Indian landmass moved **northward** 1× 1998
- The **western coast** is not depositional; it is a **faulted (emergent-submergent)** coast 1× 2004
- Latest of the old rock systems: **Gondwana** (98% of India's coal) 1× 2016
- Deccan Trap intertrappean beds have **freshwater** (not sea) fossils 1× 2024

QUANTUM IAS

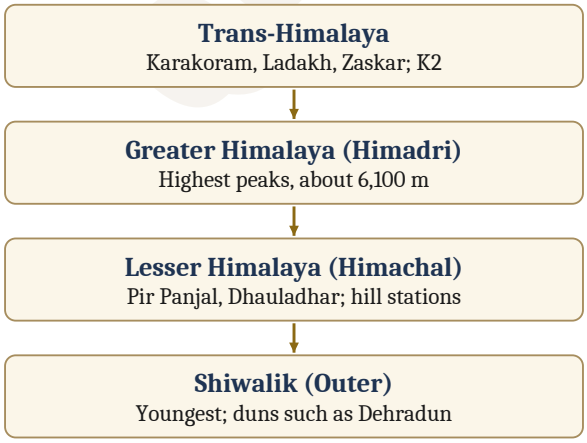
ROCK SYSTEMS, OLDEST TO YOUNGEST



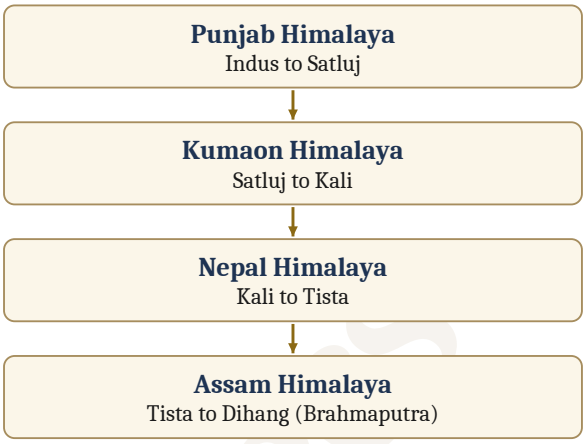
2.2 The Himalaya

- Himalaya is **still rising**: the Indian plate is pushing into the Eurasian plate (A and R true, R explains A) 1× 2024
- Most seismic zones are in the Himalaya because of **longitudinal thrusts** 1× 2019
- **Lesser Himalaya**: marine fossils; **Shiwalik**: remains of early humans; Greater Himalaya sediments are not fossil-less 1× 2019
- **Himachal** = the **Lesser Himalaya** 1× 2008
- Youngest range: **Shiwalik**; if only Himalayan vs Peninsular ranges are offered: **Himadri** 2× 2010, 2020
- **Kumaon Himalaya** lies between the **Satluj and Kali** 1× 2016

HIMALAYAN RANGES, NORTH TO SOUTH



REGIONAL DIVISIONS, WEST TO EAST (BURRARD)



2.3 Ranges, Valleys and Tunnels

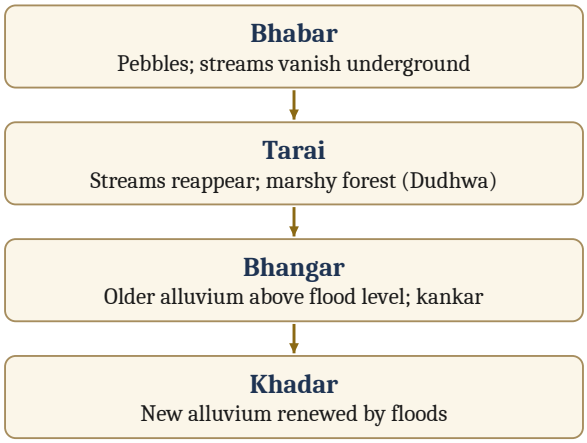
- **Pir Panjal** range is in **Jammu and Kashmir** 1× 2007
- **Atal Tunnel** (Rohtang): BRO, Manali to Lahaul-Spiti, under the **Pir Panjal** 1× 2016
- Atal Tunnel is the longest highway tunnel **above 10,000 ft**, not the world's longest (only statement 2 true) 1× 2025
- **Kashmir valley**: between the **Pir Panjal and Great Himalaya** 1× 2020
- **Kullu valley**: between the **Dhauladhar and Pir Panjal** 2× 1999, 2002
- **Nelang valley** is in **Uttarakhand** (Uttarkashi) 1× 2016
- **Imphal basin** is a **lacustrine plain** 1× 2019
- **Aksai Chin** is part of the **Ladakh plateau** 1× 1995
- **Shaksgam valley**: PoK area given by Pakistan to China (1963); India protests China's construction 1× 2024
- **Tripura** does not lie along the **Patkai** hills 1× 2015

2.4 The Northern Plains

- Gravelly plain at the foot of the Shiwalik: **Bhabar**; it is a **piedmont** region 2× 2007, 1994
- **Kuttanad** (Kerala) is India's **lowest** region, below sea level 1× 2015

QUANTUM IAS & PCS

NORTHERN PLAIN BELTS, FROM THE HILLS SOUTHWARD



2.5 Old Mountains and Peninsular Hills

- **Aravalli** = oldest mountains; a **residual** mountain; mainly in **Rajasthan** 5× 1991, 2010, 2017, 2005, one-liner
- **Satpura** lies between the **Narmada and Tapti** 1× 2007
- Central India hills, **west to east**: **Satpura, Mahadeo, Maikal, Chotanagpur** 1× 2019
- **Nilgiri** hills: junction of **Karnataka, Kerala and Tamil Nadu** 1× 2015
- **Cardamom hills** (**Yela Mala**): **Kerala-Tamil Nadu** border 3× 2008, 2010, 2023
- **Mandav hills** are in **Gujarat**, not Maharashtra 2× 2011, 2011
- Hills match: **Garo-Meghalaya, Miri-Arunachal, Kollimalai-Tamil Nadu, Dalma-Jharkhand** 1× 2017
- Hills match: **Bateshwar-Bihar, Bilari-Karnataka, Chin-Manipur, Dhosi-Haryana** 1× 2022
- **Western Ghats**: a **block mountain**, gentle eastern slope, **basalt** in the north 1× 2017
- **Kodaikanal** is in the **Palani** hills, not the Anaimalai 1× 2013

2.6 Peaks

- Highest peak of the **Peninsula**: **Anaimudi** 4× 2005, 2012, 2013, one-liner
- Highest peak claimed by India: **K2** (PoK); highest in Indian control: **Kanchenjunga**; wholly in India: **Nanda Devi** 2× 1990, 2009
- Peaks east to west: **Namcha Barwa, Kanchenjunga, Everest, Nanda Devi** 1× 2008
- **Cho Oyu, Lhotse** are Himalayan; Annamalai, Sirumalai are not 1× 2022
- Not in India: **Gosainthan** (Tibet); **Gurla Mandhata, Namcha Barwa** 2× 2005, 2019
- **Trishul** peak: Uttarakhand (Garhwal) 1× 2021
- State peaks: **Kerala-Anaimudi, Nagaland-Saramati, Uttarakhand-Nanda Devi, TN-Doddabetta** 1× 2018
- State peaks: **TN-Doddabetta, Rajasthan-Guru Shikhar, Nagaland-Saramati, MP-Dhupgarh** 2× 2021, 2025
- First Indian woman on Everest: **Bachendri Pal** (1984) 1× 1996

State	Highest peak (m)
Sikkim	Kanchenjunga (8,586)
Uttarakhand	Nanda Devi (7,816)
Himachal	Reo Purgyil (6,816)
Nagaland	Saramati (3,826)
Kerala	Anaimudi (2,695)
Tamil Nadu	Doddabetta (2,637)
Karnataka	Mullayanagiri (1,930)
Rajasthan	Guru Shikhar (1,722)
Odisha	Deomali (1,672)
MP / Jharkhand	Dhupgarh / Parasnath (1,350)

2.7 Mountain Passes

- **Se La** is in **Arunachal**, not Uttarakhand 1× 2015
- **Bhor Ghat** is in **Maharashtra**, not HP 1× 1998
- Western Ghat passes north to south: **Thal Ghat, Bhor Ghat, Pal Ghat** 1× 2023
- **Khyber** is not a Western Ghats pass (it is in Pakistan-Afghanistan) 1× 2016
- **Niti** = **Uttarakhand**; **Lipulekh** = **Uttarakhand** (not UP or Ladakh) 5× 2023, 2001, 2008, 2014, one-liner
- **Nathu La** = **Sikkim** (not Meghalaya or Arunachal) 4× 2010, 2013, 2017, one-liner
- **Zoji La** leads to **Leh** (now in Ladakh, not Arunachal) 2× 2016, 2023
- **Rohtang** = **Himachal Pradesh** 1× one-liner

QUANTUM IAS & PCS

- Match: **Banihal-J&K, Nathu La-Sikkim, Niti-Uttarakhand, Shipki-HP** 2× 2004, 2011
- Match: **Bum La-Arunachal, Jelep La-Sikkim, Muling La-Uttarakhand, Shipki La-HP** 1× 2006
- Match: **Mana-Uttarakhand, Nathu La-Sikkim, Zoji La-J&K (Ladakh), Shipki La-HP** 1× 2016
- Heights: **Debsa 5,360, Bara-lacha 4,850, Rohtang 3,978, Banihal 2,832 m** 1× 2024

HIMALAYAN PASSES, WEST TO EAST

Zoji La, Banihal
Ladakh; J&K

↓

Shipki La, Rohtang
Himachal Pradesh

↓

Mana, Niti, Lipulekh
Uttarakhand

↓

Nathu La, Jelep La
Sikkim

↓

Bum La, Se La, Bomdi La
Arunachal Pradesh

TRAP ALERT

- Order asked in 2009: **Shipki La, Lipulekh, Nathu La, Bomdi La** (west to east).
- **Bomdila** is the town; the pass is **Bomdi La**. **Jelep La** and Nathu La both lead to the Chumbi valley.

2.8 Glaciers

- Largest: **Siachen**; without Siachen in the options: **Gangotri** 3× 2013, 2017, 1995
- **Chaurabari** glacier is **north of Kedarnath** 2× 2004, 2010
- **Milam** glacier is in **Kumaon** (Pithoragarh) 1× 2014

Glacier	Where	Feeds
Siachen (76 km)	Karakoram, Ladakh	Nubra
Gangotri (30 km)	Uttarkashi	Bhagirathi
Zemu	Sikkim	Tista
Milam, Pindari	Kumaon	Gori Ganga, Pindar
Chaurabari	Kedarnath	Mandakini

2.9 The Peninsular Plateau

- **Meghalaya plateau** is part of the **Peninsular plateau** 1× 2013
- **Dhanbad** is densely populated because of **mining and industry** 1× 2007
- **Dandakaranya** lies in **Chhattisgarh and Odisha** 1× 2008

UP LINK

Bundelkhand plateau (Jhansi, Banda, Chitrakoot) is the Peninsular part of UP. The **Tarai** of Pilibhit, Lakhimpur Kheri and Bahraich holds **Dudhwa National Park**. UP has no high peaks or passes; **Lipulekh** is often wrongly given as UP but is in Uttarakhand.

LAST-MINUTE 10

1. **Nathu La, Jelep La** = Sikkim; **Se La, Bum La, Bomdi La** = Arunachal
2. **Lipulekh, Niti, Mana** = Uttarakhand; **Shipki La, Rohtang** = HP; **Zoji La** = Ladakh
3. **Anaimudi** = highest in the Peninsula; **Doddabetta** = Tamil Nadu
4. **Aravalli** oldest; **Shiwalik** youngest
5. **Kumaon** Himalaya: Satluj to Kali
6. **Kullu**: Dhauladhar-Pir Panjal; **Kashmir**: Pir Panjal-Great Himalaya
7. **Bhabar** at the foot of the hills, then Tarai
8. **Satpura** between Narmada and Tapti; **Mandav** in Gujarat
9. **Siachen** largest glacier; **Chaurabari** above Kedarnath
10. **Gondwana** = coal; Kuttanad = lowest land

Chapter 3

Coastal Regions and Islands

Built from 21 UPPSC questions (2001–2016). Focus: longest coastline, named coasts, island channels, Barren and Lakshadweep.

ASKED MOST · REVISE FIRST

- ■ **Gujarat** has the **longest coastline** 4× 2006, 2007, 2016, 2018
- • **Ten Degree Channel** separates the **Andaman** group from the **Nicobar** group 2× 2005, one-liner
- • **Barren Island** (Andaman Sea, Bay of Bengal) is India's only **active volcano** 2× 2010, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Sea Level and Maritime Zones

- India's **mean sea level** is measured at **Chennai** 1× 2013
- **Territorial waters**: **12 nautical miles** from the coast 1× 2005
- ■ Longest coastline: **Gujarat** (about 2,340 km) 4× 2006, 2007, 2016, 2018
- Coastal states: **9** (plus 4 coastal UTs) 1× one-liner

MARITIME ZONES, FROM THE COAST OUTWARD

Territorial sea
12 nautical miles; full sovereignty

↓

Contiguous zone
24 nm; customs, immigration

↓

Exclusive Economic Zone
200 nm; fishing, mining, drilling

↓

Continental shelf
To the margin edge (up to 350 nm)

TRAP ALERT

- Coastline length changed: old figure **7,516.6 km**, new count **11,098.81 km**. Read the options. **Goa** has the shortest state coastline.

3.2 Named Coasts and Coastal Towns

- **Coromandel** coast: **Krishna delta to Cape Comorin** 1× 2015
- Northern part of the western coastal plain: **Konkan** 1× 2016
- West-coast towns north to south: **Janjira, Sindhudurg, Kannur, Nagercoil** 1× 2001
- West-coast towns among the options: **Janjira, Udupi** (Auroville and Tuticorin are on the east coast) 1× 2002
- Beaches: **Digha-WB, Gopalpur-Odisha, Calangute-Goa, Marina-Tamil Nadu** 1× 2008

WEST COAST, NORTH TO SOUTH

Kathiawar coast
Gujarat: gulfs of Kutch, Khambhat

↓

Konkan
Daman to Goa; Mumbai

↓

Kannada coast
Goa to Mangaluru

↓

Malabar
Mangaluru to Kanyakumari; backwaters (kayals)

EAST COAST, NORTH TO SOUTH

Utkal coast
Odisha; Chilika lake

↓

Northern Circars
Mahanadi to Krishna

↓

Coromandel
Krishna delta to Kanyakumari; Pulicat

3.3 Andaman and Nicobar Islands

- Highest peak: **Saddle Peak** (North Andaman) 1× 2009
- **Ten Degree Channel**: **Andaman** from **Nicobar** 2× 2005, one-liner
- **Barren Island** is in the **Bay of Bengal** (Andaman Sea); India's **volcanic** island 2× 2010, one-liner

Channel	Separates
Ten Degree	Andaman group from Nicobar group

QUANTUM IAS & PCS

Channel	Separates
Duncan Passage	South Andaman from Little Andaman
Coco Channel	North Andaman from Coco Is. (Myanmar)
Nine Degree	Minicoy from main Lakshadweep
Eight Degree	Minicoy from the Maldives
Palk Strait	Tamil Nadu from Sri Lanka

3.4 Lakshadweep and Other Islands

- **Lakshadweep** is a **coral** island group 1× 2013
- **Minicoy** is one of the remotest islands from the Indian coast 1× 2013
- **Rameswaram** lies between India and Sri Lanka 1× 2006
- **Adam's Bridge** begins at **Dhanushkodi** 1× 2010
- Most populous island: **Salsette** (Mumbai) 1× 2016
- Match: **Viyant Syodhar-Gulf of Kutch, Piram-Gulf of Khambhat, Dwarka-Arabian Sea coast, Diu-Kathiawar coast** 1× 2010
- **Kori Creek** is in the **Rann of Kutch** 2× 2008, 2011

TRAP ALERT

- **Ten Degree** = Andaman-Nicobar. **Nine and Eight Degree** belong to the Lakshadweep side. Barren is **active**; Narcondam is **dormant**.

UP LINK

UP is landlocked, but UPPSC asks island and coast facts as standard map knowledge. Remember **Salsette** (Mumbai), **Majuli** (largest river island) and **Sriharikota** (ISRO launch site).

LAST-MINUTE 10

1. **Gujarat**: longest coastline; **Goa**: shortest
2. Mean sea level measured at **Chennai**
3. Territorial sea **12 nm**; EEZ **200 nm**
4. **Konkan** = north of the west coast; **Coromandel** = Krishna to Kanyakumari
5. Janjira, Sindhudurg, Kannur, Nagercoil (north to south)
6. **Saddle Peak** = highest in Andaman and Nicobar
7. **Ten Degree Channel** = Andaman | Nicobar
8. **Barren** = active volcano; **Lakshadweep** = coral
9. **Salsette** = most populous island; **Minicoy** = remotest
10. Adam's Bridge from **Dhanushkodi**; **Kori Creek** in Kutch

Chapter 4

States and Union Territories

Built from 18 UPPSC questions (1990–2019). Focus: area ranks, shared borders, capitals, NCR, Union Territories.

ASKED MOST · REVISE FIRST

- ■ Area ranks: **Rajasthan > MP > Maharashtra > UP**; UP is **4th** by area 3× 2016, 2019, one-liner
- ■ **NCR** includes parts of **Haryana, UP and Rajasthan** only; not Punjab or MP 3× 2011, 2017, one-liner
- • **Sonbhadra** touches **four states** (MP, Chhattisgarh, Jharkhand, Bihar) 1× 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Places and their States

- **Palanpur** is in **Gujarat** 1× 2004
- **Malnad** belongs to **Karnataka**, not Andhra Pradesh 1× 2000

4.2 Size, Population and Districts

- UP is **4th** by area 1× 2016

- About **30%** of India's area lies in **Rajasthan, MP and Maharashtra** 1× one-liner
- **Maharashtra**: **2nd** in population, **3rd** in area (2011) 1× 2019
- Most districts: **Uttar Pradesh** (75) 1× 2006
- **Thar**: densely populated desert, about 10,000 years old, spreading partly through human misuse; less grazing land as farming spreads (all four true) 1× 2008

Rank	By area	By population (2011)
1	Rajasthan	Uttar Pradesh
2	Madhya Pradesh	Maharashtra
3	Maharashtra	Bihar
4	**Uttar Pradesh**	West Bengal
Smallest	Goa	Sikkim

4.3 Shared Borders

- In 1993 undivided MP touched **7 states**; today MP touches **5** 1× 1993
- **Chhattisgarh** does **not** border **Bihar** 1× 2013
- **Sonbhadra** touches the boundaries of **four states** 1× 2015
- **Telangana** borders **AP, Karnataka, Maharashtra, Chhattisgarh** 1× 2015

UP LINK

UP borders **8 states + Delhi**, the most of any state: Uttarakhand, **Himachal** (Saharanpur), Haryana, Rajasthan, MP, **Chhattisgarh** and Jharkhand (both via Sonbhadra), Bihar. It does **not** border Punjab or Gujarat. It has **75 districts**, the most in India.

4.4 Capitals and the NCR

- **Manipur-Aizawl** is wrong; Manipur's capital is **Imphal** (Aizawl is Mizoram's) 1× 1990
- **Assam-Dispur, Nagaland-Kohima, Arunachal-Itanagar, Meghalaya-Shillong** 1× 1994
- NCR towns among the options: **Khurja, Karnal, Rohtak** (not Ambala) 1× 2011
- NCR does **not** include **Punjab** or **Madhya Pradesh** 2× one-liner, 2017

North-east state	Capital
Assam	Dispur
Arunachal Pradesh	Itanagar
Meghalaya	Shillong
Nagaland	Kohima
Manipur	Imphal
Mizoram	Aizawl
Tripura	Agartala
Sikkim	Gangtok

4.5 Union Territories

- Largest UT among the options in 2014: **Delhi**; now **Ladakh** is the largest UT 1× 2014
- **Puducherry** is spread over **Tamil Nadu, Kerala and Andhra Pradesh** (Puducherry, Karaikal, Mahe, Yanam) 1× 2006

UNION TERRITORIES SINCE 2019 (NOW 8)

31 October 2019
J&K state split into UTs J&K and Ladakh (UTs: 9)

↓

26 January 2020
Dadra and Nagar Haveli merged with Daman and Diu (UTs: 8)

↓

UTs with a legislature
Delhi, Puducherry, Jammu and Kashmir

↓

Largest / smallest UT
Ladakh / Lakshadweep

TRAP ALERT

- The NCR question repeats with different wrong options: **Punjab**, **Himachal**, **MP**. Only Haryana, UP and Rajasthan are in the NCR.
- Older keys call **Delhi** the largest UT; after 2019 it is **Ladakh**.

LAST-MINUTE 10

1. Area: Rajasthan, MP, Maharashtra, **UP (4th)**
2. Population: **UP** first; Maharashtra 2nd (and 3rd by area)
3. UP: most districts (75); borders 8 states + Delhi
4. **Sonbhadra** touches 4 states
5. Chhattisgarh does not touch Bihar
6. **Imphal** (Manipur), **Aizawl** (Mizoram), **Dispur** (Assam)
7. NCR = Haryana, UP, Rajasthan
8. **Ladakh** largest UT; 8 UTs today
9. Puducherry in TN, Kerala, AP
10. **Malnad** = Karnataka; **Palanpur** = Gujarat

Chapter 5

Tribes of India

Built from 35 UPPSC questions (1991–2022). Focus: tribe-state matching, the Tharu and Bhotia, Toda polyandry, Andaman tribes.

ASKED MOST · REVISE FIRST

- **Tribe-state matching** is asked almost every year: **Toda = Tamil Nadu (Nilgiris), Lepcha = Sikkim, Birhor/Santhal = Jharkhand, Gaddi = Himachal, Jarawa = Andaman** 10× 2002, 2002, 2013, 2014, 2015, 2016, 2017, 2019, 2020, 2022
- **Tharu** live in **Uttarakhand and UP**; largest ST of Uttarakhand; **Diwali is mourning** for them 6× 1991, 2005, 2005, 2015, 2017, one-liner
- **Toda** (Nilgiri hills) and **Jaunsari** practise **polyandry** 5× 1993, 2005, 2006, 2009, 2013
- **Gaddi** = tribe of the **Dhauladhar** range (Himachal) 3× 2001, 2002, 2003
- **Jarawa** live in the **Andaman and Nicobar Islands** 3× 2013, 2016, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Racial Groups and Identity

- North-east hill tribes are **Mongoloid** 1× 2003
- **Santhal** are **Proto-Australoid** 1× 2002
- Tribes are identified by **distinct culture and separate habitat** 1× 1992

Racial group	Example tribes
Negrito	Onge, Jarawa, Sentinelese, Great Andamanese
Proto-Australoid	Santhal, Munda, Ho, Oraon, Bhil
Mongoloid	Naga, Khasi, Garo, Lepcha, Bhutia

5.2 Tribes of UP and Uttarakhand

- **Tharu**: **Uttarakhand and Uttar Pradesh** 2× 1991, 2017
- **Diwali** is a festival of **mourning** for the **Tharu** 3× 2005, 2015, one-liner
- Largest ST of Uttarakhand: **Tharu** 1× 2005
- **Transhumance** is practised by the **Bhotia** 1× 1997
- "Bhotia-UP" pairs have **no wrong option** (Bhotia are notified in UP too), so the source marks them (*) 2× 2004, 2010

TRAP ALERT

- **Bhotia** (UP, Uttarakhand) and **Bhutia** (Sikkim) are **different** tribes. A "Bhutia" option in a Sikkim pair is correct; a "Bhotia-UP" pair is also correct.

5.3 Customs and Festivals

- **Polyandry**: **Toda**; **Jaunsari and Toda** (not Gond or Naga) 3× 2005, 2006, 2009
- **Jhum** (shifting cultivation) is practised by the **Khasi** 1× 2010
- **Sarhul** (sal-blossom festival): **Oraon, Munda, Santhal** of Jharkhand 1× 2005
- **Lohasur** (iron god) is worshipped by the **Agaria** (iron smelters) 1× 2016
- **Onam-Andhra Pradesh** is wrong; Onam is **Kerala's** 1× 2005
- **Sikkim-Tibetan** is wrong as the main language; it is **Nepali** 1× 2006
- **Khasi and Garo** speakers live in **Meghalaya** 1× 2002

Shifting cultivation name	Region
Jhum	North-east
Podu	Andhra Pradesh

Shifting cultivation name	Region
Bewar, Dahiya	Madhya Pradesh
Penda, Dippa	Chhattisgarh (Bastar)
Kumari	Western Ghats
Milpa / Ladang	Mexico / Indonesia

5.4 Tribes and their States

- Largest tribe of India: **Bhil** 1× 2009
- **Jarawa-Andaman, Kuki-Manipur, Tharu-UP, Gond-MP** 1× 2019
- Correct pairs: **Keria-Odisha, Paliyan-Tamil Nadu** (not Kuki-UP, Yanadi-Rajasthan) 1× 2022
- **Bhil-Rajasthan, Santhal-Jharkhand, Raji-Uttarakhand, Lepcha-Sikkim** 1× 2002
- **Jaunsari-Uttarakhand, Santhal-Jharkhand, Bhil-MP, Lepcha-Sikkim** 1× 2002
- **Birhor-Jharkhand, Bhutia-Sikkim, Toda-Tamil Nadu, Sentinelese-Andaman** 2× 2013, 2014
- **Toda-Tamil Nadu, Lepcha-Sikkim, Birhor-Jharkhand, Garo-Meghalaya** 1× 2015
- **Gaddi-Himachal, Toda-Tamil Nadu, Khasi-Meghalaya, Naga-Manipur** 1× 2017
- **Bhil-Rajasthan, Gond-Chhattisgarh, Limboo-Sikkim, Konyak-Nagaland** 1× 2016
- **Khas-Arunachal** is wrong; Khas live in the **western Himalaya** 1× 2020
- **Toda** live in the **Nilgiri hills** 2× 1993, 2013
- **Gaddi**: **Dhauladhar** range 3× 2001, 2002, 2003
- **Tamaria**: **Jharkhand** 1× 2021
- None of Chenchu, Lepcha, Dafla, Dafar is a Kerala tribe (marked *) 1× 2013

Tribe	State
Toda, Kota, Badaga, Paliyan	Tamil Nadu
Chenchu, Yanadi	Andhra Pradesh
Gaddi	Himachal Pradesh
Jaunsari, Tharu, Bhotia, Raji	Uttarakhand
Lepcha, Bhutia, Limboo	Sikkim
Konyak, Ao, Angami, Rengma	Nagaland
Khasi, Garo, Jaintia	Meghalaya
Apatani, Adi, Nyishi (Dafla)	Arunachal

Tribe	State
Kuki	Manipur
Santhal, Munda, Ho, Birhor	Jharkhand
Bhil	Rajasthan, MP, Gujarat
Gond, Baiga, Agaria	MP, Chhattisgarh

5.5 Andaman and Nicobar Tribes

- **Jarawa** live in the **Andaman and Nicobar Islands**; called India's **most primitive** tribe 3× 2013, 2016, one-liner
- **Rengma** is a **Naga** tribe, not native to the islands 1× 2020

ISLAND TRIBES, NORTH TO SOUTH



UP LINK

UP's main tribes are the **Tharu** (Tarai: Lakhimpur Kheri, Bahraich, Balrampur), **Buksa** and **Bhotia**. The Tharu mourn on **Diwali**. **Sonbhadra** has Gond, Kharwar and Cheru communities.

LAST-MINUTE 10

1. **Bhil** = largest tribe
2. **Toda** = Nilgiris, polyandry; **Jaunsari** = polyandry
3. **Tharu** = UP and Uttarakhand; Diwali mourning
4. **Bhotia** = transhumance; not the same as Bhutia (Sikkim)
5. **Gaddi** = Dhauladhar (HP)
6. **Lepcha** = Sikkim; **Birhor, Santhal** = Jharkhand
7. **Khasi, Garo** = Meghalaya; Khasi practise jhum
8. **Sarhul** = Oraon, Munda; **Agaria** worship Lohasur
9. **Jarawa, Onge, Sentinelese** = Andaman (Negrito)
10. **Rengma** = Naga, not Andaman

Chapter 6

Drainage I: The Himalayan Rivers

Built from 36 UPPSC questions (1990–2025). Focus: the Ganga system, Satluj, Bundelkhand rivers, Brahmaputra and Indus.

ASKED MOST · REVISE FIRST

- **Satluj** is the most-asked river: rises at **Rakas Tal** (Tibet), enters India at **Shipki La**, cuts across **all Himalayan ranges**, is **trans-Himalayan** and **does not rise in India** 6× 2010, 2012, 2013, 2014, 2020, 2023
- **Devprayag** = **Alaknanda + Bhagirathi**; the river is called **Ganga** from here 4× 2002, 2016, 2023, 2024
- **Betwa** (Bundelkhand): joins the **Yamuna**; **Rajghat dam** is on it; **Ken-Betwa** (UP-MP) is India's first river link 4× 2006, 2006, 2014, 2015
- Himalayan rivers are **perennial** because of **snow and glacier melt**, not the monsoon (Assertion-Reason) 2× 2011, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Nature of Himalayan Rivers

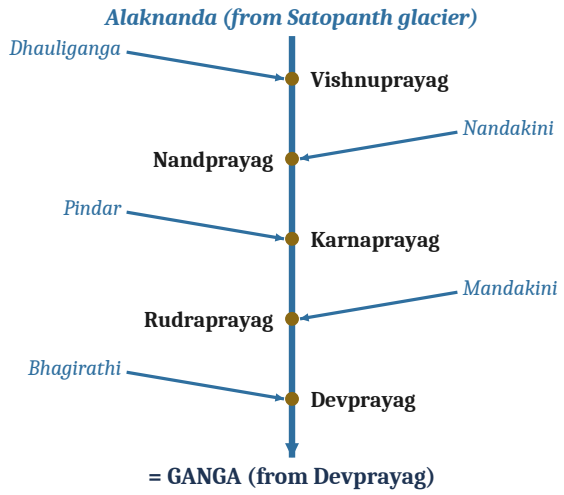
- Himalayan rivers are **perennial**: summer **snow-melt** keeps them flowing; the monsoon explains only the rainy-season flood 2× 2011, 2025
- **Ganga** is an **antecedent** river: older than the mountains, it kept cutting down as they rose 1× 2013
- **Chambal** is a **superimposed** river (also Son, Banas, Subarnarekha) 1× 2007
- **Satluj** cuts across **all ranges** of the Himalaya (Zaskar, Greater, Lesser, Shiwalik) 2× 2012, 2014
- **Trans-Himalayan** rivers (rise north of the Himalaya): **Satluj**, Indus, Brahmaputra 1× 2020

Drainage type	Key idea	Example
Antecedent	Older than the mountain	Ganga, Indus, Satluj, Brahmaputra
Consequent	Follows the original slope	Godavari, Krishna, Kaveri
Subsequent	Along weak rock bands	Strike-valley tributaries
Superimposed	Let down onto older rocks	Chambal, Son, Banas

6.2 The Ganga and its Headwaters

- Ganga rises as the **Bhagirathi** at **Gaumukh** (Gangotri glacier); called **Ganga** from **Devprayag** 4× 2002, 2016, 2023, 2024
- **Badrinath** is on the **Alaknanda**; **Kedarnath** on the **Mandakini** 2× 2002, 2013
- **Rudraprayag** = Alaknanda + **Mandakini**; **Karnaprayag** = Alaknanda + **Pindar**; **Tehri** = Bhagirathi + **Bhilangna** 2× 2002, 2024
- Adi Shankaracharya's four mathas: **Jyotirmath, Sringeri, Dwarka, Puri**; **Kedarnath is not** one of them 1× 2002
- In Bangladesh the Ganga is the **Padma**; the Bhagirathi-Hooghly branch flows past Kolkata 1× 2016
- Ganga pollution comes from **sewage and industrial waste** (Kanpur tanneries); "holier = more polluted" is false 1× 2005

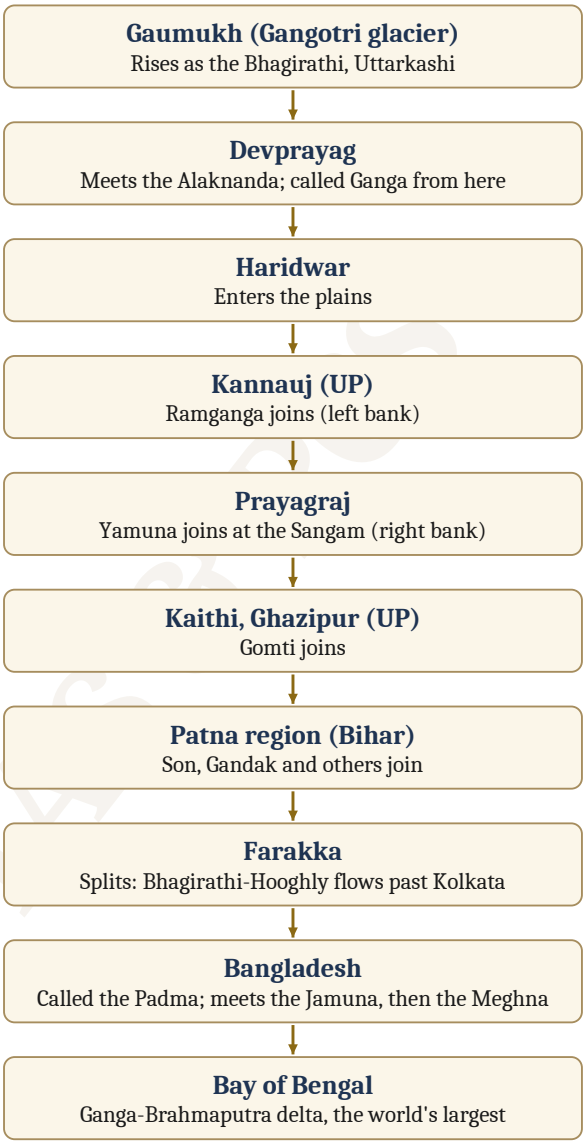
PANCH PRAYAG: WHERE THE ALAKNANDA'S TRIBUTARIES JOIN



TRAP ALERT

- **Devprayag** (Bhagirathi) and **Rudraprayag** (Mandakini) are the usual swap. Tehri is **Bhagirathi + Bhilangna**, not an Alaknanda confluence.

COURSE OF THE GANGA, SOURCE TO SEA



6.3 Tributaries of the Ganga

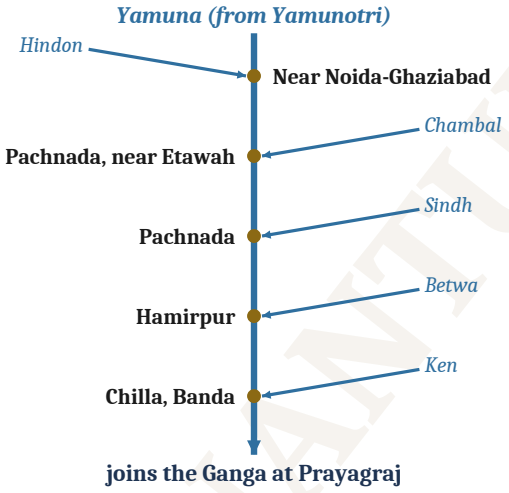
- **Gomti** is the only Ganga tributary that **rises in the plains**: Phulhar Tal, **Pilibhit**; flows past Lucknow and Jaunpur **1×** 2008
- **Yamuna** is **perennial**; it is "dead" below Delhi because water is diverted and sewage added (A true, R false) **1×** 2015
- **Ramganga** joins the **Ganga** (at Kannauj), **not** the Yamuna **1×** 2010
- Yamuna's right-bank tributaries, **west to east**: **Chambal, Sindh, Betwa, Ken** **1×** 2020
- **Son** has the largest catchment among Son, Gandak, Ramganga and Mahananda (about 70,000 sq km) **1×** 2017
- **Gandak** does **not** flow through UP; it flows through Bihar **1×** 2021

QUANTUM IAS & PCS

- **Jonk** belongs to the **Mahanadi** basin, not the Ganga **1×** 2021
- **Kosi**, the "Sorrow of Bihar", **shifts its course** the most **1×** 2013
- **Mahatma Gandhi Setu** (Patna-Hajipur) is in **Bihar** **1×** 1993
- **Moribund delta** = western part of the **Bengal (Ganga-Brahmaputra) delta** **1×** 2019
- Tributary match: **Ganga-Son, Godavari-Manjira, Krishna-Bhima, Yamuna-Ken** **1×** 2003

Tributary	Bank	Joins at
Ramganga	Left	Kannauj (UP)
Gomti	Left	Kaithi, Ghazipur (UP)
Ghaghara	Left	Near Chhapra (Bihar)
Gandak	Left	Sonepur-Hajipur (Bihar)
Kosi	Left	Kursela (Bihar)
Yamuna	Right	Prayagraj (Sangam)
Son	Right	Near Patna

YAMUNA'S TRIBUTARIES, WEST TO EAST (ASKED 2020)



UP LINK

Gomti (Pilibhit to Lucknow to Jaunpur to Kaithi), the Yamuna's Bundelkhand tributaries (Chambal near Etawah, Betwa at **Hamirpur**, Ken at Chilla in **Banda**) and the Kanpur-Prayagraj-Varanasi stretch of the Ganga are UP's most-asked river facts.

6.4 Ken-Betwa and Bundelkhand

- **Ken-Betwa Link**: **UP and MP**, MoU **2005**; India's **first** river-interlinking project; foundation stone **25 Dec 2024** **1×** 2006
- **Betwa** joins the **Yamuna** (at Hamirpur) **1×** 2014
- **Rajghat dam** (Lalitpur, joint UP-MP project) is on the **Betwa** **2×** 2006, 2015

KEN-BETWA LINK: DIRECTION OF WATER

Ken river
Daudhan dam, Panna (MP)
↓
Link canal (221 km)
Carries surplus Ken water north-west
↓
Betwa river
Feeds Bundelkhand: Jhansi, Lalitpur, Mahoba, Banda (UP)
↓
Yamuna
Betwa joins it at Hamirpur

TRAP ALERT

- Water goes **from the Ken to the Betwa** (Daudhan dam on the Ken). Both are tributaries of the **Yamuna**, not the Ganga.

6.5 The Brahmaputra System

- **Indus and Brahmaputra** rise at almost the same point, near **Mansarovar-Kailash** **1×** 2006
- Brahmaputra basin: **Tibet, India, Bangladesh** (and Bhutan) **1×** 1990
- **Teesta** was once a Ganga tributary; after the **1787 flood** it joined the **Brahmaputra**: **river capture** (A and R true, R explains A) **1×** 2021
- **Sankosh** forms the **Assam-West Bengal** boundary **1×** 2007

ONE RIVER, FOUR NAMES: THE BRAHMAPUTRA

Yarlung Tsangpo
Tibet; rises at Chemayungdung glacier near Mansarovar
↓
Siang / Dihang
Great bend round Namcha Barwa into Arunachal Pradesh
↓
Brahmaputra
Assam, after the Dibang and Lohit join near Sadiya; Majuli island
↓
Jamuna
Bangladesh; joins the Padma, then the Meghna

6.6 The Indus System

- **Satluj** rises at **Mansarovar-Rakas Tal** and enters India at **Shipki La** 1× 2023
- **Satluj does not rise in India**; Beas, Ravi and Chenab do 2× 2010, 2013
- **Kishanganga** (Neelum) is a tributary of the **Jhelum** 1× 2013
- **Doodh Ganga** flows in **Jammu and Kashmir** (also a stream of that name in Uttarakhand) 1× 2015
- World's **highest railway bridge**: on the **Chenab** (Reasi, 359 m; opened June 2025) 1× 2006

River	Source
Jhelum	Verinag spring
Chenab	Chandra + Bhaga at Tandi
Ravi	Near Rohtang
Beas	Beas Kund
Satluj	Rakas Tal (Tibet)

LAST-MINUTE 10

1. Perennial Himalayan rivers = **snow-melt**
2. Ganga = **antecedent**; Chambal = **superimposed**
3. **Satluj**: Rakas Tal, Shipki La, cuts all ranges, trans-Himalayan
4. **Devprayag** = Alaknanda + Bhagirathi
5. **Badrinath** on Alaknanda; **Kedarnath** on Mandakini
6. **Gomti** = only plains-born tributary (Pilibhit)
7. Yamuna tributaries west to east: **Chambal, Sindh, Betwa, Ken**
8. **Ken to Betwa** link, UP-MP; Rajghat dam on Betwa
9. Teesta = **river capture** (1787); Kosi = Sorrow of Bihar
10. Kishanganga to **Jhelum**; Chenab bridge = world's highest

Chapter 7

Drainage II: Peninsular Rivers and River Cities

Built from 39 UPPSC questions (1992–2025). Focus: rift valleys and estuaries, river sources, Godavari, cities on rivers.

ASKED MOST · REVISE FIRST

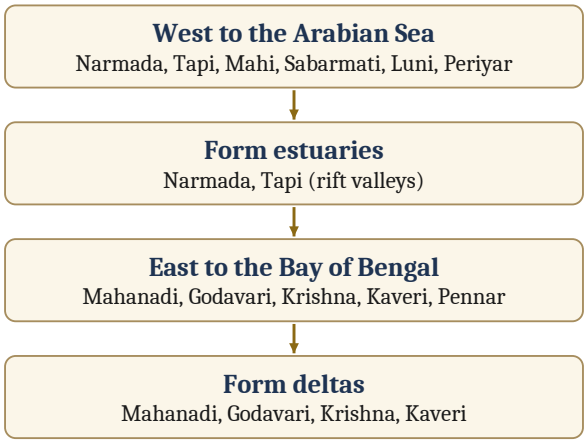
- **Narmada and Tapi** flow west through **rift valleys** and form **estuaries**, not deltas 8× 1999, 2002, 2005, 2006, 2010, 2012, 2013, 2017
- Western Ghat rivers form **no deltas**: short, fast courses over hard rock 3× 2010, 2013, 2014
- **Godavari** = longest Peninsular river with the largest basin 3× 2009, 2013, 2014
- **Bhubaneswar is not on the Mahanadi**; **Hyderabad is on the Musi**, not the Krishna 4× 1996, 1999, 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 West-flowing Rivers and Rift Valleys

- Western Ghat rivers make **no delta**: short, swift, hard rock (A and R true, R explains A) 3× 2010, 2013, 2014
- Narmada and Tapi reach the **Arabian Sea** because they flow in **rift valleys** 2× 1999, 2002
- The Narmada forms **no delta** (A false, R true) 1× 2005
- River in a **fault trough**: **Narmada**; rift valley: **Tapi** 3× 2010, 2013, 2017
- **Narmada** flows **between two ranges** (Vindhya and Satpura) 1× 2012
- **Tapti** forms an **estuary** 1× 2006
- **Damodar** is famous for **fault-valley (rift) drainage** on the east side 1× 2019
- Of Periyar, Pennar and Palar, only the **Periyar** falls into the Arabian Sea 1× 2025
- **Luni**: fresh in the upper course, **saline** lower down 1× 2019

WHICH WAY DO THEY FLOW?



7.2 Sources of the Rivers

- **Amarkantak**: source of the **Narmada** (and Son) 1× 2008
- **Son**'s actual source: **Son Bachharwar**, Bilaspur district 1× 2007
- **Tapti** does **not** rise in the Western Ghats (it rises at **Multai**, Satpura); it falls into the **Arabian Sea** 2× 2018, 1992
- Sources: **Yamuna-Yamunotri, Krishna-Mahabaleshwar, Godavari-Nashik, Mahanadi-Sihawa** 1× 2023
- **Kaveri** rises in the **Brahmagiri** hills (Talakaveri) 2× 2011, 2013

River	Source
Narmada, Son	Amarkantak (MP)
Tapi	Multai, Betul (MP)
Mahanadi	Sihawa, Dhamtari (Chhattisgarh)
Godavari	Trimbakeshwar, Nashik
Krishna	Mahabaleshwar
Kaveri	Talakaveri, Brahmagiri
Luni	Near Ajmer (Aravalli)

7.3 Basins and Tributaries

- Longest Peninsular river: **Godavari**; largest catchment: **Godavari** 3× 2013, 2014, 2009
- Largest basin among the options: **Mahanadi** 2× 2021, 2022
- "Ganga of the South": **Kaveri** in the UPPSC key (NCERT calls the Godavari Dakshin Ganga) 1× 2015
- **Pranhita** joins the **Godavari**, not the Mahanadi 1× 2023
- **Tel** is a Mahanadi tributary, **not** the Krishna's 1× 2010
- **Hagari** is a tributary of the **Tungabhadra** 1× 2009
- South Indian rivers are mainly **dendritic** 1× 2014
- **Mahanadi** forms its delta in **Odisha** 1× 2014
- **Krishna dispute**: Maharashtra, Karnataka, AP (and Telangana); no option complete (*) 1× 2013
- Water-deficient basins: options 1 and 2 (per the source key) 1× 2002

River	Main tributaries
Mahanadi	Seonath, Hasdeo, Ib, **Tel**, Jonk
Godavari	Manjra, Penganga, Wardha, Wainganga, **Pranhita**, Indravati
Krishna	Koyna, Bhima, Tungabhadra, Musi
Tungabhadra	Tunga, Bhadra, **Hagari**
Kaveri	Hemavati, Kabini, Bhavani, Amaravati

7.4 The Damodar and Others

- **Indravati** is **not** a Damodar tributary (it joins the Godavari) 1× 2019
- Damodar joins the **Hooghly** 2× 2006, 2008
- **Eden Canal** is taken from the **Damodar** 1× 2015
- Most polluted among the options: **Damodar** ("Sorrow of Bengal", coal belt) 1× 2013
- Mumbai's **Mithi** river rises from **Vihar Lake** 1× 2014

7.5 Cities on River Banks

- **Leh** is on the **right bank of the Indus** 1× 2013
- **Bhubaneswar-Mahanadi** is wrong (Cuttack is on the Mahanadi) 2× 1999, 2010
- **Hyderabad-Krishna** is wrong; Hyderabad is on the **Musi** 2× 1996, 2016
- **Cuttack-Mahanadi, Ludhiana-Satluj, Nashik-Godavari, Ujjain-Kshipra** 1× 1992
- **Ujjain** is on the **Kshipra** 1× 2008
- **Govindghat** is **not** on the Mandakini (it is on the Alaknanda) 1× 2013

City	River
Leh	Indus

QUANTUM IAS & PCS

City	River
Srinagar	Jhelum
Ludhiana	Satluj
Jabalpur	Narmada
Surat	Tapi
Ahmedabad	Sabarmati
Ujjain	Kshipra
Nashik, Rajahmundry	Godavari
Vijayawada	Krishna
Hyderabad	Musi
Cuttack, Sambalpur	Mahanadi
Tiruchirappalli	Kaveri
Kota	Chambal

UP LINK

UP river cities: **Lucknow, Jaunpur-Gomti**; **Kanpur, Varanasi-Ganga**; **Agra, Mathura-Yamuna**; **Ayodhya-Saryu**; **Gorakhpur-Rapti**; **Bareilly, Moradabad-Ramganga**. **Amarkantak** (source of the Son) feeds Sonbhadra's river.

TRAP ALERT

- **Bhubaneswar** is not on any big river; **Cuttack** is the Mahanadi city.
- **Tapi** rises in the **Satpura** (Multai), not the Western Ghats.

LAST-MINUTE 10

1. Narmada, Tapi: **rift valley**, west, **estuary**
2. Western Ghat rivers: **no delta**
3. **Damodar**: rift on the east; joins the Hooghly; Eden Canal
4. **Amarkantak** = Narmada, Son; **Sihawa** = Mahanadi
5. **Mahabaleshwar** = Krishna; **Nashik** = Godavari; **Brahmagiri** = Kaveri
6. **Godavari** = longest Peninsular river
7. Pranhita to Godavari; Tel to Mahanadi; Hagari to Tungabhadra
8. Hyderabad = **Musi**; Ujjain = **Kshipra**; Leh = **Indus**
9. **Luni**: fresh above, saline below
10. **Periyar** = the Arabian-Sea river among Periyar, Pennar, Palar

Chapter 8

Waterfalls, Lakes and Hot Springs

Built from 25 UPPSC questions (1997–2022). Focus: waterfall-river matches, Chilika, lake-state matches, Wular.

ASKED MOST · REVISE FIRST

- ■ **Waterfall-river/state matching: Jog-Sharavati, Gokak-Ghataprabha, Dudhsagar-Mandovi (Goa), Duduma-Machkund, Chitrakote-Indravati** 6× 2005, 2014, 2016, 2017, 2021, 2022
- ■ **Lake-state matching: Loktak-Manipur, Kolleru-AP, Pulicat-AP/TN, Ashtamudi-Kerala, Roopkund-Uttarakhand, Surajkund and Badkhal-Haryana** 5× 2005, 2010, 2012, 2015, 2020
- ■ **Chilika** (Odisha, **Northern Circars** coast): largest **brackish lagoon** on the east coast 4× 2008, 2016, 2021, 2021
- • Highest waterfall (UPPSC key): **Kunchikal, 455 m** (Karnataka) 2× 2015, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Waterfalls

- Highest waterfall: **Kunchikal**; height **455 m** 2× 2018, 2015
- **Jog-Krishna** is wrong; Jog is on the **Sharavati** 1× 2005
- **Budha Ghagh-Kanchi** is wrong; Budha Ghagh is on the **Burha**; **Dassam** is on the Kanchi 1× 2021
- Match: **Dudhsagar-Mandovi, Duduma-Machkund, Gokak-Ghataprabha, Jog-Sharavati** 1× 2017
- Match: **Dudhsagar-Goa, Barkana-Karnataka, Khandadhar-Odisha, Palani-Himachal** 1× 2022
- Waterfall in **Goa**: **Dudhsagar** 1× 2014
- **Chitrakote** (the "Niagara of India") is on the **Indravati**, in **Bastar** 2× 2016, 2017

Waterfall	River	State
Kunchikal	Varahi	Karnataka
Jog	Sharavati	Karnataka
Gokak	Ghataprabha	Karnataka
Barkana	Sita	Karnataka
Sivasamudram	Kaveri	Karnataka
Dudhsagar	Mandovi	Goa
Duduma	Machkund	Odisha-AP
Khandadhar	Korapani	Odisha
Chitrakote	Indravati	Chhattisgarh
Dhuandhar	Narmada	MP
Hundru	Subarnarekha	Jharkhand
Dassam	Kanchi	Jharkhand
Palani	Local stream	Himachal

TRAP ALERT

- **Palani waterfall** is in **Himachal** (Kullu), not the Palani hills of Tamil Nadu.
- **Nohkalikai** (Meghalaya) is the tallest single plunge in many sources; pick Kunchikal when it is offered.

8.2 Coastal Lagoons and Southern Lakes

- **Vembanad** lake is in **Kerala** 1× 1997
- Largest saline (brackish) lake: **Chilika**; largest **inland** salt lake: **Sambhar** 1× 2021
- **Chilika** lies on the **Northern Circars** coast (Odisha) 2× 2008, 2016
- Largest lagoon on the east coast: **Chilika** 1× 2021
- Lake shared by two states: **Pulicat** (Andhra Pradesh and Tamil Nadu) 1× 2008
- **Loktak-Meghalaya** is wrong; Loktak is in **Manipur** 1× 2005
- **Kolleru-Andhra Pradesh** is correct 2× 2010, 2012
- Match: **Ashtamudi-Kerala, Pulicat-Tamil Nadu, Roopkund-Uttarakhand, Surajkund-Haryana** 1× 2015
- Match: **Sala-Arunachal, Badkhal-Haryana, Loktak-Manipur, Kaliveli-Tamil Nadu** 1× 2020

8.3 Himalayan and Inland Lakes

- Largest **freshwater** lake: **Wular** (J&K; Jhelum) 1× 2021
- **Anchar** lake is in **Jammu and Kashmir** (Srinagar) 1× 2015
- "Mysterious lake" (skeletons): **Roopkund** (Uttarakhand) 1× 2015
- **Ghepan** lake is in **Himachal Pradesh** (Lahaul) 1× 2016
- **Chapanala** lake is in **Assam** 1× 2014
- **Fulhar** lake is in **Uttar Pradesh** (Pilibhit; source of the Gomti) 1× 2009

Lake	State / UT
Wular, Dal, Anchar	Jammu and Kashmir
Pangong Tso, Tso Moriri	Ladakh
Roopkund, Nainital	Uttarakhand
Ghepan, Chandratal	Himachal Pradesh
Sala (Sela)	Arunachal Pradesh
Chapanala	Assam
Loktak	Manipur
Sambhar, Pushkar	Rajasthan
Badkhal, Surajkund	Haryana
Lonar	Maharashtra
Kolleru, Pulicat	Andhra Pradesh
Kaliveli	Tamil Nadu

Lake	State / UT
Vembanad, Ashtamudi	Kerala
Chilika	Odisha
Fulhar	Uttar Pradesh

8.4 Hot Springs

- **Gujarat-Narayani** is wrong; **Narayani** is in **Rajasthan**; Gujarat has Tulsishyam and Unai **1×** 2021

Hot spring	State
Manikaran, Tattapani	Himachal Pradesh
Puga, Panamik	Ladakh
Taptapani, Atri	Odisha
Tulsishyam, Unai	Gujarat
Bakreshwar	West Bengal
Rajgir	Bihar
Sohna	Haryana
Narayani	Rajasthan

UP LINK

Fulhar (Phulhar) Tal in Pilibhit is UP's only lake asked so far: it is the source of the **Gomti**. UP has Ramsar sites such as **Nawabganj, Sandi, Samaspur, Parvati Arga, Saman, Sarsai Nawar, Sur Sarovar (Keetham)** and **Haiderpur**.

LAST-MINUTE 10

1. **Kunchikal** = highest waterfall (455 m)
2. **Jog** = Sharavati; **Gokak** = Ghataprabha
3. **Dudhsagar** = Mandovi, Goa
4. **Chitrakote** = Indravati, Bastar
5. **Chilika** = largest brackish lagoon, Northern Circars
6. **Sambhar** = largest inland salt lake
7. **Wular** = largest freshwater lake
8. **Loktak** = Manipur; **Pulicat** = AP-TN
9. **Roopkund** = mysterious lake; **Fulhar** = UP (Gomti)
10. **Narayani** hot spring = Rajasthan

Chapter 9

Climate and the Monsoon

Built from 25 UPPSC questions (1993–2025). Focus: monsoon branches, winter rain, rainfall extremes, local storms.

ASKED MOST · REVISE FIRST

- ■ **Winter rain**: north-west India from **western disturbances**; **Tamil Nadu** from the **north-east monsoon** **7×** 1993, 2008, 2009, 2012, 2019, 2021, one-liner
- ■ **Leh** gets the **least rainfall** among the options **3×** 2008, 2009, 2010
- India is a **monsoon** land; the **Himalaya** gives it climatic distinctiveness **2×** 1996, 1998
- **Tamil Nadu** coast stays **dry** in the south-west monsoon (parallel to the Bay branch, rain shadow of the Arabian branch) **1×** 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

9.1 Monsoon Land and the Himalaya

- India is **monsoonal** because of the **Himalaya** (A and R true, R explains A) **2×** 1996, 1998
- Himalayan vegetation varies because **climate changes with altitude** **1×** 2025
- **Southern** Himalayan slopes are **wetter and greener**; the claim that northern slopes are greener is false **1×** 2023
- **Agra** (16°C) is warmer than **Darjeeling** (4°C) in January because temperature **falls with altitude** **1×** 2022

9.2 Arrival and Branches of the Monsoon

- Monsoon reaches **Kerala** first on the mainland (Andaman and Nicobar earliest overall) **1×** 2006
- The Indian monsoon comes from the **south-west winds** **1×** 2016
- **Tamil Nadu** coast is **dry** in June-September (A and R true, R explains A) **1×** 2023

HOW THE SOUTH-WEST MONSOON SPREADS



9.3 Rainfall

- Highest average annual rainfall among the options: **Sikkim** 1× 2016
- Humid places among the options: **Kochi and Tezpur** 1× 2010
- Least rainfall: **Leh** 3× 2008, 2009, 2010
- "200 mm average" had no correct group (probably meant 200 cm) (*) 1× 1994
- India is a "thirsty land" despite heavy rain for **all the reasons given** (seasonal, uneven, run-off) 1× 1994
- Cost-effective water strategy: **rainwater harvesting** 1× 2011
- Inland waterways are weak because rain falls in **only four months** (R explains A) 1× 2008

Annual rainfall	Regions
Over 200 cm	West coast, north-east, Andaman and Nicobar
100-200 cm	East India, middle Ganga plain
50-100 cm	West UP, Punjab, Haryana, inner Deccan
Below 50 cm	West Rajasthan, Kutch, Ladakh

9.4 Winter Weather

- Winter is **dry** for most of India; NE trade winds (source key: R true but not the explanation) 1× 2023
- Most **north-east monsoon** rain: **Tamil Nadu**; Tamil Nadu's winter rain comes from it 2× 2009, 2008
- Winter rain in the **northern plains**: **western disturbances** 2× 1993, 2021
- Western-disturbance rain **decreases from west to east** 1× 2019
- Area with winter rain: the **north-west** 1× 2012
- Northern plains get winter rain, but **not because of the NE monsoon** (R not the explanation) 1× one-liner

TRAP ALERT

- **North-west** winter rain = **western disturbance**. **South-east** (Tamil Nadu) winter rain = **north-east (retreating) monsoon**. Never mix them.

9.5 Local Storms and Seasons

- Match: **Mango shower-Kerala, Blossom shower-coastal Karnataka, Kal Baisakhi-West Bengal, Loo-Punjab to Bihar** 1× 2024
- Local storm of **Bengal and Assam**, good for tea and rice: **Norwesters (Kal Baisakhi)** 1× 2023
- Hindu seasons in order: **Vasanta (spring), Grishma (summer), Varsha (rain), Sharad (autumn)** 1× 2021

Local wind	Region	Effect
Loo	Punjab to Bihar	Hot, dry, May-June
Kal Baisakhi	West Bengal, Assam	Thunder squalls; tea, jute, rice
Mango shower	Kerala, coastal Karnataka	Ripens mangoes
Blossom shower	Kerala, Karnataka	Coffee flowers
Western disturbance	North-west	Winter rain and snow

SIX HINDU SEASONS (RITU)

Vasanta
Spring: March-May

↓

Grishma
Summer: May-July

↓

Varsha
Rains: July-September

↓

Sharad
Autumn: September-November

↓

Hemanta
Pre-winter: November-January

↓

Shishira
Winter: January-March

UP LINK

UP's rainfall falls from **east to west**: eastern UP gets about 100-150 cm, western UP about 60-80 cm, because the Bay of Bengal branch weakens as it moves up the Ganga plain. The **Loo** blows across UP in May-June; western UP gets **western-disturbance** rain in winter.

LAST-MINUTE 10

1. **Himalaya** makes India a monsoon land
2. Monsoon reaches **Kerala** first (Andamans earliest)
3. Monsoon = **south-west** winds
4. **Tamil Nadu** dry in the SW monsoon; wet from the **NE monsoon**
5. North-west winter rain = **western disturbance**, decreasing west to east
6. **Leh** = least rain; **Mawsynram** = most
7. **Loo** = Punjab to Bihar; **Kal Baisakhi** = Bengal-Assam
8. **Mango shower** = Kerala; **blossom shower** = coffee
9. Temperature falls with **altitude** (Agra vs Darjeeling)
10. Seasons: Vasanta, Grishma, Varsha, Sharad, Hemanta, Shishira

Chapter 10

Natural Disasters

Built from 15 UPPSC questions (2000–2021). Focus: tsunami and cyclones, seismic zones, flood-prone areas, disaster bodies.

ASKED MOST · REVISE FIRST

- **East coast** has more cyclones than the west, but **not** because of the NE trade winds (R not the explanation) 1× 2003
- **Bihar** is the most flood-prone state; in UP the **eastern** region floods most 2× 2000, 2012
- India has **4 seismic zones** (II to V); zone **V** is the most dangerous 3× 2006, 2013, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Tsunamis and Cyclones

- Worst hit by the 2004 tsunami: **Coromandel coast** (Tamil Nadu) 1× 2014
- Tsunami Warning Centre: **Hyderabad** (INCOIS) 1× 2012
- **IMD** headquarters: **New Delhi** (est. 1875) 1× 2012
- **East coast** is more cyclone-prone (A true; R true but not the reason) 1× 2003
- Cyclone **Tauktae** formed in the **Arabian Sea** 1× 2021

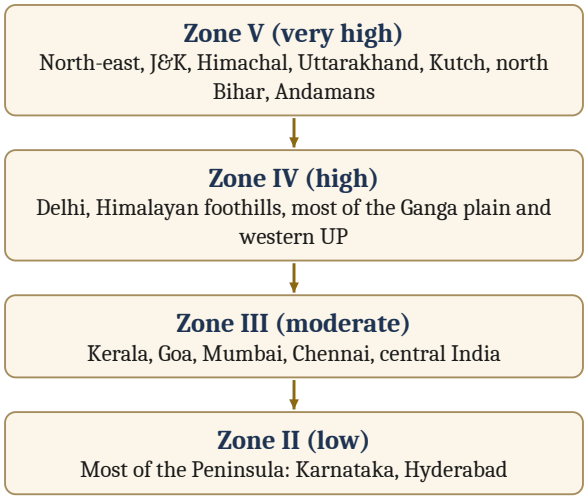
Cyclone	Year	Sea	Landfall
Fani	2019	Bay of Bengal	Odisha
Amphan	2020	Bay of Bengal	West Bengal
Tauktae	2021	Arabian Sea	Gujarat
Yaas	2021	Bay of Bengal	Odisha
Biparjoy	2023	Arabian Sea	Kutch
Michaung	2023	Bay of Bengal	Andhra Pradesh
Remal	2024	Bay of Bengal	West Bengal
Fengal	2024	Bay of Bengal	Puducherry

10.2 Earthquakes and Seismic Zones

- **Karnataka plateau** is **not** a high-seismic zone 1× 2006

- India has **4 seismic zones** 1× 2013
- City-zone pairs: Bhuj V, Srinagar V, Chennai III, Hyderabad II (three pairs wrong, *) 1× 2014
- **Koyna** is becoming quake-prone because the dam sits on an **old fault** (reservoir-induced) 1× 2001

SEISMIC ZONES, HIGHEST RISK TO LOWEST



TRAP ALERT

- **Zone I no longer exists**; zone II is the lowest. A Peninsular city like Hyderabad is never zone IV or V.

10.3 Landslides and Floods

- Himalayan **landslides** have increased because of **large-scale mining** (R explains A) 1× 2003
- Most flood-prone state: **Bihar** 1× 2000
- Most flood-hit part of UP: the **eastern** region 1× 2012
- State facing the most natural disasters among the options: **Odisha** 1× 2002
- True: damage is worst in **developing** countries; **Bhopal** gas tragedy was man-made; **mangroves** reduce cyclone damage (not "India is disaster-free") 1× 2018

10.4 Disaster Management Bodies

- India's first disaster management training institute: **Latur** (Maharashtra) 1× 2004

Body	Year	Role
NDMA	2005	Policy; chaired by the PM
NDRF	2006	Rescue battalions
NIDM	1995	Training, New Delhi
INCOIS	2007	Tsunami warning, Hyderabad
IMD	1875	Weather and cyclone warning

UP LINK

Eastern UP (Gorakhpur, Bahraich, Basti, Kushinagar) floods most, from the **Ghaghara, Rapti and Gandak** rising in Nepal. Western UP and **Delhi-NCR** lie in seismic **zone IV**; the Tarai belt near Nepal is high-risk too.

LAST-MINUTE 10

1. 2004 tsunami: **Coromandel coast**
2. Tsunami warning: **Hyderabad (INCOIS)**; IMD: **New Delhi**
3. East coast = more cyclones; **Tauktae** = Arabian Sea
4. **4 zones** (II-V); V most dangerous; Karnataka plateau is low
5. **Koyna** quakes: reservoir on an old fault
6. Himalayan landslides: **mining**
7. **Bihar** most flood-prone; **eastern UP** floods most
8. **Odisha** faces most disasters
9. Mangroves cut cyclone damage
10. First DM training institute: **Latur**

Chapter 11

Soils

Built from 29 UPPSC questions (1998–2024). Focus: black (regur) soil, laterite, usar and saline soils, erosion.

ASKED MOST · REVISE FIRST

- **Black (regur)** soil: from **Deccan basalt**, best for **cotton**, holds moisture; it is **not** rich in humus 10× 1998, 1999, 2004, 2010, 2010, 2013, 2014, 2015, 2016, 2018
- **Laterite**: heavy rain and leaching; **poor in lime**; Malabar coast, Maharashtra 4× 2000, 2001, 2006, 2023
- **Alkaline (usar)** soil: largest area in **UP**; **saline** soil: largest in **Gujarat** 4× 2007, 2012, 2017, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 Black (Regur) Soils

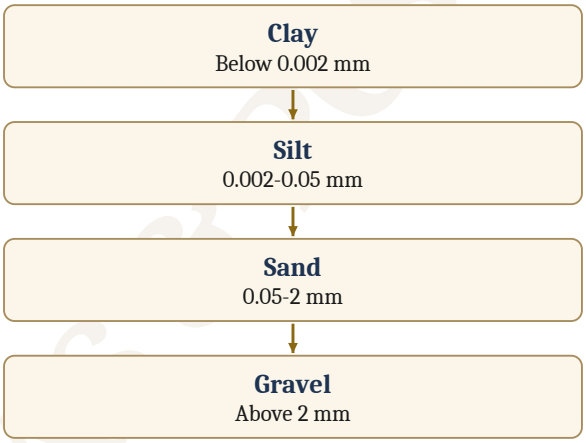
- Formed from weathered **basaltic lava**: **regur** 1× 2015
- Black cotton soil = **regur** 1× 2016
- Regur is **black**, but **not** because of humus (A true, R false) 1× 1999
- ■ Best for cotton: **regur**; good for cotton, not rich in humus 3× 2014, 2004, 2013
- Lava soils: **Malwa plateau**; Malwa's main soil: **black** 2× 1998, 2010
- Needs little irrigation because it **retains water**: **black** 1× 2010
- Wrong statement: "black soil is found in the **Himalaya**" 1× 2018

11.2 Red, Laterite and Alluvial Soils

- **Red soil**: second-largest group; it surrounds the black soil (both true, R not the explanation) 1× 2024
- Red colour comes from **diffused iron**, not high iron content; poor in N, P and humus (both true, R not the explanation) 1× 2023
- Laterite: only statement 1 correct 1× 2023
- Laterite is **not rich in lime** 1× 2001
- Laterite found in **Maharashtra**; mainly on the **Malabar coast** 2× 2006, 2000
- Most productive soil: **alluvial** 1× 2014
- Match: **Alluvial-UP, Black-Maharashtra, Red-Andhra Pradesh, Desert-Rajasthan** 1× 2021
- Tea needs **acidic** soil 1× 2002
- Most deficient micronutrient: **zinc** 1× 2004
- Himalayan soils are **not** rich in humus overall (A false, R true) 1× 2004
- Particle under **0.002 mm**: **clay** 1× 2016

Soil	Rich in	Poor in	Crop
Alluvial	Potash, lime	N, P	Wheat, rice, sugarcane
Black	Lime, iron, potash	N, P, humus	Cotton, soybean
Red	Iron (spread thin)	N, P, humus	Millets, groundnut
Laterite	Iron, aluminium	**Lime**, N, potash	Tea, coffee, cashew
Desert	Salts, phosphate	N, humus	Bajra
Saline/alkaline	Sodium salts	Calcium, N	Salt-tolerant crops

SOIL PARTICLE SIZES, FINEST TO COARSEST



11.3 Saline and Alkaline (Usar) Soils

- Usar soil: **pH above 7**; **paddy can grow** after reclamation; it is reclaimed with **gypsum**, not lime 1× 2019
- Largest **alkali** area: **Uttar Pradesh** 1× 2007
- Largest **saline** area: **Gujarat** 2× 2012, 2017
- Saline soils are unfit for crops because of **dry climate and poor drainage** (R explains A) 1× 2023

TRAP ALERT

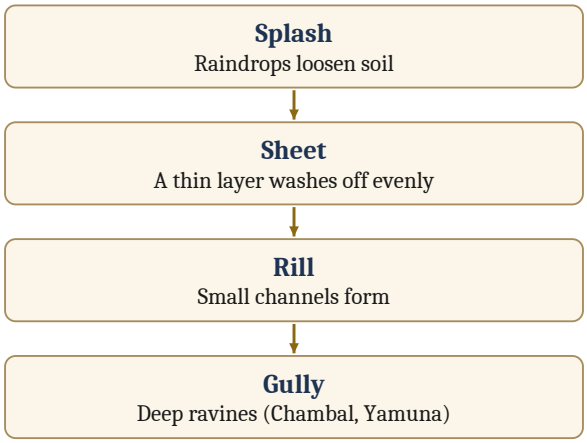
- **Saline** = Gujarat (Rann of Kutch). **Alkaline / usar** = Uttar Pradesh. **Gypsum** reclaims usar, not lime.

11.4 Soil Erosion

- Soil loss, smallest to largest: **splash, sheet, rill, gully** 1× 2023
- Critical erosion: **Shiwalik foothills**; worst affected: **Chambal valley** (ravines) 2× 2004, 2004
- Erosion is prevented by **afforestation** 1× 2014

QUANTUM IAS & PCS

EROSION STAGES (LEAST TO MOST SOIL LOSS)



UP LINK

UP has India's largest **usar (alkaline)** area: Raebareli, Unnao, Pratapgarh, Kanpur, Etawah. It is reclaimed with **gypsum**. The **Chambal-Yamuna ravines** (Etawah, Agra, Jalaun) are UP's worst gully-erosion zone. Most of UP is **alluvial** (Bhangar and Khadar); Bundelkhand has red and mixed soils.

LAST-MINUTE 10

1. **Regur** = basalt = cotton = holds water; not humus
2. Lava soil = **Malwa**
3. Red colour = **diffused** iron
4. **Laterite** = no lime; Malabar
5. **Alluvial** = most productive; UP
6. Tea = **acidic** soil; **zinc** most deficient
7. **Usar** = UP, pH > 7, gypsum
8. **Saline** = Gujarat
9. Erosion: splash, sheet, rill, **gully** (Chambal)
10. Clay < 0.002 mm

Chapter 12

Natural Vegetation and Forests

Built from 12 UPPSC questions (1991–2023). Focus: forest types and their trees, tree line, mangroves.

ASKED MOST · REVISE FIRST

- ■ **Teak and sal** belong to **tropical deciduous** forests (central India, Tarai) **3×** 1992, 2007, 2012
- • Forest-region matching: **moist deciduous-Tarai, dry deciduous-middle Ganga plain, alpine-Arunachal, evergreen-Sahyadri** **2×** 2001, 2017
- • Mangroves: **Coringa = Godavari delta**; **Sundarbans** = Ganga-Brahmaputra delta **2×** 2017, 2021

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

12.1 Forest Types and Trees

- **Teak and sal**: **tropical dry deciduous** forests **1×** 2007
- Teak grows in **MP** (largest), Karnataka, Maharashtra, UP; more than one option fits (*) **1×** 2012
- **Deodar-Nilgiri hills** is wrong; deodar grows in the **Himalaya** **1×** 2023
- Match: **Teak-central India, Deodar-high Himalaya, Sundari-Sundarbans, Cinchona-Himalayan Tarai** (per key) **1×** 1992
- **Cinchona** does **not** grow in **Chhattisgarh** **1×** 2010
- Match: **moist deciduous-Tarai, dry deciduous-middle Ganga plain, alpine-Arunachal, evergreen-Sahyadri** **1×** 2001
- Match: **tropical-Silent Valley, conifer-Himachal, mangrove-Sundarbans, deciduous-Rajasthan** (per key) **1×** 2017
- Plant with **no flowers**: **fern** **1×** 1991

FOREST BELTS, WETTEST TO DRIEST

Tropical evergreen
Over 200 cm: Western Ghats, north-east, Andamans

↓

Moist deciduous
100-200 cm: Tarai, sal belt

↓

Dry deciduous
70-100 cm: teak; middle Ganga plain, Deccan

↓

Thorn forest
Below 70 cm: khejri, babool; Rajasthan

12.2 The Himalayan Tree Line

- The tree line is **lower** in the **Western** Himalaya than in the Eastern (cooler and drier in the west) 1× 2005

HIMALAYAN VEGETATION WITH ALTITUDE

Foothills
Tropical moist deciduous: sal

↓

1,000-2,000 m
Subtropical pine (chir), oak

↓

1,500-3,000 m
Temperate: deodar, blue pine, fir, spruce

↓

3,000-3,600 m
Birch, rhododendron, juniper (tree line)

↓

Above 3,600 m
Alpine meadows (bugyals)

12.3 Social Forestry and Mangroves

- Multipurpose tree for **social forestry**: **khejri** 1× 2002
- **Vembanad-Karnataka** is wrong; Vembanad is in **Kerala** 1× 2002
- Mangrove in the **Godavari** delta: **Coringa** 1× 2021

Mangrove	State	Delta / setting
Sundarbans	West Bengal	Ganga-Brahmaputra; largest
Bhitarkanika	Odisha	Brahmani-Baitarani
Coringa	Andhra Pradesh	Godavari
Pichavaram	Tamil Nadu	Kaveri (Coleroon)
Gulf of Kutch	Gujarat	Second largest

TRAP ALERT

- **Coringa** = Godavari; **Pichavaram** = Kaveri; **Bhitarkanika** = Brahmani. **Vembanad** is a Kerala lake.

UP LINK

UP's forests are mostly **tropical dry deciduous** (sal and teak) in the **Tarai** (Dudhwa, Pilibhit) and the **Vindhyan** south (Sonbhadra, Mirzapur). UP has a small forest cover (about 6-7% of its area); **Sonbhadra** has the most forest.

LAST-MINUTE 10

1. **Teak, sal** = tropical deciduous
2. **Deodar** = Himalaya, not Nilgiris
3. Evergreen = **Sahyadri** / Silent Valley
4. Moist deciduous = **Tarai**
5. Tree line **lower in the west**
6. **Khejri** = social forestry
7. **Coringa** = Godavari delta
8. **Sundarbans** = largest mangrove
9. Cinchona not in Chhattisgarh
10. Fern = no flowers

Chapter 13

Irrigation and Canals

Built from 17 UPPSC questions (1992–2024). Focus: tube wells in UP, Indira Gandhi Canal, UP canals, canal-river matches.

ASKED MOST · REVISE FIRST

- **Uttar Pradesh** has the largest area under **tube-well** irrigation 4× 1995, 2006, 2017, one-liner
- **Indira Gandhi Canal** (Rajasthan): from **Harike barrage**; water of the **Satluj, Beas and Ravi**; uses the **Pong** dam on the Beas 5× 1992, 1999, 2010, 2013, 2024
- **Sharda Sahayak** project aims at more output, multiple cropping and better land management 3× 2001, 2002, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Sources of Irrigation

- **Ponds (tanks)** are a major source in the Peninsula because its rivers are **seasonal** (R explains A) 1× 2015
- Highest share of cropped area irrigated (2011) among the options: **Haryana** (Punjab tops overall) 1× 2015
- Largest tube-well area: **Uttar Pradesh** 3× 2006, 2017, one-liner
- Tube-well growth was most spectacular in the **Saryupar plain** (eastern UP) 1× 1995
- Life-saving (protective) irrigation means **PWP irrigation** 1× 2010

Source	Share	Leading states
Tube wells and wells	About 64%	UP, Punjab, Haryana
Canals	About 24%	Punjab, Haryana, UP, Rajasthan
Tanks	About 3%	Tamil Nadu, Telangana, AP

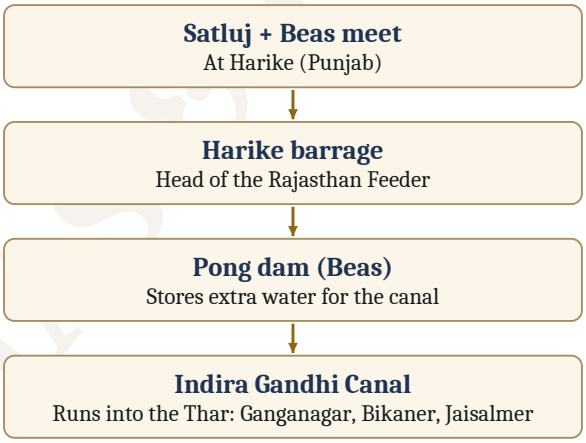
TRAP ALERT

- Highest **percentage** irrigated = **Punjab** (then Haryana). Largest **area** and **tube-well** area = **UP**. Tanks = the **south**.

13.2 Canals and Projects

- **Lower Ganga Canal** starts at **Narora** 1× 2010
- **Sharda Sahayak** aims: more production, multiple cropping, land reform (all three) 3× 2001, 2002, 2008
- Match: **Upper Bari Doab-Ravi, Sirhind-Satluj, Gurgaon-Yamuna, Indira Gandhi-Satluj, Ravi, Beas** 1× 2024
- Indira Gandhi Canal: from **Beas and Satluj**; source at **Harike barrage**; water from **Beas, Ravi, Satluj** 3× 1992, 1999, 2010
- Project using the **Pong** barrage on the Beas: **Indira Gandhi Canal** 1× 2013
- **Jawai project-Tamil Nadu** is wrong; Jawai is in **Rajasthan** 1× 2019
- **Mangalam** irrigation project: **Kerala** 1× 1998
- **Garland Canal** was proposed by **Dinshaw J. Dastur** 1× 1999

INDIRA GANDHI CANAL: WHERE ITS WATER COMES FROM



UP canal	Takes off from
Upper Ganga	Ganga at Haridwar (1854)
Lower Ganga	Ganga at **Narora**
Eastern Yamuna	Yamuna at Hathnikund
Agra Canal	Yamuna at Okhla
Sharda Canal	Sharda at Banbasa
Sharda Sahayak	Ghaghara-Sharda link
Saryu Nahar	Ghaghara-Saryu-Rapti (2021)
Betwa Canal	Betwa at Parichha

UP LINK

UP leads India in **tube-well** irrigation and has the **largest irrigated area**. Its big canal systems are the **Upper and Lower Ganga**, **Sharda Sahayak** (largest in UP) and the **Saryu Nahar** (completed 2021, nine eastern districts).

LAST-MINUTE 10

1. **UP** = largest tube-well area; **Saryupar** plain boom
2. **Punjab/Haryana** = highest % irrigated
3. Tanks = Peninsula (seasonal rivers)
4. **Lower Ganga Canal** = Narora; Upper = Haridwar
5. **Indira Gandhi Canal**: Harike; Satluj, Beas, Ravi; Pong dam
6. **Sirhind** = Satluj; **Upper Bari Doab** = Ravi; **Gurgaon** = Yamuna
7. **Jawai** = Rajasthan; **Mangalam** = Kerala
8. **Garland Canal** = Dastur
9. Protective irrigation = **PWP**
10. **Sharda Sahayak** = UP's largest canal

Chapter 14

Multipurpose River Valley Projects and Hydropower

Built from 52 UPPSC questions (1991–2025). Focus: dam-river and dam-state matches, Nagarjuna Sagar, Sardar Sarovar, Indus projects, UP projects.

ASKED MOST · REVISE FIRST

- **Nagarjuna Sagar** is on the **Krishna** (not the Godavari or Kaveri), in **Andhra Pradesh-Telangana** 7× 1991, 1993, 2004, 2004, 2008, 2011, 2012
- **Dam-river matches**: **Hirakud-Mahanadi, Matatila-Betwa, Pochampad-Godavari, Idukki-Periyar, Gandhi Sagar-Chambal, Ukai and Kakrapar-Tapi, Rihand-Rihand** 7× 1995, 2002, 2003, 2014, 2016, 2017, 2019
- **Tulbul** = **Jhelum**; **Baglihar** = **Chenab** (both raised by Pakistan) 6× 2000, 2007, 2008, 2009, 2016, 2025
- **Sardar Sarovar** (Narmada): benefits **Gujarat, MP, Maharashtra, Rajasthan**; MP gets the largest share of power 4× 1992, 1993, 1999, 2004
- First hydropower station: **Sidrapong, Darjeeling** (1897); **Sivasamudram** (Karnataka, 1902) is the oldest big one 6× 1998, 2008, 2008, 2009, 2011, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Hydropower

- Huge hydro potential in the Himalaya; the Nepal border does **not** stop eastern UP and Bihar (only statement 1 true) 1× 2025
- Oldest hydropower station among the options: **Sivasamudram** 1× 2008
- Oldest power plant: **Sidrapong**; first station at **Darjeeling** 2× 1998, 2013
- **Sivasamudram** is in **Karnataka** (not Kerala) 3× 2008, 2009, 2011
- Bhutan plant opened by PM Modi (2019): **Mangdechhu** 1× 2019

14.2 Bhakra and the Indus Projects

- **Bhakra-Nangal**: joint project of **Punjab, Haryana and Rajasthan** 1× 1994
- **Kol Dam** is on the **Satluj** 1× 2017
- **Siul (Baira Siul)** project: **Himachal Pradesh** 1× 2010
- **Baglihar** is on the **Chenab** 2× 2008, 2009
- **Tulbul** is on the **Jhelum** (Wular barrage) 3× 2000, 2007, 2016

QUANTUM

- Wrong pairs: **Pandoh-Ravi** (it is on the Beas) and **Srisailam-Tungabhadra** (it is on the Krishna) 1× 2025

Project	River
Bhakra, Nangal, Kol, Nathpa Jhakri	Satluj
Pong, Pandoh	Beas
Ranjit Sagar, Baira Siul	Ravi (system)
Baglihar, Dulhasti, Salal	Chenab
Tulbul, Kishanganga	Jhelum

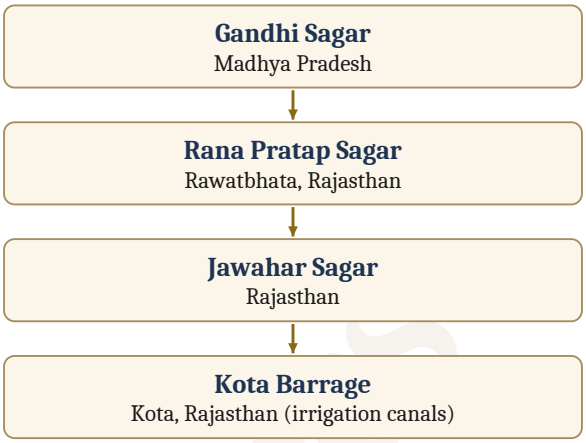
14.3 The Narmada Projects

- **Sardar Sarovar** beneficiaries: **Gujarat, Maharashtra, MP, Rajasthan** 2× 1999, 2004
- Sardar Sarovar is on the **Narmada** 1× 1992
- Most benefit (by water and power share): **Madhya Pradesh** 1× 1993
- **Harsud** town was submerged by the **Indira Sagar** reservoir 1× 2003
- **Tawa project-Krishna** is wrong; the Tawa is a **Narmada** tributary 1× 2015

14.4 Chambal, Damodar and Others

- **Gobind Sagar** (Bhakra) and **G.B. Pant Sagar** (Rihand) are **not** Chambal projects 1× 2011
- **Govind Ballabh Pant Sagar** is in **Uttar Pradesh** (Sonbhadra) 1× 2013
- Benefits shared by more than one state: **Chambal Valley** (MP and Rajasthan) 2× 2007, 2009
- **Kaveri dispute**: **Tamil Nadu, Karnataka, Kerala, Puducherry** 1× 2010
- **Damodar** was Bengal's "River of Sorrow": fast upstream, sluggish downstream (R explains A) 1× 2010
- **Tilaiya-Konar** is wrong; Tilaiya is on the **Barakar** 1× 2013
- Match: **Tilaiya-Barakar, Panchet-Damodar, Rana Pratap Sagar-Chambal, Matatila-Betwa** 1× 2014
- All true: Rihand on a **Son** tributary; Hirakud on the **Mahanadi**; Tungabhadra is **AP-Karnataka**; Maithon on the **Barakar** 1× 2016

CHAMBAL VALLEY PROJECT, UPSTREAM TO DOWNSTREAM



TRAP ALERT

- **Rana Pratap Sagar** is in **Rajasthan**, not MP; **Gandhi Sagar** is in MP.
- **Tilaiya and Maithon** = Barakar; **Panchet** = Damodar; **Konar** = Konar.

14.5 Peninsular Projects

- **Nagarjuna Sagar**: **Andhra Pradesh**; on the **Krishna**; not Godavari; not Kaveri 7× 1993, 1991, 2004, 2008, 2011, 2012, 2004
- Match: **Bhakra-Satluj, Hirakud-Mahanadi, Idukki-Periyar, Nagarjuna Sagar-Krishna** 1× 1995
- Match: **Idukki-Periyar, Matatila-Betwa, Nagarjuna Sagar-Krishna, Pochampad-Godavari** 2× 2002, 2003
- Match: **Rihand-Rihand, Nagarjuna Sagar-Krishna, Pochampad-Godavari, Matatila-Betwa** 1× 2016
- Match: **Mettur-Tamil Nadu, Mayurakshi-West Bengal, Nagarjuna Sagar-AP, Hirakud-Odisha** 1× 2008
- Match: **Farakka-WB, Ghataprabha-Karnataka, Hirakud-Odisha, Kakrapar-Gujarat** 1× 2003
- Match: **Dulhasti-Chenab, Gandhi Sagar-Chambal, Ukai-Tapi, Tawa-Tawa** 1× 2017
- Match: **Betwa-Matatila, Mahanadi-Hirakud, Chambal-Gandhi Sagar, Tapi-Kakrapar** 1× 2014
- Match: **Bhadra-Karnataka, Bhavani Sagar-Tamil Nadu, Gandhi Sagar-MP, Rana Pratap Sagar-Rajasthan** 1× 2006

- **Rana Pratap Sagar-MP** is wrong (Rajasthan) 1× 2004
- **Ukai-Tapi, Jayakwadi-Godavari, Khadakwasla-Mutha** 1× 2019
- **Telugu Ganga**: joint project of **TN, AP, Karnataka**; supplies **Chennai** 2× 1999, 1996
- **Babli** project (Godavari dispute): **Maharashtra** 1× 2012
- **Teesta Low Dam III**: **West Bengal** 1× 2003
- **Kalpong** hydro project: **Andaman and Nicobar** 1× 2002
- **Kalpasar** freshwater reservoir: **Gujarat** (Gulf of Khambhat) 1× 2010
- **Malampuzha** dam is on the **Malampuzha** river; **Cheruthoni** dam on the **Periyar** 2× 2017, 2017

Project	River	State
Hirakud	Mahanadi	Odisha
Nagarjuna Sagar, Srisailam	Krishna	AP-Telangana
Pochampad, Jayakwadi, Polavaram	Godavari	Telangana, Maharashtra, AP
Mettur, Krishna Raja Sagar	Kaveri	TN, Karnataka
Idukki, Cheruthoni	Periyar	Kerala
Ukai, Kakrapar	Tapi	Gujarat
Bhadra	Bhadra	Karnataka
Bhavani Sagar	Bhavani	Tamil Nadu
Mayurakshi	Mayurakshi	West Bengal

14.6 UP and Neighbourhood Projects

- **Tehri** dam is on the **Bhagirathi** 1× 2002
- Ramganga project: the **825.8 m length** pair is wrong (about 715 m) 1× 2024
- Match: **Rihand-Son, Rani Laxmibai (Rajghat)-Betwa, Tehri-Bhagirathi, Ramganga-Ramganga** 1× 2010
- **Singrauli** small hydro project: **Uttar Pradesh** 1× 2017

UP LINK

UP projects: **Rihand dam** (G.B. Pant Sagar, Sonbhadra) on the Rihand, a **Son** tributary; **Obra**; **Rajghat** (Rani Laxmibai Sagar) and **Matatila** on the **Betwa** (UP-MP); **Ramganga (Kalagarh)** irrigates western UP; **Tehri** was built when Uttarakhand was part of UP.

LAST-MINUTE 10

1. **Nagarjuna Sagar** = Krishna, AP-Telangana
2. **Hirakud** = Mahanadi; **Pochampad** = Godavari; **Idukki** = Periyar
3. **Matatila, Rajghat** = Betwa; **Rihand** = Son tributary (UP)
4. **Tulbul** = Jhelum; **Baglihar** = Chenab; **Pandoh** = Beas
5. **Bhakra** = Satluj; Punjab, Haryana, Rajasthan
6. **Sardar Sarovar** = Narmada; MP most power; Indira Sagar drowned Harsud
7. **Gandhi Sagar** MP; **Rana Pratap Sagar** Rajasthan
8. **Tilaiya, Maithon** = Barakar; Damodar = Sorrow of Bengal
9. **Telugu Ganga** = water to Chennai
10. **Sidrapong** (Darjeeling) first; **Sivasamudram** Karnataka

Chapter 15

Agriculture and the Green Revolution

Built from 33 UPPSC questions (1994–2024). Focus: colour revolutions, Green Revolution facts, agricultural regions, Pantnagar.

ASKED MOST · REVISE FIRST

- **Yellow Revolution** = **oilseeds** 4× 2002, 2004, 2014, 2017
- **Colour revolutions** (match): Green-food grains, White-milk, Blue-**fish** (not poultry), Grey-fertilisers, Pink-onion 5× 2010, 2010, 2014, 2015, 2016
- **Wheat** gained most from the Green Revolution 3× 1994, 2001, 2015
- India has **20 agro-ecological regions** 3× 2010, 2013, 2016
- First agricultural university: **Pantnagar** (GBPUAT, **1960**) 4× 2010, 2013, 2014, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Regions, Scholars and Institutions

- *The History of Indian Agriculture*: **M.S. Randhawa** 1× 2015
- Agro-ecological regions: **20** 3× 2010, 2013, 2016
- Sengupta and Sdasyuk (1968): **60** micro agricultural regions 1× 2017
- First agricultural university: **Pantnagar, 1960** (true); the "six colleges in 1901-05" claim is treated as false 1× 2023
- Year: **1960**; name: **GBPUAT, Pantnagar** 2× 2013, 2014
- **Zero-till seed-cum-fertiliser drill** was developed at **Pantnagar** 1× 2010

Scheme	By	Number
Agro-ecological regions	NBSS&LUP (ICAR)	**20**
Agro-climatic zones	Planning Commission (1989)	**15**
Micro agricultural regions	Sengupta and Sdasyuk	**60**
NARP zones	ICAR	127

15.2 Structure and Problems

- Indian agriculture is mainly **a means of livelihood** 1× 2004
- True: India has more **cultivated area** than the USA, China, Japan; tropical monsoon allows **year-round cropping** 1× 2024
- Low productivity has **all the given causes** (small holdings, rain dependence, old methods) 1× 2007
- **Cooperative farming** is **not** a cause of low productivity 1× 2003
- Largest average holding among the options: **Punjab** (Nagaland largest overall) 1× 2010
- Dry zone is agrarian; it has second Green Revolution potential (both true, R not the explanation) 1× 2010
- High productivity in the south: **Tamil Nadu coast** 1× 2004
- Richest in replenishable **groundwater**: **Uttar Pradesh** 1× 2006
- Pioneer in **contract farming**: **Punjab** 1× 2005

15.3 Fertilisers and Green Manure

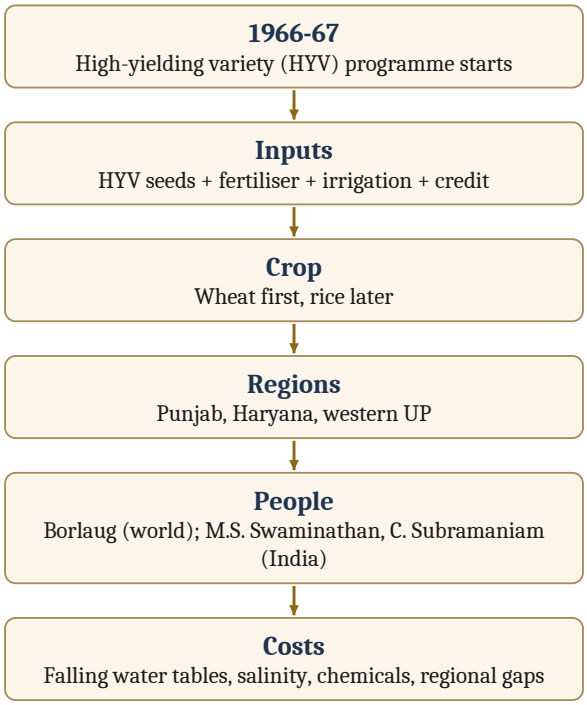
- Largest fertiliser users (at that time): **UP and Andhra Pradesh**; per hectare: **Punjab** 1× 2009
- **Balanced fertilisers** serve all the stated purposes 1× 2008
- Green manure with the most nitrogen among the options: **cowpea (lobia)** 2× 2016, 2008
- **Green agriculture** = integrated pest management, nutrient supply and resource management 1× 2016

15.4 The Green Revolution

- New Agricultural Strategy: **1966-67** (key says "seventies", meaning the seventh decade) 1× 2015
- World "Father of the Green Revolution": **Norman Borlaug** 1× 2015
- "**Evergreen Revolution**": **M.S. Swaminathan** 1× 2015
- Green Revolution = **wheat** production 1× 2015
- Biggest gain in output and yield: **wheat** 2× 1994, 2001
- Wheat yield reached about **2,000 kg per hectare** 1× 2003
- It modernised farming; the claim of little social or environmental cost is debatable (key: both true) 1× 2014

QUANTUM IAS & PCS

GREEN REVOLUTION IN BRIEF



15.5 Colour Revolutions

- Match: **food grains-Green, milk-White, fisheries-Blue, fertilisers-Grey** 1× 2014
- **Pink** Revolution: **onion** (UPPSC key) 1× 2010
- **Blue-poultry** is wrong; Blue = **fish**; Blue Revolution = **fisheries** 3× 2015, 2016, 2010
- **Yellow** = **oilseeds** 4× 2002, 2004, 2014, 2017

Revolution	Product
Green	Food grains
White	Milk
Blue	Fish
Yellow	Oilseeds
Golden	Fruits, honey
Silver	Eggs, poultry
Pink	Onion, prawns, meat
Red	Meat, tomato
Grey	Fertilisers
Brown	Leather, cocoa
Black	Petroleum
Round	Potato
Sweet	Honey

UP LINK

Pantnagar university (GBPUAT, 1960) was set up in UP (now Uttarakhand). **Western UP** was a Green Revolution core region with Punjab and Haryana. UP is the largest fertiliser consumer and the richest in **replenishable groundwater**. Average holding in UP: about **0.73 ha**.

LAST-MINUTE 10

1. **20** agro-ecological regions; **15** agro-climatic zones; **60** micro regions
2. **Pantnagar 1960** = first agricultural university; zero-till drill
3. **Randhawa** = History of Indian Agriculture
4. Green Revolution: **1966-67**, **wheat**, Punjab-Haryana-west UP
5. **Borlaug** (world), **Swaminathan** (India; Evergreen Revolution)
6. **Yellow** = oilseeds; **Blue** = fish; **Grey** = fertilisers; **Pink** = onion
7. **UP** richest in groundwater; **Punjab** largest holdings, contract farming pioneer
8. Green manure: **cowpea (lobia)**
9. Cooperative farming is not a cause of low yield
10. Tamil Nadu coast = high productivity in the south

Chapter 16

Major Crops I: Crop Seasons, Wheat and Rice

Built from 44 UPPSC questions (2001–2024). Focus: kharif and rabi crops, wheat varieties and diseases, rice area, yield and production.

ASKED MOST · REVISE FIRST

- **Wheat varieties**: Kalyan Sona, Sonalika, Raj 3077, Pusa Sindhu Ganga, UP-308 are **wheat** 4× 2005, 2015, 2016, 2016
- **Rice**: largest **area** = **UP**; highest **yield** = **Punjab**; production leader = West Bengal (UP in 2024-25) 4× 2006, 2007, 2015, 2017
- **Rabi** crops: wheat, **mustard**, **gram**; **kharif**: rice, **soybean** 4× 2004, 2011, 2012, 2016
- **Rice varieties**: Jaya, Padma, Krishna, Pusa Sugandha-5, Mahi Sugandha, Barani Deep, Pusa RH-10 5× 2006, 2006, 2013, 2015, 2015
- Wheat diseases: **rust**, **Karnal bunt** 3× 2003, 2004, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Crop Seasons

- **Mustard** is a **rabi** crop; not kharif 2× 2016, 2012
- **Soybean** is a **kharif** crop 1× 2004
- **Gram** is **not** a kharif crop (it is rabi) 1× 2011
- **Zaid** crops in irrigated areas: **moong and urad** 1× 2011
- Rabi crops lead in Punjab-Haryana, but winter rain there comes from **western disturbances**, not cyclones (A true, R false) 1× 2023
- **Rice** is a **transplanted** crop 1× 2017

Season	Sown	Harvested	Crops
Kharif	June-July	Sept-Oct	Rice, maize, bajra, arhar, cotton, soybean, groundnut
Rabi	Oct-Nov	March-April	Wheat, barley, gram, mustard, pea, potato
Zaid	March	June	Melons, cucumber, moong, urad

16.2 Wheat: States and Output

- Top three wheat states: **UP, Punjab, Haryana** (then; MP now second) 1× 2006
- Largest wheat producer (2015-16): **Uttar Pradesh** 1× 2016
- UP ranks first in **wheat and sugarcane** (and rice since 2024-25) (*) 1× 2008
- UP's biggest cropping pattern is **rice-wheat**; wheat's most critical irrigation is **crown root initiation**, not flowering 1× 2013
- Wheat output **increased** from 2006-07 to 2009-10 1× 2009

16.3 Wheat Varieties and Diseases

- Induced mutation: **Sonora-64** in the key (strictly **Sharbati Sonora**, from Sonora-64) 1× 2015
- Dwarfing gene in wheat: **Norin-10** 1× 2013
- **Kalyan Sona**, **Raj 3077**, **Pusa Sindhu Ganga**, **UP-308** are **wheat** 4× 2005, 2015, 2016, 2016
- **Macaroni (durum)** wheat suits **unirrigated** conditions 1× 2004
- **Triticale** = **wheat × rye** 1× 2014
- Wheat disease: **rust**; **Karnal bunt** 3× 2003, 2004, 2015
- Arhar for double cropping with wheat: **UPAS-120** 1× 2010

Disease	Crop
Rust, Karnal bunt, loose smut	Wheat
Blast; Khaira (zinc lack)	Rice
Tikka	Groundnut
Red rot	Sugarcane
Wilt	Arhar, gram

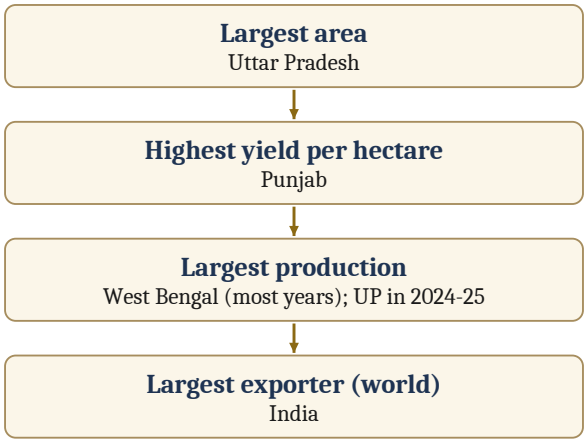
16.4 Rice: Area and Production

- Rice originated in **South-East Asia** 1× 2015
- Chief food crop of India: **rice** 1× 2004
- Largest crop area: **rice** 1× 2011
- Rice needs over **100 cm** of rain 1× 2007
- Largest rice **area**: **UP**; largest producer (then): UP per the key 2× 2007, 2006
- Wrong pair: Bihar's share of rice output is **not 15.7%** 1× 2024
- **Aman** paddy is sown in **June-July** (harvested in winter) 1× 2011
- More than half of rice output: the leading states named in the key (*) 2× 2004, 2009
- Four main rice producers (then): **WB, Punjab, UP, AP** (*) 1× 2004

QUANTUM IAS & PCS

- Ascending order: **Tamil Nadu, Punjab, Andhra Pradesh, West Bengal** 1× 2003
- Highest rice **yield**: **Punjab** 2× 2015, 2017
- Punjab **exports** rice, but it does **not** lead in production (A true, R false) 1× 2001

RICE: THREE DIFFERENT "LARGEST" ANSWERS



16.5 Rice Varieties

- **Jaya, Padma, Krishna**: **rice** 1× 2006
- **Pusa Sugandha-5**: aromatic **rice** 1× 2006
- **Mahi Sugandha**: **rice** 1× 2015
- **Barani Deep**: **paddy** 1× 2013
- Hybrid basmati: **Pusa RH-10** 1× 2015
- Seed rate for transplanted basmati: **15-20 kg/ha** 1× 2014
- Largest area under **hybrid rice**: **Uttar Pradesh** 1× 2014

TRAP ALERT

- **Gram and mustard are rabi**; **soybean is kharif**. **Moong and urad** can be kharif or zaid.
- **Norin-10** = wheat dwarfing gene; **Dee-geo-woo-gen** = rice.

UP LINK

UP is India's **largest wheat producer**, has the **largest rice area** and the largest **hybrid rice** area, and became the top **rice producer** in 2024-25. Its main cropping pattern is **rice-wheat**. Wheat variety **UP-308** and arhar **UPAS-120** come from UP-Pantnagar breeding.

LAST-MINUTE 10

1. Rabi: **wheat, mustard, gram**; kharif: **rice, soybean**
2. Zaid: **moong, urad**, melons
3. **Norin-10** = wheat dwarfing gene; **triticale** = wheat × rye
4. Kalyan Sona, Raj 3077, UP-308 = wheat
5. Jaya, Padma, Pusa Sugandha = rice
6. **Rust, Karnal bunt** = wheat
7. Rice: **UP** area, **Punjab** yield, **WB** production (UP 2024-25)
8. Rice needs **100 cm+** rain; originated in SE Asia
9. **Aman** paddy sown June-July
10. UP = largest wheat producer; rice-wheat pattern

Chapter 17

Major Crops II: Cotton, Sugarcane, Oilseeds and Pulses

Built from 44 UPPSC questions (2001–2023). Focus: UP and sugarcane, the southward shift of the sugar industry, soybean, oilseed and pulse varieties.

ASKED MOST · REVISE FIRST

- **UP** leads in sugarcane **area and production**; it is the **"Sugar Bowl"** of India 4× 2008, 2013, 2014, 2015
- Sugar industry shifted **south**: higher cane yield, more sucrose, longer crushing season (R explains A) 2× 2007, 2017
- **MP** = largest **soybean** area and (until 2023-24) production 4× 2001, 2003, 2014, 2015
- **Mustard** varieties: **Varuna, Pusa Bold, Pitambari** 3× 2007, 2013, 2015
- **Arhar** varieties: **Bahar, Malaviya Chamatkar**; Neelam is **not** arhar (it is a mango) 4× 2008, 2008, 2013, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

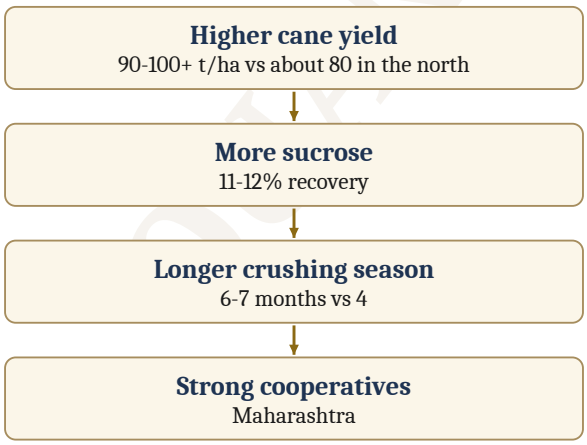
17.1 Cotton

- "**White gold**" of Maharashtra: **cotton** 1× 2002
- Cotton fibre comes from the **seed** (seed hairs) 1× 2009

17.2 Sugarcane and Sugar

- Largest sugarcane area: **UP**; leading producer: **UP** 2× 2015, 2013
- Production order (2013): **UP, Maharashtra, Karnataka, Haryana** 1× 2014
- "**Sugar Bowl**" of India: **UP** 1× 2008
- Crop with the highest **irrigated share** of its area: **sugarcane** 1× 2013
- First sugar mill (1903): **Pratappur** (Deoria, UP) 1× 2013
- Sugarcane breeding: **Coimbatore** (Sugarcane Breeding Institute) 2× 2009, 2010
- **Adsali** cane takes about **18 months** to ripen 1× 2010
- India: over 15% of world sugar, sugar is the 2nd largest agro-industry, India is the **largest consumer**; not the largest producer (Brazil) 1× 2009
- Shift to the south: **higher yield, more sucrose, longer season** (not cheap labour) 1× 2007
- South has more sugar industry because of **higher yield and juice** (R explains A) 1× 2017
- **Sakkar Nagar** (Bodhan) was in AP, now **Telangana** (*) 1× 2008

WHY SUGAR MOVED SOUTH



17.3 Oilseeds

- Leading soybean producer: **Madhya Pradesh** (Maharashtra in 2024-25) 3× 2001, 2003, 2015
- Largest soybean **area**: **MP** 1× 2014
- Largest oilseed producer (2006): **MP**; now **Rajasthan** 1× 2006
- Largest **groundnut** producer: **Gujarat** 1× 2006
- Low area but very high groundnut yield: **West Bengal** 1× 2014
- **Pegging** happens in **groundnut**; groundnut needs much **gypsum** 2× 2010, 2007
- **Kaushal**: groundnut 1× 2002
- **Varuna**, **Pusa Bold**, **Pitambari**: **mustard** 3× 2007, 2013, 2015
- Highest oil content: **sesame** 1× 2017
- **Kusum** (safflower) = *Carthamus tinctorius* 1× 2016

Oilseed	Oil %	Leading state
Sesame	46-52 (highest)	WB, Gujarat
Groundnut	45-50	Gujarat
Mustard	38-42	Rajasthan
Soybean	18-20	MP, Maharashtra
Safflower	30-35	Maharashtra
Castor	45-50	Gujarat

17.4 Pulses

- India is the largest **producer and consumer** of pulses 1× 2014
- India does **not** usually export **pulses** (it imports them) 1× 2013
- Largest pulse producer: **Madhya Pradesh** 1× 2004
- India lacks pulses but not protein (R explains A, per key) 1× 2009
- **Green gram (moong)** is **not** a rabi pulse 1× 2023
- **Legumes** fix nitrogen from the air 1× 2006
- Nutrient essential for nodules: **cobalt** 1× 2004
- Poor nitrogen fixer: **rajma** (*) 1× 2009
- NPK ratio for legumes: **1:2:2** 1× 2013
- Pulses complement cereals in intercropping (both true, R not the explanation per key; the real reason is nitrogen fixation) 1× 2023
- **Red gram (arhar)** originated in **India** 1× 2007
- **Malaviya Chamatkar**: **arhar**; **Bahar**: **arhar** 3× 2014, 2008, 2013
- **Arhar-Neelam** is wrong 1× 2008

- Leafless pea: **Aparna** 1× 2013

Kharif pulses	Rabi pulses
Arhar (tur)	Gram (largest pulse)
Moong (green gram)	Masoor (lentil)
Urad (black gram)	Pea
Lobia (cowpea)	Rajma, khesari

UP LINK

UP is the **Sugar Bowl**: top in cane area and output; India's first sugar mill was at **Pratappur** (Deoria, 1903). UP-bred varieties: arhar **Bahar** and **Narendra Arhar** (Ayodhya), **Malaviya Chamatkar** (BHU), pea **Azad** (Kanpur), groundnut **Kaushal, Chandra** (Kanpur).

LAST-MINUTE 10

1. Cotton = "white gold"; fibre from the **seed**
2. **UP** = Sugar Bowl; first mill **Pratappur 1903**
3. Cane breeding at **Coimbatore**; Adsali = **18 months**
4. Sugar moved south: yield, sucrose, longer season
5. **MP** = soybean and pulses; **Gujarat** = groundnut; **Rajasthan** = oilseeds now
6. **Sesame** = highest oil; groundnut = pegging, gypsum
7. **Varuna, Pusa Bold, Pitambari** = mustard
8. **Bahar, Malaviya Chamatkar** = arhar; **Aparna** = leafless pea
9. Moong is **not** rabi; legumes fix nitrogen; **cobalt** for nodules
10. India: largest producer, consumer and importer of pulses

Chapter 18

Silk, Plantation Crops and Spices

Built from 34 UPPSC questions (1994–2023). Focus: silk types and states, tea, coffee, rubber, spices, commodity boards.

ASKED MOST · REVISE FIRST

- **Karnataka** = largest **coffee** producer (over 70%) and largest coffee area 6× 2002, 2002, 2004, 2006, 2010, one-liner
- **Kerala** = largest **rubber** producer; rubber is never Himachal 4× 2004, 2014, 2017, 2022
- **Karnataka** leads in **mulberry silk**, silk yarn and fabric 4× 2010, 2013, 2017, 2017
- **Kerala** = "**Garden of Spices**"; leads in cardamom and pepper 4× 2010, 2011, 2013, 2018
- **Muga** silk comes only from **Assam** (India); **Eri** = Assam 3× 2008, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Silk

- Over three-quarters of raw silk came from **Karnataka and AP** in 2006 (no longer true, *) 1× 2006
- Leading **mulberry** silk state: **Karnataka** 1× 2017
- **Eri silk-Assam** is correct 1× 2008
- True: Karnataka = largest raw silk producer; **Jharkhand** = largest **tasar** producer 1× 2010
- **Muga** silk is found only in India, in **Assam** 1× 2008
- Largest silk **fabric** producer: **Karnataka**; most silk **yarn**: **Karnataka** 2× 2013, 2017

Silk	Silkworm food	Leading state
Mulberry (70%)	Mulberry	Karnataka
Tasar	Arjun, sal	Jharkhand
Eri	Castor	Assam
Muga (golden)	Som, soalu	Assam (India only)

18.2 Tea

- Largest tea producer: **Assam** 1× 1994
- Over 200 cm rain on sloping hills: ideal for **tea** 1× 2010
- India was **first** in tea output and use in 2007-08; **China** leads now, India second 1× 2008
- India exports much tea, but home consumption is **not** poor (A true, R false) 1× 2007
- "**Green Gold**" = **tea** 1× 2008
- Tea is a **stimulant** crop 1× 2017

18.3 Coffee and Rubber

- Largest coffee producer: **Karnataka**; over **70%** of output 5× 2002, 2004, 2006, 2010, 2016
- Largest coffee **area**: **Karnataka** 1× 2002
- Largest rubber producer: **Kerala**; **Kerala-Rubber** correct 3× 2004, 2014, 2022
- **Rubber-Himachal** is wrong 1× 2017

18.4 Plantations, Tobacco and Coconut

- Main plantation crops: **tea, rubber, coconut, coffee** 1× 2001
- True: **tea = Assam**, **tobacco = Andhra Pradesh**; coffee is not Tamil Nadu's main crop; jute is not Chhattisgarh's 1× 2001
- Over a third of tobacco: **Andhra Pradesh** (then); **Gujarat** now 1× 2006
- Largest tobacco producer per the 2007 key: **Uttar Pradesh** 1× 2007
- Largest tobacco area (2010): **Andhra Pradesh**; **Gujarat** now 1× 2012
- Largest coconut producer: **Kerala** (then); **Karnataka** since 2023-24 2× 2006, 2014

18.5 Spices, Cashew and Boards

- India leads in **ginger** (and turmeric, chilli); not in cardamom or cloves now (*) 1× 2010
- **Odisha** is **not** a cardamom state 1× 2014
- "**Garden of spices**": **Kerala** 3× 2010, 2011, 2013
- Largest cardamom and pepper producer: **Kerala** 1× 2018
- Biggest spice producer (2012 key): **Madhya Pradesh** 1× 2012
- Cloves grow in Tamil Nadu, Kerala, Karnataka (*) 1× 2007
- Major **cashew** producer among the options: **Maharashtra** 1× 2013
- **Coffee Board-Hyderabad** is wrong; it is in **Bengaluru** 1× 2023
- **National Horticulture Board**: **1984** 1× 2014

Board	Headquarters
Tea Board	Kolkata
Coffee Board	Bengaluru
Rubber Board	Kottayam
Spices Board	Kochi
Coconut Board	Kochi
Central Silk Board	Bengaluru
Tobacco Board	Guntur
National Horticulture Board	Gurugram (1984)

TRAP ALERT

- Leaders that **changed**: coconut (Kerala to **Karnataka**), tobacco (AP to **Gujarat**), world tea (India to **China**).
- Leaders that **did not change**: coffee **Karnataka**, rubber **Kerala**, tea **Assam**, mulberry silk **Karnataka**, muga **Assam**.

UP LINK

UP is not a plantation state. It appears in tobacco questions (an old key named UP) and in silk weaving: **Varanasi (Banarasi)** silk sarees use mulberry silk mostly from Karnataka.

LAST-MINUTE 10

1. **Assam** = tea, muga, eri; **Jharkhand** = tasar
2. **Karnataka** = coffee, mulberry silk, silk yarn and fabric
3. **Kerala** = rubber, cardamom, pepper; Garden of Spices
4. Tea = Green Gold, stimulant; 200 cm rain on slopes
5. Coconut: Kerala, now **Karnataka**
6. Tobacco: AP, now **Gujarat**
7. India leads world in **ginger**, turmeric, chilli
8. **Coffee Board = Bengaluru**; Rubber Board = Kottayam; Tea Board = Kolkata
9. National Horticulture Board **1984**
10. **Maharashtra** = major cashew producer

Chapter 19

Farming Practices, Horticulture and Livestock

Built from 67 UPPSC questions (1990–2024). Focus: crop-state matching, maize, seed classes, fruit varieties, livestock and milk.

ASKED MOST · REVISE FIRST

- **Crop-state matching**: **jute-West Bengal, tea-Assam, rubber-Kerala, sugarcane-UP, groundnut-AP/Gujarat, mustard-Rajasthan, soybean-MP** 7× 2008, 2008, 2010, 2011, 2014, 2016, 2017
- **Maize** is grown **all year round**; it is a **C4** plant; Shaktiman-I and II are maize 4× 2006, 2007, 2014, 2015
- **West Bengal** = jute industry and largest jute area 2× 2004, 2007
- **UP** = top in **milk**, foodgrains, potato (2013) 3× 2008, 2014, 2014
- **Amrapali** = **Dasheri × Neelam** mango 2× 2013, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

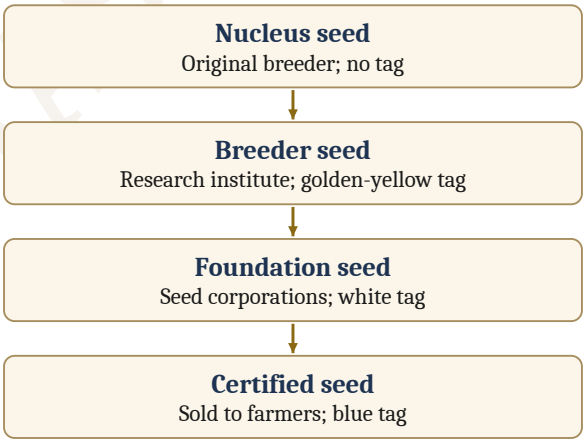
19.1 Shifting Cultivation

- **Jhum** is practised mostly in **Nagaland** 1× 2005
- Jhum is a problem in the hills of **Odisha and MP** (per the 1998 key) 1× 1998
- Match: **Western Ghats-Kumari, SE Rajasthan-Waltre, North-east-Jhum, MP and Chhattisgarh-Dahiya** 1× 2017

19.2 Seeds, Diseases and Weeds

- Seed stages: **Nucleus, Breeder, Foundation, Certified** 1× 2023
- **Certified** seed tag: **blue** 1× 2016
- Match: **sugarcane-red rot, paddy-khaira, arhar-wilt, potato-late blight** 1× 2014
- **Rice-green hair** is wrong 1× 2015
- **Paddy-bathua** is wrong (bathua is a wheat weed) 1× 2016

SEED MULTIPLICATION CHAIN



19.3 Fibre Crops

- Jute industry: **West Bengal**; largest jute area: **West Bengal** 2× 2004, 2007
- India was the largest jute producer (Bangladesh now leads); **Yellow Revolution** is about oilseeds, not jute (*) 1× 2023
- Jute is **native to India** and a **kharif** crop (both true, R not the explanation) 1× 2024
- Fibre crops: **cotton, jute, sunhemp** 1× 2017
- Cotton is a **kharif** crop needing **50-75 cm** rain (both true) 1× 2024

19.4 Maize, Saffron and Cropping

- Grown **all year round**: **maize** 2× 2007, 2015
- **C4 plant**: **maize** 1× 2014

- Maize (then): **Karnataka, Bihar, Rajasthan, Andhra Pradesh** (descending) 1× 2010
- **Shaktiman-I and II**: **maize** (quality protein) 1× 2006
- **Saffron**: **Jammu and Kashmir** (Kashmir, Pampore) 2× 1990, 2007
- Best crop cycle for **eastern UP**: **paddy-maize-wheat** 1× 2008
- Subsistence crops: **coarse cereals and rice** 1× 2007
- Cereal product: **oatmeal** 1× 2002
- First fully **organic** state: **Sikkim** 1× 2016

19.5 Leading States and Varieties

- **Banana-Maharashtra** was the wrong pair then (Tamil Nadu led); now Maharashtra leads 2× 2008, 2011
- Largest potato producer (2013): **UP**; "potato-MP" is wrong 2× 2014, 2016
- Largest share of **foodgrains**: **UP** 1× 2008
- **Gujarat** leads in **bajra** (with Rajasthan) 1× 2008
- Match: **groundnut-AP, mustard-Rajasthan, soybean-MP, coconut-Kerala** 1× 2008
- Match: **jute-WB, tea-Assam, rubber-Kerala, sugarcane-UP** 3× 2010, 2014, 2017
- Season: **sunflower-all seasons, muskmelon-zaid, cotton-kharif, linseed-rabi** 1× 2016
- **Paddy-Radhey** is wrong (Radhey is a gram variety) 1× 2016
- **Wheat-Shusk Samrat** is wrong (it is a pea/lentil variety, not wheat) 1× 2016
- Best potato for processing: **Kufri Chipsona-2** 1× 2008

19.6 Horticulture

- India is the largest producer of **mango and papaya** (2020-21) 1× 2021
- India is **second** in vegetables; **China** is first 2× 2006, 2015
- India gives about **15%** of world fruit 1× 2013
- Top fruit in India by output: **banana**; top banana state: **Maharashtra** (per key) 2× 2023, 2013
- National Research Centre for Banana: **Trichy** 1× 2014
- Seedless mango: **Sindhu** 1× 2007
- **Amrapali** = Dasheri × Neelam; regular-bearing variety: none of the options 3× 2013, 2016, 2014
- **Lalit**: **guava**; **Kanchan**: **aonla** 2× 2008, 2015
- "Ganga Varanasi" is not a listed variety (*) 1× 2015
- **Sultana, Gulabi, Kali Champa**: **grapes** 1× 2018

Variety	Crop
Amrapali, Mallika, Sindhu	Mango
Lalit, Allahabad Safeda	Guava
Kanchan, Banarasi	Aonla
Sultana, Gulabi	Grapes
Kufri Chipsona	Potato
Shaktiman	Maize
Radhey	Gram

19.7 Mixed Statements

- Wrong: cotton fibre length 1-5 cm 1× 2013
- Wrong: linseed grows all year 1× 2013
- Wrong: Bihar is the largest lac producer (Jharkhand is) 1× 2013
- Wrong: **Aman** rice is sown in March-April (it is June-July) 1× 2013
- Rajma is the poor nitrogen fixer 1× 2013
- More than one statement untrue (Kaushal is groundnut; MP leads in pulses) (*) 1× 2013
- India is the largest producer of none of sugarcane, tobacco, coffee, oilseeds (*) 1× 2010

19.8 Livestock, Milk and Fish

- Stock farming = **breeding animals** 1× 2005
- About one-third of cattle: **West Bengal, UP, MP** (2019 census) (*) 2× 2006, 2009
- **UP** has the most **buffaloes**; WB the most cattle; Telangana the most sheep (*) 1× 2008
- High-milk cow breed: **Sahiwal** 1× 2004
- India is **first** in milk; top milk state (2011-12): **UP** 2× 2008, 2014
- **White Revolution** = **milk** 1× 2015
- West Bengal led in fish (mostly **inland**), now **Andhra Pradesh** leads (*) 2× 2007, 2015

QUANTUM IAS & PCS

Livestock (2019 census)	Leading state
Cattle	West Bengal
Buffalo	Uttar Pradesh
Sheep	Telangana
Goat	Rajasthan
Pig	Assam
Poultry	Tamil Nadu
Total livestock	Uttar Pradesh

UP LINK

UP leads India in **milk, buffaloes, total livestock, foodgrains, potato, sugarcane and wheat**. Best eastern-UP cycle: **paddy-maize-wheat**. UP cattle breeds: **Ponwar**, **Gangatiri**; buffalo: **Bhadawari**. Guava **Allahabad Safeda** and **Lalit** come from UP.

LAST-MINUTE 10

1. Jhum = **Nagaland**; Kumari = Western Ghats; Dahiya = MP
2. Seeds: nucleus, breeder, foundation, **certified (blue tag)**
3. **Maize** = all year, C4, Shaktiman
4. Saffron = **Kashmir**; first organic state = **Sikkim**
5. Jute = **West Bengal**; native to India; kharif
6. **UP** = milk, buffalo, foodgrains, potato
7. **Sahiwal** = top milk cow; India first in milk
8. Banana = top fruit; NRC Banana = **Trichy**
9. **Amrapali** = Dasheri × Neelam; **Lalit** = guava; **Kanchan** = aonla
10. Fish: **Andhra Pradesh** now leads

Chapter 20

Mineral Resources I: Metallic and Non-metallic Minerals

Built from 60 UPPSC questions (1991–2025). Focus: mine-mineral and mine-state matching, Dharwar rocks, iron ore, mica, uranium.

ASKED MOST · REVISE FIRST

- **Mine-mineral matching** is the chapter's main format: **Khetri-copper, Kodarma-mica, Dalli-Rajhara-iron, Ankleshwar-oil, Zawar-zinc/lead/silver, Jaduguda-uranium, Talcher-coal, Kolar-gold** 13× 1998, 2001, 2002, 2005, 2008, 2008, 2008, 2009, 2012, 2013, 2016, 2016, 2020
- **Dharwar** rocks are the most mineralised and hold **iron ore** 7× 1994, 1995, 2005, 2012, 2014, 2017, 2019
- Iron-ore mines and states: **Gorumahisani, Badampahar-Odisha; Bailadila, Dalli-Rajhara-Chhattisgarh; Noamundi-Jharkhand; Kudremukh-Karnataka** 4× 2004, 2010, 2015, 2025
- **Mica**: Rajasthan now (AP earlier); MP does not produce it; belt in **Hazaribagh, Gaya, Munger** 5× 2001, 2003, 2004, 2009, 2010
- **Rajasthan**: near monopoly in **zinc**; also limestone, gypsum, salt; largest copper reserves 3× 2011, 2013, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Rock Systems

- Most mineralised rock system: **Dharwar** 3× 1995, 2005, 2017
- Iron ore occurs in **Dharwar** rocks 4× 1994, 2012, 2014, 2019
- **Vindhyan** rocks are rich in **limestone** 1× 2004
- **Gondwana** = about 95% of coal; ferrous and non-ferrous minerals are in Dharwar (both true, R not the explanation) 1× 2017

20.2 Critical Minerals

- **Lithium** inferred resources at **Reasi** (J&K); India imports lithium, nickel, cobalt (both true) 1× 2023

20.3 Iron Ore

- Iron-ore regions: **Kudremukh and Singhbhum** (*) 1× 1992
- Match: **Gurumahisani-Odisha, Bailadila-Chhattisgarh, Noamundi-Jharkhand, Kudremukh-Karnataka** 1× 2025
- Match: **Badam Pahar-Odisha, Dalli-Rajhara-Chhattisgarh, Kudremukh-Karnataka, Noamundi-Jharkhand** 1× 2010
- **Bailadila-MP** is wrong; it is in **Chhattisgarh** 1× 2004
- Largest mechanised mine: **Bailadila** 1× 2015
- Top iron-ore states (2020-21): **Odisha and Chhattisgarh** 1× 2022
- True: richest deposits in **Karnataka** (per key); **Odisha** is the largest producer 1× 2010
- **Iron ore-Guntur** is wrong 1× 2017

Iron-ore mine	State
Gorumahisani, Badampahar, Keonjhar	Odisha
Noamundi, Gua, Chiria	Jharkhand
Bailadila, Dalli-Rajhara	Chhattisgarh
Kudremukh, Bellary	Karnataka
North and South Goa	Goa

20.4 Manganese, Chromite, Nickel

- Leading **manganese** producer: **Maharashtra** (MP for many years; Odisha has most reserves) 2× 2008, 2012
- **Srikakulam** is **not** in Karnataka (it is in AP) 1× 2023
- Monopoly in **chromite**: **Odisha** (Sukinda) 2× 2008, 2019

20.5 Copper, Zinc, Bauxite, Tin

- Largest **copper** reserves: **Rajasthan** 1× 2011
- Match: **Chandrapur-Maharashtra, Hassan-Karnataka, Khammam-AP, Khetri-Rajasthan** 1× 2011
- Near monopoly in **zinc**: **Rajasthan** 1× 2013
- **Bauxite** = ore of **aluminium** 1× 1997
- Top bauxite states: **Odisha and Gujarat** 2× 2021, one-liner
- **Tin**: reserves and production in **Chhattisgarh** (Dantewada) 2× 2009, 2015

20.6 Mica, Asbestos and Stones

- Largest mica producer: **Rajasthan** now (Andhra Pradesh in 2001-04) 2× 2001, 2004
- **MP** does **not** produce mica 2× 2003, 2010
- Largest mica belt: **Hazaribagh, Gaya, Munger** 1× 2009
- India led the world in **sheet mica** in 2001 (*) 1× 2001
- Asbestos: no state produces it now (*) 1× 2015
- Self-sufficiency in 1991: **graphite** (*) 1× 1991
- Best **marble**: **Makrana** (Rajasthan); marble is a **metamorphic** rock 2× 1992, 1991
- **Garnet** is linked with metamorphic rock 1× 2010
- **Granite** slabs and slates: **Lalitpur** (UP) 1× 2002
- Sonbhadra minerals: **andalusite, pyrite, dolomite** (not uranium) 1× 2001

20.7 Atomic Minerals, Gold, Diamonds

- **Monazite** is abundant in **Kerala**; also Tamil Nadu 2× 2013, 2002
- **Jaduguda** (Jharkhand): **uranium** 1× 2007
- Match: **Domiasiat-Meghalaya, Lambapur-AP (Telangana), Rohil-Rajasthan, Gogi-Karnataka** 1× 2008
- Diamond mines: **Madhya Pradesh** (Panna) 1× 2014

20.8 Mixed Mine Matching

- World: **silver-Mexico, mica-India, iron ore-Russia, copper-Chile** 1× 2017
- **Manganese-MP, nickel-Odisha, lead-zinc-Rajasthan, asbestos-AP** 1× 1998
- **Iron ore-Odisha, copper-Rajasthan, gold-Karnataka, mica-AP** 1× 2008
- **Mineral oil-Gujarat, gypsum-Rajasthan, gold-Karnataka, bauxite-Odisha** 2× 2006, 2010

QUANTUM IAS & PCS

- **Makum-coal, Dalli-Rajhara-iron, Koraput-bauxite, Chitradurga-manganese** 1× 2016
- **Kamptee-copper** is wrong (Kamptee = coal) 1× 2009
- **Badam Pahar-iron, Kodarma-mica, Mosabani-copper, Rava-petroleum** 1× 2008
- **Nellore-manganese** is wrong (Nellore = mica) 1× 2002
- **Ankleshwar-petroleum, Dalli-Rajhara-iron, Kodarma-mica, Khetri-copper** 1× 2001
- **Iron-Keonjhar, copper-Khetri, coal-Korba, mica-Kodarma** 1× 2005
- **Gold-Kolar, coal-Jharia, copper-Khetri, iron-Kudremukh** 1× 2008
- **Coal-Talcher, copper-Khetri, iron-Kudremukh, silver-Zawar** 1× 1998
- **Manganese-Koraput** is wrong (Koraput = bauxite) 1× 2013
- **Copper-Khetri, gas plant-Auraiya, aluminium-Korba, petroleum-Kochi** 1× 2002
- **Palamu-bauxite, Hazaribagh-mica, Khetri-copper, Keonjhar-manganese** 1× 2008
- **Gurumahisani-iron, Talcher-coal, Jaduguda-uranium, Zawar-lead** 2× 2012, 2016
- Wrong: **copper-Chitradurga** and **manganese-Bhilwara** 1× 2020
- **Limestone-cement, copper-electrical, bauxite-aircraft, manganese-steel** 1× 2021
- **Rajasthan**: limestone, gypsum, salt, zinc 1× 2023

Place	Mineral
Khetri	Copper
Kodarma, Hazaribagh	Mica
Zawar	Zinc, lead, silver
Jaduguda	Uranium
Kolar, Hutti	Gold
Panna	Diamond
Makrana	Marble
Koraput, Panchpatmali	Bauxite
Talcher, Jharia, Korba	Coal
Ankleshwar	Petroleum
Sukinda	Chromite

TRAP ALERT

- Distractor places: **Guntur** (tobacco), **Nellore** (mica), **Kamptee** (coal), **Koraput** (bauxite), **Bhilwara** (zinc, not manganese).
- **Bailadila** is now in **Chhattisgarh**, not MP.

UP LINK

UP's minerals are in the south: **Sonbhadra** (coal of the Singrauli field, limestone at Churk-Dalla-Obra, andalusite, pyrite, dolomite), **Lalitpur** (granite, rock phosphate), **Mirzapur** (Chunar sandstone) and **Shankargarh** (Prayagraj, silica sand for glass).

LAST-MINUTE 10

1. **Dharwar** = most minerals, iron ore; **Vindhyan** = limestone
2. **Odisha** = iron, chromite, bauxite leader
3. **Bailadila** = Chhattisgarh, largest mechanised mine
4. **Rajasthan** = zinc, copper reserves, gypsum, marble, mica now
5. **Khetri** = copper; **Kodarma** = mica; **Zawar** = zinc-silver
6. **Jaduguda** = uranium; monazite = Kerala beaches
7. **Panna** = diamond; **Makrana** = marble
8. Tin = **Chhattisgarh**; Lithium = **Reasi**
9. Bauxite = aluminium; Odisha and Gujarat lead
10. **Lalitpur** = granite; **Sonbhadra** = andalusite, pyrite

Chapter 21

Mineral Resources II: Coal, Petroleum and Natural Gas

Built from 59 UPPSC questions (1998–2025). Focus: coalfield-state and coalfield-valley matching, reserves vs production, oilfields, refinery-state matching, natural gas.

ASKED MOST · REVISE FIRST

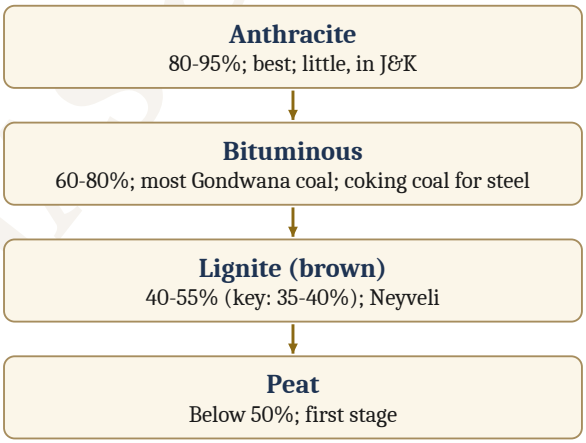
- **Coalfield matching** is the main format: **Talcher-Odisha, Singareni-Telangana, Korba/Bisrampur-Chhattisgarh, Sohagpur/Umaria-MP, Jharia/Karanpura-Jharkhand, Raniganj-WB, Namchik-Arunachal, Makum-Assam** 10× 2008, 2010, 2010, 2016, 2016, 2017, 2019, 2023, 2024, 2025
- **Refinery matching**: **Numaligarh/Noonmati-Assam, Koyali/Jamnagar-Gujarat, Tatipaka-AP, Haldia-WB, Barauni-Bihar, Manali/Nagapattinam-TN** 7× 2005, 2008, 2010, 2013, 2014, 2016, 2025
- **Gondwana** rocks give over **90%** of India's coal 4× 2006, 2008, 2016, 2017
- Largest coal **producer**: **Odisha** (then Chhattisgarh) 5× 2002, 2009, 2009, 2010, 2014
- **Gujarat** = leading onshore petroleum state; **Bombay High** gives most natural gas 5× 2003, 2004, 2010, 2012, 2013

■ asked 3 or more times · asked once or twice 4× times asked, followed by the exam years

21.1 Coal: Reserves and Production

- Main source of coal: **Gondwana** system; over **90%** of coal 4× 2006, 2008, 2016, 2017
- India is a leading coal producer with vast reserves in **Jharkhand and Chhattisgarh** (both true) 1× 2023
- Over half of proved reserves in **Jharkhand and Odisha**; key says both true, R not the explanation (resources are now about 361 bt, not 293.5) 1× 2019
- Reserves order in 2014: Jharkhand first; **Odisha** leads in total resources now (*) 1× 2014
- Key's order **Chhattisgarh, Jharkhand, Odisha, Maharashtra** fits **production**; proved reserves (2014) = Jharkhand, Odisha, Chhattisgarh, Maharashtra 1× 2016
- Two leading producers: **Chhattisgarh and Odisha** 1× 2010
- Production order 2008: Jharkhand, Chhattisgarh, Odisha, MP; now **Odisha, Chhattisgarh, MP, Jharkhand** (*) 1× 2008
- Largest coal producer: **Odisha** 3× 2002, 2009, 2014
- Top three producers: **Odisha, Chhattisgarh, MP** 1× 2009

COAL GRADES BY CARBON CONTENT



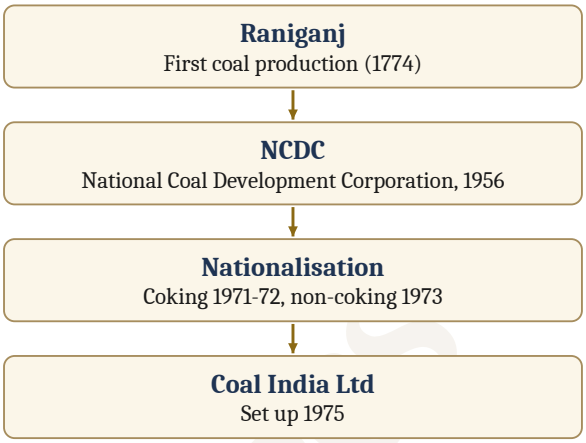
21.2 Coalfields and their States

- **Chota Nagpur** industry grew around **coal**; it gave about **60%** of coal (then) 2× 2004, 1998
- Valleys: **Damodar-Barakar, Son-Umaria, Godavari-Singareni, Mahanadi-Talcher** 1× 2010
- Valleys: **Talcher-Mahanadi, Karanpura-Damodar, Singrauli-Son, Singareni-Godavari** 1× 2019
- **Talcher**: **Odisha** 1× 2010
- **Singareni-MP** is wrong (it is **Telangana**) 1× 2023
- Correct: **Raniganj-WB, Ramgarh-Jharkhand, Talcher-Odisha**; **Maolong** is not Arunachal (it is in Meghalaya) 1× 2025
- **Hutar-Jharkhand, Ib River-Odisha, Tatapani-Ramkola-Chhattisgarh, Umaria-MP** 1× 2024

- **Karanpura-Jharkhand, Singareni-AP (now Telangana), Neyveli-TN, Korba-Chhattisgarh** 1× 2010
- **Rajmahal-Jharkhand, Sohagpur-MP, Bisrampur-Chhattisgarh, Raniganj-WB** 1× 2017
- **Makum-Assam, Namchik-Arunachal, Giridih-Jharkhand, Sohagpur-MP** 1× 2019
- **Namchik-Namphuk**: **Arunachal Pradesh** 1× 2008
- **Jhilmili-Chhattisgarh, Karanpura-Jharkhand, Rampur Himgir-Odisha, Sohagpur-MP** 1× 2016
- **Rangit Valley-Sikkim, Tandur-Telangana, Warora-Maharashtra, Lalmatia-Jharkhand** 1× 2016
- **Bisrampur** is known for **coal** 2× 2004, 2008
- Coalfield with the largest reserves: **Raniganj** 1× 1999
- **Jharia**: **Jharkhand** (best coking coal) 1× 2014

Coalfield	State	Valley
Raniganj	West Bengal	Damodar
Jharia, Bokaro, Giridih, Karanpura	Jharkhand	Damodar
Rajmahal (Lalmatia), Hutar	Jharkhand	Rajmahal, North Koel
Talcher, Ib River, Rampur-Himgir	Odisha	Mahanadi
Korba, Bisrampur, Jhilmili, Tatapani	Chhattisgarh	Mahanadi, Son
Singrauli, Sohagpur, Umaria	Madhya Pradesh	Son
Singareni, Tandur	Telangana	Godavari
Wardha, Kamptee, Warora	Maharashtra	Wardha
Makum; Namchik-Namphuk	Assam; Arunachal	Tertiary coal

COAL HISTORY IN ORDER



- Order: **Raniganj, NCDC, nationalisation, CIL** 1× 2023

21.3 Lignite and the Use of Coal

- Lignite has about 35-40% carbon (true); **Jharkhand** is **not** the lignite leader (A true, R false) 1× 2002
- Largest lignite deposit: **Tamil Nadu**; largest lignite field: **Neyveli** 2× 2013, 2019
- **Jharkhand** does **not** have large lignite reserves 1× 2017
- Most coal is used for **power (energy) generation** 1× 2008
- Coal is a major part of inter-state rail traffic; Bengal-Jharkhand coal feeds the north-west (R explains A) 1× 2010

21.4 Petroleum: Oilfields

- First oil well: **Makum** (1867) per the original key; revised key **Digboi** (oil struck 1889, refinery 1901) 1× 2013
- Leading petroleum producer (onshore): **Gujarat** 3× 2003, 2004, 2010
- **Ankleshwar**: **petroleum** 1× 2004
- **Navgam** oilfield: **Gujarat** 1× 2013
- Wrong fact: Bombay High is **not** 60 km off Mumbai (it is about 176 km) 1× 2014

Region	Oil and gas fields
Assam	Digboi, Naharkatiya, Moran, Lakwa, Rudrasagar
Gujarat	Ankleshwar, Kalol, Navagam, Mehsana, Lunej

Region	Oil and gas fields
Rajasthan (Barmer)	Mangala, Bhagyam, Aishwariya
Offshore Mumbai	Mumbai High, Bassein (gas), Panna-Mukta
KG basin	KG-D6 (gas), Ravva

21.5 Refineries

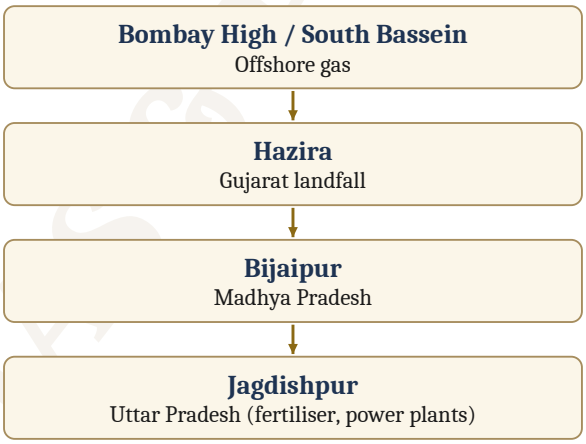
- Biggest refinery: **Jamnagar** (Gujarat) 1× 2016
- **Hatia** has no refinery (it is the Ranchi heavy-engineering town) 1× 2010
- **Noonmati** (Guwahati): **Assam** 1× 2008
- **Panagudi-Tamil Nadu** is the correct pair 1× 2006
- **Nayara-Gujarat, Manali-TN, Numaligarh-Assam, Tatipaka-AP** 1× 2025
- **Haldia-WB, Jamnagar-Gujarat, Kochi-Kerala, Numaligarh-Assam** 1× 2005
- **Numaligarh-Assam, Tatipaka-AP, Koyali-Gujarat, Barauni-Bihar** 1× 2010
- **Tatipaka**: **Andhra Pradesh** 1× 2016
- **Noonmati-Assam, Koyali-Gujarat, Mangalore-Karnataka, Panipat-Haryana** 1× 2014
- **Tatipaka-AP, Koyali-Gujarat, Nagapattinam-TN, Numaligarh-Assam** 1× 2013
- **Koyali**: **Gujarat** (near Vadodara) 1× 2016

Refinery	State
Digboi (1901), Noonmati, Bongaigaon, Numaligarh	Assam
Barauni	Bihar
Haldia	West Bengal
Koyali, Jamnagar, Vadinar (Nayara)	Gujarat
Mathura	Uttar Pradesh
Panipat	Haryana
Mangaluru	Karnataka
Kochi	Kerala
Manali, Nagapattinam	Tamil Nadu
Visakhapatnam, Tatipaka	Andhra Pradesh

21.6 Strategic Reserves and Natural Gas

- Strategic reserves at **Visakhapatnam, Mangaluru, Padur**; **Ankleshwar** not chosen 1× 2008
- "Hydrocarbon Vision 2025" (key: "Hydrogen") = **storage/warehousing of petroleum** 1× 2004
- Oil exploration: **Oil India Ltd** (with ONGC) 1× 2004
- **HBJ** pipeline carries gas from the **South Basin** (Bassein); built by **GAIL** 2× 2006, 2014
- Reliance's huge gas find: **coastal Andhra** (KG basin) 1× 2002
- **KG-D6**: **gas** 1× 2011
- Most natural gas: **Bombay High** 2× 2012, 2013

HBJ GAS PIPELINE (GAIL, 1987)



TRAP ALERT

- Coal "first" depends on the word: **resources** = Odisha; **production** = Odisha; **coking coal** = Jharkhand (Jharia); **field with the largest reserves** = Raniganj.
- **Hatia** is not a refinery; **Haldia** is West Bengal, not Odisha; **Singareni** is Telangana, not MP; **Maolong (Mawlong)** is Meghalaya.
- **Ankleshwar** is an oilfield, not a strategic reserve site.

UP LINK

UP has the **Mathura** refinery (IOC, 1982), the **Singrauli** coalfield's UP part in **Sonbhadra** (feeding Anpara, Obra, Rihand, Singrauli power plants), and the end of the **HBJ pipeline at Jagdishpur** (Amethi), which feeds gas plants such as **Auraiya** and fertiliser units in UP.

LAST-MINUTE 10

1. **Gondwana** = over 90% of coal; Tertiary coal = Assam, Arunachal
2. Coal production: **Odisha, Chhattisgarh**; best coking coal: **Jharia**
3. **Raniganj** = first coal (1774), largest reserves
4. Order: Raniganj, NCDC, nationalisation, **CIL 1975**
5. Lignite = **Neyveli** (Tamil Nadu); not Jharkhand
6. First well **Makum 1867**; Digboi oil 1889, refinery 1901
7. **Gujarat** = onshore oil; **Bombay High** = most gas
8. **Jamnagar** = biggest refinery; **Koyali** = Gujarat; **Tatipaka** = AP
9. **Hatia** = no refinery; **Noonmati, Numaligarh** = Assam
10. HBJ = **GAIL**, Hazira to **Jagdishpur**; KG-D6 = gas

Chapter 22

Power and Energy Resources

Built from 38 UPPSC questions (1991–2023). Focus: energy mix, power-plant-state matching, nuclear stations, wind, solar, tidal and hot springs.

ASKED MOST · REVISE FIRST

- **Thermal (coal)** is the largest source of energy and of power generation **4×** 2002, 2006, 2008, 2011
- **Nuclear station-state matching**: **Narora-UP, Rawatbhata (Kota)-Rajasthan, Kakrapar-Gujarat, Kaiga-Karnataka, Tarapur-Maharashtra** **5×** 1999, 2002, 2007, 2015, 2017
- **Thermal plant-state matching**: **Ukai-Gujarat, Patratu-Jharkhand, Kahalgaon-Bihar, Korba-Chhattisgarh, Kudgi-Karnataka** **5×** 2006, 2010, 2013, 2016, 2023
- India is self-sufficient in **thorium** **3×** 2002, 2003, 2005
- **Gujarat** = first in **wind** power and **tidal** potential (Khambhat, Kachchh) **3×** 1991, 2013, 2022

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

22.1 The Energy Mix

- Largest source of energy: **coal** **2×** 2002, 2008
- Largest share in power generation: **thermal** **2×** 2006, 2011
- Order in the 2000s: thermal, hydro, nuclear, wind; now wind has overtaken hydro and nuclear (*) **2×** 2010, 2014
- Installed capacity (2019): **hydro, gas, nuclear, diesel** **1×** 2016
- **Biogas** is **not** a commercial source **1×** 2013

Source (end 2025)	Capacity	Share
Coal and lignite	226 GW	44%
Solar	136 GW	26%
Wind	55 GW	11%
Large hydro	51 GW	10%
Gas and diesel	21 GW	4%
Nuclear	8.8 GW	2%

22.2 Thermal Power Stations

- **Singrauli**: NTPC's **oldest** unit; it is in **UP** (Sonbhadra), not Chhattisgarh **1×** 2021
- **Obra** thermal plant was built with **Russian** (Soviet) help **1×** 2008

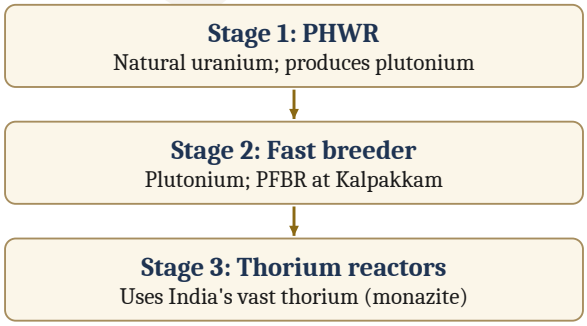
QUANTUM IAS

- **Ukai-Gujarat, Patratu-Jharkhand, Pench-MP, Dabhol-Maharashtra** 2× 2010, 2013
- **Uran** gas plant: **Maharashtra** 1× 2010
- **Ramagundam**: AP then, **Telangana** now 1× 2004
- **Kahalgaon-Bihar, Khargone-MP, Korba-Chhattisgarh, Kudgi-Karnataka** 1× 2023
- **Paras-AP** is wrong (Paras is in **Maharashtra**) 1× 2006
- Thermal capacity rank 2008: **Maharashtra** first; now **UP** leads (*) 1× 2008
- Plant Load Factor then: **65-70%** 1× 2004
- **Korba-wind** is wrong (Korba = thermal) 1× 2016
- **Chamera-hydro, Gwal Pahari-solar, Kudankulam-atomic, Patratu-thermal** 1× 2019

22.3 Nuclear Power

- Self-sufficient in **thorium** 3× 2002, 2003, 2005
- Nuclear energy is promising, but nuclear minerals are **not** everywhere (A true, R false) 1× 2006
- First nuclear reactor: **Apsara** (1956, Trombay) 1× one-liner
- **Kota-Rajasthan, Tarapur-Maharashtra, Kakrapar-Gujarat, Narora-UP** 1× 1999
- **Gujarat-Kakrapar, Karnataka-Kaiga, Rajasthan-Rawatbhata, UP-Narora** 1× 2002
- Commissioning-year pairs: none correct (*) 1× 2007
- **Ennore-Meghalaya** is wrong (Ennore = Tamil Nadu, thermal) 1× 2015
- **Mithi Virdi** was planned with the **USA** (moved to **Kovvada**, AP) 1× 2014
- **Haripur-Haryana** is wrong; Haripur is **West Bengal**; Haryana's is **Gorakhpur** 1× 2017
- **Anushakti Vidyut Nigam** = NPCIL + **NTPC** 1× 2015

INDIA'S THREE-STAGE NUCLEAR PROGRAMME



Nuclear station	State	First unit
Tarapur	Maharashtra	1969
Rawatbhata (Kota)	Rajasthan	1973
Kalpakkam	Tamil Nadu	1984
Narora	Uttar Pradesh	1991
Kakrapar	Gujarat	1993
Kaiga	Karnataka	2000
Kudankulam	Tamil Nadu	2013 (Russia)
Gorakhpur	Haryana	Under construction

22.4 Hydro, Wind, Solar and Tidal

- Hydro station: **Koyna** (Maharashtra) 1× 1992
- Correct: **Sabarigiri-hydro, Ghatprabha-irrigation, Ramganga-multipurpose**; Idukki is **hydro**, not thermal 1× 2017
- **Shivanasamudra** = hydel (1902); **Kalpakkam** = nuclear 1× 2023
- First in wind power: **Gujarat** (Tamil Nadu earlier) 1× 2013
- Asia's biggest 150 MW wind project (1999): **Tamil Nadu** (Muppandal) 1× 1999
- First in installed capacity: **Gujarat** per key (Maharashtra in 2006); Gujarat leads now with renewables 1× 2006
- **Rampura**, first solar-powered village: **UP** 1× 2010
- First UT on 100% solar: **Diu** 1× 2017
- Largest tidal energy area: **Khambhat** coast; most potential: **Bhavnagar** (Gulf of Khambhat) 2× 2022, 1991

22.5 Hot Springs

- **Lasundra-Rajasthan** is wrong (Lasundra is in **Gujarat**) 1× 2014
- **Jwalamukhi-J&K** is wrong (it is in **Himachal Pradesh**) 1× 2015

QUANTUM IAS

TRAP ALERT

- **Singrauli** NTPC plant is in **UP**; **Paras** is Maharashtra; **Ennore** is Tamil Nadu; **Haripur** is West Bengal.
- **Idukki, Koyna, Sabarigiri, Shivanasamudra** = hydro; **Korba, Patratu, Obra** = thermal.
- Lasundra (Gujarat), Jwalamukhi (HP), Manikaran (HP), Puga (Ladakh): geothermal sites.

UP LINK

UP has **Narora** atomic plant (Bulandshahr, 1991), **Singrauli** (NTPC's oldest), **Obra** (Soviet aid), **Anpara**, **Rihand** in Sonbhadra (India's "power capital"), the gas plant at **Auraiya**, and **Rampura**, India's first solar village. UP now leads in thermal capacity.

LAST-MINUTE 10

1. Coal/thermal = largest energy and power source
2. India self-sufficient in **thorium**; first reactor **Apsara**
3. **Narora-UP, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan**
4. Haripur = WB; Gorakhpur nuclear = **Haryana**; Mithi Virdi to Kovvada (USA)
5. Singrauli = NTPC's oldest (UP); **Obra** = Russia
6. Ukai = Gujarat; Patratu = Jharkhand; Kahalgaon = Bihar; Kudgi = Karnataka
7. **Gujarat** = wind, tidal (Khambhat), largest capacity now
8. **Rampura** (UP) = first solar village; **Diu** = first solar UT
9. Shivanasamudra = hydel; Idukki = hydro
10. Jwalamukhi = HP; Lasundra = Gujarat

Chapter 23

Industries and Industrial Centres

Built from 42 UPPSC questions (1994–2024). Focus: steel plants, aluminium centres, PSU headquarters, textile and agro centres, fertilizer, paper and cement towns.

ASKED MOST · REVISE FIRST

- ■ **Place-industry matching** dominates: **Renukoot-aluminium, Korba-BALCO, Muri-aluminium (Jharkhand), Sindri-fertilizer, Phulpur-fertilizer, Titagarh-paper, Bhadohi-carpet, Chunar-cement** 13× 1994, 1998, 2001, 2002, 2003, 2006, 2008, 2008, 2010, 2010, 2013, 2017, 2024
- ■ **Wrong-pair** traps: **Bhilai-MP, BALCO-Raipur, Muri-Chhattisgarh, Amlai-Chhattisgarh, Dharmapuri-Odisha, Sindri-copper, Pipri-pesticides** 8× 2002, 2003, 2004, 2004, 2009, 2010, 2010, 2024
- • **Gujarat** = largest petrochemical centre (**Jamnagar**) 2× 2006, 2013
- ■ **UP**: most paper mills; **Phulpur** = biggest cooperative fertilizer plant; **Chunar** = cement 3× 2008, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

23.1 Iron and Steel

- **Bhilai-MP** is wrong now (it is in **Chhattisgarh**) 1× 2004
- **Rourkela** gets iron ore from **Keonjhar** 1× 2005
- Plant far from coalfields: **Bhadravati** (charcoal, hydel) 1× 2013
- **SAIL** was formed in **1973** (not in the options, *) 1× 2006

Steel plant	State	Helped by
TISCO, Jamshedpur (1907)	Jharkhand	Tata (private)
Bhadravati (1923)	Karnataka	Mysore State
Bhilai	Chhattisgarh	Soviet Union

Steel plant	State	Helped by
Rourkela	Odisha	West Germany
Durgapur	West Bengal	Britain
Bokaro	Jharkhand	Soviet Union
Visakhapatnam (RINL)	Andhra Pradesh	Indigenous, shore-based

23.2 Aluminium and Copper

- **Korba** (Chhattisgarh): **aluminium** (BALCO) 2× 1994, 2001
- **BALCO-Raipur** is wrong (BALCO is at **Korba**) 2× 2002, 2003
- **BALCO-Korba, Hindalco-Renukoot, Indian Aluminium-Hirakud, NALCO-Koraput** 1× 2017
- **Alupuram-Kerala, Angul-Odisha, Belgaum-Karnataka, Korba-Chhattisgarh** 1× 2010
- **Muri-Chhattisgarh** is wrong (Muri is in **Jharkhand**) 1× 2024
- **Aluminium-J.K. Nagar, copper-Malanjkhand, zinc-Tundoo, jute-Bhatpara** 1× 2013

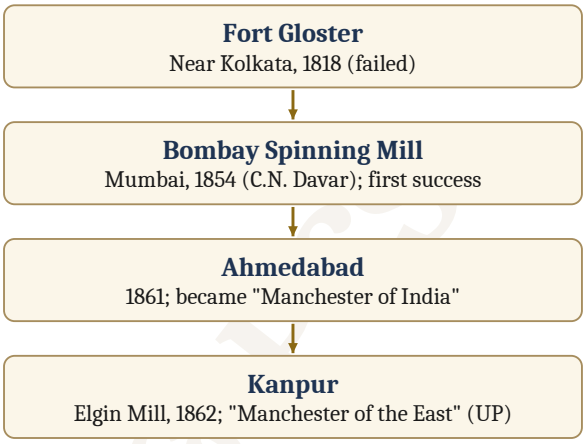
23.3 PSUs and Headquarters

- **ONGC-New Delhi, MECL-Nagpur, NMDC-Hyderabad, NALCO-Bhubaneswar** 1× 2021
- **SAIL-Delhi, Hindustan Zinc-Udaipur, HEC-Ranchi, Instrumentation Ltd-Kota** 1× 2010
- **Heavy engineering-Ranchi, machine tools-Pinjore, aluminium-Renukoot, fertilizer-Sindri** 1× 1998
- **BEL-Noida** is wrong (BEL's UP unit is at **Ghaziabad**) 1× 2006

23.4 Textiles, Carpets, Rubber

- Ancient industry of India: **cotton textile** 1× 2004
- First cotton mill (1818): **Fort Gloster**, near Kolkata 1× 2013
- **Ahmedabad** is not the largest cotton centre now (Mumbai historically; A false, R true) 1× 2020
- **Kakinada-jute, Virudhunagar-cotton textile, Channapatna-silk, Bhadohi-carpet** 2× 2010, 2013
- **Silk-Mysore, petrochemicals-Jawahar Nagar, fertilizer-Talcher, pharma-Rishikesh** 1× 2010
- First synthetic rubber plant: **Panipat** per key (many texts say **Bareilly**, 1963) 1× 2016
- Rubber industry at **Panjim** (per key) 1× 2007

FIRST COTTON MILLS



23.5 Fertilizers and Petrochemicals

- **Pipri-pesticides** is wrong (Pipri = Rihand hydel, Sonbhadra) 1× 2004
- **Sindri-copper** is wrong (Sindri = fertilizer) 1× 2010
- Biggest cooperative fertilizer plant: **Phulpur** (IFFCO, Prayagraj) 1× 2010
- **Visakhapatnam-shipbuilding, Muri-aluminium, Gurgaon-automobile, Panki-fertilizer** 1× 2006
- **Brajrajnagar-paper, Kaimur-cement, Haldia-petrochemical, Phulpur-fertilizer** 1× 2008
- Largest petrochemical factory: **Gujarat**; largest centre: **Jamnagar** 2× 2006, 2013
- **Nagapattinam-AP** is wrong (it is **Tamil Nadu**) 1× 2011
- **Dharmapuri-Odisha** is wrong (it is in **Tamil Nadu**) 1× 2010

23.6 Paper, Cement and Others

- **Amlai-Chhattisgarh** is wrong (Amlai paper mill is in **MP**, Shahdol) 1× 2009
- Most paper mills: **Uttar Pradesh** 1× 2008
- **Paper-Titagarh, cement-Lakheri, steel-Bhilai** 1× 2013
- **Chunar** (Mirzapur): **cement** 1× 2008
- India is **second** in world cement; largest cement state (2018-19): **Rajasthan** 2× 2013, 2016

QUANTUM IAS & PCS

- MP pesticide town: **Bhopal** 1× 2008
- **Sugar-Silvassa** is wrong 1× 2014
- **Pithampur** (MP): **automobile** hub 1× 2015
- **Diamond Parks**: zones to make and export diamonds and jewellery 1× 2004

Aluminium centre	State
Renukoot (Hindalco)	Uttar Pradesh
Korba (BALCO)	Chhattisgarh
Angul, Damanjodi (NALCO), Hirakud	Odisha
Muri	Jharkhand
J.K. Nagar	West Bengal
Belagavi	Karnataka
Alupuram	Kerala
Mettur	Tamil Nadu

23.7 Industrial Regions

- **Sivakasi** is **not** in Chota Nagpur; it is in the **Madurai-Coimbatore-Bengaluru** region 2× 2015, 2014

TRAP ALERT

- States changed: **Bhilai** (MP to Chhattisgarh), **Ramagundam** (AP to Telangana).
- **Muri** = Jharkhand; **Amlai** = MP; **Dharmapuri, Nagapattinam** = Tamil Nadu; **Sivakasi** = Tamil Nadu (matches, fireworks).
- Rourkela = Germany; Durgapur = Britain; Bhilai and Bokaro = Soviet Union.

UP LINK

UP centres asked: **Renukoot** (Hindalco aluminium), **Phulpur** (IFFCO), **Panki** (Kanpur, fertilizer), **Chunar** (cement), **Pipri** (Rihand hydel), **Bhadohi** (carpets), **Kanpur** (Manchester of the East), **Bareilly** (synthetic rubber, 1963). UP has the most paper mills. New: semiconductor unit at **Jewar**, mobiles at **Noida**, Defence Corridor nodes.

LAST-MINUTE 10

1. Bhilai = **Chhattisgarh** (Soviet); Rourkela = Germany; Durgapur = Britain
2. Rourkela ore from **Keonjhar**; Bhadravati far from coal
3. BALCO = **Korba**; Hindalco = **Renukoot**; NALCO = Koraput/Angul
4. **Muri** = Jharkhand; Alupuram = Kerala
5. HEC = **Ranchi**; Hindustan Zinc = **Udaipur**; NMDC = Hyderabad
6. First cotton mill **Fort Gloster 1818**
7. **Phulpur** = cooperative fertilizer; Sindri = fertilizer
8. **Jamnagar** = petrochemical capital; Gujarat leads
9. **UP** = most paper mills; **Chunar** = cement; **Rajasthan** = most cement
10. **Sivakasi** = Madurai-Coimbatore-Bengaluru region

Chapter 24

Research Centres and Institutes

Built from 34 UPPSC questions (1990–2021). Focus: ICAR institutes in UP, Hyderabad and Karnal institutes, institute-city matching, polar stations.

ASKED MOST · REVISE FIRST

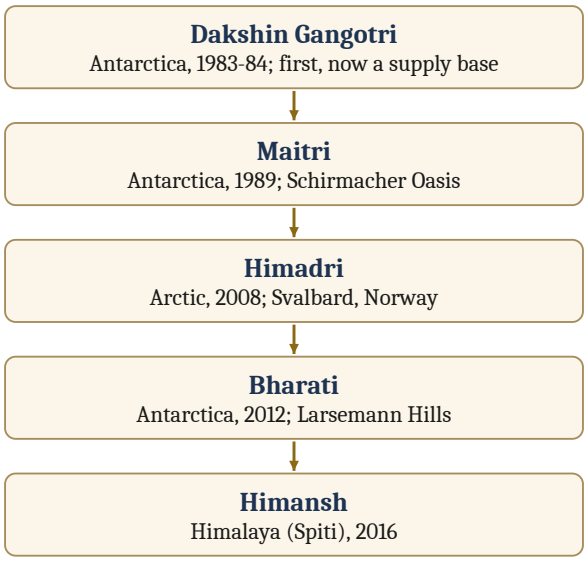
- **UP institutes**: pulses **Kanpur**, sugarcane **Lucknow**, vegetables **Varanasi**, subtropical horticulture **Lucknow**, grassland and agroforestry **Jhansi** 11× 2004, 2010, 2010, 2010, 2013, 2015, 2015, 2016, 2017, 2017, 2021
- **Hyderabad**: **CRIDA**, ICRISAT, NAARM, rice directorate, NPPTI 6× 2008, 2010, 2013, 2013, 2014, 2015
- **Karnal**: **NDRI**, CSSRI, wheat research 3× 2013, 2013, 2016
- **Nagpur**: Indian Bureau of Mines, **NEERI** 2× 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Polar Research Stations

- Third Indian Antarctic station: **Bharati** (2012) 1× 2011

INDIA'S POLAR STATIONS



24.2 ICAR Institutes in Uttar Pradesh

- Indian Institute of **Pulses** Research: **Kanpur** 3× 2004, 2010, 2015
- Indian Institute of **Vegetable** Research: **Varanasi** 2× 2016, 2017
- Central Institute of **Subtropical Horticulture**: **Lucknow** 1× 2010
- Indian Institute of **Sugarcane** Research: **Lucknow** 1× 2013
- **Subtropical Horticulture-Srinagar** is wrong (Srinagar has **Temperate** Horticulture) 1× 2017
- Agroforestry research centre: **Jhansi**; **Agroforestry-Banda** is wrong 2× 2010, 2021
- Indian **Grassland and Fodder** Research Institute: **Jhansi** 1× 2015
- **Goat** research: **Makhdoom** (Mathura) 1× 2017
- National Centre of **Organic** Farming: **Ghaziabad** 1× 2015
- Centre for **Social Forestry and Eco-Rehabilitation**: **Prayagraj** 1× 2019

UP city	Institute
Lucknow	Sugarcane Research; Subtropical Horticulture; NBRI; CDRI; CIMAP
Kanpur	Pulses Research; National Sugar Institute
Varanasi	Vegetable Research; IRRI South Asia centre
Jhansi	Grassland and Fodder; Agroforestry
Makhdoom (Mathura)	Goat research
Meerut (Modipuram)	Farming Systems Research
Ghaziabad	Organic and Natural Farming
Prayagraj	Eco-Rehabilitation (ICFRE)
Bareilly (Izatnagar)	IVRI; Avian research

24.3 Other Agricultural Institutes

- **ICRISAT**: **Hyderabad** (Patancheru) 1× 2014
- **CRIDA** (dryland agriculture): **Hyderabad** 3× 2008, 2013, 2015
- **NAARM**: **Hyderabad** 1× 2010
- National Institute of Agricultural Marketing: **Jaipur** 1× 2013
- Central Institute of Agricultural Engineering: **Bhopal** 1× 2009
- National Dairy Research Institute: **Karnal** 2× 2013, 2016
- National Dairy Development Board: **Anand** 1× 2006
- Horticulture university: **Solan** (Nauni, HP) 1× 1990
- **Rice-Cuttack, farming systems-Meerut, soil science-Bhopal, agroforestry-Jhansi** 1× 2013
- **CSSRI-Karnal, CTCRI-Thiruvananthapuram, IRRI-Manila, CAZRI-Jodhpur** 1× 2013
- **Rice-Hyderabad, wheat-Karnal, maize-New Delhi, pulses-Kanpur** 1× 2013
- **Sugarcane-Lucknow, NBPGR-New Delhi, NPPTI-Hyderabad**; wheat is **Karnal**, not Kanpur (*) 1× 2014
- Plant Protection, Quarantine and Storage: **Faridabad** 1× 2013
- First **Krishi Vigyan Kendra**: **1974** (Puducherry) 1× 2017

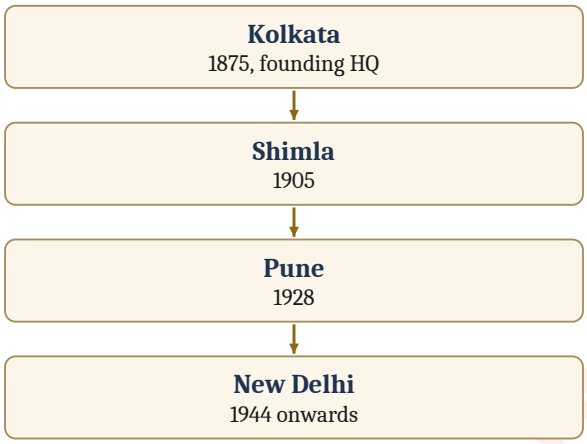
24.4 Scientific and Mining Institutes

- **NEERI-Nagpur, CFTRI-Mysuru, CPRI-Shimla, CTRI-Rajahmundry** 1× 2016

QUANTUM IAS & PCS

- Indian Bureau of Mines: **Nagpur** 1× 2010
- Mining institute: **Dhanbad** (ISM, now IIT-ISM) 1× 1994
- Indian Diamond Institute: **Surat** 1× 2013
- No Museum of Natural History: **Hyderabad** (it is in Delhi, Mysuru, Bhopal, Bhubaneswar, Sawai Madhopur) 1× 2013
- IMD HQ order: **Kolkata, Shimla, Pune, New Delhi** 1× 2018
- International Solar Alliance secretariat: **Gurugram** 1× 2019
- Central Institute of Indian Languages: **Mysuru** 1× 2019

IMD HEADQUARTERS OVER TIME



City	Institutes
Hyderabad	ICRISAT, CRIDA, NAARM, rice, NPPTI, NIRD
Karnal	NDRI, CSSRI, wheat and barley
Nagpur	NEERI, Indian Bureau of Mines, MECL, soil survey
Mysuru	CFTRI, Indian Languages
Bhopal	Soil Science, Agri Engineering, Forest Management
Cuttack	National Rice Research Institute
Jodhpur	CAZRI (arid zone)
Shimla	Potato Research
Rajahmundry	Tobacco Research

TRAP ALERT

- Same-state swaps: sugarcane **Lucknow** (not Kanpur); pulses **Kanpur** (not Lucknow); vegetables **Varanasi**; agroforestry **Jhansi** (not Banda).
- Subtropical horticulture = Lucknow; **temperate** horticulture = Srinagar.
- NDRI = **Karnal**; NDDB = **Anand**.

UP LINK

UP is the most asked state in this chapter: **Lucknow** (sugarcane, subtropical horticulture, CDRI, NBRI, CIMAP), **Kanpur** (pulses, sugar), **Varanasi** (vegetables, IRRI centre), **Jhansi** (fodder, agroforestry), **Makhdoom** (goats), **Ghaziabad** (organic farming), **Prayagraj** (eco-rehabilitation), **Agra** (potato centre).

LAST-MINUTE 10

1. Antarctic: Dakshin Gangotri, Maitri, **Bharati**; Arctic: **Himadri**
2. Pulses = **Kanpur**; sugarcane = **Lucknow**; vegetables = **Varanasi**
3. Fodder and agroforestry = **Jhansi**; goats = **Makhdoom**
4. Organic farming = **Ghaziabad**; eco-rehabilitation = **Prayagraj**
5. CRIDA, ICRISAT, NAARM = **Hyderabad**
6. NDRI = **Karnal**; NDDB = **Anand**; CAZRI = **Jodhpur**
7. Rice = **Cuttack**; potato = **Shimla**; tobacco = **Rajahmundry**
8. IBM, NEERI = **Nagpur**; CFTRI = **Mysuru**
9. IMD: Kolkata, Shimla, Pune, New Delhi
10. ISA = **Gurugram**; first KVK = **1974**

Chapter 25

Transport: Roads, Railways, Ports and Waterways

Built from 61 UPPSC questions (1990–2025). Focus: NHDP corridors, railway zones and factories, port types and port-state matching, NW-1.

ASKED MOST · REVISE FIRST

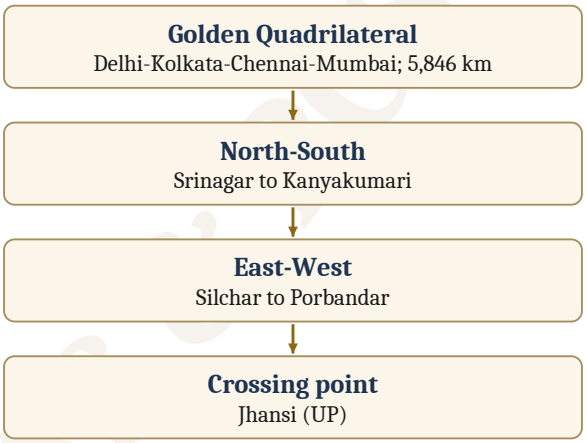
- **Port-state matching**: **Kandla/Mundra-Gujarat, Nhava Sheva-Maharashtra, Paradip-Odisha, Tuticorin-TN, Kakinada/Krishnapatnam-AP** 7× 2008, 2008, 2010, 2010, 2013, 2015, 2025
- **Port types**: deepest/landlocked **Visakhapatnam**, open sea and artificial **Chennai**, outer port **Haldia**, natural **Kochi, Visakhapatnam** 8× 1997, 2007, 2012, 2012, 2013, 2017, 2023, 2023
- **NW-1** = **Prayagraj-Haldia** on the Ganga, the longest waterway 4× 2009, 2010, 2014, 2016
- NS and EW corridors cross at **Jhansi** (UP) 2× 1999, 2001
- **Rail factories**: ICF **Perambur**, RCF **Kapurthala**, Modern Coach **Raebareli**, BLW **Varanasi** 3× 1995, 2008, 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

25.1 Roads and National Highways

- Road share of total transport: about **80%** 1× 2014
- Old **NH-4** (Mumbai-Chennai) did not touch MP, Rajasthan or UP (*) 1× 2010
- Most NH length (then): **UP**; now **Maharashtra** leads 1× 2000
- Most **state highway** length: **Maharashtra** 2× 2010, 2013
- NH length order (per key): Arunachal first, Sikkim last 1× 2004
- Old **NH-3** (Agra-Mumbai) did not touch **Bhopal** 1× 2001
- Golden Quadrilateral joins **Delhi, Kolkata, Chennai, Mumbai**; EW corridor ends at **Porbandar** 1× 2008
- **Golden Quadrilateral** = highways 1× 2004
- First six-lane expressway: **Mumbai-Pune** 2× 2001, 2003
- **PMGSY**: roads for unconnected villages 1× 2001

NHDP CORRIDORS



25.2 Railways

- First train: **Bombay-Thane, 1853** 2× 1990, 1996
- **Hyderabad**-SCR HQ is wrong (it is **Secunderabad**) 1× 2002
- Correct: **Northern-New Delhi, North Eastern-Gorakhpur**; South Eastern is **Kolkata**, not Cuttack 1× 2025
- Shortest Gorakhpur-Mumbai route: via **Prayagraj** 1× 1993
- **RCF-Kapurthala, Wheel and Axle-Bengaluru, DLW-Varanasi, ICF-Perambur** 1× 2008
- Coach-making states: **Punjab and Tamil Nadu** 1× 2008
- Third coach factory: **Raebareli** (Lalganj) 1× 1995
- Railway Staff College: **Vadodara** 1× 1991
- Bullet train: **Mumbai-Ahmedabad** 1× 2001
- First CNG train: **Rewari-Rohtak** 1× 2016
- First rail track on PPP: **Gujarat** 1× 2016
- Konkan Railway: **Roha to Mangalore**; it does **not** touch **Belgaum** 2× 2013, 2010
- State without rail in 2013: **Meghalaya** (line 2014); Sikkim is the one left now (*) 1× 2013

- Longest tunnels then: **Pir Panjal, Karbude, Nathuwadi, Berdewadi** 1× 2015
- New **Pamban** bridge: built by RVNL; India's first **vertical-lift** sea bridge (both true) 1× 2025

Zone	HQ
Northern	New Delhi
North Eastern	Gorakhpur
North Central	Prayagraj
South Eastern	Kolkata
East Coast	Bhubaneswar
South Central	Secunderabad
East Central	Hajipur
NE Frontier	Maligaon

25.3 Ports and Harbours

- Biggest port (2007 key): **Mumbai** 1× 2007
- **Kochi** is on the **Malabar** coast 1× 2013
- Landlocked harbour: **Visakhapatnam** 1× 2023
- **Chennai-deepest port** is wrong; **Kolkata-deepest** is wrong 2× 1997, 2023
- Deepest among options: **Paradip** (Visakhapatnam is the deepest landlocked) 1× 2017
- Not a natural harbour: **Chennai**; natural ports: **Kochi and Visakhapatnam** 2× 2007, 2013
- Open seaport: **Chennai** 1× 2012
- Outer port: **Haldia** (of Kolkata) 1× 2012
- Not on the east coast: **Kandla**; Odisha coast: **Paradip** 2× 2008, 2013
- **Kandla** is in **Gujarat**; on the **Gulf of Kachchh** 2× 1993, 2004
- **Paradip-Odisha, Tuticorin-TN, Kakinada-AP, Alappuzha-Kerala** 1× 2025
- **Kandla-Gujarat, Nhava Sheva-Maharashtra, Paradip-Odisha, Tuticorin-TN** 1× 2010
- **Mundra-AP** is wrong (Mundra is **Gujarat**) 1× 2015
- AP port city: **Kakinada**; **Krishnapatnam** helps **AP** 2× 2010, 2008
- 12th major port for coal: **Ennore**, near **Chennai** 2× 2004, 2010
- **Jamnagar** is not a port town (per key) 1× 2011
- Built to relieve Mumbai: **Nhava Sheva (JNPT)** 1× 2013
- No LNG terminal: **Kandla** (then) 1× 2005
- **Sethusamudram** links the **Gulf of Mannar** with **Palk Bay** 2× 2004, 2008
- **Karaikal** is not on the west coast 1× 2008

Port type	Example
Landlocked, deepest	Visakhapatnam
Oldest artificial, open sea	Chennai
Tidal	Kandla (Deendayal)
Riverine	Kolkata
Outer port	Haldia
Satellite, container	Nhava Sheva (JNPT)
Lagoon (natural)	Kochi
First corporatised	Ennore (Kamarajar)

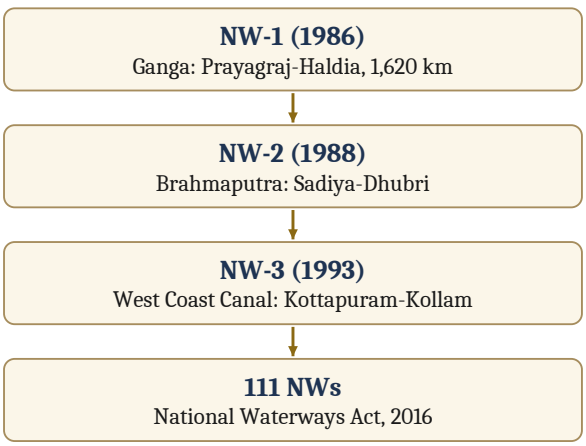
25.4 Shipyards and Ship-breaking

- **Garden Reach-WB, Hindustan Shipyard-AP, Mazagon-Maharashtra, Cochin-Kerala** 1× 2011
- Largest ship-breaking yard: **Alang** (Gujarat) 2× 2006, 2009

25.5 Waterways and Airports

- National Waterway on the Ganga: **Prayagraj to Haldia**; longest waterway 4× 2010, 2016, 2009, 2014
- National Inland Navigation Institute: **Patna** 1× 2009
- International status (24 June 2020): **Kushinagar** airport 1× 2020

NATIONAL WATERWAYS 1-3



TRAP ALERT

- **Perambur = ICF; Kapurthala = RCF.** Wheel and Axle = **Bengaluru**.
- **Visakhapatnam** = deepest landlocked; **Chennai** = artificial; **Kolkata** = shallow riverine.
- **Secunderabad**, not Hyderabad, is the SCR HQ; **Mundra** and **Kandla** are Gujarat.

UP LINK

UP links: NS-EW crossing at **Jhansi**; railway zones at **Gorakhpur** (NER) and **Prayagraj** (NCR); **Banaras Locomotive Works** (Varanasi) and **Modern Coach Factory** (Raebareli); **NW-1** starts at **Prayagraj** with a terminal at Varanasi; international airports at **Kushinagar**, Lucknow, Varanasi, Ayodhya and **Jewar** (Noida); Eastern DFC runs through UP.

LAST-MINUTE 10

1. Road = about **80%** of transport; Maharashtra = most NH and SH now
2. GQ: Delhi, Kolkata, Chennai, Mumbai; NS-EW cross at **Jhansi**
3. First train **1853 Bombay-Thane**
4. NER = **Gorakhpur**; NCR = Prayagraj; SCR = **Secunderabad**
5. ICF **Perambur**; RCF **Kapurthala**; MCF **Raebareli**
6. Konkan: **Roha-Mangalore**; bullet train Mumbai-Ahmedabad
7. **Visakhapatnam** = landlocked; **Haldia** = outer port; Chennai = artificial
8. **Kandla, Mundra** = Gujarat; Ennore = 12th major port
9. Sethusamudram = **Gulf of Mannar-Palk Bay**; Alang = ship-breaking
10. **NW-1 Prayagraj-Haldia**; Kushinagar = international airport (2020)

Chapter 26

Tourism and Miscellaneous

Built from 18 UPPSC questions (1996–2018). Focus: hill station-state matching, grasslands, temple hills, odd-one-out pairs.

ASKED MOST · REVISE FIRST

- **Hill station-state matching: Chakrata, Auli-Uttarakhand; Kufri, Kasauli, Keylong, Khajjiar-HP; Gulmarg, Pahalgam-J&K; Haflong-Assam; Kalimpong-WB; Ooty-TN; Chikmagalur-Karnataka** 5× 1997, 2002, 2003, 2015, 2017
- **Wrong-pair** traps: **Jaisalmer-City of Lakes, BARC-Karnataka, IGKV-Rajasthan, Swaminathan-geology** 4× 1996, 2004, 2005, 2016
- Temple hills: **Biligiri Ranga** (Ranganatha, Karnataka); Tirupati = **Seshachalam** 2× 2015, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

26.1 Hill Stations and Tourist Centres

- **Chakrata-Uttarakhand, Haflong-Assam, Kalimpong-WB, Kufri-HP** 2× 2003, 2015
- **J&K-Gulmarg, HP-Kasauli, Gujarat-Udvada, TN-Point Calimere** 1× 1997
- **HP-Keylong, Uttarakhand-Auli, Karnataka-Chikmagalur, TN-Ooty** 1× 2002
- **Banni-Gujarat, Bugyal-Uttarakhand, Khajjiar-HP, Pahalgam-J&K** 1× 2017
- **Jaisalmer-City of Lakes** is wrong (that is **Udaipur**; Jaisalmer is the Golden City) 1× 2005
- Tourism campaign: **Incredible India** 1× 2010
- Not a ghost town: **Charkhari** (Mahoba, UP); ghost towns = Kuldhara, Dhanushkodi 1× 2015

Hill station	State
Chakrata, Auli, Mussoorie, Nainital	Uttarakhand
Kufri, Kasauli, Keylong, Khajjiar	Himachal Pradesh

Hill station	State
Gulmarg, Pahalgam, Sonamarg	Jammu and Kashmir
Haflong	Assam (only hill station)
Kalimpong, Darjeeling	West Bengal
Chikmagalur, Coorg	Karnataka
Ooty, Kodaikanal, Yercaud	Tamil Nadu
Mount Abu	Rajasthan (only hill station)
Pachmarhi	Madhya Pradesh

26.2 Temple Hills

- Tirupati (Venkateswara) is on the **Seshachalam (Tirumala)** hills, AP; none of the options fitted (*) 1× 2018
- Lord Ranganatha temple: **Biligiri Ranga** Hills (Karnataka) 1× 2015

26.3 Miscellaneous

- Both power and manure: **biogas plants** 1× 2015
- **Speed Post** launched: **1986** 1× 2013
- Locusts enter India from **Pakistan** (Thar side) 1× 2010
- **BARC-Karnataka** is wrong (BARC is in **Mumbai**) 1× 2004
- Himalayan Mountaineering Institute: **Darjeeling** 1× 2008
- **Diamond Harbour, Salt Lake City**: **Kolkata** 1× 1997
- **Jangal Mahal**: **West Bengal** (Jhargram, Purulia, Bankura) 1× 2012
- **Indira Gandhi Krishi Vishwavidyalaya-Rajasthan** is wrong (it is in **Raipur, Chhattisgarh**) 1× 2016
- **M.S. Swaminathan-geology** is wrong (he was an **agricultural** scientist) 1× 1996

TRAP ALERT

- **Udaipur** = City of Lakes; **Jaisalmer** = Golden City; **Jodhpur** = Blue City; **Jaipur** = Pink City.
- **Tirupati** = Seshachalam hills (AP); **BR Hills** = Karnataka; **Shevaroy** (Yercaud) and **Javadi** = Tamil Nadu.
- **Haflong** (Assam) and **Mount Abu** (Rajasthan) are each the only hill station of their state.

UP LINK

UP tourism links: **Charkhari** (Mahoba, "Kashmir of Bundelkhand"), the **Buddhist circuit** (Sarnath, Kushinagar, Kapilvastu-Piprahwa, Shravasti, Sankisa, Kaushambi), the **Ramayana circuit** (Ayodhya, Chitrakoot, Shringverpur), **Krishna circuit** (Mathura-Vrindavan), and the **Kashi Vishwanath Corridor**.

LAST-MINUTE 10

1. Chakrata, Auli = **Uttarakhand**; Bugyal = Uttarakhand meadows
2. Kufri, Kasauli, Keylong, **Khajjiar** = **HP**
3. Gulmarg, Pahalgam = **J&K**; **Haflong** = Assam
4. Kalimpong = WB; Ooty = TN; Chikmagalur = Karnataka
5. **Banni** grassland = Gujarat; Udvada = Gujarat
6. **Udaipur** = City of Lakes, not Jaisalmer
7. Tirupati = **Seshachalam**; Ranganatha = **BR Hills**
8. **BARC** = Mumbai; HMI = **Darjeeling**
9. **Jangal Mahal** = West Bengal; IGKV = Chhattisgarh
10. Speed Post **1986**; locusts from **Pakistan**

QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

World Geography

PYQ-Based Revision Pointers

Chapters 1 to 20

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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QUANTUM IAS & PCS

Chapter 1

The Universe, the Solar System and the Sun

Built from 43 UPPSC questions (1990–2019). Focus: Chandrasekhar and black holes, planet records, Venus, Saturn, the Sun's energy and distance.

ASKED MOST · REVISE FIRST

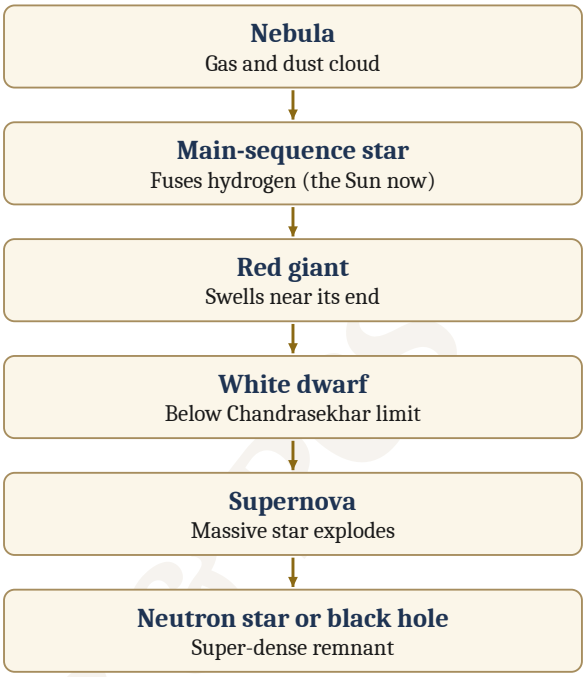
- **Black hole** theory: **S. Chandrasekhar**; Chandrasekhar limit = collapse limit of stars **8×** 1996, 1997, 1997, 2001, 2003, 2015, 2016, 2019
- **Mercury**: nearest the Sun, shortest year, smallest planet now **3×** 1991, 2002, 2004
- **Venus**: Morning Star, Earth's twin; with Mercury has **no moons** **4×** 2007, 2008, 2015, 2016
- **Saturn**: rings, Titan, Cassini (2004) **4×** 1990, 2004, 2006, 2010
- Solar energy = **nuclear fusion** **3×** 1996, 2001, 2006

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

1.1 Universe, Galaxies, Stars

- **Big Bang** explains the origin of the **universe** **1×** 2016
- Milky Way: a **spiral galaxy** **1×** 2001
- Group of stars in a shape: **constellation** **1×** 2013
- Hubble's first detailed image of a star's surface: **Betelgeuse** **1×** 1995
- Star-collapse limit: **Chandrasekhar limit** (1.4 solar masses) **1×** 1997
- Black hole theory: **S. Chandrasekhar**; first information **4×** 1996, 2015, 2019, 2016
- Star-related phenomenon: **black hole** **1×** 1997
- Black hole unseen because even **light cannot escape** (R explains A) **2×** 2001, 2003
- **Supernova**: a **dying star** **1×** 2015
- Not in space: **brittle star** (a sea animal) **1×** 1996
- Not a space term: **byte** **1×** 1997

LIFE OF A STAR



1.2 Planets

- Number of planets: **8** since Pluto became a dwarf planet (2006); 9 then (*) **1×** 1992
- Planets **shine but are non-luminous** (reflect light) **1×** 1992
- Order from Sun: **Earth, Jupiter, Uranus, Pluto** **2×** 2002, 2003
- Between Earth and Sun: **Mercury and Venus** **1×** 2010
- Earth lies between **Venus and Mars** **1×** 2012
- Nearest the Sun: **Mercury**; shortest orbit: **Mercury** **2×** 2004, 2002
- Smallest planet: **Mercury** (Pluto then) **1×** 1991
- Largest: **Jupiter** **1×** 1990
- Wrong pair: slowest planet-**Mars** (it is **Neptune**) **1×** 2011
- Longest revolution among options: **Pluto** (Neptune among planets) **1×** 2004
- No satellites: **Mercury and Venus** **1×** 2016
- Morning Star: **Venus**; Earth's twin: **Venus** **3×** 2015, 2007, 2008
- **Magellan** spacecraft: **Venus** **1×** 1991
- Day and tilt like Earth's: **Mars**; possible life: **Mars** **2×** 2001, 2000
- Jupiter's ring: **silicates** (dust) **1×** 2004

QUANTUM IAS & PCS

- Rings: **Saturn**; Cassini orbited **Saturn** (2004) 2× 1990, 2004
- Saturn is **warmer than Neptune** 1× 2010
- **Titan**: Saturn's largest moon 1× 2006
- **Nix and Hydra**: moons of **Pluto** 1× 2010
- H, He, CH_4 together: giant planets (no option exact, *) 1× 2004
- **Asteroids**: between **Mars and Jupiter** 1× 2008

Planet	Record
Mercury	Nearest, smallest, 88-day year; no moon
Venus	Brightest, hottest; spins backwards; no moon
Mars	Red planet; Earth-like day and tilt
Jupiter	Largest; fastest spin; Ganymede (largest moon)
Saturn	Rings; least dense; Titan; most moons
Uranus	Rolls on its side
Neptune	Farthest; longest year (165 years)

1.3 The Sun

- Sun's core: **plasma** (no option exact, *) 1× 2001
- ■ Solar energy: **nuclear fusion** 3× 1996, 2001, 2006
- Solar eclipse on **new moon** day; when the **Moon is between Earth and Sun** 2× 1991, 1990
- Earth-Sun distance: **149.6 million km** 1× 2010
- Nearest the Sun (perihelion): **3 January** 1× 2010

TRAP ALERT

- **Hottest = Venus** (not Mercury); **largest moon = Ganymede** (not Titan).
- Pluto options: check if the question is **before 2006**.
- Solar eclipse = **new moon**; lunar eclipse = **full moon**.

UP LINK

No UP item. Indian link: **Aditya-L1** (2023) studies the Sun from the L1 point; the Indian Standard Meridian passes near **Mirzapur**, UP.

LAST-MINUTE 10

1. Black hole, limit = **Chandrasekhar**
2. Supernova = dying star; Milky Way = spiral
3. 8 planets since **2006**
4. Mercury: nearest, smallest; Jupiter: largest
5. Venus: Morning Star, twin, hottest
6. No moons: **Mercury, Venus**
7. Mars: Earth-like day; Saturn: rings, **Titan**
8. Asteroids: **Mars-Jupiter**
9. Sun: **fusion**; 149.6 million km; perihelion **3 Jan**
10. Solar eclipse on new moon

Chapter 2

The Earth and the Moon

Built from 15 UPPSC questions (1992–2016). Focus: Earth's crust and axis, the Moon's facts, blue moon, lunar missions.

ASKED MOST · REVISE FIRST

- Only one face of the Moon is seen: rotation = revolution period 1× 2006
- **Blue moon**: two full moons in one month 2× 2007, 2009
- Match: **planet-Uranus, satellite-Moon, comet-Halley, spacecraft-Mariner** 2× 1992, 1999

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 The Earth

- Earth's nickname: **Green Planet** (per key) 1× 1999
- Most abundant element in the crust: **oxygen** 1× 2013
- Sunlight reaches Earth in about **8 minutes** 1× 2005
- Axis tilted **23.5°** (66.5° to orbital plane) 1× 2016
- True: northern end of axis = North Pole; axis stays parallel (1 and 3) 1× 2000
- A ball thrown from a satellite keeps **orbiting in the same path** 1× 1995

Crust element	Share
Oxygen	46.6%

Crust element	Share
Silicon	27.7%
Aluminium	8.1% (most common metal)
Iron	5.0% (most in the whole Earth)

2.2 The Moon

- One face seen because rotation = revolution (R explains A) 1× 2006
- Moon's core: **viscous liquid** per key (now thought solid inner + fluid outer) 1× 2004
- No sound on the Moon: **no atmosphere** 1× 2003
- Half moon: Sun-Earth-Moon angle **90°** 1× 2005
- Match: **planet-Uranus, satellite-Moon, comet-Halley, spacecraft-Mariner** 2× 1992, 1999
- Blue moon: **two full moons in a month** 2× 2007, 2009
- On the Moon an object's **weight decreases** (mass same) 1× 2006
- Earth-Moon distance: about **3,84,400 km** (key 364 thousand, perigee) 1× 2008
- SELENE: **Japan**, first Asian lunar orbiter of its kind 1× 2009

MILESTONES ON THE MOON

Luna 2 (1959)
USSR; first impact

↓

Apollo 11 (1969)
USA; first humans

↓

Chandrayaan-1 (2008)
India; water molecules

↓

Chang'e 4 (2019)
China; far side

↓

Chandrayaan-3 (2023)
India; south pole; 23 Aug = National Space Day

TRAP ALERT

- Most abundant in **crust** = oxygen; in **whole Earth** = iron.
- Weight changes on the Moon; **mass does not**.
- Blue moon is about **two full moons**, not colour.

UP LINK

No UP item. **Chandrayaan-3** landing day, 23 August, is now **National Space Day**.

LAST-MINUTE 10

1. Crust: **oxygen**, silicon, aluminium
2. Light from Sun: **8 min 20 s**
3. Axis tilt **23.5°**
4. Moon: same face, rotation = revolution
5. No air, so no sound on the Moon
6. Half moon: angle **90°**
7. Blue moon = 2 full moons in a month
8. Moon distance **3,84,400 km**
9. **SELENE** = Japan
10. Chandrayaan-3 = south pole, 2023

Chapter 3

Latitudes, Longitudes, Tropics and Seasons

Built from 25 UPPSC questions (1990–2020). Focus: time calculation, Prime Meridian and Date Line, countries on the Equator and Tropics, solstices and equinoxes.

ASKED MOST · REVISE FIRST

- Time: **15° = 1 hour**; 5 hours ahead = **75° E**; 3 hours ahead = **45° E** 2× 2006, 2008
- ■ Equator crosses **Kenya, Colombia, Gabon, Somalia**; not Rwanda or Equatorial Guinea 4× 2007, 2008, 2010, 2020
- ■ **21 June**: longest day, smallest noon shadow, summer solstice 3× 1992, 2006, 2010
- Equinoxes: **21 March and 23 September** 2× 1994, 2004
- Prime Meridian crosses **France and Algeria** 2× 2008, 2010

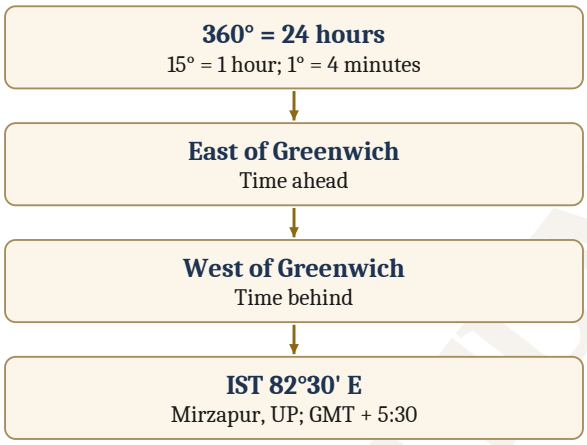
■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Latitude, Longitude, Time

- Equator meets Prime Meridian in the **Atlantic Ocean** (Gulf of Guinea) 1× 2009

- Farthest north of the options: **Rome** 1× 2011
- Noon at Greenwich, 5 pm local: **75° E** 1× 2008
- GMT 3 am, local 6 am: **45° E** 1× 2006
- Great circle with Prime Meridian: **180°** 1× 2000
- First sunrise of the millennium in India: **93°30' E** (Arunachal) 1× 2001
- Prime Meridian crosses **Algeria and France** 2× 2008, 2010
- Date Line 12 hours from Greenwich, at 180° (R explains A) 1× 1992
- Date Line has no minimum distance (per key) 1× 1997
- Ship moving at Earth's spin speed: still **10:00 Monday** 1× 2000
- Shortest path: along a **meridian** (great circle) 1× 1992

TIME FROM LONGITUDE



3.2 Equator and Tropics

- Equator does not pass: **Equatorial Guinea and Rwanda** 1× 2020
- On the Equator: **Colombia and Kenya**; **Kenya** 3× 2010, 2007, 2008
- Equator and both Tropics cross: **Africa** 1× 2004
- Tropic of Cancer does not cross **Iran** 1× 2005
- Tropic of Capricorn does not cross **Bolivia** 1× 1997
- Capricorn crosses **Australia, Namibia, Brazil, Chile** 1× 2003

Line	Key countries
Equator	Gabon, DR Congo, Uganda, Kenya, Somalia, Indonesia, Ecuador, Colombia, Brazil
Tropic of Cancer	Mexico, Egypt, Saudi Arabia, UAE, Oman, India, Bangladesh, Myanmar, China
Tropic of Capricorn	Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Madagascar, Australia

3.3 Solstices and Seasons

- Equinoxes: **21 March and 23 September** 2× 1994, 2004
- Longest day: **21 June**; summer solstice **21 June** 2× 1992, 2010
- Smallest noon shadow: **21 June** 1× 2006
- 12-hour day at summer solstice: **Equator** 1× 1997
- Noon Sun at Tropic of Cancer on winter solstice: **43°** 1× 1997
- Midnight sun: **North Pole** (and Arctic) 1× 1990

TRAP ALERT

- **Iran** is not on the Tropic of Cancer; **Bolivia** not on Capricorn.
- **Equatorial Guinea** (mainland) and **Rwanda** are off the Equator.
- Earth is **nearest the Sun in January**, yet it is northern winter.

UP LINK

IST is based on **82°30' E**, which passes near **Mirzapur** (UP). The Tropic of Cancer crosses 8 Indian states but **not UP**.

LAST-MINUTE 10

1. 15° = 1 hour; 1° = 4 minutes
2. 5 h ahead = **75° E**; 3 h ahead = **45° E**
3. Prime Meridian: **France, Algeria**
4. Date Line **180°**
5. Equator: **Kenya, Colombia**; not Rwanda
6. All three lines: **Africa**
7. Not on Cancer: **Iran**; not on Capricorn: **Bolivia**
8. Equinox **21 Mar, 23 Sep**
9. Solstice **21 June**: longest day
10. IST: 82°30' E near **Mirzapur**

Chapter 4

Geological History, Volcanoes and Earthquakes

Built from 25 UPPSC questions (2004–2021). Focus: Pangaea and Gondwana, active volcanoes and their countries, Ring of Fire, epicentre and seismograph.

ASKED MOST · REVISE FIRST

- **Volcano-country matching**: **Etna-Sicily (Italy), Vesuvius-Italy, Erebus-Ross Island, Cotopaxi-Ecuador, Merapi and Krakatoa-Indonesia, Colima and Paricutin-Mexico, Apo-Philippines, Rainier-USA** 4× 2010, 2015, 2016, 2021
- Active volcanoes: **Cotopaxi, Etna, Fuji, Vesuvius**; Aconcagua and Kilimanjaro are not 2× 2007, 2011
- **Mauna Loa, Kilauea** = Hawaii; Kilauea most active 4× 2005, 2006, 2009, 2014
- **Ring of Fire** = earthquakes, volcanoes, **Pacific** 1× 2008

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Geological History

- Age of the Earth: **uranium (radiometric) dating** 1× 2006
- Oldest fossils: about **3.5 billion** years (all options wrong, *) 1× 2005
- Original landmass: **Pangaea** 1× 2016
- Not part of Gondwana: **North America** 1× 2016
- Order: **shelled animals, insects, reptiles, mammals** 1× 2018
- Little Ice Age: **1650-1870 AD** 1× 2015
- Son valley formations order: **Sihawal, Patpara, Baghor, Khetaunhi** 1× 2017
- Thar desert: **Pleistocene and recent deposits** 1× 2018

PANGAEA BREAKS UP

Pangaea
One supercontinent (Wegener, 1912)
↓
Laurasia
North America, Europe, Asia
↓
Gondwana
S. America, Africa, India, Australia, Antarctica
↓
Tethys Sea
Between them; sediments became the Himalayas and Alps

4.2 Volcanoes

- Highest volcano among options: **Cotopaxi** (overall: Ojos del Salado) 1× 2015
- **Mauna Loa**: **Hawaii** 2× 2005, 2014
- Most active: **Kilauea** 2× 2006, 2009
- Active: **Cotopaxi, Etna, Fujiyama** (not Aconcagua) 1× 2007
- Active: **Etna and Vesuvius** 1× 2011
- **Kilimanjaro**: **Tanzania** 1× 2009
- "Lighthouse of the Mediterranean": **Stromboli** 1× 2004
- **Etna-Sicily, Vesuvius-Italy, Erebus-Ross Island, Cotopaxi-Ecuador** 1× 2010
- **Kinabalu-Malaysia, Elburz-Iran, Aconcagua-Argentina, Kilimanjaro-Tanzania** 1× 2015
- **Sabancaya-Peru, Etna-Italy, Colima-Mexico, Merapi-Indonesia** 1× 2015
- **Krakatoa**: **Indonesia** 1× 2016
- **Rainier-USA, Etna-Italy, Paricutin-Mexico, Apo-Philippines** 1× 2021

Volcano	Place
Etna / Vesuvius / Stromboli	Sicily / Naples / Aeolian Is. (Italy)
Kilauea, Mauna Loa	Hawaii
Cotopaxi, Chimborazo	Ecuador
Paricutin, Colima, Popocatepetl	Mexico
Krakatoa, Merapi, Tambora	Indonesia
Pinatubo, Taal, Apo	Philippines

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Volcano	Place
Erebus	Ross Island, Antarctica
Barren Island	India (only active)

4.3 Earthquakes

- Ring of Fire: **earthquakes, volcanoes, Pacific** (1, 2, 3) 1× 2008
- Measured on the **Richter scale** 1× 2015
- Recorded by a **seismograph** 1× 2017
- First to record: **epicentre** (surface point above focus) 1× 2017
- NW India quake-prone due to **plate tectonics** 1× 2005

TRAP ALERT

- **Etna** is on **Sicily**; Vesuvius on the mainland.
- **Aconcagua, Kilimanjaro** are volcanic but **not active**.
- Focus = inside the Earth; **epicentre** = surface above it.

UP LINK

The **Son valley** formations (Sihawal, Patpara, Baghor, Khetaunhi) lie in the middle Son basin just south of UP; India's seismic Zone IV covers western UP and the Terai.

LAST-MINUTE 10

1. **Pangaea** (Wegener); Gondwana + Laurasia
2. North America not in Gondwana
3. Little Ice Age **1650-1870**
4. Kilauea most active; Mauna Loa = Hawaii
5. **Stromboli** = lighthouse of the Mediterranean
6. Etna = Sicily; Erebus = Antarctica
7. Krakatoa, Merapi = Indonesia; Paricutin = Mexico
8. Ring of Fire = **Pacific**
9. Richter scale; seismograph; epicentre
10. Barren Island = India's active volcano

Chapter 5

Continents, Mountains, Plateaus and Valleys

Built from 42 UPPSC questions (1992–2024). Focus: continent areas, mountain-country matching, mountain types, Death Valley and basins.

ASKED MOST · REVISE FIRST

- **Mountain-country matching**: **Pyrenees-France/Spain, Apennines-Italy, Balkan-Bulgaria, Alps-Switzerland, Atlas and Toubkal-Morocco, Kilimanjaro-Tanzania, Elburz-Iran, Black Forest-Germany** 9× 2004, 2004, 2010, 2012, 2014, 2014, 2020, 2023, 2024
- **Africa** has the **most countries**; largest is **Asia** (key once said Africa) 2× 2014, 2015
- **Andes** = longest range 2× 2008, 2016
- **Death Valley**: California, USA; rift valley; extreme heat 4× 2005, 2009, 2012, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Continents

- Largest continent: **Africa** per key (actually **Asia**; Africa second) 1× 2014
- Descending: **Africa, South America, Europe, Australia** 1× 2015
- Descending: **Asia, Africa, North America, Europe** 1× 2017
- Smallest: **Australia**; two smallest: **Australia and Europe** 2× 1992, 2008
- Most countries: **Africa** (54) 1× 2015
- Most plains by share: **Europe** 1× 2013

5.2 Mountain Ranges

- Longest range: **Andes** 1× 2016
- Length order: **Andes, Rockies, Himalaya, Great Dividing Range** 1× 2008
- **Andes-South America, Altai-Asia, Alps-Europe, Appalachian-North America** 1× 2023

- West to east: **Pontic, Zagros, Hindu Kush, Karakoram** 1× 2024
- Not Tertiary: **Appalachians** (old fold) 1× 2014
- **Alps-folded, Vosges-block, Vindhya-residual, Fuji-volcanic** 1× 2013
- "Sea of mountains": **British Columbia** 1× 2015
- **White Mountains** and **Black Mountain**: **USA** 2× 2001, 2009
- **Allegheny-USA, Cantabrian-Spain, Elburz-Iran, Mackenzie-Canada** 1× 2010
- **Darling Range**: **south-western** coast of Australia 1× 2020
- India-Myanmar ranges: none of the options (Patkai, Naga, Lushai/Arakan) 1× 1998
- **Black Forest**: **Germany**; along the **Rhine** 3× 2004, 2012, 2014
- **Titlis**: **Switzerland** 1× 2004
- In Italy: **Apennines** 1× 2014
- **Spain-France-Pyrenees, Switzerland-Alps, Bulgaria-Balkan, Italy-Apennine** 1× 2024
- **Atlas-Morocco, Kilimanjaro-Tanzania, Chimborazo-Ecuador, Pyrenees-France/Spain** 1× 2020
- **Kilimanjaro-Tanzania, Toubkal-Morocco, Stanley-Uganda, Hoggar-Algeria** 1× 2023
- **Drakensberg**: **South Africa** 2× 2007, 2008

Type	Examples
Young fold	Himalaya, Alps, Andes, Rockies, Atlas
Old fold	Appalachians, Urals, Aravalli
Block	Vosges, Black Forest, Sierra Nevada
Volcanic	Fuji, Kilimanjaro, Cotopaxi
Residual	Vindhya, Nilgiri, Aravalli

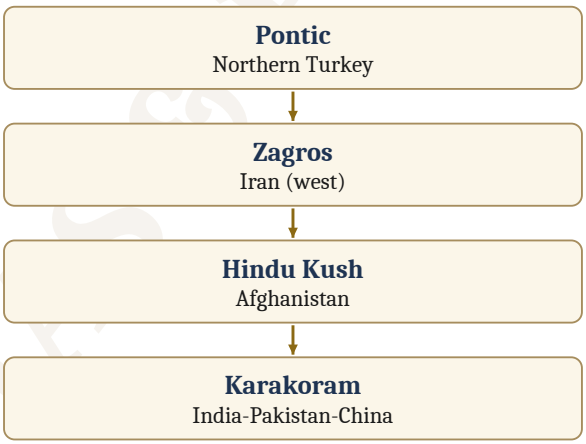
5.3 Plateaus

- Tibetan Plateau average height: about **5 km** 1× 2013
- Largest plateau among options: **Colorado** (per key) 1× 2017
- Telegraphic Plateau: part of the **North Atlantic Ridge** 1× 2019

5.4 Valleys and Basins

- **Death Valley**: a **rift valley**; "Devil's Golf Course", **USA**; known for **heat**; in **California** 4× 2012, 2014, 2005, 2009
- **Panjshir Valley**: **Afghanistan** 1× 2002
- **Silicon Valley**: **California** 1× 2002
- **Taylor Valley**: **Antarctica** 1× 1998
- **Great Artesian Basin**: **Australia** 2× 2002, 2003
- **Great Basin**: **USA** 1× 2004
- **Valley of the Kings**: **Egypt** 1× 1998
- **Blind valley**: **karst** region 1× 2013

WEST TO EAST ACROSS WEST ASIA



TRAP ALERT

- **Great Basin** = USA; **Great Artesian Basin** = Australia.
- **Darling Range** = WA coast; **Darling River** = NSW.
- **Appalachians** are old fold, not Tertiary.

UP LINK

UP links: the **Vindhyan** hills (residual mountains) reach Mirzapur and Sonbhadra; UP lies on the Indo-Gangetic **plain**, built from Himalayan alluvium.

LAST-MINUTE 10

1. Asia largest; **Africa** most countries; Australia smallest
2. **Andes** longest; then Rockies
3. Pyrenees = France-Spain; Apennines = Italy; Balkan = Bulgaria
4. Black Forest = Germany (Rhine); Titlis = Switzerland
5. Atlas, Toubkal = Morocco; Drakensberg = South Africa
6. Elburz = Iran; Mackenzie = Canada; Allegheny = USA
7. Appalachians = old fold; Vosges = block; Vindhya = residual
8. Death Valley = rift, hottest, California
9. Great Artesian Basin = **Australia**
10. Valley of the Kings = **Egypt**

Chapter 6

Deserts and Grasslands

Built from 31 UPPSC questions (1993–2025). Focus: desert-country matching, Atacama and Kalahari, temperate vs tropical grasslands, Pampas.

ASKED MOST · REVISE FIRST

- **Pampas** = **Argentina** (temperate grassland, main farm belt) 5× 2005, 2012, 2016, 2017, 2018
- **Grassland matching**: **Llanos-Venezuela/S. America, Steppe-Eurasia, Prairies-N. America, Veld-South Africa, Downs-Australia, Puszta-Hungary** 6× 2002, 2010, 2014, 2015, 2016, 2017
- **Savanna** = tropical, mostly **Africa**, "land of big game" 5× 1993, 2011, 2012, 2013, 2019
- **Kalahari** = Botswana, Namibia; Southern Hemisphere 3× 2008, 2010, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Definition and Location

- Desert: rainfall under **25 cm** a year 1× 2009
- Hot deserts on **western margins at 15°-30°**; trade winds lose moisture moving west (R explains A) 1× 2019
- Deserts can supply solar energy; they get a year's human energy use in 6 hours of sun (R explains A) 1× 2013

6.2 Deserts of the World

- Biggest desert: **Sahara** (largest hot desert) 1× 1994
- Smallest to largest: **Great Sandy, Gobi, Arabian, Sahara** 1× 2023
- Most populated desert: **Thar**; largest in South Asia: **Thar** 2× 2012, 1996
- **Gobi**: **Mongolia and China** only 2× 1993, 2025
- **Taklamakan**: **China** 1× 2008
- Wrong: **Gibson-Brazil** (Gibson is in **Australia**) 2× 2016, 2018
- **Karakum**: **Turkmenistan** 1× 2022
- **Dasht-e-Lut**: **Iran** 1× 2015
- Not true: Atacama lies in **southern** Chile (it is **northern**) 1× 2006
- Driest place: **Atacama** 1× 2009
- Temperate desert: **Patagonian** 1× 2011
- **Kalahari**: **Botswana and Namibia**; in the **Southern Hemisphere** 3× 2008, 2010, 2015

Desert	Country
Sahara	North Africa
Arabian (Rub al Khali)	Saudi Arabia
Gobi	Mongolia, China
Taklamakan	China (Xinjiang)
Karakum / Kyzylkum	Turkmenistan / Uzbekistan-Kazakhstan
Dasht-e-Lut	Iran
Thar	India, Pakistan
Kalahari / Namib	Botswana, Namibia / Namibia
Atacama	Northern Chile (driest)
Patagonian	Argentina (temperate)
Gibson, Great Sandy, Great Victoria	Australia

6.3 Grasslands

- Wrong: **Veld-Australia** (Veld = South Africa) 1× 2015

QUANTUM IAS & PCS

- **Llanos-South America, Steppe-Europe/Eurasia, Prairies-North America, Veld-Africa** 1× 2016
- **Pampas-Argentina, Puszta-Hungary, Veld-South Africa, Downs-Australia** 1× 2017
- Wrong: **Downs-tropical** (Downs are **temperate**) 1× 2002
- Not temperate: **Savanna** 2× 2012, 2013
- Eurasian temperate grassland: **Steppes** 1× 2014
- Savanna most widespread in **Africa** 2× 1993, 2011
- "Land of big game": **tropical savanna** 1× 2019
- Treeless South American plains: **Pampas**; in **Argentina** 3× 2005, 2012, 2018
- Pampas main farm region: fertile soil, temperate climate (R explains A) 1× 2016
- Correct: **Llanos-Venezuela** 1× 2010

GRASSLANDS BY TYPE

Temperate
Prairies, Steppes, Pampas, Veld, Downs, Puszta, Canterbury

↓

Tropical
Savanna (Africa), Llanos (Venezuela), Campos (Brazil)

TRAP ALERT

- **Gibson** = Australia; **Atacama** = northern Chile.
- **Downs** and **Veld** are temperate; **Llanos** and **Campos** are tropical.
- Largest desert overall = **Antarctica**; largest hot = **Sahara**.

UP LINK

No UP item. Indian link: the **Thar**, the most densely populated desert, is fed by the Indira Gandhi Canal, while UP's Bundelkhand faces semi-arid conditions.

LAST-MINUTE 10

1. Desert: under **25 cm** rain
2. Hot deserts on **west coasts**, 15°-30°
3. Sahara largest hot; Thar most populated
4. Gobi = Mongolia, China; Karakum = Turkmenistan
5. Gibson = **Australia**; Lut = **Iran**
6. Atacama = northern Chile, driest
7. Patagonian = temperate desert
8. Kalahari = Botswana, Namibia
9. Pampas = Argentina; Puszta = Hungary; Downs = Australia
10. Savanna = Africa, tropical; Llanos = Venezuela

Chapter 7

Countries and their Borders

Built from 42 UPPSC questions (1994–2023). Focus: areas and coastlines, regional groups, landlocked countries, dependent territories, boundary lines.

ASKED MOST · REVISE FIRST

- **Landlocked**: **Laos** (only one in SE Asia), **Bolivia, Chad**, Afghanistan, Hungary, Switzerland; **Liberia** and **Azerbaijan** (per key) are not 7× 2001, 2005, 2009, 2010, 2013, 2015, 2022
- **North America** has **no** landlocked country 1× 2005
- Boundary lines: **Radcliffe-India/Pakistan, Durand-Afghanistan/Pakistan, Maginot-France/Germany, 38th parallel-Koreas, Rio Grande-USA/Mexico** 5× 1994, 2005, 2008, 2014, 2015
- Longest land border with India: **Bangladesh** 1× 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 Size and Coastline

- Largest country: **Russia** (not in options, *) 1× 1996
- Ascending area: **Argentina, India, Australia, Brazil** 1× 2012

- Not larger than India: **France** 1× 2013
- Longest coastline: **Canada** 1× 2015
- Longest border with India: **Bangladesh** 1× 2013

7.2 Regions and Neighbours

- Not in the Arabian Peninsula: **Syria** (only 4) 1× 2023
- Not in the Middle East: none clearly (Turkiye least wrong, *) 1× 2021
- Israel borders **Lebanon, Syria, Jordan, Egypt** 1× 2017
- Gaza borders **Egypt and Israel** (both) 1× 2023
- Horn of Africa: **Somalia, Ethiopia, Djibouti**; not **Sudan** 2× 1999, 2009
- Not Scandinavian: **Finland** 1× 2001
- No Baltic coast (per key): **Sweden and Norway** option (Norway has none) 1× 2021
- Not Central American: **Sierra Leone** 1× 2014
- Not in Africa: **Guyana** 1× 2017
- Not on Africa's west coast: **Botswana** 1× 2016
- Zambia borders all four, Botswana by 150 m (*) 1× 2013
- Afghanistan does not border **Russia** 2× 2002, 2012

7.3 South-East Asia and Islands

- Indonesia has no land border with **Brunei** 1× 2021
- Indonesia: over 15,000 islands, over 150 active volcanoes: key says neither 1× 2021
- **Borneo** divided among **three** countries (key also says volcanic) 1× 2022
- Galapagos: **volcanic** (true); part of Peru (false; **Ecuador**) 1× 2021
- **Australia-Christmas Island, Denmark-Greenland, France-Martinique, Spain-Santa Cruz** 1× 2000
- In America but politically European: **Greenland** 1× 2002
- **Aleutian-USA, Bear Island-Norway, Greenland-Denmark, Franz Josef Land-Russia** 1× 2014

7.4 Landlocked and Transcontinental

- Not landlocked: **Liberia** 1× 2005
- Landlocked in Africa: **Chad** 1× 2010
- Only landlocked in SE Asia: **Laos** 2× 2001, 2015
- Landlocked: **Bolivia** 1× 2022
- No landlocked country: **North America** 1× 2005
- Landlocked: **Afghanistan, Hungary, Switzerland** (not Malaysia) 1× 2009
- Not landlocked (Caspian coast): **Azerbaijan** per key 1× 2013
- Landlocked: **Armenia, Botswana, Ethiopia**; Montenegro is coastal (*) 1× 2016
- South Sudan: **landlocked**, main river **White Nile** (I and III); capital Juba 1× 2010
- Transcontinental: **Georgia and Turkey** 1× 2009
- Zimbabwe was **Southern Rhodesia** 1× 2004

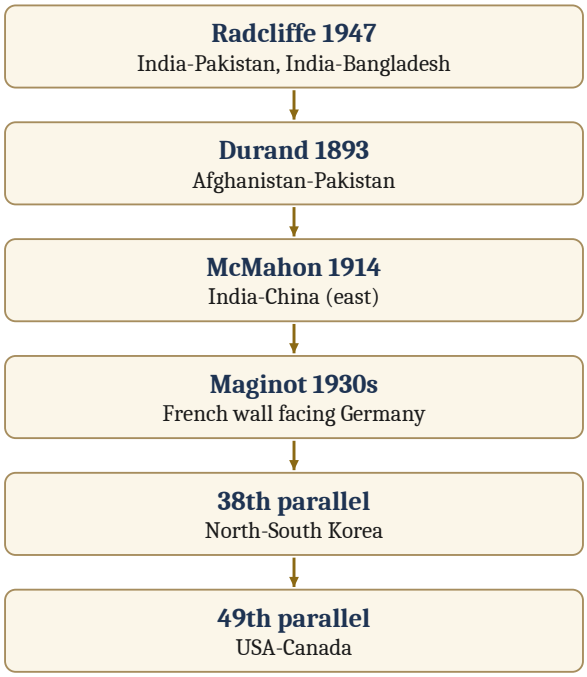
Continent	Landlocked examples
Africa (16, most)	Ethiopia, Chad, Niger, Mali, Zambia, Zimbabwe, Botswana, Uganda
Asia	Afghanistan, Nepal, Bhutan, Laos, Mongolia, Central Asian states
Europe	Switzerland, Austria, Hungary, Czechia, Serbia
South America	Bolivia, Paraguay
North America, Australia	None

7.5 Boundary Lines

- Wrong: **Durand Line-Bangladesh/India** (it is **Afghanistan-Pakistan**) 1× 2015
- **Radcliffe Line**: **India and Pakistan** 1× 2014
- **Maginot Line**: **France-Germany** 1× 1994
- Rivers as borders: **Rio Grande** and Rhine (two right, *) 2× 2002, 2003
- **Rio Grande**: **Mexico and USA** 1× 2008
- Koreas divided at **38° N** 1× 2005

QUANTUM IAS & PCS

FAMOUS LINES



TRAP ALERT

- **Guyana** = South America; **Guinea** = Africa.
- **Finland** is Nordic, not Scandinavian.
- Uzbekistan and Liechtenstein are **doubly landlocked**.

UP LINK

UP borders **Nepal** (open border, 1,751 km for India overall), with border districts such as Maharajganj, Siddharthnagar, Bahraich, Lakhimpur Kheri and Pilibhit.

LAST-MINUTE 10

1. Russia largest; India 7th; Canada longest coast
2. Bangladesh = longest border with India
3. Horn of Africa: Somalia, Ethiopia, Eritrea, Djibouti
4. Finland not Scandinavian; Guyana not African
5. Laos = only landlocked in SE Asia
6. Bolivia, Paraguay = landlocked in S. America
7. North America: no landlocked country
8. Greenland = Denmark; Christmas Island = Australia
9. Durand = Afghanistan-Pakistan; Radcliffe = India-Pakistan
10. 38° N = Koreas; Rio Grande = USA-Mexico

Chapter 8

Capital Cities, Towns and Geographical Sobriquets

Built from 50 UPPSC questions (1990–2024). Focus: capital matching by region, largest city vs capital, Central Asian and African capitals, nicknames.

ASKED MOST · REVISE FIRST

- **Capital matching** is the format in almost every question here **11×** 1992, 1995, 1997, 2003, 2009, 2009, 2011, 2015, 2016, 2019, 2023
- **Central Asia**: **Tashkent, Dushanbe, Bishkek, Ashgabat, Astana**; Bishkek is **not** Tajikistan's **4×** 2007, 2009, 2011, 2019
- **Africa** wrong pairs: **Nigeria-Rabat, Cape Verde-Bamako, Khartoum-Ethiopia** **3×** 2007, 2020, 2024
- **Finland**: Helsinki; "land of a thousand lakes"; Oslo is **Norway's** **4×** 1993, 2001, 2012, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

8.1 Europe

- Not capitals: **Rotterdam, Zurich** (Riga, Zagreb are) **1×** 2023
- All pairs correct; key took **Berlin-Germany** (*) **1×** 2008
- **Bulgaria-Sofia, Georgia-Tbilisi, Iceland-Reykjavik, Norway-Oslo** **1×** 2003
- **Helsinki**: **Finland**; **Finland-Oslo** is wrong **2×** 1993, 2001
- Wrong: **Christchurch-New Zealand** (capital **Wellington**) **1×** 2010
- Not a national capital: **Bonn** **1×** 2006

8.2 Asia

- **Myanmar-Yangon (then; now Nay Pyi Taw), Cambodia-Phnom Penh, Vietnam-Hanoi, Laos-Vientiane** **1×** 2016
- Wrong: **Khartoum-Ethiopia** (Khartoum = Sudan; Ethiopia = Addis Ababa) **1×** 2024

- Wrong: **Bishkek-Tajikistan** (Bishkek = Kyrgyzstan) 1× 2007
- **Uzbekistan-Tashkent, Tajikistan-Dushanbe, Kyrgyzstan-Bishkek, Turkmenistan-Ashgabat** 1× 2019
- **Azerbaijan-Baku, Gambia-Banjul, Kazakhstan-Astana, Latvia-Riga** 1× 2011
- Correct: **Lebanon-Beirut** 1× 2010
- Wrong: **China-Shanghai** (Beijing) 1× 2005
- **Sri Lanka-Colombo, Maldives-Male, Burma-Rangoon, Zimbabwe-Harare, Mauritius-Port Louis** 1× 1992
- **Palestine-Ramallah, Kosovo-Pristina, Turkish Cyprus-Nicosia, Taiwan-Taipei** 1× 2015
- Not in the Sunni Triangle: **Basra** 1× 2022
- Not in Afghanistan: **Fergana, Samarkand** (Uzbekistan) 1× 2023
- Highest of the options: **Lhasa** 1× 1993
- Wrong: **Baikonur-Ukraine** (it is in **Kazakhstan**) 1× 1992

Country	Capital
Kazakhstan	Astana
Uzbekistan / Tajikistan	Tashkent / Dushanbe
Kyrgyzstan / Turkmenistan	Bishkek / Ashgabat
Azerbaijan / Georgia / Armenia	Baku / Tbilisi / Yerevan
Myanmar	Nay Pyi Taw (Yangon earlier)
Laos / Cambodia / Vietnam	Vientiane / Phnom Penh / Hanoi

8.3 Africa

- Wrong: **Nigeria-Rabat** (Nigeria = Abuja; Rabat = Morocco) 1× 2007
- **Windhoek-Namibia, Accra-Ghana, Nairobi-Kenya, Lusaka-Zambia** 1× 2009
- **Burundi-Bujumbura (Gitega now), Kazakhstan-Astana, Laos-Vientiane, Mali-Bamako** 1× 2009
- **Kampala-Uganda, Kigali-Rwanda, Kinshasa-DR Congo, Khartoum-Sudan** 1× 2023
- **Brazil-Brasilia, Cuba-Havana, Kenya-Nairobi, Zambia-Lusaka** 1× 1995
- **Australia-Canberra, Tajikistan-Dushanbe, Tanzania-Dodoma, Switzerland-Bern** 1× 1997
- Wrong: **Cape Verde-Bamako** (Praia; Bamako = Mali) 1× 2020
- Wrong: **Maldives-Malabo** (Male; Malabo = Equatorial Guinea) 1× 2017

8.4 Americas and Oceania

- Correct: **Cyprus-Nicosia** 1× 2012
- Wrong: **Dominica-Santo Domingo** (Roseau; Santo Domingo = Dominican Republic) 1× 2021
- **NSW-Sydney, Queensland-Brisbane, Victoria-Melbourne, Western Australia-Perth** 1× 2021

8.5 Towns

- **Pisa** (leaning tower): **Italy** 1× 1993
- Mega cities on coasts as gateways to ocean routes (R explains A) 1× 2005
- Not in Silicon Valley: **Santa Rosa** 1× 2013
- Wrong: **Tasmania-Uganda** (Tasmania = Australia) 1× 1990
- South of the Arctic Circle: **Iceland** 1× 2013
- **Sun City**: **South Africa** 1× 2011
- **Davos-Switzerland, Barcelona-Spain, Auckland-New Zealand, Kandy-Sri Lanka** 1× 2014

LARGEST CITY IS NOT THE CAPITAL



8.6 Sobriquets

- Continent of birds: **South America** 1× 1994
- Blue Grass State: **Kentucky** 1× 2004
- Gateway to the West (Turkey): **Istanbul** 2× 2005, 2010
- Land of the rising sun: **Japan** 1× 1991
- Garden of lakes: **Finland**; thousand lakes: **Finland** 2× 2012, 2015
- City of the Golden Gate: **San Francisco** 1× 2013
- Manchester of the East: **Osaka** 1× 2014
- Pearl of Siberia: **Lake Baikal** 1× 2015
- Land of morning calm: **Korea** 1× 2015
- Oil capital of Europe: **Aberdeen** 1× 2017
- Land of Promise: **Mindanao** 1× 2021

Nickname	Place
Land of the Midnight Sun	Norway
Land of the Thunder Dragon	Bhutan
Land of White Elephants	Thailand
Land of a Thousand Hills	Rwanda
Sugar Bowl / Pearl of the Antilles	Cuba
Playground of Europe	Switzerland
Emerald Isle	Ireland
Big Apple / Windy City	New York / Chicago

TRAP ALERT

- **Dominica** (Roseau) vs **Dominican Republic** (Santo Domingo).
- **Malabo** = Equatorial Guinea; **Bamako** = Mali; **Rabat** = Morocco.
- **Baikonur** cosmodrome is in **Kazakhstan**.

UP LINK

UP's own sobriquets: **Kanpur** = Manchester of the East (Indian usage; UPPSC keys use Osaka), **Varanasi** = oldest living city, **Agra** = city of the Taj, **Lucknow** = city of nawabs.

LAST-MINUTE 10

1. Tashkent, Dushanbe, **Bishkek** (Kyrgyzstan), Ashgabat, Astana
2. Oslo = Norway; Helsinki = Finland
3. Nicosia = Cyprus; Ramallah = Palestine; Pristina = Kosovo
4. Abuja = Nigeria; Rabat = Morocco; Bamako = Mali
5. Khartoum = Sudan; Addis Ababa = Ethiopia
6. Kampala, Kigali, Kinshasa, Dodoma, Windhoek, Accra
7. Baikonur = Kazakhstan; Basra not in Sunni Triangle
8. Osaka = Manchester of the East; Aberdeen = oil capital
9. Finland = thousand lakes; Korea = morning calm
10. Kentucky = Blue Grass; Mindanao = Land of Promise

Chapter 9

Oceans, Seas and Ocean Currents

Built from 35 UPPSC questions (1991–2025). Focus: seas and their neighbours, warm and cold currents by ocean, salinity, tides, deepest points.

ASKED MOST · REVISE FIRST

- **Benguela** = cold current of the **South Atlantic**; **Agulhas** = warm, **Indian Ocean** 6× 2005, 2014, 2017, 2020, 2023, 2025
- Deepest points: **Pacific-Mariana (Challenger), Atlantic-Puerto Rico, Indian-Java (Sunda), Arctic-Molloy/Eurasian Basin** 3× 1991, 2017, 2022
- Cold currents: **Humboldt (Peru)**, California; warm: **Brazil**, Agulhas 4× 1995, 1998, 2017, 2025
- Salinity highest near the **Tropics**; most of it is **sodium chloride** 2× 2000, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 The World's Water

- **Datum line**: the sea-level reference for heights and depths 2× 1991, 1994
- Water stores, descending: **oceans, ice and glaciers, groundwater, rivers** 1× 2025
- Most fresh water: **glaciers and ice** 1× 2015
- Gravitational soil water tension: **below 1/3 atmosphere** 1× 2010

9.2 Oceans and Seas

- Indian Ocean: Greeks' **Erythraean Sea**; spans **both sides of the Equator** (both) 1× 2025
- **Red Sea**: an **axial trough** (rift) 1× 2017
- **Baku**: on the **Caspian Sea** 1× 2019
- West to east: **Mediterranean, Black, Caspian, Aral** 1× 2016
- **Aral Sea**: **Kazakhstan and Uzbekistan** 1× 2012
- Not on the Black Sea: **Syria** 1× 2021
- Sea between Italy, Sicily, Sardinia, Corsica: **Tyrrhenian** 1× 2010
- Not on the Mediterranean: **Chad** 1× 2010
- Deepest of the options (per key): **South China Sea** 1× 2015
- No coastline: **Sargasso Sea**; known for **seaweed (sargassum)** 2× 2016, 2014

9.3 Ocean Currents

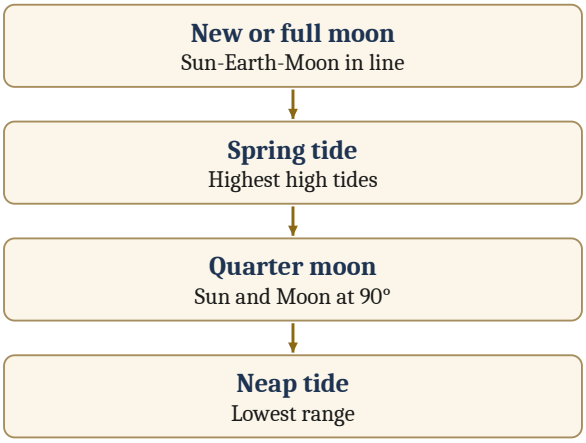
- Indian Ocean current: **Agulhas** 1× 2020
- Not cold: **Agulhas** 1× 2017
- Not in the Indian Ocean: **Benguela** 1× 2014
- Atlantic current: **Benguela** 1× 2023
- Cold, South Atlantic: **Benguela** 1× 2005
- Not in the South Atlantic loop: **Canary** (North Atlantic) 1× 2000
- Not cold: **Agulhas and Brazil** (1 and 2) 1× 2025
- Cold: **Peruvian (Humboldt)** 1× 1995
- Wrong: **Humboldt-North Pacific** (it is **South Pacific**) 1× 1998
- Warm: **Brazil Current** 1× 2017

Ocean	Warm	Cold
North Atlantic	Gulf Stream, N. Atlantic Drift	Labrador, Canary
South Atlantic	Brazil	Benguela, Falkland
North Pacific	Kuroshio	Oyashio, California
South Pacific	East Australian	Humboldt (Peru)
Indian	Agulhas, Mozambique	West Australian

9.4 Salinity and Tides

- Main salt in sea water: **sodium chloride** 1× 2000
- Highest salinity (per key): **Lake Van** (Dead Sea is actually saltier) 1× 1995
- Saltiest sea: Red Sea among seas; Dead Sea overall (*) 2× 1998, 2012
- Salinity is highest near the **Tropics of Cancer and Capricorn** 1× 2023
- Salinity gradient: **halocline** 1× 2015
- Tides: **gravity, centripetal and centrifugal forces** 1× 1991
- **Spring tide**: Sun, Earth, Moon in a **straight line** 1× 1999

SPRING AND NEAP TIDES



9.5 Ocean Deeps

- Deepest trench: **Challenger Deep** (Mariana) 1× 1991
- **Pacific-Mariana, Atlantic-Puerto Rico, Indian-Java, Arctic-Eurasian Basin** 1× 2017
- **Pacific-Mariana, Arctic-Molloy, Indian-Sunda, Atlantic-Puerto Rico** 1× 2022

QUANTUM IAS & PCS

TRAP ALERT

- **Benguela** (Atlantic) vs **Agulhas** (Indian), both near South Africa.
- **Canary** is North Atlantic; **Humboldt** is South Pacific.
- Warm + cold meet = fog and fish (Grand Banks, Japan).

UP LINK

No UP item. India link: the **Deep Ocean Mission** and **Samudrayaan** (Matsya-6000) target the Indian Ocean floor.

LAST-MINUTE 10

1. Oceans 96.5%; fresh water mostly in **ice**
2. Red Sea = rift; Sargasso = no coast
3. Baku = Caspian; Aral = Kazakhstan-Uzbekistan
4. Tyrrhenian = Italy-Sicily-Sardinia-Corsica
5. Benguela = cold, S. Atlantic; Agulhas = warm, Indian
6. Humboldt = cold, S. Pacific; Brazil = warm
7. Canary not in the South Atlantic loop
8. Salinity: NaCl; highest near the Tropics; halocline
9. Spring tide: Sun-Earth-Moon in a line
10. Challenger Deep = deepest; Puerto Rico = Atlantic; Java = Indian

Chapter 10

Rivers of the World

Built from 33 UPPSC questions (1992–2025). Focus: Congo and Limpopo, Nile, Asian rivers and their outflows, Rhine and Mississippi, city-river matching.

ASKED MOST · REVISE FIRST

- **City-river matching**: Berlin-Spree, Rome-Tiber, Paris-Seine, London-Thames, Hamburg-Elbe, Antwerp-Scheldt, Rotterdam-Rhine, Vienna-Danube, Kinshasa and Brazzaville-Congo, Asuncion-Paraguay, Basra-Shatt al-Arab, Niamey-Niger **13×** 1998, 1999, 2002, 2004, 2004, 2009, 2015, 2015, 2016, 2016, 2017, 2019, 2025
- **Congo (Zaire)** crosses the Equator twice; **Limpopo** crosses the Tropic of Capricorn twice **2×** 2010, 2015
- **Mississippi**: source **Lake Itasca**; **bird's-foot** delta **2×** 2015, 2018
- **Rhine**: rift valley; **Rotterdam** at its mouth **3×** 1998, 2021, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

10.1 Africa

- Crosses the Equator twice: **Zaire (Congo)** **1×** 2015
- Crosses the Tropic of Capricorn twice: **Limpopo** **1×** 2010
- Not in the Nile basin: **Chad** **1×** 2013
- **Blue Nile** rises in **Lake Tana** **1×** 2016
- Egypt "gift of the Nile", an oasis in the Sahara (R explains A) **1×** 2005

THE NILE



10.2 Asia and Europe

- Largest river of Asia: **Yangtze** **1×** 2016

- Flows south: **Salween** 1× 2010
- Wrong: **Mekong-south-west** (it flows south/south-east) 1× 2024
- **Lena-Arctic, Amur-Pacific, Tigris-Persian Gulf, Mahi-Arabian Sea** 1× 2016
- Mekong rises in **Tibet** (true); delta in southern Cambodia (false, it is **Vietnam**) 1× 2021
- **Volga** falls into the **Caspian Sea** 1× 1992
- Rhine statements: only 2 correct; **Rotterdam at its mouth** (only 2) 2× 2021, 2023
- Rift-valley river: **Rhine** 1× 1998
- Wrong: **Rhone-Spain** (Rhone = France/Switzerland) 1× 2008
- Best radial drainage: **Sri Lanka**; **Mahaweli Ganga** = Sri Lanka 2× 2008, 2013
- **Chisapani Gorge**: **Nepal** (Karnali) 1× 2015
- **Nobi and Kanto** plains: **Japan** 1× 2021

10.3 Americas

- **Lake Itasca**: source of the **Mississippi** 1× 2015
- Bird's-foot delta: **Mississippi** 1× 2018
- USA-Mexico border river: **Rio Grande** (key also Colorado) 1× 2023

10.4 Cities and Rivers

- Wrong: **New Orleans-Missouri** (Mississippi) and **Hyderabad-Paleru** (Musi) 1× 2025
- **Shatt al-Arab-Basra, Paraguay-Asuncion, Niger-Niamey, Danube-Vienna** 1× 2016
- **Paris-Seine, Kinshasa-Congo, Bangkok-Chao Phraya, Asuncion-Paraguay** 1× 2019
- Wrong: **Sydney-Seine** 1× 2015
- **Berlin-Spree, Lahore-Ravi, New York-Hudson, Rome-Tiber** 2× 1999, 2002
- Wrong: **Vienna-Volga** (Danube) 3× 1998, 2004, 2016
- Correct: **Hamburg-Elbe** 1× 2004
- **Potomac-USA, Nile-Sudan, Tigris-Iraq, Thames-England** 1× 2003
- Wrong: **Lisbon-Tiber** (Tagus) 1× 2009
- **London-Thames, Antwerp-Scheldt, Rotterdam-Rhine, Hamburg-Elbe** 1× 2015
- **Khartoum-Nile, Brazzaville-Congo, Rotterdam-Rhine, Paris-Seine** 1× 2017

City	River
London / Paris / Rome	Thames / Seine / Tiber
Berlin / Hamburg	Spree / Elbe
Vienna, Budapest, Belgrade	Danube
Lisbon / Warsaw	Tagus / Vistula
Kinshasa, Brazzaville	Congo
Bamako, Niamey	Niger
Baghdad / Basra	Tigris / Shatt al-Arab
Bangkok	Chao Phraya
New York / Washington	Hudson / Potomac
Asuncion	Paraguay

TRAP ALERT

- **Vienna** is on the Danube, not the Volga; **Lisbon** on the Tagus, not the Tiber.
- Mekong delta is in **Vietnam**, not Cambodia.
- **Bird's foot** = Mississippi; **arcuate** = Nile, Ganga.

UP LINK

UP link: India's **Ganga** forms an arcuate delta (with the Brahmaputra); the 2025 question used **Hyderabad-Musi** as a home trap. Varanasi is UP's Ganga city, as Vienna is the Danube's.

LAST-MINUTE 10

1. Congo crosses Equator twice; Limpopo crosses Capricorn twice
2. Blue Nile from **Lake Tana**; meets White Nile at **Khartoum**
3. Yangtze = Asia's largest; Volga → Caspian
4. Lena → Arctic; Amur → Pacific; Tigris → Persian Gulf
5. Rhine: rift valley; Rotterdam at mouth
6. Mississippi: Itasca; bird's-foot delta
7. Rio Grande = USA-Mexico border
8. Sri Lanka = radial drainage (Mahaweli)
9. Kinshasa, Brazzaville = Congo
10. Berlin-Spree; Hamburg-Elbe; Antwerp-Scheldt

Chapter 11

Islands, Lakes and Waterfalls

Built from 33 UPPSC questions (1990–2023). Focus: largest islands, Japan and Indonesia island order, Pacific island groups, lake records, Great Lakes cities, waterfalls.

ASKED MOST · REVISE FIRST

- ■ Lake records: **Caspian** largest, **Baikal** oldest and deepest, **Tanganyika** second deepest and longest **3×** 2008, 2014, 2015
- • **Lake Michigan** lies **entirely in the USA** **2×** 2015, 2017
- ■ Japan north to south: **Hokkaido, Honshu, Shikoku, Kyushu**; Honshu largest **3×** 1991, 2017, 2019
- ■ Lake-country wrong pairs: **Onega-Canada, Reindeer-Russia, Five Flower-Mongolia, Malawi-Asia** **4×** 2014, 2015, 2017, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

11.1 Islands

- Largest island: **Greenland** **1×** 1993
- Largest of options: **New Zealand's South Island** **1×** 2015
- Correct: **Nuuk** (Greenland) is about 3 m above sea level **1×** 2012
- Divided among three countries: **Borneo** **1×** 2010
- South Indian Ocean: Chagos, Mauritius, Seychelles all fit (*) **1×** 2004
- Not correct: **Falklands in the Indian Ocean** (they are South Atlantic) **1×** 1993
- Socotra: belongs to **Yemen** (not Oman); UNESCO site 2008 (only 2) **1×** 2021
- Salt-pan island: **Saint-Martin** **1×** 2017
- **Macau** returned to China in **1999** **1×** 1990

11.2 Island Groups

- Largest island of Japan: **Honshu** **1×** 1991
- North to south: **Hokkaido, Honshu, Shikoku, Kyushu** **2×** 2017, 2019
- Indonesia west to east: **Sumatra, Java, Bali, Lombok** **1×** 2021
- Melanesia: **Solomon Islands** **1×** 2012
- Not Polynesia: **Tinian** (Micronesia) **1×** 2021
- Not in the Caribbean: **Madeira** (Atlantic, Portugal) **1×** 2021
- "Sheep Islands": **Faroe Islands** **1×** 2015

Pacific group	Islands
Melanesia	New Guinea, Solomons, Vanuatu, Fiji, New Caledonia
Micronesia	Guam, Marshalls, Kiribati, Tinian, Palau, Nauru
Polynesia	Hawaii, Samoa, Tonga, Tuvalu, Tahiti, New Zealand

11.3 Lakes

- Largest lake by area and volume: **Caspian Sea** **1×** 2008
- Oldest and deepest: **Baikal** **1×** 2015
- Second deepest and longest: **Tanganyika** **1×** 2014
- By area: **Superior, Victoria, Great Bear, Baikal**; key order 4, 3, 2, 1 **1×** 2016
- Wrong: world's tallest waterfall-**Tugela** (it is **Angel**) **1×** 2015
- Lake Victoria statements: neither correct **1×** 2021
- Wrong: **Malawi-Asia** (Africa) **1×** 2017
- Wholly in the USA: **Michigan** **2×** 2015, 2017
- On Lake Superior: **Duluth** **1×** 2014
- **Erie-Detroit, Michigan-Gary, Ontario-Hamilton, Superior-Duluth** **1×** 2011
- Wrong: **Onega-Canada** (Russia) **1×** 2023
- **Balkhash**: **Kazakhstan** **1×** 2021
- Wrong: **Reindeer Lake-Russia** (Canada) **1×** 2014
- Wrong: **Five Flower Lake-Mongolia** (China, Jiuzhaigou) **1×** 2015
- World Heritage lake: **Plitvice** (Croatia) **1×** 2014

QUANTUM IAS & PCS

GREAT LAKES, WEST TO EAST

Superior
Largest; Duluth; Soo Canal to Huron

↓

Michigan
Only one wholly in USA; Chicago, Gary

↓

Huron
Joined to Michigan at Mackinac

↓

Erie
Shallowest; Cleveland, Buffalo; near Detroit

↓

Ontario
Smallest; Toronto, Hamilton; to St. Lawrence

11.4 Waterfalls

- In Venezuela: **Angel Falls** (highest) 1× 2014
- **Victoria Falls**: on the **Zambezi** (true); Zambia-Mozambique border (false; **Zambia-Zimbabwe**) 1× 2023

TRAP ALERT

- **Socotra** = Yemen; **Madeira** = Portugal (Atlantic).
- **Victoria Falls** = Zambia-Zimbabwe; **Lake Victoria** = Uganda-Kenya-Tanzania.
- Highest fall = **Angel**; Tugela is second.

UP LINK

No UP item. UP has no natural great lakes; its large water bodies are reservoirs such as **Rihand (Govind Ballabh Pant Sagar)** in Sonbhadra.

LAST-MINUTE 10

1. Greenland largest island; Borneo = 3 countries
2. Japan N-S: Hokkaido, Honshu, Shikoku, Kyushu
3. Indonesia W-E: Sumatra, Java, Bali, Lombok
4. Tinian = Micronesia; Solomons = Melanesia
5. Caspian largest lake; Baikal oldest, deepest
6. Tanganyika: longest, 2nd deepest
7. Michigan wholly in USA; Duluth on Superior
8. Balkhash = Kazakhstan; Onega = Russia
9. Angel = highest fall (Venezuela)
10. Victoria Falls = Zambezi, Zambia-Zimbabwe

Chapter 12

Straits, Gulfs, Canals and Dams

Built from 24 UPPSC questions (1991–2025). Focus: straits and what they separate, the Suez Canal, Soo and Ludwig canals, dam-river matching.

ASKED MOST · REVISE FIRST

- **Suez Canal** joins the **Red Sea and Mediterranean**; ends at **Port Said and Suez**; cut India-Europe route by about **7,000 km** 8× 1992, 1992, 1995, 2005, 2010, 2013, 2018, 2025
- Straits: **Gibraltar** (Europe-Africa), **Bering** (Asia-N. America), **Bass** (Tasmania), **Hormuz** (Persian Gulf-Gulf of Oman), **Dover** (Channel Tunnel) 6× 1999, 2004, 2005, 2007, 2010, 2023
- Dams: **Hoover-Colorado, Kariba-Zambezi, Aswan-Nile, Panchet Hill-Damodar** 3× 2013, 2015, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Straits

- Europe from Africa: **Gibraltar** 1× 1999
- Asia from North America: **Bering Strait** 1× 2004
- Tasmania from Australia: **Bass Strait** 1× 2023
- **Hormuz**: between the **Persian Gulf and Gulf of Oman** 1× 2005

- UK-France tunnel: **Strait of Dover**; Dover joins the **English Channel and North Sea** 2× 2007, 2010
- **Lombok Strait**: between **Bali and Lombok** 1× 2017
- **Ten Degree Channel**: **Andaman from Nicobar** 1× 2001

Strait	Separates
Gibraltar	Spain and Morocco
Bosphorus, Dardanelles	Europe and Asia (Turkiye)
Bab-el-Mandeb	Yemen and Djibouti
Hormuz	Iran and Oman
Malacca	Malaysia and Sumatra
Bering	Russia and Alaska
Palk	India and Sri Lanka
Ten Degree Channel	Little Andaman and Car Nicobar

12.2 Gulfs

- Persian Gulf does not border **Oman** (main coast on Gulf of Oman) 2× 2007, 2008
- Correct: **Gulf of Tonkin-Vietnam** 1× 2003

12.3 Canals

- Suez joins the **Red Sea and Mediterranean**; (2 and 3) 2× 1992, 2025
- Ports at the ends: **Port Said and Suez** 1× 2013
- Suez shortened India-West distance; joins Mediterranean and Red Sea (R explains A) 1× 1992
- Largest navigation canal: **Suez** 3× 1995, 2005, 2010
- India-Europe saving: about **7,000 km** 1× 2018
- Lakes north to south: **Manzala, Timsah, Great Bitter, Little Bitter** 1× 2019
- **Soo Canal**: **Superior and Huron** 1× 1991
- Inland waterway in southern Germany: **Ludwig Canal** 1× 2017
- Longest sea bridge then: **Jiaozhou Bay** (Hong Kong-Zhuhai-Macau now) 1× 2009

THE SUEZ CANAL, NORTH TO SOUTH



12.4 Dams

- **Colorado-Hoover, Damodar-Panchet Hill, Nile-Aswan, Zambezi-Kariba** 1× 2013
- **Kariba Dam**: **Zambezi** 1× 2016
- Wrong: **Hoover-Tennessee** (Hoover is on the **Colorado**) 1× 2015
- China's first Brahmaputra dam (2010): **Zangmu** 1× 2013

Dam	River	Country
Three Gorges	Yangtze	China
Hoover	Colorado	USA
Aswan High / GERD	Nile / Blue Nile	Egypt / Ethiopia
Kariba, Cahora Bassa	Zambezi	Zambia-Zimbabwe, Mozambique
Itaipu	Parana	Brazil-Paraguay
Zangmu, Medog	Yarlung Tsangpo	China

TRAP ALERT

- **Hoover** = Colorado; **TVA** dams = Tennessee.
- **Soo** = Superior-Huron; **Welland** = Erie-Ontario.
- Suez has **no locks**; Panama uses locks.

QUANTUM IAS & PCS

UP LINK

UP link: India's **Damodar Valley Corporation** copied the US **Tennessee Valley Authority**; UP's own big dams include **Rihand** (Sonbhadra) and **Matatila** (Betwa, Lalitpur).

LAST-MINUTE 10

1. Gibraltar = Europe-Africa; Bering = Asia-America
2. Bass = Tasmania; Hormuz = Persian Gulf-Gulf of Oman
3. Dover = Channel Tunnel; Lombok = Bali-Lombok
4. Ten Degree Channel = Andaman-Nicobar
5. Suez: Red Sea-Mediterranean; Port Said, Suez; 1869
6. Suez saves ~7,000 km; lakes Manzala-Timsah-Bitter
7. Soo = Superior-Huron; Ludwig = southern Germany
8. Hoover = Colorado; Kariba = Zambezi
9. Aswan = Nile; Panchet Hill = Damodar
10. Zangmu = China's Brahmaputra dam

Chapter 13

The Atmosphere

Built from 18 UPPSC questions (1993–2024). Focus: nitrogen, layers of the atmosphere, ozone and ionosphere, albedo, horse latitudes, Coriolis force, jet streams, clouds.

ASKED MOST · REVISE FIRST

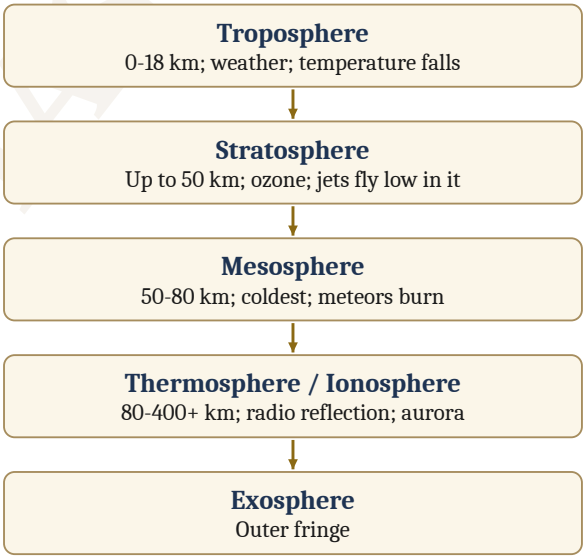
- ■ **Ozone layer** lies in the **stratosphere** 4× 1997, 2002, 2003, 2005
- ■ **Ionosphere** reflects **radio waves**; aurora 4× 2002, 2005, 2013, one-liner
- • **Nitrogen** = largest share of air (78%) 2× 1993, 2017
- • Lowest layer: **troposphere** (weather) 2× 2002, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

13.1 Composition and Layers

- Largest share: **nitrogen**; permanent gas: **nitrogen** 2× 1993, 2017
- Ascending order: **troposphere, stratosphere, mesosphere, ionosphere** (key gave descending, a) 1× 2005
- Lowest layer: **troposphere** 1× 2016
- ■ Ozone layer: **stratosphere** 3× 1997, 2003, 2005
- Deflects radio waves: **ionosphere**; reflects wireless signals 2× 2005, 2013
- Northern Lights: **ionosphere** (thermosphere) 1× one-liner
- Correct: **troposphere-weather, stratosphere-ozone, ionosphere-radio waves**; not mesosphere-aurora (1, 2, 3) 1× 2002

LAYERS, GROUND UP



13.2 Heat and Pressure

- Air heated mainly **from the Earth's surface** (long-wave); conversion happens at the surface (R explains A) 1× 1997
- **Albedo**: **reflecting power** 1× 2019
- Not true: maximum-temperature zone lies **on the Equator** (it is around 10-20° N, the thermal equator) 1× 2002
- Humid summer heat: **muggy** weather 1× 2002
- **Horse latitudes**: about **30°** 1× 2008
- Air pressure lowest in **summer** 1× 2014

Belt	Latitude
Equatorial low (doldrums)	0-5°
Subtropical high (horse latitudes)	30-35°
Subpolar low	60-65°
Polar high	90°

13.3 Winds, Jets, Clouds

- Leftward deflection in the Southern Hemisphere: **Coriolis force** 1× 2023
- Jet streams are **westerly** (A false); speed 300-500 km/h (R true) 1× 2024
- Highest clouds (of options): **cirrostratus** 1× 2009

TRAP ALERT

- Key used "ascending" for a top-down list; upward order is **4, 3, 2, 1**.
- Jet streams are **westerlies**, not easterlies (except the tropical easterly jet over India).
- Aurora and radio reflection = **ionosphere**, not mesosphere.

UP LINK

UP sits under the **subtropical westerly jet** in winter, which steers **western disturbances** bringing winter rain to western UP.

LAST-MINUTE 10

1. Air: N2 78%, O2 21%, Ar 0.93%
2. Lowest = troposphere (weather)
3. Ozone = **stratosphere**
4. Radio waves, aurora = **ionosphere**
5. Coldest layer = mesosphere
6. Albedo = reflecting power
7. Horse latitudes = **30°**
8. Coriolis: right in N, **left in S**
9. Jet streams: westerly, 300-500 km/h
10. Cirrus family = highest clouds

Chapter 14

Cyclones, Tornadoes and Local Winds

Built from 18 UPPSC questions (1992–2021). Focus: names of tropical cyclones by region, tornado and hurricane scales, local winds and their countries.

ASKED MOST · REVISE FIRST

- Cyclone names: **Willy-willy-Australia, Hurricane-USA, Typhoon-China, Baguio-Philippines, Cyclone-India** 5× 2001, 2005, 2014, 2019, 2020
- Local wind wrong pairs: **Mistral-Australia, Shamal-Austria, Bora-Poland, Blizzard-Chile, Sirocco-France** 5× 2005, 2013, 2016, 2019, 2021
- Tornadoes: **USA**, "Tornado Alley" in the **Mississippi plains**; **Fujita scale** 3× 2005, 2009, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Cyclones and Tornadoes

- **Australia-Willy-willy, China-Typhoon, India-Cyclone, USA-Hurricane** 1× 2005
- **Willy-willy-Australia, Hurricane-USA, Typhoon-China, Baguio-Philippines** 3× 2014, 2019, 2020
- Willy-willy: cyclone of **NW Australia** 1× 2001
- Tornado origin (key): **Caribbean Sea** 1× 1996
- Tornadoes associated with the **USA** 1× 2009
- **Tornado Alley**: **Mississippi plains** 1× 2005
- Tornado intensity: **Fujita scale** 1× 2009
- Hurricane damage: **Saffir-Simpson scale** 1× 2016
- Did not hit India: **Nargis** (Myanmar, 2008) 1× 2013
- Global warming raises frequency and severity of all listed extremes 1× 2018

Region	Name
Atlantic, Caribbean (USA)	Hurricane
NW Pacific (China, Japan)	Typhoon

Region	Name
Philippines	Baguio
NW Australia	Willy-willy
Indian Ocean	Cyclone

14.2 Local Winds

- Winds to the Mediterranean: westerlies (winter) or trades (summer) (*) 1× 1992
- **Foehn**: **Switzerland** (Alps) 1× 2010
- Wrong: **Bora-Poland** (Adriatic) 1× 2016
- Wrong: **Blizzard-Chile** (Canada, USA) 1× 2005
- "Blood rain" (Sirocco): **Italy** 1× 1998
- Wrong: **Sirocco-France** only (Bora-Italy, Blizzard-Canada right) 1× 2013
- Wrong: **Mistral-Australia** (France) 1× 2019
- Wrong: **Shamal-Austria** (Iraq, Gulf) 1× 2021

WARM VS COLD LOCAL WINDS

Warm, descending
Foehn (Alps), Chinook (Rockies), Santa Ana (California)

↓

Hot, desert
Sirocco (Sahara to Italy), Khamsin (Egypt), Harmattan (W. Africa), Shamal (Iraq), Loo (N. India)

↓

Cold
Mistral (Rhone, France), Bora (Adriatic), Blizzard (Canada), Purga (Siberia)

TRAP ALERT

- **Mistral** = Rhone valley, France; **Bora** = Adriatic.
- **Sirocco** blows from the Sahara **to Italy**; red dust = "blood rain".
- **Fujita** = tornadoes; **Saffir-Simpson** = hurricanes; **Richter** = earthquakes.

UP LINK

UP's own local wind is the **Loo**, the hot dry westerly of May-June; pre-monsoon **dust storms** and **Norwesters** strike eastern India.

LAST-MINUTE 10

1. Willy-willy = Australia; Baguio = Philippines
2. Typhoon = China/Japan; Hurricane = USA
3. Tornado Alley = Mississippi plains
4. Fujita = tornado; Saffir-Simpson = hurricane
5. Nargis hit **Myanmar**
6. Foehn = Switzerland; Chinook = Rockies
7. Sirocco = blood rain, Italy
8. Mistral = France; Bora = Adriatic
9. Shamal = Iraq/Gulf; Blizzard = Canada
10. Loo = north India

Chapter 15

World Climate and Forests

Built from 37 UPPSC questions (1996–2025). Focus: forest types and trees, Mediterranean vegetation names, climate-region matching, Mediterranean and West European climates, records.

ASKED MOST · REVISE FIRST

- ■ **Mediterranean climate**: rain in **winter** from the **westerlies**; dry summer 4× 2004, 2004, 2022, 2024
- Forest-tree matching: **monsoon-teak and sal, equatorial-mahogany, Mediterranean-olive, coniferous-pine and spruce** 2× 2013, 2021
- Mediterranean scrub names: **maquis, chaparral (California), fynbos (S. Africa), matorral (Chile)** 1× 2019
- ■ Climate-region matching: **equatorial-Congo, Mediterranean-California, monsoon-Bangladesh, savanna-Sudan** 3× 1999, 2002, 2003
- ■ Records: wettest **Mawsynram**, coldest **Verkhoyansk** 4× 1997, 1999, 2001, 2004

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Forests

- Highest forest share (of options): **Japan** 1× 2010

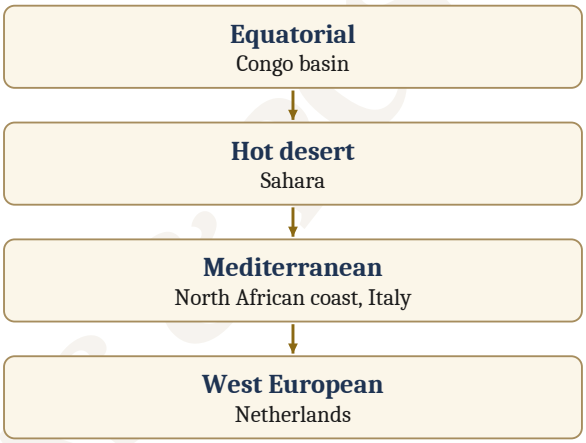
- Constitution requires **60%** forest cover: **Bhutan** 1× 2016
- Largest and densest forests: **Brazil** 1× 2004
- Tropical rainforest: **Congo Basin** 1× 1996
- Dense hot forest in Africa: **Ivory Coast** 1× 1998
- **Selva**: **broadleaf evergreen** 1× 2019
- Brazil: rainforests are **selvas**; fertile **terra roxa** (both) 1× 2025
- Soil erosion a big problem in **tropical rainforests** 1× 1998
- **Taiga**: **temperate (cold)** region 1× 2016
- No conifers in **Amazonia** 1× 2001
- **Monsoon-teak and sal, equatorial-mahogany and rosewood, Mediterranean-plum and olive, coniferous-pine and spruce** 1× 2013
- Not Mediterranean: **teak** 1× 2021
- **Maquis-Mediterranean coast, fynbos-southern Africa, chaparral-California, matorral-Chile** 1× 2019
- **Epiphytes-equatorial, acacia-Sahara, baobab-savanna, cedars-Mediterranean** 1× 2019

Forest	Trees
Equatorial (selva)	Mahogany, rosewood, ebony, rubber
Tropical monsoon	Teak, sal, bamboo
Savanna	Acacia, baobab
Mediterranean	Olive, cork oak, cedar, citrus
Taiga	Pine, spruce, fir, larch

15.2 Climate Types

- Main factor: **latitude** 1× 2010
- Rain depends on **humidity** 1× 2017
- Largest share of land: **arid regions** 1× 2002
- All climate zones: **Asia** 1× 2016
- Not equatorial: **Ethiopia** (highland) 1× 2013
- Equator suits plants not people; small annual range (both true, R not the explanation) 1× 2010
- **California-Mediterranean, West Australia-hot desert, Bangladesh-monsoon, Siberia-cold temperate** 1× 1999
- **Equatorial-Congo, Mediterranean-California, monsoon-Bangladesh, savanna-Sudan** 2× 2002, 2003
- Interior Australia is desert (true); northern Australia temperate (false, it is **tropical**) 1× 2025
- Linked to shifting pressure belts: **Mediterranean and monsoon** 1× 2010
- Zaire to Netherlands: **equatorial, hot desert, Mediterranean, West European** 1× 2002

ZAIRE TO THE NETHERLANDS



15.3 Mediterranean and West European

- Correct: **wet winter, dry summer-Italy** 1× 2004
- Winter rain because of the **westerlies** (R explains A) 1× 2004
- Winter rain (true); summer under dry trades (true, R not the explanation) 1× 2022
- Winter rain; belts shift south in winter (both) 1× 2024
- NW European ports open all year; SW winds all year (both true, R not the explanation; warm North Atlantic Drift) 1× 2002
- Western Europe: rain in **all months**, in the **westerlies** (both) 1× 2024

15.4 Records and El Nino

- Wettest place: **Mawsynram** 2× 1997, 2001
- Coldest place (key): **Verkhoyansk** 2× 1999, 2004
- **El Nino** develops in the **Pacific** 1× 2015
- El Nino statements: only 1 correct 1× 2023
- **Hekistotherms**: plants of **very low temperature** 1× 2013
- "Vegetation is the true index of climate": **Koppen** 1× 2018

TRAP ALERT

- **Teak** is a monsoon tree, never Mediterranean.
- Northern Australia is **tropical**, not temperate.
- Open winter ports of NW Europe owe more to the **North Atlantic Drift** than to winds.

UP LINK

UP has **tropical monsoon (deciduous)** forests of **sal** in the Terai and teak-dry deciduous in Bundelkhand; Dudhwa's sal forests are the State's best.

LAST-MINUTE 10

1. Bhutan: **60%** forest by Constitution
2. Selva = broadleaf evergreen; Amazon, Congo
3. Taiga = pine, spruce; no conifers in Amazonia
4. Maquis, chaparral, fynbos, matorral = Mediterranean scrub
5. Teak and sal = monsoon forest
6. Mediterranean: winter rain from **westerlies**
7. West Europe: rain all year; westerlies
8. Asia has every climate zone
9. Mawsynram wettest; Verkhoyansk coldest (key)
10. Koppen: vegetation indexes climate

Chapter 16

Major Tribes of the World

Built from 23 UPPSC questions (1996–2025). Focus: tribe-homeland matching, Bushmen, Maasai, Bedouin, Semang, Arctic peoples.

ASKED MOST · REVISE FIRST

- ■ **Bushmen (San)** = **Kalahari**, Botswana, Namibia **8×** 1996, 2005, 2008, 2008, 2009, 2012, 2016, 2025
- ■ **Maasai** = **Kenya** (East Africa); never West Africa or Zambia **5×** 2005, 2006, 2009, 2009, 2025
- ■ **Bedouin** = Arabia; **Eskimo/Inuit** = Canada; **Pygmies** = Congo basin **5×** 2001, 2005, 2009, 2016, 2025
- • **Semang** = **Malaysia**, not Indonesia **2×** 2002, 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

16.1 Tribes and Homelands

- Correct: **Eskimo-Canada** **1×** 2001
- Correct: **Bushman-Kalahari** **1×** 2012
- Wrong: **Maasai-West Africa** only **1×** 2025
- **Eskimo-Canada, Maasai-Kenya, Bedouin-Saudi Arabia, Bushman-Botswana** **1×** 2009
- **Khirghiz-Central Asia, Bushman-Kalahari, Ainu-Japan, Bedouin-Arab** **1×** 2016
- **Bushman-Namibia, Bedouin-Saudi Arabia, Pygmies-Congo, Maasai-Kenya** **1×** 2005
- **Jews-Israel, Teda-Libya, Beja-Egypt, Lur-Iran** **1×** 2020
- All pairs acceptable (Bushmen also in South Africa) (*) **1×** 1996
- Bushmen: **Kalahari**; correct: **Bushmen-Botswana** **2×** 2008, 2008
- Wrong: **Maasai-Western Africa**; **Maasai-Zambia** **2×** 2006, 2009
- Wrong: **Bantu-Sahara** **2×** 2010, 2012
- Wrong: **Khirghiz-Sudan** **1×** 2019
- Wrong: **Congo basin-Negrito** (Pygmies) **1×** 2004

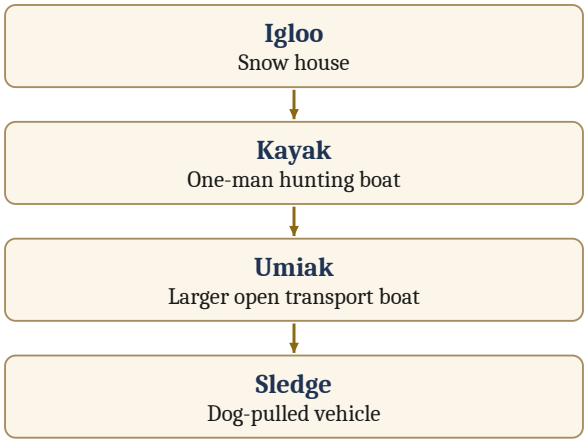
People	Homeland
Inuit (Eskimo)	Alaska, Canada, Greenland
Sami (Lapps)	Scandinavia
Kirghiz	Central Asia
Bedouin	Arabian desert
Maasai	Kenya, Tanzania
Pygmies	Congo basin
Bushmen (San)	Kalahari
Semang	Malaysia
Ainu	Japan (Hokkaido)
Maori	New Zealand

16.2 Asia and the Arctic

- Semang live in **Malaysia**; **Semang-Indonesia** is wrong **2×** 2002, 2010
- Wrong: **Alaska-Koryak** (Koryak = Kamchatka, Russia) **1×** 2021

- Inuit not found in **Sweden** 1× 2009
- **Lapps (Sami)**: **Scandinavia** 1× 2010
- **Kayak-narrow hunting boat, "Inuit" (igloo)-snow house, Umiak-transport boat, Sledge-dog vehicle** 1× 2014

ARCTIC WAY OF LIFE



16.3 Ways of Life

- Not pastoral nomads: **Pygmies** (hunter-gatherers) 1× 2005
- Native North Americans: **Mongoloid** 1× 2015

TRAP ALERT

- **Maasai** = East Africa (Kenya, Tanzania).
- Sweden and Finland have **Sami**, not Inuit.
- **Koryak** = Russia; **Kirghiz** = Central Asia.

UP LINK

UP's own tribes are few: **Tharu** (Terai: Lakhimpur Kheri, Bahraich), **Buksa**, **Gond**, **Kharwar** (Sonbhadra). The Tharu, like the Maasai and Bedouin, are a classic "tribe-region" pair.

LAST-MINUTE 10

1. Bushmen (San) = **Kalahari**, Botswana
2. Maasai = **Kenya**, Tanzania
3. Bedouin = **Arabia**; Kirghiz = Central Asia
4. Pygmies = **Congo**; hunter-gatherers
5. Eskimo/Inuit = Canada, Alaska, Greenland
6. Sami (Lapps) = Scandinavia
7. Semang = **Malaysia**; Ainu = **Japan**
8. Teda = Libya; Beja = Egypt; Lur = Iran
9. Koryak = Kamchatka
10. Kayak = hunting; umiak = transport

Chapter 17

World Agriculture and Livestock

Built from 65 UPPSC questions (1990–2023). Focus: leading producers of crops, crop-region matching, plantation crops, shifting cultivation names, milk.

ASKED MOST · REVISE FIRST

- ■ **India** = top **milk** producer 5× 2003, 2008, 2010, 2011, 2012
- ■ **Brazil** = top **coffee** producer; **Vietnam** second (and second exporter); estates called **fazendas** 5× 2009, 2012, 2013, 2016, 2017
- ■ **Thailand** = top **natural rubber** producer (about one-third) 4× 2001, 2005, 2006, 2010
- ■ **Brazil** = top **sugarcane**; India second; **Cuba** = "Sugar Bowl" 4× 2008, 2013, 2015, 2016
- • **China** = top tea exporter (by value) 2× 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

17.1 Cereals

- Leading producer of both wheat and rice: none now (India leads rice, China wheat) (*) 2× 2004, 2006
- India in rice: **first** (2023, 2024); second earlier 1× 2013
- Largest rice area: **India** 1× 2016
- Wheat, descending: **China, India, Russia, USA** 1× 2008

- Highest rice yield (of options): **China** 1× 2015
- Hybrid rice most popular in **China** 1× 2013
- World rice gene bank: **Philippines** (IRRI) 1× 2015
- Rice two-thirds of exports: none (Thailand nearest; India now top exporter) (*) 2× 2010, 2013
- Second in corn: **China** (USA first) 1× 2013
- Maize as staple: **Middle Africa** 1× 2007

17.2 Sugar, Fibres, Plantations

- Largest sugarcane producer: **Brazil**; second: **India** 2× 2008, 2016
- Top beet producers: Russia and France (later data reshuffle, *) 1× 2010
- Sugar from beet only: **Ukraine** 1× 2013
- Exports most of its sugar: **Mauritius** 1× 2011
- "Sugar Bowl": **Cuba** 2× 2013, 2015
- Highest cotton yield (2014): **Australia**; of options: **China** 2× 2015, 2013
- Largest coconut producer: **Indonesia** 1× 2008
- Largest rubber producer: **Thailand**; top two: **Thailand and Indonesia**; one-third from **Thailand** 4× 2001, 2006, 2005, 2010
- Top coffee producers: **Brazil and Vietnam**; top: **Brazil** 3× 2009, 2012, 2013
- Second coffee exporter: **Vietnam** 1× 2016
- Brazil's coffee estates: **fazendas** 1× 2017
- US coffee state: **Hawaii** 1× 2015
- Largest tea exporter: **China** 2× 2016, 2017
- Not a cocoa producer: **Latvia** 1× 2021
- Wrong: **coffee-Uzbekistan** 1× 2023
- Not correct: sugarcane is Malaysia's main plantation crop (it is **rubber, oil palm**) 1× 2008

17.3 Crop Regions

- **Coffee-Sao Paulo, jute-Ganges delta, rice-Yangtze, wheat-Prairies** 1× 2003
- **Coffee-Sao Paulo, rice-Yangtze, wheat-Huang He, tea-Kandy** 1× 2010
- **Cocoa-Ghana, coffee-Ivory Coast, tea-Kenya, sugarcane-South Africa** 1× 2013
- **Rubber-Malaysia, coffee-Colombia, olive-Italy, sugarcane-Mauritius** 2× 2010, 2013
- **China-iron ore, India-milk, Saudi Arabia-petroleum, Thailand-rubber** 1× 2000
- **Chad-petroleum, Niger-uranium, Rwanda-coffee, Benin-cotton** 1× 2017
- Producer statements: now China tobacco and Brazil soybean true (*) 1× 2000
- Coconut and sugarcane in the Philippines: **Spanish and Americans** 1× 2023
- Peanuts main crop of **Gambia** 2× 2012, 2013
- Groundnut native to **Brazil** 1× 2010

Crop	Leader
Rice / Wheat	India / China
Maize / Soybean	USA / Brazil
Sugarcane / Coffee	Brazil
Tea	China
Cocoa	Cote d'Ivoire
Rubber	Thailand
Coconut	Indonesia
Milk	India

17.4 Fruits and Vines

- India's rank in fruit: **second** 1× 2016
- Not tropical: **apple** 1× 1990
- Sacramento-San Joaquin valley: **California** 1× 1998
- Commercial viticulture is Mediterranean; 85% of grapes go to wine (R explains A) 1× 2005
- Citrus well developed in **Mediterranean regions** 1× 2020

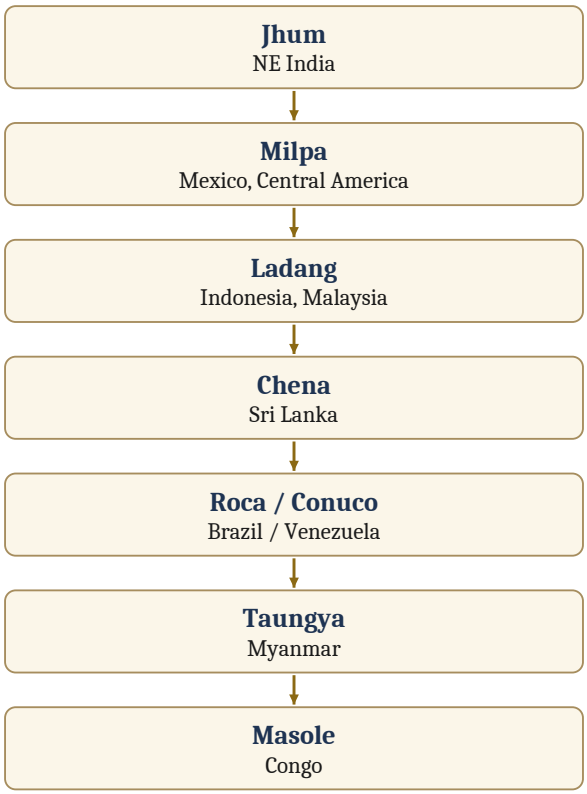
17.5 Farming Systems

- Plantations clustered on coasts; Amazon lacks workers (2 and 3) 1× 1998
- Plantation crops: definitions differ (*) 1× 2004
- Crops between tree rows: **alley cropping** 1× 2007
- Monoculture: **commercial grain farming** 1× 2010
- Not shifting cultivation: **fazenda** 2× 2006, 2010
- **Jhum-SE Asia, Milpa-Mexico** correct; Ladang is not Sri Lanka (1 and 2) 1× 2023
- Blue-green algae biofertiliser: **rice** 1× 2010
- Not in tea: **morphine** 1× 2010
- First HYV seeds: **Mexico** (Borlaug, wheat) 1× 2008

QUANTUM IAS & PCS

- WTO Green Box: **research and plant protection** (1 and 4) 1× 2009

SHIFTING CULTIVATION NAMES



17.6 Narcotics and Livestock

- Heroin from **opium** 1× 2007
- Golden Crescent: **Afghanistan, Iran, Pakistan** 1× 2007
- Shahtoosh: **chiru** (Tibetan antelope) 1× 2006
- Hurt beef's image: **mad cow disease** 1× 2005
- Largest milk producer: **India** 4× 2003, 2008, 2010, 2012
- Milk order: **India, USA, China, Russia** (then) 1× 2011
- Highest milk per cow (of options): **Denmark** 1× 2018

TRAP ALERT

- **Fazenda** = coffee estate, not shifting cultivation; **Chena** = Sri Lanka.
- Tea exports: China by **value**, Kenya by **volume**.
- No one country now leads in both rice and wheat.

UP LINK

UP is India's top producer of **milk, wheat and sugarcane**; India itself leads the world in milk. Coffee and rubber are absent in UP, which grows mango, guava and potato.

LAST-MINUTE 10

1. India = milk, rice (now), second in fruit
2. China = wheat, tea export, hybrid rice
3. Brazil = sugarcane, coffee (fazendas); Vietnam second
4. Thailand = rubber; Indonesia = coconut
5. Cuba = Sugar Bowl; Ukraine = beet sugar
6. IRRI gene bank = Philippines; HYV = Mexico
7. Gambia = peanuts; groundnut native to Brazil
8. Milpa = Mexico; Ladang = Indonesia; Chena = Sri Lanka
9. Golden Crescent = Afghanistan, Iran, Pakistan
10. Shahtoosh = chiru

Chapter 18

World Minerals and Energy Resources

Built from 71 UPPSC questions (1991–2025). Focus: coalfield and iron-ore region matching, leading producers (which change over time), mining centres, oilfields and reserves, uranium.

ASKED MOST · REVISE FIRST

- **Coalfield matching**: **Ruhr, Saar-Germany; Donbas/Donetsk-Ukraine; Kuzbas-Russia; Lancashire, Bristol-UK; Appalachian-USA; Red Basin-China; NSW-Australia** 8× 1999, 2007, 2010, 2013, 2013, 2022, 2023, 2025
- **Mining centres**: **Mesabi-iron, Kalgoorlie-gold, Chuquicamata-copper, Kimberley-diamond, Witwatersrand-gold, Katanga-copper, Butte-silver, Kinta-tin** 6× 1991, 1999, 2004, 2010, 2017, 2020
- **Mexico** = top **silver**; **Chile** = top **copper** ore and **iodine** 6× 1993, 1996, 2011, 2014, 2018, 2022
- **Kazakhstan** = top **uranium** producer; **Australia** = largest reserves; **Canada** = Uranium City 5× 2006, 2009, 2009, 2010, 2014
- Largest oil reserves: **Venezuela** 3× 2010, 2012, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Coal

- All true: China leads coal; Donetsk-Ukraine; Saar-Germany; Appalachian-USA 1× 1999
- **Donbass**: **coal**; largest Ukrainian coal centre: **Donetsk** 2× 2007, 2023
- **Kuzbass**: coal and iron ore; **Novosibirsk** a centre (both) 1× 2023
- **Appalachian-USA, Lancashire-England, Ruhr-Germany, Kuzbas-Russia** 1× 2025
- **Kuzbas-Russia, Red Basin-China, Bristol-UK, New South Wales-Australia** 1× 2013
- **Donetsk-Ukraine, Kuznetsk-Russia, Lancashire-UK, Saar-Germany** 1× 2013
- **Ruhr**: **Germany** 2× 2010, 2022
- Over half the world's coal: **China** (USA per older key) 1× 2006
- **Rukwa** (Tanzania): **coal** 1× 2010

18.2 Iron Ore

- Producer order: **Australia, Brazil, China, India** now (*) 1× 2008
- Largest producer of options: **Brazil**; **India** among old options (*) 2× 2013, 1991
- France no longer a European iron producer (Lorraine mines closed 1997) (*) 2× 2010, 2013
- **Lorraine-France, Midlands-UK, Kiruna-Sweden, Kostanay-Kazakhstan** 1× 2010
- Wrong: **Germany-Normandy** (Normandy = France) 2× 2016, 2020
- **Mount Newman**: **iron ore** (Pilbara, Australia) 1× 2023
- Lake Superior region: **iron ore** (Mesabi) 1× 2017
- Great Lakes region industry: **steel and engineering** 1× 2016

Iron-ore region	Country
Pilbara (Mount Newman)	Australia
Carajas, Itabira	Brazil
Mesabi Range	USA
Kiruna	Sweden
Lorraine, Normandy	France
Krivoi Rog	Ukraine
Kostanay	Kazakhstan

18.3 Metals

- Largest copper ore producer: **Chile**; key **China** (refined copper) 2× 1996, 1998
- Chile leads in copper due to Andean **porphyry copper** (R explains A) 1× 2022
- **Chuquicamata**: **copper** 1× 2017
- Africa's top copper producer (then): **Zambia**; now **DR Congo** 1× 2010
- Japan nearly self-sufficient in **copper** (per key) 1× 2014
- Gold order: now **China, Russia, Australia** (*) 1× 2008
- **Kalgoorlie**: **gold**; **Coolgardie**: **Western Australia** 2× 1991, 2011
- Largest gold exporter to India: **Switzerland** 1× 2013
- Largest silver producer: **Mexico** 3× 1993, 2011, 2014
- Aluminium among options: **India** (2023 data) 1× 1993
- Bauxite: **Australia** per key; **Guinea** now leads 2× 2013, 2015
- Tin: **Indonesia** (then); China now 1× 2010
- **Kinta Valley** (Malaysia): **tin** 1× 2020

- Titanium: **Russia** (per key) 1× 2009
- **Postmasburg** (South Africa): **manganese** 1× 2020
- Diamond: **Russia** 2× 2012, 2013
- Iodine: **Chile** 1× 2018
- Afghanistan's $1 trillion find: **lithium** 1× 2008
- Wrong: **copper-cinnabar** (cinnabar = mercury ore) 1× 2017
- All pairs broadly correct (tin-Bolivia, iron-Brazil, silver-Mexico, nitrate-Peru) (*) 2× 2010, 2013
- **Silver-Mexico, mica-India, iron ore-Russia, copper-Chile** 1× 2017

18.4 Mineral Regions

- **Coal-Karaganda, gold-Highveld, iron-Krivoi Rog, oil-San Joaquin Valley** 1× 2010
- Wrong: **Mesabi-copper** (Mesabi = iron) 1× 2004
- **Iron-Mesabi, petroleum-Baku, copper-Bingham, uranium-Port Radium** 1× 1999
- **Baku**: **petroleum** 1× 1993
- Correct: **copper-Arizona** 1× 2007
- Wrong: **Saar-iron ore** (Saar = coal) 1× 2007
- **Copper-Katanga, diamond-Kimberley, gold-Witwatersrand, silver-Butte** 1× 2010

LEADERS THAT CHANGED

Iron ore
China/India (old keys) to Australia

↓

Gold
South Africa to China (since 2007)

↓

Bauxite
Australia to Guinea (2022-23)

↓

Petroleum
Saudi Arabia/Russia to USA (since 2018)

↓

Copper in Africa
Zambia to DR Congo

18.5 Petroleum and Gas

- Producer order: now **USA, Russia, Saudi Arabia** (*) 3× 1998, 2008, one-liner
- Largest producer (key, then): **Russia** 1× 2010
- Middle East order: **Saudi Arabia, Iran, Iraq, Kuwait** 1× 2008
- West Asia's top producer: **Saudi Arabia** 1× 2017
- South-East Asia's top producer: **Indonesia** 1× 2004
- Largest reserves: **Venezuela**; **Iran** among older options 4× 2010, 2012, 2013, 2002
- All true: Middle East reserves, Alaska, USA producer-importer 1× 1998
- Alaska statements: both correct 1× 2021
- Wrong: **Kuwait-Kashagan** (Kashagan = **Kazakhstan**) 1× 2024
- **Pegu Yoma** (Myanmar): **mineral oil** 1× 2022
- Largest gasohol producer and user: **USA** 1× 2011
- "Peace Pipeline" (Iran-Pakistan-India): none of the options 1× 2007
- Top crude supplier to India 2017-18: **Iraq** 1× 2017

Oilfield	Country
Ghawar	Saudi Arabia
Burgan	Kuwait
Kirkuk, Rumaila	Iraq
Kashagan, Tengiz	Kazakhstan
Baku	Azerbaijan
Prudhoe Bay	Alaska
Maracaibo	Venezuela
Pegu Yoma	Myanmar

18.6 Uranium and Others

- Top uranium producer: **Kazakhstan** 2× 2009, 2010
- Largest reserves: **Australia** 1× 2009
- **Uranium City**: **Canada**; important uranium ore: **Canada** 2× 2006, 2014
- Not a major Russian resource (per key): **uranium** 1× 2017
- Natural camphor: from a tree native to **China and Japan** 1× 2014

TRAP ALERT

- **Mesabi** = iron; **Saar** = coal; **cinnabar** = mercury; **Kashagan** = Kazakhstan.
- "Largest producer" keys age fast; check the year.
- Chile leads **mine** copper; China leads **refined** copper.

UP LINK

UP's minerals are minor (Sonbhadra coal, limestone, Lalitpur rock phosphate); the **Singrauli** coalfield straddling UP-MP is India's equivalent of a Ruhr-style energy hub.

LAST-MINUTE 10

1. Ruhr, Saar = Germany; Donbas = Ukraine; Kuzbas = Russia
2. Karaganda = Kazakhstan; Red Basin = China
3. Mesabi = iron (USA); Kiruna = Sweden; Krivoi Rog = Ukraine
4. Mount Newman (Pilbara) = iron, Australia
5. Chile = copper, iodine; Mexico = silver
6. Kalgoorlie = gold; Kimberley = diamond; Witwatersrand = gold
7. Katanga = copper, cobalt; Kinta = tin
8. Kazakhstan = uranium; Canada = Uranium City
9. Venezuela = largest oil reserves; USA top producer now
10. Kashagan = Kazakhstan; Baku = oil

Chapter 19

Industrial Cities and Regions of the World

Built from 25 UPPSC questions (1990–2022). Focus: city-industry matching, shipbuilding traps, industrial regions, footloose industry.

ASKED MOST · REVISE FIRST

- ■ **City-industry matching**: **Pittsburgh-steel, Detroit-automobiles, Osaka-cotton, Dundee-jute, Sheffield-cutlery, Glasgow and St. Petersburg-ships, Anshan-steel, Havana-cigars, Abadan-oil refinery** 6× 2002, 2005, 2005, 2010, 2011, 2022
- ■ Wrong shipbuilding pairs: **Birmingham, Chelyabinsk, Moscow, Venice**; Camden is New Jersey, not Florida 5× 1996, 1998, 2001, 2005, 2016
- • **Footloose** industry: **software, electronics** 2× 2013, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Iron and Steel

- **Hamilton-Canada, Birmingham-UK, Essen-Germany, Anshan-China** 1× 2011
- Wrong: **Birmingham-shipbuilding** 1× 1998
- Wrong: **Chelyabinsk-shipbuilding** 1× 2001
- Wrong: **Moscow-shipbuilding** 1× 2005
- Wrong: **Mesabi-coalfield** (iron ore) 1× 2013
- Market-based steel locations: **Japan** 2× 2010, 2013

19.2 Textiles, Automobiles, Ships

- **Pittsburgh-iron and steel, Shanghai-cotton textiles, Dundee-jute, Leningrad-shipbuilding** 1× 2005
- **Nagoya-cotton textiles, Salem-iron and steel, Los Angeles-aircraft, Abadan-oil refinery** 1× 2010
- **Osaka-cotton, Detroit-automobiles, Cuba-cigars, St. Petersburg-shipbuilding** 1× 2022
- **Anshan-iron and steel, Detroit-automobiles, Havana-cigars, Nagoya-cotton** 1× 2002
- Cotton textile centre: **Osaka** 1× 1990
- Wrong: **Venice-shipbuilding** (per key) 1× 1996
- **Anshan-steel, Glasgow-shipbuilding, Ottawa-paper, Sheffield-cutlery** 1× 2005
- Wrong: **Houston-automobiles** (oil, space) 1× 2010
- Wrong: **Camden-Florida** (New Jersey) 1× 2016
- Wrong: **cotton textiles-Detroit** 1× 2015

City	Industry
Pittsburgh, Gary	Steel (USA)
Detroit	Automobiles
Houston	Oil, space
Sheffield / Birmingham	Cutlery / Steel (UK)
Glasgow, Belfast	Ships
Dundee / Manchester	Jute / Cotton
Essen (Ruhr)	Steel (Germany)
Magnitogorsk, Chelyabinsk	Steel (Urals)
St. Petersburg	Ships

TATTVA IAS & PCS

City	Industry
Osaka / Nagoya	Cotton / Cars, textiles
Anshan	Steel (China)
Abadan / Havana	Refinery / Cigars

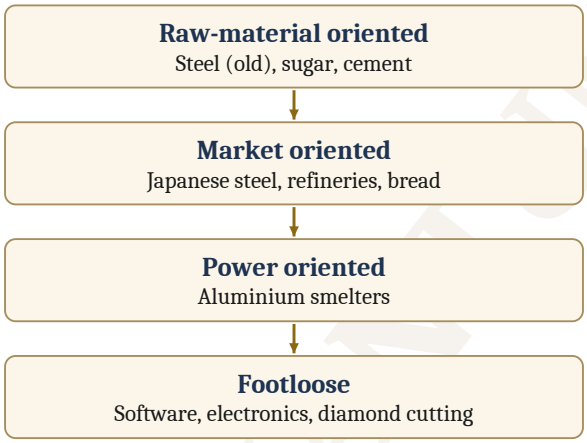
19.3 Industrial Regions

- **Kinki-Japan, Canton-China, Lorraine-France, Belo Horizonte-Brazil** 1× 2016
- Correct: **Ruhr-Germany** 1× 2016
- Wrong: **Scotland industrial region-Sweden** (UK) 1× 2013

19.4 Textiles and Location

- Biggest wool producer (key): **China** 1× 2013
- German wool centre: **Wuppertal** 1× 2013
- Leading cotton textile producer: **China** 1× 2005
- Footloose: **software** 1× 2013
- Raw material not primary: **electronics** 1× 2015
- Japan leads through **high technology** 1× 2008

HOW INDUSTRIES CHOOSE A SITE



TRAP ALERT

- **Birmingham** = steel; **Glasgow, Belfast** = ships.
- Moscow and Chelyabinsk are inland: **no shipbuilding**.
- **Camden** = New Jersey; **Houston** = oil, not cars.

UP LINK

UP's footloose-industry hubs are **Noida** (electronics, mobiles, software) and Lucknow; Kanpur is UP's "Manchester", like Osaka in Japan and Manchester in the UK.

LAST-MINUTE 10

1. Pittsburgh = steel; Detroit = cars; Houston = oil
2. Sheffield = cutlery; Glasgow = ships; Dundee = jute
3. Essen = steel; Wuppertal = wool
4. St. Petersburg = ships; Chelyabinsk = steel
5. Osaka = cotton (Manchester of Japan); Nagoya = cars, textiles
6. Anshan = steel; Havana = cigars; Abadan = refinery
7. Kinki = Japan; Canton = China; Belo Horizonte = Brazil
8. Japan steel = market-based
9. Footloose = software, electronics
10. Camden = New Jersey

Chapter 20

World Transport and Miscellaneous

Built from 41 UPPSC questions (1991–2024). Focus: sea routes, railways and tunnels, ports, geographers and their ideas, HQs of organisations, assorted facts.

ASKED MOST · REVISE FIRST

- **North Atlantic** = busiest sea route; links two industrial zones 4× 1998, 2002, 2003, 2004
- **Bermuda Triangle** = **North Atlantic** 5× 2001, 2002, 2003, 2008, 2015
- Busiest port by cargo: **Shanghai**; Japan's busiest: **Yokohama** 3× 2004, 2010, 2015
- Geographers: **Taylor-neo-determinism, Sauer-planting hearths, Engler-floral regions, Lee-migration, Spencer and Thomas-11 cultural worlds** 4× 1994, 2016, 2017, 2024

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Routes, Railways, Tunnels

- North Atlantic busiest; joins two industrial zones (R explains A) 2× 2002, 2003
- Busiest sea route: **North Atlantic** 2× 1998, 2004

- Not on the Trans-Siberian: **Sochi** 1× 2019
- Rail length: **USA**, China, Russia, India, Canada (no code fits, *) 2× 2008, 2014
- Longest rock tunnel: Delaware Aqueduct, USA (none of options, *) 1× 2013
- **Seikan** rail tunnel: **Japan** 1× 2012

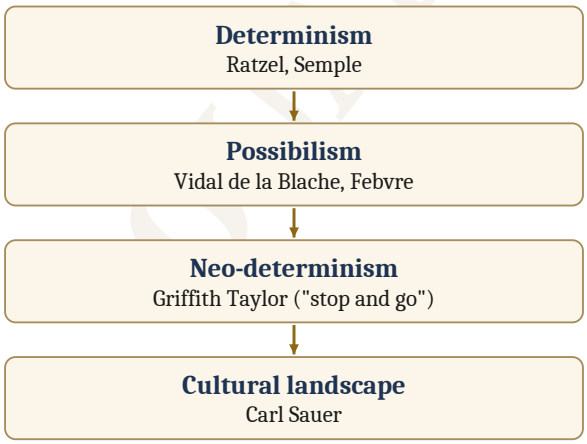
20.2 Ports and Shipbuilding

- Busiest seaport by cargo: **Shanghai**; second biggest (of options): **Shanghai** 2× 2010, 2004
- Japan's busiest port: **Yokohama** 1× 2015
- **Alexandria**: **Egypt** 1× 2007
- **Port Diamond**: **South Africa** 1× 2010
- **Gwadar**: **Balochistan** 1× 2021
- Wrong: **Igarka-China** (Igarka = Russia, Yenisei) 1× 2016
- Largest shipbuilder by value: **China** 1× 2010

20.3 Geographers and Concepts

- **Planting hearths-C.O. Sauer, cryophyte steppe-Arctic tundra, world floral regions-Adolf Engler, Puszta-Hungary** 1× 2024
- **Neo-determinism**: **Griffith Taylor** 1× 1994
- Spencer and Thomas: **11** cultural worlds 1× 2016
- Migration theory: **Lee** 1× 2017
- **Umland**: the **area surrounding a town** 1× 2017
- **Isogonic** lines: equal **magnetic declination** 1× 1998

SCHOOLS OF THOUGHT



20.4 Places and Facts

- **Grand Banks**: **eastern coast of North America** (Newfoundland) 1× 2010
- **Bermuda Triangle**: **North Atlantic** 4× 2001, 2002, 2003, 2008
- **Golden Triangle-opium (SE Asia), Bermuda Triangle-North Atlantic, rice bowl of the Far East-Myanmar, Red Basin-China** 1× 2015
- Correct: **Morocco-Atlas** 1× 2002
- Wrong: "Queen of the Adriatic"-**Rome** (it is **Venice**) 1× 2015
- "Country of 40 tribes": **Kyrgyzstan** 1× 2015
- True: Cuba sugar bowl, Hong Kong SAR, Australia federal (1, 2, 4); USA not top milk producer 1× 2005
- Two wrong: Bosnia not Baltic; Nova Scotia = Canada (*) 1× 2013
- Not true: species diversity **increases** towards the poles (it decreases) 1× 2019
- Indian Ocean bases due to geopolitical value (key b; source a) 1× 2019

20.5 Organisations, Energy, History

- **IMO**: **London** 1× 2016
- Wrong: **FAO-New York** (FAO = **Rome**) 1× 2002
- Not in OPEC: **China** 1× 2002
- Nuclear plant count order: USA, France, Japan, Russia then (*) 1× 2013
- 99% energy from hydro: **Norway** 1× 2021
- **Chernobyl**: **Ukraine** 1× 2013
- Great Leap Forward famine: **China** 1× 2013
- US launch centre: **Cape Kennedy** (Canaveral) 1× 1991
- Largest wine producer: **Italy** 1× 2010
- Asia's biggest gold market (then): **China** 1× 2017
- Tree age: **counting rings** 1× 2006

Body	HQ
IMO	London
FAO, WFP, IFAD	Rome
WHO, WTO, ILO	Geneva
ICJ	The Hague
UNESCO, OECD	Paris
IAEA, OPEC	Vienna
UNEP	Nairobi
ISA	Gurugram

TRAP ALERT

- **FAO** = Rome, not New York; **IMO** = London.
- **Igarka** = Russia; **Gwadar** = Balochistan (Pakistan).
- Species diversity is highest at the **Equator**.
- Queen of the Adriatic = **Venice**.

UP LINK

UP links: the **Golden Quadrilateral** and **NW-1 (Prayagraj-Haldia)** are India's busiest land and inland water routes; **Jewar** airport and the **Eastern Dedicated Freight Corridor** put UP on new trade routes.

LAST-MINUTE 10

1. North Atlantic = busiest sea route
2. Shanghai = busiest port; Yokohama = Japan's busiest
3. Seikan tunnel = Japan; Sochi not on Trans-Siberian
4. Gwadar = Balochistan; Igarka = Russia
5. Taylor = neo-determinism; Sauer = planting hearths
6. Engler = floral regions; Lee = migration
7. Bermuda Triangle = N. Atlantic; Golden Triangle = opium
8. Kyrgyzstan = 40 tribes; Venice = Queen of the Adriatic
9. IMO = London; FAO = Rome
10. Norway = 99% hydro; Chernobyl = Ukraine

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TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Environment and Ecology

PYQ-Based Revision Pointers

Chapters 1 to 16

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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Chapter 1

Environment, Sustainable Development and Ecology

Built from 44 UPPSC questions (2001–2025). Focus: sustainable development and Brundtland, Silent Spring, Environment Protection Act 1986, ecosystem facts, ecological succession.

ASKED MOST · REVISE FIRST

- **Sustainable development**: meet present needs without compromising future generations (Brundtland, **Our Common Future**, 1987) 4× 2012, 2019, 2020, 2025
- **Ocean** = largest and most stable ecosystem 4× 2003, 2013, 2014, 2018
- Ecosystems are **open** systems, not closed 3× 2014, 2015, 2017
- **Deforestation** disturbs ecological balance 2× 2013, 2014
- **Ecomark** = label for eco-friendly goods 3× 2004, 2012, 2021
- Succession: **nudation, migration, ecesis, reaction, stabilisation**; the sequence = **sere** 3× 2016, 2020, 2024

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

1.1 Environment

- Environment refers to **all** the listed components 1× 2006
- Not a category of environment: **operational environment** 1× 2020
- Environmental degradation means **all** the listed harms 1× 2006
- Not related to environmental protection (key): **poverty alleviation** 1× 2013
- Urbanisation and industrialisation are harmful in **all** listed ways 1× 2002
- Mid-1980s debate on development: **politics and development** was not a theme 1× 2020
- **Plachimada** (Coca-Cola, groundwater): **Kerala** 1× 2009
- **Greenwash**: **false promises** of protecting the environment 1× 2017

1.2 Sustainable Development

- Important for well-being; Brundtland definition (R explains A) 1× 2019
- Report that started the debate (1987): **Our Common Future** 1× 2020
- Sustainable development (true); Agenda 21 signed in 1995 (false, **1992**) 1× 2025
- Order: **Limits to Growth (1972), Brundtland (1987), Montreal Protocol in force (1989), Rio (1992)** 1× 2020
- Inter-generational use of **natural resources** 1× 2012
- Important for sustainable development: **all** listed 1× 2004
- Carrying capacity: waste within absorption capacity (only 2); extraction must stay **below** regeneration 1× 2025
- Sustainable agriculture: use land so its **quality stays intact** 1× 2001

GLOBAL ENVIRONMENT MILESTONES



1.3 Books, Laws, Labels

- **Silent Spring** (1962): **Rachel Carson** 1× 2020
- **The Population Bomb**: **Paul R. Ehrlich** 1× 2023
- **Green Development**: **W.M. Adams** 1× 2011
- **Environment (Protection) Act**: **1986** 1× 2022
- Ecological balance helped by forest, environment, industrial, education policies (all) 1× 2004

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- **Ecomark: safe goods for the environment** 3× 2004, 2012, 2021

Indian law	Year
Wildlife (Protection) Act	1972
Water Act (set up CPCB)	1974
Forest (Conservation) Act	1980 (Van Adhiniyam 2023)
Air Act	1981
Environment (Protection) Act	1986 (after Bhopal)
Biological Diversity Act	2002
Forest Rights Act	2006
National Green Tribunal Act	2010

1.4 Ecology and Ecosystem

- Ecology: study of **organisms and environment** 1× 2014
- True: ecosystem has biotic and abiotic parts; **autotrophs** make food (1 and 2) 1× 2002
- Not true: an ecosystem is a **closed system** 2× 2014, 2015
- Biosphere is open (true); no new substances can enter (false) 1× 2017
- Order: **population, community, ecosystem, landscape** (then biome, biosphere) 1× 2017
- True: **producers outnumber primary consumers** 1× 2019
- Components not interdependent (false); humans affect environment (true) 1× 2021
- Largest ecosystem: **ocean**; most stable: **marine** 4× 2003, 2014, 2013, 2018
- Artificial ecosystem: **rice field** 1× 2016
- Cycling of elements: **biogeochemical cycles** 2× 2020, 2021

1.5 Ecological Balance

- Does not help balance: **deforestation**; disturbs it 2× 2013, 2014
- Not related to balance: **industrial management** 1× 2005
- Vegetation is the index of climate; water-loving plants in moist climate (R explains A) 1× 2009

1.6 Succession and Niche

- Phases: **nudation, migration, ecesis, reaction, stabilisation**; from migration if nudation missing 2× 2016, 2024
- Whole sequence of communities: **sere** 1× 2020
- "Ecological transition": **Bennett** 1× 2020
- Niche: tolerance range, resources, role; each species distinct (both) 1× 2022
- Not true: species become **fewer** in a niche when resources are enough 1× 2024

TRAP ALERT

- **Agenda 21** = **1992**, not 1995.
- Ecosystems are **open** for energy.
- **Sere** = sequence; succession = the process.

UP LINK

UP links: the **Environment Protection Act 1986** is enforced in UP by the **UP Pollution Control Board** (Lucknow); UP's **Ganga** and Yamuna cases at the NGT test these laws.

LAST-MINUTE 10

1. Brundtland **1987**: Our Common Future
2. Rio **1992**: Agenda 21; SDGs **2015**
3. Silent Spring = **Rachel Carson**; Population Bomb = **Ehrlich**
4. EPA **1986**; Ecomark = eco-friendly goods
5. Greenwash = false green claims; Plachimada = Kerala
6. Ocean = largest, most stable ecosystem
7. Ecosystem = open system; rice field = artificial
8. Biogeochemical cycles = element cycling
9. Succession: nudation → migration → ecesis → reaction → stabilisation
10. Sere = whole sequence

Chapter 2

Food Chain, Energy Flow and Ecosystem Concepts

Built from 32 UPPSC questions (2001–2025). Focus: trophic levels, producers, 10% law, energy flow, ecological pyramids, productivity, ecotone and footprint.

ASKED MOST · REVISE FIRST

- **Solar energy** = driving force of ecosystems; energy flow is **one-way** 4× 2015, 2015, 2016, 2019
- Energy **decreases** at each step; **10% law** by **Lindeman** 4× 2017, 2019, 2023, 2025
- Producers = **green plants**; **phytoplankton** in the sea 3× 2005, 2013, 2017
- **Eucalyptus** is not eco-friendly 3× 2005, 2010, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Food Chains

- Trophic levels: **organisms linked in a food chain** 1× 2023
- Order: **grass, grasshopper, frog, snake** 1× 2010
- Primary consumers: **ants and deer** 1× 2006
- Man: **primary and secondary** consumer 1× 2016
- Producers: **green plants** 1× 2013
- Main marine producers: **phytoplankton** 1× 2005
- Link between biotic and abiotic: **producers** 1× 2017
- Primary productivity = autotrophs; secondary = heterotrophs (both) 1× 2017
- Correct: **green plants-ecosystem** (producers) 1× 2015

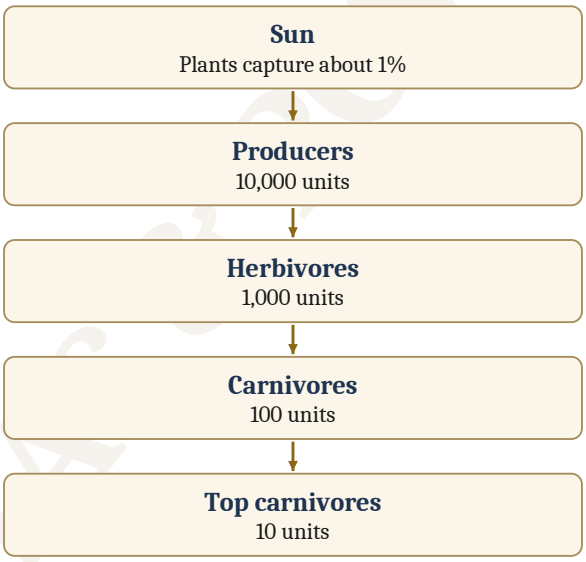
2.2 Aquatic Ecosystems

- **Lotic**: **rivers** (flowing water) 1× 2023
- Phytoplankton thrive in the **limnetic** (open-water) zone (R explains A) 1× 2022

2.3 Energy Flow

- Primary energy source: **solar**; driving force 3× 2015, 2016, 2019
- Energy flow is **unidirectional** 1× 2015
- Energy **decreases** up the levels 2× 2017, 2019
- **Ten Percent Rule**: **Lindeman** 1× 2023
- Plants capture about **1%** of sunlight; about **10%** passes up (both) 1× 2025
- Highest level (key): **omnivores** (top carnivores in most texts) 1× 2002

10% LAW OF ENERGY



2.4 Pyramids and Productivity

- Inverted biomass pyramid: **pond** 1× 2010
- Equatorial forest densest biomass; rain and heat all year (R explains A) 1× 2023
- Least biomass production: **deep sea** 1× 2002
- Decreasing productivity: **mangroves, grasslands, lakes, oceans** 1× 2021

Pyramid	Inverted?
Numbers	Yes in a tree ecosystem
Biomass	Yes in pond and ocean
Energy	Never

2.5 Terms and Thinkers

- Bionomics: synonymous with ecology; values natural systems (2 and 3) 1× 2001
- Land needed to sustain a person: **ecological footprint**; measured in **global hectares** 2× 2012, 2016
- Zone between two communities: **ecotone** 1× 2012
- "Deep ecology": **Arne Naess** 1× 2014

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- Monoclimax theory: **F.E. Clements** 1× 2021
- Niche concept: **Grinnell** 1× 2005
- Not eco-friendly: **eucalyptus** 3× 2005, 2010, 2011
- "Ecology is permanent economy": **Chipko**; Chipko = **plant (tree) conservation** 2× 2007, 2015

Term	Thinker
Ecology	Haeckel (1866)
Ecosystem	Tansley (1935)
Niche	Grinnell
10% law	Lindeman
Monoclimax	Clements
Deep ecology	Arne Naess

TRAP ALERT

- **Energy flows** one way; **nutrients cycle**.
- Energy pyramid **never** inverts; pond biomass pyramid does.
- **Lotic** = flowing; **lentic** = still.

UP LINK

UP link: **Chipko** began in the Garhwal hills (then in UP, now Uttarakhand) with Sunderlal Bahuguna and Chandi Prasad Bhatt; slogan "Ecology is permanent economy".

LAST-MINUTE 10

1. Sun = energy source; flow is one-way
2. 10% law = **Lindeman**; plants catch ~1%
3. Producers: green plants; phytoplankton at sea
4. Grass → grasshopper → frog → snake
5. Man = primary + secondary consumer
6. Pond = inverted biomass pyramid
7. Mangroves most productive; deep sea least
8. Ecotone = transition zone; footprint = global hectares
9. Deep ecology = **Arne Naess**; niche = **Grinnell**
10. Eucalyptus not eco-friendly; Chipko = permanent economy

Chapter 3

Biodiversity, Its Loss and Conservation

Built from 29 UPPSC questions (2001–2023). Focus: meaning, tropical rainforests, Western Ghats and hotspots, habitat destruction, in-situ and ex-situ conservation.

ASKED MOST · REVISE FIRST

- Biggest cause of biodiversity loss: **habitat destruction** 5× 2002, 2015, 2016, 2017, 2023
- Maximum biodiversity: **tropical rainforests** 4× 2012, 2014, 2016, 2018
- India's richest region: **Western Ghats**; Vindhyas are not a hotspot 4× 2014, 2015, 2016, 2019
- Coral bleaching: **rising sea temperature** (global warming) 2× 2012, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Meaning and Distribution

- Biodiversity: **range of species in an environment**; diverse plants and animals of an area 2× 2014, 2014
- Term "biodiversity" first used by **Walter G. Rosen** (1985) 1× 2013
- Most significant aspect: **maintaining the ecosystem** 1× 2015
- UN decade 2011-20: **Biodiversity** 1× 2012
- Most biodiversity: **tropical rainforest** 4× 2012, 2014, 2016, 2018
- Not true: diversity **increases towards the poles** (it decreases) 1× 2019
- Most butterflies in the tropics; they can't bear cold (R explains A) 1× 2019

3.2 India and Hotspots

- Richest biodiversity (of options): **Silent Valley**; Silent Valley is in **Kerala** 2× 2003, 2008
- **Valley of Flowers**: **Uttarakhand** 1× 2002
- Hotspot region (of options): **Andaman and Nicobar** (Sundaland) 1× 2009

- Most biodiverse area: **Western Ghats** 3× 2014, 2015, 2019
- Not a hotspot: **Vindhyas** 1× 2016

India's hotspots	Area
Himalaya	Indian Himalaya
Indo-Burma	North-east India
Western Ghats and Sri Lanka	Western Ghats
Sundaland	Nicobar Islands

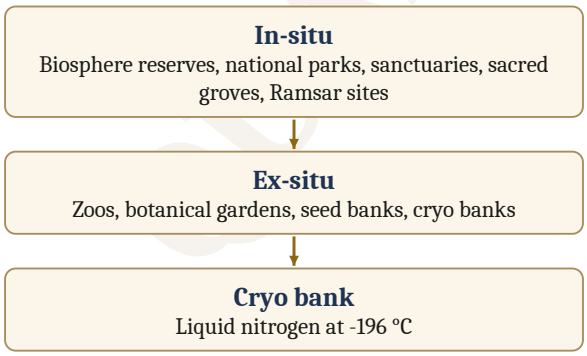
3.3 Causes of Loss

- Destruction due to **all** listed causes 1× 2002
- Maximum danger: **habitat and vegetation destruction** 1× 2017
- Main reason: **natural habitat destruction** 3× 2015, 2016, 2023
- Bird decline: **habitat loss and pesticide use** (3 and 4) 1× 2001
- Not adding biodiversity: **fewer trophic levels** 1× 2015
- Coral bleaching: **rise in sea temperature**; **global warming** 2× 2012, 2015

3.4 Conservation

- Important strategy: **biosphere reserves** 1× 2013
- Biosphere reserves conserve **genetic variation** 1× 2004
- Not in-situ: **botanical garden** 1× 2018
- Cryo-bank gas: **liquid nitrogen** (-196 °C) 1× 2009

CONSERVATION METHODS



TRAP ALERT

- **Habitat destruction** beats pollution and hunting as the main cause.
- **Botanical gardens, zoos** = ex-situ; **sacred groves** = in-situ.
- **Vindhyas** are not a hotspot.

UP LINK

UP's biodiversity sits mostly in the **Terai** (Dudhwa, Katarniaghat, Pilibhit) and the **Himalayan foothill** belt it shares with Uttarakhand; UP has no hotspot area of its own.

LAST-MINUTE 10

1. Term "biodiversity": **Walter G. Rosen**
2. UN Biodiversity Decade **2011-20**
3. Tropical rainforests richest; diversity falls toward poles
4. Western Ghats richest in India
5. Hotspots: Himalaya, Indo-Burma, W. Ghats, Sundaland
6. Silent Valley = Kerala; Valley of Flowers = Uttarakhand
7. Main loss cause: **habitat destruction**
8. Coral bleaching = warmer seas
9. Botanical garden = ex-situ
10. Cryo bank = liquid nitrogen

Chapter 4

Endangered Species, Ramsar Convention and Biodiversity Miscellany

Built from 21 UPPSC questions (2002–2021). Focus: Red Data Book, vultures and diclofenac, Ramsar wetlands, CBD Nagoya, Biodiversity Act and NBA.

ASKED MOST · REVISE FIRST

- Vultures nearly wiped out by **diclofenac** given to cattle 3× 2008, 2015, 2018
- **Ramsar Convention** = **wetlands**; World Wetlands Day **2 February** 3× 2008, 2008, 2016
- **Red Data Book** lists endangered species 2× 2002, 2013
- **Biological Diversity Act**: **December 2002**; **NBA** 2003, **Chennai** 2× 2010, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Endangered Species

- Endangered species listed in the **Red Data Book**; species facing extinction 2× 2013, 2002
- Highly endangered Indian bird: **Great Indian Bustard** 1× 2005
- Wrong: **desert cat-endangered** 1× 2014
- Extinct if unseen for **50 years** (old rule; IUCN now has no fixed period) 1× 2005
- Vulture killer: **diclofenac**; painkillers to cattle 3× 2008, 2018, 2015

IUCN category	Meaning
Critically Endangered	Extremely high risk
Endangered	Very high risk
Vulnerable	High risk
Near Threatened	Close to threatened
Least Concern	Widespread

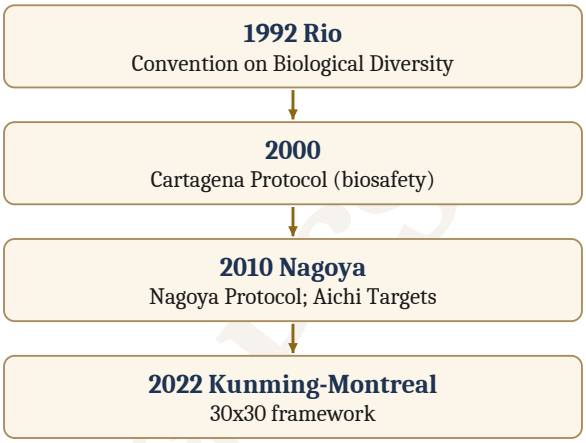
4.2 Ramsar and Wetlands

- Ramsar Convention (1971): **wetlands** 2× 2008, 2016
- Wetlands Day: **2 February** 1× 2008
- Ramsar sites: **Chilika, Keoladeo, Wular, Loktak** (all) 1× 2013
- Ramsar site of the options: **Sundarbans** 1× 2015
- Ramsar site in a State: **Point Calimere** (Tamil Nadu) 1× 2021
- Wetlands useful for **all** listed services 1× 2012

4.3 Laws and Institutions

- CBD COP10: **Nagoya** (2010) 1× 2010
- Biodiversity Act passed: **December 2002** 1× 2010
- NBA set up: **2003, Chennai** 1× 2014
- Centre for Biodiversity Policy and Law (with Norway): **Chennai** 1× 2017

CBD TIMELINE



4.4 Miscellany

- Not a cause of extinction: **broad niche** 1× 2014
- **Sea buckthorn**: **Ladakh** 1× 2004
- Largest ecosystem: **biosphere** 1× 2017
- "Tomorrow's Biodiversity": **Vandana Shiva** 2× 2004, 2005

TRAP ALERT

- Wetlands Day **2 Feb**; Environment Day **5 June**.
- **CBD COP10 = Nagoya**; climate COP15 = Copenhagen.
- A **broad** niche protects; a narrow one endangers.

UP LINK

UP's Ramsar sites include **Nawabganj, Parvati Agra, Saman, Samaspur, Sandi, Sarsai Nawar, Sur Sarovar, Upper Ganga (Brijghat-Narora), Bakhira** and **Haiderpur** wetlands.

LAST-MINUTE 10

1. Red Data Book = endangered species
2. Diclofenac killed vultures
3. Great Indian Bustard = highly endangered
4. Ramsar = wetlands (1971); Wetlands Day **2 Feb**
5. Chilika, Keoladeo, Wular, Loktak = Ramsar sites
6. Point Calimere = Tamil Nadu
7. CBD COP10 = **Nagoya**
8. Biodiversity Act **2002**; NBA **2003 Chennai**
9. Sea buckthorn = Ladakh
10. Tomorrow's Biodiversity = **Vandana Shiva**

Chapter 5

Greenhouse Effect and Climate Change

Built from 40 UPPSC questions (2001–2024). Focus: which gases are greenhouse gases, CO2 and methane, effects of warming, emitters, net zero, Milankovitch.

ASKED MOST · REVISE FIRST

- **Not greenhouse gases**: **N2, O2, argon, H2, SO2, NO2, propane** 11× 2007, 2012, 2012, 2016, 2016, 2016, 2017, 2017, 2019, 2020, 2022
- **CO2** = main cause of global warming 6× 2004, 2004, 2006, 2008, 2013, 2015
- **Methane** from **paddy fields** and cattle 3× 2017, 2018, 2019
- **China** = largest CO2 emitter 3× 2010, 2015, 2018
- Warming effects: sea-level rise, melting glaciers, disease, coral bleaching, early mango flowering 4× 2001, 2002, 2003, 2020

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Greenhouse Effect

- Concept of greenhouse gases: **Joseph Fourier** 1× 2011
- Greenhouse effect: **rise in global temperature** 1× 2018
- Without it Earth would average **-18 °C** 1× 2020
- Enhanced by **CO2**; main gases **CO2 and methane** 3× 2015, 2004, 2013
- Largest natural share: **water vapour** 1× 2011
- Not responsible: **argon** 2× 2016, 2022
- Not directly warming: **oxides of sulphur and nitrogen** 1× 2013
- Not a natural GHG (of options): **nitrogen oxide** 2× 2019, 2020
- Not GHG: **SO2**; **oxygen**; **NO2**; **propane**; **nitrogen**; **hydrogen** 7× 2007, 2016, 2017, 2016, 2017, 2012, 2012
- Both harmful and useful: **CO2** per key (ozone in most texts) 1× 2013

Gas	Main source	GWP (100 yr)
CO2	Fossil fuels, deforestation	1
Methane	Paddy, cattle, wetlands, landfills	~28
Nitrous oxide (N2O)	Fertilisers	~265
CFCs, HFCs, SF6	Industry	Thousands

5.2 CO2 and Methane

- Global warming from **CO2**; worry grows with CO2 5× 2004, 2013, 2004, 2006, 2008
- Natural optimum CO2: **0.03%** (now about 0.042%) 1× 2017
- Does not add CO2: **photosynthesis** 1× 2012
- Methane from **both** listed sources 1× 2018
- Paddy-field gas: **methane** 2× 2017, 2019

5.3 Causes and Effects

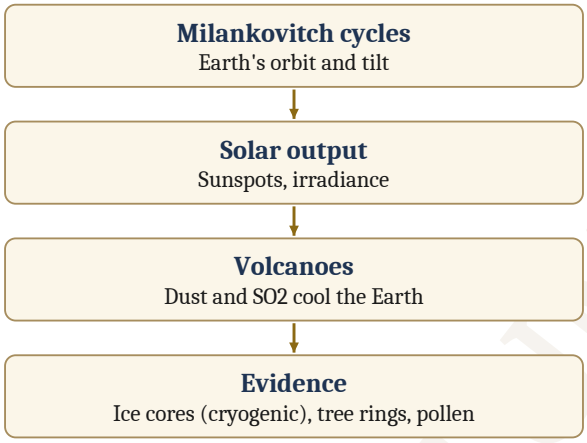
- Climate change from **all** listed causes 1× 2013
- Primary causes: **fossil fuels, cars, deforestation**; not solar flares (1, 2, 4) 1× 2017
- Effect of greenhouse effect: unclear question (*) 1× 2017
- Effects: **melting glaciers, early mango flowering, health harm** (2, 3, 4) 3× 2001, 2002, 2003
- Effects: sea-level rise, glaciers, disease, coral bleaching (all) 1× 2020
- More frequent: **cyclones, storms, hurricanes** 1× 2018

- Andaman coral bleaching: **2 °C warmer Indian Ocean** 1× 2024
- Warming last century: about **0.9 °C** 1× 2012
- **Urban heat island**: cities hotter than nearby rural areas 1× 2023
- Climate change raises social tension via food insecurity (R explains A) 1× 2017

5.4 Emitters and Net Zero

- Largest CO2 emitter: **China** 3× 2010, 2015, 2018
- Carbon-negative country: **Bhutan** 1× 2015
- Net zero = emissions **balanced by removals**, not zero (only 2) 1× 2021
- IPCC: **deep, rapid, sustained** cuts essential (only 2) 1× 2024

NATURAL CAUSES OF CLIMATE CHANGE



5.5 Natural Causes

- Not astronomical theory: **solar irradiance** 1× 2015
- Axis shift as climate factor: **Milutin Milankovitch** 1× 2015
- Cryogenic indicator: **ice core** 1× 2015

TRAP ALERT

- **N2O** is a GHG; **NO2** is not.
- N2, O2, Ar make up air but are **not** GHGs.
- **Net zero** is not zero emissions.

UP LINK

UP links: UP's large **paddy and livestock** sector makes it a big methane source; **Lucknow and Kanpur** show urban heat-island effects in heatwaves.

LAST-MINUTE 10

1. Greenhouse idea: **Fourier**; without it **-18 °C**
2. CO2 = main human driver; water vapour = largest natural share
3. Methane = paddy, cattle
4. Not GHG: N2, O2, Ar, H2, SO2, NO2, propane
5. CO2 natural level **0.03%**
6. Warming ~0.9 °C last century
7. China = largest emitter; Bhutan = carbon negative
8. Net zero = balance, not zero
9. Milankovitch = orbit; ice core = cryogenic proof
10. Urban heat island = hotter cities

Chapter 6

Climate Change Conferences, Treaties and Action Plans

Built from 25 UPPSC questions (2001–2025). Focus: Agenda 21, UNFCCC, Kyoto and carbon credits, Paris COP21, GCF and GEF, NAPCC missions.

ASKED MOST · REVISE FIRST

- **Agenda 21** (Rio 1992) = action plan for **sustainable development** 3× 2013, 2014, 2023
- **Kyoto Protocol** = **climate change**; source of **carbon credits** 4× 2015, 2017, 2019, 2021
- **COP21 = Paris** (2015) 2× 2015, 2018
- **NAPCC** released **2008**; 8 missions, **no nuclear** mission 5× 2010, 2016, 2016, 2018, 2021

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Earth Summits and UNFCCC

- Earth Summit + 5: **1997** 1× 2016
- Agenda 21: action plan for the **21st-century environment**; **sustainable development** 3× 2013, 2014, 2023
- Wrong: **carbon trading-Montreal Protocol** (it is Kyoto) 1× 2015

- UNFCCC in force: **21 March 1994** 1× 2012
- Paris conference: **COP21** 2× 2015, 2018

6.2 Protocols and Funds

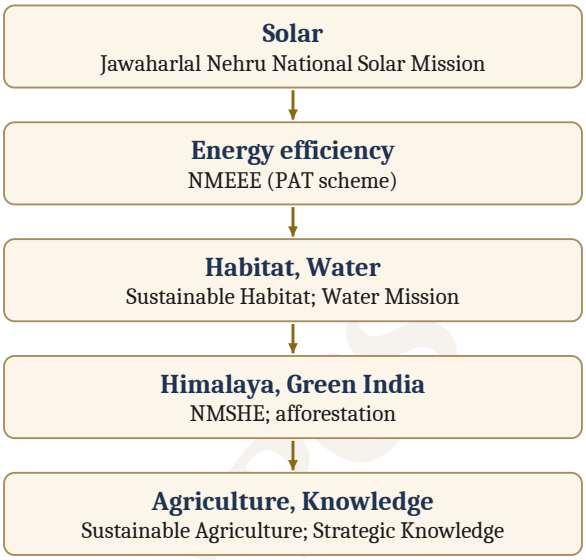
- Wrong: **Kyoto-save water**; Kyoto = **climate change** 2× 2019, 2017
- Incorrect: GCF set up at **Durban** (it was **Cancun**, 2010) 1× 2013
- GEF set up **1991** (true); only rich countries donate (false) (only 1) 1× 2025
- Not true of **CDM**: it bars rich countries investing in poor ones 1× 2001
- Carbon credits come from the **Kyoto Protocol** 1× 2021
- Not correct: World Bank's 2011 ERPA was signed with the **Government of India** 1× 2010
- Carbon tax from 2019: **Singapore** 1× 2017
- UN Climate Action Summit 2019: **New York** 1× 2019

COP	Place	Outcome
COP3 1997	Kyoto	Kyoto Protocol
COP15 2009	Copenhagen	$100 bn pledge
COP16 2010	Cancun	Green Climate Fund
COP21 2015	Paris	Paris Agreement
COP26 2021	Glasgow	India net zero 2070
COP27 2022	Sharm el-Sheikh	Loss and Damage fund
COP28 2023	Dubai	Global Stocktake
COP29 2024	Baku	$300 bn finance goal

6.3 NAPCC

- First NAPCC: **2008** 3× 2016, 2018, 2021
- Not among the 8 missions: **nuclear power** 2× 2010, 2016
- Mission start years: sources differ (*) 1× 2016

NAPCC'S 8 MISSIONS (2008)



6.4 Miscellany

- **Great Barrier Reef**: NE Australia (true); over 2,500 km (false per key) (only 1) 1× 2023
- **Deep Carbon Observatory**: carbon research and deep microbes (both) 1× 2019
- Cloud brightening uses **sea water** 1× 2009
- Ozone layer blocks UV; depletion linked to CFCs (both) 1× 2025
- No global atmosphere watch station then: **India** 1× 2012
- **SAFAR** inaugurated (2018) by **Dr. Harsh Vardhan** 1× 2018

TRAP ALERT

- **GCF = Cancun**, not Durban; **Kyoto = 1997**.
- **Montreal** = ozone; **Kyoto** = carbon trading.
- Nuclear power is **not** an NAPCC mission.

UP LINK

UP has its own **State Action Plan on Climate Change** under the NAPCC; it targets solar parks in **Bundelkhand** and Green India plantation drives.

LAST-MINUTE 10

1. Agenda 21 = Rio 1992; sustainable development
2. Earth Summit + 5 = **1997**
3. UNFCCC in force **21 March 1994**
4. Kyoto **1997** = carbon credits, CDM
5. Paris = **COP21** (2015)
6. GCF = **Cancun 2010**; GEF = 1991
7. Singapore = carbon tax 2019
8. NAPCC **2008**; 8 missions; no nuclear
9. Cloud brightening = sea water spray
10. SAFAR = air-quality forecasting (2018)

Chapter 7

The Ozone Layer

Built from 19 UPPSC questions (1996–2023). Focus: stratospheric ozone, UV protection, CFCs and other depleters, the ozone hole, Ozone Day.

ASKED MOST · REVISE FIRST

- Ozone layer: **stratosphere** 6× 2008, 2010, 2012, 2018, 2018, 2023
- Ozone absorbs **ultraviolet** rays; protects life 5× 2004, 2006, 2010, 2013, 2014
- **CFCs** deplete ozone 3× 2009, 2014, 2016
- Ozone Day: **16 September** (Montreal Protocol, 1987) 4× 1996, 2014, 2015, 2022

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

7.1 Ozone and Location

- UV light forms **O3** 1× 2018
- Ozone layer: **stratosphere** 5× 2008, 2010, 2012, 2018, 2023
- Wrong: **ozone layer-troposphere** 1× 2018
- Natural regulator of stratospheric ozone (per key): **nitrogen dioxide** (NOx) 1× 2016

7.2 What Ozone Does

- Absorbs **ultraviolet** radiation; protects from **UV rays** 2× 2013, 2014
- Shields life from UV 2× 2004, 2006
- Ozone essential for life; blocks high-energy radiation (R explains A) 1× 2010
- UV causes **skin cancer** 1× 2014

7.3 Depleting Substances

- Main depleter: **CFCs** 3× 2009, 2014, 2016
- Not true: CFCs are **not** greenhouse gases (they are) 1× 2004
- Depleters: **CFCs, halons, carbon tetrachloride** (all) 1× 2012
- Not a depleter (key): **methyl chloroform** (actually it is one) 1× 2002
- Wrong: **CFC-automobile** (CFCs from refrigerants, sprays) 1× 2017
- Does not deplete ozone: **carbon monoxide** 1× 2008

Substance	Use
CFCs	Refrigerators, ACs, aerosols
Halons	Fire extinguishers
Carbon tetrachloride	Solvent
Methyl bromide	Fumigant
HFCs	No ozone harm; strong GHGs (Kigali 2016)

7.4 The Ozone Hole

- Antarctic ozone hole found by a **British team** (Farman, 1985) 1× 2013
- Ozone "halo" over Tibet (2005): **G.W. Kent Moore** 1× 2013
- Ozone Day: **16 September** 4× 1996, 2014, 2015, 2022

QUANTUM IAS & PCS

SAVING THE OZONE LAYER

1974
Molina and Rowland warn about CFCs

↓

1985
Vienna Convention; British Antarctic Survey finds the hole

↓

16 Sept 1987
Montreal Protocol (Ozone Day)

↓

2016
Kigali Amendment: HFC phase-down

↓

~2066
Antarctic hole expected to heal

TRAP ALERT

- Ozone absorbs **UV**, not infrared.
- Ground-level ozone is a **pollutant**; stratospheric ozone protects.
- CFCs **are** greenhouse gases too.

UP LINK

UP's refrigeration and foam industries (Noida, Ghaziabad) switched from CFCs under India's **Montreal Protocol** phase-out; HCFCs end by 2030.

LAST-MINUTE 10

1. Ozone layer = **stratosphere** (15-35 km)
2. Absorbs **UV-B**; UV causes skin cancer, cataract
3. CFCs, halons, CCl4 deplete ozone
4. CFCs also greenhouse gases
5. CO does not deplete ozone
6. Ozone hole: British team, 1985
7. Tibet ozone halo: Kent Moore
8. Montreal Protocol **1987**
9. Ozone Day **16 September**
10. Kigali **2016** = HFCs

Chapter 8

Forests and Their Types

Built from 17 UPPSC questions (2002–2023). Focus: forest types and their regions, tropical rainforest records, mangroves, forests as a resource.

ASKED MOST · REVISE FIRST

- Most plant diversity and biomass: **tropical evergreen / rain forest** 2× 2017, 2017
- Mangroves: **Sundarbans**; **Bhitarkanika** is in **Odisha** 3× 2004, 2016, 2016
- **Western Ghats** = evergreen forests 1× 2015
- Forest types: **tropical-Silent Valley, coniferous-Himachal, mangrove-Sundarbans, deciduous/thorn-Rajasthan** 1× 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Types and Productivity

- Maximum plant diversity: **tropical evergreen forests** 1× 2017
- Maximum plant biomass: **tropical rainforest** 1× 2017
- Lowest net primary productivity: **temperate grasslands** 1× 2015
- Tropical deciduous forests most extensive in **Asia** 1× 2014
- Evergreen forest: **Western Ghats** 1× 2015
- Wrong: **Eastern Deccan Plateau-moist forests** (dry deciduous) 1× 2013
- **Tropical forest-Silent Valley, coniferous-Himachal Pradesh, mangroves-Sundarbans, deciduous-Rajasthan** 1× 2017

Forest type	Rainfall	Example region
Tropical evergreen	Over 200 cm	Western Ghats, NE, A&N
Moist deciduous	100-200 cm	Sal, teak belt
Dry deciduous	70-100 cm	Central India, UP
Thorn	Under 70 cm	Rajasthan, Gujarat

Forest type	Rainfall	Example region
Montane coniferous	Altitude	Himalaya (deodar, pine)
Mangrove	Tidal deltas	Sundarbans, Bhitarkanika

8.2 Mangroves

- Famous for mangroves: **Sundarbans** 1× 2004
- Protected mangrove region (of options): **Goa** 1× 2016
- Bhitarkanika: not Vansadhara-Subarnarekha delta, not West Bengal (**neither**; it is Brahmani-Baitarani, Odisha) 1× 2016
- Coastal Odisha cyclone-prone; mangrove loss in Mahanadi delta (key: R explains A) 2× 2006, 2010

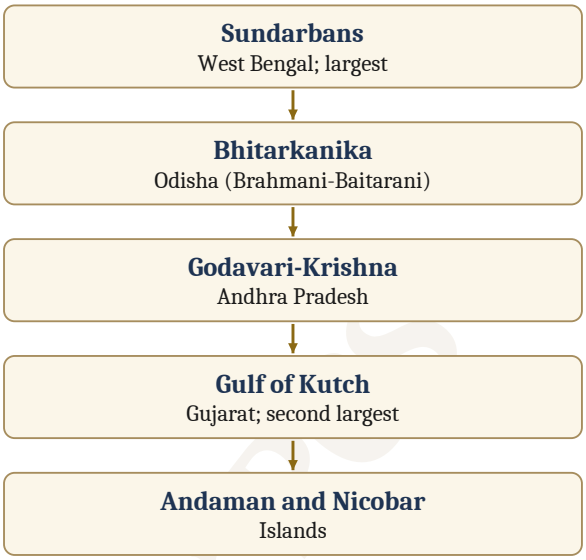
8.3 Forests as a Resource

- Forests renewable; improve the environment (both true, R not the explanation) 2× 2002, 2004
- Hydrological cycle controlled by **forests** (per key) 1× 2023
- Natural resources usable later; development depends on technology (both true, R not the explanation) 1× 2016
- Aluminium a green metal (true); substitutes wood (false) 1× 2009

8.4 Causes of Loss

- Forest loss linked to demographic transition; inverse to population (R explains A) 1× 2016
- Main cause (key): **industrial development** (FAO: farm expansion) 1× 2011

MANGROVE BELTS OF INDIA



TRAP ALERT

- **Bhitarkanika** = Odisha, not West Bengal.
- **Evergreen** = Western Ghats; **coniferous** = Himalaya.
- Lowest NPP of options = **temperate grassland**.

UP LINK

UP's forests are mostly **tropical dry and moist deciduous** (sal in the Terai; teak, tendu in Bundelkhand and Sonbhadra); UP has no mangroves.

LAST-MINUTE 10

1. Tropical evergreen: most plant diversity and biomass
2. Western Ghats = evergreen
3. Coniferous = Himalaya; thorn = Rajasthan
4. Mangroves: **Sundarbans** largest; Bhitarkanika = Odisha
5. Gulf of Kutch = 2nd mangrove area
6. Asia = most tropical deciduous forest
7. Temperate grassland = lowest NPP
8. Forests control the water cycle
9. Aluminium = green metal
10. Forest loss rises with population

QUANTUM IAS

Chapter 9

India State of Forest Report (ISFR)

Built from 20 UPPSC questions (1991–2022). Focus: national forest percentages by report year, state rankings by area and percentage, very dense forest, UP.

ASKED MOST · REVISE FIRST

- ■ Highest forest **percentage**: **Mizoram** 5× 2002, 2008, 2010, 2012, 2016
- • Largest forest **area**: **Madhya Pradesh** 1× 2000
- ■ Most **very dense** forest: **Arunachal Pradesh** 3× 2004, 2008, 2011
- • National cover by report: **21.54%** forest (2017); **24.16%** forest + tree (2015) 2× 2016, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 National Figures

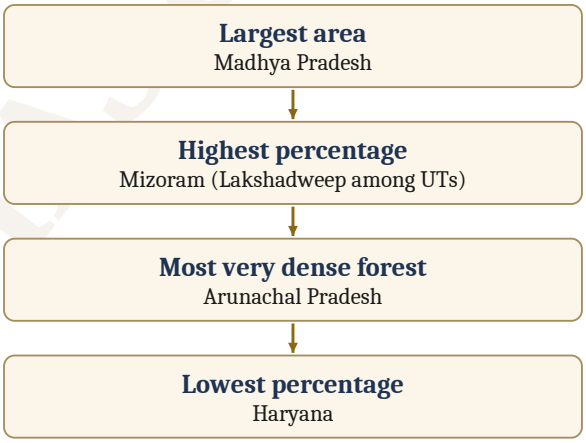
- Forest share (1991 key): **22%** 1× 1991
- ISFR 2017: **21.54%** under forest 1× 2018
- ISFR 2015: **24.16%** forest and tree cover 1× 2016
- Dense forest share: about **12%** 1× 2008

ISFR 2023	Figure
Forest cover	21.76%
Forest + tree cover	25.17%
Mangroves	~4,992 sq km
Target (Forest Policy 1988)	33%

9.2 State Rankings

- ■ Largest forest percentage: **Mizoram**; among States/UTs: **Mizoram** 5× 2002, 2010, 2012, 2016, 2008
- • Most forest of HP, Arunachal, Meghalaya, Sikkim: **Arunachal** 1× 2004
- • Over 75% forest group: none fits now (*) 1× 1995
- • Over 80% forest (then): **Arunachal**; now **Mizoram** 1× 2008
- • Over 33% forest (ISFR 2021): **17** states/UTs 1× 2021
- • Largest forested area: **Madhya Pradesh** 1× 2000
- • Largest very dense forest: **Arunachal Pradesh** 1× 2011
- • Not correct: **Nagaland** is the most forested State 1× 2010
- • Most forest gain (ISFR 2021): **Andhra Pradesh** 1× 2022
- • Second largest mangroves: **Gujarat** 1× 2008
- • Tree cover maximum (key): **East Deccan** 1× 2009
- • Descending cover: **Manipur, Odisha, Maharashtra, Haryana** 1× 2002
- • Forest area order: Arunachal, Chhattisgarh, Odisha, AP (no code, *) 1× 2011
- • Most tribal districts (ISFR 2019 options): **Manipur** 1× 2021

THREE DIFFERENT "FIRSTS"



9.3 Uttar Pradesh

- UP's recorded forest area (2011): about **7%** of its area 1× 2012

TRAP ALERT

- **Area** (MP) vs **percentage** (Mizoram) vs **very dense** (Arunachal).
- "Recorded forest area" (legal) differs from "forest cover" (satellite).
- Answers change by report year; check the ISFR edition.

UP LINK

UP has only about **6%** forest cover in recent ISFRs, far below the 33% goal; the State's mass plantation drives (crores of saplings each July) aim to raise green cover.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. ISFR by **Forest Survey of India** (Dehradun), every 2 years
2. 2023: forest **21.76%**; forest + tree **25.17%**
3. Target **33%** (two-thirds in hills)
4. Mizoram = highest %; MP = largest area
5. Arunachal = most very dense forest
6. Haryana = lowest %
7. West Bengal, Gujarat = top mangroves
8. 17 States/UTs over 33% (2021)
9. Andhra gained most (2021)
10. UP ~7% recorded forest

Chapter 10

Forest and Wildlife Conservation Efforts

Built from 26 UPPSC questions (1998–2025). Focus: Forest Policy and the 33% goal, Wildlife and Forest Acts, schemes and operations, institutions, wildlife facts.

ASKED MOST · REVISE FIRST

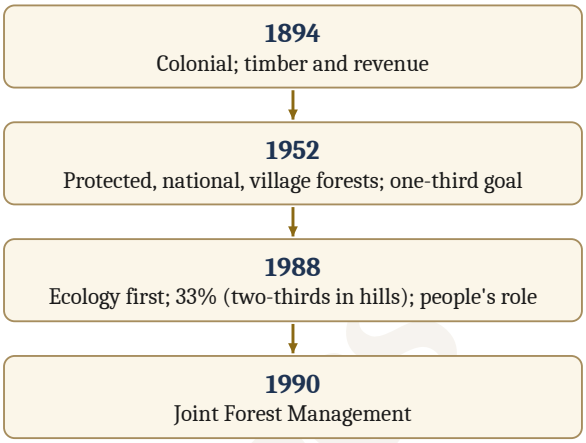
- Forest goal: **33%** (one-third) of area 2× 2002, 2014
- **Wildlife (Protection) Act 1972**; **Forest (Conservation) Act 1980** 4× 2013, 2015, 2017, 2021
- Order: **Wildlife Act (1972), Project Tiger (1973), Project Elephant (1992), Biodiversity Act (2002)** 2× 2019, 2020
- **Forest Research Institute**: **Dehradun** 2× 2008, 2016
- WWF symbol: **giant panda** 3× 2001, 2002, 2003

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Forest Policy

- Forest area for ecological balance: **33%**; one-third 2× 2014, 2002
- Not a forest class in the 1952 policy: **national park** 1× 2016
- Policy aims (key): **ecological balance, people's participation** (1 and 4) 1× 2001

FOREST POLICIES



10.2 Laws

- Wildlife Protection Act: **1972** 1× 2015
- Forest (Conservation) Act: **1980** 2× 2013, 2017
- **Forest Conservation-1980, Environment Protection-1986, Air-1981, Water-1974** 1× 2021
- Order: **Wildlife Act, Project Tiger, Project Elephant, Biodiversity Act** 2× 2019, 2020
- Wildlife Act protects only animals (false); Project Elephant only Asian elephants (false): **neither** 1× 2023

10.3 Schemes and Operations

- **MISHTI**: restores mangroves (true); begun 2019 (false, **2023**) 1× 2025
- **Operation Olivia**: Coast Guard guards olive ridleys off Odisha (true); in August (false) 1× 2025
- "Apna Van Apna Dhan": **Himachal Pradesh** 1× 2012
- Plantation town: **Valparai** 1× 2013
- **SAVE** (India-Nepal): **vultures** 1× 2011
- New Forest Inventory (2016) revisit: **5 years** 1× 2017

10.4 Institutions

- **Forest Research Institute**: **Dehradun** 2× 2008, 2016
- **NBPGR**: **New Delhi** 1× 2013
- Rajiv Gandhi Wildlife Conservation Award: **all** listed 1× 1998

Institution	Location
FRI, FSI, Wildlife Institute of India	Dehradun

Institution	Location
Forest Management Institute	Bhopal
NBPGR	New Delhi
BSI, ZSI	Kolkata
NBA	Chennai
SACON	Coimbatore

10.5 Wildlife Facts

- India's share of world wildlife: about **6.5-7.5%** (*) 1× 2008
- India's largest fish: **whale shark** 1× 2008
- Not a major extinction cause (key): **fast population growth** 1× 2020
- **Pugmark** technique: **counting wild animals** 1× 2008
- WWF symbol: **giant panda** 3× 2001, 2002, 2003
- Statements 1, 2, 4 correct; eucalyptus is not eco-friendly 1× 2002
- Largest forest share (of options): **Suriname**; **Japan** 2× 2009, 2013

TRAP ALERT

- **MISHTI = 2023**; olive ridleys nest **Jan-March**.
- Wildlife Act also protects **plants**.
- National parks are a Wildlife Act category, not a 1952 forest class.

UP LINK

UP links: **Project Tiger** covers **Dudhwa, Pilibhit, Amangarh** and **Ranipur** tiger reserves; UP runs the giant one-day **Van Mahotsav** plantation drives.

LAST-MINUTE 10

1. Forest goal **33%**; hills two-thirds
2. Wildlife Act **1972**; Forest Conservation Act **1980**
3. Water **1974**, Air **1981**, EPA **1986**
4. Project Tiger **1973**; Elephant **1992**; Biodiversity Act **2002**
5. MISHTI (2023) = mangroves
6. Operation Olivia = olive ridleys, Odisha
7. SAVE = vultures
8. FRI = **Dehradun**; NBPGR = **New Delhi**
9. Pugmark = animal census
10. WWF symbol = **giant panda**

Chapter 11

National Parks

Built from 41 UPPSC questions (1996–2024). Focus: park-state matching, park-animal matching, first parks, marine parks.

ASKED MOST · REVISE FIRST

- **Park-state matching**: **Bandipur-Karnataka, Similipal-Odisha, Pin Valley and Rohla-Himachal, Indravati-Chhattisgarh, Dachigam-J&K, Intanki-Nagaland, Betla-Jharkhand, Guindy-TN** 12× 1998, 2002, 2002, 2002, 2010, 2015, 2016, 2016, 2017, 2018, 2020, 2024
- **Corbett** = first national park (1936); fed by the **Ramganga** 2× 2006, 2017
- Animals: **Kaziranga-rhino, Gir-lion, Sundarbans-tiger, Periyar-elephant, Dachigam-hangul** 4× 2006, 2014, 2018, 2022
- **Dudhwa** (UP) = rhino rehabilitation 2× 2004, 2022
- Marine park: **Gulf of Kutch** (first), **Gulf of Mannar** 3× 2013, 2015, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 First Parks

- First national park: **Jim Corbett**; water from **Ramganga** 2× 2017, 2006

- **Yellowstone**: **USA** 3× 2004, 2010, 2011
- Order: **Corbett, Kanha, Kaziranga, Silent Valley** 1× 2016
- Genetic diversity in parks: **in-situ** conservation 1× 2023
- Most national parks: **Madhya Pradesh** 2× 2004, 2008
- Most parks in MP; most sanctuaries in A&N (both) 1× 2021

11.2 Park and State

- **Rajiv Gandhi NP** (Nagarhole): **Karnataka** 1× 2002
- **Keibul Lamjao**: on **Loktak** lake 1× 2022
- Wrong: **Pin Valley-J&K** (Himachal) 2× 2016, 2024
- Wrong: **Similipal-MP** (Odisha) 1× 2018
- Wrong: **Bandipur-Tamil Nadu** (Karnataka) 1× 2010
- True: Similipal in Odisha; Siam = Thailand (1 and 2) 1× 2002
- **Bandhavgarh-MP, Bandipur-Karnataka, Rohla-Himachal, Gir-Gujarat** 1× 2002
- **Uttarakhand-Rajaji, Assam-Manas, Odisha-Similipal, Karnataka-Bandipur** 1× 1998
- **Dachigam-J&K, Papikonda-AP, Sariska-Rajasthan, Bandipur-Karnataka** 1× 2015
- **Assam-Kaziranga, Gujarat-Gir, MP-Kanha, Uttarakhand-Corbett** 1× 2002
- **Indravati**: **Chhattisgarh**; in **Bastar** 2× 2016, 2015
- **Intanki-Nagaland, Betla-Jharkhand, Shiroi-Manipur, Guindy-Tamil Nadu** 1× 2017
- **Indravati-Chhattisgarh, Mollem-Goa, Kalesar-Haryana, Betla-Jharkhand** 1× 2020
- Correct: **Keoladeo-Bharatpur** 1× 2009
- **Kaziranga-Golaghat, Kudremukh-Chikmagalur, Silent Valley-Palghat, Pench-Nagpur** 1× 2013
- **Dhoopgarh** peak: **Satpura NP** 1× 2021

11.3 Parks and Animals

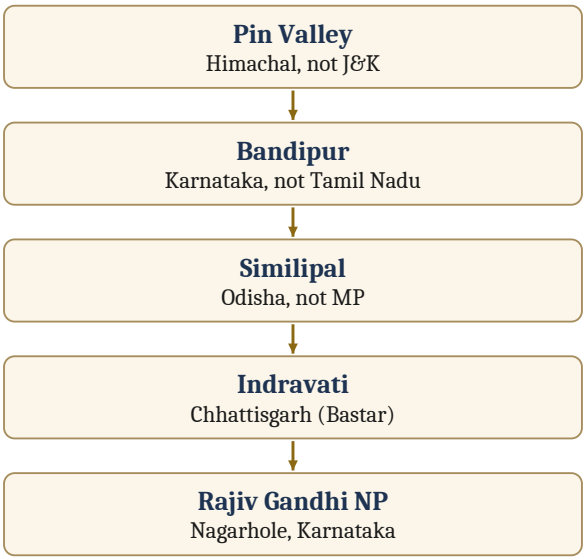
- Wrong: **Ghana (Keoladeo)-lion** (birds) 1× 2009
- **Kaziranga-rhino, Gir-lion, Sundarbans-tiger, Periyar-elephant** 1× 2006
- Wrong: **Periyar-hangul** 1× 2014
- **Dachigam**: **hangul** (Kashmir stag); hangul in **J&K** 2× 2018, 2022
- Rhino rehabilitation: **Dudhwa** 2× 2004, 2022
- Siberian crane habitat: **Rajasthan** (Keoladeo) 2× 1996, 2010
- Nilgiri "cloud goats" (tahr): **Eravikulam** 1× 2005

Park	Animal
Kaziranga	One-horned rhino
Gir	Asiatic lion
Sundarbans	Tiger
Periyar	Elephant
Dachigam	Hangul
Eravikulam	Nilgiri tahr
Keibul Lamjao	Sangai deer
Keoladeo	Birds
Dudhwa	Swamp deer, rhino

11.4 Marine Parks and Miscellany

- Marine park: **Gulf of Mannar** 1× 2015
- Oceanic (first marine) park: **Kutch** 2× 2013, 2016
- Rani Jhansi Marine NP (A&N): **1996** 1× 2017
- UNESCO site July 2016: **Khangchendzonga** 1× 2016
- National Biological Garden: **New Delhi** 1× 2014
- Project Elephant: **1992** (no option, *) 1× 2007
- Garden of Five Senses: near **Mehrauli** 1× 2009
- Over 30% under national parks: **Bhutan** 1× 2015

PARK-STATE TRAPS



TRAP ALERT

- **Keoladeo** = birds, not lions; **Periyar** = elephants, not hangul.
- First **marine** park = Gulf of **Kutch**.
- Most national parks = **MP**; most sanctuaries = **A&N**.

UP LINK

Dudhwa National Park (Lakhimpur Kheri) is UP's only national park: swamp deer (barasingha), tiger, and **rhino rehabilitation** since 1984. UP's other big reserves are **Pilibhit, Amangarh, Ranipur** (tiger reserves) and **Katarniaghat** (sanctuary).

LAST-MINUTE 10

1. Corbett = first NP (1936), Ramganga
2. MP = most national parks
3. Kaziranga = rhino; Gir = lion
4. Dachigam = hangul; Eravikulam = Nilgiri tahr
5. **Dudhwa (UP)** = rhino rehabilitation, swamp deer
6. Keoladeo = birds, Siberian crane (Bharatpur)
7. Pin Valley = Himachal; Bandipur = Karnataka
8. Similipal = Odisha; Indravati = Chhattisgarh
9. Keibul Lamjao = Loktak, floating
10. Gulf of Kutch = first marine park

Chapter 12

Biosphere Reserves, Wildlife Sanctuaries and Project Tiger

Built from 47 UPPSC questions (1994–2025). Focus: biosphere reserve-state matching, UNESCO MAB, sanctuaries, Project Tiger, tiger reserve-state matching.

ASKED MOST · REVISE FIRST

- ■ Biosphere reserves: **Nokrek-Meghalaya (Garo Hills), Manas-Assam, Similipal-Odisha, Agasthyamalai-Kerala, Dihang-Dibang-Arunachal, Pachmarhi-MP, Nanda Devi-Uttarakhand, Cold Desert-Himachal** 11× 2007, 2010, 2013, 2013, 2014, 2019, 2021, 2022, 2024, 2025, one-liner
- ■ Tiger reserves: **Bandipur-Karnataka, Melghat-Maharashtra, Buxa-West Bengal, Pakke and Namdapha-Arunachal, Dampa-Mizoram, Sariska-Rajasthan; Amangarh is UP** 7× 2003, 2006, 2008, 2010, 2012, 2012, 2021
- ■ **Project Tiger** = **1973**; Tiger Day **29 July** 4× 1994, 2002, 2014, 2018
- • **Periyar** (Kerala) = wild elephants 2× 2011, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Biosphere Reserves

- Preserve **genetic diversity** 1× 2004
- Correct: **biosphere-Edward Suess** (coined "biosphere") 1× 2013
- Not in UNESCO network (then): **Rann of Kutch** 1× 2009
- Largest UNESCO-listed: **Gulf of Mannar** 1× 2013
- **Nilgiri-1986, Nanda Devi-1988, Sundarbans-1989, Kanchenjunga-2000** 1× 2012
- **Nokrek-Meghalaya, Manas-Assam, Dihang-Dibang-Arunachal, Agasthyamalai-Kerala** 1× 2013
- **Similipal-Odisha, Nokrek-Meghalaya, Agasthyamalai-Kerala, Manas-Assam** 2× 2019, 2024
- Correct: **Nokrek-Meghalaya**; wrong: **Nokrek-Manipur** 2× 2007, 2025
- **Similipal-Odisha, Dihang-Dibang-Arunachal, Nokrek-Meghalaya, Cold Desert-Himachal** 1× one-liner
- **Similipal-Odisha, Pachmarhi-MP, Nanda Devi-Uttarakhand, Nokrek-Meghalaya** 1× 2021
- Wrong: **Manas-Meghalaya** (Assam) 1× 2013

- **Manas-Assam, Sundarbans-West Bengal, Nanda Devi-UP (then), Pachmarhi-MP** 1× 2022
- Spread over Garo Hills: **Nokrek** 1× 2014
- 16th reserve, Cold Desert: **Himachal Pradesh** 1× 2010
- World Heritage site: **Nanda Devi** 2× 2005, 2008
- **Bhitarkanika**: **Odisha** (Ramsar; only on UNESCO tentative list) 1× 2008

Biosphere reserve	State
Nilgiri (first, 1986)	TN, Kerala, Karnataka
Nanda Devi	Uttarakhand
Nokrek	Meghalaya
Gulf of Mannar	Tamil Nadu
Sundarbans	West Bengal
Manas	Assam
Similipal	Odisha
Dihang-Dibang	Arunachal Pradesh
Pachmarhi, Panna	Madhya Pradesh
Agasthyamalai	Kerala, TN
Kachchh (largest)	Gujarat
Cold Desert	Himachal Pradesh

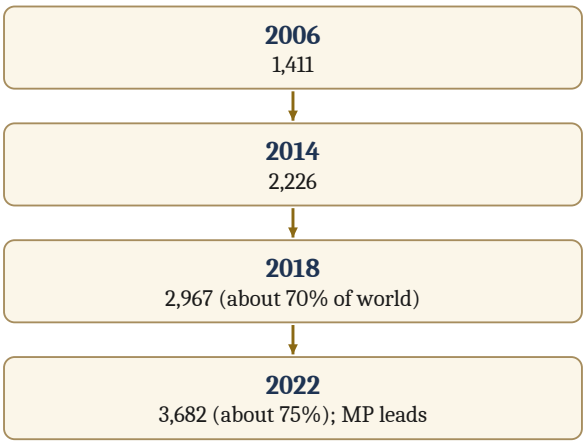
12.2 Sanctuaries

- **Periyar**: **wild elephants**; in **Kerala** 2× 2011, 2015
- **Chinnar** sanctuary: **Kerala** 1× 2005
- Wild ass sanctuary: **Gujarat** (Little Rann) 1× 2010
- **Kaziranga**: **rhinoceros** 1× 2012
- **Garampani-Assam, Namdapha-Arunachal, Pakhal-Andhra, Sariska-Rajasthan** 1× 2003
- **Namdapha-Arunachal, Bandipur-Karnataka, Periyar-Kerala, Lamjao-Manipur** 1× 2015
- **Dachigam-J&K, Keoladeo-Rajasthan, Kanha-MP, Periyar-Kerala** 1× 2014
- Wrong: **Similipal-MP** (Odisha) 1× 2009
- Tamil Nadu bird sanctuary: **Karikili** 1× 2008
- **National Chambal Sanctuary** is not in **Haryana** (it is MP, Rajasthan, UP) 1× 2020
- UP sanctuary of 1975: **Katarniaghat** (dates differ, *) 1× 2022
- **Shuklaphanta**: **Nepal** 1× 2016
- Largest botanical garden: **Kew** (England) 1× 2015

12.3 Project Tiger

- Aim: **save the Indian tiger** from extinction 1× 1994
- Started: **1973** 2× 2002, 2018
- International Tiger Day: **29 July** 1× 2014
- World Tiger Summit 2010: **St. Petersburg** 1× 2011
- Tiger census 2018: India has **70%** of world tigers; about **30%** outside reserves (both) 1× 2020
- Largest tiger habitat: **Andhra Pradesh** (Nagarjunsagar-Srisailam) 1× 2011
- **Bandipur-Karnataka, Melghat-Maharashtra, Buxa-West Bengal, Pakke-Arunachal** 1× 2008
- **Bandipur-Karnataka, Manas-Assam, Namdapha-Arunachal, Similipal-Odisha** 1× 2006
- Mizoram's tiger reserve: **Dampa** 1× 2012
- Wrong: **Buxa-Bihar** (West Bengal); Buxa = **West Bengal** 2× 2010, 2003
- Lies in two States: **Pench** (MP, Maharashtra) 1× 2008
- **Sariska**: **Rajasthan** 1× 2012
- Not in MP: **Amangarh** (UP) 1× 2021
- Newly declared (2015): **Rajaji** 1× 2015
- First to use drones: **Bandipur** 1× 2014

TIGER CENSUS (CENTRAL ESTIMATE)



TRAP ALERT

- **Buxa** = West Bengal; **Dampa** = Mizoram; **Amangarh** = UP.
- **Nokrek** = Meghalaya; **Manas** = Assam.
- **Chinnar** = Kerala, not Kashmir.

UP LINK

UP's tiger reserves: **Dudhwa, Pilibhit** (won TX2 award 2020 for doubling tigers), **Amangarh** (Bijnor) and **Ranipur** (Chitrakoot, 2022). The **National Chambal Sanctuary** covers UP, MP and Rajasthan; **Katarniaghat** guards gharials.

LAST-MINUTE 10

1. Nilgiri = first biosphere reserve (1986)
2. Nokrek = Meghalaya (Garo Hills); Manas = Assam
3. Cold Desert = Himachal; Dihang-Dibang = Arunachal
4. Gulf of Mannar = largest UNESCO-listed
5. Project Tiger **1973**; Tiger Day **29 July**
6. 2022 census **3,682**, ~75% of world
7. Nagarjunsagar-Srisailam = largest reserve
8. Buxa = WB; Dampa = Mizoram; Pakke = Arunachal
9. Pench = MP + Maharashtra; Amangarh = UP
10. Chambal sanctuary: MP, Rajasthan, UP

Chapter 13

Non-Conventional Energy

Built from 23 UPPSC questions (2000–2025). Focus: solar energy, India's energy mix, solar firsts, ISA, biofuels, hydrogen.

ASKED MOST · REVISE FIRST

- ■ **Solar** = most eco-friendly, no air pollution, greatest renewable potential in India **5×** 2004, 2006, 2011, 2015, 2018
- ■ **Hydrogen** = least polluting fuel **4×** 2010, 2013, 2015, 2017
- ■ Solar firsts: **Cochin** airport; **Diu** first 100% solar UT **3×** 2017, 2017, 2019
- • **Jatropha** = biodiesel crop **1×** 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

13.1 Sources

- Solar radiation's main role: **water cycle** **2×** 2000, 2004
- Ultimate source of fossil-fuel energy: **Sun** **1×** 2016
- Non-renewable: **petroleum** **1×** 2021
- **Minerals-exhaustible, solar-inexhaustible, biogas-methane and CO2, deforestation-rainfall** **1×** 2015
- Not biomass energy: **nuclear** **1×** 2022
- Used most in organic form: **solar energy** (via plants) **1×** 2004

13.2 India's Mix and Solar

- Second source after fossil fuels: **solar** **1×** 2021
- Greatest renewable potential: **solar** **1×** 2011
- Best for sustainable power (key): **hydro-electricity** **1×** 2012
- ■ Most eco-friendly: **solar cell** **3×** 2004, 2006, 2015
- No air pollution: **solar** **1×** 2018
- First fully solar airport: **Kochi** **1×** 2017
- First 100% solar UT: **Diu** **2×** 2017, 2019

Installed power (2025)	Share
Fossil	~48-50%
Solar	~26%
Hydro	~11%
Wind	~11%
Nuclear	~1.7%
Target 2030	500 GW non-fossil

13.3 International

- Decade of Sustainable Energy for All: **United Nations** **1×** 2017
- ISA: World Solar Bank and OSOWOG under it (true); 38 members (false; over 100) (only 2) **1×** 2025
- Largest offshore wind farm (2024): disputed (*) (UK's Hornsea 2 is largest) **1×** 2023

13.4 Biofuels and Others

- Not correct: biofuel is **cost-effective** **1×** 2014
- Biodiesel crop: **jatropha** **1×** 2015
- Green fuel source: karanja/ethanol (*) **1×** 2010
- ■ Least polluting fuel: **hydrogen** **4×** 2010, 2013, 2015, 2017
- First 100% organic state: **Sikkim** **1×** 2019

- Nuclear projects must study **all** listed effects 1× 2013
- No geothermal sources: **Ganga delta** 1× 2016

INDIA'S CLEAN-FUEL MISSIONS

E20 ethanol
20% blend nationwide (2025)
↓
SATAT / GOBARdhan
Compressed biogas
↓
Green Hydrogen Mission (2023)
5 Mt/year by 2030
↓
PM Surya Ghar
Rooftop solar
↓
PM-KUSUM
Solar pumps for farmers

TRAP ALERT

- **Diu** = first solar UT; **Cochin** = first solar airport.
- **ISA** has 100+ members; HQ Gurugram.
- **Hydrogen** burns to water; least polluting.

UP LINK

UP's solar push: **Bundelkhand** solar parks (Jhansi, Lalitpur, Jalaun), **Ayodhya** as a model solar city, and UP Solar Policy 2022 targeting 22 GW; **Rampura** (2010) is the first solar village.

LAST-MINUTE 10

1. Sun = ultimate energy source
2. Solar = most eco-friendly, top potential
3. Solar is 2nd after fossil fuels
4. Kochi = first solar airport; Diu = first solar UT
5. Hydrogen = least polluting
6. Jatropha, karanja = biodiesel
7. Sikkim = first organic state
8. ISA: Gurugram; World Solar Bank, OSOWOG
9. Ganga delta: no geothermal
10. Nuclear energy is not biomass

Chapter 14

Pollution: Air, Acid Rain, Water, Sound and Soil

Built from 75 UPPSC questions (2000–2025). Focus: biodegradable pollutants, primary and secondary pollutants, CO, lichens, acid rain, arsenic and fluoride, BOD, decibels, Bhopal.

ASKED MOST · REVISE FIRST

- **Bhopal gas**: **methyl isocyanate** (MIC), 1984 6× 2001, 2002, 2008, 2014, 2017, 2022
- **Lichens** = bio-indicators of air pollution 7× 2010, 2012, 2013, 2015, 2019, 2021, 2025
- **Acid rain**: **SO2 and nitrogen oxides**; Norway and Canada hit hardest 10× 2000, 2001, 2001, 2002, 2003, 2007, 2010, 2011, 2013, 2024
- **Carbon monoxide** binds haemoglobin; most dangerous 4× 2006, 2013, 2014, 2016
- **Fly ash** comes from **burning coal** 3× 2004, 2004, 2021
- **BOD** measures water pollution; high BOD = sewage 4× 2007, 2011, 2021, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Basic Ideas

- Human-made pollutants: **anthropogenic**; human-modified environment: anthropogenic 2× 2005, 2019
- Biodegradable: **sewage**; **rubber** (of options) 3× 2014, 2016, 2014
- **DDT**: **non-biodegradable** 1× 2022
- Not biodegradable: **pesticide** 1× 2023
- Polythene lasts because it is a **polymer** 1× 2007
- Takes longest to decay: **plastic bag** 1× 2008

14.2 Air Pollution

- Burning coal, petrol, diesel: **air pollution** 1× 2011
- Fossil-fuel air pollutant: **SO2** 1× 2011
- Not a water pollutant: **SO2** 1× 2011
- Not secondary: **SO2** (primary) 1× 2014

- Secondary pollutant: **smog** 1× 2018
- Eye irritant in smoke: **PAN** (peroxyacetyl nitrate) 1× 2018
- Photochemical smog always has **ozone** 1× 2022
- Common atmospheric pollutant: **CO** 1× 2006
- Affects blood, can kill: **CO**; binds haemoglobin faster than O2; most dangerous 3× 2013, 2016, 2014
- Fly ash: **burning coal** 3× 2004, 2004, 2021
- Lichens: good indicators (true); grow well in pollution (false) (only 1) 1× 2025
- Air-pollution indicator: **lichen**; bio-indicator 6× 2010, 2013, 2021, 2012, 2015, 2019
- Most polluted city (then): **Ankleshwar** 1× 2009
- Greenpeace 2020 most polluted: **Jharia** 1× 2019
- Cleans polluted air: **rain** 1× 2013
- Delhi winter pollution peaks (true); more vehicle combustion in winter (false) 1× 2004
- Cigarette smoke: **CO and benzene** (per key) 1× 2015
- Not an air-pollution control device: **cyclone divider** 1× 2013
- Bio-decomposer: free to farmers; made of fungi (both) 1× 2025
- **NCAP**: launched **2019** 1× 2020
- **PRANA** portal: monitors **NCAP** 1× 2023

Primary	Secondary
SO2, CO, NOx	Ground-level ozone
Hydrocarbons	PAN
Particulates, lead	Photochemical smog, acid rain

14.3 Acid Rain

- CO2, SO2, NO2 dissolve in rain; polluted air (R explains A) 1× 2024
- Acid rain: **nitrogen oxides and SO2**; clouds react with SO2 5× 2001, 2007, 2010, 2013, 2001
- Main gas: **SO2** 2× 2002, 2003
- Suffer acid rain: **Canada and Norway** 1× 2011
- Most acid rain: **Norway** 1× 2000

14.4 Water Pollution

- Non-biotic groundwater pollutant: **arsenic** 1× 2012
- Arsenic highest in **Kolkata**; **West Bengal**; natural arsenic in **Bangladesh** 3× 2005, 2015, 2022
- **Knock-knee**: **fluoride** 1× 2016
- Ground Water Quality Report 2024: **Central Ground Water Board** (only 3) 1× 2025
- Worst chemical effluents: **leather** industry 1× 2005
- Not air pollution: **eutrophication** 1× 2013
- Eutrophication: nutrient enrichment; depletes oxygen (both) 1× 2019
- High BOD: **sewage** contamination 1× 2007
- River pollution measured by **dissolved oxygen** 2× 2011, 2021
- BOD: **pollution assay** of water 1× 2023
- Ganga BOD highest: **Kanpur to Allahabad** 1× 2008
- "Biological desert": **Damodar** 1× 2012
- UP's "biological disaster" river: **Yamuna** 1× 2018
- Largest solid waste per head: **Mumbai** 1× 2010

14.5 Noise and Soil

- Noise unit: **decibel** 1× 2017
- Loudest: **jet flight** 1× 2003
- Rustling leaves: **20 dB** 1× 2018
- **Green muffler**: **noise pollution** (tree belts) 2× 2014, 2017
- Overuse of fertilisers causes **all** listed harms 1× 2016
- Soil pollution worse; fertilisers enter the food chain (R explains A) 1× 2010
- Crop rotation helps in **all** listed ways 1× 2016

14.6 Pollutants and Diseases

- Bhopal (1984): **methyl isocyanate** 6× 2001, 2002, 2008, 2014, 2017, 2022
- Wrong: **fluoride-indigestion** (fluorosis) 1× 2017
- Most toxic metal in exhaust: **lead** 2× 2006, 2009
- **Asbestos-lung, lead-brain, mercury-stomach, CO-bloodstream** 1× 2012
- **CO-liver and kidney, NOx-cancer, soil particles-breathing, lead-nervous system** (per key) 1× 2008
- **Acid rain-SO2, smog-unsaturated hydrocarbons, haemoglobin-CO, ozone depletion-CFCs** 1× 2019

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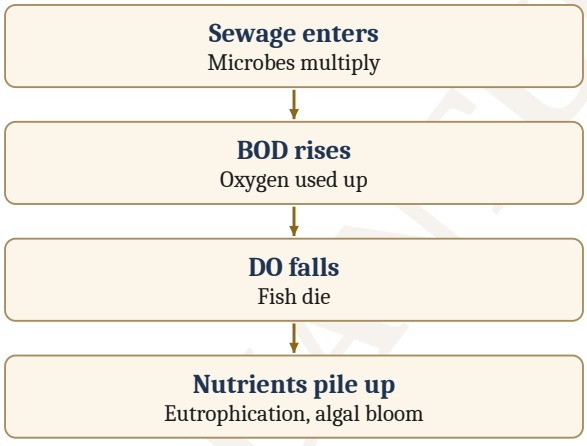
- Radiation: **heredity change, blood circulation, cancer** (1, 3, 4) 1× 2009
- RSPM size: **less than 5 µm** (per key; PM10 and PM2.5 today) 1× 2017

Pollutant	Disease
Mercury	Minamata
Cadmium	Itai-itai
Nitrate	Blue baby
Fluoride	Fluorosis, knock-knee
Arsenic	Black-foot, skin cancer
Asbestos	Asbestosis
Silica	Silicosis
CO	Blocks haemoglobin

14.7 Vehicle Norms and Miscellany

- Euro norms cover **all** listed gases 1× 2013
- Euro-II diesel sulphur: **0.05% or less** 1× 2013
- Cold wave of January 2012: **La Nina** 1× 2010
- Mustard seed adulterant: **argemone** seed 1× 2010
- **Incinerators**: **burning waste** 1× 2018

READING WATER POLLUTION



TRAP ALERT

- **SO2 is primary**; smog, ozone, PAN are secondary.
- High **BOD**, low **DO** = polluted.
- **Eutrophication** is water pollution.
- Lichens **vanish** where air is dirty.

UP LINK

UP links: **Kanpur** tanneries push Ganga BOD highest between **Kanpur and Prayagraj**; the **Yamuna** is called a "biological disaster"; UP cities (**Ghaziabad, Noida, Lucknow**) are NCAP non-attainment cities; Sonbhadra's **Singrauli** belt has fly-ash and mercury problems.

LAST-MINUTE 10

1. Bhopal 1984 = **MIC**
2. Lichens = air-pollution indicator
3. Acid rain = SO2 + NOx; Norway, Canada
4. CO binds haemoglobin
5. Fly ash = coal; SO2 = primary pollutant
6. PAN = eye irritant; smog has ozone
7. Arsenic = West Bengal, Bangladesh; fluoride = knock-knee
8. BOD high = sewage; Damodar = biological desert
9. Noise in **decibels**; leaves 20 dB; green muffler
10. NCAP **2019**; PRANA portal

Chapter 15

Water Conservation and River Cleaning

Built from 13 UPPSC questions (2002–2023). Focus: World Water Day, Water Act and Cess, chlorine, lake conservation, Ganga and Yamuna action plans.

ASKED MOST · REVISE FIRST

- **Ganga Action Plan** and **Central Ganga Authority**: **1985**; aim: pollution-free water 2× 2014, 2022
- **Water Act 1974**; Water Cess **1977** 2× 2013, 2023
- **World Water Day**: **22 March** 1× 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Water as a Resource

- World Water Day: **22 March** 1× 2011
- Not a biotic resource: **freshwater** 1× 2016

- CGWA under the **Environment Protection Act**; India has the largest groundwater-irrigated area (2 and 3) 1× 2005

15.2 Laws

- Water (Prevention and Control of Pollution) Act: **1974** 1× 2023
- Water Cess enforced: **1977** 1× 2013

15.3 Treatment

- Drinking-water purification gas: **chlorine** 1× 2013
- Disinfectants: **all** listed (chlorine, ozone, UV) 1× 2016

15.4 Lakes and Rivers

- Urban lakes under NLCP: **Bhoj, Sukhna, Pichola** (not Chilika) 1× 2002
- Added to NLCP (2010): **Bhimtal** 1× 2010
- Central Ganga Authority: **1985** 1× 2014
- Ganga Action Plan aim: **pollution-free water only** 1× 2022
- **NGRBA**: **February 2009** 1× 2010
- Yamuna Action Plan: **1993** 1× 2016

CLEANING THE GANGA

1985
Central Ganga Authority; Ganga Action Plan I

↓

1993
GAP-II; Yamuna and Gomti Action Plans

↓

2008
Ganga declared National River

↓

2009
NGRBA (Feb)

↓

2014
Namami Gange (NMCG)

↓

2016
National Ganga Council

Day	Date
World Wetlands Day	2 February
World Water Day	22 March
World Environment Day	5 June
Ozone Day	16 September

TRAP ALERT

- **Water Act 1974**; **Cess Act 1977**; Air Act 1981.
- **NGRBA** = 2009; **Namami Gange** = 2014.
- **Chilika** is not an urban lake in the NLCP list.

UP LINK

The Ganga and its tributaries (**Yamuna, Gomti, Ramganga, Kali**) flow through UP; Kanpur's tanneries and Varanasi's ghats are the core of GAP and **Namami Gange**. The **Gomti Action Plan** (1993) covers Lucknow.

LAST-MINUTE 10

1. World Water Day **22 March**
2. Water Act **1974**; Cess **1977**
3. CGWA under **EPA 1986**
4. Chlorine purifies drinking water
5. Bhoj, Sukhna, Pichola = NLCP urban lakes
6. Ganga Action Plan & CGA **1985**
7. Yamuna Action Plan **1993**
8. NGRBA **Feb 2009**
9. Namami Gange **2014**
10. Freshwater = abiotic resource

Chapter 16

Organisations, Awards, Days and Miscellaneous

Built from 54 UPPSC questions (2000–2025). Focus: institute-city matching, UN body headquarters, environment awards, important days, Mission LiFE, environmental movements.

ASKED MOST · REVISE FIRST

- **NEERI: Nagpur** (1958) 6× 2004, 2005, 2011, 2012, 2014, 2017
- **World Environment Day: 5 June** 5× 2004, 2011, 2012, 2014, 2025
- Day matching: **Wetlands 2 Feb, Wildlife 3 March, Water 22 March, Health 7 April, Heritage 18 April, Biodiversity 22 May, No Tobacco 31 May, Population 11 July, Tiger 29 July, Ozone 16 Sept, Soil 5 Dec** 7× 2008, 2017, 2018, 2022, 2023, 2023, 2025
- HQs: **UNEP-Nairobi, WMO-Geneva, IMO-London, Greenpeace-Amsterdam** 4× 2005, 2010, 2010, 2016
- **CAZRI: Jodhpur** 2× 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Organisations

- NEERI: **Nagpur**; set up **1958** 6× 2004, 2005, 2011, 2012, 2014, 2017
- **WMO: Geneva** 1× 2010
- **UNEP: Nairobi** 1× 2010
- **Greenpeace International: Amsterdam** 1× 2005
- **IMO: London** 1× 2016
- **Nutrition-Hyderabad, Wildlife Institute-Dehradun, Ayurveda-Jaipur, Naturopathy-Pune** 1× 2004
- Wrong: **IIFM-Kolkata** (it is **Bhopal**) 1× 2012
- ICFRE pairs: Arid-Jodhpur, Rain-Jorhat, Tropical-Jabalpur right; Himalayan is **Shimla**, not Dehradun (3 correct) 1× 2023
- **CAZRI: Jodhpur** 2× 2008, 2010
- Indian Institute of Ecology and Environment: **New Delhi** 2× 2014, 2016
- ENVIS centre on population and environment: **IIPS Mumbai** 1× 2017
- India's working Antarctic stations: **Bharati and Maitri** 1× 2021
- First National Centre for Marine Biodiversity: **Jamnagar** 2× 2018, 2023

Body	HQ
UNEP	Nairobi
WMO, IPCC	Geneva
IMO	London
IUCN, WWF	Gland
Greenpeace	Amsterdam
UNFCCC	Bonn
ISA	Gurugram

16.2 Awards

- **Indira Gandhi Paryavaran Puraskar**: contribution to the **environment** 1× 2004
- **Rajiv Gandhi Environment Award: clean technology** 1× 2006
- **Borlaug Award: agriculture**; Borlaug's Nobel: **Peace** (1970) 2× 2001, 2018
- **Tyler Prize: environmental protection** 1× 2004
- World Biodiversity Day: **22 May** 1× 2018
- **Global 500: environment conservation** 1× 2019
- **Champions of the Earth** 2018: **Narendra Modi** (with Macron) 1× 2017

16.3 Days

- World Habitat Day: first Monday of October (*) 1× 2000
- Anti-Tobacco Day: **31 May**; "5 May" is wrong 2× 2008, 2017
- **19 November: World Toilet Day** 1× 2015
- World Environment Day: **5 June** 4× 2004, 2011, 2012, 2014
- WED 2018 theme: **Beat Plastic Pollution** (India host) 1× 2017
- World Wetlands Day: **2 February** 1× 2023
- Wrong: **World Wildlife Day-22 March** (it is **3 March**) 1× 2025
- **Population-11 July, Wildlife-3 March, Ozone-16 September, Health-7 April** 1× 2023
- **18 April-Heritage, 22 May-Biodiversity, 29 July-Tiger, 5 December-Soil** 1× 2022
- National Dolphin Day: **5 October** 1× 2022
- Cleanliness Day on the last Saturday monthly: **Sierra Leone** 2× 2002, 2003

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ENVIRONMENT CALENDAR



16.4 Miscellany

- **Mission LiFE**: launched June 2022 (true); proposed at COP25 (false; **COP26**) (only 1) 1× 2025
- Eco-friendly lifestyle initiative: **Mission LiFE** 1× 2023
- India's World Heritage sites by Aug 2010: **28** 1× 2009
- Newly listed (2002): **Mahabodhi Temple** 1× 2002
- Wrong: **Silent Valley-Baba Amte** (Narmada) 1× 2009
- **Appiko**: Karnataka (key also Kerala) 1× 2025
- "Rally for the Valley": **resettlement** (Narmada) 1× 2003
- **Mitti Bachao**: **Hoshangabad**, MP 1× 2020
- Land degradation neutrality by **2030** 1× 2019
- Anthropogenic biome: **cropland** 1× 2018
- Collects and releases water: **watershed** 1× 2022
- Not eco-friendly: **eucalyptus** 1× 2022
- Dust-pollution tree: **Sita Ashok** 1× 2007
- Evergreen fruit plant: **loquat** 1× 2008
- Mega-diverse country: **Australia** 1× 2015
- **SODAR**: **Kaiga and Kalpakkam** 1× 2004
- **Skin cancer-UV, noise-decibel, global warming-CO2, ozone hole-CFC** 1× 2015
- "Father of Economic Ecology" (UNEP): **M.S. Swaminathan** 1× 2007
- Peninsular glaciation: **Pleistocene** 1× 2010
- India's largest charkha: **Noida**, made of **plastic waste** (both) 1× 2019

Movement	Leader / place
Chipko (1973)	Bahuguna, Bhatt; Chamoli (then UP)
Appiko (1983)	Panduranga Hegde; Karnataka
Silent Valley	KSSP, Sugathakumari; Kerala
Narmada Bachao	Medha Patkar, Baba Amte
Bishnoi (1730)	Amrita Devi; Khejarli
Mitti Bachao (1977)	Hoshangabad, MP

TRAP ALERT

- **UNEP** = Nairobi, not Geneva.
- **IIFM** = Bhopal; **Himalayan FRI** = Shimla.
- **Wildlife Day 3 March**; Water Day 22 March.
- **Borlaug** won the Nobel **Peace** Prize.

UP LINK

UP links: **Chipko** began in Chamoli when it was part of UP; ICFRE's **Forest Research Centre for Eco-Rehabilitation** is at **Prayagraj**; India's largest **charkha** of plastic waste stands in **Noida**.

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LAST-MINUTE 10

1. NEERI = **Nagpur** (1958); CAZRI = **Jodhpur**
2. UNEP = Nairobi; WMO = Geneva; IMO = London
3. Greenpeace = Amsterdam; IIFM = Bhopal
4. NCMB = **Jamnagar**; Antarctic: Bharati, Maitri
5. Environment Day **5 June**; Ozone Day **16 Sept**
6. Wildlife Day **3 March**; Tobacco Day **31 May**
7. Dolphin Day **5 Oct**; Toilet Day **19 Nov**
8. Tyler Prize = environment; Champions of the Earth 2018 = Modi
9. Mission LiFE = June 2022 (proposed COP26)
10. Swaminathan = Father of Economic Ecology

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TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

Economics and Social Development

PYQ-Based Revision Pointers

Chapters 1 to 27

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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Chapter 1

Nature of the Economy and National Income

Built from 46 UPPSC questions (1990–2024). Focus: mixed economy, growth vs development, who estimates national income, growth records, savings.

ASKED MOST · REVISE FIRST

- India = **mixed economy** (public and private sectors co-exist) **8×** 1990, 1990, 1999, 2006, 2006, 2013, 2013, 2016
- Highest growth year (in key): **2006-07** **11×** 1997, 1999, 2000, 2001, 2002, 2003, 2005, 2006, 2008, 2013, 2014
- National income estimated by **CSO**, now the **NSO** (2019) **7×** 1995, 2004, 2006, 2008, 2010, 2012, 2023
- First per capita income estimate: **Dadabhai Naoroji**, Rs. 20 (1867-68) **2×** 2007, 2013
- **Hindu rate of growth** = growth of **national income** (~3.5%) **3×** 1996, 2004, 2006
- **Household sector** saves the most **3×** 1998, 2001, 2004

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

1.1 Nature of the Economy

- Mixed economy: **public and private sectors co-exist** **4×** 1990, 1999, 2006, 2013
- India: **mixed economy** **4×** 1990, 2006, 2013, 2016
- India underdeveloped and mixed (true); Keynes "father of economics" (false; Adam Smith) **1×** 2024
- Underdeveloped economy: **low per capita income, low capital formation** **1×** 2017
- Developing signs: **agriculture, disguised unemployment, poor human capital** **1×** 2017
- India: a **developing** economy **1×** 2011
- Features: **agriculture dominant, low per capita income, mass unemployment** **1×** 2004
- Not a feature: **lack of natural resources** **1×** 2004
- Not a second-generation reform: **legal system reforms** **1×** 2009
- Indian model serves **state and person both** **1×** 2013
- Not an economic activity: **voluntary social service** **1×** 2023

1.2 Growth vs Development

- Prices and money income both double: real income **unchanged** **1×** 2004
- Amartya Sen: **people-oriented**, security for the poorest, safeguards on global integration **1×** 1999
- "Plan for people, not commodities": **Amartya Sen** **1×** 2004
- Hindu rate of growth: **national income** growth **3×** 1996, 2004, 2006
- Labour's GNP share falls because **wages lag prices** **1×** 2008
- **Development-constructive change, growth-GDP, sustainable-environment, quality of life-health** **1×** 2010
- 1991 NEP strategy: **all** (LPG) **1×** 2015
- Not a growth factor: **technocrats and bureaucrats** **1×** 2021
- Best growth measure: **per capita product** **1×** 2013
- Development = better living; per capita income alone not enough (R explains A) **1×** 2024
- Standard of living: **per capita income** **1×** 2013

Term	Meaning
GDP	Output within the country
GNP	GDP + net factor income from abroad
NNP	GNP - depreciation
National income	NNP at factor cost
Per capita income	National income ÷ population

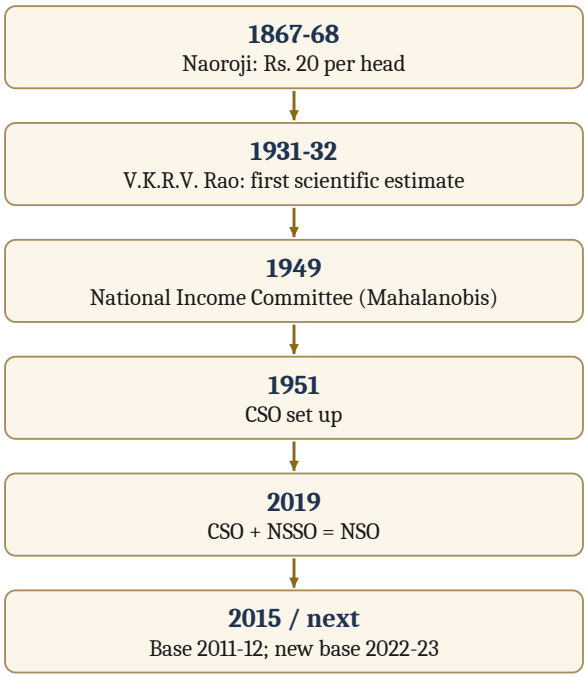
1.3 Who Measures

- Rs. 20 per capita (1867-68): **Dadabhai Naoroji** **2×** 2007, 2013
- National Income Committee (1949): **P.C. Mahalanobis** **1×** 2015
- Estimated by **CSO** (now **NSO**); **NSO** **7×** 1995, 2004, 2006, 2008, 2010, 2012, 2023
- Capital formation data: **CSO/NSO** (with RBI) (*) **1×** 2008
- New base year (2015): **2011-12** **1×** 2015

QUANTUM IAS & PCS

- Development needs many angles (true); govt focuses on micro issues (false) 1× 2015

MEASURING INDIA'S INCOME



1.4 Growth and States

- Per capita income 2009-10: about **Rs. 46,500** 2× 2011, 2013
- Fastest 1990s year: **1999-2000** 2× 2002, 2003
- Highest growth (key): **2006-07** 11× 1997, 1999, 2000, 2001, 2002, 2003, 2005, 2006, 2008, 2013, 2014
- RBI estimate for 2002-03: **5-5.5%** 1× 2002
- ADB forecast for 2005: **6.9%** 1× 2004
- Largest state economy: **Maharashtra** 1× 2023
- Per capita 1998-99: **Punjab, Maharashtra, Gujarat, UP** 1× 2004
- Lowest per capita income: **Bihar** 3× 1993, 2004, 2008
- Highest per capita (then): **Punjab** 1× 1994
- Highest GSDP growth 2010-11: **Bihar** 1× 2009

1.5 Savings

- Capital formation depends on savings and institutions; investment essential (both) 1× 2021
- Slow per capita growth: **population, prices, slow farm and industry** (1, 2, 3) 1× 2001
- Highest savings rate of the 2000s: **2007-08** 1× 2009
- Savings ratio 2006-07: **above 32%** of GDP 1× 2008
- Biggest saver: **household sector** 2× 2001, 2004
- Household savings mostly in **physical assets** 1× 2009
- Savings order: **household, private corporate, public enterprises, government** 1× 1998

TRAP ALERT

- **Naoroji** = first estimate; **Mahalanobis** = 1949 committee.
- **NSO** (2019) now estimates national income.
- **Hindu rate** (Raj Krishna) = slow national income growth.

UP LINK

UP is India's **third-largest state economy** (after Maharashtra and Tamil Nadu in recent data) and aims at a **USD 1 trillion** economy; its per capita income is still well below the national average.

LAST-MINUTE 10

1. India = mixed, developing economy
2. Growth = GDP; development = growth + change
3. Per capita income = living standard
4. Naoroji: Rs. 20 (1867-68)
5. 1949 committee: **Mahalanobis**
6. NSO (2019) estimates income; base **2011-12**
7. Hindu rate = national income (~3.5%)
8. Maharashtra = largest state economy; Bihar = lowest per head
9. Households save most; mostly physical assets
10. Amartya Sen: plan for people

Chapter 2

Sustainable Development and Human Capital

Built from 29 UPPSC questions (2008–2025). Focus: SDG numbers, SDG India Index, Limits to Growth and Brundtland, human capital, inclusive growth.

ASKED MOST · REVISE FIRST

- SDG numbers: **1-poverty, 3-health, 4-education, 6-water, 10-inequality, 13-climate, 14-life below water, 15-life on land** 3× 2019, 2025, one-liner
- **NITI Aayog** = nodal body for SDGs; **Kerala** tops SDG India Index 3× 2019, 2019, 2021
- Sustainable development: use resources now while **conserving for the future** 3× 2018, 2023, 2023
- Skills, education, health build **human capital** 3× 2015, 2017, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 The SDGs

- Not an SDG: **space research** 1× 2023
- SDG 4 (education): **free compulsory schooling, better infrastructure** (1 and 2) 1× 2024
- **Goal 1-end poverty, Goal 3-health, Goal 4-education, Goal 6-water** 1× 2025
- Water for all: **SDG 6** 1× 2019
- **SDG 10-reduced inequalities, 13-climate action, 14-life below water, 15-life on land** 1× one-liner
- UN Secretary-General presents an annual SDG report (only 1 true) 1× 2023

SDG	Goal
1 / 2	No poverty / Zero hunger
3 / 4	Health / Quality education
5 / 6	Gender equality / Clean water
7 / 8	Clean energy / Decent work
10 / 13	Reduced inequality / Climate action
14 / 15	Life below water / Life on land
17	Partnerships

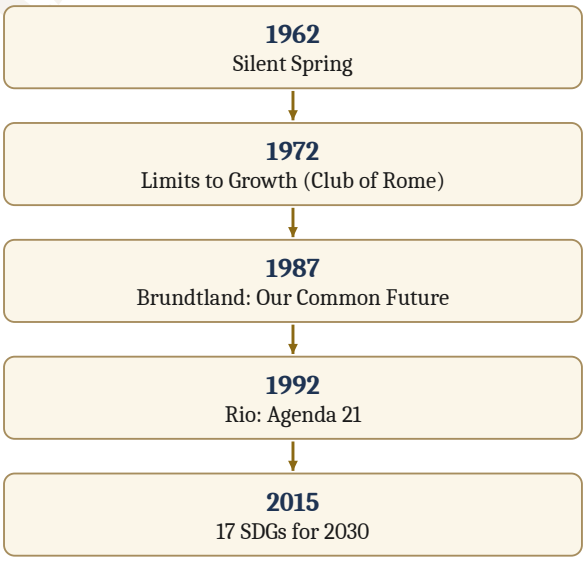
2.2 SDGs in India

- Nodal institution: **NITI Aayog** 1× 2019
- India's SDG Index rank 2017: **116th** 1× 2017
- UP in SDG India Index 2019-20: **Performer** (now Front Runner) 1× 2020
- Not in top five (2020-21): **Gujarat** 2× 2019, 2021
- First in SDG India Index 2019: **Kerala** 1× 2019
- Not in top three of SDG Urban Index 2021-22: **Indore** (Shimla, Coimbatore, Chandigarh led) 1× 2021
- UP's SDG progress report: **Department of Planning** 1× 2025
- India's success matters globally; one-sixth of humanity (R explains A) 1× 2025

2.3 History of the Idea

- Brundtland popularised SD (true); it is "Limits to Growth" (false; **Our Common Future**) 1× 2024
- "Limits to Growth": **Club of Rome** (1972) 1× 2019
- Environmental Kuznets Curve: **inverted U** 1× 2019
- Balancing use and conservation: **sustainable development** 3× 2018, 2023, 2023
- **Neemrana** model: **Rajasthan** 1× 2010
- Sustainable tourism: manage tourism with cultural and ecological integrity 1× 2020

FROM LIMITS TO GOALS



2.4 Human Capital

- Skill programmes raise **human capital** 1× 2017
- No direct effect on human capital: **irrigation** 1× 2017
- Human capital pays back; education and health raise productivity (R explains A) 1× 2025
- More human capital: **all** listed gains 1× 2015
- Inclusive growth needs **all** listed measures 1× 2008

QUANTUM IAS & PCS

- Inclusive growth not ensured by **high national income growth** alone 1× 2013
- First Economic Survey chapter on SD and climate: **2011-12** 1× 2015

TRAP ALERT

- **Limits to Growth** = Club of Rome; **Our Common Future** = Brundtland.
- SDG **14** = water life; **15** = land life.
- EKC is an **inverted U**.

UP LINK

UP moved from **Performer** (2019-20) to **Front Runner** in the SDG India Index 2023-24; its **Planning Department** prepares the State's SDG indicator reports.

LAST-MINUTE 10

1. 17 SDGs, 169 targets, 2030
2. SDG 1 poverty; 4 education; 6 water
3. SDG 10 inequality; 13 climate; 14 sea; 15 land
4. NITI Aayog = nodal; Kerala tops index
5. Club of Rome = Limits to Growth (1972)
6. Brundtland = Our Common Future (1987)
7. EKC = inverted U
8. Human capital = education, health, skills
9. Irrigation is not human capital
10. UP: Performer to Front Runner

Chapter 3

Agriculture I: Structure, Land Reforms and Production

Built from 52 UPPSC questions (1996–2024). Focus: land reforms, farm sizes, agriculture's share, leading crop states, food self-sufficiency, early schemes.

ASKED MOST · REVISE FIRST

- Agriculture's share of the economy is **decreasing** (now about 18% of GVA) 8× 1996, 1998, 1999, 2004, 2010, 2012, 2013, 2017
- Land reforms: **abolition of intermediaries, tenancy reform, ceilings**; multiple cropping is not one 3× 2015, 2017, 2019
- **National Horticulture Mission**: **May 2005** (Tenth Plan); UP is covered 4× 2009, 2010, 2010, 2016
- **Kisan Diwas**: **23 December** (Charan Singh) 1× 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

3.1 Basic Ideas

- **Kisan Divas**: **23 December** 1× 2017
- Genetic industry includes **agriculture** 1× 2016
- **Cereal yield**: output per unit harvested area (only 1) 1× 2024
- **Bullock capitalist**: farmers with some resources but **not rich** 1× 2017
- Did not favour co-operative farming: **Charan Singh** 1× 1996

3.2 Land Reforms and Holdings

- Not a land reform: **multiple-cropping scheme** 1× 2015
- Tenancy reform: **rent regulation, security of tenure, ownership** (all three) 1× 2019
- 1-2 hectare holder: **small farmer** 1× 2021
- Agrarian reform: **abolition of intermediaries, tenancy reform, land ceilings** 1× 2017

Holding	Area
Marginal	Below 1 ha (~68%)
Small	1-2 ha
Semi-medium	2-4 ha
Medium	4-10 ha
Large	Over 10 ha
Average	1.08 ha

3.3 Agriculture in the Economy

- Slow farm growth: **rural poverty** 1× 2016

- Agriculture's income share: depends on year (about 18% now) (*) 5× 1996, 1998, 2010, 2012, 2013
- Agriculture's contribution is **decreasing** 3× 1999, 2004, 2017
- Negative farm growth: **2014-15**; **Third Plan** 2× 2017, 2004
- Rural households in farming (NSSO 2014): **57.8%** 1× 2016
- Cropping intensity: about **155%** 1× 2012
- Not a cause of low productivity: **co-operative farming** 2× 2002, 2007
- Agriculture matters in **all** listed ways 1× 2007
- Agricultural income tax: **State Governments** 1× 2009
- Most UP workers: **agriculture** 1× 2012

3.4 Crops and States

- Largest soybean area: **Madhya Pradesh** 1× 2012
- Highest cotton yield (2014): **Australia** 1× 2015
- Black soil suits **cotton** 1× 2008
- Critical wheat irrigation: **CRI stage** 1× 2005
- Wheat order: now MP, Haryana, Rajasthan, Bihar among options (*) 1× 2012
- Net sown area: about **14 crore hectares** 1× 2016
- Largest vegetable producer: **China** 1× 2015
- Onion exports 2015-16: about **12 lakh tonnes** 1× 2016
- **Mango-UP, Litchi-Bihar, Coconut-Kerala, Apple-J&K** 1× 2021
- Wrong: **coffee-Mongolia** 1× 2021
- Coffee order (2016): **Brazil, Vietnam, Colombia, Indonesia** 1× 2018
- Cash crops: **sugarcane, cotton, rapeseed** (and potato) 1× 2023
- Niger seed oil: **37-47%** 1× 2021
- Mulberry silk: **Karnataka** 1× 2017

Crop	Leading state
Wheat, sugarcane, mango	Uttar Pradesh
Rice	West Bengal, UP
Soybean	Madhya Pradesh
Cotton	Gujarat
Litchi	Bihar
Apple	J&K
Mulberry silk	Karnataka

3.5 Food and Trade

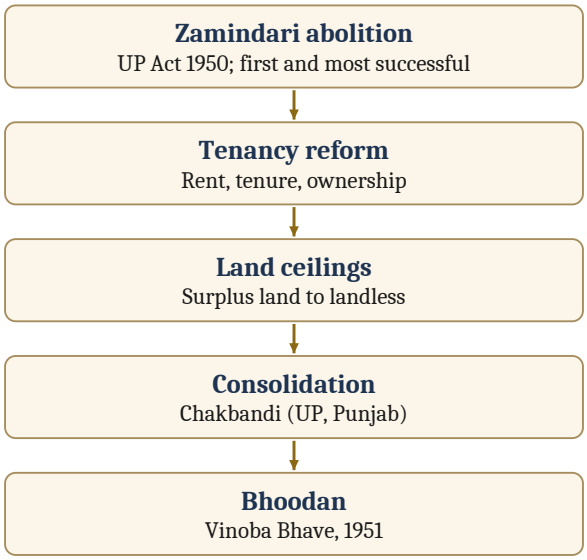
- Foodgrain fell in **2004-05** 1× 2008
- Economy stagnant? unclear (*) 1× 2001
- Near self-sufficiency; no bulk imports (R explains A) 2× 2002, 2003
- National but not household food security: **stock exists, access lacking** 1× 2008
- Reforms needed; many farmers rely on monsoon (both true, R not the explanation) 1× 2023
- Largest pulse import 2015-16: **lentil** 1× 2016
- Top farm export earner: **basmati rice**; most valuable: **rice** 2× 2010, 2010
- Export mango: **Alphonso** 1× 2012

3.6 Early Schemes

- **Kisan Bahi** (UP): **1992** 1× 2010
- RKVY: wrong that aid is an **80% grant** (it is 100%) 2× 2009, 2010
- Sengupta and Sdasyuk: **60** micro regions 1× 2017
- NHM: **Tenth Plan**; **May 2005**; covers **UP** 3× 2009, 2016, 2010
- Not an NHM aim (key): **financial assistance** 1× 2010
- Krishi Shramik Samajik Suraksha: **pension and insurance** 1× 2001
- Small Farmers Development Agency: **1971** (no option, *) 1× 2014

QUANTUM IAS & PCS

LAND REFORM STEPS



TRAP ALERT

- **Small** = 1-2 ha; **marginal** = below 1 ha.
- Agricultural income is taxed by **States**.
- RKVY grant is **100%**, not 80%.

UP LINK

UP leads India in **wheat, sugarcane, potato, mango and milk**; its **Zamindari Abolition Act (1950)** was the model land reform, **Kisan Bahi** began in 1992, and most UP workers are in farming.

LAST-MINUTE 10

1. Kisan Diwas **23 Dec**
2. Land reform: zamindari, tenancy, ceilings
3. Small farmer **1-2 ha**
4. Farm share falling (~18% of GVA)
5. Negative farm growth: Third Plan, **2014-15**
6. 57.8% rural households farm (2014)
7. MP = soybean; Australia = top cotton yield
8. Mango UP; Litchi Bihar; Apple J&K
9. NHM **May 2005**; Kisan Bahi **1992**
10. Basmati = top farm export

Chapter 4

Agriculture II: Food Security, Revolutions, Prices and Credit

Built from 70 UPPSC questions (1992–2025). Focus: NFSA and NFSM, colour revolutions, ICAR and institutes, crop insurance, NAFED and MSP, farm credit and KCC.

ASKED MOST · REVISE FIRST

- ■ **Kisan Credit Card** = **1998-99** 4× 2008, 2010, one-liner, one-liner
- ■ **Commercial banks** = largest source of farm credit; foreign banks the least 3× 2004, 2006, 2010
- • **NFSA 2013**: 75% rural, 50% urban; focus on women and children 2× 2019, 2024
- ■ Revolutions: **blue-fish, yellow-oilseeds, golden-horticulture, grey-fertilisers, black-petroleum; Operation Flood-milk** 5× 1992, 1997, 2007, 2012, 2022
- ■ **CACP** recommends MSP; **CCEA** approves FRP 3× 2002, 2019, one-liner
- • **NAFED** = apex agricultural marketing cooperative 2× 1995, 2007

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

4.1 Food Security

- Food security needed even in surplus years; starvation from unequal access (R explains A) 1× 2025
- Not a dimension: **accountability** 1× 2025
- Entitlement concept: **Amartya Sen** 1× 2025
- NFSA: 75% rural, 50% urban; nutrition for women and children (1 and 2); not enforced in 2010 1× 2024
- NFSA: coverage, nutrition, **eldest woman** head of household (all) 1× 2019
- First e-ration card: **Delhi** 1× 2021
- Not an NFSM aim: **credit facilities** 1× 2016
- NFSM crops: coarse cereals added later (*) 1× 2010

- NFSM began **rabi 2007-08**; covers **wheat, pulses, rice, coarse cereals** 2× 2008, 2013
- PDS price rise: to **cut subsidy burden** 1× 1995
- FCI's economic cost = MSP + bonus + **procurement incidentals and distribution cost** 1× 2019

4.2 Revolutions

- **Operation Flood**: **milk** 1× 1992
- Nandini Krishak Samriddhi (UP): Sahiwal, Gir, Tharparkar, Gangatiri only (true); experience rule (false) 1× 2023
- Largest fish producer: **China** 1× 2021
- **Blue revolution**: **fish** 1× 1997
- **Yellow revolution**: **oilseeds**; **rapeseed-mustard** 2× 2007, 2012
- **Golden-horticulture and honey, Grey-fertilisers, Yellow-oilseeds, Black-petroleum** 1× 2022
- Green Revolution gains: **Punjab, Haryana, western UP** 1× 2008
- Eco-friendly practice: **organic farming** 1× 2008
- NATP funded by the **World Bank** 1× 2002

Revolution	Product
Green	Foodgrains
White	Milk (Operation Flood, Kurien)
Blue	Fish
Yellow	Oilseeds
Golden	Horticulture, honey
Grey	Fertilisers
Black	Petroleum
Pink	Meat, prawns
Round	Potato

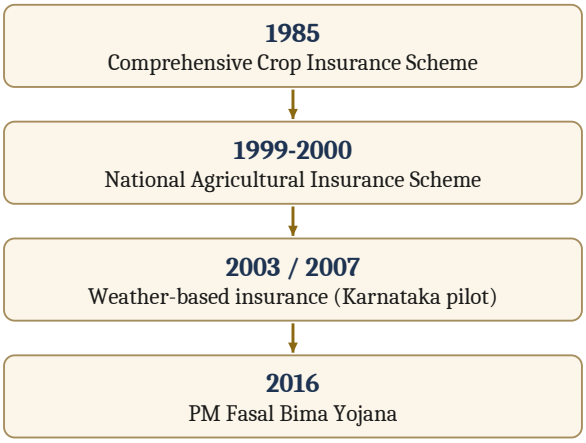
4.3 Research and Institutes

- First KVK: **1974** (Puducherry) 1× 2017
- ICAR order: **KVK (1974), Lab to Land (1979), NATP (1998), NAIP (2006)** 1× 2016
- *Handbook of Agriculture*: **ICAR** 1× 2012
- Kisan Call Centre: **Ministry of Agriculture** (2004) 1× 2023
- IIVR: **Varanasi** 1× 2015
- CFTRI: **Mysore** 1× 2012
- NAARM: **Hyderabad** 1× 2012
- IGFRI: **Jhansi** 1× 2012
- International Potato Centre: **Lima, Peru** 1× 2021
- **Neeru-Meeru**: **Andhra Pradesh** (2000) 1× 2016

4.4 Crop Insurance and Schemes

- Crop insurance began: **1985** 1× 2012
- NAIS: **1999-2000** 1× 2003
- NAIS extended to kharif in Budget **2004-05** 2× 2004, 2005
- Weather-based insurance first in **Karnataka** 1× 2010
- Crop insurance monopoly: none now (*) 1× 2008
- True: insurance 1985, UP has 9 agro-climatic zones, Food for Work 1977 (1, 2, 3) 1× 2010
- **Hariyali Yojana**: **water management** (watersheds) 2× 2010, 2016
- Rainfed Area Development: **2011-12** 1× 2015
- Vishesh Krishi and Gram Udyog Yojana: **farm exports** 3× 2010, one-liner, one-liner
- NLRMP: **2008** 1× 2016

CROP INSURANCE



4.5 Marketing and Prices

- **NAFED**: **agricultural marketing**; apex marketing cooperative 2× 1995, 2007
- Not an agri-export agency: **IFFCO** 1× 2017
- Markets regulated by State **APMC Acts** 1× 2011
- Farm demand is **inelastic** 1× 2013
- Sugarcane FRP approved by the **CCEA** 1× 2019
- Agricultural Prices Commission: **1965** 1× 2002

- MSP: fixed price, helps food security (II and III) 1× 2016
- Wheat MSP 2013: **Rs. 1,350**; paddy MSP 2018-19: **Rs. 1,750** 2× 2013, 2018
- MSP for fruits and vegetables: **Kerala** 1× 2016
- Groundnut MSP 1995-96: **Rs. 900** 1× 1995
- Government buying price: **procurement price** 1× 2008
- MSP recommended by **CACP** 1× one-liner
- Price Stabilisation Fund crop: **potato** 1× 2021

4.6 Credit

- Effective farm interest (2011-12): **4%** 1× 2011
- Not running cost: **land revenue** 1× 2014
- Not helping rural economy: **private money-lending** 1× 2013
- Agri-finance principles: **all** 1× 2016
- Long-term farm credit: **Land Development Bank** 1× 2014
- Largest source: **commercial banks** 2× 2004, 2006
- Least: **foreign private banks** 1× 2010
- ■ KCC: **1998-99**; gives timely crop credit (only 2) 4× 2008, 2010, one-liner, one-liner
- Order: **RRB (1975), NABARD (1982), SHG linkage (1992), KCC (1998)** 2× 2020, 2021

TRAP ALERT

- NFSA came in **2013**, not 2010; women 18+ head the card.
- **CACP** recommends MSP; **CCEA** approves.
- **KCC = 1998-99**, not 2001.

UP LINK

UP links: **western UP** gained from the Green Revolution; **Nandini Krishak Samriddhi Yojana** (UP) supports desi cow breeds (Sahiwal, Gir, Tharparkar, Gangatiri); UP has **9 agro-climatic zones**; **IIVR (Varanasi)** and **IGFRI (Jhansi)** are in UP.

LAST-MINUTE 10

1. NFSA 2013: 75% rural, 50% urban
2. NFSM from rabi 2007-08
3. Operation Flood = milk; blue = fish; yellow = oilseeds
4. Grey = fertilisers; black = petroleum; golden = horticulture
5. First KVK **1974**; IIVR Varanasi; IGFRI Jhansi
6. Crop insurance 1985; NAIS 1999; PMFBY 2016
7. NAFED = marketing; APMC = mandis
8. CACP 1965; CCEA approves FRP
9. Commercial banks = largest farm lenders
10. Order: RRB, NABARD, SHG, KCC (1998-99)

Chapter 5

Industry I: Policy, MSMEs, Start-ups and Reforms

Built from 58 UPPSC questions (1990–2025). Focus: industrial policies, LPG 1991, disinvestment, MSME definitions, committees, Make in India, start-ups, infrastructure plans.

ASKED MOST · REVISE FIRST

- ■ First industrial policy **1948**; joint sector **1956**; licensing from **IDR Act 1951**; liberal policy **1991** 4× 2004, 2007, 2008, 2008
- ■ **Micro enterprise**: investment up to **Rs. 1 crore**, turnover up to **Rs. 5 crore** (2020 rule) 3× 2011, 2016, 2021
- ■ **Golden handshake** = **Voluntary Retirement Scheme** 3× 1992, 2002, 2007
- ■ **National Renewal Fund** = **social security** for displaced workers 3× 2004, 2008, 2010

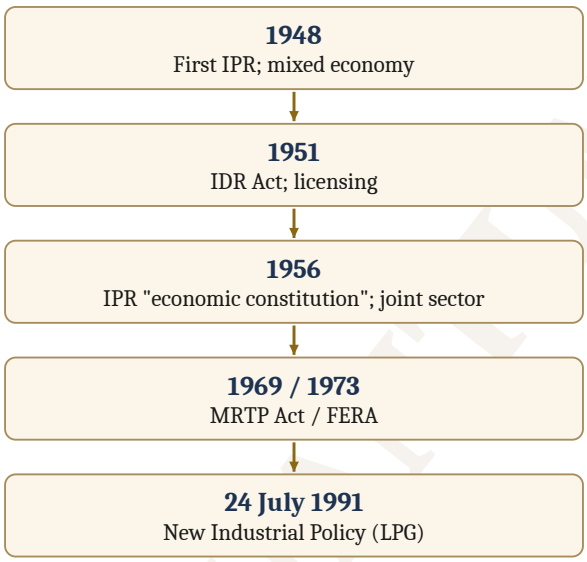
■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Industrial Policy

- First policy of free India: **1948** 1× 2008
- Joint sector: **IPR 1956** 1× 2004
- Licensing based on the **Industries Act 1951** 1× 2007

- Liberal policy: **1991** 1× 2008
- Order: **Minimum Wages (1948), IDR (1951), MRTP (1969), FERA (1973)** 1× 2017
- UP minimum wage for unskilled (April 2017): **Rs. 7,400** a month 1× 2017
- ISO certifications and licence liberalisation (both true, R not the explanation) 1× 1997
- 1991 licensing kept for **18** industries 1× 2004
- LPG announced by PM **Narasimha Rao** 1× 2013
- Disinvestment part of liberalisation; proceeds used per policy (both true, R not the explanation) 2× 2002, 2004
- Disinvestment (true); PSUs created too few jobs (false as the reason) 1× 2000
- Disinvestment began **1991-92** 1× 2008
- National Renewal Fund: **social security** 3× 2004, 2008, 2010
- Golden handshake: **VRS** 3× 1992, 2002, 2007

INDUSTRIAL POLICIES



5.2 MSMEs

- New MSME definition: announced in Atmanirbhar package; same formula for manufacturing and services (1 and 3) 1× 2016
- Investment to Rs. 1 crore, turnover to Rs. 5 crore: **micro** 3× 2011, 2016, 2021
- Most registered factories (2011): **Tamil Nadu** 1× 2011
- Small and cottage units mainly **generate jobs** 1× 2004
- UP's largest employer industry: **handloom** 1× 2008
- SSI problems: **all** listed; not one: **strikes and lockouts** 2× 1991, 2008
- Committee order: Karve, Nayak, Abid Hussain, Kohli (no code, *) 1× 2019
- End SSI reservation: **Abid Hussain Committee** 1× 2006
- Fast co-op growth: **cotton textile** 1× 2007
- First industry in India: **cottage industry** 1× 2008
- **Udyami** helpline: **MSMEs** 1× 2015
- Wrong: **Tarapore-customer services** (Tarapore = convertibility) 1× 2010
- **Meera Seth**: **handlooms** 1× 1997
- **Sathyam**: **textile policy** 1× 2000
- Unorganised sector commission: **Arjun Sengupta** 1× 2005

MSME (2020 rule)	Investment	Turnover
Micro	Rs. 1 crore	Rs. 5 crore
Small	Rs. 10 crore	Rs. 50 crore
Medium	Rs. 50 crore	Rs. 250 crore

5.3 Make in India and Start-ups

- **Make in India** logo: **lion** 1× 2014
- Launched 2014; promotes manufacturing (both) 1× 2022
- Start-up needs met by **all** listed 1× 2017
- New-age financing for start-ups: **Pranab Mukherjee** 1× 2016
- **Stand-Up India**: **women, SC and ST** 1× 2021
- **PM Mudra**: **Ministry of Finance** 1× 2025
- **e-Biz**: **single window** for government services 1× 2016

5.4 Infrastructure and CSR

- Biggest company (Fortune India 2019): **Reliance Industries** 2× 2016, 2019
- CSR: Act is **2013** and spend is **2%** (neither statement right) 1× 2019
- Railways: 100% electrification and net-zero by 2030 (both) 1× 2021
- First rubber-tyre metro: **Nashik** 1× 2021

- PPP revitalisation committee: **Vijay Kelkar** 1× 2017
- NIP is 2019-25, not 2020-30 (A false); equitable access aim (R true) 1× 2020
- Gati Shakti has **7 engines** (true); launched 2021, not 2022 (only 2) 1× 2023
- India 2nd largest mobile gaming market (false); 950 million+ internet users (true) 1× 2025
- Largest e-commerce firm (key): **M-junction** 1× 2015

5.5 Statistics and Relations

- Industry's GDP share 2006-07: **25-30%** 1× 2006
- Not in IIP: **construction** 1× 2016
- IIP base changed (2011) to **2004-05** (now 2011-12) 1× 2010
- Apex industrial finance (then): **IDBI** 1× 2017
- **BIFR**: sick units 1× 2002
- **Mega Food Parks**: food-processing infrastructure 1× 2009
- Social objective: **employment** 1× 2016
- **Sole trader-unlimited liability, partnership-contract, co-operative-weaker sections, public company-many risk bearers** 1× 2008
- **Kandla**: **export processing zone** 2× 2013, 2015
- Labour-intensive: **employs more hands** 1× 2006
- Not a party to industrial relations: **consumers** 1× 2014
- Workers in management: **policy decisions** 1× 1990

TRAP ALERT

- **1956** = joint sector; **1948** = first policy.
- **Abid Hussain** ended SSI reservation; **Tarapore** = convertibility.
- CSR = **2%** under the **2013** Act.

UP LINK

UP links: **handloom** is UP's largest employer industry; UP's minimum wage for unskilled labour was **Rs. 7,400** (2017); **ODOP** (One District One Product, 2018) and the **Defence Corridor** are UP's MSME pillars.

LAST-MINUTE 10

1. IPR **1948**; IDR Act **1951**; IPR **1956** (joint sector)
2. New policy **1991**; licensing for 18 industries
3. Disinvestment from **1991-92**; VRS = golden handshake
4. NRF = social security
5. Micro: Rs. 1 cr / Rs. 5 cr
6. Abid Hussain = end SSI reservation
7. Make in India: lion, 2014; Stand-Up = women, SC, ST
8. Mudra under Finance Ministry
9. CSR 2% (2013 Act); IIP excludes construction
10. Gati Shakti 2021, 7 engines; NIP 2019-25

Chapter 6

Industry II: Public Sector, Major Industries and Services

Built from 48 UPPSC questions (1990–2023). Focus: PSUs and Ratna status, refineries, steel plants, industry-centre matching, services sector.

ASKED MOST · REVISE FIRST

- **Service sector** = largest source of national income 5× 2004, 2010, 2014, 2017, 2021
- Tertiary sector: **trade, transport, finance, real estate**; not forestry and fishing 6× 2002, 2003, 2004, 2008, 2015, 2017
- Centre-industry matching: **Visakhapatnam-ships, Muri-aluminium, Gurgaon-cars, Panki-fertiliser, Coimbatore-cotton, Kapurthala-coaches, Barauni-refinery, Titagarh-jute, Ludhiana-hosiery, Bhadohi-carpets** 4× 1994, 1999, 2001, 2003
- **UP** = largest sugar producer 2× 2011, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Public Sector

- PSU roles: utilities, overhead capital, balanced growth, equality (all) 1× 2019

- Odd one out (private): **Essar Oil** 1× 1997
- **ONGC**: **1956** 1× 2014
- **OIL**: **oil exploration** 1× 2001
- Largest government trading company: **MMTC** 1× 2008
- **HAL**: **aircraft** 1× 2008
- Tourism and hotels: **ITDC** 1× 2004
- **BSNL**: **2000**; **100%** government equity 2× 2009, 2008
- GAIL pipeline: **Vijayawada to Vijaipur** 1× 2008
- Oil prices now set by **oil companies** 1× 2015
- **Navratna**: selected PSUs 1× 2008
- Not Maharatna in 2013: **Oil India** (Maharatna since 2024, *) 2× 2010, 2013
- Navratna then: **MTNL** 1× 2012
- Not Navratna: **Indian Aluminium Company** 1× 2006
- Navratna aluminium firm: **NALCO** 1× 2013

Status	Examples
Maharatna	ONGC, NTPC, SAIL, IOCL, Coal India, BHEL, GAIL, Oil India
Navratna	HAL, BEL, NALCO
Fully government	BSNL (2000)

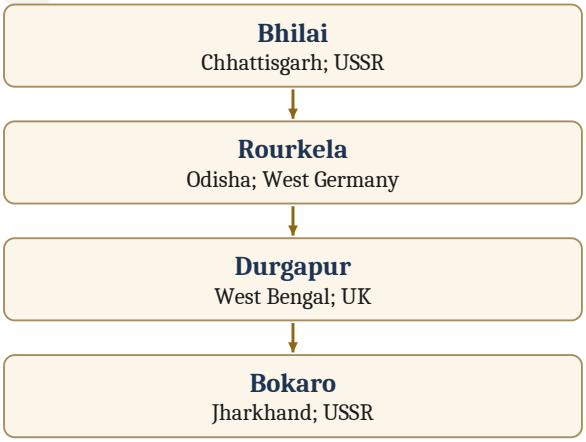
6.2 Refineries

- Private sector (1998): Mangalore and Essar then (*) 1× 1998
- Private among Digboi, Barauni, Numaligarh, Mathura: none (*) 1× 2004
- **Barauni-Bihar, Bongaigaon-Assam, Bina-MP, Koyali-Gujarat** 1× 2023
- Ideal for petrochemicals: **Gujarat** 1× 2002
- Coastal Gujarat an industrial workshop (R explains A) 1× 2001

6.3 Major Industries

- Steel plant not on a river: **Bhilai** 1× 2021
- **Rourkela**: with **Germany** 1× 2012
- **Bokaro**: **Soviet Union** 1× 1991
- **Bhilai-Chhattisgarh, Bokaro-Jharkhand, Durgapur-West Bengal, Rourkela-Odisha** 1× 2001
- **Indian Diamond Institute**: **Surat** 1× 2012
- **Visakhapatnam-shipbuilding, Muri-aluminium, Gurgaon-automobiles, Panki-fertiliser** 1× 2003
- **Coimbatore-cotton, Rourkela-steel, Kapurthala-rail coaches, Barauni-refining** 1× 1999
- **Singrauli-coal, Kajarhat-cement, Koyali-oil, Anand-milk** 1× 1994
- Largest sugar producer: **UP** 1× 2011
- Wrong: **Katni-paper** (Katni = cement) 1× 1990
- **Jute-Titagarh, silk-Bangalore, woollen hosiery-Ludhiana, carpets-Bhadohi** 1× 2001
- Sugar order: **UP, Maharashtra, Karnataka, Tamil Nadu** 1× one-liner
- Wrong: **MALCO-Chennai** (MALCO is at **Mettur**) 1× 2022
- **Nepanagar**: **newsprint** 1× 2001
- **Hazira** fertiliser: **natural gas** 1× 1991

PUBLIC STEEL PLANTS AND PARTNERS



6.4 History and Companies

- Order (key): **Indian sugar industry, Rishra jute mill, Bombay textile mill, first steel** 1× 2021
- MNCs (of options): **Samsung and LG** 1× 2014
- KVIC HQ: **Mumbai** 1× 2014
- Not FMCG: **automobiles** 1× 2012
- Largest software company: **TCS** 1× 2011

6.5 Tertiary Sector

- Tertiary: **trade and transport, finance and real estate** (1 and 2) 6× 2002, 2003, 2004, 2008, 2015, 2017

QUANTUM IAS & PCS

- Largest source of national income: **services** 5× 2004, 2010, 2014, 2017, 2021
- Largest services share in GSDP (2008-09): **Delhi** 1× 2010
- Steady growth 2001-05: **services** 1× 2007
- Not correct (Survey 2023): fashion, grocery, merchandise two-thirds of e-commerce by 2030 1× 2023
- Tertiary share **rises** as economies develop 1× 2006
- **TRAI**: **telecom** 1× 2017

TRAP ALERT

- **Bhilai** is not on a river; Rourkela = Germany, Durgapur = UK.
- **MALCO** = Mettur; **Katni** = cement.
- Forestry and fishing are **primary**, not tertiary.

UP LINK

UP centres: **Panki** (fertiliser), **Bhadohi** (carpets), **Renukoot** (aluminium), **Singrauli** (coal and power), **Churk** (cement), **Mathura** (IOCL refinery); UP is India's **top sugar** producer.

LAST-MINUTE 10

1. ONGC **1956**; BSNL **2000**, 100% govt
2. MMTC = largest govt trader; HAL = aircraft; ITDC = hotels
3. NALCO = Navratna; Oil India = Maharatna (2024)
4. Barauni Bihar; Bongaigaon Assam; Bina MP; Koyali Gujarat
5. Rourkela Germany; Bokaro USSR; Bhilai not on a river
6. Surat = diamonds; Muri = aluminium; Panki = fertiliser
7. Titagarh jute; Ludhiana hosiery; Bhadohi carpets
8. UP = top sugar producer
9. Services = largest share of GDP
10. TRAI = telecom; KVIC = Mumbai

Chapter 7

Fiscal Policy, the Union Budget and Deficits

Built from 39 UPPSC questions (1993–2024). Focus: who frames fiscal policy, Budget facts, receipts and spending, the four deficits, FRBM, deficit financing, budgeting methods.

ASKED MOST · REVISE FIRST

- Fiscal deficit target 2019-20: **3.3% of GDP** 9× 1996, 2000, 2001, 2002, 2005, 2008, 2010, 2015, 2019
- **Primary deficit** = fiscal deficit − **interest payments** 3× 2004, 2008, 2010
- Deficit order: **fiscal > revenue > primary** 4× 2002, 2003, 2004, 2008
- Largest outlay in Budget 2011-12 (key): **energy** 5× 2001, 2002, 2009, 2010, 2011
- **Economic Survey** and fiscal policy: **Ministry of Finance** 4× 2010, 2010, 2012, 2015
- **Deficit financing** adds money and leads to **inflation** 3× 1993, 2003, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

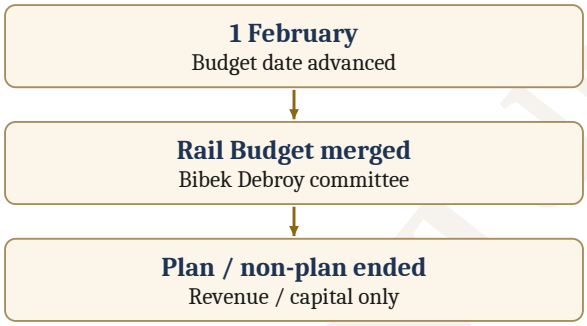
7.1 Fiscal Policy

- Fiscal policy framed by the **Ministry of Finance** 1× 2012
- Long-term fiscal policy (1985): **V.P. Singh** 1× 2014
- Not a fiscal-policy aim: **regulating inter-State trade** 1× 2006
- Response to supply shocks: **cut taxes on fuel and imports** (only 3) 1× 2024
- Subsidy a big deficit source (true); farm subsidy higher than rich countries (false) 2× 2002, 2004
- Survey 2015-16 "Chakravyuha": from socialism with limited entry to **marketism without exit** 1× 2017
- Economic Survey prepared by the **Ministry of Finance**; published by it 3× 2010, 2010, 2015

7.2 The Budget

- Not among 2017-18's ten themes: **export performance** 1× 2017
- Budget 2023: States allowed **3.5% of GSDP** fiscal deficit 1× 2023
- AIDC (2021-22): on **29** products 1× 2021
- Did not happen in 2017-18: **more centrally sponsored schemes** 1× 2017
- Health Ministry 2023-24: about **Rs. 89,155 crore** 1× 2023
- Not true of 2023-24 outlay shares: **subsidies 9%** 1× 2023
- Largest outlay (2011-12): **energy** 5× 2001, 2002, 2009, 2010, 2011
- 2018-19 allocations: **defence Rs. 2.82, food subsidy Rs. 1.69, rural development Rs. 1.38, agriculture Rs. 0.63 lakh crore** 1× 2017
- PSB disinvestment target 2020-21: **Rs. 90,000 crore** 1× 2021
- Railway research body (2016-17): **SRESTHA** 1× 2016
- New luxury class (2013-14): **Anubhuti** 1× 2014

BUDGET REFORMS OF 2017-18



7.3 Receipts and Spending

- 2010-11: capital receipts **less than** revenue receipts 1× 2009
- Not public revenue: **subsidy** 1× 2009
- Current-account income: **corporation tax, PSU profits, excise** (I, II, V) 1× 1998
- Largest Budget head (now): **revenue expenditure** (non-plan in 2006) 1× 2006
- Largest revenue expenditure item: **interest payments** 1× 2017

7.4 Deficits

- Targets: 2017-18 at **3.5%**, 2018-19 at **3.3%** (1 and 2) 1× 2017
- Fiscal deficit target 2019-20: **3.3%** 9× 1996, 2000, 2001, 2002, 2005, 2008, 2010, 2015, 2019
- States' fiscal deficit record then: about Rs. 1 lakh crore (*) 1× 2000
- Order: **fiscal, revenue, primary** 4× 2002, 2003, 2004, 2008
- Fiscal deficit minus interest: **primary deficit** 3× 2004, 2008, 2010
- Primary deficit 2010-11: **below 2%** of GDP 1× 2009
- Revenue deficit minus capital-asset grants: **effective revenue deficit** 1× 2015
- Fiscal deficit mostly met by **domestic borrowing** 1× 2005
- Deficit financing causes inflation by raising money supply (R explains A) 1× 2003
- Failed deficit financing: **inflation**; raises **money supply** 2× 1993, 2016
- **FRBM Act**: **2003** 1× 2008

Deficit	Formula
Revenue	Revenue spending – revenue receipts
Effective revenue	Revenue deficit – capital-asset grants
Fiscal	Total spending – (revenue + non-debt capital receipts)
Primary	Fiscal deficit – interest

7.5 Budgeting Methods

- Zero-based budgeting first in the **USA** 1× 2017
- ZBB in India; reviews every scheme (R explains A) 1× 2011
- Performance budget from the **USA** 1× 2010

TRAP ALERT

- **Primary** = fiscal – interest; **effective revenue** = revenue – capital grants.
- **Interest** is the biggest revenue spend.
- **FRBM 2003**; fiscal policy = Finance Ministry, monetary = RBI.

UP LINK

UP links: States may run a fiscal deficit of about **3-3.5% of GSDP**; UP passed its own **Fiscal Responsibility and Budget Management Act (2004)** and presents India's largest State Budget.

LAST-MINUTE 10

1. Fiscal policy & Survey = **Finance Ministry**
2. Long-term fiscal policy = **V.P. Singh** (1985)
3. Fiscal deficit target 2019-20 = **3.3%**
4. Fiscal > revenue > primary
5. Primary = fiscal − interest
6. Effective revenue = revenue − capital grants
7. Interest = biggest revenue spend
8. FRBM **2003**
9. Deficit financing leads to inflation
10. ZBB & performance budget = USA

Chapter 8

Taxation, GST and the Finance Commission

Built from 67 UPPSC questions (1991–2025). Focus: GST, direct vs indirect taxes, who levies what, VAT and excise, Finance Commissions and chairmen, tax committees.

ASKED MOST · REVISE FIRST

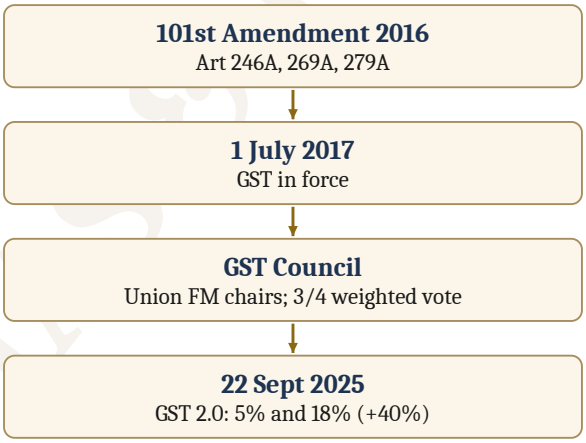
- ■ **Finance Commission** shares taxes between Centre and States 4× 2003, 2007, 2017, 2021
- ■ **Income tax** = direct and progressive; not State-collected 4× 1991, 1993, 2004, 2010
- ■ FC chairmen: **1st Neogy, 9th Salve, 10th Pant, 11th Khusro, 12th Rangarajan, 13th Kelkar, 14th Y.V. Reddy, 16th Panagariya** 4× 2007, 2021, 2023, 2025
- ■ **Kelkar** committee = tax reform; ended Section 88 rebates 4× 2004, 2004, 2008, 2016
- • **GST** = indirect tax; **101st Amendment** 2× 2017, 2021
- ■ **CENVAT/MODVAT** = central excise 3× 2004, 2008, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 GST

- GST is an **indirect tax** 1× 2017
- Enabled by the **101st Amendment** 1× 2021
- GST creates one market and ends cascading (both) 1× 2017
- GST Council: State finance ministers as members; decides rates and thresholds (1 and 2); States cannot keep VAT 1× 2016
- GST in October 2020: **over Rs. 1 lakh crore** 1× 2016
- ITC is input-tax credit (true); available on non-taxable supplies (false) (only 1) 1× 2023
- **Revenue Neutral Rate**: **GST** 1× 2016
- **Project Saksham**: **indirect tax IT network** 1× 2017

GST AT A GLANCE



8.2 Direct and Indirect

- Direct Tax Code: **income tax** 1× 2018
- Expenditure tax suggested by **Kaldor** 1× 2010
- Direct taxes' GNP share then about 5% (*) 1× 2004
- Tax-GDP ratio: depends on year (*) 1× 2004
- Indirect taxes regressive: **same rate for all incomes** 1× 2019
- Income tax: **progressive and direct** (I and II) 2× 2004, 2010
- Income tax introduced by **James Wilson** (1860) 1× 2014
- Does not raise the buyer's price: **income tax** 1× 1995
- Tax on imports and exports: **customs duty** 1× 2006
- Direct: **income tax**; **wealth tax** 2× 1991, 2014
- Not indirect: **income tax**; not direct: **sales tax** 2× 1993, 1993

8.3 Who Levies What

- 15% surcharge (2016-17): income **above Rs. 1 crore** 1× 2016
- "Very senior citizens": **80 years and above** 1× 2010
- Central Sales Tax: (i) and (ii) correct; not in SEZs 1× 2010
- Two largest Central taxes then: excise and customs; income tax now (*) 1× 1995
- 2020-21 revenue: **corporate tax 18%, income tax 17%, excise 7%, customs 4%** 1× 2016
- Top Central tax 2009-10: **corporation tax** 2× 2008, 2009
- Not a Central levy: **road tax** 1× 2013
- Union-only: **customs and corporation tax** 1× 2005
- Largest Central tax (then): excise (*) 1× 2004
- Not from corporate income: **capital gains tax** 1× 2005
- Not a Union tax: **entertainment tax** 1× 1992
- Not a State tax: **corporation tax** 1× 2008
- Liquor excise: **State Governments** 1× 2014

Union	States	Local bodies
Income tax, corporation tax	Agricultural income tax, land revenue	Property tax
Customs, central excise	Liquor excise, stamp duty	Tax on fairs
CGST, IGST	SGST, road tax	Water charges

8.4 Excise, VAT, Service Tax

- **Capital gains-sale of property, excise-factory produce, customs-imports, corporate tax-income** 1× 2016
- **MODVAT**: **excise duty** 1× 2011
- **CENVAT**: **central excise** 2× 2004, 2008
- VAT: on **all stages** from production to sale 2× 2006, 2012
- Not on VAT (2005): **Chhattisgarh and UP** 1× 2005
- VAT nationwide: **2005** 1× 2015
- Service tax: **indirect**, from **1994-95** (2 and 3) 1× 2013

8.5 Finance Commission

- FC fixes States' share of taxes and grants; primary duty: tax distribution 4× 2003, 2007, 2017, 2021
- Not correct: PC and FC advice **binding** 1× 2009
- Not shared with UP: **agricultural income tax** 1× 2000
- 16th FC: chaired by **Arvind Panagariya** (true); set up 31 Dec 2024 (false; **2023**) (only II) 1× 2023
- Not correct of 15th FC: covers only 2020-25 (it is 2020-26) 1× 2019
- 13th FC's top weight: **fiscal capacity distance** 1× 2011
- First FC chairman: **K.C. Neogy** 1× 2021
- **11th-Khusro, 12th-Rangarajan, 13th-Kelkar, 14th-Y.V. Reddy** 1× 2025
- **9th-Salve, 10th-Pant, 11th-Khusro, 12th-Rangarajan** 1× 2007
- 10th FC share: **29%** 2× 1995, 1997

FC	Chairman	States' share
1st	K.C. Neogy	-
10th	K.C. Pant	29%
13th	Vijay Kelkar	32%
14th	Y.V. Reddy	42%
15th	N.K. Singh	41%
16th	Arvind Panagariya	41% (2026-31)

8.6 Reforms and Local Finance

- **Kelkar Task Force**: **taxes**; latest tax committee then 2× 2004, 2004
- Abolish Section 88 rebates: **Kelkar** 2× 2008, 2016
- Early-1990s tax reform: **R.J. Chelliah** 1× 1995
- Different for tax relief: **Indira Vikas Patra** 1× 1997
- No tax rebate: **Kisan Vikas Patra** 1× 2008
- PAN 4th letter **P** = **individual** 1× 2014
- PAN not valid as **proof of address** 1× 2007
- AGR includes interest, dividend, forex gain (all) 1× 2016
- Gratuity ceiling: Rs. 10 lakh then; now Rs. 20-25 lakh (*) 1× 2014
- Panchayat funds: **FC grants, cooperative banks, CSS allocations, SFC** (1, 2, 3, 5) 1× 2007
- Gram Panchayat tax: **tax on local fairs** 1× 2018

- NHAI's main fund: **cess** 1× 2011

TRAP ALERT

- **16th FC** set up **31 Dec 2023**; covers 2026-31.
- Liquor excise and agricultural income tax = **States**.
- **VAT 2005** (UP 2008); **service tax 1994-95**.

UP LINK

UP got the largest share (about **17.9%** under the 15th FC, about 17.6% under the 16th) of devolved taxes; UP joined **VAT** only on **1 January 2008**; UP Gram Panchayats can tax **local fairs**.

LAST-MINUTE 10

1. GST: indirect; **101st**; 1 July 2017; Art 279A
2. Income tax: direct, progressive; James Wilson 1860
3. Kaldor = expenditure tax
4. Union: customs, corporation tax; States: liquor, farm income
5. MODVAT 1986; CENVAT 2000; VAT 2005
6. Service tax 1994-95; indirect
7. First FC **Neogy**; 13th **Kelkar**; 14th **Y.V. Reddy** (42%)
8. 16th FC **Panagariya**, 41%, 2026-31
9. Kelkar = Section 88; Chelliah = 1991 reform
10. PAN "P" = individual

Chapter 9

Economic Planning and NITI Aayog

Built from 57 UPPSC questions (1990–2024). Focus: Planning Commission, NDC, NITI Aayog, pre-independence plans, the twelve Five Year Plans and their themes.

ASKED MOST · REVISE FIRST

- **NDC** = approval of Five Year Plans; formed **6 August 1952** 7× 2001, 2003, 2004, 2008, 2008, 2014, 2016
- 11th Plan: **2007-12**, "faster and more inclusive growth", 9% target 5× 2006, 2008, 2008, 2011, 2015
- **Fifth Plan** = poverty removal (Garibi Hatao), self-reliance, zero net aid 3× 2007, 2014, 2016
- **Economic planning** = **Concurrent List** 2× 2002, 2006
- **NITI Aayog**: no full-time chairman; first Vice-Chairman **Arvind Panagariya** 3× 2015, 2017, 2018

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 Planning Bodies

- Planning: **Concurrent List** 2× 2002, 2006
- **DMEO**: attached to **NITI Aayog** 1× 2020
- "Economic Cabinet" (Ashok Chanda): **Planning Commission** 1× 2018
- NDC: **approval of Five Year Plans**; formed **6 August 1952** 7× 2001, 2003, 2004, 2008, 2008, 2014, 2016
- Planning Commission set up: **1950**; by a **Cabinet resolution** 2× 2009, one-liner
- Planning Commission: a **quasi-political** body 1× 2014
- Not correct of NITI: **full-time chairman** (PM chairs) 1× 2015
- SDG nodal institution: **NITI Aayog** 1× 2019
- First NITI Vice-Chairman: **Arvind Panagariya** 2× 2017, 2018
- India Innovation Index 2019 top: **Karnataka** 1× 2019
- Fastest-growing State 2014-15: **Bihar** 1× 2016

9.2 Ideas Behind Planning

- Planning for balanced development, even spread, regional equity, resource use (all four) 2× 2009, 2016
- Gadgil-Mukherjee formula: highest weight to **population** 1× 2008
- **Gandhian Plan** (1944): **Shriman Narayan Agarwal** 1× 2015
- Opposed planned development: **Mahatma Gandhi** 1× 2019

- Rolling plan: annual, multi-year and perspective features (all four) 1× 2016
- Rolling plan in operation: **1978-79** 1× 2007

Early plan	Year	By
Planned Economy for India	1934	M. Visvesvaraya
National Planning Committee	1938	Congress (Nehru)
Bombay Plan	1944	Industrialists
Gandhian Plan	1944	Shriman Narayan Agarwal
People's Plan	1945	M.N. Roy
Sarvodaya Plan	1950	Jayaprakash Narayan

9.3 Five Year Plans

- 12th Plan: **2012-17**; agriculture growth about **4%** 2× 2012, 2014
- 12th Plan vision target: **9%** 1× 2012
- Sustainable growth focus: **12th Plan** 1× 2018
- Largest 12th Plan allocation: **social services** 2× 2013, 2014
- 11th Plan: **2007-12**; largest spend: **social sector** 4× 2008, 2011, 2015, 2010
- 11th Plan industry target: **10.5%**; agriculture **4%**; GDP **9%** 4× 2008, 2007, 2009, 2006
- 10th Plan mid-term growth target: **7%**; original target **8%** 2× 2004, 2002
- Largest 10th Plan outlay: **energy** 1× 2005
- 10th Plan investment rate **higher than** savings rate 1× 2008
- Growth with social justice and equity: **9th Plan**; 9th target **6.5%** 2× 2015, 1996
- Women's Component Plan: **9th Plan** 1× 2016
- April 1992-March 1997: **8th Plan** 1× 1998
- Human development at the core: **8th Plan** 1× 2016
- IRDP in the 7th Plan: **total household approach** 1× 1994
- Self-reliance and zero net foreign aid: **5th Plan** 1× 2014
- Poverty abolition slogan: **5th Plan**; Garibi Hatao programmes: **5th** 2× 2007, 2016
- Order: **Green Revolution, bank nationalisation, Garibi Hatao** 1× 2016
- Growth Centre approach: **4th Plan** 1× 1996
- Industrialisation strategy: **2nd Plan** 1× 1991
- First Plan priority: **agriculture** 1× 2011
- **1st-Harrod-Domar, 2nd-Mahalanobis, 3rd-Chakravarty, 4th-Ashok Rudra** 1× 2021
- **1st-community development, 2nd-rapid industrialisation, 3rd-basic industries, 4th-growth with stability and self-reliance, 5th-minimum needs** 1× 1994
- **1st-1951-56, 3rd-1961-66, 4th-1969-74, 6th-1980-85** 1× 2014
- 11th Plan approach: **faster and more inclusive growth** 1× 2008
- Years outside any plan: **7** 1× 1999
- "Jobless growth" decade: **1991-2000** 1× 2014
- Twenty Point Programme: **1975** 2× 2008, 2009
- Bottom-up planning still a goal; village a viable unit (R explains A) 1× 1998
- Order: **IMF membership, first devaluation (1949), First Plan to Parliament, NDC** 1× 2003

QUANTUM IAS & PCS

FIVE YEAR PLANS IN ONE LINE EACH



9.4 Planned Cities and Yojana

- **Kanchipuram-ancient, Jaisalmer-medieval, Kodaikanal-colonial, Bhilai-post-independence** 1× 2024
- *Yojana* magazine: **Publications Division** 1× 1990

TRAP ALERT

- Planning Commission **1950** (Cabinet resolution); NDC **1952**; NITI **2015**.
- **Garibi Hatao** = 5th Plan; **human development** = 8th; **social justice** = 9th.
- **7** plan-free years: 1966-69, 1978-80 (rolling), 1990-92.

UP LINK

UP links: UP's **State Transformation Commission** (2023) replaced the State Planning Commission on NITI lines; UP's planning department prepares SDG and district plans.

LAST-MINUTE 10

1. Planning = Concurrent List
2. Planning Commission 15 Mar **1950**; NDC 6 Aug **1952**
3. NITI Aayog 1 Jan **2015**; first VC Panagariya
4. Gandhian Plan = Shriman Narayan Agarwal
5. 1st Harrod-Domar; 2nd Mahalanobis
6. 5th = Garibi Hatao, self-reliance
7. 8th = human development; 9th = social justice
8. 11th = inclusive growth, 9%; 12th = sustainable, last plan
9. Rolling plan **1978-80**; 7 plan-free years
10. Twenty Point Programme **1975**

Chapter 10

Money, Inflation and the Reserve Bank of India

Built from 68 UPPSC questions (1990–2021). Focus: money measures, currency facts, inflation types and indices, RBI functions and history, CRR, SLR, repo, OMO.

ASKED MOST · REVISE FIRST

- **RBI** = banker's bank, government's adviser, forex keeper, credit controller; HQ **Mumbai** 7× 2001, 2009, 2010, 2010, 2010, 2012, 2017
- **Monetary policy** = **RBI**; complements fiscal policy 5× 2002, 2006, 2010, 2016, 2017
- **Repo** = RBI lends to banks; **reverse repo** = banks lend to RBI; **bank rate** = RBI rediscount rate 5× 2008, 2008, 2008, 2009, 2011
- **CRR** rise cuts liquidity and credit 3× 2004, 2010, 2013
- Headline inflation measured by **CPI (Combined)** 4× 2004, 2008, 2012, 2021
- **Decimal coinage**: **1957** 3× 1995, 2005, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

10.1 Money and Currency

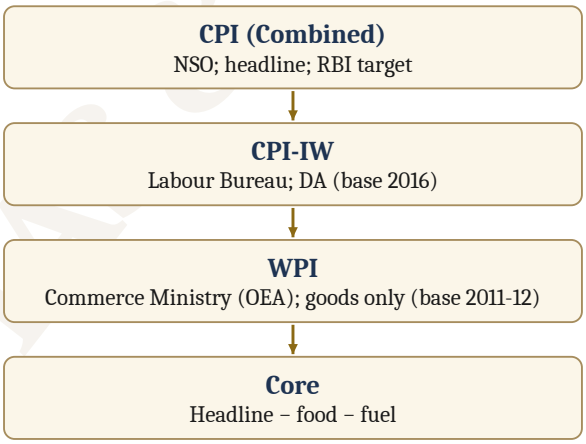
- Broad money (M3): **currency, demand deposits, time deposits** (all) 1× 2008
- Money multiplier: **broad money ÷ reserve money** 2× 2008, 2009
- **Black money**: illegal income on which tax is unpaid 2× 2002, 2003
- Not a demonetisation aim: **raising GDP growth** 1× 2017
- Paper currency began: **1862** (per key; Act 1861) 1× 2011
- ■ Decimal coinage: **1957** 3× 1995, 2005, 2010
- RBI can print notes up to **Rs. 10,000** 1× 2014
- Minimum gold stock: **Rs. 115 crore** 1× 2008
- ■ Note issue: **minimum reserve system** 3× 2004, 2007, 2010
- Coins issued by the **Ministry of Finance** 1× 2010
- Mints: **Mumbai, Kolkata, Hyderabad** (and Noida) 1× 2008

Measure	Includes
M0	Currency + bank deposits with RBI
M1	Currency with public + demand deposits
M2	M1 + post office savings
M3	M1 + time deposits (broad money)

10.2 Inflation

- Fall in inflation rate: prices rise **more slowly** (II only) 1× 2017
- Inflation benefits **debtors** 2× 1995, 2016
- Bottleneck inflation from supply gaps (R explains A) 1× 2021
- **Core inflation**: headline minus **food and fuel** 1× 2017
- ■ Common measure: **CPI** (WPI earlier) 3× 2004, 2008, 2012
- Headline inflation: **CPI (Combined)** 1× 2021
- CPI: covers goods and services; used by RBI (1 and 3); not compiled by OEA 1× 2019
- CPI-IW base shifted from 1982 to **2001**; earlier 1960 to **1982** 2× 2008, 1990
- Central employees' wages compensated by **CPI-IW**; DA based on **CPI** 2× 2001, 2008
- WPI published by the **Ministry of Commerce and Industry** 1× 2021
- WPI's largest weight: **manufactured products** 1× 2008
- WPI 2004-05 series released **14 September 2010** 1× 2009
- **PPI**: average change in producers' prices 1× 2017
- Not an anti-inflation step: **cutting interest rates** 1× 1996
- WPI revision group (2004): **Abhijit Sen** 1× 2004

PRICE INDICES



10.3 The RBI

- RBI: apex bank, regulates money, oversees NABARD (I, II, IV); not loans to businesses 1× 2001
- Government's financial agent and adviser: **RBI** 1× 2017
- Treasury bills sold by the **RBI** 1× 2009
- J&K banking (then) not done by RBI (*) 1× 2014
- Not correct: RBI was **established in 1949** (it was **1935**; nationalised 1949) 1× 2011
- HQ: **Mumbai** 1× 2010
- Bankers' bank: **RBI** 1× 2010
- Forex reserves kept by the **RBI** 1× 2012
- Credit creation controlled by the **RBI** 1× 2010
- Not correct: RBI issues **all** notes (Re. 1 note by Government) 1× 2007
- Re. 1 note: **Finance Secretary's** signature 1× 2004
- Governors in order: **I.G. Patel, Manmohan Singh, C. Rangarajan, D. Subbarao** 1× 2021

QUANTUM IAS & PCS

- Never RBI Governor: **Raja J. Chelliah** 1× 2005
- RBI financial year: **April-March** (from 2020-21; earlier July-June) 1× 2013

10.4 Monetary Tools

- Not an RBI function: **credit creation** (commercial banks do it) 1× 2008
- Monetary policy **complements** fiscal policy 1× 2016
- Not a monetary aim: **equitable distribution of income** 1× 2011
- Monetary policy by the **RBI**; money and credit controlled by the RBI 2× 2006, 2010
- Not quantitative control: **credit rationing** 1× 2017
- **OMO**: trading in securities; a **quantitative** tool 3× 2008, 2010, 2017
- **Operation Twist**: sell short-term, buy long-term to boost long-term investment (R explains A) 1× 2016
- Anti-inflation steps: **raise bank rate, CRR, SLR** (1, 2, 3); not buying securities 1× 2016
- Cash and liquid assets ratio: **SLR** 1× 2007
- Not set by the RBI: **PLR** 1× 2008
- Since reforms **both SLR and CRR reduced** 1× 1999
- Deposits kept with RBI: **CRR** 1× 2013
- CRR rise: **less liquidity**; **less credit creation** 2× 2010, 2004
- Repo rate: **monetary policy** 1× 2017
- "Repurchase option" (key): **none of the above** 1× 1998
- RBI's short-term lending rate: **repo** 1× 2008
- Banks lend to RBI at **reverse repo** 1× 2008
- **Bank rate**: RBI's rediscount rate; charged by RBI 3× 2008, 2009, 2011
- Repo-reverse repo gap (2010-11): **1%** 1× 2010
- 5 Dec 2019: repo **5.15%**, reverse repo **4.9%** 2× 2014, 2019
- Repo regulated by the **RBI** 1× 2002

Tool	Meaning
CRR	Cash kept with RBI
SLR	Cash, gold, G-secs kept by banks
Repo	RBI lends to banks
Reverse repo / SDF	Banks park money with RBI
Bank rate	RBI rediscounts bills
OMO	Buy or sell G-secs

TRAP ALERT

- **RBI 1935**, nationalised **1949**.
- Coins and Re. 1 note = **Government**.
- **Credit creation** is done by commercial banks, not RBI.
- Disinflation ≠ deflation.

UP LINK

UP link: India's fourth mint is at **Noida** (1988); UP's government employees' DA follows **CPI-IW** like the Centre's.

LAST-MINUTE 10

1. M3 = M1 + time deposits; multiplier = M3/M0
2. Decimal coins **1957**; minimum reserve, Rs. 115 cr gold
3. Coins, Re. 1 note = Govt (Finance Secretary)
4. Core = headline − food − fuel; debtors gain from inflation
5. CPI = headline (NSO); WPI = Commerce Ministry; DA = CPI-IW
6. RBI **1935**, Mumbai; nationalised 1949
7. RBI year April-March from 2020-21
8. Repo: RBI lends; reverse repo: RBI borrows
9. CRR up means liquidity down; OMO = securities
10. Operation Twist: buy long, sell short

Chapter 11

Banking in India

Built from 83 UPPSC questions (1992–2024). Focus: bank history and nationalisation, types of banks, development banks, financial inclusion, NPAs and safety, cooperative and rural banking, NABARD, RRBs.

ASKED MOST · REVISE FIRST

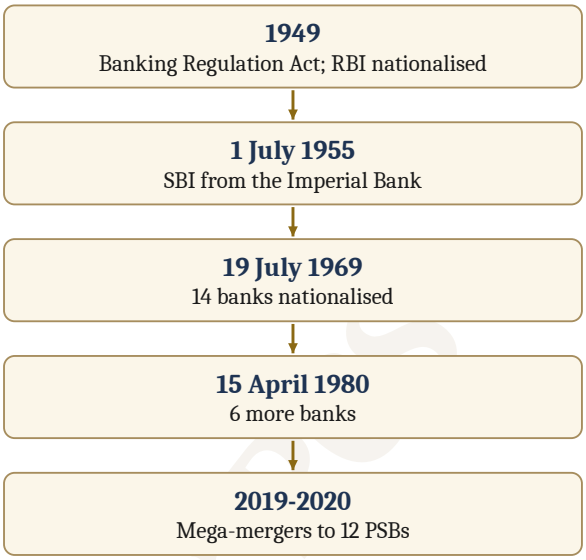
- **14 banks** nationalised in **July 1969**; 6 more on **15 April 1980** 5× 1996, 2004, 2006, 2007, 2014
- **NABARD**: **1982**; apex rural credit; runs **RIDF** 8× 2003, 2004, 2004, 2005, 2008, 2009, 2014, 2014
- **SBI** = largest commercial bank; first Indian bank in China 5× 2003, 2004, 2005, 2006, 2007
- **SIDBI**: small industry; HQ **Lucknow** 5× 2004, 2012, 2014, 2016, 2021
- **Land Development Bank** = long-term farm loans; part of the cooperative structure 4× 1992, 2008, 2008, 2016
- **Lead Bank Scheme** (December 1969) handles **district credit planning** 5× 2002, 2003, 2007, 2022, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 History and Nationalisation

- **Allahabad Bank-1865, PNB-1894, Awadh Commercial-1881, Alliance Bank of Shimla-1874** 1× 2023
- Earliest event: **Banking Regulation Act** (1949) 1× 2016
- 14 banks nationalised: **July 1969** 4× 1996, 2006, 2007, 2014
- Gorwala committee led to SBI; 6 banks nationalised 15 April 1980 (both) 1× 2004
- Number of PSBs (2011): depends on IDBI count; **12** now (*) 1× 2011
- Bank mergers: scale, capital, reach, global size (all) 1× 2019
- **EASE Index**: measures **PSB performance** (only 1) 1× 2021

BANK NATIONALISATION AND MERGERS



11.2 Types of Banks

- Public sector bank: **Indian Overseas Bank** 1× 2006
- Private bank: **Bank of Punjab** 1× 2001
- Not nationalised: **ICICI Bank** 1× 2004
- Largest commercial bank: **SBI** 3× 2003, 2004, 2005
- "Farmers' clubs": **SBI** 2× 2008, 2010
- SBI chairman from April 2011: **Pratip Chaudhuri** 2× 2006, 2012
- "Simply Click" card: **SBI** 1× 2016
- First Indian bank branch in China: **SBI**; first private: **Axis Bank** 3× 2006, 2007, 2014
- First moving ATM: **ICICI** 1× 2014
- Merged with PNB: **New Bank of India** (1993; OBC in 2020) 1× 2010
- Long-term industrial finance: **all** listed 1× 1998
- **Indian Bank-public, ICICI-private, Citibank-foreign, Saraswat-cooperative** 1× 2016
- Foreign bank with most branches: **Standard Chartered** 1× 2008

11.3 Development Banks and Inclusion

- Order: **1st nationalisation, RRBs, 2nd nationalisation, NABARD** 1× 2023
- Order: **SBI, Lead Bank Scheme, RRBs, NABARD** 1× 2005
- **LBS**: **Lead Banking Scheme**; started **December 1969** 2× 2022, one-liner
- Lead bank functions by a **designated bank** 1× 2007

- District credit planning: **lead bank** 2× 2002, 2003
- Small-industry bank: **SIDBI**; HQ **Lucknow**; finances small industry 4× 2012, 2016, 2021, 2004
- Wrong: **SIDBI-Mumbai** (Lucknow) 1× 2014
- **ICICI-1955, IDBI-1964, IFCI-1948, SIDBI-1990** 1× 2010
- **IFCI**: a **development bank** 1× 2012
- Marketing institution (key): **SAIL** 1× 2002
- Not an apex institution: **SBI** 1× 2008
- Financial inclusion helped by **all** listed 1× 2008
- Not an inclusion aim: **shrinking banking infrastructure** 1× 2016
- Hindrances: **all** listed 1× 2013
- Inclusion steps: **nationalisation, RRBs, village adoption** (all) 1× 2024
- Financial inclusion committee (2005): **C. Rangarajan** 2× 2013, 2014
- Payment banks: **Nachiket Mor** 1× 2014
- **Shadow banking**: non-bank financial intermediaries 1× 2014
- Microfinance committee: **Y.H. Malegam** 2× 2010, 2014

Institution	Year	HQ / role
IFCI	1948	First development bank
ICICI	1955	Industrial credit
IDBI	1964	Industrial finance
NABARD	1982	Rural credit (Mumbai)
EXIM Bank	1982	Trade finance
SIDBI	1990	MSMEs (Lucknow)

11.4 Balance Sheets and Safety

- Bank asset: **credit to farmers** 1× 2007
- Largest liability: **time deposits** 2× 2008, 2009
- **NPA**: loans with unpaid interest or principal (90 days) 1× 2009
- Under RBI control (then): NABARD; now fully government-owned (*) 1× 1998
- **Basel II**: **capital adequacy** standards 1× 2005
- **DICGC**: RBI subsidiary; insures up to **Rs. 5 lakh** (both) 1× 2016

11.5 Cooperative and Rural Banking

- First Land Development Bank (1920): **Jhang** 1× 2016
- LDB: **long-term farm loans**; part of the **cooperative credit structure** 3× 1992, 2008, 2008
- Central Cooperative Bank: **district level** 2× 2004, 2008
- Urban cooperative banks (2012): 1,618 (no option, *) 1× 2012
- Consumer cooperative stores set up **by members** 1× 2016
- **NABARD**: **1982**; set up by an Act (true), only farm credit (false) (only 1) 3× 2003, 2014, one-liner
- NABARD refinances **agriculture and rural development**; apex farm-credit body 2× 2004, 2005
- NABARD does not refinance the **EXIM Bank** 1× 2002
- RIDF nodal agency: **NABARD** 4× 2004, 2008, 2009, 2014
- RRBs: **1975** 1× 2010
- Not an RRB function: **take over ARDC's work** 1× 2005
- Order: **RRB, NABARD, SHG linkage, KCC** 1× 2020
- Does not implement SHG linkage (it promotes it): **NABARD** 1× 2015
- **Narasimham**: **banking reforms** 1× 2008
- **Raghuram Rajan**: **financial sector reforms** 1× 2007
- **Bimal Jalan** panel: **new bank licences** 1× 2014
- First Disinvestment Commission chairman: **G.V. Ramakrishna** 1× 2004
- **Swabhimaan**: banks to the **rural poor's doorstep**; rural banking 2× 2010, 2012
- KCC covers **consumption and investment credit**; KCC from **1998-99** 2× 2010, 2014

Tier	Level
State Cooperative Bank	State
Central Cooperative Bank	District
PACS	Village
Land Development Bank	Long-term (first at Jhang, 1920)

11.6 Instruments

- **Negotiable Instruments Act**: **1882** 1× 2016
- IFSC set up at **Gandhinagar** (GIFT City) 1× 2016

- "Smart money": **credit card** 1× 2008
- "Plastic money": **bank credit cards** 1× 2006
- Not prepaid: **credit card** 1× 2014
- **PIN**: Personal Identification Number 1× 2023

TRAP ALERT

- **NPA** = overdue **90 days**.
- **DICGC** insures **Rs. 5 lakh** per depositor per bank.
- **SIDBI** HQ = Lucknow, not Mumbai.
- NABARD **promotes** SHG linkage; banks implement it.

UP LINK

UP links: **SIDBI** is headquartered in **Lucknow**; the first RRB group included **Prathama Bank, Moradabad** (1975); **Allahabad Bank** (1865, Prayagraj) merged into Indian Bank in 2020; UP's RRBs merged into one under "One State, One RRB".

LAST-MINUTE 10

1. Allahabad Bank **1865**; PNB 1894
2. Banking Regulation Act **1949**; SBI **1955**
3. 14 banks **19 July 1969**; 6 banks **1980**
4. Lead Bank Scheme **Dec 1969**
5. IFCI 1948; ICICI 1955; IDBI 1964
6. RRB 1975, NABARD 1982, SHG 1992, KCC 1998
7. SIDBI **1990**, Lucknow
8. Rangarajan = inclusion; Nachiket Mor = payment banks; Malegam = MFIs
9. NPA 90 days; DICGC Rs. 5 lakh; Basel = capital
10. First LDB **Jhang 1920**; District = Central Coop Bank

Chapter 12

Capital Market, Insurance and Market Basics

Built from 48 UPPSC questions (1996–2025). Focus: IRDAI and insurance, SEBI and mutual funds, Sensex and indices, market terms, credit rating agencies, market basics.

ASKED MOST · REVISE FIRST

- **Sensex** = **30** companies; **Greenex** = **25** 4× 2005, 2014, 2015, 2021
- **Gilt-edged** = government securities market 3× 2002, 2004, 2008
- **Insider trading** = share market offence 3× 2007, 2009, 2017
- **Broker** is not a speculator 2× 2004, 2017
- **IRDAI** regulates insurance; **SEBI** regulates mutual funds 4× 2002, 2005, 2007, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 Insurance and Savings

- Order: **LIC (1956), UTI (1964), IDBI (1964), SEBI (1988/92)** (no exact code, *) 1× 2004
- Insurance regulator: **IRDA**; IRDAI regulates **insurance companies** 3× 2002, 2005, 2007
- **Actuaries**: **insurance** 1× 2008
- Indemnity does not apply to **life insurance** 1× 2009
- Provident fund: **contractual savings** 1× 1998
- Lost most importance: **pension funds** (per key) 1× 2010

12.2 SEBI and Exchanges

- Mutual funds regulated by **SEBI** 1× 2013
- **SEBI Act**: **1992** 1× 2016
- Private mutual funds: **1993** 1× 2011
- UTI's aims: **all** listed 1× 2004
- NSE promoters: **all** listed (IDBI, SBI, LIC, GIC) 1× 2013
- Sensex: **30** blue-chip companies 2× 2005, 2021
- **BSE Greenex**: **25** companies 2× 2014, 2015

Index	Market
Sensex (30)	BSE, Mumbai
Nifty 50	NSE
Nikkei	Tokyo
Dow Jones, Nasdaq	New York
FTSE	London
Hang Seng	Hong Kong
Brent	Crude oil price

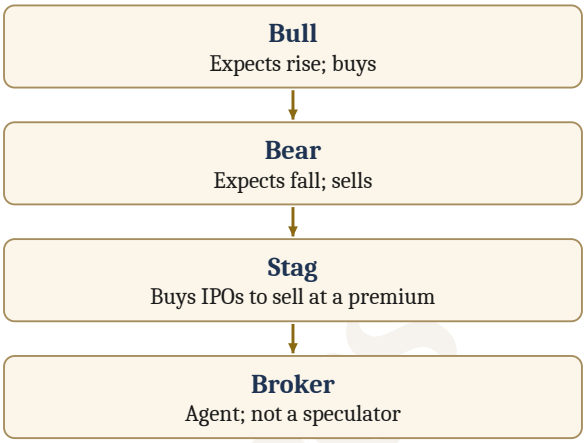
12.3 Market Terms

- **Dalal Street**: **Mumbai** 1× 2012
- **Bull and bear**: **share market** 1× 2002
- Not a speculator: **broker** 2× 2004, 2017
- **Insider trading**: share market 3× 2007, 2009, 2017
- Capital market: **share market** 1× 2008
- Not a money-market participant (key): **RBI** 1× 1999
- Commercial paper: credit for **corporates** 1× 2004
- **Blue chip**: shares with steady high returns 1× 2008
- **Gilt-edged**: **government securities** 3× 2002, 2004, 2008
- Irrelevant to share market: **SAPs** 1× 2005
- **Nikkei**: **Tokyo** share index 1× 2008
- **Brent**: **crude oil prices** 1× 2021
- Market cap above GDP (2007): India; varies now (*) 1× 2009
- Dividend declared on **subscribed capital** (per key) 1× 1996
- Different ownership: **debenture** (a loan, not ownership) 1× 2011

12.4 Credit Rating

- Indian rating agencies: **all** listed 1× 2008
- Order: **CRISIL (1987), ICRA (1991), CARE (1993)** 1× 2019
- CRISIL: rates **company creditworthiness** 1× 2008

MARKET PLAYERS



12.5 Market Basics

- Market tendency leads to **consumerism** 1× 2013
- **Buyer's market**: supply exceeds demand 1× 2010
- Vegetables: **very short-term** market 1× 2013
- Total product constant means marginal product **zero** 1× 2013
- Fixed demand, more supply: price **falls** 1× 2016
- Businessmen profit from price rise; buyers cut needs (both true, R not the explanation) 2× 2002, 2003
- Profit-volume ratio improved by **all** listed means 1× 2017
- **Shadow prices** for public goods: **J. Tinbergen** 1× 2010
- Can bargain below MRP (true); MRP must be charged (false) 1× 2025
- Trade covers **demand and supply** (only 1) 1× 2021
- **Merchandise turnover-COGS/average stock, current ratio-CA/CL, acid test-liquid assets/CL, capital employed ratio-NPBIT/total assets** 1× 2023
- A condition is essential (true); breach gives only damages (false, it allows repudiation) 1× 2023
- Commerce removes exchange hurdles and delivers goods (both) 1× 2021
- Accounting is the language of business; gives all needed information (R explains A) 1× 2023

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TRAP ALERT

- Life insurance has **no indemnity**.
- **Sensex 30**; Greenex **25**; Nifty **50**.
- Order: **CRISIL, ICRA, CARE**.
- **Buyer's market** = supply > demand.

UP LINK

UP links: SEBI-registered investors are growing fastest in UP; **LIC** (1956) and the **UP State** insurance schemes like the **Mukhyamantri Krishak Durghatna Kalyan Yojana** sit alongside IRDAI rules.

LAST-MINUTE 10

1. LIC 1956; UTI 1964; SEBI 1988 (Act 1992)
2. IRDAI = insurance (Hyderabad)
3. No indemnity in life insurance
4. SEBI = mutual funds; private MFs 1993
5. Sensex 30; Greenex 25; Nikkei Tokyo; Brent oil
6. Bull buys; bear sells; broker not speculator
7. Gilt-edged = govt securities; CP = corporates
8. CRISIL 1987, ICRA 1991, CARE 1993
9. Buyer's market: supply > demand
10. Shadow prices: Tinbergen

Chapter 13

Human Development

Built from 26 UPPSC questions (2001–2025). Focus: PQLI, HDI makers and indicators, India's HDI rank, Gross National Happiness, Kerala.

ASKED MOST · REVISE FIRST

- **HDI**: built by **Mahbub ul Haq**, published by **UNDP** since **1990** **3×** 2006, 2018, 2025
- HDI uses **health, education, income**; not social inequality, environment, urbanisation, nutrition **6×** 2004, 2008, 2009, 2013, 2020, 2024
- **Gross National Happiness**: **Bhutan** **4×** 2001, 2002, 2003, 2009
- **Kerala** = highest HDI among States **3×** 2005, 2008, 2011

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

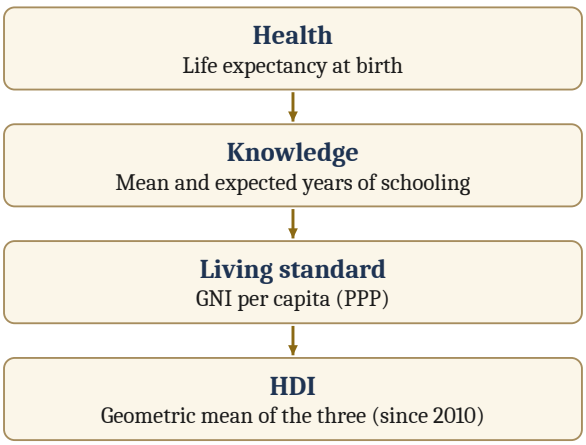
13.1 Ideas of Development

- Three stages of social development: **Auguste Comte** **1×** 2021
- Physical quality of life: **infant mortality and literacy** **1×** 2004
- **PQLI**: **Morris D. Morris** **1×** 2019
- First to stress the human side of development: **UNDP** **1×** 2013

13.2 The HDI

- Developed by **Mahbub ul Haq**; first by **UNDP** **2×** 2006, 2018
- First HDR 1990 with HDI (true); led by Haq and Priesner (false) (only 1) **1×** 2025
- Life expectancy goalposts: 85 and 25 then; **85 and 20** now (*) **1×** 2006
- Not in HDI (2004): social inequality; adult literacy also dropped since 2010 (*) **1×** 2004
- Not used: **social inequality** **1×** 2020
- HDI indicators: **income, education, life expectancy** (1, 2, 4) **1×** 2024
- Not a constituent: **health and nutrition** **2×** 2008, 2009
- Not in 1990 HDI: **urbanisation** **1×** 2013
- HDI and SDG ranks of States go together; SDGs stress health and education (R explains A) **2×** 2019, 2024

HOW HDI IS BUILT



13.3 India and the World

- HDI order (HDR 2021-22): **Hong Kong, Australia, Japan, Russia** (*) 1× 2022
- India 130 (2020) to 132 (2021): neither "upgraded" nor "slipped" wording correct per key 1× 2023
- India's 2014 rank: mixed years (135/130) (*) 3× 2001, 2004, 2015
- Better HDI from **more schooling and adult literacy** (3 and 4) 1× 2004
- "Jobless, rootless, ruthless, voiceless, futureless" growth: **UNDP** 1× 2015
- **Gross National Happiness**: **Bhutan** 4× 2001, 2002, 2003, 2009

HDR	India's rank
2021-22	132 of 191
2023-24	134 (revised 133)
2025 (2023 data)	130 of 193

13.4 States

- Highest HDI (2001): **Kerala** 2× 2005, 2008
- Order (2001): **Kerala, Punjab, UP, Bihar** 1× 2005
- Kerala first in HDI; highest unemployment (both true, R not the explanation) 1× 2011
- Highest HDI in 2008 (per key): **Uttar Pradesh** 1× 2013

TRAP ALERT

- **PQLI** = Morris D. Morris; **HDI** = Mahbub ul Haq.
- HDI never includes **inequality, environment, urbanisation**.
- **GNH** = Bhutan.

UP LINK

UP ranks low on HDI among States (below the national average) despite the 2013 key; it scores better on the SDG Index (Front Runner, 2023-24).

LAST-MINUTE 10

1. Comte = three stages
2. PQLI = Morris D. Morris (IMR, literacy, life at 1)
3. HDI = Mahbub ul Haq, UNDP, 1990
4. Health, education, income only
5. Geometric mean since 2010; goalposts 20-85
6. India 130th (HDR 2025)
7. UNDP = jobless, ruthless growth critique
8. GNH = Bhutan
9. Kerala tops States
10. HDI ties with SDG ranks

Chapter 14

Welfare Schemes I: Health, Education, Women and Social Security

Built from 93 UPPSC questions (1992–2025). Focus: health missions and launch years, ICDS, skills, women's schemes, insurance and pensions, education schemes, Aadhaar and welfare bodies.

ASKED MOST · REVISE FIRST

- **NRHM**: **12 April 2005** (Tenth Plan); village worker **ASHA** 6× 2005, 2006, 2008, 2010, 2013, 2017
- **Mid-day Meal**: **1995**, for universal primary education; renamed **PM POSHAN** (2021) 5× 2006, 2008, 2010, 2016, 2023
- **Sarva Shiksha Abhiyan**: **2001**, ages **6-14** 5× 2006, 2008, 2014, 2015, 2016
- **Aadhaar**: identity for residents; first Aadhaar village **Tembhli** 6× 2008, 2008, 2009, 2009, 2010, 2011
- **Swadhar** = women in difficult circumstances 2× 2010, 2013
- Launch years: **ICDS 1975, blindness 1976, leprosy 1983, NRHM 2005; OBB 1987, NLM 1988, SSA 2001, Saakshar Bharat 2009** 3× 2013, 2014, 2019

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Health and Nutrition

- **Ayushman Bharat**: Rs. 5 lakh per **family** (not person); Centre-State **60:40** (only 2) 1× 2020
- **RSBY** (2008 Act): **unorganised sector workers** 1× 2017
- **NRHM**: **12 April 2005**; not launched in the 11th Plan; **Tenth Plan** 5× 2005, 2006, 2008, 2010, 2013
- NRHM village worker: **ASHA** 1× 2017
- National Health Policy: **1983** 1× 1993
- Public health spend 2009-10: **1.09%** of GDP 1× 2013
- NFHS-3: private sector primary for **70% of urban** households 1× 2017
- Life expectancy 2006-10: **males 65.8, females 68.1** 1× 2008
- **Shabri Sankalp Abhiyan**: **National Nutrition Mission** 1× 2021
- Not under ICDS: **family planning**; **free books and uniforms** 2× 2020, 2017
- **Blindness control-1976, ICDS-1975, leprosy eradication-1983, NRHM-2005** 1× 2019
- **PM Jan Arogya-2018, PM Vaya Vandana-2017, PM Rojgar Protsahan-2015 (key), Swachh Bharat-2014** 1× 2020
- **PM Garib Kalyan Package-2020, Muskaan-2021, Mission Indradhanush-2014, PM-JAY-2018** 1× 2023

14.2 Skills and Innovation

- Skill schemes: **STRIVE, SANKALP, SHREYAS** (not STRIDE) 1× 2021
- Skill Development Initiative: **May 2007** 1× 2010
- Skill India launched **July 2015** (true); only IT training (false) (only 1) 1× 2020
- Skill India International Centres (Budget 2023-24): **30** 1× 2023
- Biggest Global Skill Park: **Bhopal** 1× 2017
- World Youth Skills Day 2021 theme: **Reimagining Youth Skills Post-Pandemic** 1× 2021
- **DHRUV** (10 Oct 2019): talented children 1× 2019
- **Atal Innovation Mission**: **NITI Aayog** 1× 2020
- **NIDHI**: student start-ups; supports 20 students a year (both) 1× 2020

14.3 Women and Girls

- **Beti Bachao Beti Padhao**: **January 2015** 1× 2015
- **Bhagyashree**: **Maharashtra** 1× 2014
- **Swavalambini**: **Ministry of Skill Development** (only 1) 1× 2025
- **Swawlamban**: **skills and training for women** (per key); launched **2010** 2× 2006, 2016
- **Swadhar**: **women in difficult circumstances** 2× 2010, 2013
- SHG loans (2013): **7%** 1× 2013
- Wrong: **Mahila Samakhya-1989** (it was 1988-89) 1× 2021
- **Mahila Samriddhi Yojana**: **1993** 1× 2015
- **Janani Suraksha** replaced the **National Maternity Benefit Scheme** 1× 2005
- JSY aim: cut **maternal and neonatal mortality** 1× 2023
- Free care for pregnant women and infants (June 2011): **Janani Shishu Suraksha Karyakram** 2× 2010, 2011
- Not a women's income programme: **Integrated Rural Energy Programme** 1× 2013
- **ABC index**: **health** 2× 2014, 2018

14.4 Social Security

- NPS for traders: turnover **up to Rs. 1.5 crore** 1× 2020
- **Ashraya Bima**: workers who **lost jobs** 1× 2001
- **Aam Aadmi Bima**: **rural landless BPL labour**; launched **2 October 2007** 2× 2008, 2013
- Unorganised Workers' Social Security Act: **2008** 2× 2012, 2015
- Not social security: **SJSRY** 1× 2013
- Not a social-protection measure: **skill development** 1× 2009
- Not true of PMSBY: age 18-50 (it is **18-70**); premium now Rs. 20 1× 2015
- Not an employment programme: **NSSF** 1× 2014
- **AABY-2007, PM Aatmanirbhar Swasth Bharat-2021, PM-JAY-2018, PMSBY-2016 (key; launched 2015)** 1× 2021
- **Shram Yogi Maan-dhan-Rs. 55-200/month, PMJJBY-Rs. 436/year, PMSBY-Rs. 20/year, Atal**

QUANTUM IAS & PCS

Pension-beneficiary's choice 1× 2024

- NSAP does not include **Mahila Kisan Sashaktikaran Pariyojana** 1× 2019
- Minimum Needs Programme: **investment in human resources** 1× 2004
- Not under MNP: **social forestry** 1× 2013
- Not reducing inequality: **liberalisation** 2× 2006, 2009
- **NSAP**: **old-age pension** for the very poor 1× 2008

Insurance / pension	Premium	Cover
PMJJBY (2015)	Rs. 436/year	Life Rs. 2 lakh, 18-50
PMSBY (2015)	Rs. 20/year	Accident Rs. 2 lakh, 18-70
Atal Pension (2015)	By age and pension	Rs. 1,000-5,000
PM-SYM (2019)	Rs. 55-200/month	Rs. 3,000 pension at 60
PM-JAY (2018)	Free	Rs. 5 lakh per family

14.5 Education

- **Kasturba Gandhi Balika Vidyalaya**: **2004** 1× 2016
- **SSA**: **2001**; ages **6-14** 5× 2014, 2006, 2008, 2015, 2016
- **Saakshar Bharat**: **female literacy** 1× 2015
- RTE: free education up to **elementary level** 1× 2015
- Not a social programme: **Look East Policy** 1× 2015
- Mid-day Meal (1995) for **universal primary education**; started **1995** 3× 2008, 2006, 2016
- Renamed **PM POSHAN** in **2021** 1× 2023
- Run by the **Ministry of HRD** (Education) 1× 2010
- Does not run mid-day meals: **contractor** 1× 2013
- **Akshaya Patra**: **mid-day meals** 1× 2015
- UP "Go to School" campaign: **2000** 1× 2012
- **Kalp Yojana**: **primary education** 2× 2001, 2010
- UP **Shiksha Mitra**: village youth teach (true); teacher-pupil norms (true, not the explanation) 1× 2012
- **DPEP**: **1994** 1× 2004
- **Prarambhik Shiksha Kosh**: receives the **education cess** 1× 2005
- **Operation Blackboard**: **primary education** 1× 1993
- IEDSS (2009-10) under **RMSA** 1× 2016
- **SSA-2001, Saakshar Bharat-2009, Operation Blackboard-1987, NLM-1988** 2× 2013, 2014
- **Roshni**: skills for **tribal youth** 1× 2013
- **Nalanda Project**: **Ministry of Minority Affairs** 2× 2014, 2015

EDUCATION SCHEMES IN ORDER



14.6 Other Bodies

- **Bonded Labour (Abolition) Act**: **1975** (Ordinance; Act 1976) 1× 2002
- **National Children's Fund**: **1979** 1× 1993
- **CAPART**: helps and evaluates rural welfare schemes; under **Rural Development Ministry** 2× 1997, 2006
- **Aadhaar**: identity for residents; 12-digit number brand 3× 2008, 2009, 2009
- **Tembhli**: first Aadhaar village 3× 2008, 2010, 2011
- **PURA**: **A.P.J. Abdul Kalam** 1× 2014
- PM's Shram Awards: **PSU and selected private-unit employees** 1× 2013
- National Handicapped Finance Corporation: **Social Justice Ministry** 1× 2015

- National Institute for the Visually Handicapped: **Dehradun** 1× 2012
- First open university: **Andhra Pradesh** (1982) 1× 1992
- **Tarun Bharat Sangh**: **Rajasthan** 1× 2003

TRAP ALERT

- PM-JAY = **Rs. 5 lakh per family**, not per person.
- **STRIDE** is a UGC research scheme, not skills.
- **ICDS** has no family planning or free books.
- Mid-day Meal = **PM POSHAN** since 2021.

UP LINK

UP schemes asked: **Shiksha Mitra** (village youth as teachers), **School Chalo / Go to School** campaign (2000), **Kalp Yojana**; UP is the biggest user of PM-JAY and mid-day meals by numbers.

LAST-MINUTE 10

1. NRHM **12 Apr 2005**; ASHA; NHP 1983
2. ICDS 1975; Indradhanush 2014; POSHAN Abhiyaan 2018
3. PM-JAY 2018: Rs. 5 lakh per family, 60:40
4. Skill India **15 July 2015**; Global Skill Park Bhopal
5. BBBP **Jan 2015**; Swadhar = women in difficulty
6. JSY replaced Maternity Benefit Scheme; JSSK 2011
7. PMSBY Rs. 20; PMJJBY Rs. 436; AABY 2007
8. Mid-day Meal 1995 to PM POSHAN 2021
9. SSA 2001 (6-14); RTE 2009; Saakshar Bharat = women
10. Aadhaar village Tembhli; PURA = Kalam

Chapter 15

Welfare Schemes II: Rural Development, Employment and Basic Services

Built from 65 UPPSC questions (2001–2025). Focus: early rural programmes, the order of employment schemes, MGNREGA, Jan Dhan and energy schemes, sanitation, housing and urban missions.

ASKED MOST · REVISE FIRST

- **SGSY**: **1999**; did not include JRY; became **NRLM** (2011) 7× 2001, 2002, 2002, 2003, 2004, 2006, one-liner
- **MGNREGA**: **100 days** a year per household; 200 districts first, whole country **1 April 2008** 5× 2005, 2006, 2006, 2008, 2008
- **Sangam Yojana** = welfare of the **disabled** 4× 2004, 2006, 2008, 2021
- Scheme-year matching: **JRY 1989, EAS 1993, SGSY 1999, PMGY 2000, SGRY 2001, KSSY 2001, NRHM 2005, NSAP 1995, SJSRY 1997** 3× 2002, 2016, 2022
- **Swachh Bharat (Gramin)**: **October 2014**, target **2019** 3× 2014, 2015, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Early Rural Development

- **Etawah Pilot Project**: **Albert Mayer**; started **1948** 3× 2016, 2023, 2013
- Order: **Key Village Scheme, CDP, democratic decentralisation, DRDA** 1× 2021
- **CDP**: **2 October 1952**; architect **S.K. Dey** 2× 2011, 2014
- Order: **CDP, Drought Prone Area Programme, Food for Work, TRYSEM** 1× 2025
- **TRYSEM**: **rural development** programme 1× 2014
- Order: **IRDP, NREP, DWCRA, JRY** 1× 2025
- Wrong: **JRY-April 1995** (JRY began **1989**) 1× 2005
- Not in SGSY: **JRY** 3× 2001, 2002, 2003

- SGSY: **1999**; became **NRLM** (June 2011) 4× 2002, 2004, 2006, one-liner
- Two wrong pairs: Bharat Nirman (infrastructure), SJSRY (urban) (*) 1× 2015
- Order: **PM Rozgar Yojana (1993), REGP (1995), PMGY (2000), VAMBAY (2001)** 1× 2024
- Order: **IRDP, SGSY, NREGA, Jan Dhan** 1× 2024
- **Krishi Shramik Samajik Suraksha-2001, SGSY-1999, EAS-1993, PM Gramodaya-2000** 1× 2002
- **SGSY-1999, SJSRY-1997, JRY-1989, NRHM-2005** 1× 2016
- **NRHM-2005, NSAP-1995, SGSY-1999, SGRY-2001** 1× 2022
- Wrong: **MGNREGA-2003** (Act 2005, scheme 2006) 1× 2015
- Not employment-related: **RMSA** 1× 2015

RURAL EMPLOYMENT SCHEMES IN ORDER



15.2 MGNREGA

- Extended (2007-08) to **330 districts** 1× 2008
- First in **200 districts** 1× 2006
- Whole country from **1 April 2008** 1× 2006
- Guarantee: **100 days** a year 2× 2005, 2008
- Budget 2020-21 cut: **Ayushman** (per key) 1× 2016
- Budget 2023-24 statements on MGNREGA and JJM: **neither** correct 1× 2023
- NREGA does not guarantee 100 days to **every adult member** (it is per household) 1× 2008
- MGNREGA gains: **more purchasing power, less migration** (1 and 3) 1× 2015

15.3 Inclusion and Energy

- **Aspirational Districts Programme**: **January 2018** 1× 2021
- **Saubhagya**: **free electricity** connections for poor households 1× 2017
- **PM Jan Dhan**: **28 August 2014** 1× 2015
- Order: Jan Dhan, Make in India, Sukanya, Atal Pension (no code, *) 1× 2017
- Order: **Jan Dhan, Digital Gender Atlas, Mudra, PMSBY** 1× 2017
- Guinness-record DBT: **PAHAL** 1× 2019
- **Ujjwala** launched at **Ballia** (UP), 1 May 2016 1× 2016
- DDU Gram Jyoti: rural electricity (no single option, *) 1× 2015
- **RIDF**: under **NABARD**; administered by NABARD 2× 2014, 2013
- Not under RIDF: **rural industries** 2× 2010, 2013

15.4 Water and Sanitation

- National Rural Drinking Water Programme: part of **Bharat Nirman** 1× 2015
- **Jal Jeevan Mission**: Budget **2019-20** 1× 2022
- **PMAY-G-November 2016, JJM-August 2019, SBM-G-October 2014, Ujjwala-May 2016** 1× 2024
- Order: **Swachh Bharat, PMKVY, PM Garib Kalyan, Mahila Shakti Kendra** 1× 2024
- Wrong: **Swachh Bharat-safe drinking water** (it is sanitation) 1× 2017
- Clean India target: **2019** 1× 2014
- SBM (Grameen): **October 2014** 1× 2015
- Rural Sanitation Strategy: **2019-2029** 1× 2019
- SBM is about community behaviour; open defecation tied to routine (both) 1× 2019
- Not in the top three (Swachh Survekshan 2021): **Tirupati** (Indore, Surat, Vijayawada were) 1× 2021
- UP in Swachh Survekshan 2021: **sixth** 1× 2021
- People did not see open defecation as a threat; purity norms kept toilets out (both true, R not the explanation) 1× 2019

QUANTUM IAS & PCS

- Nirmal Bharat Abhiyan: none of the options (key) 1× 2012
- Nirmal Gram Puraskar: **sanitation** (ODF villages) 1× 2008
- Toilet revolution: **subsidy and loans for flush toilets** 1× 2001
- ILCS: shared funding; **75% central subsidy** (I and II) 1× 2016

Scheme	Launch
Swachh Bharat	2 Oct 2014
PM Jan Dhan	28 Aug 2014
PMAY (Urban) / AMRUT	25 June 2015
Ujjwala	1 May 2016 (Ballia)
PMAY (Gramin)	20 Nov 2016 (Agra)
Saubhagya	Sept 2017
Jal Jeevan Mission	Aug 2019

15.5 Housing and Urban Missions

- Wrong: **Rurban Mission-2015** (it began **2016**) 1× 2023
- AMRUT: covers about **60%** of urban people (true); 400 cities (false, **500**) (only 2) 1× 2016
- Order: **JNNURM (2005), Urban Housing Policy (2007), AMRUT (2015), JJM (2019)** 1× 2025
- **Sugamya Bharat**: **persons with disabilities** 1× 2021
- **Sangam Yojana**: **welfare of the handicapped** 4× 2004, 2006, 2008, 2021
- **Sankalp**: **HIV/AIDS** 2× 2008, 2012
- **PM Gramodaya**: rural infrastructure (schools, health, roads, housing, water) 2× 2004, 2013
- **AMRUT**: **urban infrastructure** 2× 2015, 2016
- PMAY span: Housing for All by 2022, since extended to 2029 (*) 1× 2016
- **Rajiv Awas Yojana**: **slum-free India** 1× 2010

TRAP ALERT

- NREGA guarantee is per **household**, not per adult.
- **Swachh Bharat** = sanitation, not drinking water.
- **AMRUT** = **500** cities.
- **Ujjwala** began at **Ballia**, UP.

UP LINK

UP links: the **Etawah Pilot Project** (1948) was the seed of community development; **Ujjwala** began at **Ballia** (2016) and **PMAY-Gramin** at **Agra** (2016); UP ranked **sixth** in Swachh Survekshan 2021.

LAST-MINUTE 10

1. Etawah 1948 = Albert Mayer; CDP 2 Oct 1952 = S.K. Dey
2. TRYSEM 1979; IRDP 1978-80; JRY 1989; EAS 1993
3. SGSY 1999 (not JRY) to NRLM 2011
4. NREGA: 200 districts (2006) to all India 1 Apr 2008; 100 days/household
5. Jan Dhan **28 Aug 2014**; PAHAL = Guinness DBT
6. Ujjwala Ballia 2016; Saubhagya = electricity
7. RIDF = NABARD; JJM 2019
8. SBM Oct 2014, target 2019; strategy 2019-29
9. AMRUT 500 cities; Rurban 2016; RAY = slum-free
10. Sangam & Sugamya = disabled; Sankalp = HIV

Chapter 16

Poverty and Unemployment

Built from 62 UPPSC questions (1997–2025). Focus: poverty-line committees and calorie norms, state-wise poverty, the MPI, disguised unemployment and NSSO measures.

ASKED MOST · REVISE FIRST

- Poverty line rests on **household consumption expenditure**; calorie norm **2,400 rural, 2,100 urban** 6× 2001, 2008, 2008, 2009, 2013, 2020
- **Committees**: **Alagh** (calories only), **Lakdawala** (official line then), **Tendulkar** (**37.2%**), **Hashim** (urban BPL); order **Dandekar-Rath, Lakdawala, Tendulkar, Rangarajan** 7× 2012, 2013, 2013, 2013, 2018, 2020, 2025
- **MPI**: **OPHI + UNDP**, 2010, replaced **HPI** (1997); **10 indicators** in health, education, living standard 7× 2004, 2010, 2015, 2019, 2020, 2024, 2025
- **Disguised unemployment**: more workers than needed; **agriculture, rural areas** 6× 2004, 2004, 2006, 2008, 2010, 2017
- State poverty: **Odisha** highest (1999-2000); **UP** most poor (2011-12); **MP** = "Ethiopia of India" 7× 2002, 2003, 2005, 2010, 2010, 2014, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 What Poverty Is

- Day for the Eradication of Poverty theme 2022-23: **"Dignity for all in practice"** 1× 2023
- Types of poverty: **absolute, relative, subjective** (not "functional") 1× 2023
- UN defined **absolute poverty** in **1995** (Copenhagen Summit) 1× 2017
- Poverty line differs across time and countries because basic needs vary (A and R true, R explains A) 1× 2025
- Not a social indicator of poverty: **size of the house** 1× 2025
- "The Challenge of World Poverty": **Gunnar Myrdal** 1× 2023

16.2 The Poverty Line

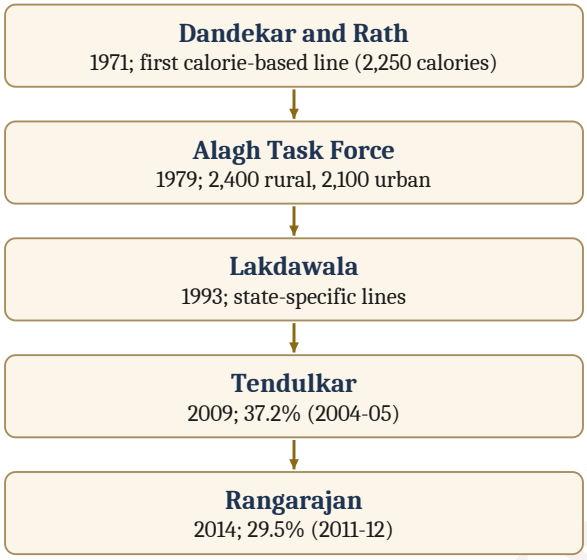
- Methods used: **head count ratio, calorie intake, household consumption expenditure** (not per capita income) 1× 2020
- Poverty level is fixed on **household consumption expenditure** 2× 2009, 2013
- Defined in terms of **calorie intake** 1× 2008
- Line based only on nutrition: **Alagh** Committee (1979) 1× 2020
- Calories: **2,100 urban** and **2,400 rural** per person a day 2× 2001, 2008
- New BPL standards committee: **Suresh Tendulkar** 1× 2013
- Tendulkar estimate: **37.2%** below the line (2004-05) 2× 2012, 2013
- Lines used for estimates (then): **Lakdawala** Committee 1× 2013
- Urban BPL identification: **S.R. Hashim** Committee 1× 2018
- Order: **Dandekar-Rath, Lakdawala, Tendulkar, Rangarajan** 1× 2025
- Fixes the poverty line: **Planning Commission** (now **NITI Aayog**); NITI is the nodal agency 2× 2014, 2018
- **Lorenz curve** measures **income inequality** 1× 2018
- **Cultural poverty**: **Oscar Lewis**; **vicious circle of poverty**: **Nurkse** 2× 2020, 2014
- BPL people as an identified group helps **targeting** (A and R true, R explains A) 1× 1998
- Rich and poor both malnourished: wrong diet and poor diet (A and R true, R explains A) 1× 2010
- Poverty in 2009-10 (Planning Commission, April 2011): **30%** (29.8%) 1× 2011
- 11th Plan aim: cut the poverty ratio by **10 percentage points** by 2012 1× 2013

Committee	Year	Basis	Estimate
Task Force (Y.K. Alagh)	1979	Calories only: 2,400 rural, 2,100 urban	-
Lakdawala	1993	Calorie line with state prices	36% (1993-94)

QUANTUM IAS & PCS

Committee	Year	Basis	Estimate
Tendulkar	2009	Consumption basket incl. health, education	37.2% (2004-05); 21.9% (2011-12)
Rangarajan	2014	Food and non-food; Rs 32 rural, Rs 47 urban a day	29.5% (2011-12)
S.R. Hashim	2012	Identifying urban BPL families	-

POVERTY COMMITTEES IN ORDER



16.3 Poverty Across States and Hunger

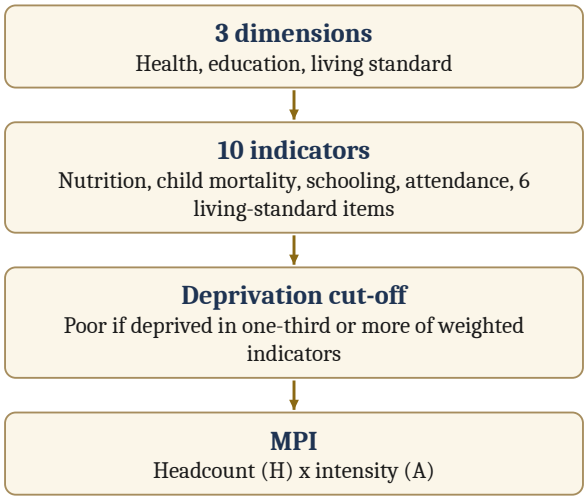
- India's **Global Hunger Index 2017** rank: **100th** (of 119) 1× 2019
- Highest share of poor (NCAER survey): **Bihar** 1× 2009
- Bihar is backward; regional disparity is true but not the reason (A and R true, R does not explain A) 1× 2015
- Highest share of poor (Planning Commission, 1999-2000): **Odisha** 4× 2002, 2003, 2005, 2014
- Largest **number** of poor (2011-12): **Uttar Pradesh** 1× 2014
- Least share of poor: **Himachal Pradesh** (2011), **Goa** (2010), **Kerala** (2004, 2010), per the options given 4× 2004, 2010, 2010, 2011
- **Madhya Pradesh** = **"Ethiopia of India"** for malnutrition and infant deaths (A and R true, R explains A) 2× 2010, 2010
- No clear picture of poverty extent; strategy shifts are true but not the reason (A and R true, R does not explain A) 1× 1997
- PDS: aim was **subsidised essentials** and it began as **price support** (1 and 2); food subsidy is paid by the **Centre** 1× 2013
- Rural poverty programmes: **all of the above** 1× 2010

State fact	Answer
Highest ratio, 1999-2000	Odisha (about 47%)
Highest ratio, 2011-12 (Tendulkar)	Chhattisgarh (39.9%)
Largest number of poor, 2011-12	Uttar Pradesh (about 6 crore)
Lowest ratios	Goa, Kerala, Himachal Pradesh
"Ethiopia of India"	Madhya Pradesh
Global Hunger Index	100th of 119 (2017); 105th of 127 (2024)

16.4 Multidimensional Poverty

- UNDP poverty categories: **both** income poverty and human poverty 1× 2022
- **Human Poverty Index**: HDR **1997**; replaced by **MPI** in **2010** 2× 2004, 2020
- Global MPI report: **OPHI and UNDP** (3 and 4) 1× 2024
- MPI in the HDR every year since **2010** is correct; it does not measure only BPL people (only 1) 1× 2025
- MPI has **10 indicators** 1× 2010
- MPI dimensions: **health, education, living standard** 1× 2019
- Best index for the **intensity** of poverty: **MPI** 1× 2015
- India's MPI-poor share (HDR 2013): **53.7%** 1× 2014
- Not a cause of poverty: **rising rate of investment** 1× 2013
- Not a social indicator of poverty: **fewer means of transport** 1× 2017
- Indian social structure: **(i), (ii), (iii)**; poverty is not a structural feature 1× 2014

HOW THE MPI IS BUILT



16.5 Unemployment

- Educated unemployment rises with fast growth when professional education is lacking (A and R true, R explains A, per key) 2× 2005, 2006
- Urban poverty is rooted in rural areas; low rural education (A and R true, R explains A) 1× 2013
- **Disguised unemployment**: more people on a job than needed 3× 2004, 2008, 2010
- Found mainly in the **primary (agricultural)** sector, in **rural** areas (I and II) 3× 2004, 2006, 2017
- Unemployed for the major part of the year: **usual status** unemployment 1× 2019
- Not an NSSO measure: **current monthly status** 2× 2005, 2016
- Unemployment data collected and published by **NSSO** 1× 2014
- Not a cause of unemployment: **growth of per capita income** 1× 2014
- Rate between Jul-Sep 2019 and Jul-Sep 2022: key says **increased**; PLFS urban data show about 8% to about 7%, so treat with care (*) 1× 2024

Measure (NSSO / PLFS)	Reference period
Usual Principal Status	Major part of the year (365 days); chronic unemployment
Current Weekly Status	The week before the survey
Current Daily Status	Each day of the week; best for underemployment
PLFS	Since 2017-18; monthly estimates since 2025

TRAP ALERT

- Calories: **2,400 rural, 2,100 urban** (not the reverse). Poverty line = **consumption**, not income.
- **Odisha** = highest ratio (1999-2000); **UP** = most poor in number; **MP** = "Ethiopia of India".
- MPI = **OPHI + UNDP**, not World Bank; **10** indicators; replaced HPI in **2010**.
- NSSO has no "current monthly status"; disguised unemployment = **rural agriculture**, not urban manufacturing.

UP LINK

UP link: UP had the **largest number of poor** in 2011-12 (about 6 crore); in NITI Aayog's national MPI, UP saw the **largest fall**, with about **5.94 crore** people leaving multidimensional poverty (2013-14 to 2022-23).

LAST-MINUTE 10

1. Poverty line = **consumption expenditure**; 2,400 rural, 2,100 urban
2. Alagh = calories only; **Tendulkar 37.2%**; Hashim = urban BPL
3. Order: Dandekar-Rath, Lakdawala, Tendulkar, Rangarajan
4. Absolute poverty defined **1995**; Myrdal = Challenge of World Poverty
5. Lorenz = inequality; **Oscar Lewis** = culture of poverty; **Nurkse** = vicious circle
6. **Odisha** highest (1999-2000); **UP** most poor; **MP** = Ethiopia of India
7. GHI 2017 = **100th**
8. HPI **1997** to MPI **2010**; OPHI + UNDP; **10 indicators**
9. Disguised = agriculture, rural; UPS = major part of the year
10. No "current monthly status" in NSSO

Chapter 17

India's Foreign Trade

Built from 51 UPPSC questions (1994–2023). Focus: trade size and share, trading partners, balance of trade versus balance of payments, top imports and exports.

ASKED MOST · REVISE FIRST

- **Trading partners**: **USA** = top export market and largest surplus; **China** = largest import source; **UAE** led exports in 2009-10 **6×** 2007, 2010, 2010, 2010, 2013, 2021
- **Asia (and ASEAN)** = largest trade region and over half of imports **3×** 2006, 2009, 2010
- **Petroleum and crude oil** = biggest import; with **capital goods** the top two **4×** 1995, 2004, 2004, 2008
- **Engineering goods** = largest export group now **2×** 2017, 2023
- **Balance of trade** = **goods** only; BoP = all transactions with the rest of the world **5×** 1994, 2009, 2012, 2016, 2017

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

17.1 Size of India's Trade

- Commerce Department's vision: a major world-trade player by **2020** **2×** 2014, 2017
- Total trade 2011-12: far above every option (*); about Rs 38 lakh crore **2×** 2014, 2015
- Trade-GDP ratio statements: none exactly right (*) **1×** 2016
- Merchandise trade-GDP: **29.5% (2004-05) to 40.6% (2012-13)** **1×** 2014
- Goods trade-GDP: 14.2% (1990-91) to **almost three times** by 2011-12 **1×** 2013
- Goods and services exports: about **22%** of GDP (2010-11) **1×** 2010
- India's share of world exports: 1.85% (1950), about 0.6% in 1998 (*); about **1.8%** now **1×** 1998
- Largest share of world trade: **China** now (USA when asked) **1×** 2013

Trade fact (latest)	Figure
Share of world goods exports	About 1.8%
Share of world services exports	About 4.3%
Top goods exporter	China
Total exports 2024-25	About US$ 825 billion (goods + services)
FTP 2023 target	US$ 2 trillion exports by 2030

17.2 Trading Partners

- Largest trade region (2008-09): **Asia and ASEAN** **1×** 2009
- Largest share of imports (2009-10): **Asia and ASEAN** **1×** 2010
- More than 50% of imports from **Asian** countries **1×** 2006
- Largest import group when asked: **OPEC** (oil); top single sources now China, Russia, UAE, USA **1×** 2001
- Top two partners in 2011: **China and UAE** **1×** 2013
- Largest trading partner: **USA** **1×** 2007
- Largest trade surplus (2019-20): **USA**; surplus on current account with **USA** **2×** 2004, 2021
- Largest import source by value: **China** **1×** 2010
- Largest export market: **USA** now (UAE in 2009-10) **2×** 2010, 2010
- USA's share of India's exports: about **20%** **1×** 2002

Partner (2024-25)	Fact
USA	Largest export market; largest trade surplus
China	Largest import source; largest trade deficit
UAE	2nd export market; 3rd import source
Russia	2nd import source (crude since 2022)

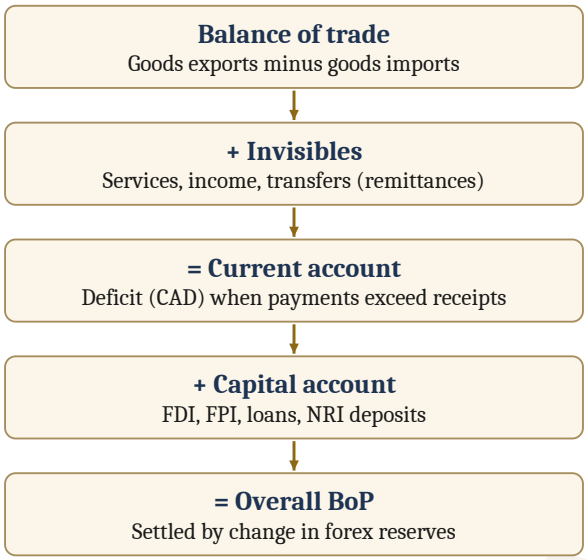
17.3 Balance of Trade and Payments

- Trade deficit peaked 2012-13, then fell as POL and non-POL deficits fell (A and R true, R explains A) **1×** 2017
- Balance of trade includes **goods** **1×** 2016
- Net exports = **exports minus imports** **1×** 2017

QUANTUM IAS & PCS

- Balance of payments relates to **exports and imports** 1× 2012
- BoP includes **all of the above** 1× 2009
- BoP = complete record of all economic transactions of residents with the **rest of the world** 1× 1994
- CAD definition is true; CAD comes from **importing** more, not exporting more (A true, R false) 1× 2016

BALANCE OF PAYMENTS



17.4 What India Imports

- Most important import over the years: **petroleum and crude oil** 1× 2004
- Two largest imports: **capital goods and petroleum** 1× 1995
- 2004-05 import order: **fuel, capital goods, gold and silver, chemicals** 1× 2008
- Most important imports: **petroleum and edible oil** (per key) 1× 2004
- Biggest **LNG** supplier: **Qatar** 1× 2011
- Import procedure begins with an **indent** 1× 2011
- Letter of Credit is produced by the **importer** 3× 2009, 2011, 2013

17.5 What India Exports

- Largest export share 2004-05: **manufactured goods** 1× 2006
- Greatest export share: **engineering goods** 2× 2017, 2023
- Largest export earner: key says **agricultural and allied products** (gems and jewellery when asked); treat with care (*) 1× 2007
- Single largest export item: key says **chemicals** (gems and jewellery when asked) (*) 1× 2012
- Largest export share: key says **electronic goods** (gems and jewellery when asked) (*) 2× 2003, 2004
- Gems and jewellery share (2010): about **18%** 1× 2008
- World textile export rank: **fourth** then, about sixth now (*) 1× 2007
- Largest buyer of Indian textiles and leather: **USA** 3× 1994, 1998, 2002
- Top exporting state 2007-09: **Maharashtra**; **Gujarat** leads now 1× 2008
- Largest steel buyer 2003-04: **China** 1× 2003
- Highest export growth year: **2004-05** 1× 2008
- Largest rice exporter (2015), replacing **Thailand**: **India** 1× 2016
- Largest tea exporter: **Kenya** 1× 1998
- Long-staple cotton leader: **Egypt** 1× 2002
- Largest forex-earning service: **IT and software** 1× 2008

Export fact	Leader
Largest export group	Engineering goods (about US$ 117 billion)
Top exporting state now	Gujarat (about 30%)
Largest services export	IT and software
Textiles and leather buyer	USA
Rice	India, largest exporter since 2012
Tea	Kenya, largest exporter

TRAP ALERT

- **Balance of trade** = goods only; current account adds services, income, transfers.
- **USA** = export market and surplus; **China** = import source and deficit.
- **Importer** produces the L/C; import starts with an **indent**.
- Old keys (gems and jewellery, UAE) changed; read the year of the question.

QUANTUM IAS & PCS

UP LINK

UP link: UP's top exports are **carpets (Bhadohi), brassware (Moradabad), leather (Kanpur, Agra), handicrafts and engineering goods**; **ODOP** backs them, and UP is among the top exporting states, with goods exports of roughly **Rs 1.7 lakh crore** a year (soft figure).

LAST-MINUTE 10

1. Asia = largest trade region; over 50% of imports
2. **USA** = top export market and surplus; **China** = top import source
3. Petroleum + capital goods = top imports; **Qatar** = LNG
4. Engineering goods = largest export group
5. Balance of trade = **goods**; net exports = exports minus imports
6. BoP = all transactions with the rest of the world
7. **Importer** produces the Letter of Credit
8. India passed **Thailand** in rice (2015); **Kenya** = tea exporter
9. **Egypt** = long-staple cotton; IT software = top service
10. Merchandise trade-GDP: 29.5% to **40.6%**

Chapter 18

Trade Policy, SEZs and Export Promotion

Built from 45 UPPSC questions (1993–2023). Focus: trade terms, globalisation, Foreign Trade Policy, EPZs and SEZs, export bodies, tourism and trade facts.

ASKED MOST · REVISE FIRST

- **SEZ**: policy **April 2000**; Act passed **2005**, in force **2006** 6× 2007, 2009, 2010, 2014, 2015, 2017
- **First EPZ in Asia**: **Kandla, 1965**; first private EPZ: **Surat** 3× 2005, 2009, 2011
- **Free trade zones**: to promote **export industries** 3× 2002, 2003, 2007
- **Invisible export** = **services**; **entrepot trade** = goods imported for re-export 3× 2002, 2006, 2007
- **ECGC** = export credit insurance; **EXIM Bank** = **1982** 3× 2001, 2005, 2009

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Trade Concepts

- **Invisible export**: export of **services** 2× 2006, 2007
- **Entrepot trade**: goods **imported for export** 1× 2002
- A **tariff reduces the volume of trade** 1× 2017
- **Duty drawback**: **refund of import duty** to exporters 1× 2011
- Slow growth of exports: **all of the above** causes 1× 2017

Term	Meaning
Invisible exports	Services (IT, tourism, shipping) and remittances
Entrepot trade	Imports for re-export (Singapore, Dubai)
Duty drawback	Refund of import duty on inputs of exports
RoDTEP (2021)	Refund of embedded taxes on exports; replaced MEIS
Anti-dumping duty	On imports sold below home price; India is a top user
Super 301	US law against countries with unfair trade barriers

18.2 Globalisation and Trade Policy

- **Globalisation**: **dismantling barriers** to international trade 1× 2017
- Globalisation of India: **minimum restrictions** on economic ties with other countries 1× 2017
- Not correct about globalisation: it created **optimism among small entrepreneurs** 1× 2010

QUANTUM IAS & PCS

- More MNCs serve **all of the above** policies (liberalisation, privatisation, globalisation) 1× 2004
- After 1991: export share, **FDI** and **forex reserves** rose; agriculture's share did **not** (2, 3 and 4) 1× 2017
- **Free trade policy**: **absence of tariffs** 1× 2005
- FTP 2004-09 target: **1.5%** of world trade by 2009 1× 2007
- High-performing exporters: **Star Trading Houses** 1× 2011
- Towns of Export Excellence: all four options were named (*) 1× 2013

STATUS HOLDERS UNDER THE FTP



18.3 EPZs and SEZs

- Free trade zones: for **export industries**, free of excise, mainly export production 3× 2002, 2003, 2007
- SEZ policy announced: **April 2000** 3× 2014, 2015, 2017
- SEZ Act passed **2005**; in force **2006** (10 February) 3× 2007, 2009, 2010
- Not an SEZ aim: **discouraging foreign investment** 1× 2009
- First EPZ in Asia: **Kandla, 1965** 1× 2011
- City without an SEZ, place without an FTZ: no option fits (*) 2× 1999, 2001
- First EPZ 1965, SEZ policy 2000 and Patola-Vadodara correct per key (1, 2, 3); Panna is **diamonds**, not gold (Patola is chiefly Patan) 1× 2009
- An EPZ: **Falta** (West Bengal) 1× 2023
- First private EPZ: **Surat** 1× 2005
- Highest SEZ export growth: **2007-08** 1× 2007
- Most SEZs (June 2012): **Andhra Pradesh** 1× 2012
- SEZ India app (2017): **Ministry of Commerce and Industry** 1× 2017

EPZ / SEZ fact	Detail
First EPZ in Asia	Kandla (Gujarat), 1965
Other EPZs	SEEPZ Mumbai, Falta, Noida, Chennai, Kochi, Visakhapatnam
First private EPZ	Surat
SEZ policy	April 2000
SEZ Act	2005; in force 10 February 2006
Most SEZs (2012)	Andhra Pradesh

18.4 Export-Promotion Bodies

- Channel exports and imports: **MMTC, EXIM Bank, STC, FCI** (all four) 1× 1999
- **EXIM Bank**: **1982** 1× 2005
- Promote foreign trade: **all of the above** 1× 2011
- **ECGC**: **export financing and insurance** against risks 2× 2001, 2009
- **India Brand Equity Fund**: **1996** 1× 1997
- **MPEDA**: set up **1972**; HQ is **Kochi**, not Kolkata (only 1) 1× 2023

Body	Year	Role
STC	1956	State trading
ECGC	1957	Export credit insurance
MMTC	1963	Minerals and metals
MPEDA	1972	Marine exports; HQ Kochi
EXIM Bank	1982	Export-import finance
IBEF	1996	Brand India

18.5 Tourism and Trade Relations

- Most-visited country: **France** 2× 2006, 2013
- Favourite destination of Indian tourists (2010): **Singapore** 1× 2013
- Foreign tourist arrivals fell in **2009-10** 1× 2010
- Best tourist city (Travel and Leisure, 2009): **Udaipur** 1× 2010
- **Nathu La** reopened (2006) for border trade with **China** 1× 2006

- Most anti-dumping measures (1995-2005): **India** 1× 2005
- **Super 301**: against **barriers to free trade** 1× 1993
- IITF 2021: theme **Atmanirbhar Bharat**; held in **New Delhi**, not Jaipur (only 2) 1× 2021

TRAP ALERT

- **Invisible** = services; **entrepot** = re-export; duty drawback = **refund**, not a tax.
- SEZ: policy **2000**, Act **2005**, in force **2006**; do not mix the three.
- **Kandla** = first EPZ in Asia; **Surat** = first private EPZ; **Falta** = EPZ in West Bengal.
- MPEDA HQ = **Kochi**; IITF = **Pragati Maidan, New Delhi**.

UP LINK

UP link: **Noida EPZ** (1985) is now the **Noida SEZ**; UP's **ODOP** (2018) and the **Districts as Export Hubs** plan push exports from districts such as Bhadohi (carpets) and Moradabad (brass); the **UP International Trade Show** is held at Greater Noida.

LAST-MINUTE 10

1. Invisible = **services**; entrepot = **import for export**
2. Tariff **reduces** trade; duty drawback = refund of import duty
3. Free trade = **no tariffs**; globalisation = dismantling barriers
4. SEZ policy **April 2000**; Act **2005**; in force **2006**
5. **Kandla 1965** = first EPZ in Asia; **Surat** = first private EPZ
6. **Falta** = EPZ; AP had most SEZs (2012)
7. EXIM Bank **1982**; ECGC = export insurance; IBEF **1996**
8. MPEDA **1972**, Kochi
9. Nathu La = **China** trade (2006); France = most visited
10. India = most anti-dumping cases; Super 301 = USA

Chapter 19

Exchange Rate, FDI and External Debt

Built from 41 UPPSC questions (1991–2024). Focus: devaluation, rupee convertibility, FDI sources and limits, external debt, forex reserves, the euro and FEMA.

ASKED MOST · REVISE FIRST

- **Devaluation**: cut in a currency's value; **exports cheaper, imports costlier**; India: **1949** (first), 1966, **1991** (twice) 9× 1991, 1994, 1998, 2001, 2003, 2004, 2008, 2008, 2012
- Rupee **convertible on current account** (**1994**); capital account studied by **Tarapore** 3× 2004, 2007, 2017
- **FDI source**: **Mauritius** earlier, **Singapore** now (tax treaty change) 3× 2006, 2010, 2024
- **Forex reserves**: **foreign currency assets** are the largest part; **SDRs are included** 3× 2001, 2002, 2023
- **FERA** replaced by **FEMA**; fully in force **2002** 2× 2007, 2013

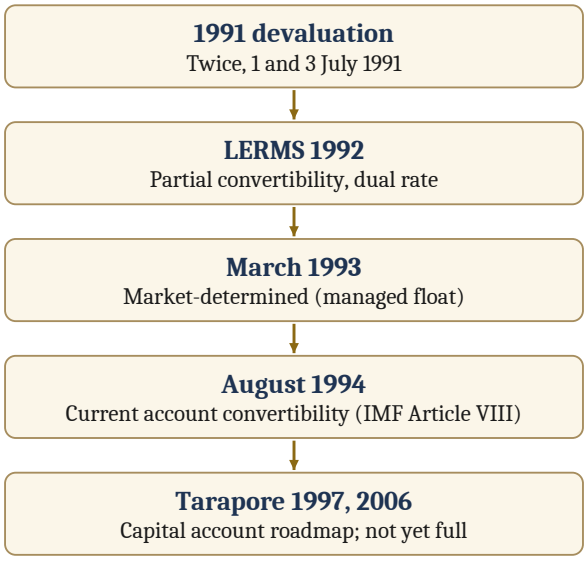
■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Exchange Rate and Convertibility

- Flexible rate is set by demand and supply; that supply is **not** set by the central bank (A true, R false) 1× 2024
- Rupee convertibility now: for **current account** transactions only 1× 2004
- Fully convertible on current account: **1994** 1× 2017
- Full capital account convertibility: **Tarapore** Committee 1× 2007
- **Devaluation**: **reduction in value** against major currencies 3× 1991, 1994, 1998
- First devaluation: **1949** 1× 2008
- Devalued twice in **1991-92** 1× 2012
- Devaluation may promote exports as prices abroad fall (A and R true, R explains A) 2× 2001, 2004
- Effect: **imports costlier, exports cheaper** 1× 2008

- Aim: to fix the **balance of trade** 1× 2003
- Remittances 2022: **India 1st, Mexico 2nd, China 3rd, Philippines 4th** 1× 2024

RUPEE CONVERTIBILITY STEPS



19.2 Foreign Investment

- Most FDI in the last decade: **services** sector (telecom was taken as the answer in 2006) 1× 2006
- Large foreign investment came in **life insurance, banking, automobiles** (1, 2 and 3) 1× 2009
- Forex earners: **tourism, medical care, garments, leather** (all four) 1× 2008
- FPIs issue **participatory notes** to unregistered foreign investors 1× 2019
- FPI limit in corporate bonds (Budget 2020-21): **15%** 1× 2020
- Largest equity source: **Mauritius** when asked; **Singapore** now 2× 2006, 2010
- Singapore top FDI source in 2023-24 because of the Mauritius treaty change (A and R true, R explains A) 1× 2024
- FDI recently opened in **retail trade** 1× 2010
- Insurance FDI (2012 proposal): **26% to 49%** 1× 2012
- Foreign equity in small-scale units (1995-96): **24%** 1× 1995

FDI fact	Detail
Largest source now	Singapore, then Mauritius, USA
Top sector	Services
Insurance limit	26% to 49% (2015) to 74% (2021) to 100% (Budget 2025-26)
Multi-brand retail	51% with conditions; single-brand 100%
Banned	Lottery, gambling, chit funds, Nidhi, real estate business, atomic energy

19.3 External Debt

- External debt end-2010: over **US$ 300 billion** 1× 2010
- Largest component of external debt: **commercial borrowings** 1× 2017
- **ECBs** are regulated by the **RBI** 1× 2010
- Most indebted (World Bank 2010): **Russia, China, Turkey, Brazil, India** 1× 2011
- World Bank class for India: **less indebted** 1× 2009
- Not true of green bonds: that they are **short-maturity** loans 1× 2016
- **Oil Pool Account** closed: **1 April 2002** 1× 2010

19.4 Forex Reserves, the Euro and FEMA

- Not true: **SDRs are not included** (they are included) 1× 2001
- Largest share of reserves: **foreign currency assets** 1× 2002
- Reserves end-December 2024: **US$ 640.3 billion** 1× 2023
- Rupee value hit by **inflation, fiscal deficit, crude prices** (all three) 1× 2013
- New single European currency: **Euro** 1× 1995
- Not true of the euro in 1999: **in circulation** (notes came in 2002) 1× 1999
- Euro is the currency of **20** EU states (per key; 20 since Croatia joined, 2023) 1× 2001
- FERA replaced by **FEMA** 1× 2007
- FEMA finally implemented: **2002** (passed 1999, in force June 2000) 1× 2013
- Sanctions forex for imports: **exchange bank** (authorised dealer) 1× 2011

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Forex reserves	Detail
Parts	Foreign currency assets (largest), gold, SDRs, IMF reserve tranche
Kept by	RBI
End-December 2024	US$ 640.3 billion
Record	Over US$ 700 billion (September 2024)

TRAP ALERT

- **Devaluation** = government act; **depreciation** = market fall.
- **Current account** convertible since 1994; **capital account** still not full.
- SDRs **are** part of forex reserves; ECBs are regulated by the **RBI**, not SEBI.
- FEMA: passed **1999**, in force **2000**, FERA fully gone **2002**.

UP LINK

UP link: UP draws FDI through **Noida-Greater Noida** (electronics, mobiles, data centres) and the **Defence Corridor**; UP Investors Summit 2023 drew pledges of over **Rs 33 lakh crore** (soft figure); remittances matter most in eastern UP districts with Gulf migrants.

LAST-MINUTE 10

1. Devaluation: **1949, 1966, 1991 (twice)**; exports cheaper
2. Current account convertible **1994**; **Tarapore** = capital account
3. India = **top remittance receiver** (2022)
4. **Services** = top FDI sector; **Singapore** = top source now
5. P-notes issued by **FPIs**; FPI bond limit **15%**
6. ECBs regulated by **RBI**; commercial borrowing = largest debt part
7. India = **less indebted** (World Bank)
8. Forex: FCAs largest; **SDRs included**; Dec 2024 = **US$ 640.3 bn**
9. Euro notes from 2002; **20** euro states
10. FEMA replaced FERA; final **2002**

Chapter 20

International Organisations and Global Rankings

Built from 73 UPPSC questions (1996–2025). Focus: who publishes which report, IMF and World Bank, WTO boxes and TRIPS, WEF and ILO, and membership of regional groups.

ASKED MOST · REVISE FIRST

- **Report-publisher matching**: WEO and GFSR = **IMF**; WDR = **World Bank**; HDR = **UNDP**; WIR = **UNCTAD**; Emissions Gap = **UNEP**; TTCI = **WEF**; Wage and Social Protection = **ILO** 7× 2004, 2016, 2020, 2023, 2024, 2025, 2025
- **WTO**: **1995**, Geneva, from **GATT** (Uruguay Round, **Arthur Dunkel**); Doha = **Qatar** 8× 1996, 2002, 2006, 2008, 2008, 2010, 2017, 2017
- **IMF**: India a member from **1945**; lends for **temporary BoP deficits**; **SDR** = "paper gold" 5× 2002, 2004, 2004, 2015, 2016
- **Membership traps**: not in SAARC **Cambodia**; not in BIMSTEC **Indonesia**; not in GCC **Iran, Iraq**; not in ASEAN **India**; not in OPEC **China** 6× 1999, 2003, 2010, 2013, 2025, 2025
- **SCO**: India full member at **Astana, 2017**; BRICS bank = **New Development Bank** (first head **K.V. Kamath**) 3× 2015, 2017, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 G-20 and Global Reports

- India's G-20 logo: **lotus with 7 petals** for **7 continents** (both) 1× 2023
- G-20 2023 wrong pair: **Brazil-Joko Widodo** (Widodo led Indonesia; Brazil = Lula) 1× 2023
- First World Summit for Social Development: **Copenhagen** (1995) 1× 2017
- **HDR-UNDP, WEO-IMF, WIR-UNCTAD, Emissions Gap-UNEP** 1× 2024
- **Global Financial Stability Report**: **IMF** only 1× 2025

- **World Development Report**: **IBRD** (World Bank) 1× 2004
- Highest per capita income (WDR 2000): **Luxembourg** 2× 2001, 2003
- Ease of Doing Business 2017: India **100th** 1× 2017
- EoDB: **77th** (2019), **63rd** (2020) (both); the report was dropped in 2021 1× 2020

Report	Publisher
World Economic Outlook, Global Financial Stability	IMF
World Development Report, Global Economic Prospects	World Bank
Human Development Report, Gender Social Norms Index	UNDP
World Investment Report	UNCTAD
Emissions Gap Report	UNEP
Gender Gap, Travel and Tourism, Global Talent	World Economic Forum
Global Wage, World Social Protection	ILO

20.2 The World's Largest Economies

- Second largest economy: **China** now (Japan in 2005) 1× 2005
- PPP top four then: USA, China, Japan, India (*); India is **third** by PPP now 2× 2005, 2009
- 2019: GDP Rs 209 lakh crore, **5th largest**, passed **UK and France** (1, 2, 3) 1× 2016
- WDR 2008 order: **USA > China > UK > India** 2× 2008, 2009
- **US$ 5 trillion** economy target: **2024** (2024-25) 2× 2019, 2020
- Third largest economy then: **Germany** (Japan earlier) 1× 2007

Rank (nominal GDP, IMF 2025)	Country
1	USA
2	China
3	Germany
4	Japan
5	India (about US$ 4.1 trillion)
By PPP	China, USA, India

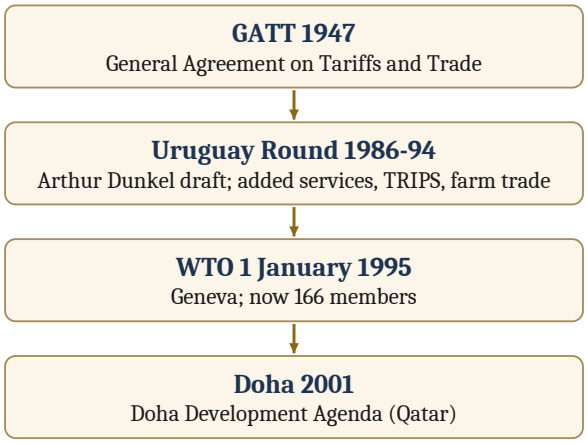
20.3 IMF and World Bank

- Loans and grants for state infrastructure: **World Bank** 1× 2002
- ICAR's National Agricultural Innovation Project (World Bank): **four** components 1× 2009
- **Green Index**: **World Bank** 1× 2014
- Not correct: heads from **Italy and Germany** (custom: IMF head European, World Bank head American); both are Bretton Woods bodies (only 2) 1× 2025
- World Bank HQ: **Washington** 1× 2007
- IMF loans: for **temporary BoP deficits** 1× 2016
- India joined the IMF: **1945** (founder member) 2× 2004, 2015
- **Paper gold** = **SDRs** of the IMF 2× 2002, 2004

20.4 WTO, GATT and TRIPS

- Guard of world trade: **WTO** 1× 2017
- WTO: set up by the members of **GATT** 1× 2008
- WTO set up: **1995** 1× 1996
- **Doha** Ministerial (2001): **Qatar** 1× 2002
- Pascal Lamy was followed by **Roberto Azevedo** 1× 2014
- Not a WTO member: **Serbia** 1× 2016
- Not an Agreement on Agriculture pillar: **anti-dumping duties** 1× 2010
- MSP-market price gap paid directly: **blue box** (per key) 1× 2016
- Farm subsidy statements: **1 and 4** correct per key; WTO does not ban indirect subsidies 1× 2022
- Subsidy cuts are needed under WTO; India is developing (A false, R true) 1× 1998
- **GATT** = General Agreement on Tariffs and Trade; **Arthur Dunkel** = GATT; **Uruguay Round** = GATT 3× 2008, 2010, 2017
- **TRIPS** covers **trademarks, industrial designs, GIs**; not labour standards 1× 2001
- Not in TRIPS protection: **plant breeding** (left to sui generis law) 1× 2010
- National IPR Policy: backs Doha and TRIPS; nodal body **DIPP** (both) 1× 2024

FROM GATT TO WTO



WTO box	Meaning
Amber box	Trade-distorting support (MSP, input subsidies); capped at 10% for developing nations
Blue box	Payments tied to production limits
Green box	Non-distorting support (research, income support)

20.5 WEF, ILO and ADB

- WEF founder: **Klaus Schwab** 1× 2008
- Global Talent Index 2020 wrong pair: **USA-4** (USA was 3rd) 1× 2019
- Not a Gender Gap parameter: **leisure** 1× 2017
- First **Gender Social Norms Index**: **UNDP** 1× 2016
- **Travel and Tourism Competitiveness Index**: **WEF** 1× 2020
- ADB's India Resident Mission: **New Delhi** 1× 2010
- 100th ILO conference (2011): **domestic workers** 1× 2010
- **World Social Protection Report**: **ILO** 1× 2023
- **Global Wage Report 2024-25**: **ILO** only 1× 2025
- **WTO-no quantitative curbs, IMF-BoP finance, SAARC-South Asian cooperation, IDA-soft loans** 1× 2015
- Correct HQ: **WTO-Geneva** 1× 2006

Body	HQ / fact
World Economic Forum	1971, Klaus Schwab; Geneva (Davos meet)
ILO	1919, Geneva
ADB	1966, Manila; India mission in New Delhi
IMF, World Bank	Washington, D.C.
WTO	Geneva
FAO	Rome

20.6 Regional Groupings

- Not in **APTA**: **Myanmar** 1× 2010
- Not in **OPEC**: **China** (Indonesia also left) 1× 1999
- Not in **SAARC**: **Cambodia** 2× 2010, 2013
- Not in **BIMSTEC**: **Indonesia** (only 4) 1× 2025
- Odd one out: **SAARC** 1× 2015
- 9th BRICS Summit host: **China** (Xiamen) 1× 2016
- BRICS bank in China: **New Development Bank** (Shanghai); first president **K.V. Kamath** 2× 2015, 2017
- Highest per capita income in BRICS: **Russia** 1× 2010
- India a full SCO member at **Astana** (2017) 1× 2017
- Not in the **Shanghai Five**: **Vietnam** 1× 2001
- Not in SCO in 2010: India (then an observer) (*) 1× 2010
- **Varanasi**: first SCO **tourism and cultural capital** (Samarkand, 2022) 1× 2023
- Not in **Mekong-Ganga** cooperation: **Bangladesh** 1× 2001
- **Benelux** = **Belgium, Netherlands, Luxembourg** 1× 2004
- 37th G-8 Summit (2011): **Deauville** 1× 2010
- Voice of Global South: India held the 3rd in 2024; Global South = developing nations (neither) 1× 2024
- India-China-Brazil WTO bloc: **G-77** (per key; the farm bloc is G-20) 1× 2008
- Not in ASEAN: **India**; **Brunei** is a member 2× 1996, 2003
- Not in GCC: **Iran and Iraq** 1× 2025
- **Ind-Aus ECTA**: first-of-its-kind; attended by Commerce Secretary Sunil Barthwal (both) 1× 2024
- India-Australia MATES: bilateral; graduates aged **18-30** work up to **two years** (both) 1× 2024

QUANTUM IAS & PCS

Group	Fact
SAARC	1985; HQ Kathmandu; 8 members
BIMSTEC	1997; 7 members; not Indonesia
ASEAN	1967; 10 members incl. Brunei; Timor-Leste 11th (2025)
BRICS	New Development Bank, Shanghai; K.V. Kamath first head
SCO	2001 from Shanghai Five; India full member 2017
GCC	Saudi, UAE, Qatar, Kuwait, Bahrain, Oman

TRAP ALERT

- IMF and World Bank are both in **Washington**; WTO, ILO, WEF in **Geneva**; ADB in **Manila**.
- **WEO** is IMF, not World Bank; **WIR** is UNCTAD, not UNDP.
- India joined the IMF in **1945** (founder), not 1947.
- Indonesia is in ASEAN, not BIMSTEC; Iran and Iraq are not in the GCC.

UP LINK

UP link: **Varanasi** was named the SCO's first **Tourism and Cultural Capital** (2022-23); Varanasi also hosted a **G-20** development ministers' meeting (2023), and Lucknow, Agra and Greater Noida hosted other G-20 meetings.

LAST-MINUTE 10

1. WEO, GFSR = **IMF**; WDR = World Bank; HDR = **UNDP**; WIR = **UNCTAD**
2. TTCI, Gender Gap = **WEF** (Klaus Schwab); Wage, Social Protection = **ILO**
3. G-20 logo: lotus, **7 petals = 7 continents**
4. India in IMF **1945**; SDR = **paper gold**
5. WTO **1995**, Geneva; Doha = **Qatar**; Azevedo after Lamy
6. TRIPS: trademarks, designs, GIs; not labour or plant breeding
7. NDB = BRICS bank; **K.V. Kamath**
8. SCO full member **Astana 2017**; **Varanasi** = SCO cultural capital
9. Not SAARC: Cambodia; not BIMSTEC: **Indonesia**; not GCC: Iran, Iraq
10. Benelux = Belgium, Netherlands, Luxembourg

Chapter 21

Miscellaneous Economy

Built from 84 UPPSC questions (1994–2025). Focus: committees and their subjects, economists and their books, Gandhi's trusteeship, institutes and their cities, quality marks, energy and steel facts.

QUANTUM IAS

ASKED MOST · REVISE FIRST

- **Committee-subject matching**: **Dutt-licensing, Wanchoo-direct taxes, Rajamannar-Centre-State, Chakravarty-monetary policy**; **Tarapore** is not bank service; **Rekhi** is not export-import **3×** 2008, 2010, 2015
- **Trusteeship** = **Mahatma Gandhi**; basis of the Gandhian economy **4×** 2009, 2012, 2013, 2013
- **Book-author matching**: **Wealth of Nations-Adam Smith**; Myrdal, Hirschman, Kaldor; *Principles* is **Ricardo**, not Marx **3×** 2012, 2023, 2024
- **AGMARK**: quality mark for farm goods; Act **1937**; not issued by FCI **5×** 2002, 2004, 2009, 2013, 2023
- **Noida** institutes: **V.V. Giri National Labour Institute, NIESBUD**; agricultural economics institute (NIAP) = **New Delhi** **4×** 2013, 2015, 2017, 2023

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

21.1 Education and Society

- UN year **2023**: International Year of **Millets** **1×** 2021
- **NEP 2020**: drafted under **K. Kasturirangan**; it replaced a **34-year** policy, not 38 (only 1) **1×** 2020
- NEP panel head (June 2017): **K. Kasturirangan** **1×** 2017
- Not part of NEP aims: **Bharat Nirman** (ULLAS and DIKSHA are) **1×** 2025
- Tamil Nadu corporal-punishment guidelines (2024): key says both; date of statement 1 is doubtful (*) **1×** 2024
- **Marginal man**: **Robert E. Park** **1×** 2020
- **Missing women**: **Amartya Sen** **1×** 2015
- **Global village**: based on **transport and communication** **1×** 2018

21.2 Institutions, People and Committees

- **Competition Commission**: **2003** (Act 2002) **1×** 2004
- CCI chair (2010): **Ashok Chawla** **1×** 2010
- Milk marketing: **GCMMF** (Amul) **1×** 2009
- IMF Managing Director (2019): **Kristalina Georgieva** **1×** 2019
- Wrong pair: **A.M. Khusro-UPSC** (he chaired the 11th Finance Commission) **1×** 2008
- **Swaminathan-Green Revolution, L.K. Jha-administrative reforms, Kurien-milk, Morarji Desai-social control on banks** **2×** 2003, 2022
- Wrong: **Rekhi-export-import** (Rekhi = indirect taxes) **1×** 2008
- Wrong: **Tarapore-customer service** (that is Goiporia) **1×** 2010
- **Dutt-licensing, Wanchoo-direct taxes, Rajamannar-Centre-State, Chakravarty-monetary policy** **1×** 2015
- RBI Governor who became Finance Minister: **C.D. Deshmukh** **1×** 2008
- "From Dependence to Self Reliance": **Bimal Jalan** **1×** 2023
- Indians in Forbes top 50 (2000): **Azim Premji, Dhirubhai Ambani** (*) **1×** 2001
- **OGL, FOB-external trade; SJRY, TRYSEM-employment; WPI, CPI-index numbers; CRR, SLR-banking** **1×** 2007
- **Pressure group**: influences **policy decisions** **1×** 2004

Committee	Subject
Goiporia (1990)	Customer service in banks
Rekhi	Indirect taxes
Goswami (1993)	Industrial sickness
Janakiraman (1992)	Securities scam
Malhotra (1993)	Insurance reform
Tarapore (1997, 2006)	Capital account convertibility
Dutt (1969)	Industrial licensing
Wanchoo (1971)	Direct taxes
Rajamannar (1969-71)	Centre-State relations
Chakravarty (1985)	Monetary policy

21.3 Books, Prizes and Indices

- Nobel in Economics 2015: **Angus Deaton** **1×** 2015

QUANTUM IAS & PCS

- "India's Economic Policy: The Gandhian Blueprint": **Charan Singh** 1× 2012
- PM when gold was pledged abroad (1991): **Chandra Shekhar** 1× 2010
- Corruption Index ranks: **2010-87, 2011-95, 2012-94, 2014-85** 1× 2014
- **RESIDEX** (residential prices, NHB): **2007**; tracks **land/housing prices** 2× 2011, 2015
- Trade and Merchandise Marks Act: **1958** 1× 2016
- Indian Patents Act in force: **1972** 1× 2014
- Patents (Second Amendment): **2002** 1× 2002
- "Bahini Darbar" women's paper: **Madhya Pradesh** 1× 2014
- Most greenhouse gas emissions (2013-14): **China** 1× 2014

Recent Nobel in Economics	Winners
2015	Angus Deaton
2023	Claudia Goldin
2024	Acemoglu, Johnson, Robinson
2025	Joel Mokyr, Philippe Aghion, Peter Howitt

21.4 Institutes and Schemes

- Indian Institute of Carpet Technology: **Bhadohi**, not in options (*) 1× 2006
- **V.V. Giri National Labour Institute**: **Noida** 1× 2017
- Agricultural Economics and Policy Research institute: **New Delhi** 2× 2013, 2023
- None correct: Finance Commission is **constitutional**, Planning Commission was **non-statutory**, RBI nationalised **1 January 1949** 1× 2009
- **Entrepreneurship Development Institute**: **Ahmedabad** 1× 2011
- **NIESBUD**: **Noida** 1× 2015
- Swadesh Darshan circuits: no option fits (*) 1× 2017
- Portal for small traders: **e-lala** 1× 2016
- **Project Arrow**: modernising **post offices** 2× 2009, 2010
- Not economic infrastructure: **coal mines** 1× 2004
- **Super Bazar**: retail-sale organisation 1× 2003

21.5 Economic Thought

- **Myrdal-Economic Theory and Underdeveloped Regions, Hirschman-Strategy of Economic Development, Kaldor-Strategic Factors, Adam Smith-Wealth of Nations** 1× 2024
- "Wealth of Nations": **Adam Smith** 1× 2012
- Wrong pair: **Karl Marx-Principles of Political Economy and Taxation** (that is **Ricardo**) 1× 2023
- Not part of a country's wealth: **money supply** 1× 2004
- Father of scientific management: **F.W. Taylor** 1× 2017
- **Trusteeship** = **Gandhi**; Gandhian economy rests on it 3× 2009, 2012, 2013
- Correct: trusteeship and UP Zamindari Act in the Ninth Schedule (1 and 2); voting age cut by the **61st** Amendment; Shetkari Sanghatana = **Sharad Joshi** 1× 2013
- Not correct: **"age-quake"** means child population (it means **ageing**) 1× 2013
- Not an achievement equated with women's empowerment: **strategic choices** 1× 2013
- Most household waste per head: **USA** 1× 2008

Book	Author
The Wealth of Nations (1776)	Adam Smith
Principles of Political Economy and Taxation	David Ricardo
Das Kapital	Karl Marx
General Theory (1936)	J.M. Keynes
Asian Drama	Gunnar Myrdal
The Strategy of Economic Development	Albert Hirschman
Strategic Factors in Economic Development	Nicholas Kaldor

21.6 Energy, Infrastructure and Industry

- Infrastructure spending 2019-23: **UP 1st, Maharashtra 2nd, Tamil Nadu 3rd, Karnataka 4th** 1× 2024
- Electricity generated order: changes yearly (*); without renewables now **UP, Maharashtra,**

QUANTUM IAS & PCS

Gujarat, Tamil Nadu 1× 2000

- Per capita electricity: a few hundred kWh then (*); about **1,460 kWh** now 1× 1998
- Renewable share of capacity (Dec 2011): about **11%** 1× 2012
- DISCOM interest relief: **UDAY** 1× 2021
- Wrong pair: **primary energy-tidal power** (per key) 1× 2021
- First PPP railway line: **Gujarat** 1× 2015
- **Golden Triangle**: **Delhi, Agra, Jaipur** 1× 2016
- Over a third (now over half) of world crude steel: **China** 2× 2011, 2022
- **Kamaiya**: hereditary **bonded labour** in **Nepal** 1× 2002
- Not correct: India **supplied manufactured goods** under British rule 1× 2013
- Not true of the Arthashastra: it **describes the economic life** of the time (per key) 1× 1994
- Non-domestic LPG cylinder: **19 kg** 1× 2009
- PVR: **Priya Village Roadshow** 1× 2011

21.7 Quality Marks and Miscellany

- **AGMARK**: **quality**; a seal of quality guarantee 3× 2002, 2004, 2013
- AGMARK Act: **1937** 1× 2009
- AGMARK: quality of farm goods; **not** issued by FCI (only 1) 1× 2023
- **ISO 14001**: certificate for an **environmental management** system 1× 1999
- GI tag for black pottery in UP: **Nizamabad** (Azamgarh) 1× 2021
- **Ecomark**: **environment-friendly** products (1991); **medicines** are not covered 2× 2013, 2022
- Packaging matters for **all the above** reasons 1× 2009
- Air India and Indian Airlines merged as **Air India** 1× 2007
- **Universal Service Obligation Fund**: telecom liabilities (rural telecom) 1× 2003

Mark	For
AGMARK	Farm products (1937 Act; Directorate of Marketing)
ISI	Industrial goods (BIS)
Ecomark	Environment-friendly goods (1991)
Hallmark	Gold and silver purity
ISO 14001	Environmental management

TRAP ALERT

- *Principles of Political Economy and Taxation* = **Ricardo**, not Marx.
- AGMARK is not issued by FCI; Ecomark does not cover medicines.
- Voting age to 18 = **61st** Amendment; Shetkari Sanghatana = **Sharad Joshi**, not Medha Patkar.
- Tarapore = convertibility; Goiporia = bank service; Rekhi = indirect taxes.

UP LINK

UP link: **Nizamabad black pottery** (Azamgarh) has a GI tag; the **Indian Institute of Carpet Technology** is at **Bhadohi**; **V.V. Giri Labour Institute** and **NIESBUD** are in **Noida**; UP ranked **first** in infrastructure spending (2019-23) and leads in thermal power generation.

LAST-MINUTE 10

1. Year of Millets **2023**; NEP 2020 = **Kasturirangan**
2. Marginal man = **Park**; missing women = **Sen**
3. CCI **2003**; Khusro = Finance Commission
4. Morarji Desai = social control on banks; Kurien = milk
5. Deaton 2015; Charan Singh = Gandhian Blueprint; Jalan = Dependence to Self Reliance
6. Trusteeship = **Gandhi**; Taylor = scientific management
7. Ricardo = *Principles*; Hirschman = *Strategy*; Kaldor = *Strategic Factors*
8. V.V. Giri, NIESBUD = **Noida**; EDI = Ahmedabad
9. AGMARK **1937**; Ecomark **1991**; ISO 14001 = environment
10. Golden Triangle = Delhi, Agra, Jaipur; China = top steel

Chapter 22

Population Theories, Census and Growth

Built from 58 UPPSC questions (1998–2025). Focus: Malthus and other theories, the first census, decadal growth (1921, 1961-71, 2001-11), state rankings, Nagaland and Kerala growth facts.

ASKED MOST · REVISE FIRST

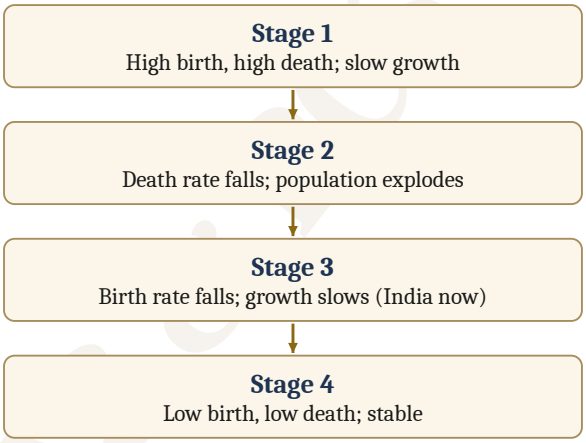
- **1921** = **"Year of Great Divide"** (only fall in population before it) **5×** 2007, 2009, 2010, 2013, 2016
- **2001-11 growth**: **17.70%**, the **lowest** decade since independence **6×** 2012, 2013, 2014, 2015, 2016, 2017
- **UP** = most populous (about **16.5-17%** of India); **Maharashtra** second **9×** 2001, 2004, 2008, 2010, 2011, 2012, 2014, 2016, 2021
- **Nagaland**: highest growth 1981-2001; **only fall** in 2001-11; **Kerala** lowest in 1991-2001 **13×** 1999, 2000, 2002, 2004, 2005, 2006, 2007, 2008, 2012, 2013, 2013, 2017, 2017
- **Malthus**: population **geometric**, food **arithmetic**; first census **1872, Lord Mayo** **6×** 2001, 2017, 2022, 2023, 2023, 2024

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

22.1 Population Theories

- Population control is needed for the environment; growth does **not** keep balance (A true, R false) **1×** 2023
- Natural growth = **crude birth rate and crude death rate** (A and B) **1×** 2018
- Gradual long-run change in growth pattern: **demographic transition** **1×** 2016
- Stable population: **constant birth and death rates** **1×** 2014
- India's phase now: **high growth with definite signs of slowing** **1×** 2015
- Malthus's best check: **birth control** (per key; Malthus stressed moral restraint) **1×** 2017
- Malthusian theory: about **population** **1×** 2022
- Population **geometric**, food **arithmetic** **1×** 2024
- **Optimum-Cannan, social maladjustment-Henry George, demographic transition-Thompson, food supply-Malthus** **1×** 2023
- Not a demographic feature: **standard of living** **1×** 2015

DEMOGRAPHIC TRANSITION



22.2 The Census of India

- First **regular** census: **1881** (Lord Ripon) **1×** 2007
- First census in British times: **Lord Mayo** (1872) **3×** 2001, 2023, 2023
- First census year: **1881** per key (1872 was non-synchronous) **1×** 2023
- Census every **10 years** is true; population does not stay unchanged (A true, R false) **1×** 2016
- Census 2011 motto: **"Our Census, Our Future"** **1×** 2016

Census fact	Detail
First census	1872 (Lord Mayo); first regular 1881 (Lord Ripon)
Law	Census Act, 1948; every 10 years; Union List
Census 2011	15th overall, 7th after independence
Census 2027	16th; first digital census with caste count; reference date 1 March 2027
Run by	Registrar General and Census Commissioner (MHA)

22.3 Growth of India's Population

- Growth order: **Haryana, Punjab, Uttarakhand, Himachal** 1× 2025
- Lowest decadal growth: **2001-11** 1× 2012
- Wrong pair: **2001-11 = 19.05%** (it is **17.70%**) 1× 2016
- 2001-11 growth: **17.70%** 4× 2013, 2014, 2015, 2017
- Highest percentage change: **1971** (1961-71 decade) 1× 2014
- Highest decadal growth: **1961-71** 2× 2005, 2010
- Highest annual exponential growth: **1971-81** 1× 2006
- **1921** = Year of Great Divide 5× 2007, 2009, 2010, 2013, 2016
- Fast growth since 1951 is true; 1921, not 1951, is the great divide (A true, R false) 1× 2000

Decade	Growth (%)	Note
1911-21	-0.31	Only fall; 1921 = Great Divide
1961-71	24.80	Highest decadal growth
1971-81	24.66	Highest annual exponential growth
2001-11	17.70	Lowest since independence
2011 total	121.09 crore	17.5% of world population
Now	About 146 crore	India passed China in April 2023

22.4 Population of the States

- Females in India (2011, provisional): **48.46%** 1× 2010
- Most populous state (2011): **Uttar Pradesh** 4× 2012, 2014, 2016, 2021
- UP outnumbers each of **Japan, Russia, Brazil, Nigeria** (and UK, Germany, Pakistan) 1× 2001
- UP 2001: literacy 56.3%, density 690, most SCs in **Sitapur** (1, 2, 4); most populous district is **Allahabad**, not Kanpur Nagar 1× 2006
- UP most populous is true; order behind it is **Maharashtra, Bihar, West Bengal** (A true, R false) 1× 2010
- UP share of India: about **17%** (16.5%) 1× 2010
- Order: **UP, Maharashtra, Bihar, Andhra Pradesh** (4, 3, 2, 1) 1× 2013
- Order: **UP, Maharashtra, Bihar, West Bengal** 1× one-liner
- Second most populous: **Maharashtra** 4× 2001, 2004, 2008, 2011
- Order: **UP, Bihar, West Bengal, Andhra Pradesh** (3, 1, 4, 2) 1× 2024
- Second in population and third in area: **Maharashtra** 1× 2013
- Over a third of India: **UP, Maharashtra, Bihar** 1× 2006
- Least populous among options: **Karnataka** 1× 2004
- Chhattisgarh larger than West Bengal in area; WB more populous (both) 1× 2007
- 1990s order: **Maharashtra, West Bengal, MP, Tamil Nadu** 1× 1998
- 2001 order: **Puducherry, Chandigarh, Mizoram, Sikkim** 1× 2009

State (2011)	Population (crore)	Share
Uttar Pradesh	19.98	16.5%
Maharashtra	11.24	9.3%
Bihar	10.41	8.6%
West Bengal	9.13	7.5%
Madhya Pradesh	7.26	6.0%
Least populous	Sikkim (state); Lakshadweep (UT)	-

22.5 Growth Across States

- Highest growth 1991-2001: **Nagaland** 3× 2002, 2004, 2005
- Highest growth 1981-2001: **Nagaland**; highest in the 1991 Census too 2× 2000, 2013
- Lowest growth 1991-2001: **Kerala** 4× 1999, 2006, 2007, 2008
- Lowest growth (negative) 2001-11: **Nagaland** 1× 2013
- Population fell in Census 2011: **Nagaland** 3× 2012, 2017, 2017
- Lowest growth among options (2011): **Andhra Pradesh** 1× 2012
- Highest growth 2001-11 among options: **Bihar** 1× 2015
- UP growth 2001-11 (provisional): **20.09%** 1× 2010

- UP has the most people; it is **not** the densest (Bihar is) (A true, R false) 1× 2008
- UP's fast growth: better **health and disease control** 1× 2006
- UP districts 2011: **Allahabad, Moradabad, Ghaziabad, Lucknow** 1× 2014
- Correct: lowest population **Lakshadweep**, lowest density **Arunachal**, highest growth **Dadra and Nagar Haveli** (1, 3, 4); highest density is **Delhi**, not Chandigarh 1× 2011

TRAP ALERT

- **1921** = Great Divide; **1951** = demographic divide (start of fast growth). Do not swap.
- Highest decadal growth = **1961-71**; highest annual exponential = **1971-81**.
- UP is the most **populous**, not the most **dense** (Bihar); most populous UP district = **Prayagraj**.
- Optimum population = **Cannan**; transition = **Thompson**; maladjustment = **Henry George**.

UP LINK

UP link: UP had **19.98 crore** people in 2011 (**16.5%** of India), more than Brazil; its 2001-11 growth was **20.2%**; **Prayagraj** was its most populous district, followed by Moradabad, Ghaziabad and Azamgarh; Census 2027 will be the first digital count.

LAST-MINUTE 10

1. Malthus: population **geometric**, food **arithmetic**
2. Cannan = optimum; Thompson = transition; Henry George = maladjustment
3. First census **1872 (Mayo)**; regular **1881 (Ripon)**
4. 2011 motto: **Our Census, Our Future**
5. **1921** = Great Divide; **1961-71** = highest decadal growth
6. **2001-11 = 17.70%**, lowest since independence
7. UP 1st, Maharashtra 2nd, Bihar 3rd, WB 4th
8. UP = about 16.5% of India; females 48.5%
9. **Nagaland** = highest 1991-2001, negative 2001-11
10. **Kerala** = lowest 1991-2001; Bihar fastest-growing big state 2001-11

Chapter 23

Population Density and Literacy

Built from 59 UPPSC questions (1991–2023). Focus: densest and sparsest states and UTs, literacy rates (2001, 2011), female literacy and gender gaps, UP district literacy.

ASKED MOST · REVISE FIRST

- **Densest state**: **Bihar**; order **Bihar, West Bengal, Kerala, UP** 11× 2000, 2004, 2009, 2010, 2012, 2014, 2016, 2017, 2019, 2023, 2023
- **Sparsest state**: **Arunachal Pradesh**; UTs: densest **Delhi** (Puducherry among options), sparsest **Andaman and Nicobar** 4× 2001, 2017, 2017, 2021
- **Literacy 2011**: **74.04%**; male 82.14, female 65.46; gap **16.68**; counted from **age 7** 5× 2004, 2010, 2011, 2012, 2014
- **Kerala** first, **Mizoram** second; **Bihar** lowest but the **biggest rise** (2001-11) 8× 2006, 2008, 2010, 2011, 2014, 2015, 2016, 2017
- **Female literacy**: highest **Kerala**, lowest **Rajasthan** (2011 provisional); smallest gap **Meghalaya** 5× 2009, 2010, 2013, 2016, 2017

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

23.1 Population Density

- Kerala dense and highly literate; literacy does not explain density (A and R true, R does not explain A) 1× 2001
- India's density: **has steadily increased** 1× 2014
- Highest density (2011): **Bihar** 6× 2004, 2012, 2014, 2016, 2017, 2023
- Order: **Bihar, West Bengal, Kerala, Uttar Pradesh** 4× 2009, 2010, 2019, 2023
- Densest BIMARU state: **Bihar** 1× 2000
- Lowest density: **Arunachal Pradesh** 2× 2001, 2021
- Ascending (density): **Odisha, AP, Tripura, Punjab** 1× 2003

- Lowest-density order (2001): **Arunachal, Mizoram, Sikkim, Nagaland** 1× 2006
- Sparsest in 1991 options: **Jammu and Kashmir** 1× 1991
- Below 100 per sq km (2011): **Mizoram** 1× 2010
- Densest UT among options: **Puducherry** (Delhi overall) 1× 2017
- Sparsest UT: **Andaman and Nicobar** 1× 2017
- Lowest density among options: **Rajasthan** 1× 2010
- Ascending: **Gujarat, Karnataka, Assam, Haryana** 1× 2010
- True: UP has **more than twice** India's density 1× 2005

Density (2011)	Persons per sq km
India	382 (325 in 2001; 117 in 1951)
Bihar, West Bengal, Kerala, UP	1,106; 1,028; 860; 829
Arunachal, Mizoram, Sikkim	17; 52; 86
Delhi, Chandigarh, Puducherry	11,320; 9,258; 2,547
Andaman and Nicobar	46

23.2 Literacy in India

- **Effective literacy**: population aged **7 and above** 1× 2014
- Biggest rise in literacy: **1991-2001** 1× 2006
- Literacy 2011 (provisional): **74.04%** 2× 2004, 2012
- Most literate district (provisional 2011): **Serchhip** (Mizoram) 1× 2010
- First fully literate district outside Kerala: **Bardhaman** (West Bengal) 1× 2008
- Biggest literacy rise 2001-11: **Bihar** 2× 2011, 2016
- Highest literacy among options: **Manipur** 1× 2017
- Second in female literacy after Kerala: **Mizoram** 1× 2014
- Second in literacy and in urbanisation: **Mizoram** 1× 2017
- Highest literacy among options (2001): **Mizoram**; top in the North-East 3× 2004, 2006, 2008
- 2001 order: **Kerala, Himachal, Tamil Nadu, Manipur** 1× 2006
- Order: **Kerala, Mizoram, Goa, Maharashtra** 1× 2002
- Lowest four (ascending, 2011): **Bihar, Arunachal, Rajasthan, Jharkhand** 1× 2010
- 2001 ascending: **Bihar, UP, Rajasthan, MP** 1× 2002
- Highest UT literacy (2001): **Lakshadweep** 1× 2013
- 2011 order: **Tripura, Himachal, Sikkim, Nagaland** 1× 2010
- Male-female gap (2011): **16.68** points 1× 2010
- Female literacy (2011): about **65.5%** 1× 2011

Literacy (2011)	Figure
India	74.04% (M 82.14, F 65.46)
Highest states	Kerala 94.0, Mizoram 91.3, Goa 88.7, Tripura 87.2
Lowest states	Bihar 61.8, Arunachal 65.4, Rajasthan 66.1, Jharkhand 66.4
Highest UT	Lakshadweep 91.8
Lowest UT	Dadra and Nagar Haveli 76.2

23.3 Female Literacy and Gaps

- Smallest male-female gap (2011): **Meghalaya** 2× 2013, 2016
- Largest gap among options: **Uttar Pradesh** (Rajasthan overall) 1× 2015
- Top male and female literacy statements (UTs and states): none fits exactly (*) 1× 2010
- Highest female literacy among options (2001): **Chhattisgarh** 1× 2005
- Highest and lowest female literacy (2011): **Kerala and Rajasthan** 1× 2010
- Highest female literacy among options (2001): **Mizoram** 1× 2013
- Lowest female literacy (2001): **Bihar** 2× 2002, 2003
- Lowest female literacy (2011): **Rajasthan** 2× 2009, 2017
- Highest female literacy among UTs: **Lakshadweep** 1× 2013
- Highest literacy among options: **Uttarakhand** 1× 2009
- Lowest literacy (2011): **Bihar** 2× 2010, 2015
- Highest UT literacy (2011): **Lakshadweep** 1× 2023

- Lowest UT literacy: **Dadra and Nagar Haveli** 1× 2017
- Highest rural literacy among options: **Goa** 1× 2010

23.4 Literacy in Uttar Pradesh

- UP literacy 2011: **67.7%**, not among the options (*) 1× 2015
- UP rank in literacy (2001): **31st** 1× 2010
- District order: **Gautam Buddha Nagar, Kanpur Nagar, Auraiya, Etawah** 2× 2012, 2017
- Most literate UP district: **Gautam Buddha Nagar** 1× 2012
- Lowest female literacy among options: **Rampur** 1× 2012
- Correct: lowest literacy **Shravasti**, lowest density **Lalitpur** (1 and 3); highest sex ratio is **Jaunpur**, most populous city **Lucknow** 1× 2012

UP (2011)	Fact
Literacy	67.7% (M 77.3, F 57.2)
Highest district	Gautam Buddha Nagar (80.1%)
Lowest district	Shravasti (46.7%)
Highest sex ratio	Jaunpur
Lowest density	Lalitpur
Most populous city	Lucknow

TRAP ALERT

- Densest state = **Bihar**, not West Bengal or Kerala; densest UT = **Delhi**.
- Literacy is counted from **age 7**, not age 6 or 5.
- Female literacy lowest: **Bihar** in 2001, **Rajasthan** in 2011 (provisional).
- UP: lowest literacy **Shravasti**; highest sex ratio **Jaunpur**, not Deoria.

UP LINK

UP link: UP's density (829) is more than twice India's (382); literacy rose to **67.7%** (2011) but the male-female gap stayed about **20 points**; **Gautam Buddha Nagar** tops and **Shravasti** trails; the **ULLAS** scheme now targets adult literacy in UP.

LAST-MINUTE 10

1. Density India **382**; rises every census
2. **Bihar** densest; **Arunachal** sparsest
3. Delhi densest UT; **Andaman** sparsest UT
4. Literacy from **age 7**; 2011 = **74.04%**
5. Male 82.14, female 65.46; gap **16.68**
6. Kerala > **Mizoram** > Goa; Bihar lowest
7. Biggest rise 1991-2001; Bihar biggest rise 2001-11
8. Meghalaya = smallest gap; Rajasthan = lowest female literacy (2011)
9. Lakshadweep = top UT; Serchhip = top district; Bardhaman = first outside Kerala
10. UP: **GB Nagar** top, **Shravasti** bottom

Chapter 24

Sex Ratio, Social Groups and Population Dynamics

Built from 86 UPPSC questions (1990–2023). Focus: sex ratio and child sex ratio by census and state, SC, ST and religion data, National Population Policy 2000, TFR, mortality and the demographic dividend.

ASKED MOST · REVISE FIRST

- **Kerala** = highest sex ratio and females outnumber males; **Puducherry** among UTs 8× 1995, 2001, 2004, 2006, 2009, 2012, 2014, 2023
- **Lowest sex ratio**: **Haryana** (state), **Daman and Diu** (UT) 7× 2010, 2011, 2011, 2013, 2017, 2018, 2021
- Sex ratio by census: **1901 highest (972)**, 1951 = **946**, 1991 lowest (927), **2011 = 943** (provisional 940) 6× 2008, 2011, 2013, 2013, 2014, 2015
- **NPP 2000**: stable population goal moved from **2045 to 2070**; replacement **TFR 2.1** 6× 2001, 2010, 2014, 2017, 2020, 2023
- **SC**: largest number **UP**, highest share **Punjab**, none in **Nagaland**; **ST**: largest **MP and Chhattisgarh**, none in **Punjab** 11× 2002, 2003, 2006, 2007, 2008, 2008, 2010, 2010, 2014, 2016, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Sex Ratio in India

- Highest sex ratio of any census: **1901** 1× 2008
- Wrong pair: **2011-925** 1× 2013
- Sex ratio **946**: **1951** 1× 2013
- Wrong pairs: **1991-938** (it was 927) and 2011-940 (final 943) 1× 2014
- Sex ratio 2011 (provisional): **1000 : 940** 2× 2011, 2015
- Order (provisional): **Kerala, Puducherry, Tamil Nadu, AP** 1× 2010
- Most females per 1,000 males: **Kerala** 5× 1995, 2001, 2012, 2014, 2023
- Females outnumber males (UT among options): **Puducherry** 3× 2004, 2006, 2009
- Highest among options: **Tamil Nadu** 1× 2014
- Above the national average among options: **Jharkhand** 2× 2013, 2014
- Males outnumber females (aged 7+): **Bihar, Maharashtra, UP, West Bengal** (all four) 1× 2010
- Lowest state: **Haryana**; rich but lowest sex ratio: **Haryana** 3× 2013, 2017, 2018
- Lowest among options (2001): **Gujarat** 1× 2009
- Lowest UT: **Daman and Diu** 4× 2010, 2011, 2011, 2021
- Biggest fall 1991-2001: **Daman and Diu**; biggest rise: **Arunachal** 2× 2005, 2005
- UP's improvement 2001-11: 10 (provisional) or 14 (final) (*) 1× 2013
- UP sex ratio 2011: **912** 1× 2017
- Highest district (provisional): **Mahe** (Puducherry) 1× 2010

Census	Sex ratio
1901	972 (highest)
1951	946
1991	927 (lowest)
2001	933
2011	943 (provisional 940)
NFHS-5 (2019-21)	1,020

24.2 Child Sex Ratio

- Causes of low sex ratio: **maternal deaths, girl-child deaths, foeticide, more boys born** (all four) 1× 2005
- Child sex ratio since 1961: **continuously decreasing** 1× 2011
- Child sex ratio 2011 (provisional): **914** (final 918) 1× 2010
- Correct pair: **Madhya Pradesh-912** 1× 2010
- Most boys per 100 girls: **Haryana** 1× 2019
- Highest child sex ratio among options: **Chhattisgarh** 1× 2012
- Order: **Mizoram, Meghalaya, Punjab, Haryana** 1× 2011
- India's CSR fell, Haryana's rose, rural sex ratio above urban (i, ii, iii) 1× 2017

Child sex ratio (0-6)	Figure
1961 to 2011	976 to 918; falling every census
Highest (2011)	Mizoram, Meghalaya (970); Chhattisgarh (969)
Lowest (2011)	Haryana (834), Punjab (846)

24.3 Scheduled Castes, Tribes and Religion

- Highest SC share: **Punjab** 3× 2002, 2003, 2006
- Largest SC number: **Uttar Pradesh** 3× 2008, 2010, 2014
- Zero SC population: **Nagaland** 1× 2016
- Few SCs in the North-East; low cultivated area does not explain it (A and R true, R does not explain A) 1× 2017
- ST share of India (2001): **8.20%** 1× 2006
- No scheduled tribes: **Punjab** 1× 2008
- ST population ascending: **Arunachal, Mizoram, Nagaland, Meghalaya** 1× 2021
- Largest tribal population: **MP and Chhattisgarh** 3× 2007, 2010, 2016
- ST share ascending: **Maharashtra, Rajasthan, MP, Meghalaya** 1× 2023
- Lowest ST share: **Uttar Pradesh** 1× 2017
- Religions by size (ascending): **Jain, Buddhist, Sikh, Christian** 1× 2010
- Largest Muslim population: **Uttar Pradesh** 1× 2007
- Highest Christian share (1991): **Nagaland** 1× 1998
- Kerala's top literacy is not due to Christians (A true, R false) 1× 1990

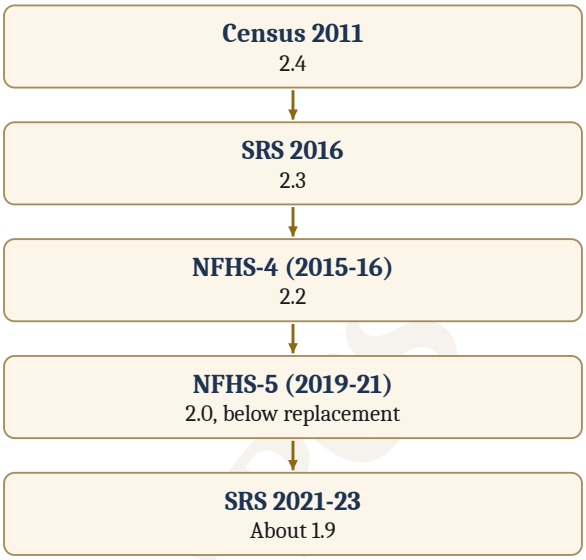
- Sex ratio by religion: **Christian, Buddhist, Jain, Muslim** 1× 2010
- Lowest sex ratio by religion: **Sikh** 1× 2016

SC / ST (2011)	Fact
SC share of India	16.6%
Largest SC number / share	Uttar Pradesh / Punjab (31.9%)
No SCs	Nagaland, Lakshadweep
ST share of India	8.6%
Largest ST number	Madhya Pradesh
Highest ST share	Mizoram (94.4%)
No STs	Punjab, Haryana, Delhi, Chandigarh, Puducherry
Lowest ST share	Uttar Pradesh (0.6%)

24.4 Population Policy and Fertility

- New National Population Policy: **2000** 1× 2017
- Not in the 10-year action plan: **special reward for men** taking terminal methods 1× 2009
- Population stabilisation goal: **2045 to 2070** 3× 2010, 2014, 2023
- Birth rate target by 2010: **21 per 1,000** 1× 2002
- Replacement TFR: **2.1** 2× 2001, 2020
- TFR (SRS 2016): **2.3** 1× 2017
- TFR (NFHS-4, 2015-16): **2.2** 1× 2018
- TFR (Census 2011): **2.4** 1× 2016
- Highest TFR: **Bihar** now (UP when asked) 1× 2001
- Population shapes social change; working wives have fewer children (A and R true, R explains A) 1× 2000
- Highest female life expectancy (2001-06) among options: **Punjab** 1× 2004
- Equal birth and death rates, no migration: **stability of population** 1× 2013

INDIA'S FERTILITY PATH (TFR)



24.5 Mortality

- Crude death rate: **below 9** per 1,000 1× 2007
- Death rate 2000: **9 per 1,000** 1× 2001
- Lowest death rate (2015) among options: **Maharashtra** 1× 2017
- Register births and deaths within **21 days** 1× 2019
- First death census: **Karnataka** 1× 2013
- IMR about 70 when asked (*); now **25** (SRS 2023) 1× 2000
- Lowest IMR: **Kerala**; lowest urban IMR: **Kerala** 3× 2005, 2014, 2015
- IMR below the national average (2011): **Tamil Nadu** 1× 2013
- Eleventh Plan IMR target: **28** 1× 2017
- Highest child mortality (UNICEF, then): **Odisha** 1× 2001

24.6 Age Structure and Demographic Dividend

- Average age of Indians in 2020: **29 years** 1× 2016
- Under 14 (2011): about **29%** 1× 2013
- Aged 20 and above: **59.29%** 1× 2016
- Below 35 years: about **65%** 1× 2012
- Old-age dependency ratio (2011): **14.2%** 1× 2016
- **Demographic dividend**: working population aged **15-59** 1× 2022
- "Demographic bonus": a **surge in productive-age population** 1× 2009

- Dependent group: **0-14 years** (and 60+) 1× 2013
- Dependency ratio falling: the **15-59** share is high 1× 2019
- Human resources and development: both effects (1 and 2) 1× 2023
- Not an adverse effect of population growth: **larger working population** 1× 2015
- Self-dependent disabled people: about **2%** (per key) 1× 2015
- National Literacy Mission 75% target year: **2007** 1× 2006
- Progressive population-resource areas: **Ganga-Yamuna Doab, Tamil Nadu** (1 and 4) 1× 2010

TRAP ALERT

- Highest census sex ratio = **1901**; lowest = **1991**; 1951 = **946**.
- Kerala and **Puducherry** are the only places with more females; Haryana and **Daman and Diu** lowest.
- **Punjab** = highest SC share and **no STs**; **Nagaland** = no SCs; UP = most SCs, fewest STs by share.
- NPP 2000 stabilisation year: **2070**, not 2045 (old) or 2050.

UP LINK

UP link: UP's sex ratio rose from **898 (2001)** to **912 (2011)**; UP has the **most SCs** (4.14 crore) and **most Muslims**, but the **lowest ST share** (0.6%); its TFR fell to about **2.4-2.6**, and **Jaunpur** has the state's highest sex ratio.

LAST-MINUTE 10

1. Sex ratio: 1901 **972**; 1951 **946**; 1991 **927**; 2011 **943**
2. Kerala **1,084**; Puducherry; Mahe top district
3. Haryana lowest state; **Daman and Diu** lowest UT
4. Child sex ratio falling since 1961: **918**
5. SC: UP most, **Punjab** highest share, **Nagaland** none
6. ST: **MP** most, **Mizoram** highest share, **Punjab** none, UP lowest share
7. Jain < Buddhist < Sikh < Christian; Sikh lowest sex ratio
8. NPP **2000**; stabilisation **2070**; TFR 2.1
9. Register births and deaths in **21 days**; Kerala lowest IMR
10. Demographic dividend = **15-59** age group

Chapter 25

Urbanisation I: Concepts and Levels

Built from 61 UPPSC questions (1998–2023). Focus: what makes a town, phases of urban growth, the 2011 urban share, most and least urbanised states, and rural India.

ASKED MOST · REVISE FIRST

- Urban share 2011: **31.16%** (about **37.7 crore**); 2001: **27.8%** 9× 2002, 2008, 2009, 2010, 2010, 2014, 2014, 2017, 2023
- **Census town**: **5,000+** people, **75%** male workers in non-farm work, **400+** per sq km; **3,894** census towns (2011) 3× 2008, 2009, 2010
- **Fastest urban growth**: **1971-81**; moderate phase **1931-61** 6× 2001, 2002, 2003, 2003, 2007, 2009
- **Most urbanised**: **Goa** (Tamil Nadu among big states); **least**: **Himachal Pradesh** 12× 1998, 2002, 2004, 2005, 2006, 2010, 2010, 2012, 2012, 2017, 2021, 2023
- **Largest urban population**: **Maharashtra**, then **UP** 4× 2004, 2006, 2008, 2010

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

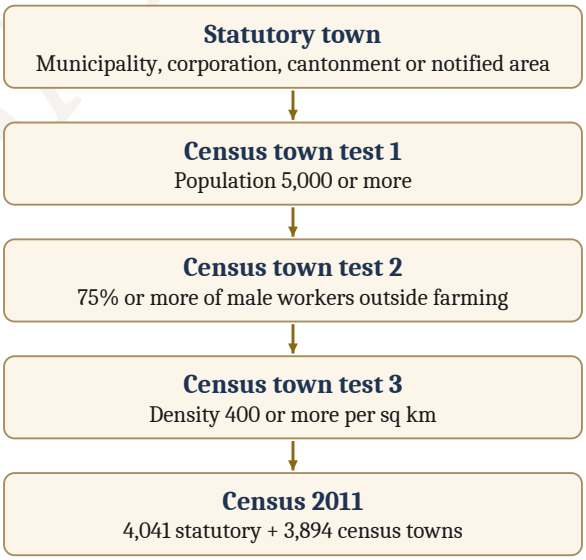
25.1 Village, Town and Urban Growth

- City differs from village in **values, household make-up, way of living, economic activity** (all) 1× 2004
- Urban growth shows **all of the above** (more urban people, more towns, more income) 1× 2004
- Not an element of a rural community: **cultural diversity** 1× 2021
- **T.K. Oommen** on urban families: **all of the above** 1× 2014
- Causes of urbanisation: **migration, educational institutions, industrialisation** (1, 2, 3) 1× 2015
- Not in inclusive development (per key): **semi-urban** people 1× 2016
- Urban population by 2030 (key): **575 million** 1× 2017
- India: a **moderately low** urbanised country 1× 1998
- India is **not** over-urbanised; cities do lack infrastructure (A false, R true) 1× 2017
- India less urbanised than China; poor planning is not the reason (A and R true, R does not explain A) 1× 2010
- 2001 statements: urban 285 million, 27.78%, more than USA's population, self-reliant (all) 2× 2002, 2003
- India's urban people outnumber the USA's; US urbanisation is higher (A and R true, R does not explain A) 1× 2000
- India's urban population exceeds the USA's total (A and R true, R explains A, per key) 1× 2002
- Urban India is a platform for **all of the above** 1× 2015
- Fast urbanisation after 2001 and mobile revolution are unrelated (A and R true, R does not explain A) 1× 2016
- Not true: urbanisation has been **uniform** across regions 1× 2002
- Urbanisation and literacy both cut the **birth rate** (both) 1× 2021
- Urbanisation **reduced both birth and death rates** 1× 2007

25.2 Phases and Definitions

- Moderate urbanisation phase: **1931-61** 1× 2003
- Largest decadal urban growth: **1971-81** 5× 2001, 2002, 2003, 2007, 2009
- Not a criterion for a town: **physical expansion** 1× 2008
- Census-town tests: **5,000 people, 75% male non-farm workers, 400 per sq km** (1, 2, 3); no area test 1× 2009
- Census towns (2011): **3,894** 1× 2010
- Urban share 2011: **31.16%**, not in the options (none of the above) 7× 2002, 2008, 2009, 2010, 2014, 2017, 2023
- Most urban religion: **Jains** 1× 2022
- Fastest urban growth 2001-11 among options: **Sikkim** 1× 2023
- City dwellers (2011): about **37 crore** 1× 2014

WHAT MAKES A TOWN (CENSUS)



Year	Urban share
1901	10.8%
1951	17.3%
2001	27.8%
2011	31.16% (37.7 crore)
2030 (projected)	About 40%

25.3 Urbanisation by State

- Over a quarter of urban India: **Maharashtra and UP** 2× 2004, 2008
- Top three (2001): **Maharashtra, UP, Tamil Nadu** 1× 2006

QUANTUM IAS & PCS

- Urban population order: **Maharashtra, UP, Tamil Nadu, West Bengal** 1× 2010
- Ascending urban population: **Sikkim, Arunachal, Nagaland, Mizoram** 1× 2010
- Ascending (2001): **West Bengal, Tamil Nadu, UP, Maharashtra** 1× 2005
- Correct: **Himachal Pradesh** is the least urbanised 1× 2002
- **Most urbanised-Tamil Nadu, largest urban population-Maharashtra, densest-Delhi, least dense-Arunachal** (per key) 3× 2009, 2013, 2016
- Most urbanised (2011): **Goa** 2× 2010, 2023
- Most urbanised big state (key): **Tamil Nadu** 5× 1998, 2005, 2006, 2010, 2012
- Two most urbanised: **Goa and Mizoram** 1× 2008
- Then (key): **Maharashtra, Gujarat, Karnataka** 3× 2001, 2003, 2004
- Order: **Tamil Nadu, Maharashtra, Gujarat, West Bengal** 1× 2007
- Most urbanised landlocked state: **Punjab** 1× 2013
- Key order **Maharashtra, West Bengal, AP, Gujarat** fits urban **numbers**, not percentage (*) 1× 2017
- Least urbanised among options: **West Bengal** 1× 2016
- Order: **Tamil Nadu, Kerala, Maharashtra, Gujarat** 1× 2009
- Highest urban density among options: **West Bengal** 1× 2013
- Least urbanised among options: **Bihar** 1× 2017
- Least urbanised (2011): **Himachal Pradesh** 4× 2004, 2012, 2017, 2021
- Least urbanised UT: **Andaman and Nicobar** 1× 2017
- UT order: **Delhi, Chandigarh, Lakshadweep, Daman and Diu** 1× 2010

Urbanisation (2011)	States / UTs
Most urbanised	Goa 62.2%, Mizoram 52.1%, Tamil Nadu 48.4%
Least urbanised	Himachal 10.0%, Bihar 11.3%, Assam 14.1%
Largest urban population	Maharashtra 5.08 crore, UP 4.45 crore
Most urbanised UT	Delhi 97.5%, Chandigarh 97.3%
Least urbanised UT	Andaman and Nicobar 37.7%

25.4 Rural India

- Fewest villages (UT): **Daman and Diu** 1× 2014
- Correct: urban increase exceeded rural, and 2001 urbanisation **27.81%** (1 and 3); 2011 rural share is **68.84%** 1× 2010
- Rural share (2001): **72.2%** 1× 2008
- Highest rural share: **Himachal Pradesh** 1× 2015
- Largest rural population: **Uttar Pradesh** 1× 2018
- Lowest rural child share: **Kerala** 1× 2014
- Lowest child sex ratio in both rural and urban areas: **Haryana** 1× 2018
- Most rural pucca houses (2001): **Punjab** 1× 2008
- Traders and makers mark the village-town line; they do not wholly depend on villages (A true, R false) 1× 2009
- **Occupational structure**: distribution of workers among **occupations** 1× 2015

TRAP ALERT

- **Goa** = most urbanised state; **Tamil Nadu** = most urbanised **large** state; old keys often say Tamil Nadu.
- **Maharashtra** = most urban **people**; **Delhi** = most urbanised UT.
- Census town: no **area** test; 75% of **male** workers.
- 2001-11 = first decade when urban numbers grew more than rural.

UP LINK

UP link: UP had the **second-largest urban population** (4.45 crore, 2011) but only **22.3%** urbanisation; it has the **largest rural population** (15.5 crore); UP has **17 municipal corporations** and many census towns around Noida-Ghaziabad.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Urban **31.16%** (2011); 27.8% (2001); 37.7 crore
2. Census town: **5,000**, **75%** male non-farm, **400**/sq km
3. **3,894** census towns; moderate phase **1931-61**
4. Fastest urban growth **1971-81**
5. Jains most urban; India = moderately low, not over-urbanised
6. **Goa** most urbanised; **Himachal** least; Bihar next lowest
7. **Maharashtra** largest urban population, then UP
8. **Delhi** most urbanised UT; **Andaman** least
9. Rural share **68.84%**; UP largest rural population
10. Punjab = most urban among landlocked states

Chapter 26

Urbanisation II: Cities, Migration and Urban Missions

Built from 53 UPPSC questions (1993–2021). Focus: town classes, million-plus cities and megacities, UP's million cities, migration terms, slums, JNNURM and the Smart Cities Mission.

ASKED MOST · REVISE FIRST

- **Million-plus cities**: **35 (2001)**, **53 (2011)**; they hold **42.6%** of urban people **5×** 2006, 2013, 2017, 2017, 2019
- **UP million cities**: **6** (2001, most of any state), **7** (2011) **4×** 2004, 2005, 2005, 2015
- **Town classes**: **6** classes; small towns = **Class IV, V, VI**; Class I grows fastest and holds over 60% **5×** 2000, 2007, 2008, 2010, 2019
- **Largest cities**: **Greater Mumbai, Delhi, Kolkata**; first million city (1901) = **Kolkata** **4×** 1993, 2002, 2004, 2016
- **JNNURM** = named after **Nehru**; for **urban infrastructure**, not electrification; Smart Cities fund water and sewerage from **property tax** **5×** 2004, 2010, 2010, 2013, 2015

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

26.1 Classes of Towns

- Classes of urban centres: **6** **1×** 2008
- Small towns: **Class IV, V and VI** **1×** 2007
- Fastest population growth: **Class I** towns **1×** 2000
- Class I share of urban population (2001): **62.32%** **1×** 2010
- Over 60% of urban people in Class I cities; **53** million-plus UAs in 2011 (both) **1×** 2019

Town class	Population
Class I	1,00,000 and above
Class II	50,000-99,999
Class III	20,000-49,999
Class IV	10,000-19,999
Class V	5,000-9,999
Class VI	Below 5,000

26.2 Million-Plus Cities and Megacities

- Million cities (2011): **53** **2×** 2013, 2017
- Million cities (2001): **35**, not in options (*) **1×** 2006
- Metropolitan share of urban people (2011): **42.61%** **1×** 2017
- Million cities' share (2001): about **38%** **1×** 2006
- Megacities grow at different rates; India had **six**, not four, in 2001 (A true, R false) **1×** 2010
- Not a megacity (2001): **Ahmedabad** **1×** 2005
- Order (2001): **Greater Mumbai, Delhi, Kolkata, Bengaluru** **1×** 2004
- Most populous pair (2011): **Greater Mumbai and Delhi** **1×** 2016
- Order (1991): **Bombay, Calcutta, Delhi, Madras** **1×** 1993
- Most million cities (2001): **Uttar Pradesh** **1×** 2005
- No million-plus UA (2011): **Odisha**; no million city: **Odisha and Assam** **2×** 2016, one-liner
- Not a million city in MP: **Gwalior** **1×** 2008
- UP's share of the 35 million cities (2001): **6** **1×** 2004
- UP's million cities (2011): **7** **1×** 2015
- Ascending: **Allahabad, Meerut, Agra, Lucknow** **1×** 2010

- Order: **Ghaziabad, Agra, Varanasi, Meerut** 1× 2015
- Last-ranked million city (2011): **Kota** 1× 2016
- Smallest million city (then): **Rajkot** 1× 2007
- State with a million city (2001): **Jharkhand** 1× 2006
- First million city (1901): **Kolkata** 1× 2002
- Metro with highest sex ratio: **Chennai** 2× 2002, 2003
- UP's fastest urban growth: **1971-81** 1× 2009
- Densest UP city (2011): **Ghaziabad** 1× 2012
- Not true: UP has **only 5** million cities 1× 2005

UP million-plus cities (2011)	Note
Kanpur, Lucknow	About 2.9 million each
Ghaziabad	Densest UP city
Agra, Varanasi	-
Meerut, Prayagraj	Smallest of the seven

26.3 Migration and Urban Society

- Correct: northern cities' skewed sex ratio, scattered eastern cities, fewer rural migrants in western cities (1, 2, 4) 1× 2010
- Correct: 1, 2 and 3; urbanisation = rural-to-urban migration; the **Indus Project** was India-**USA**, not Canada 1× 2013
- Rural-to-urban migration: **population implosion** (Philip M. Hauser) 1× 2003
- Migration theory: **Lee** (Everett Lee, push-pull) 1× 2017
- Cities expand as village-city flow rises (A and R true, R explains A) 1× 1998
- Largest stream is **rural to rural**, not rural to urban; it is mostly women (A false, R true) 1× 2017
- **Urban corridors**: **expanding urban functions** 1× 2009

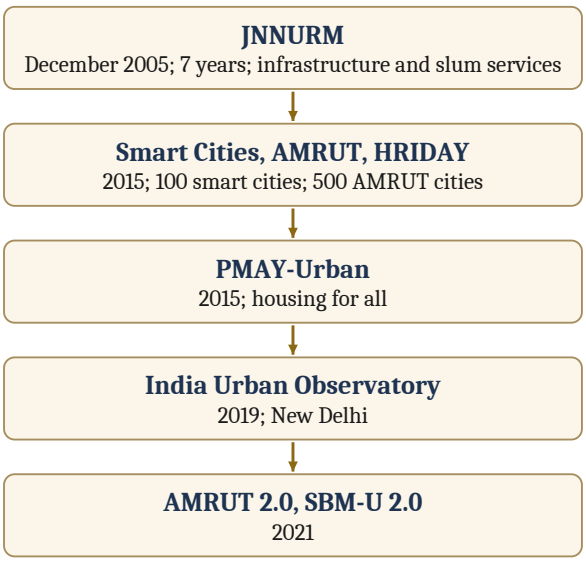
26.4 Urban Problems

- Crime rose with urbanisation and industry; family breakdown explains it (A and R true, R explains A) 1× 2002
- Rapid urbanisation brings **slums, crowding, falling services** (1, 2, 3); not fast infrastructure 1× 2001
- Correct: concrete coasts, a big city between Mumbai and Ahmedabad, slums and pollution (2, 3, 4); Maharashtra is not the most urbanised 1× 2008
- Urban poverty is rooted in rural areas with low literacy (A and R true, R explains A) 1× 2010
- Largest slum population among options: **Delhi** 1× 2011
- Slum population order: **Maharashtra, AP, West Bengal, UP** 1× 2009
- Most slum-reporting cities: **Tamil Nadu** 1× 2007

26.5 Urban Missions

- **India Urban Observatory**: **New Delhi** 1× 2021
- Not urban infrastructure: **housing** 1× 2004
- Not an urban infrastructure scheme: **Digital India** 1× 2016
- National Urban Renewal Mission named after **Jawaharlal Nehru** 1× 2004
- Not a JNNURM aim: **urban electrification** 1× 2010
- Not true of JNNURM: a **10-year** programme (it was 7 years) 1× 2013
- JNNURM improves **urban infrastructure** 1× 2010
- Smart Cities: water and sewerage funded by **property tax** 1× 2015
- Not a smart-city aim: **stabilising** quality of life (it aims to improve it) 1× 2017

URBAN MISSIONS IN ORDER



QUANTUM IAS & PCS

TRAP ALERT

- Small towns = **Class IV-VI**; Class I = 1 lakh+.
- Megacities 2001 = **6** (no Ahmedabad); million cities 35 (2001), **53** (2011).
- Largest migration stream = **rural to rural** (women, marriage).
- JNNURM = **7** years, not 10; no electrification.

UP LINK

UP link: UP had **7 million-plus cities** in 2011 (Kanpur, Lucknow, Ghaziabad, Agra, Varanasi, Meerut, Prayagraj); **10 UP cities** were picked for the Smart Cities Mission (Lucknow, Kanpur, Agra, Varanasi, Prayagraj, Aligarh, Bareilly, Moradabad, Saharanpur, Jhansi), and **Ghaziabad** is the densest UP city.

LAST-MINUTE 10

1. **6** town classes; small = Class IV-VI
2. Class I = **62.3%** of urban people (2001), fastest growing
3. Million cities **35 (2001), 53 (2011)**; 42.6% of urban people
4. 6 megacities in 2001; **Ahmedabad** not one
5. Mumbai > Delhi > Kolkata; Kolkata first million city
6. UP: **6** (2001), **7** (2011) million cities; Ghaziabad densest
7. Kota last (2011); Rajkot smallest (2001); Odisha none
8. **Hauser** = population implosion; **Lee** = migration
9. Most slums: Maharashtra; most slum towns: Tamil Nadu
10. JNNURM = Nehru, 7 years; Smart Cities = property tax

Chapter 27

World Population and Urbanisation

Built from 66 UPPSC questions (1998–2023). Focus: population milestones, India's share, density and sex ratio across countries, world urbanisation, migration, fertility and mortality.

ASKED MOST · REVISE FIRST

- **World Population Day**: **11 July 1987**, when the world hit **5 billion** 3× 2010, 2012, 2015
- **India's share** of world population: about **17.5%** (2011) 3× 2010, 2012, 2016
- **Densest in SAARC**: **Maldives**; **lowest density**: **Mongolia** (Canada among options) 6× 2002, 2003, 2008, 2009, 2013, 2017
- **World urbanisation**: about **55%** (2018); **70%** by **2050**; most urbanised continent **Australia** 5× 2002, 2008, 2010, 2017, 2018
- **Population growth**: lowest in **Europe**, highest in **Africa** (1990-2000) 4× 2002, 2003, 2008, 2014

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

27.1 Urban Sociology

- **Folk-urban continuum**: studies in **Mexico** (Robert Redfield) 1× 2013
- Primary urbanisation (Redfield and Singer): a **great tradition** 1× 2013
- **Megalopolis**: **Jean Gottmann** 1× 2023

27.2 World Population

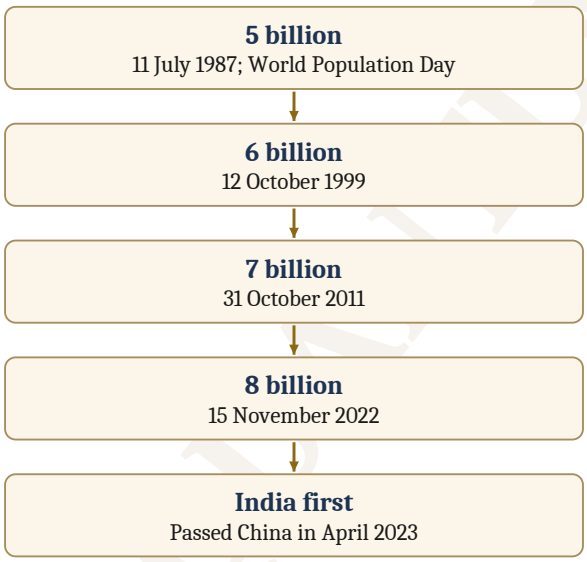
- World Population Report 2021: **UNFPA** 1× 2021
- Seven-billionth baby (31 October 2011): the UN named no single child (*); UP's **Nargis** was one symbolic baby 1× 2009

■ World Population Day marks **5 billion**: **11 July 1987** 3× 2010, 2012, 2015

- Theme 2017: **"Family Planning: Empowering People, Developing Nations"** 1× 2017
- About half of humanity lives between **20 N and 40 N** 1× 1998

- India's share (2011 estimate): **17.5%** 2× 2010, 2012
- India's share (Census 2011): **17.31%** 1× 2016
- World population in 2050 (key): **10 billion** (UN projection now about 9.7 billion) 1× 2010
- Asia's share (2008): about **61%** 1× 2008
- Largest population among options: **Indonesia** 1× 2010
- Order Brazil, Indonesia, Nigeria, Pakistan: changed since (*) 1× 2008
- Asian countries in the top ten (key): **5** (six when asked) 1× 2008
- Only correct: **Brazil** is the most populous Southern Hemisphere country (only I) 1× 2023
- Africa's most populous: **Nigeria** 1× 2011
- Largest population among options (UN 2012): **Dominica** 1× 2015
- Slowest growth 1990-2000: **Europe** 3× 2002, 2003, 2014
- Fastest growth: **Africa** 1× 2008
- Highest growth among options: **Philippines** 1× 1999

WORLD POPULATION MILESTONES



27.3 Density, Sex Ratio and Religion

- Low-density areas are **not** usually over-populated; their carrying capacity is low (A false, R true) 1× 2010
- Lowest density among options: **Canada** 1× 2013
- Lowest density: **Mongolia** 1× 2017
- Densest SAARC country: **Maldives** 4× 2002, 2003, 2008, 2009
- Densest South Asian country among options: **India** 1× 2007
- Densest in East Asia: **South Korea** 1× 2014
- Densest in South America: **Ecuador** 1× 2010
- Most favourable sex ratio: **Finland** 1× 2011
- Sex ratio order (2011): **Bangladesh, Pakistan, India, China** 1× 2011
- Fear of falling sex ratio: **sex-determination tests** 1× 2004
- Religions by size: **Christians, Muslims, Hindus, Buddhists** 1× 2004

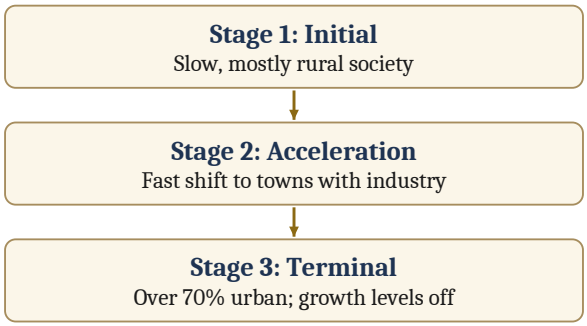
Density fact	Country
Lowest density	Mongolia (Canada among big countries)
Densest in SAARC	Maldives
Densest in East Asia	South Korea
Densest in South America	Ecuador

27.4 World Urbanisation

- Urban share (UN-DESA 2018): **55%** 1× 2018
- Urban share (2016): **54.5%** 1× 2017
- Acceleration stage: **second** stage of the urbanisation curve 1× 2021
- 70% urban by **2050** 1× 2008
- Not a city-prosperity test (UN-Habitat): **optimum population** 1× 2013
- World turned half urban in **2008**, not in options (*) 1× 2009
- Largest urban population: **China** 1× 2008
- Most urbanised continent: **Australia** 2× 2002, 2010
- Most urbanised region among options: **Europe** 1× 2003
- Country bigger than India's urban population: USA then; none now (*) 1× 2002
- Most urbanised country: **Singapore** 2× 2002, 2005
- Most urbanised in South America: **Uruguay** 1× 2008
- South Asia order: **Pakistan, India, Bangladesh, Sri Lanka** 2× 2010, 2014

- Most urbanised in South Asia: **Bhutan** now (Pakistan when asked) 2× 2008, 2014
- Most urbanised in West Asia: **Kuwait** 1× 2011
- Largest urban agglomeration: **Tokyo-Yokohama** 1× 2003
- Urbanisation follows industry; in developing nations it is a movement in itself (A and R true, R does not explain A) 2× 2002, 2016
- Asian cities over-urbanised as growth outpaced the economy (A and R true, R explains A) 1× 2010
- Western cities have more females (A and R true, R explains A, per key) 1× 2016

URBANISATION CURVE



27.5 Migration

- Not 2% of people living abroad; **Oceania** has the highest migrant share (A false, R true) 1× 1999
- Highest migrant share: **UAE** now (Australia when asked) 1× 1999
- Largest migrant population: **Europe** 1× 1999

27.6 Fertility, Mortality and Welfare

- China's growth slowed due to the **one-child policy** (A and R true, R explains A) 2× 1999, 2001
- Lowest fertility (UN, then): **Italy** 1× 1998
- Lowest fertility now (key): **China** (Italy when asked) 1× 2000
- Highest birth rate: **Afghanistan** 1× 2007
- Highest life expectancy: **Japan** 1× 2006
- More deaths than births in Asia: **Japan** 1× 2008
- Highest maternal mortality in Asia (then): **Nepal** 1× 2001
- Largest aged share in South Asia: **Sri Lanka** 1× 2013
- About 80% of people lack **social security** 1× 2010
- Poorest in South Asia (WDR 2006): **Bangladesh** 1× 2005
- First official family planning programme: **India (1952)** 2× 2011, 2012

TRAP ALERT

- World Population Day = **5 billion** (1987), not 6 or 7 billion.
- **Maldives** is the densest SAARC country, not Bangladesh.
- Most urbanised continent = **Australia/Oceania**; most urbanised region among options = **Europe**.
- **India** was the first country with official family planning (**1952**).

UP LINK

UP link: on 31 October 2011, baby **Nargis**, born in **Mal (Lucknow district)**, was symbolically named the world's seven-billionth baby by Plan India; UP alone (about 24 crore now) would be the world's 5th or 6th most populous country.

LAST-MINUTE 10

1. Folk-urban continuum = **Redfield, Mexico**; megalopolis = **Gottmann**
2. World Population Day: **11 July 1987** (5 billion)
3. 8 billion: 15 November 2022; UNFPA = population report
4. Half of humanity: **20-40 N**; Asia about **60%**
5. India = 17.5% of world; **Nigeria** = Africa's most populous
6. **Maldives** densest SAARC; **Mongolia** least dense
7. Urban: **55%** (2018); **70% by 2050**; China largest urban population
8. **Australia** most urbanised continent; **Singapore** country; **Uruguay** S. America
9. **Tokyo** = largest agglomeration; Japan = longest life, more deaths than births
10. India: first family planning programme, **1952**

QUANTUM IAS & PCS

TATTVA SERIES

UPPSC PRELIMS | FAST REVISION

General Science

PYQ-Based Revision Pointers

Chapters 1 to 29

Every pointer comes from an actual UPPSC question · Frequency marks · Year tags
Asked-Most Boxes · Trap Alerts · Quick Tables · Last-Minute 10

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QUANTUM IAS & PCS

Chapter 1

Units and Measurement

Built from 27 UPPSC questions (1990–2023). Focus: SI units and unit-quantity matching, light year and parsec, small lengths, odd units such as barrel, cusec and Dobson.

ASKED MOST · REVISE FIRST

- **Unit-quantity matching**: **joule-work, newton-force, watt-power, pascal-pressure, newton second-impulse** 6× 1990, 2001, 2002, 2005, 2006, 2013
- **Light year** and **parsec** are units of **distance**, not time 6× 1997, 2010, 2010, 2013, 2023, one-liner
- **Celsius and centigrade** are temperature scales, **not units of heat** 3× 2001, 2004, 2017
- Small lengths: **1 nm = 10^{-7} cm = 10^{-9} m**; **1 micron = 10^{-4} cm**; smallest: **fermi** 4× 2005, 2011, 2013, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

1.1 Work, Power and Energy Units

- Unit of power: **watt** 1× 2010
- Unit of work: **joule** 1× 1996
- Megawatt measures power **generated** 1× 1998

1.2 Astronomical and Small Lengths

- **Light year**: distance light travels in **one year**; a unit of **distance** 3× 2010, 2013, one-liner
- **Parsec**: unit of **distance** 1× 1997
- Distance at which Earth's orbit radius subtends **1 arc second**: **1 parsec** 1× 2023
- Multiply by 0.39 to get inches: **centimetre** 1× 2010
- **1 nanometre = 10^{-7} cm** 1× 2013
- Correct: **1 metre = 10^{9} nanometre** 1× 2023
- 5 feet in nanometres: none of the options (about 1.524 x 10^{9} nm) 1× 2023
- **1 micron = 10^{-4} cm** 1× 2011
- Smallest unit of length: **fermi** (femtometre, 10^{-15} m) 1× 2005

Unit	Value	Use
Light year	9.46 x 10^{12} km	Stars, galaxies
Astronomical unit	About 1.5 x 10^{8} km	Solar system
Parsec	3.26 light years	Largest of the three
Angstrom	10^{-10} m	Wavelength, atomic size
Nanometre	10^{-9} m	Nanotechnology
Micron	10^{-6} m (10^{-4} cm)	Cells, dust
Fermi	10^{-15} m	Nuclear size

1.3 SI Units and Matching

- **Acceleration-m/s^2, force-newton, work-joule, impulse-newton second** 2× 2001, 2005
- Wrong SI pair: **pressure-dyne** (dyne is a CGS unit of force) 1× 2013
- **Watt-power, knot-speed of a ship, nautical mile-navigation, calorie-heat** 1× 2002
- **Joule-work, ampere-current, watt-power, volt-potential, calorie-heat** 1× 1990
- Wrong pair: **light year-time** (it is distance) 1× 2010
- **Mach-high speed, angstrom-wavelength, pascal-pressure, joule-energy** 1× 2006
- **Decibel**: **sound intensity** 1× 2011
- Wrong pair: **Celsius-unit of heat** 2× 2001, 2004
- Not a unit of heat: **centigrade** 1× 2017
- Wrong SI pair: **radioactivity-curie** (SI unit is the **becquerel**) 1× 2022
- Electric potential is a **scalar** quantity (only 1) 1× 2021

Quantity	SI unit	Older units
Force	Newton	Dyne, kg-wt
Work, energy, heat	Joule	Erg, calorie (4.18 J), kWh
Power	Watt	Horsepower (746 W)
Pressure	Pascal	Bar, atmosphere, mm of Hg
Temperature	Kelvin	Celsius, Fahrenheit
Radioactivity	Becquerel	Curie
Power of lens	Dioptre	-

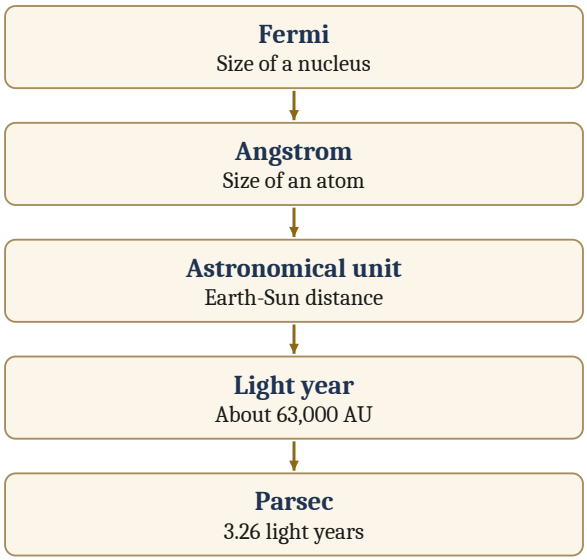
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Quantity	SI unit	Older units
Illuminance	Lux	Phot

1.4 Odd Units

- **1 barrel** of oil = **159 litres** 1× 2009
- **Cusec-rate of flow, byte-computer, Richter-earthquake, bar-pressure** 1× 2008
- **Dobson unit**: **ozone** concentration 1× 2021

DISTANCE UNITS, SMALL TO LARGE



TRAP ALERT

- **Parsec > light year > AU**; all three measure **distance**.
- **1 micron = 10^{-4} cm**, not 10^{-6} cm (that is in metres).
- SI unit of radioactivity = **becquerel**; curie is older. Dyne is force, not pressure.
- **Knot** = speed (nautical mile per hour); **nautical mile** = distance at sea.

UP LINK

UP link: **Indian Standard Time** is reckoned from **82.5 E**, which passes near **Mirzapur** (UP); the time signal itself is kept by the **National Physical Laboratory**, New Delhi.

LAST-MINUTE 10

1. Watt = power; joule = work; newton = force; pascal = pressure
2. Light year, parsec = **distance**; parsec = 3.26 light years
3. **1 nm = 10^{-9} m = 10^{-7} cm**; 1 micron = 10^{-4} cm
4. Fermi = smallest length unit
5. Impulse = **newton second**
6. Decibel = sound intensity; Mach = high speed
7. Celsius, centigrade = temperature scales
8. Becquerel = SI unit of radioactivity
9. 1 barrel = **159 litres**; cusec = rate of flow
10. Dobson unit = ozone

Chapter 2

Measuring Instruments and Scales

Built from 31 UPPSC questions (1990–2021). Focus: instrument-use matching, weather instruments, pyrometer and pyrheliometer, Richter scale and seismograph, SONAR and RADAR.

ASKED MOST · REVISE FIRST

- ■ **Instrument matching**: **anemometer-wind, barometer-pressure, hygrometer-humidity, ammeter-current, seismograph-earthquake** 5× 1992, 2001, 2008, 2012, 2015
- ■ **Radar**: finds objects with **radio waves**; **SONAR**: finds submerged objects with **ultrasonic** waves 7× 1996, 2000, 2001, 2004, 2008, 2016, one-liner
- ■ **Pyrometer** = very **high temperature** 6× 1998, 2004, 2008, one-liner, one-liner, one-liner
- ■ **Hygrometer** = **relative humidity** 3× 1995, 1996, 2014
- ■ **Richter scale** and **seismograph** = earthquakes 6× 1990, 2004, 2014, 2015, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

2.1 Navigation and Detection

- Locating submerged objects: **SONAR** 2× 2000, 2016

- SONAR uses **ultrasonic waves** 1× one-liner
- **Radar**: presence and location of objects with **radio waves** 4× 1996, 2001, 2004, 2008

2.2 Weather Instruments

- Wind velocity: **anemometer** 1× 2016
- Wrong pair: **tacheometer-pressure difference** (it is for surveying) 1× 1997
- **Pyrometer**: **high temperature** 4× 1998, 2004, 2008, one-liner
- Pyrometer is a **radiation thermometer**; for 2000 C: **total radiation pyrometer** 2× one-liner, one-liner
- **Pyrheliometer**: **solar radiation** 1× 2015
- **Manometer**: **gas pressure** 1× 1990
- **Ammeter-current, hygrometer-humidity, spring balance-weight, barometer-pressure** 1× 2001
- Wrong pair: **ammeter-electric power** (ammeter measures current) 1× 2015
- Atmospheric pressure: **barometer** 1× 2010
- Relative humidity: **hygrometer** 3× 1995, 1996, 2014
- **Ringelmann scale**: density of **smoke** 1× 2021
- **Stethoscope-heart sound, sphygmomanometer-blood pressure, caratometer-gold purity, luxmeter-light** 1× 2008
- Blood pressure: **sphygmomanometer** 1× one-liner

Instrument	Measures
Anemometer	Wind speed
Barometer	Atmospheric pressure
Hygrometer	Relative humidity
Hydrometer	Density of liquids
Lactometer	Purity of milk
Pyrometer	Very high temperature
Pyrheliometer	Solar radiation
Manometer	Gas pressure
Altimeter	Height
Tacheometer	Survey distance and elevation

2.3 Temperature and Light

- No negative values: **Kelvin** scale 1× 2017
- Correct pair: **thermoresistor-electronic thermometer** 1× 2016
- Intensity of light: **luxmeter** 1× 2004

2.4 Earthquakes, Radiation and Others

- Earthquake intensity: **Richter scale** 2× 2004, 2015
- Richter scale: amplitude of waves, i.e. **magnitude** 1× one-liner
- Seismometer device: **seismograph** 3× 1990, 2014, one-liner
- **Earthquake-seismograph, height-altimeter, current-ammeter, resistance-ohm** 1× 1992
- **Anemometer-wind, seismograph-earthquakes, barograph-pressure, hygrometer-humidity** 1× 2012
- Wrong pair: **seismometer-curvature of surface** 2× 1999, 2010
- Radioactivity: **Geiger counter** 1× 2010
- **Phonometer**: sound intensity, not in the options (*) 1× 2014
- Lie detector: **polygraph** 1× 2013

Instrument	Use
Geiger-Muller counter	Radioactivity
Seismograph	Records earthquake waves
Crescograph	Plant growth (J.C. Bose)
Kymograph	Muscle movement, pulse (graph)
Polygraph	Lie detection
Sphygmomanometer	Blood pressure
Audiometer	Hearing
Luxmeter	Light intensity
Caratometer	Purity of gold
Odometer	Distance travelled by a vehicle

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TRAP ALERT

- **Hygrometer** = humidity; **hydrometer** = liquid density.
- **Pyrometer** = high temperature; **pyrheliometer** = solar radiation.
- **Barometer** = air pressure; **manometer** = gas pressure; **ammeter** = current, not power.
- Kelvin has no negative values; SONAR = sound, RADAR = radio waves.

UP LINK

UP link: IMD's **Meteorological Centre, Lucknow** runs the UP network of rain gauges, automatic weather stations and a **Doppler weather radar** (soft detail); the **seismograph** network covers UP because its Terai belt lies in seismic zone IV.

LAST-MINUTE 10

1. Anemometer = wind; barometer = air pressure
2. Hygrometer = humidity; hydrometer = liquid density
3. Pyrometer = high temperature; pyrheliometer = solar radiation
4. Manometer = gas pressure; ammeter = current
5. SONAR = ultrasonic, submerged objects; RADAR = radio waves
6. Richter = magnitude; seismograph records quakes
7. Geiger counter = radioactivity; polygraph = lie detector
8. Sphygmomanometer = BP; caratometer = gold
9. Luxmeter = light; Ringelmann = smoke
10. Kelvin scale has no negative values

Chapter 3

Mechanics and Motion under Gravity

Built from 27 UPPSC questions (1990–2024). Focus: vectors, inertia and friction, energy conversion, weight at the poles, pendulum clocks, inverse-square gravity, satellites and escape velocity.

ASKED MOST · REVISE FIRST

- ■ Weight is **maximum at the poles** (g is largest there) **3×** 2006, 2009, 2015
- • Heavy and light balls dropped together **reach the ground together** **2×** 1992, 2015
- • Pendulum clocks **lose time in summer**: the pendulum lengthens **2×** 1994, 2012
- • **Inertia of rest**: a passenger's head falls back as the train starts **1×** 2010
- • Satellite stays in orbit by **centripetal force** supplied by **Earth's gravity** **2×** 2006, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

3.1 Quantities, Motion and Inertia

- Vector quantity: **displacement** **1×** 2014
- Man inhaling deeply on a sensitive balance: reading **increases** (per key) **1×** 2013
- Acceleration: $\mathbf{a = (v - u)/t}$ **1×** 2003
- Average speed for two equal halves: $\mathbf{2v_1v_2/(v_1 + v_2)}$ **1×** 2024
- Washing machine dryer works on **centrifugation** **1×** 2004
- Head leans back as a train starts: **inertia of rest** **1×** 2010
- Hard to walk on ice: **less friction** than on a road **1×** 1994
- Harder to start a loaded cart than keep it moving: **less friction once moving** (static > kinetic) **1×** 2013

3.2 Energy

- Not from the Sun: **nuclear** (and **geothermal**) energy **1×** 2022

- Wind power converts **kinetic energy** into electricity 1× 2016
- Non-conventional sources are **renewable** 1× 2023
- **Light to electric-solar cell, electric to sound-loudspeaker, mass to heat-nuclear reactor, chemical to heat and light-fuel combustion** 1× 2002
- Kinetic energies 4 : 9 (same mass): velocities **2 : 3** 1× 2017

Device	Energy change
Solar cell	Light to electrical
Loudspeaker	Electrical to sound
Microphone	Sound to electrical
Nuclear reactor	Mass to heat
Dynamo / generator	Mechanical to electrical
Electric motor	Electrical to mechanical
Battery	Chemical to electrical

3.3 Gravity, Weight and Mass

- Leaning Tower of Pisa stands: the vertical through its **centre of gravity falls within the base** 1× 2009
- 10 kg and 1 kg balls dropped from 20 m: **land together** 2× 1992, 2015
- Weight of a body: **maximum at the poles** 3× 2006, 2009, 2015
- Apparent weight highest in **hydrogen** (least buoyancy among the options) 1× 2017
- Weight feels more in a lift **accelerating upward** 1× 1990
- Pendulum clocks lose time in summer: **longer pendulum, longer period** 2× 1994, 2012
- Earth-Sun distance doubled: gravity becomes **one-fourth** 1× 2009
- We do not feel Earth's orbital speed: our **relative speed** is zero 1× 2013
- If gravity vanished: **weight zero, mass unchanged** 1× 2012

WEIGHT VERSUS MASS



3.4 Satellites and Escape Velocity

- Astronauts cannot stand erect: key says **no gravity**; the real reason is **weightlessness** in free fall 1× 2007
- Apple released from an orbiting spaceship: **moves along with it** 1× 2010
- Geosynchronous satellite: centripetal force from **Earth's gravity** 1× 2006
- Satellite kept in orbit by **centripetal force** 1× 2013
- No atmosphere on the Moon: **escape velocity is less than** the gas molecules' rms speed 1× 2012

Fact	Value
Earth's escape velocity	11.2 km/s
Moon's escape velocity	About 2.4 km/s
Orbital speed near Earth	About 7.9 km/s
Geostationary height	About 35,786 km; period 24 h
g on the Moon	About one-sixth of Earth's

TRAP ALERT

- **Displacement, velocity, force, weight** = vectors; distance, speed, work = scalars.
- Average speed over equal distances is the **harmonic**, not simple, mean.
- Gravity follows the **inverse square**: double the distance, one-fourth the force.
- Orbit "weightlessness" is **free fall**, not zero gravity.

UP LINK

UP link: **IIT Kanpur** built and launched **Jugnu** (2011), a nano-satellite on PSLV-C18, among the first student satellites from India.

LAST-MINUTE 10

1. Displacement = vector; a = (v - u)/t
2. Inertia of rest: head goes back as the train starts
3. Washing machine = centrifugation; ice = low friction
4. Nuclear and geothermal energy are not from the Sun
5. Weight **maximum at poles**; mass never changes
6. Heavy and light balls fall together
7. Pendulum clock **loses** time in summer
8. Double distance = **1/4** gravity
9. Satellite orbit = centripetal force by gravity
10. Escape velocity **11.2 km/s**; Moon has no air

Chapter 4

Properties of Matter

Built from 23 UPPSC questions (1992–2023). Focus: surface tension, pressure in bubbles and at height, water's maximum density at 4 C, Archimedes' principle and flotation.

ASKED MOST · REVISE FIRST

- ■ **Surface tension**: spherical raindrops, floating needle, oil spreading on water **4×** 1994, 2005, 2010, 2012
- ■ Water is **densest at 4 C** (277 K); fish survive under ice **5×** 1997, 2005, 2008, one-liner, one-liner
- ■ A ship **rises a little** on entering the sea; sea water is denser **4×** 1992, 2002, 2004, one-liner
- • Pressure inside a soap bubble is **more than** outside **2×** 1995, 2014

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

4.1 Elasticity and Surface Tension

- Rubber's Young's modulus is **less** than steel's; coil stretching depends on **shear modulus** (only 2) **1×** 2023
- Raindrops are spherical due to **surface tension** **1×** 2005
- Oil spreads on water: its surface tension is **less** than water's **2×** 1994, 2010
- Pressure inside a soap bubble: **more than atmospheric** **2×** 1995, 2014
- Charging a soap bubble: its **radius increases** **1×** 2019
- Iron needle floats due to **surface tension** **1×** 2012

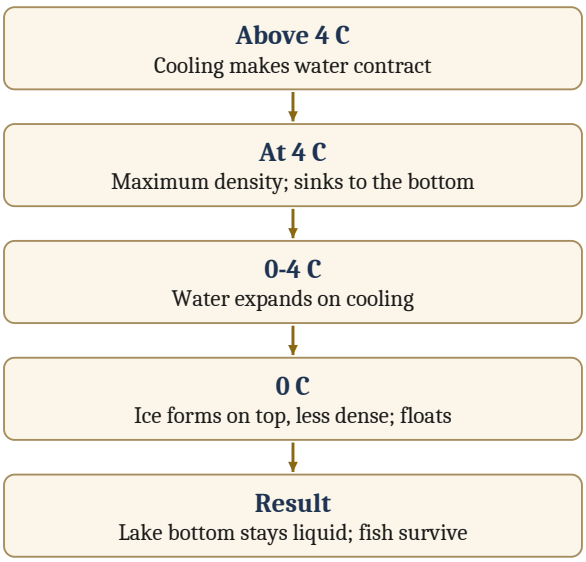
4.2 Pressure and Flow

- Pen leaks in a plane: **lower air pressure** **1×** 1992
- Sudden fall in barometer reading: **stormy** weather **1×** 1996
- Streamlined flow in a pipe: **slowest near the walls, fastest along the axis** **1×** 2003

4.3 Density and Water at 4 C

- More of a substance does not change its **density** **1×** 1992
- Cooling water from 9 C to 3 C: volume **first decreases, then increases** **1×** 1997
- Maximum density of water: **4 C** (277 K) **2×** 2008, one-liner
- Aquatic life survives freezing: water is **densest at 4 C**, so liquid water stays below the ice **2×** 2005, one-liner

ANOMALOUS EXPANSION OF WATER



QUANTUM IAS & PCS

4.4 Archimedes' Principle and Flotation

- Iron nail floats on mercury, sinks in water: iron is **denser than water, lighter than mercury** 1× 1994
- Needle sinks, iron ship floats: needle's density is **greater than the water it displaces** 1× 2004
- Bucket feels **heavier above the water surface** 1× 2009
- 200 g of water in a weightless balloon weighs **zero** in water 1× 2014
- Iceberg above the sea: about **1/9** 1× 2005
- Easier to swim in the sea: sea water is **salty** (denser) 1× 2002
- ■ Ship from river to sea: it **rises a little**; sea water is denser (A and R true, R explains A) 3× 1992, 2004, one-liner

Principle	Everyday example
Surface tension	Raindrops, floating needle, soap bubbles
Capillarity	Lamp wick, ink in blotting paper, sap in plants
Archimedes	Ships, submarines, hydrometer, icebergs
Bernoulli	Aircraft lift, roofs blown off, atomiser
Pascal's law	Hydraulic lift, brakes, press

TRAP ALERT

- Water's density peak is **4 C**, not 0 C; ice is **less** dense and floats.
- Soap bubble pressure is **higher** inside; oil spreads because its surface tension is **lower**.
- Steel is **more** elastic (higher Young's modulus) than rubber.

UP LINK

UP link: the **Ganga** at Prayagraj meets sea water nowhere, so river barges on **NW-1** float lower than they would at sea (fresh water is less dense), a direct case of Archimedes' principle.

LAST-MINUTE 10

1. Raindrops spherical = surface tension
2. Needle floats = surface tension
3. Oil spreads: lower surface tension
4. Bubble pressure **more** inside
5. Pen leaks in a plane = low pressure
6. Barometer falls fast = storm
7. Water densest at **4 C** (277 K)
8. 1/9 of an iceberg above water
9. Ship rises in the sea; sea swimming easier
10. Iron floats on mercury, sinks in water

Chapter 5

Light and Optics

Built from 60 UPPSC questions (1990–2023). Focus: total internal reflection, scattering and red signals, mirrors in vehicles, myopia and hypermetropia, the eye's sensitivity and the speed of light.

ASKED MOST · REVISE FIRST

- ■ **Total internal reflection**: **mirage, diamond's sparkle, optical fibre** 11× 1995, 1996, 2008, 2010, 2012, 2012, 2013, 2014, 2015, 2017, one-liner
- ■ **Scattering**: blue sky, red sunset; **red** for danger as it scatters **least** 6× 2007, 2013, one-liner, one-liner, one-liner, one-liner
- ■ **Convex mirror** for rear view; **concave mirror** for searchlights and headlights 6× 1991, 2009, 2010, 2011, 2015, 2023
- ■ **Myopia** is corrected with a **concave lens**; hypermetropia with a **convex** lens 5× 2002, 2008, 2010, 2011, 2013
- ■ Air bubble in water = **concave lens**; parallel mirrors give **infinite** images 7× 1994, 2002, 2008, 2010, 2013, 2013, 2015

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

5.1 Nature and Speed of Light

- Visible light: about **390-780 nm** 1× 2005
- Eye most sensitive at **5500 Å** (yellow-green) 2× 2008, 2014

- Not correct: eye is most sensitive to **red** 1× 2021
- Light seems to go straight: its **wavelength is very small** 1× 2014
- Highest velocity: **light**; light is faster than sound 2× 2002, 2018
- Light is fastest in a **vacuum** 1× 2014
- Sunlight reaches Earth in about **8 minutes** (8 min 20 s) 1× 2006
- Not luminous: **Moon** (it reflects sunlight) 1× 2023

5.2 Refraction and Total Internal Reflection

- Unchanged when light changes medium: **frequency** 1× 2003
- Pool looks shallower: **refraction** 1× 2023
- Sun and Moon look oval near the horizon: **refraction** 1× 2008
- Optical fibre: **total internal reflection** 3× 2008, 2010, one-liner
- Optical fibre: high capacity, low power, **no radio interference** (1, 2, 3) 3× 2001, 2002, 2003
- **Mirage**: **total internal reflection** 4× 1995, 2012, 2013, 2014
- Diamond shines: **total internal reflection** 4× 1996, 2012, 2015, 2017
- 3-D pictures: **holography** 2× 1990, 2008

Phenomenon	Cause
Mirage, diamond sparkle, optical fibre	Total internal reflection
Shallow-looking pool, bent stick, twinkling stars	Refraction
Rainbow, prism spectrum	Dispersion (with refraction and reflection)
Blue sky, red sunset	Scattering
Colours of soap film and oil	Interference

5.3 Colour, Scattering and Dispersion

- Eclipse retinal burn: key says **ultraviolet** (visible blue light and infrared also harm) 1× 2003
- A star's colour shows its **temperature** 3× 2003, 2005, 2008
- Photosynthesis with invisible light: some **bacteria** 1× 2013
- Photosynthesis: **light to chemical** energy 1× 1998
- Diffusion of light in air: **dust particles** 1× 2021
- Colour is set by **wavelength** 2× 2012, 2014
- Shortest wavelength in the visible spectrum: **violet** 1× one-liner
- Least deviated by a prism: **red** 1× 2016
- Red traffic signals: **longest wavelength** 1× 2013
- Red for danger: it **scatters least** (A and R true, R explains A) 2× 2007, one-liner
- Blue sky and red sunset: **scattering** (short wavelengths scatter more) 3× one-liner, one-liner, one-liner
- Sky seen by an astronaut or from great height: **black** 2× 2005, one-liner
- Colours in sunlight: **7** 1× 2015
- Grass looks green: it **reflects green** light 1× 2017
- Expanding universe: **Doppler effect** (red shift) 1× 2010
- Black hole unseen: light cannot escape its gravity (A and R true, R explains A) 1× 2001
- Middle colour of a rainbow: **green**; extremes: **violet and red** 2× 2012, one-liner

VIBGYOR: WAVELENGTH AND SCATTERING



5.4 Mirrors and Lenses

- Air bubble in water acts as a **concave lens** 4× 2002, 2008, 2010, 2013

- Images between two parallel mirrors: **infinite** 3× 1994, 2013, 2015
- Mirror to see your full image: **half your height**; for 1.50 m, **0.75 m** 4× 2010, 2011, 2014, 2014
- Radius of curvature of a plane mirror: **infinity** 1× 2015
- Not correct: doctors use a **convex** mirror for ENT checks (they use **concave**) 1× 2010
- Searchlight and car headlight: **concave mirror** 2× 2009, 2023
- Rear-view mirror: **convex mirror** 4× 1991, 2010, 2011, 2015
- Power of sunglasses: **0 dioptre** 1× 1995
- Astronomical telescope image: **virtual and magnified** 1× 2013
- Equal convex and concave lenses together: a **plane glass sheet** 1× 2017

Mirror / lens	Use
Concave mirror	Searchlight, headlight, shaving, dentist and ENT, solar cooker
Convex mirror	Rear view, street-corner safety
Plane mirror	Periscope, kaleidoscope
Concave lens	Myopia
Convex lens	Hypermetropia, magnifier, camera
Cylindrical lens	Astigmatism

5.5 The Human Eye

- Image on the retina: **smaller and inverted** (real) 2× 1995, 2004
- Retina is like a camera's **film** 1× 2011
- Eye focuses by **changing the lens's convexity** (accommodation) 1× 2014
- Least distance of distinct vision: **25 cm** 1× 2008
- Not true of hypermetropia: corrected by a **concave** lens (it needs convex) 1× 2010
- Myopia corrected by a **concave lens** 4× 2002, 2008, 2011, 2013

TRAP ALERT

- Mirage is **total internal reflection** (per keys), not simple reflection.
- Frequency never changes in refraction; speed and wavelength do.
- **Convex** = rear view; **concave** = headlight; ENT doctors use **concave**.
- Myopia = **concave**; hypermetropia and old-age presbyopia = **convex**.

UP LINK

UP link: the **Aryabhatta Research Institute of Observational Sciences (ARIES)** is in Nainital (Uttarakhand, carved from UP), and the **Indian Institute of Astrophysics** runs telescopes elsewhere; in UP, **IIT Kanpur** and **BHU** run optics and photonics labs (soft detail).

LAST-MINUTE 10

1. Visible light **390-780 nm**; eye most sensitive at **555 nm** (yellow-green)
2. Light fastest in vacuum; Sun to Earth **8 min 20 s**
3. Frequency unchanged in refraction
4. Mirage, diamond, optical fibre = **TIR**
5. Blue sky = scattering; red signal = least scattered
6. Sky black from space; 7 colours; green in middle
7. Air bubble in water = **concave lens**
8. Full-length mirror = **half your height**; parallel mirrors = infinite images
9. Convex mirror = rear view; concave = headlight
10. Myopia = **concave lens**; near point 25 cm

Chapter 6

Heat and Thermodynamics

Built from 40 UPPSC questions (1993–2025). Focus: latent heat, Celsius-Fahrenheit conversion, boiling point and pressure, pressure cookers, evaporative cooling, humidity, light clothes and cloudy nights.

ASKED MOST · REVISE FIRST

- **-40** is the same on Celsius and Fahrenheit; **F = (9/5)C + 32** 6× 2003, 2015, 2015, 2016, 2019, 2024
- **Pressure cooker**: higher pressure raises the **boiling point** 3× 1995, 2007, 2015
- Water boils **below 100 C** on hills: **lower air pressure** 2× 1994, 2023
- **Light/white clothes** in summer: they **reflect** heat and absorb little 3× 1993, 2007, 2016
- **Earthen pitcher** cools by **evaporation**; sweat cools the body 3× 2012, 2019, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

6.1 Latent Heat and Temperature Scales

- Steam burns more than boiling water: steam carries **latent heat** 1× 1993
- **F = (9/5)C + 32** 1× 2024
- Celsius and Fahrenheit read the same at **-40** 2× 2003, 2015
- 98.4 F = **36.89 C** 1× 2015
- 200 F = **93.3 C** 1× 2019
- Averages follow the same rule: **$F_{av} = (9/5)C_{av} + 32$** 1× 2016
- Alcohol thermometers in cold countries: **lower freezing point** 1× 1997
- Antifreeze in car engines: **ethylene glycol** 2× 2008, 2010

Point	Celsius	Fahrenheit	Kelvin
Absolute zero	-273.15	-459.67	0
Same reading	-40	-40	233
Water freezes	0	32	273
Body temperature	37	98.6	310
Water boils (sea level)	100	212	373

6.2 Expansion and Cooling Devices

- Heating expands a body and **lowers its density** 1× 2006
- Glass rod in steam: width also **increases** 1× 1994
- Ice melting: volume **decreases** 1× 2012
- **Thermostat**: keeps **temperature** steady automatically 3× 1993, 1996, one-liner
- Safest fridge temperature for food: **4 C** 1× 2007
- Soggy biscuits turn crisp in a fridge: **low humidity** inside 1× 2009
- Ice cools a drink by taking **latent heat** (A and R true, R explains A) 1× 2009
- Ice in the freezer cuts cooling: ice is a **poor conductor** (A and R true, R explains A) 1× 2010
- Not true: AC and cooler **both keep humidity** (a cooler adds it, an AC removes it) 1× 2007

6.3 Humidity and Evaporation

- Relative humidity falls as temperature rises; evaporation raises absolute humidity (A and R true, R does not explain A) 1× 2024
- Clothes dry fastest at **20% RH and 60 C** 1× 2008
- Earthen pitcher keeps water cold by **evaporation** 1× 2012
- Sweating depends on **temperature and humidity** both 1× 2019
- Sweat evaporation removes heat; humidity adds discomfort (A and R true, R does not explain A) 1× 2025
- Humid heat felt in **humid weather** 1× 2003
- **Xeric** condition: **low humidity** 1× 2014

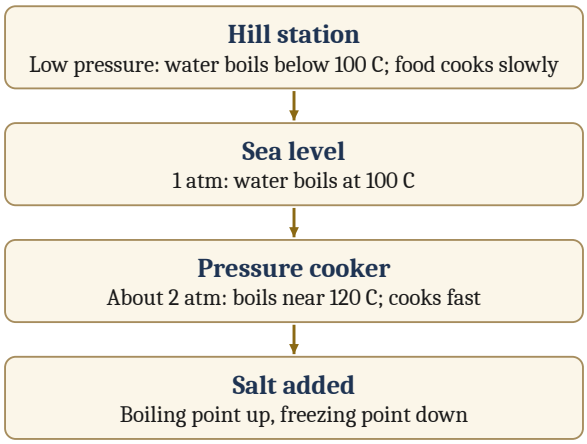
6.4 Pressure and Boiling Point

- We do not feel air pressure: **blood pressure** slightly exceeds it 1× 2010
- Water boils below 100 C at height: **air pressure falls** 1× 1994
- Boiling point falls with height; pressure **decreases** with height (A true, R false) 1× 2023

QUANTUM IAS & PCS

- Pressure cooker cooks fast: water boils at a **higher temperature** 2× 2007, 2015
- Cooker's top temperature depends on the **vent area and weight** 1× 1995
- Salt in water: boiling point **rises**, freezing point **falls** 1× 2006
- Bottle bursts on freezing: water **expands** as it freezes 1× 1995

PRESSURE AND BOILING POINT



6.5 Conduction, Convection and Radiation

- Thermos flask: no heat loss by **conduction and convection** (both; silvered walls stop radiation) 1× 2019
- Best umbrella: **white on top, black below** 1× 1993
- White clothes in summer: they **reflect** heat 1× 2007
- Light clothes in summer: light colours absorb little heat (A and R true, R explains A) 1× 2016
- Cloudy nights are warmer: clouds **stop heat radiating** away 1× 2008
- Reflects most sunlight: **fresh snow** (highest albedo) 1× 2021
- Cold storage lengthens fruit life: **lower respiration** 1× 2021
- Wool keeps us warm, but it works by **trapping air**; the protein reason is false (A true, R false) 1× 2001

TRAP ALERT

- Steam burns more because of **latent heat**, not higher temperature alone.
- On hills water boils **below** 100 C; in a cooker **above** 100 C.
- Salt raises boiling point but **lowers** freezing point.
- Ice melting **shrinks** in volume; water freezing **expands**.

UP LINK

UP link: Kanpur, Lucknow and Prayagraj often cross **45 C** in May-June, and the dry **loo** wind lowers relative humidity, which is why earthen pots (matkas) from **Khurja** and Nizamabad keep water cool by evaporation.

LAST-MINUTE 10

1. Steam burns more = latent heat
2. **F = (9/5)C + 32**; equal at **-40**
3. Antifreeze = ethylene glycol; alcohol thermometer for cold lands
4. Fridge food at **4 C**; ice melting reduces volume
5. Matka cools by **evaporation**; clothes dry at low RH, high temp
6. Water boils below 100 C on hills (low pressure)
7. Pressure cooker = higher boiling point
8. Salt: BP up, FP down
9. White clothes, white-top umbrella reflect heat
10. Cloudy nights warmer; fresh snow reflects most

Chapter 7

Waves and Sound

Built from 33 UPPSC questions (1990–2025). Focus: the electromagnetic spectrum, ionosphere, sound's speed in media, echo and reverberation, ultrasound, noise limits and TV remotes.

ASKED MOST · REVISE FIRST

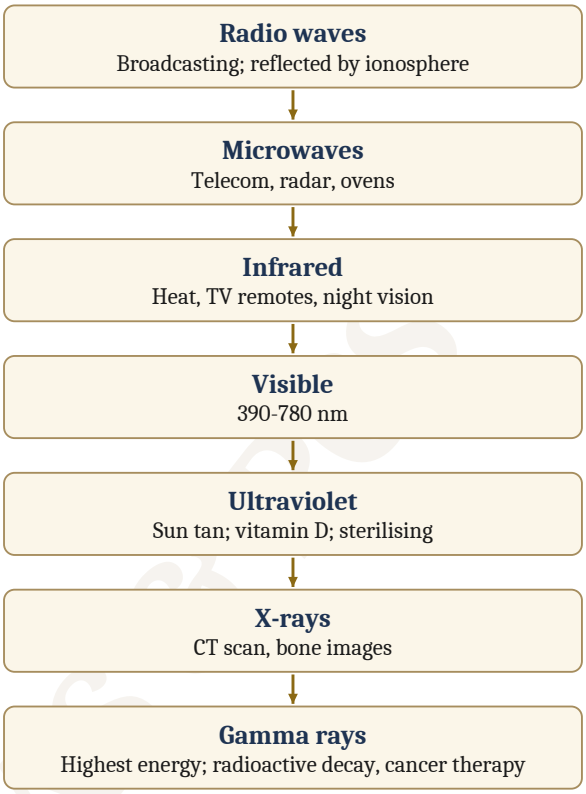
- ■ Sound is fastest in **steel** (solids); it **cannot travel in a vacuum** **8×** 1990, 2002, 2002, 2008, 2014, 2018, one-liner, one-liner
- ■ TV remote uses **infrared** **3×** 2002, 2010, 2013
- ■ **Ionosphere** reflects radio waves back to Earth **3×** 1991, 1998, one-liner
- ■ **Ultrasonic** = above **20,000 Hz**; bats navigate with it **4×** 2002, 2004, 2005, 2012
- • Hazardous noise: above **80 dB** **2×** 2008, 2013

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

7.1 The Electromagnetic Spectrum

- Increasing frequency: **radio, infrared, visible, X-rays** **1×** 2023
- Decreasing frequency: **gamma, X-rays, ultraviolet, radio** **1×** 2025
- ■ Long radio waves are reflected by the **ionosphere** **3×** 1991, 1998, one-liner
- Atmosphere is heated mainly by **longwave terrestrial radiation** **1×** 2022
- Not true of cosmic rays: they are **electromagnetic waves** (they are charged particles) **1×** 2005
- Telecommunication waves: **microwaves** **1×** 2013
- Dish signals fail in rain: rain **absorbs and scatters** radio energy (1, 2, 3 per key) **1×** 2017
- TV signals limited in range: Earth is **curved** **2×** 1994, 2010
- CT scan uses **X-rays** **1×** 2011

ELECTROMAGNETIC SPECTRUM, LOW TO HIGH FREQUENCY



7.2 Propagation and Speed of Sound

- Highest sound velocity: **steel** **2×** 2008, 2018
- Speeds: **air < water < steel** **1×** one-liner
- Sound can travel in **solids and gases** (and liquids) **1×** 2002
- Sound cannot travel in a **vacuum** **2×** 1990, 2014
- Astronauts cannot hear each other on the Moon: **no atmosphere** (no medium) **2×** 2002, one-liner
- Men's and women's voices differ: vocal cords differ in **length and thickness** (A and R true, R explains A) **1×** 2025

Medium	Speed of sound (approx.)
Air (20 C)	343 m/s
Water	1,480 m/s
Steel	About 5,000-6,000 m/s
Vacuum	Zero (no sound)

7.3 Echo, Reverberation and Ultrasound

- Reverberation in large halls: **multiple reflections** (A and R true, R explains A) **2×** 2002, 2003
- Concert hall walls should **absorb sound** **1×** 2007

QUANTUM IAS

- Clear echo needs at least **16.5 m** (17 m) 1× 2007
- Ultrasound can kill insects, clean, treat disease, run automatic doors (**all correct**) 1× 2004
- **Ultrasonic**: above **20,000 Hz** 1× 2012
- Bats hunt in the dark with **ultrasonic waves** 1× 2005
- Heart vibrates at infrasonic; **Mach number** for speed; ultrasonic above 20,000 Hz (1, 3, 4); sound is **slower** in gases 1× 2002

Range	Frequency	Example
Infrasonic	Below 20 Hz	Earthquakes, whales, heartbeats
Audible	20-20,000 Hz	Human hearing
Ultrasonic	Above 20,000 Hz	Bats, dolphins, SONAR, ultrasound scans

7.4 Noise Limits

- WHO safe city noise: **45 dB** 1× 2010
- Day limits: **residential 55, silence zone 50, industrial 75 (key option 70), commercial 65 dB** 1× 2022
- Hazardous noise: above **80 dB** 2× 2008, 2013

7.5 Musical Instruments, Remote Controls and Television

- Sitar makes **stationary transverse** waves 1× 2021
- TV remote: key says **radio waves**; most remotes use **infrared** 1× 2013
- TV remote uses **infrared** 3× 2002, 2010, 2013
- When a TV is switched on, **audio and video start together** 1× 2007
- TV audio is sent by **frequency modulation** 2× 2007, 2010

TRAP ALERT

- Sound: **steel > water > air**; light is the reverse (fastest in vacuum).
- Cosmic rays are **particles**, not EM waves.
- Remote = **infrared**; telecom = microwaves; CT = X-rays.
- Clear echo distance = **17 m** (half of 34 m at 0.1 s).

UP LINK

UP link: CPCB noise rules apply in UP cities; the UP government has enforced the **silence zones** (100 m around hospitals, schools, courts) and curbs on **loudspeakers** at religious places (2022 drive).

LAST-MINUTE 10

1. Radio, IR, visible, UV, X, gamma: rising frequency
2. **Ionosphere** reflects radio waves
3. Cosmic rays are not EM waves; CT = X-rays
4. Sound fastest in **steel**; no sound in vacuum
5. Moon: no air, so no sound
6. Clear echo needs **17 m**
7. Ultrasonic **> 20,000 Hz**; bats use it
8. WHO safe noise **45 dB**; hazard above **80 dB**
9. TV remote = **infrared**; TV audio = **FM**
10. Sitar = stationary transverse waves

Chapter 8

Electricity and Conductivity

Built from 46 UPPSC questions (1990–2025). Focus: units of electricity, bulb resistance, dynamo, motor and transformer, fuse and earthing, lamps, conductors, lightning and semiconductors.

ASKED MOST · REVISE FIRST

- **kWh sums**: units = watts x hours / 1000 (100 W for 4 h = **0.4**; 5 x 100 W for 20 h = **10**; 15 x 200 W for 24 h = **72**) 4× 2006, 2009, 2017, 2023
- **Tungsten** filament in bulbs 3× 1990, 2005, 2011
- **Dynamo** = mechanical to electrical (electromagnetic induction); **motor** = electrical to mechanical 5× 2007, 2009, 2012, 2017, one-liner
- **Fluorescent tube**: **mercury vapour and argon** at low pressure 3× 1990, 2006, 2009
- **Semiconductors**: **silicon and germanium**; resistance **falls** on heating 6× 2005, 2015, 2015, 2015, 2016, one-liner

UAITJUMP

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

8.1 Electrical Energy and the Unit

- **1 kWh = 3.6 x 10^6 J** 1× 2006
- 100 W bulb for 4 hours: **0.4 unit** 1× 2009
- Five 100 W bulbs for 20 hours: **10 units** 1× 2017
- Fifteen 200 W bulbs for 24 hours (Silkyara tunnel): **72 units** 1× 2023

8.2 Resistance, Bulbs and Circuits

- Of two bulbs, the **dim one has higher resistance** 3× 2008, 2009, one-liner
- Maximum light from two identical bulbs: connect in **parallel** 1× 2021
- 15 V across a brown-green-black resistor (15 ohm): **1 ampere** 1× 2023
- No current between bodies at the **same potential** 1× 2014
- n equal charged drops merge: potential becomes **$n^{2/3}$ V** 1× 2015

8.3 Dynamo, Motor and Transformer

- Dynamo is a **converter of energy** 1× 2009
- Works on electromagnetic induction: **dynamo** 2× 2012, 2017
- Dynamo: **mechanical to electrical** 1× one-liner
- Electrical to mechanical: **electric motor** 1× 2007
- Motors burn at low voltage: they **draw more current** 1× 2008
- Transformer: **steps AC voltage up or down** 1× 2006
- Mobile charger: a **step-down transformer** 1× 2017
- AC to DC: **rectifier** 1× 2006

Device	Does
Dynamo / generator	Mechanical to electrical (induction)
Electric motor	Electrical to mechanical
Transformer	Changes AC voltage (works only on AC)
Rectifier	AC to DC
Inverter	DC to AC
Fuse	Melts on excess current

8.4 Safety: Fuse, Earthing and Plugs

- Bulb filament: **tungsten** 3× 1990, 2005, 2011
- Longest pin of a 3-pin plug: **earth** 1× 2007
- Earthing is for **safety** 1× 2002
- Fuse wire: **high resistance, low melting point** 1× 1993
- Fuse wire is lead-tin alloy; its melting point is **lower**, not higher, than copper's (A true, R false) 1× 2025

8.5 Lamps and Lighting

- **CFL** = Compact Fluorescent Lamp 1× 2015
- Gas in advertising signs: **neon** 1× 2001
- Fluorescent tubes: **mercury vapour and argon** 1× 2006
- Tube light: **argon at low pressure** with mercury vapour 1× 2009
- Gases in a fluorescent tube: **mercury and neon** (old key) 1× 1990
- Why a tube over a bulb: **none of these** (per key) 1× 2009

8.6 Solar Cells and the Earth's Magnetism

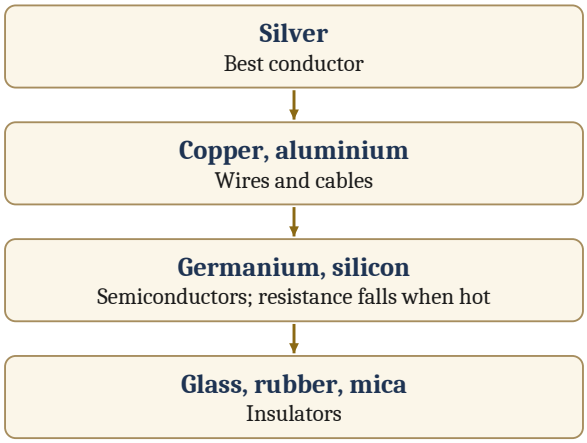
- Ascending power use: **fan, TV, electric iron, electric kettle** 1× 1992
- **Photovoltaic cells** = **solar cells**; they turn sunlight into electricity 2× 2016, 2022
- Earth's magnetic field: **currents in the core** 1× 2011

8.7 Conductors and Lightning

- Best conductor: **silver** 1× 2015
- Lightning conductors protect buildings by sending charge to earth (A and R true, R explains A) 1× 1999
- Copper rods for lightning conductors: better conductor, does not rust (A and R true, R explains A) 1× 2013
- Closed car is safe in lightning: it acts as a **hollow conductor** (A and R true, R explains A) 1× 2023

QUANTUM IAS & PCS

CONDUCTORS TO INSULATORS



8.8 Superconductors and Semiconductors

- Superconductors would pay most at **room temperature** 1× 2000
- Highest superconducting temperature: about **133-138 K** when asked (*) 1× 2013
- High-temperature superconductors: **ceramic oxides** 1× 2000
- Semiconductor resistance on heating: **decreases** 1× 2015
- Transistor component: **germanium** 1× 2016
- Commonest transistor material: **silicon** 1× 2015

■ Semiconductors: **silicon and germanium** (1 and 4) 3× 2005, 2015, one-liner

TRAP ALERT

- The **dim** bulb has the **higher** resistance ($P = V^2/R$).
- Metals' resistance rises with heat; **semiconductors' falls**.
- Fuse = **low** melting point, **high** resistance; earth pin is the **longest and thickest**.
- Transformers work only on **AC**; charger = step-down transformer + rectifier.

UP LINK

UP link: the **Silkyara tunnel** sum (2023) was set in Uttarakhand; in UP, **Jewar** (Gautam Buddha Nagar) has an approved **semiconductor** assembly unit (HCL-Foxconn, 2025), and UP's largest solar parks are in **Bundelkhand**.

LAST-MINUTE 10

1. **1 kWh = 3.6 x 10^6 J**; units = W x h / 1000
2. Dim bulb = **higher resistance**; parallel = more light
3. Tungsten filament; fuse = low melting point
4. Earth pin is longest; earthing = safety
5. Dynamo = mechanical to electrical; motor = reverse
6. Transformer = AC voltage; rectifier = AC to DC
7. Tube light = **mercury vapour + argon**; neon = signs
8. Silver = best conductor; copper lightning rods
9. Closed car = safe in lightning (hollow conductor)
10. Si, Ge = semiconductors; resistance falls on heating

Chapter 9

Nuclear Physics

Built from 45 UPPSC questions (1992–2025). Focus: fission and fusion, atom and hydrogen bombs, moderators and coolants, thorium, breeder reactors, India's nuclear plants and tests.

ASKED MOST · REVISE FIRST

- ■ **Atom bomb = fission**; **hydrogen bomb = fusion** (of deuterium); H-bomb by **Edward Teller** 6× 1994, 2007, 2015, 2016, 2021, one-liner
- ■ **Moderators**: **heavy water, light water, graphite**; heavy water also a **coolant** 4× 1993, 2011, one-liner, one-liner
- ■ **Fast Breeder Test Reactor** (Kalpakkam) coolant: **liquid sodium** 3× 1997, 2008, 2010
- ■ **Kalpakkam** = nuclear power (IGCAR, Tamil Nadu) 4× 2001, 2002, 2023, one-liner
- ■ **Plant-state matching**: **Narora-UP, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan, Kudankulam-TN** 4× 2004, 2009, 2010, 2025

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

9.1 Discoveries and Basic Ideas

- Wrong pair: **transuranic synthesis-Enrico Fermi** (per key; Seaborg made most) 3× 2001, 2002, 2003
- Not correct: **splitting a nucleus is fusion** (it is fission) 1× 1999
- Rutherford doubted nuclear power ("moonshine") (A and R true, R explains A, per key) 1× 1994
- Particle claimed to rebut relativity: **microwave photon** (per key) 1× 2009
- Cyclotrons accelerate **protons** (charged particles) 1× 1997
- Most energetic radiation among options: **X-rays** (gamma rays are higher still) 2× 2005, 2018

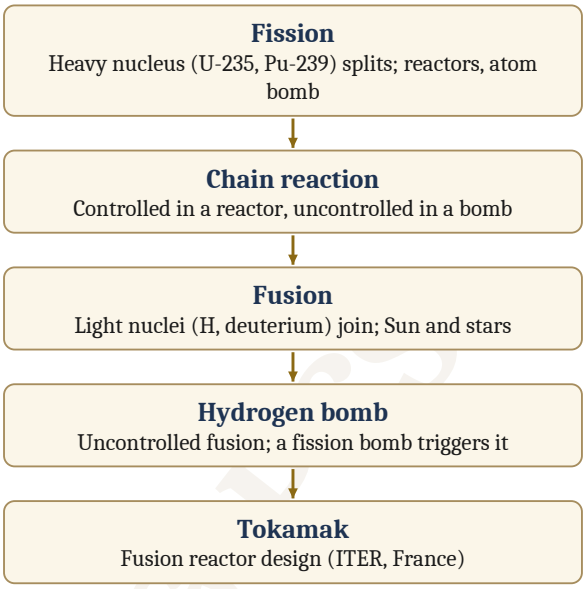
9.2 Energy of the Stars

- Stars shine by **nuclear fusion and gravitational contraction** (1 and 2) 1× 2000
- Star energy: **hydrogen turns into helium** 1× 2009

9.3 Reactors and Bombs

- Nuclear reactor: a "**molecular furnace**" (per key) 1× 1992
- Fusion reactors: **tokamak** design, **very high temperature** (ii and iii); not fission 1× 2016
- Reactor vs bomb: the reactor's chain reaction is **controlled** 1× 1999
- Radioactivity is a **nuclear property**; H-bomb is **not** fission (only 1) 2× 2016, 2021
- Atom bomb: **nuclear fission** 1× 2007
- H-bomb: **fusion of deuterium**; uncontrolled fusion 2× 1994, one-liner
- H-bomb developed by **Edward Teller** 1× 2015
- Nuclear medicines are **radioisotopes** 1× 1998

FISSION VERSUS FUSION



9.4 Fuel, Moderators and Coolants

- Nuclear fuel abundant in India: **thorium** 1× 2003
- Not a nuclear fuel: **calcium** 1× 2014
- **Enriched uranium**: natural uranium with more **U-235** 1× 2009
- Moderator: **heavy water** (and light water); **graphite** too 2× 2011, one-liner
- Heavy water: **coolant** and moderator; ordinary water too 2× 1993, one-liner
- Heavy water made at **Trombay** (among options) 1× 2015
- First heavy water plant: **Nangal** 1× 2010
- **Breeder reactor**: makes **more fissile material than it burns** 1× 2009
- Kalpakkam breeder statements: **1 and 3** per key 1× 2005
- FBTR coolant: **liquid sodium** 3× 1997, 2008, 2010

Part	Material	Job
Fuel	U-235, Pu-239, (Th-232 via U-233)	Fission
Moderator	Heavy water, light water, graphite	Slows neutrons
Control rods	Cadmium, boron	Absorb neutrons
Coolant	Heavy water, water, liquid sodium, CO2	Removes heat
Shield	Concrete, lead	Stops radiation

9.5 India's Nuclear Map

- **Dhruva, Purnima, Cirus**: Indian **research reactors** 1× 2004
- **Kalpakkam**: nuclear power plant 3× 2001, 2002, 2023
- **IGCAR**: **Tamil Nadu** (Kalpakkam) 1× one-liner
- Wrong pair: **Harish-Chandra Research Institute-Chennai** (it is in **Prayagraj**) 1× 2010
- **Kudankulam-TN, Kakrapar-Gujarat, Kaiga-Karnataka, Rawatbhata-Rajasthan** 1× 2025
- **Kalpakkam-TN, Narora-UP, Kakrapar-Gujarat, Trombay-Maharashtra** 1× 2010
- Wrong: **Rajasthan Atomic Power-Udaipur** (it is **Rawatbhata**, near Kota) 1× 2009
- Plants at **Narora and Kalpakkam** (1 and 3) 1× 2004
- In seismic zone IV: **Narora** 1× 2010
- Recently active (2000): **Kaiga** 1× 2000
- Russia to build **6** units at Kudankulam 1× 2009
- Highest installed capacity: **Kudankulam** 1× 2014
- Nuclear share of power: about **3%** (*) 1× 2008

9.6 Nuclear Tests and Security

- Pokhran 1974 code name: **Smiling Buddha** 1× 2009
- **Operation Shakti**: Pokhran tests of **1998** 1× 1999
- ICONS-2024: India rebuilt the **Counter Nuclear Smuggling Team** (only 2) 1× 2024

TRAP ALERT

- Atom bomb = **fission**; H-bomb = **fusion**; reactors = **controlled** fission.
- Heavy water is both **moderator and coolant**; FBTR uses **liquid sodium**.
- Rajasthan plant = **Rawatbhata** (Kota), not Udaipur; Harish-Chandra Institute = **Prayagraj**.
- Gamma rays carry the most energy; X-rays only if gamma is not in the options.

UP LINK

UP link: **Narora Atomic Power Station** (Bulandshahr, 1991) lies in **seismic zone IV**; the **Harish-Chandra Research Institute** (physics and mathematics) is at **Jhunsi, Prayagraj**.

LAST-MINUTE 10

1. Fission = splitting; fusion = joining (stars: H to He)
2. Atom bomb = fission; H-bomb = fusion, **Edward Teller**
3. Reactor = controlled chain reaction; fusion reactor = tokamak
4. India rich in **thorium**; enriched U = more U-235
5. Moderators: heavy water, light water, graphite
6. FBTR coolant = **liquid sodium**; breeder makes more fuel
7. Nangal = first heavy water plant; Dhruva, Cirus = research reactors
8. Narora = UP, zone IV; Kudankulam = largest, Russian
9. Rawatbhata = Rajasthan; Kakrapar = Gujarat; Kaiga = Karnataka
10. Smiling Buddha **1974**; Operation Shakti **1998**

Chapter 10

Computer and Information Technology

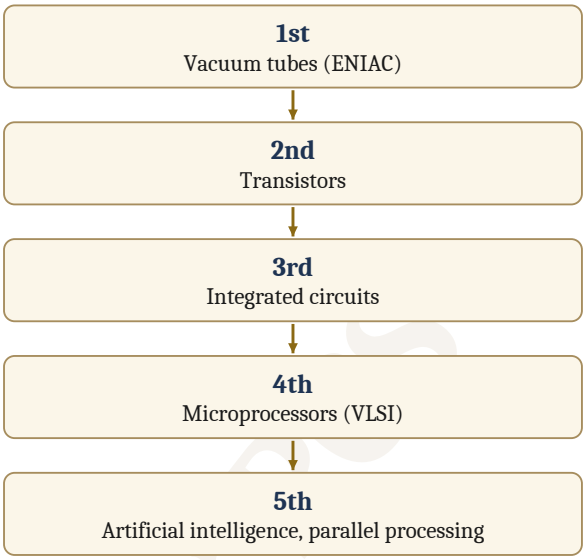
Built from 81 UPPSC questions (1993–2024). Focus: computer generations and chips, Indian supercomputers, memory and bits, operating systems and languages, the internet, modem and full forms.

ASKED MOST · REVISE FIRST

- IC chips are made of **silicon** 5× 2002, 2004, 2006, 2010, 2014
- **Modem** sends data over telephone lines 6× 2002, 2007, 2013, 2015, one-liner, one-liner
- **RAM** = Random Access Memory, **volatile**; **ROM** = permanent 6× 2002, 2003, 2012, 2013, 2024, one-liner
- Indian supercomputers: **PARAM** (first, C-DAC), **Anupam** (BARC); PARAM is not a missile 5× 1999, 2004, 2004, 2006, 2009
- **1 byte = 8 bits**; bit = **binary digit**; 1 KB = 1,024 bytes 6× 2006, 2006, 2010, 2013, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

COMPUTER GENERATIONS



10.1 History and Supercomputers

- First computer, father of the computer: **Charles Babbage** 2× 2007, one-liner
- First mouse body (about 40 years back): **wood** 1× 2010
- Third-generation computers: **integrated circuits** 1× 2016
- Miniaturisation: **IC chips** 1× 2007
- IC chip material: **silicon** 5× 2002, 2004, 2006, 2010, 2014
- ICs are classed by the **number of transistors** 1× 2015
- First Indian supercomputer: **PARAM** 1× 2004
- Wrong pair: **PARAM 10,000-missile** (it is a supercomputer) 3× 1999, 2006, 2009
- BARC supercomputer: **Anupam** 1× 2004
- World's fastest (Dec 1996): **10^{12} operations a second** (teraflop) 1× 1998
- Fastest computer (2008): none of the options (*); **El Capitan** (USA) leads now 1× 2008
- **Magic Cube** supercomputer: **China** 1× 2015
- Smallest, fastest brain-like computer: **quantum computer** 1× 2000
- **Y2K**: problem when a year's last two digits become **00** 1× 2000

10.2 Hardware and Memory

- Most important part: **CPU** 1× 2012
- Intel Core i9: a **processor** 1× 2023
- After a double click, information goes to the **CPU** 1× 2013
- Only input devices: **mouse, keyboard, scanner** 1× 2024
- Word document, video or MP3: a **file** 1× 2013
- CPU = **Central Processing Unit** 1× one-liner
- Memory and word length are measured in **bits** 2× 2006, 2010
- **1 byte = 8 bits** 1× 2006
- **1 GB = 1,024 MB**; **1 KB = 1,024 bytes** 2× one-liner, one-liner
- **Bit** = **binary digit** 1× 2013
- MAC address: **48 bits** 1× 2023
- Self-complementing code: **5211** 1× 2013
- Permanent, non-volatile memory: **ROM** 2× 2012, 2024
- **RAM** = Random Access Memory; loses data when power goes 4× 2002, 2003, 2013, one-liner
- Fastest memory: **cache** 1× 2013
- Not true of flash memory: **cheaper than disk drives** 1× 2004
- Large-storage hardware: **both** options 1× 1999
- **Index hole**: **floppy disk**; **IRQ 6**: floppy drive controller 2× 2013, 2015

QUANTUM IAS & PCS

- Keyboard cable port: **USB** 1× 2015
- Laser printers use a **semiconductor laser** 1× 2009
- Daisy wheel printer: **impact** printer 1× 2013

Memory	Fact
Cache	Fastest; between CPU and RAM
RAM	Working memory; volatile
ROM	Permanent; holds boot program
Flash	Non-volatile; pen drives, SSD (costlier per GB than hard disks)
Hard disk	Large, cheap, slower

10.3 Software and Languages

- Controls basic functions: **operating system** 1× 2013
- Not an OS: **Microsoft Office** 1× 2023
- Running many apps at once: **multitasking** 1× 2015
- Scientific language: **FORTRAN** 2× 2002, 2003
- BASIC: a **procedural** language 1× 2013
- **Assembler**: assembly to **machine language** 1× 2005
- Computer virus: a **software program** 2× 1993, 2010
- Error in a program: **bug** 1× 2010
- **Stuxnet**: malware that sabotaged Iran's **nuclear centrifuges** 1× 2010
- **Talaash**: C-DAC's Devanagari **OCR** and portal 1× 2004
- C-DAC anti-cyber-theft software: **NAYAN** 2× 2008, 2009

10.4 Internet and Networks

- Worldwide linked computers: **Internet** 1× 2004
- Internet topology: **tree** (per key) 1× 2013
- Internet works on **packet switching** 1× 2015
- Between physical and network layers: **data link layer** 1× 2013
- Rules for data exchange: **protocols** 1× 2015
- Not IT terms: **optical storage** (per key), **Pinaka** (a rocket) 2× 2001, 2002
- IT terms: **all of the above** 1× 2011
- Finds a user's e-mail address: **Finger** 1× 2010
- **WWW** = **World Wide Web**; inventor **Tim Berners-Lee** 3× 2010, 2015, one-liner
- **Modem**: data over telephone lines 6× 2002, 2007, 2013, 2015, one-liner, one-liner
- **Bluetooth**: wireless link between devices 1× 2011
- First station with Google free Wi-Fi: **Mumbai Central** 1× 2015
- Not true of Li-Fi: uses optical fibre; India's trial by the I&B Ministry (it was **MeitY**) 1× 2017
- E-mail can attach **all** file types 2× 2013, one-liner
- **SkyDrive** (OneDrive): **Microsoft** 1× 2013

10.5 Full Forms and IT in Governance

- **SIM** = Subscriber Identity Module 1× 2012
- **ICT** = Information and Communications Technology 1× 2015
- **ATM** = Automated Teller Machine 1× 2014
- **UPS** = Uninterrupted Power Supply 1× 2015
- **Bibliometry**: an information management tool 1× 2019
- World Computer Literacy Day: **2 December** 1× 2016
- **Vidya Vahini**: **computer education** in schools 1× 2013
- e-Governance **cuts cost** and **raises citizens' input** (2 and 3) 1× 2024

TRAP ALERT

- **RAM** is volatile; **ROM** is not; **cache** is fastest.
- **IC** = 3rd generation; **microprocessor** = 4th.
- MS Office is an application, not an OS; FORTRAN = scientific, COBOL = business.
- Li-Fi uses **LED light**, not fibre.

UP LINK

UP link: **C-DAC Noida** works on language computing and cyber security; **IIT Kanpur** hosts the **PARAM Sanganak** supercomputer (National Supercomputing Mission, 2020) and the C3iHub cyber security centre.

QUANTUM IAS & PCS

LAST-MINUTE 10

1. **Charles Babbage** = father of the computer
2. IC = 3rd generation; chips = **silicon**
3. **PARAM** = first Indian supercomputer (C-DAC); Anupam = BARC
4. 1 byte = **8 bits**; 1 KB = **1,024** bytes; MAC = 48 bits
5. RAM volatile; ROM permanent; cache fastest
6. OS controls the computer; MS Office is not an OS
7. FORTRAN = scientific; assembler = assembly to machine
8. Internet = packet switching; modem = phone-line data
9. WWW = **Tim Berners-Lee**; SkyDrive = Microsoft
10. Stuxnet = Iran centrifuges; NAYAN, Talaash = C-DAC

Chapter 11

Space Technology

Built from 79 UPPSC questions (1990–2025). Focus: India's early satellites, ISRO centres, Chandrayaan, Mangalyaan and Aditya-L1, astronauts, geostationary satellites, rockets and cryogenic engines.

ASKED MOST · REVISE FIRST

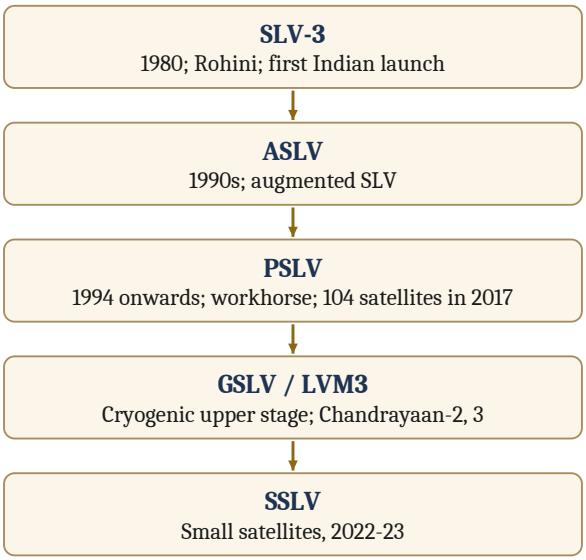
- Rockets and jet engines work on **conservation of momentum** 6× 2007, 2009, 2013, 2014, 2021, 2023
- **Cryogenic engine**: **liquid hydrogen** fuel with **liquid oxygen**; used in rockets 5× 2004, 2006, 2006, 2010, one-liner
- Satellites run on **solar cells** 3× 2000, 2006, 2014
- **Black box** of an aircraft is **orange** 3× 1996, 2004, 2009
- **ISRO places**: **Thumba-rocket launching (Kerala), Sriharikota-satellite launches (near Pulicat lake), SAC-Ahmedabad, Hassan-Master Control** 5× 1998, 2003, 2004, 2010, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

11.1 India's Early Satellites and Programmes

- First Indian satellite: **Aryabhata** (1975) 2× 2006, 2009
- Order: **SITE, STEP, APPLE, IGMDP** 1× 1998
- India's telecom is satellite-linked; not for "star war" (A true, R false) 1× 1990
- **IRS-1A** launched from **Baikonur** 1× 1994
- **Kalpana-1** = renamed **METSAT** (September 2002) 2× 2004, one-liner
- Not a satellite: **SLV-3** (a launch vehicle) 2× 2004, 2010
- GSLV-D2 is a launch vehicle, not a satellite (*) 1× 2001
- **EDUSAT**: launched **20 September 2004** 1× 2004
- PSLV-C12 (2009): RISAT-2 for border watch and **ANUSAT** by Anna University (all four correct) 1× 2008
- Launched 22 December 2005: **INSAT-4A** 1× 2004
- **INSAT-3C** and **GSAT-8** launched from **Kourou** 2× 2002, 2010
- INSAT-4C (failed) weighed **2,168 kg** 1× 2005
- Launched 29 September 2012: **GSAT-10** 1× 2012
- Launch vehicle order: **SLV, ASLV, PSLV, GSLV** 1× 2004
- **BHUVAN**: ISRO's **3-D geoportal** 1× 2008

INDIAN LAUNCH VEHICLES



11.2 ISRO and Its Centres

TUMAS PCS

- Master Control Facility (Hassan): **Karnataka** 1× 2010
- **IIRS**: **Dehradun** 1× 2006
- **ISRO-Bengaluru, IUCAA-Pune, IUAC-New Delhi, VSSC-Thiruvananthapuram** 1× 2010
- Wrong: **ISRO Satellite Centre-Sriharikota** (it is Bengaluru) 1× 2010
- **Space Applications Centre**: **Ahmedabad** 1× 1998
- **Thumba**: **Kerala** 1× 2004
- **Sriharikota** lies near **Pulicat lake** 1× 2003
- **Kakrapar-nuclear plant, Pokhran-bomb test, Thumba-rocket launching, Sriharikota-satellite launching** 1× 2016
- VSSC works on **propellants and launch vehicles** (1 and 4) 1× 2004
- **PraVaHa** CFD software: a VSSC feat for aerodynamic design (both) 1× 2024
- Tenth ISRO Chairman (January 2022): **S. Somanath** 1× 2021

Centre	Place	Work
ISRO HQ, URSC	Bengaluru	Satellites
VSSC	Thiruvananthapuram	Launch vehicles
SDSC-SHAR	Sriharikota (AP)	Launch pad
SAC	Ahmedabad	Payloads, applications
MCF	Hassan, Bhopal	Satellite control
IIRS	Dehradun	Remote sensing

11.3 Moon, Mars and Sun Missions

- **Chandrayaan-3: 14 July 2023; Aditya-L1: 2 September 2023; INSAT-3DS: 17 February 2024; RLV-LEX-01: 2 April 2023** 1× 2025
- Chandrayaan-2 project director: **M. Annadurai** (per key) 1× 2010
- Chandrayaan-2 launcher: **GSLV Mk III-M1** 1× one-liner
- Chandrayaan-3 soft landing made India the **fourth** country to land on the Moon 1× 2023
- Chandrayaan-3 payload for soil elements: **APXS** 1× 2023
- LUPEX: India-Japan Moon mission; JAXA builds the **rover**, ISRO the **lander** (only 1) 1× 2024
- Luna-25 aimed at Moon access; Russia-China lead the ILRS (A and R true, R explains A) 1× 2023
- **Mangalyaan**: launched **November 2013** (both) 1× 2023
- Evidence of life on Mars: **water, gypsum, methane** 1× 2009
- Planet that may suit life: **Mars** 1× 2018
- Methane in the atmosphere of **Mars** (and Jupiter) 2× 2010, 2016
- Saturn's atmosphere has much **methane** (per key) 1× 2022
- Water on the Moon confirmed by NASA's **SOFIA** 1× 2016
- Underground lake (2018): **Mars** 1× 2018
- **DAVINCI+** and **VERITAS**: **Venus** 1× 2021
- **NISAR**: **ISRO and NASA** 1× 2023
- **Kepler** telescope: **Earth-like planets** 1× 2008

11.4 Astronauts and Human Spaceflight

- First woman in space: **Valentina Tereshkova** 2× 2001, 2004
- "I am a citizen of the Milky Way": **Kalpana Chawla** 1× 2010
- Longest single NASA flight: none of the options (Frank Rubio, 371 days) 1× 2022
- Wilmore and Williams returned (March 2025) on SpaceX **Dragon** 1× 2023
- **Tim Peake**: astronaut (UK) 1× 2015
- NASA's last space shuttle: **Atlantis** 1× 2009
- Not correct of Shenzhou-7: **four** astronauts (it had three) 1× 2008
- **Micius**: China's **quantum satellite** communication system 1× 2016
- **Gravity Tractor**: craft to deflect **asteroids** 1× 2009
- Apollo 8 (21 December 1968): no option correct (*) 1× 1999

11.5 Satellites, Black Holes and Comets

- **Geostationary**: moves with Earth's rotation and stays fixed over one place; period **24 hours** 2× 2005, one-liner
- Satellites tracked by the **Doppler effect** 1× 2017

Quantum IAS & PCS

- Radio blackout during re-entry: heat ionises the air (A and R true, R explains A) **2×** 2002, 2003
- Communication satellites lie in the **exosphere** **1×** 2004
- Satellite power: **solar cells** **3×** 2000, 2006, 2014
- No candle flame in space: flames need **gravity** (convection) (A and R true, R explains A) **1×** 2009
- Black hole idea: **S. Chandrasekhar** **1×** 2011
- Black hole: a **collapsed star** **1×** 2019
- Comet's tail points **away from the Sun** **1×** 2011

11.6 Rockets, Jets and Cryogenics

- Jet engine uses air's oxygen; a rocket carries **liquid** oxidiser, not gas (only 1) **2×** 2007, 2009
- Jet engine: **conservation of linear momentum** **1×** 2013
- Rocket: **conservation of momentum** **3×** 2014, 2021, 2023
- Jets fly in the lower stratosphere: **no clouds, no vertical winds** (both) **1×** 2013
- Wing is convex above; faster air on top gives **lower** pressure there (A true, R false) **1×** 2013
- **Black box**: **orange** **3×** 1996, 2004, 2009
- **Stratellite**: high-altitude **airship** in the stratosphere **1×** 2016
- Liquid oxygen and hydrogen technology: **cryogenics** **1×** 2010
- Cryogenic engine: **rocket** technology; LOX oxidiser, LH2 fuel **3×** 2004, 2006, one-liner
- Rocket propellants: all options are used (*) **1×** 2010
- Discovery's tank (2006): **liquid hydrogen and oxygen** **1×** 2006
- **Curiosity-Mars rover, Messenger-Mercury probe, Rustom-1-DRDO UAV, Aakash-2-tablet** **1×** 2013

TRAP ALERT

- **SLV-3, PSLV, GSLV** are rockets, not satellites.
- ISRO Satellite Centre = **Bengaluru**; Sriharikota = launches; Thumba = sounding rockets.
- Black box is **orange**, not black.
- Chandrayaan-3 made India the **4th** to land and **1st** near the lunar south pole.

UP LINK

UP link: **Shubhanshu Shukla** of **Lucknow** flew on **Axiom-4** (June-July 2025), the first Indian on the **International Space Station**; ISRO's space technology incubation centre and Gaganyaan training links include **IIT Kanpur** (soft detail).

LAST-MINUTE 10

1. Aryabhata = first satellite (1975); Kalpana-1 = METSAT
2. SLV, ASLV, PSLV, GSLV in order
3. Thumba = Kerala; Sriharikota = Pulicat lake; SAC = Ahmedabad
4. Chandrayaan-3: 14 July 2023; India **4th** on the Moon; APXS
5. Aditya-L1: 2 September 2023; NISAR = ISRO + NASA
6. Tereshkova = first woman in space; Atlantis = last shuttle
7. Geostationary period **24 h**; satellites run on solar cells
8. Comet tail away from the Sun; Chandrasekhar = black holes
9. Rocket = conservation of momentum; cryogenic = LH2 + LOX
10. Black box = **orange**

Chapter 12

Defence Technology

Built from 49 UPPSC questions (1995–2025). Focus: IGMDP missiles and their types, Agni ranges, BrahMos, Israeli systems, Tejas and drones, ships and submarines, joint exercises.

QUANTUM

ASKED MOST · REVISE FIRST

- **Missile matching**: **Prithvi-surface to surface, Trishul-surface to air, Agni-intermediate range, Nag-anti-tank, Pinaka-rocket launcher, Nishant-UAV** 6× 2001, 2002, 2007, 2012, 2013, one-liner
- **BrahMos** = **India and Russia** (Brahmaputra + Moskva) 5× 2003, 2004, 2016, 2018, one-liner
- **Israel** partner: **Phalcon radar, Barak-8, Iron Dome**, naval pilotless helicopter 4× 2004, 2008, 2016, 2025
- **Astra** = **air-to-air** missile 3× 2004, 2010, one-liner
- **Agni-I** about **700 km**; **Agni-II** above 2,000 km 4× 2003, 2006, 2008, 2011

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

12.1 The Integrated Guided Missile Programme

- **Prithvi-surface to surface, Trishul-surface to air, Pinaka-multi-barrel rocket launcher, Nishant-remote battlefield vehicle** 1× 2002
- Correct pair: **Nag-anti-tank missile** 2× 2012, 2013
- Surface-to-surface: **Prithvi** 1× one-liner
- **Trishul-short-range SAM, Prithvi-surface to surface, Agni-intermediate range, Nag-anti-tank** 2× 2001, 2007
- Short-range missile (key): **Nag** 1× 1996
- Not correct of Astra: kills planes up to **90 km** (per key) 1× 2010
- Air-to-air missile tested in Balasore (May 2003): **Astra** 2× 2004, one-liner
- **Agni-I** range: **700-2,000 km** band; about **700 km** (March 2010) 2× 2003, 2008
- **Dhanush** range: **350 km** 1× 2008
- Long-range missile above 2,000 km: **Agni-II** (an IRBM, not a true ICBM) 2× 2006, 2011
- Agni launch statements (2002): no clean answer (*) 1× 2002
- Agni-II launched from **Chandipur** (per key) 1× 2004
- Not correct of Agni-5: "**not exclusively indigenous**" (it is indigenous) 1× 2015
- **Nirbhay** (7 November 2017): nuclear-capable, **1,000 km**, land-attack cruise missile, counters Babur (1, 2, 3); it is **subsonic** 1× 2017
- Tested 27 January 1996 at Chandipur: **Prithvi-II** 2× 1995, 2006
- Not correct of **Shaurya**: range **900 km** (it is about 700 km); a surface-to-surface missile 2× 2011, one-liner

Missile	Type	Note
Prithvi	Surface to surface	Short range; first IGMDP missile
Agni series	Ballistic	Agni-V about 5,000 km; Agni-P new generation
Akash	Surface to air	Medium range
Trishul	Surface to air	Short range (closed)
Nag	Anti-tank	Fire and forget
Astra	Air to air	Beyond visual range
BrahMos	Supersonic cruise	India-Russia
Nirbhay	Subsonic cruise	1,000 km

12.2 BrahMos

- Developed by **India and Russia** 4× 2003, 2004, 2018, one-liner
- Test of 18 October 2020: **BrahMos naval version** 1× 2016
- Not correct of BrahMos: options (c) and (d) per key; base range was **290 km** 2× 2008, 2009
- Tested March 2009 at **Pokhran** 1× 2007
- Delivered to the **Philippines** (April 2024); its range is **290 km**, not 490 (A true, R false) 1× 2024

12.3 Air Defence and Foreign Partners

- **Iron Dome**: **Israel** (only 3) 1× 2025
- **RudraM-II**: an **air-to-surface** missile tested off Odisha, not Rajasthan (neither) 1× 2024
- **Phalcon radar**: from **Israel** 1× 2004
- **Barak-8**: with **Israel** 1× 2016
- Navy's pilotless helicopter: with **Israel** 1× 2008

12.4 Aircraft and Drones

- **Hansa-2**: a **trainer aircraft** 1× 1996

QUANTUM IAS

- **Tejas**: indigenous **light combat aircraft** 2× 2004, one-liner
- **INDRA**: a **radar** 2× 2003, 2004
- **FWD-200B**: India's first indigenous **bomber UAV**; 100 kg payload (both) 1× 2024
- **NETRA**: a mini **UAV** that flies up to 300 m 1× 2009
- Not true of F-18 Super Hornet: used by **eight** air forces 1× 2002
- Not an aircraft: **Sarath (BMP-II)**, an infantry combat vehicle 1× 2009

12.5 Ships and Submarines

- **Arihant-nuclear submarine, AWACS-Israeli Phalcon, Atlas Centaur-US cryogenic rocket, Nag-anti-tank** 1× 2010
- **Admiral Gorshkov**: aircraft carrier (now **INS Vikramaditya**) 1× 2012
- Scorpene submarine commissioned January 2023: **INS Vagir** 1× 2023
- **Sarthak** (2020): Coast Guard **offshore patrol vessel** 1× 2020
- Coast Guard OPV (March 2010): **Vishwast** 1× 2008
- **INS Shakti**: a fleet tanker built in Italy, so no option fits (*) 1× 2010
- Incorrect: a virus **cannot** reduce pollution 1× 2004

12.6 Exercises and Operations

- **SLINEX II**: India-**Sri Lanka** naval exercise 1× 2011
- **Operation Pawan**: **IPKF** in Jaffna 1× 1998
- **KHANJAR-XII** (March 2025): India-**Kyrgyzstan** 1× 2025
- 2024 dates: **Dustlik April 15-28, Shakti May 13-26, Tiger Triumph March 18-25, MILAN February 19-27** 1× 2024

Exercise	With
SLINEX	Sri Lanka
Khanjar	Kyrgyzstan
Dustlik	Uzbekistan
Shakti	France
Tiger Triumph	USA (tri-service)
Malabar	USA, Japan, Australia

TRAP ALERT

- **Trishul** = surface to air; **Prithvi** = surface to surface; **Nag** = anti-tank; **Astra** = air to air.
- BrahMos = **India-Russia**; Phalcon, Barak, Iron Dome = **Israel**.
- Nirbhay is **subsonic**; BrahMos is **supersonic**.
- INDRA = radar; Tejas = LCA; Sarath = infantry vehicle, not aircraft.

UP LINK

UP link: the **UP Defence Industrial Corridor** has nodes at Lucknow, Kanpur, Agra, Aligarh, Jhansi and Chitrakoot; the **BrahMos integration and testing facility** at **Lucknow** opened in May 2025, and HAL Lucknow and Kanpur make aircraft parts.

LAST-MINUTE 10

1. Prithvi = SSM; Trishul = SAM; Nag = anti-tank; Astra = AAM
2. Agni-I 700 km; Agni-II above 2,000 km; Dhanush 350 km
3. Shaurya = surface to surface, about 700 km
4. Nirbhay = subsonic cruise, 1,000 km
5. BrahMos = India + Russia; to Philippines (2024)
6. Israel: Phalcon, Barak-8, Iron Dome
7. Tejas = LCA; Hansa = trainer; INDRA = radar; NETRA = UAV
8. Arihant = nuclear submarine; Gorshkov = Vikramaditya
9. Vagir = Scorpene (2023)
10. Khanjar = Kyrgyzstan; SLINEX = Sri Lanka; Pawan = IPKF

Chapter 13

Scientists, Inventions and Physics Miscellany

Built from 66 UPPSC questions (1990–2024). Focus: inventor-invention matching, Indian scientists, lasers, nanotechnology and MRI, magnets and cells, labs and polar stations.

ASKED MOST · REVISE FIRST

- **National Science Day (28 February)** marks the **Raman effect** (1928) **5×** 1990, 1993, 2002, 2003, one-liner
- **Discoverer matching**: **radium-Curie, penicillin-Fleming, X-ray-Roentgen, dynamite-Nobel, evolution-Darwin, revolver-Colt** **5×** 1992, 2005, 2009, 2010, 2011
- Antarctic stations: **Dakshin Gangotri** (first), **Bharati** (latest) **3×** 2006, 2009, 2010
- **Aluminium** is not attracted by an ordinary magnet; copper is non-magnetic **3×** 1990, 2002, one-liner
- **J.C. Bose** worked in biology and physics; he won **no Nobel** **2×** 2006, 2008

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

13.1 Inventions and Discoveries

- Machine gun: **James Puckle** **1×** 2014
- Telescope (astronomical use): **Galileo** **1×** 1994
- Biggest "telescope": **IceCube** (a neutrino detector at the South Pole) **1×** 2010
- X-ray machine: **Wilhelm Roentgen** **1×** 2005
- Gas engine: **Daimler** **1×** 1995
- Wrong pair: **J. Perkins-penicillin** (Fleming found penicillin) **1×** 2011
- **Radium-Curie, penicillin-Fleming, X-ray-Roentgen, measles vaccine** (smallpox: Jenner) **1×** 1992
- Radioactivity: **Becquerel** **1×** 2003
- Order: **diode, transistor, IC, SMD** **1×** 2016
- **Nobel-dynamite, Fleming-penicillin, Darwin-evolution, Curie-radium** **1×** 2010
- **Revolver-Colt, dynamite-Nobel, law of cooling-Newton, law of pressure-Pascal** **1×** 2009
- Laws of planetary motion: **Kepler** **1×** 2010
- Photoelectric effect discovered by **Hertz** **1×** 2003
- National Science Day: **Raman effect** **4×** 1990, 1993, 2002, 2003

Discovery	Scientist
Radioactivity	Henri Becquerel
Radium, polonium	Marie Curie
X-rays	Wilhelm Roentgen
Penicillin	Alexander Fleming
Smallpox vaccine	Edward Jenner
Dynamite	Alfred Nobel
Planetary laws	Johannes Kepler
Photoelectric effect	Hertz (found), Einstein (explained)

13.2 Indian Scientists

- Birth centenary in 1988: **C.V. Raman** **1×** 1991
- New theory of relativity (Hoyle-Narlikar): **J.V. Narlikar** **1×** 2016
- **Homi Bhabha Award**: **nuclear energy** **1×** 2012
- Biology and physics both: **J.C. Bose** **1×** 2008
- Designed Aryabhata: **U.R. Rao** **1×** 2017
- **Vinod Dham** (Padma Bhushan 2025): father of the **Pentium chip** **1×** 2023
- India's atom bomb: **Raja Ramanna** **2×** 2002, 2004
- Not a Nobel winner: **J.C. Bose** **1×** 2006
- **Thermal ionisation**: **M.N. Saha** **1×** 2014
- Einstein's Nobel: **photoelectric effect** **1×** one-liner
- National Science Day: **28 February** **1×** one-liner

13.3 Modern Physics and Devices

- "God particle": **Higgs boson** **1×** 2013
- Laser gives **stimulated** radiation **1×** 2012
- Lasers treat cancer, heart, eye and kidney problems (*) **1×** 2002
- First ruby laser gave **red** pulses (only 2) **1×** 2023
- Miniaturisation: **nanotechnology** **1×** 2004
- **Nano plug**: a tiny **hearing aid** **1×** 2015
- Carbon nanotubes: **Sumio Iijima** **1×** 2008

- **Nano Hummingbird**: pocket US **spy drone** 1× 2010
- Space-based solar power as a national goal: supply is nearly uninterrupted (A and R true, R explains A) 1× 2010
- Most accurate clock (1 s in 300 million years): **strontium** atoms 1× 2008
- Carbon microphone: **telephone** 1× 2009
- **MRI**: **nuclear magnetic resonance** 1× 2008

13.4 Magnetism, Cells and Everyday Devices

- Not attracted by a magnet: **aluminium** (weakly paramagnetic) 2× 2002, one-liner
- Non-magnetic element among options: **copper** 1× 1990
- Tape-recorder tape: coated with **ferromagnetic powder** 1× 1998
- Torch cell terminals: **zinc and carbon** 1× 2010
- Automatic watches run on **hand movements** 1× 2007
- Correct: liquid crystals in displays, modem on phone lines (1 and 2); NIO is in **Goa** 1× 2009
- Liquid crystals: **all of the above** uses 1× 1996
- **Soleckshaw**: a **solar rickshaw** 1× 2008
- **Cytotron**: makes **artificial climate** 1× 2016
- Water jet technology: **drilling mines** 2× 1998, 2004
- Engine cooling: **radiator** 1× 2014

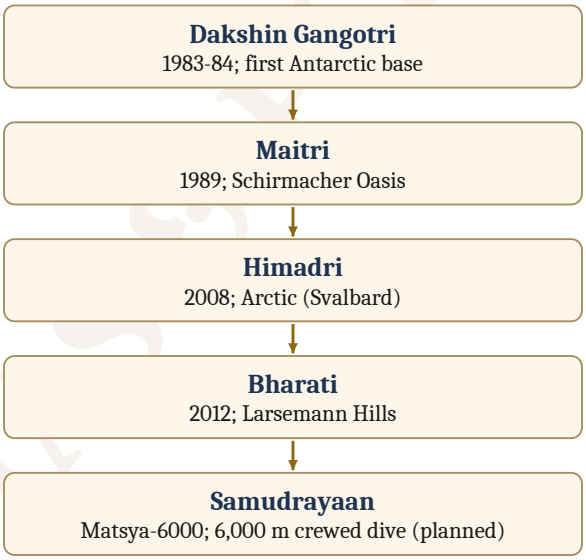
13.5 Laboratories and Institutions

- **National Physical Laboratory**: **New Delhi** 1× 2009
- **National Chemical Laboratory**: **Pune** 1× 2015
- Not a science body: **ICSSR** (social science) 1× 2008
- Wrong pair: **Antarctic Study Centre-Bangalore** (NCPOR is in **Goa**) 1× 2004
- Naturopathy and Yogic Science institute: **Bengaluru** 1× 2002
- **Hi-Tec City-Hyderabad, Science City-Kolkata, rocket centre-Thumba, CDRI-Lucknow** 1× 1999
- "Nuclear Reactor Time Bomb": **Takashi Hirose** 1× 2011

13.6 Polar and Ocean Missions

- First Antarctic station: **Dakshin Gangotri** 1× 2006
- Latest Antarctic station: **Bharati** 2× 2009, 2010
- First manned deep-ocean mission: **Samudrayaan** 1× 2023
- Wrong: **isohaline-snowfall** (isohalines join equal salinity) 1× 2001
- Correct: **pycnocline-sea water density** 1× 2024
- Salinity gradient: **halocline** 1× one-liner

INDIA'S POLAR AND OCEAN STATIONS



13.7 Technology and Society

- **Dry ice-solid CO2, gene therapy-blood diseases, cryonics-freezing bodies, cobalt-60-cancer** 2× 1997, 2004
- iPhone 4S: 8 hours 3G talk time and **Siri** (2 and 3) 1× 2009
- Mobiles and vehicles changed rural **mobility and connectivity** (1 and 2) 1× 2010
- **GPS** = Global Positioning System 1× 2015

TRAP ALERT

- Galileo improved the telescope; **Hertz found** and **Einstein explained** the photoelectric effect.
- **Jenner** = smallpox vaccine; Fleming = penicillin; Becquerel = radioactivity.
- **J.C. Bose** has no Nobel; Raman (1930) does.
- NCPOR (polar research) and NIO are both in **Goa**.

QUANTUM IAS & PCS

UP LINK

UP link: the **Central Drug Research Institute (CDRI)**, **NBRI**, **CIMAP** and **IITR** (toxicology) are CSIR labs in **Lucknow**; **Shanti Swarup Bhatnagar**, the first CSIR director-general, was born in Bhera (now Pakistan) but studied in Lahore, while **Birbal Sahni Institute of Palaeosciences** is also at Lucknow.

LAST-MINUTE 10

1. National Science Day **28 February** = Raman effect
2. Roentgen = X-ray; Becquerel = radioactivity; Curie = radium
3. Fleming = penicillin; Nobel = dynamite; Colt = revolver
4. Kepler = planetary laws; Hertz = photoelectric effect
5. J.C. Bose = biology + physics, no Nobel; M.N. Saha = thermal ionisation
6. Raja Ramanna = atom bomb; U.R. Rao = Aryabhata; Vinod Dham = Pentium
7. Higgs boson = God particle; MRI = NMR
8. Aluminium not attracted by magnets; torch cell = zinc-carbon
9. NPL = Delhi; NCL = Pune; CDRI = **Lucknow**
10. Dakshin Gangotri first, Bharati latest; Samudrayaan = deep ocean

Chapter 14

Atomic Structure, Matter and Solutions

Built from 20 UPPSC questions (1992–2025). Focus: particles of the atom, isotopes, physical and chemical change, colloids and emulsions, osmosis and sublimation.

ASKED MOST · REVISE FIRST

- Nucleus = **protons and neutrons**; neutron found by **Chadwick** 3× 1995, 1996, 1996
- **Isotopes**: same **atomic number**, different mass; protium, deuterium, tritium are isotopes of **hydrogen** 3× 2009, 2025, one-liner
- **Milk** = an **emulsion**; **honey** is not a colloid 3× 2002, 2024, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

14.1 Particles of the Atom

- Nucleus holds **protons and neutrons** 1× 1996
- Order of discovery: **electron, proton, neutron, pi meson** 1× 2023
- Neutron discovered by **James Chadwick** (1932) 2× 1995, 1996
- Not a sub-atomic particle: **deuteron** (a deuterium nucleus) 1× 2007
- Particle-antiparticle pair: **electron-positron** 1× 2005
- Zero charge: **neutrino** 1× 2021

Particle	Discoverer	Charge
Electron	J.J. Thomson (1897)	Negative
Proton	Rutherford (1919)	Positive
Neutron	Chadwick (1932)	None
Positron	Anderson (1932)	Positive (antielectron)
Neutrino	Predicted by Pauli (1930)	None

14.2 Isotopes and Isobars

- Isotopes: same atomic number, different mass; H isotopes are **not** of carbon (only 1) 1× 2025
- Same protons, different neutrons: **isotopes** 2× 2009, one-liner

Term	Same	Different	Example
Isotopes	Atomic number	Mass number	Protium, deuterium, tritium
Isobars	Mass number	Atomic number	Argon-40, calcium-40

QUANTUM IAS & PCS

Term	Same	Different	Example
Isotones	Neutron number	Protons	C-14, N-15

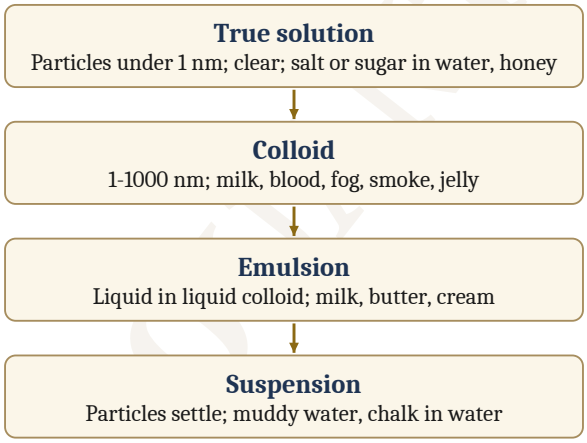
14.3 Physical and Chemical Changes

- Water to vapour: a **physical** change 1× 1992
- Wrong pair: **pasteurisation-tea** (it is for **milk**) 1× 2019

14.4 Mixtures, Colloids and Solutions

- Salt solution is homogeneous; a homogeneous mix has the **same** composition throughout (A true, R false) 1× 2022
- Salt-ice mix goes below 0 C; salt **lowers**, not raises, the freezing point (A true, R false) 1× 1998
- Milk: an **emulsion** 2× 2024, one-liner
- Not a colloid: **honey** (a true solution) 1× 2002
- Dissolving 100 g sugar in half a litre of water: **volume** stays the same (per key) 1× 2017
- Osmosis: solvent flows from **pure solvent to solution**; reversing it is reverse osmosis (only 1) 1× 2023
- Not used to purify solid impurities: **distillation** (it is for liquids) 1× 2016
- Camphor, naphthalene, dry ice vanish: **sublimation** 1× 2019

KINDS OF MIXTURES



TRAP ALERT

- Deuteron is a **nucleus**, not a fundamental sub-atomic particle.
- Salt **lowers** freezing point (salt-ice freezing mixtures).
- Honey is a true solution; milk is an emulsion.
- Sublimation = solid straight to vapour (camphor, naphthalene, dry ice, iodine).

UP LINK

UP link: **pasteurised milk** from UP's dairy cooperative **Parag** (PCDF, Lucknow) and the large private dairies of western UP; UP is India's top milk-producing state.

LAST-MINUTE 10

1. Nucleus = protons + neutrons; Chadwick = neutron
2. Electron, proton, neutron, pi meson: order of discovery
3. Electron-positron = particle-antiparticle; neutrino = no charge
4. Isotopes = same atomic number; isobars = same mass number
5. H isotopes: protium, deuterium, tritium
6. Water to vapour = physical change; pasteurise milk
7. Milk = emulsion; honey not a colloid
8. Salt lowers freezing point
9. Osmosis: solvent to solution
10. Camphor, naphthalene, dry ice = sublimation

Chapter 15

Metals, Ores and Alloys

Built from 35 UPPSC questions (1993–2025). Focus: special metals (tungsten, sodium, mercury, uranium), ores and minerals, rusting and tarnishing, alloy composition.

ASKED MOST · REVISE FIRST

- **Alloys**: **solder = tin + lead; bronze = copper + tin; brass = copper + zinc**; German silver has **no silver** 7× 1996, 2003, 2005, 2015, 2016, one-liner, one-liner
- **Sodium**: kept in **kerosene**; yellow **street lamps** 3× 2000, 2007, 2017
- **Mercury**: liquid metal, best liquid conductor of heat, used in thermometers 5× 2003, 2005, 2014, 2020, one-liner
- **Rusting** needs **oxygen and moisture** and **increases** iron's weight; galvanised (zinc) iron does not rust 3× 2002, 2008, 2016
- **Stainless steel** = iron + **chromium** (+ nickel); **silicon** is not added 4× 2004, 2017, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

15.1 Properties of Metals

- Hardest metal among options: **tungsten** 1× 1996
- Heaviest natural element: **uranium** 1× 2004
- Costliest metal "discovered recently": no fit (*); californium and rhodium are the costliest 1× 2016
- Stored in **kerosene**: **sodium** 1× 2017
- Yellow street lamps: **sodium** 1× 2000
- Sodium lamp light is **monochromatic** and does not split in fog 1× 2007
- Best liquid conductor of heat: **mercury** 2× 2005, 2014
- Mercury in thermometers: it **expands uniformly** (per key: "high circulation power") 1× 2003
- Liquid metal at room temperature: **mercury** 1× one-liner
- **Sodium-cut with a knife, mercury-liquid, silver-good conductor, lead-poor conductor of heat** 1× 2020
- **Silver-best conductor, aluminium-most abundant metal, gold-most malleable, lead-least heat conducting** 1× 2002

15.2 Ores and Minerals

- Found in native state: **gold** 1× 2016
- Contains calcium: **gypsum** 1× 2019
- Limestone: **calcium carbonate** 1× 1993
- Doctors and sculptors use **Plaster of Paris** (calcium sulphate) 1× 2004
- **Mica**: **electrical** industry 1× 2010
- **Bauxite**: ore of **aluminium** 2× 2006, 2015
- Wrong pair: **copper-cinnabar** (cinnabar is mercury's ore) 1× 2017
- **Diamond-carbon, marble-calcium, sand-silicon, ruby-aluminium** 1× 2010

Metal	Main ore
Aluminium	Bauxite
Iron	Haematite, magnetite
Copper	Chalcopyrite
Mercury	Cinnabar
Zinc	Zinc blende
Lead	Galena
Tin	Cassiterite

15.3 Corrosion

- Galvanised iron does not rust: **zinc** oxidises first (A and R true, R explains A) 1× 2002
- Rusting needs **oxygen and moisture** 1× 2016
- Rusting **increases** iron's weight 1× 2008
- Rust is brown iron oxide, not **tannin** (A true, R false) 1× 1998
- Silver turns black from **silver sulphide** (both, 2025); the 2008 key is misprinted (*) 2× 2025, 2008

15.4 Alloys

- **Amalgam**: alloy with **mercury** (mercury-zinc) 1× 2012
- **Solder**: **tin and lead** 2× 1996, 2005
- **Bronze**: **copper and tin** 2× 2015, 2016

QUANTUM IAS & PCS

- **German silver**: **0% silver** (copper, zinc, nickel) 1× 2003
- Common to brass, bronze, German silver: **copper**; brass = **copper and zinc** 2× one-liner, one-liner
- True: brass is copper-zinc; amalgams contain mercury (i and iii); magnetite is an **iron** ore 1× 2001
- Not used for stainless steel: **silicon** 2× 2004, 2017
- Stainless steel: iron with **chromium**, nickel, manganese, carbon 2× one-liner, one-liner

Alloy	Made of
Brass	Copper + zinc
Bronze	Copper + tin
German silver	Copper + zinc + nickel
Solder	Tin + lead
Stainless steel	Iron + chromium + nickel + carbon
Amalgam	Mercury + another metal
Duralumin	Aluminium + copper + magnesium

TRAP ALERT

- **Cinnabar** = mercury; **magnetite** = iron; **bauxite** = aluminium.
- German silver has **no silver**; bronze has **tin**, brass has **zinc**.
- Rust **adds** weight (iron oxide); galvanising = **zinc** coat.
- Hardest metal: tungsten among options (chromium is hardest pure metal); densest: osmium.

UP LINK

UP link: **Moradabad** ("Peetal Nagri") is famous for **brassware** (a GI product); **Renukoot** (Sonbhadra) has Hindalco's aluminium smelter, fed by bauxite from Jharkhand and Chhattisgarh.

LAST-MINUTE 10

1. Tungsten = hard, high melting point (bulb filament)
2. Uranium = heaviest natural element
3. Sodium in kerosene; sodium lamps = yellow street lights
4. Mercury = liquid metal; best liquid heat conductor
5. Gold occurs native; bauxite = aluminium; cinnabar = mercury
6. Limestone = CaCO3; Plaster of Paris = calcium sulphate
7. Rust needs O2 + moisture; weight increases
8. Galvanised = zinc; silver tarnish = silver sulphide
9. Solder = Sn + Pb; bronze = Cu + Sn; brass = Cu + Zn
10. German silver = no silver; stainless steel = chromium

Chapter 16

Non-Metals, Gases, Acids, Bases and Salts

Built from 53 UPPSC questions (1993–2025). Focus: carbon's allotropes and fullerene, dry ice, heavy water, gases for divers, balloons and tyres, laughing gas, pH and aqua regia, common salts.

ASKED MOST · REVISE FIRST

- **Fullerene** is a carbon cage (C-60), **not** a fluorine compound 6× 2001, 2002, 2003, 2006, 2010, 2017
- **Dry ice** = **solid carbon dioxide** 3× 2009, 2013, 2014
- **Heavy water**: **deuterium** in place of hydrogen; molecular weight **20** 4× 1993, 2007, 2015, 2016
- **Divers** breathe **oxygen and helium** 3× 2004, 2014, 2016
- **Hypo (sodium thiosulphate)** = photographic fixer; plates coated with **silver bromide** 5× 2010, 2014, 2015, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

16.1 Carbon and Its Allotropes

- Most compounds formed by **carbon** 1× 2016
- Most stable form of carbon: **graphite** 1× 2022
- Made of carbon only: **graphene**, a carbon **nanostructure** and the **strongest** material 3× 2015, 2015, 2018
- Not a carbon allotrope: **oxocarbon** 1× 2010
- Fundamental element among options: **diamond** (pure carbon) 1× 1995
- Hardest natural substance: **diamond** 1× 2016
- Pencil "lead": **graphite** 2× 1994, 2006
- Not in the fullerene Nobel team: **"Faimen"**; winners were **Kroto, Curl, Smalley** 1× 2009
- **Buckminsterfullerene**: **60 carbon atoms** in pentagons and hexagons 1× 2010
- Wrong pair: **fullerene-fluorine compound** 5× 2001, 2002, 2003, 2006, 2017
- First made in labs: **Kevlar, fullerene, graphene, black gold** 1× 2019
- No carbon: **sand** (silica) 1× 2013
- Contains carbon: **lignite**; highest-carbon coal: **anthracite** 2× 1993, one-liner
- Most carbon among options: **pig iron** 1× 2014
- **1 carat = 200 mg** 1× 2013
- True: calcium sulphate makes water hard; diamond is harder than copper and iron (1 and 2) 1× 2001
- **Dry ice**: **solid CO2** 3× 2009, 2013, 2014

Form of carbon	Note
Diamond	Hardest natural substance; poor conductor
Graphite	Most stable; pencil lead, lubricant, moderator
Fullerene (C-60)	Football cage; Kroto, Curl, Smalley (Nobel 1996)
Graphene	One-atom sheet; strongest material (Geim, Novoselov, 2010 Nobel)
Carbon nanotube	Rolled graphene (Iijima, 1991)

16.2 Hydrogen and Water

- Commonest element in the universe: **hydrogen** 1× 2007
- Cleanest fuel: **hydrogen** 1× 2015
- **Heavy water**: **deuterium** in place of hydrogen 3× 1993, 2007, 2016
- Heavy water molecular weight: **20** 1× 2015
- Salty to pure water: **reverse osmosis** 1× 2013
- Vanaspati ghee: oil hardened with **hydrogen** 1× 2013

16.3 Noble Gases and Nitrogen

- Not a noble gas: **actinium** 1× 2023
- Divers' breathing gas: **oxygen and helium** 3× 2004, 2014, 2016
- Very soluble in water: **ammonia** 1× 1993
- Laughing gas: **nitrous oxide** (N2O) 3× 2010, 2011, one-liner
- Not correct: **nitric oxide is laughing gas** 1× 2010
- Nitrogen cycle keeps nitrogen in land and air **unchanged** 1× 2008
- Aircraft tyres: **nitrogen** 1× 2014
- Helium over hydrogen in balloons: **no explosive mix** with air 1× 2011
- Lighter than air: **helium** only (HCl and SO2 are heavier) 1× 2025
- Liquid non-metal at room temperature: **bromine** 1× 2017

Gas	Use
Helium	Balloons, divers' mix (with oxygen), MRI cooling
Neon	Advertising signs
Argon	Bulbs, tube lights, welding
Nitrogen	Aircraft tyres, food packing
Nitrous oxide	Laughing gas, anaesthetic
Hydrogen	Vanaspati, cleanest fuel

16.4 Acids, Bases and pH

- Acids have pH **below 7**, bases **above 7** (neither statement right) 1× 2025
- **Aqua regia**: concentrated **HCl and HNO3** (3:1); dissolves gold 2× 2014, 2023

16.5 Common Salts and Their Uses

- **Washing soda-sodium carbonate, caustic soda-sodium hydroxide, neela thotha-copper sulphate, hypo-sodium thiosulphate** 1× 2002
- **Bleaching powder-calcium oxychloride, baking soda-sodium bicarbonate, washing soda-sodium carbonate, Plaster of Paris-calcium sulphate hemihydrate** 2× 2016, 2021
- Baking soda: **sodium hydrogen carbonate**, used in **fire extinguishers** (both) 1× 2025
- Baking soda formula: **$NaHCO_3$** 1× 2024
- Bleaching powder: **calcium oxychloride** 1× 2010
- Salt not for the dining table: all options are edible (*) 1× 2014
- Photography and antichlor: **sodium thiosulphate** 1× 2010
- Photographic fixer: **sodium thiosulphate** (hypo) 2× 2014, one-liner
- Photographic plate: **silver bromide** 2× 2015, one-liner
- Wrong pair: **silver bromide-artificial photosynthesis** 1× 2019

Common name	Chemical
Baking soda	Sodium bicarbonate (NaHCO3)
Washing soda	Sodium carbonate
Caustic soda	Sodium hydroxide
Bleaching powder	Calcium oxychloride
Plaster of Paris	Calcium sulphate hemihydrate
Neela thotha	Copper sulphate
Hypo	Sodium thiosulphate

TRAP ALERT

- **Nitrous** oxide (N2O) = laughing gas; nitric oxide (NO) is not.
- Fullerene is **pure carbon**, not fluorine.
- Acid pH < 7; base pH > 7.
- **Baking soda** = bicarbonate; **washing soda** = carbonate.

UP LINK

UP link: **Phulpur** (IFFCO) and other UP fertiliser plants make **ammonia** from natural gas; **Rae Bareli** (Unchahar) and other UP plants use nitrogen and hydrogen gases in industry (soft detail).

LAST-MINUTE 10

1. Carbon forms most compounds; graphite = most stable
2. Graphene = strongest; diamond = hardest natural
3. Fullerene = C-60; Kroto, Curl, Smalley
4. 1 carat = **200 mg**; anthracite = most carbon
5. Dry ice = solid **CO2**; heavy water = **D2O**, weight 20
6. Divers: **O2 + He**; aircraft tyres: **N2**
7. Laughing gas = **N2O**; helium safer than hydrogen
8. Bromine = liquid non-metal; aqua regia = HCl + HNO3
9. Bleaching powder = **calcium oxychloride**
10. Hypo = fixer; silver bromide = photographic plate

Chapter 17

Organic Chemistry, Polymers, Acids and Explosives

Built from 42 UPPSC questions (1992–2025). Focus: fruit-ripening chemicals, natural and synthetic polymers, bullet-proof polymers, Teflon and Bakelite, natural acids and their sources, explosives.

ASKED MOST · REVISE FIRST

- **Lactic acid**: muscle fatigue, sour milk, curd **5×** 1992, 2004, 2008, 2010, one-liner
- Not an explosive: **nitrochloroform**; RDX (cyclonite) is one **7×** 2001, 2002, 2003, 2014, 2015, one-liner, one-liner
- **Bullet-proof**: **Kevlar (polyamide)** vests, **Lexan (polycarbonate)** windows **5×** 2005, 2010, 2014, 2015, 2017
- **Fruit ripening**: **ethylene, ethephon, calcium carbide** **4×** 2005, 2010, 2011, 2014
- **Acid-source matching**: **acetic-vinegar/pickle, tartaric-grapes/tamarind, lactic-curd, malic-apple, carbonic-soda, methanoic-nettle sting** **7×** 2003, 2008, 2009, 2015, 2017, 2023, 2025

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

17.1 Basic Organic Compounds

- Basis of all organic compounds: **carbon** **2×** 2001, 2015
- Methane source: **paddy fields** **1×** 2014
- Ripening chemicals: **ethephon**, **calcium carbide** (banned), **ethylene** **4×** 2005, 2010, 2011, 2014
- **Wood spirit**: **methyl alcohol** **1×** 2015
- Fermented germinated barley: **beer** **1×** 2016

17.2 Polymers

- Natural rubber: polymer of **isoprene** **1×** 1992
- Not a natural polymer: **nylon** **2×** 2001, 2015
- Natural silk contains **nitrogen** (a protein) **1×** 2023
- Natural polymer: **silk** **1×** one-liner
- Not a natural polymer: **ghee** (a fat) **1×** 2009
- Not synthetic: **fibroin** (silk protein) **1×** 2020
- Polythene from **ethylene** **1×** 2021
- Not a thermoplastic: **neoprene** (an elastomer) **1×** 2020
- Bullet-proof vests: **polyamides** (Kevlar) **3×** 2005, 2010, 2014
- Bullet-proof windows: **polycarbonates** **1×** 2015
- Bullet-proof materials: **Kevlar and Lexan** (I and III) **1×** 2017
- Teflon, Dacron are polymers; neoprene is synthetic rubber; polythene is polyethylene (1, 2, 3); natural rubber is **isoprene** **2×** 2002, 2003
- Non-stick pans: **Teflon** **2×** 2015, one-liner
- Pottery glazing: **zinc oxide** (per key) **1×** 2015
- Phenol makes **Bakelite** **1×** 2010
- Not biodegradable: **PVC** **1×** 2017

Polymer	Made from / use
Natural rubber	Isoprene
Neoprene	Chloroprene; synthetic rubber
Polythene	Ethylene
PVC	Vinyl chloride; pipes; not biodegradable
Teflon	Tetrafluoroethylene; non-stick pans
Bakelite	Phenol + formaldehyde; switches
Nylon	Polyamide; first synthetic fibre
Kevlar	Aramid (polyamide); bullet-proof vests
Lexan	Polycarbonate; bullet-proof glass
Dacron / terylene	Polyester fibre

17.3 Organic Acids in Nature

- Muscle fatigue: **lactic acid** **3×** 1992, 2008, 2010
- Milk turns sour: **lactic acid** **2×** 2004, one-liner
- Wrong pair: **acetic acid-curd** (curd has lactic acid) **1×** 2015
- Wrong pair: **acetic acid-potato** (only 3) **1×** 2025
- **Pickle-acetic, sour milk-lactic, apple-malic, soda water-carbonic** **1×** 2003
- **Vinegar-acetic, tamarind-tartaric, curd-lactic, nettle sting-methanoic (formic)** **1×** 2023
- In grapes, tamarind, banana: **tartaric acid** **2×** 2008, 2017
- Photography acid: **oxalic acid** **1×** 2003
- Baking powder: **tartaric acid** **1×** 2012
- Vinegar: **acetic acid** **1×** 2009

Acid	Found in
Acetic	Vinegar, pickles
Citric	Lemon, orange

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Acid	Found in
Tartaric	Grapes, tamarind; baking powder
Lactic	Curd, sour milk, tired muscles
Malic	Apple
Oxalic	Tomato, spinach
Formic (methanoic)	Ant and nettle sting
Carbonic	Soda water, cold drinks

17.4 Explosives

- Not an explosive: **nitrochloroform** 5× 2001, 2002, 2003, 2014, 2015
- **RDX** invented by **Henning** (1898); RDX = **cyclonite**, an explosive 3× 2016, one-liner, one-liner
- No nitroglycerine: **amatol** (TNT + ammonium nitrate) 1× 2010
- Explosives made from **glycerol** (nitroglycerine) 1× 2016
- "Nobel oil": **TNG** (nitroglycerine) 1× one-liner

TRAP ALERT

- Curd = **lactic** acid, not acetic; vinegar = **acetic**.
- Natural rubber = **isoprene**; neoprene = **chloroprene**.
- Bullet-proof vest = **Kevlar**; window = **polycarbonate**.
- **Calcium carbide** ripening is banned in India (it gives acetylene and arsenic traces).

UP LINK

UP link: **mango** ripening with banned **calcium carbide** is a recurring FSDA (UP food safety) target in Lucknow, Malihabad and Saharanpur mandis; ethephon and ethylene chambers are the approved methods.

LAST-MINUTE 10

1. Carbon = basis of organic compounds; paddy = methane
2. Ethylene, ethephon ripen fruit; carbide banned
3. Wood spirit = methyl alcohol; barley = beer
4. Natural rubber = isoprene; nylon not natural
5. Kevlar vests; Lexan windows; Teflon non-stick
6. Phenol = Bakelite; PVC not biodegradable
7. Lactic = curd, fatigue; tartaric = grapes, tamarind, baking powder
8. Acetic = vinegar; formic = nettle, ant; carbonic = soda
9. RDX = cyclonite (Henning); TNG = Nobel oil
10. Nitrochloroform is not an explosive

Chapter 18

Fuels

Built from 28 UPPSC questions (1990–2025). Focus: natural gas and CNG, octane and cetane numbers, anti-knock agents, biogas, LPG, gasohol and hydrogen storage.

ASKED MOST · REVISE FIRST

- **Methane** = main part of **natural gas, CNG and biogas** 7× 2005, 2010, 2016, 2025, one-liner, one-liner, one-liner
- **Biogas** = **methane + carbon dioxide**, made by **fermentation** 5× 2009, 2010, 2013, 2016, 2025
- **LPG** = **propane and butane** (key adds ethane) 3× 2013, 2021, one-liner
- **Octane** number rates **petrol**; **cetane** rates **diesel** 2× 2009, 2015
- **Gasohol** = **petrol + ethanol** (not benzene) 2× 2009, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

18.1 Natural and Fossil Fuels

- Natural fuel: **petroleum** 1× 1990
- Cleanest fossil fuel: **natural gas** 1× 2014
- Fossil fuel among options: **natural gas** 1× 2016
- Natural gas is mainly **methane** 1× one-liner

18.2 Engines, Octane and Cetane

- **Octane number**: quality of **petrol** 1× 2009
- **Cetane number**: quality of **diesel** 1× 2015
- Diesel engine burns **diesel vapour and air** (no spark; compression ignition) 1× 1994
- **Carburettor**: in a **petrol** engine 1× 2022
- Anti-knock agent: **tetraethyl lead** (now banned) 2× 1998, one-liner
- Diesel in heavy vehicles: **more power and economy** 1× 1992

PETROL VERSUS DIESEL ENGINES

Petrol engine
Carburettor or injector; spark plug; octane number

↓

Diesel engine
No spark; compression heats air; cetane number

↓

Knocking
Early burning in petrol engines; TEL once stopped it

↓

Unleaded petrol
India from 2000; TEL is now banned

18.3 Biogas, LPG and CNG

- Fuel gas from a biogas plant: **methane** 1× 2016
- Biogas: mainly **methane and CO2** (1 and 3) 2× 2010, 2025
- Gobar gas: mainly **methane** 1× one-liner
- Biogas plant works by **fermentation** (anaerobic) 1× 2013
- Gobar gas plant: **S.V. Desai** (IARI); printed option did not match (*) 1× 2015
- LPG: **propane and butane** (key: ethane, propane, butane) 3× 2013, 2021, one-liner
- **CNG** = **Compressed Natural Gas**; mainly **CH4** 3× 2005, 2015, one-liner

Fuel	Main component
Natural gas, CNG	Methane
Biogas, gobar gas	Methane + CO2
LPG	Propane + butane (ethyl mercaptan added for smell)
Coal gas	Hydrogen + methane + CO
Water gas	CO + hydrogen
Producer gas	CO + nitrogen

18.4 Alcohol Fuels and Hydrogen

- **Gasohol**: **gasoline and ethanol** 1× 2016
- Mandatory ethanol blending: **Brazil** 1× 2015
- True: fossil fuels, geothermal as non-conventional, gobar gas = methane (1, 3, 4); gasohol is **not** benzene-alcohol 1× 2009
- Hydrogen for vehicles stored in a **hydride** 1× 2008

18.5 Petroleum Refining

- Refining by-product: **asphalt** (bitumen) 1× 2013
- **Hydrocarbon Vision 2025**: **storage of petroleum products** (energy security) 1× 2000

TRAP ALERT

- Octane = **petrol**; cetane = **diesel**.
- Carburettor is in **petrol** engines only.
- Gasohol has **ethanol**, not benzene; biogas is not hydrogen-rich.
- LPG smells because **ethyl mercaptan** is added; LPG itself is odourless.

UP LINK

UP link: UP is India's top **ethanol** (sugarcane molasses) producer, feeding the **E20** petrol programme; **CNG** buses run in Lucknow, Agra and Kanpur, and **compressed biogas** plants have come up at Gorakhpur and Budaun (soft detail).

QUANTUM IAS & PCS

LAST-MINUTE 10

1. Natural gas = **methane**; cleanest fossil fuel
2. Octane = petrol; cetane = diesel
3. Carburettor = petrol engine; diesel = compression ignition
4. Tetraethyl lead = old anti-knock (banned)
5. Biogas = CH4 + CO2; made by fermentation
6. Gobar gas = S.V. Desai (IARI)
7. LPG = propane + butane; CNG = CH4
8. Gasohol = petrol + **ethanol**; Brazil blends by law
9. Hydrogen stored as **hydride**
10. Asphalt = refining by-product

Chapter 19

Food Chemistry, Fertilisers and Chemistry Miscellany

Built from 65 UPPSC questions (1992–2021). Focus: food preservatives, urea and DAP, pesticides and rat poison, silver iodide for artificial rain, chemical-use matching, phosphorus in matches, battery acid.

ASKED MOST · REVISE FIRST

- **Preservatives**: **sodium benzoate**, benzoic acid; chips packed in **nitrogen** 6× 1992, 1996, 2010, 2013, 2016, one-liner
- **Silver iodide** makes **artificial rain** 4× 2008, 2013, 2020, one-liner
- **Zinc phosphide** = **rat poison** 4× 2011, 2012, 2019, 2020
- **Chemical-use matching**: **acetone-nail polish remover, CCl4-fire extinguisher, freon-refrigerant, TEL-anti-knock, BHC-insecticide, eosin-red ink, blue vitriol-fungicide** 6× 2010, 2011, 2013, 2013, 2014, 2020
- **Urea**: about **46% nitrogen** (amide form); best for **foliar spray**; 2.2 kg gives 1 kg N 6× 2015, 2017, 2017, 2019, one-liner, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

19.1 Food Preservation

- Preservative: **sodium benzoate** (fruit juice) 3× 2010, 2013, one-liner
- Preservative: **benzoic acid** 2× 1992, 1996
- Potato chips packed in **nitrogen** 1× 2016
- Refrigeration **slows biochemical reactions** 2× 2011, 2013
- Decolouring raw sugar: **animal charcoal** 1× 1998

19.2 Sugars and Foods

- Main sugar in honey: **fructose** 1× 2002
- Invert sugar is sweeter than sucrose; it comes from hydrolysing sucrose (A and R true, R does not explain A) 1× 2020
- Best fruit for jelly: **guava** (rich in pectin) 1× 2013
- Organic food: grown **without artificial fertilisers and pesticides** 1× 2014

19.3 Chemicals in Medicine and Daily Use

- Sedative: **potassium bromide** 1× 2010
- Irreversible complex with haemoglobin: **carbon monoxide** 1× 1996
- Clove oil extract: **eugenol** 1× 2010
- Detergents clean well, but they **lower** surface tension (A true, R false) 1× 1997

19.4 Fertilisers

- Plants need chemical fertilisers for phosphorus; manure gives little (A and R true, R explains A) 1× 1998
- Element not supplied by fertilisers: **hydrogen** 1× 2007
- Leaves most acidity in soil: **ammonium sulphate** 1× 1996
- **Urea**: a nitrogen-containing **organic** compound 1× 2017
- Urea nitrogen: about **46-47%**, in **amide** form 2× one-liner, one-liner
- Most soluble at 20 C among options: **urea** 1× 2017
- Urea for 1 kg nitrogen: **2.2 kg** 1× 2019
- **DAP**: **18% N, 46% P2O5** 1× 2008
- Commonest foliar fertiliser: **urea** 1× 2015

Fertiliser	Nutrient content
Urea	46% N (amide)
DAP	18% N, 46% P2O5
Ammonium sulphate	21% N; makes soil acidic
SSP	16% P2O5
MOP	60% K2O
Gypsum	Reclaims alkaline soil

19.5 Pesticides and Agricultural Chemicals

- Fire alarm: **photocell** 1× 2020
- Fireproof clothing: **aluminium sulphate** (per key) 1× 2015
- Mosquito repellent: **pyrethrum** 1× 2009
- Pyrethrin comes from a **seed plant** (chrysanthemum) 1× 2005
- Rat poison: **zinc phosphide** 1× 2012
- Aluminium phosphide: a grain **fumigant** (several right uses, *) 1× 2008
- Herbicide: **sodium chlorate** 1× 2019
- Correct: **ECG-heart, nicotine-tobacco** (2 and 4) 1× 2002
- Alkaline soil reclaimed by **calcium sulphate** (gypsum) 1× 2008
- Soil-related: **edaphic** 1× 2015
- **Humus**: an **organic colloid** 1× 2014

19.6 Uses of Chemicals

- Artificial clouds and rain: **silver iodide** 3× 2008, 2013, one-liner
- **Blue vitriol-fungicide, eosin-red ink, silver iodide-artificial rain, zinc phosphide-rodenticide** 1× 2020
- **Philosopher's wool**: **zinc oxide** 1× 2007
- Correct pair: **zinc phosphide-rat poison** 1× 2011
- **Iron-haemoglobin, lead-storage battery, silver-photography, copper-lightning conductor** 1× 2010
- **Morphine-analgesic, sodium-kerosene, boric acid-antiseptic, German silver-alloy** 1× 2011
- "Metal of the future": **titanium** 1× 2010
- Wrong pair: **silver-electric bulb** (bulbs use tungsten) 1× 2002
- Red ink: **eosin** 1× 2014
- Sodium chromate gives **red** (per key; it is yellow) 1× 2016
- **Acetone-nail polish remover, CCl4-fire extinguisher, H2O2-wound dressing, liquid ammonia-refrigerant** 1× 2013
- **Freon-refrigerant, TEL-anti-knock, BHC-insecticide, CCl4-fire extinguisher** 1× 2013
- **Kevlar-synthetic fibre, Taxol-anticancer, zinc phosphide-rodenticide, nitrocellulose-explosive** 1× 2019

19.7 Glass, Batteries and Matches

- Glass that cuts UV: **Crookes glass** 1× 2021
- Wrong pair: **HCl-accumulator** (car batteries use **sulphuric acid**) 1× 1998
- Safety matches: **red phosphorus** 2× 2016, one-liner
- Glows in air and darkness: **white phosphorus** 1× 2016
- Dry cell cannot be recharged; its reaction is **irreversible** (A true, R false) 1× 2004
- Car battery acid: **sulphuric acid** 2× 2015, 2016

19.8 Miscellany

- Ozone kills microbes; it is not radioactive (A and R true, R does not explain A) 1× 2013
- **Radioactive dating**: age of **rocks** 1× 2008
- Age of the Earth: **uranium** method 1× 1992
- Desalination plant: **Lakshadweep** (Kavaratti) and Chennai 1× 2007
- **Mustard gas**: **chemical warfare** 1× 2008
- Smoke screens: **titanium oxide** particles 1× 2009
- Incorrect: **nuclear splitting is fusion** 2× 2004, 2010
- Correct pair: **ozone depletion-skin cancer** 1× 2015
- **Technology Day 11 May, Photography Day 19 August, Sports Day 29 August, Nagasaki Day 9 August** 1× 2003

QUANTUM IAS & PCS

TRAP ALERT

- Car battery = **sulphuric acid**; dry cell cannot be recharged.
- Safety matches = **red** phosphorus; white phosphorus glows and is stored in water.
- Zinc phosphide = rat poison; aluminium phosphide = grain fumigant.
- Gypsum fixes **alkaline** soil; lime fixes **acidic** soil.

UP LINK

UP link: UP is the largest user of **urea**; plants at **Phulpur** (IFFCO), **Gorakhpur** (HURL, restarted 2021), **Babrala**, **Aonla** and **Shahjahanpur** make it; IIT Kanpur ran **cloud-seeding** (silver iodide) trials in 2023.

LAST-MINUTE 10

1. Sodium benzoate = preservative; chips in **nitrogen**
2. Honey = fructose; guava = jelly; eugenol = clove oil
3. CO binds haemoglobin; KBr = sedative
4. Urea 46% N; DAP 18-46; ammonium sulphate = acidity
5. Pyrethrum = mosquito coil; zinc phosphide = rat poison
6. Silver iodide = artificial rain; eosin = red ink
7. Philosopher's wool = **ZnO**; titanium = metal of future
8. Freon = refrigerant; CCl4 = fire extinguisher; acetone = nail polish remover
9. Red phosphorus = safety matches; H2SO4 = car battery
10. Technology Day 11 May; Sports Day 29 August

Chapter 20

Branches of Biology, Evolution and Classification

Built from 82 UPPSC questions (1990–2025). Focus: branches and "-cultures", evolution, odd animals (seahorse, bat, echidna, spider), insectivorous plants, edible plant parts, plant pigments.

ASKED MOST · REVISE FIRST

- **Edible parts**: potato = **stem** (tuber), turmeric and ginger = **rhizome** (stem), sweet potato = **root**, clove = **flower bud** 11× 2002, 2003, 2005, 2006, 2010, 2011, 2013, 2015, 2021, 2021, one-liner
- **Nepenthes** = **insectivorous** pitcher plant (Meghalaya); traps insects for **nitrogen** 6× 2011, 2013, 2016, 2017, 2023, one-liner
- **Gerontology** = study of **ageing**; osteology = bones; demography = population 5× 1990, 1994, 2008, 2014, 2017
- **Spider is not an insect**; bat is not a bird; seahorse **is** a fish 4× 2008, 2008, 2010, 2011
- **Pigments**: apple red = **anthocyanin**; tomato = **lycopene**; papaya = **caricaxanthin**; chilli heat = **capsaicin** 5× 2002, 2008, 2015, 2016, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

20.1 Branches of Biology

- Zoology: **living and dead** animals 1× 1995
- **Birds-ornithology, inheritance-genetics, environment-ecology, fossils-palaeobiology** 1× 1998
- Population study: **demography** 2× 1994, 2017
- Wrong pair: **vegetable growing-horticulture** (that is **olericulture**) 1× 2015
- Wrong pair: **pisciculture-lac insects** (pisciculture = fish) 1× 2004
- **Viticulture**: **grapes** 1× 2003
- **Vermiculture**: **earthworms** 1× 2005
- Ageing: **gerontology** 2× 2008, 2014

- Bones: **osteology** 1× 1990

Culture	Rearing / growing of
Apiculture	Honeybees
Sericulture	Silkworms
Lac culture	Lac insects
Pisciculture	Fish
Viticulture	Grapes
Olericulture	Vegetables
Floriculture	Flowers
Vermiculture	Earthworms

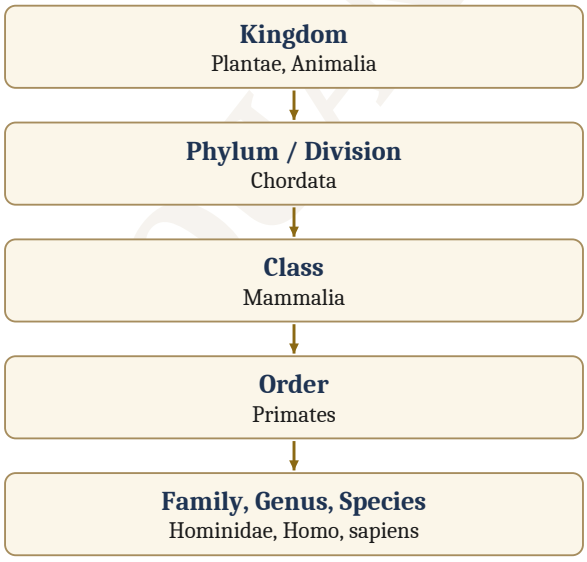
20.2 Origin and Evolution of Life

- Absent in the early atmosphere: **oxygen** 1× 2016
- Oldest organism: **blue-green algae** (cyanobacteria) 1× 2008
- **Mutation theory**: **Hugo de Vries** 1× 2005
- Dinosaurs: **Mesozoic** reptiles 2× 1998, 2001

20.3 Classification and Microbes

- Order from higher to lower: **kingdom, phylum, class, order** 1× 2024
- Bacteria seen with a **compound microscope** 1× 1995
- Some bacteria make food, but they lack **chloroplasts** (A true, R false) 1× 2000
- Statements about bacteria: **none** correct 1× 1995

TAXONOMIC RANKS



20.4 The Animal World

- A fish: **seahorse** 1× 2008
- Largest toothed whale: **sperm whale** 1× 2015
- Dolphins and whales breathe through **blowholes**; earthworms through **skin** (both) 1× 2025
- Earthworms are farmers' friends; they have **five pairs** of hearts, not two (only 1) 1× 2025
- Not a bird: **bat** (a mammal) 1× 2011
- Amphibians tolerate mixed blood; their heart has **three** chambers (A true, R false) 1× 2023
- Egg-laying mammal: **echidna** (and platypus) 1× 2022
- Smallest ape: **gibbon** 1× 2016
- **Nilgai**: key says deer family (it is an **antelope**, Bovidae) 1× 2006
- No hooves: **jackal** 1× 2017
- **Octopus**: a **mollusc** 1× 2004
- Largest invertebrate: **squid** 1× 2004

20.5 Insects and Silk

- Not an insect: **spider** (8 legs) 2× 2008, 2010
- White (colourless) blood: **cockroach** 1× 2019
- Spider silk: **gossamer** 2× 2017, 2021
- Not from insects: **pearl** (from oysters) 1× 2018
- Not a beneficial insect: **Helicoverpa armigera** (pod borer) 1× 2023
- Only **female** mosquitoes suck blood and have **bigger wings** (2 and 5) 1× 2016

20.6 Animals of Special Habitats

- Desert animals: **xericoles** 1× 2023
- Longest without water: **kangaroo rat**; near-solid urine 2× 2018, one-liner
- Snake fangs: modified **maxillary teeth** 1× 2016

20.7 Special Plants

- Wrong pairs: **Nepenthes-parasite**, **Nepenthes-hydrophyte** 2× 2016, 2017
- Insectivorous plant: **Nepenthes** 1× 2023
- **Nepenthes khasiana**: **Meghalaya** 1× 2011
- Insectivorous plants make up for **nitrogen** 2× 2013, one-liner
- **Mycorrhiza**: **fungi and plant roots** 1× 2019
- **Leghaemoglobin**: **legume root nodules** 1× 2005

20.8 Edible Parts of Plants

- **Lychee**: a **drupe** (per key) 1× 2010
- Turmeric: **stem**, a **rhizome** 2× 2006, 2021
- Modified root: **sweet potato** 2× 2005, 2010
- A stem: **ginger** 1× 2015
- Wrong pair: **potato-root** (it is a stem tuber) 2× 2011, 2013
- **Clove**: dried **flower buds** 4× 2002, 2003, 2021, one-liner
- Wrong pair: **saffron-seed** (saffron = stigma) 2× 2001, one-liner
- Garlic smell: a **sulphur compound** (allicin) 1× 2008
- Onion tears: **sulfenic acid** (syn-propanethial oxide) 1× 2015
- Chilli heat: **capsaicin** 2× 2008, 2016
- **Mulberry**: a **sorosis** fruit 1× 2006
- Not a stem fibre: **cotton** (a seed hair) 1× 2002
- Man-made cereal: **triticale** (wheat x rye) 1× 2017

Plant	Edible part
Potato	Stem tuber
Ginger, turmeric	Rhizome (stem)
Onion, garlic	Bulb (leaf bases)
Carrot, radish, sweet potato	Root
Clove	Flower bud
Saffron	Stigma
Cauliflower	Inflorescence

20.9 Ecological Terms and Pigments

- Tolerate wide temperature ranges: **eurythermal** 1× 2023
- Desert roots are long to **seek water**, not because of hot soil (A true, R false) 2× 1998, one-liner
- Desert plants: **succulent** 1× 2013
- Water absorbed mostly by **root hairs** 1× 2007
- **Hydrophyte**: aquatic plant 1× 1994
- Land dwellers: **terrestrial** 1× 2024
- Wrong pair: **herbivore-secondary consumer** (it is primary) 1× 2000
- Fish survive under ice: **bottom water does not freeze** 1× 2010
- Red apple: **anthocyanin**; red tomato: **lycopene** 2× 2002, one-liner
- Yellow papaya: **caricaxanthin** 1× 2015
- **Yeast** and **mushroom**: **fungi** 2× 2008, 2013
- Cotton is mostly **cellulose** 1× 2009

TRAP ALERT

- Potato = **stem**; sweet potato = **root**; turmeric = rhizome (stem).
- Spider (arachnid) is not an insect; bat is a mammal; seahorse **is** a fish.
- Nepenthes = insectivorous, not parasitic or aquatic.
- Saffron = **stigma**, not seed; clove = flower bud.

UP LINK

UP link: **NBRI Lucknow** keeps a large botanic garden and herbarium; **Nilgai** (blue bull) raids crops across UP; **Bhadohi-Mirzapur** carpets and **Varanasi** silk link to sericulture and wool, and **Malihabad** is famous for mango horticulture.

LAST-MINUTE 10

1. Gerontology = ageing; osteology = bones; viticulture = grapes
2. Blue-green algae = oldest; de Vries = mutation; dinosaurs = Mesozoic
3. Seahorse = fish; bat = mammal; echidna lays eggs
4. Spider not an insect; cockroach = white blood
5. Only female mosquitoes bite
6. Nepenthes = insectivorous (nitrogen); mycorrhiza = fungus + root
7. Potato = stem; sweet potato = root; clove = bud; saffron = stigma
8. Capsaicin = chilli; anthocyanin = apple; lycopene = tomato
9. Triticale = man-made cereal; cotton = cellulose
10. Kangaroo rat = never drinks; desert plants = succulent

Chapter 21

Cell, Heredity and Biofertilisers

Built from 30 UPPSC questions (1991–2024). Focus: cell organelles, Mendel and DNA, chromosome number, Azolla and blue-green algae, Rhizobium and other nitrogen fixers.

ASKED MOST · REVISE FIRST

- ■ **Azolla** (aquatic fern, with blue-green algae) = biofertiliser for **paddy** 6× 2008, 2009, 2011, 2023, one-liner, one-liner
- ■ **Blue-green algae** fix nitrogen, chiefly in **paddy** fields 4× 2006, 2010, 2019, one-liner
- ■ **Watson and Crick**: **double helix** of DNA; DNA sugar = **deoxyribose** 4× 2012, 2016, 2018, one-liner
- Humans have **46** chromosomes 2× 1991, 1992
- **Mitochondria** = power house; **cell wall** marks a plant cell 2× 1991, 2013

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

21.1 The Cell

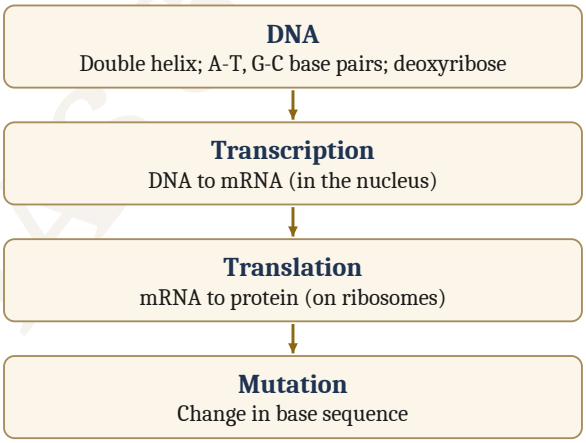
- Plant cell differs from animal cell by its **cell wall** 1× 1991
- Power house of the cell: **mitochondria** 1× 2013
- Plasma membrane: **lipids and proteins** (both) 1× 2008
- Not true: the nucleus has **no role** in reproduction (it does) 1× 2024

Organelle	Job
Nucleus	Controls the cell; holds DNA
Mitochondria	Respiration; ATP (power house)
Chloroplast	Photosynthesis (plants)
Ribosome	Protein synthesis
Lysosome	Digestion ("suicide bags")
Golgi body	Packaging and secretion
Cell wall	Plants only; cellulose

21.2 Heredity and DNA

- Mendel's laws: worked on the **garden pea** 1× 2002
- **Genome mapping**: mapping **genes** 1× 2002
- Change in a gene's base sequence: **mutation** 1× 2020
- Unit of heredity: **gene** 1× 2010
- Not a part of DNA among options: **tyrosine** (an amino acid) 1× 1994
- DNA sugar: **deoxyribose** 1× 2016
- ■ Double helix: **Watson and Crick** (1953) 3× 2012, 2018, one-liner
- Human chromosomes: **46** (23 pairs) 2× 1991, 1992
- No sex chromosomes: **lizard** (per key) 1× 2004

FROM GENE TO PROTEIN



21.3 Biofertilisers

- **Biofertilisers**: **cultures of micro-organisms** 1× 2023
- Azolla and blue-green algae are biofertilisers; alfalfa is a crop (*) 2× 2001, 2002
- Not a nitrogen fixer: **mycorrhiza** 1× 2017
- Nitrogen-fixing bacteria live in roots of **pulses** 1× 2019
- Wrong pair: **Azotobacter-bioinsecticide** (it fixes nitrogen) 2× 2002, 2003
- Blue-green algae: biofertiliser for **paddy** 2× 2006, 2010
- Biological nitrogen fixation: **blue-green algae** 1× 2019
- Azolla works because of its partner **blue-green alga** (Anabaena) 1× 2011
- Azolla is an **aquatic fern** 2× one-liner, one-liner
- Biofertiliser microbes: **cyanobacteria** 1× one-liner

QUANTUM IAS & PCS

- Water-logged rice fields: **Azolla** 1× 2023
- Biofertiliser source: **Azolla** 2× 2008, 2009
- Nitrogen-use efficiency improved by **all of the above** 1× 2009
- Phosphate biofertiliser: **VAM** (vesicular arbuscular mycorrhiza) 1× 2019
- Soybean nitrogen fixer: **Rhizobium japonicum** 1× 2006

Biofertiliser	Use
Rhizobium	Legume root nodules (pulses, soybean)
Azotobacter	Free-living nitrogen fixer
Blue-green algae (Anabaena, Nostoc)	Paddy fields
Azolla + Anabaena	Paddy fields
VAM / mycorrhiza	Phosphorus uptake

TRAP ALERT

- Mycorrhiza helps with **phosphorus**, not nitrogen fixation.
- Azotobacter is a nitrogen fixer, not a bioinsecticide.
- DNA sugar = deoxyribose; RNA sugar = ribose.
- Cell wall is in **plants**; centrioles are in **animals**.

UP LINK

UP link: UP's paddy belt in the east (Gorakhpur, Basti, Deoria) is promoted for **Azolla** and blue-green algae; the **Indian Institute of Pulses Research** at **Kanpur** works on Rhizobium-rich pulse crops.

LAST-MINUTE 10

1. Cell wall = plant cell; mitochondria = power house
2. Mendel = garden pea; gene = unit of heredity
3. Watson-Crick = double helix (1953); DNA sugar = deoxyribose
4. Humans = **46** chromosomes
5. Mutation = change in base sequence
6. Biofertilisers = microbe cultures
7. Azolla = aquatic fern + Anabaena; paddy
8. Blue-green algae = paddy nitrogen
9. Rhizobium japonicum = soybean
10. VAM = phosphate; mycorrhiza does not fix N

Chapter 22

Human Body Systems

Built from 90 UPPSC questions (1991–2025). Focus: bones and body make-up, digestion and enzymes, heart and blood, haemoglobin and blood groups, kidneys, brain parts, hormones and glands.

ASKED MOST · REVISE FIRST

- **Adrenaline** = "**fight or flight**" hormone 5× 2001, 2002, 2003, 2004, 2010
- **Haemoglobin**: has **iron**, makes blood **red**, carries **oxygen**; gives no immunity 6× 2001, 2004, 2005, 2016, 2019, one-liner
- **O** group = **universal donor** (O negative in emergencies); blood groups found by **Landsteiner** 4× 2003, 2006, 2016, one-liner
- **Enzymes are proteins**; protein digestion starts in the **stomach** (pepsin) 5× 1996, 2004, 2013, 2020, 2025
- **206** bones; smallest = **stapes** (ear) 4× 2004, 2008, 2010, one-liner

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

22.1 Composition and Skeleton

- Water in an adult body: about **65%** 1× 2007
- Most abundant element: **oxygen** 2× 2013, 2014

- Increasing amount in a 70 kg body: **copper, iron, sodium, potassium** 1× one-liner
- Bones in an adult: **206** 1× 2004
- Smallest bone: **stapes**, in the **ear** 3× 2008, 2010, one-liner
- Head-neck joint: **pivot** joint 1× 2024
- Main salt in bones: **calcium phosphate** 1× 2015
- **Arthroplasty**: **hip-joint replacement** 2× 2001, 2002
- Bones joined by **ligaments** 1× 2008
- Dental plaque: **food, saliva, acids and bacteria** 1× 2004
- Bones and teeth need **calcium and phosphorus** 1× 2011
- Not needed for bones and teeth: **iodine** 1× 2015
- Strongest muscle: **jaw** (masseter) 1× 2008

22.2 Digestion

- Most digestion: **small intestine** 1× 1991
- Enzymes are **proteins** 3× 1996, 2004, 2013
- Fat digestion needs **bile and lipase** (1 and 2) 1× 2016
- Protein digestion begins in the **stomach** 1× 2020
- Protein-digesting enzyme: **trypsin** (among options) 1× 2010
- Released in the stomach: **pepsin** only (only 3) 1× 2025
- Humans cannot digest cellulose; diastase digests starch (A and R true, R does not explain A) 1× 2017
- Extra glucose stored as **glycogen** in the **liver** 1× 2004
- Fasting over 10 days: liver **glucose falls** 1× 2021
- **Bile** is made in the **liver** 1× 1999
- Gallstones affect **fat** digestion 1× 2017
- Proteins to amino acids: **pepsin** (per key) 1× 2017
- **Vitamin-carotene, enzyme-pepsin, hormone-testosterone, protein-keratin** 2× 1998, 2021

Enzyme	Where	Acts on
Salivary amylase (ptyalin)	Mouth	Starch
Pepsin	Stomach	Proteins
Trypsin	Pancreas (small intestine)	Proteins
Lipase	Pancreas	Fats (with bile)
Bile (not an enzyme)	Liver	Emulsifies fats

22.3 Respiration

- Gas exchange in the lungs: **alveoli** 1× 2021
- Less oxygen in blood: breathing **increases** 1× 2000
- Most energy: **ATP to ADP** 1× 2015
- Energy is stored as **ATP** 1× 2016

22.4 Heart and Blood

- Human heart: **four** chambers 1× 2023
- Only vein with oxygenated blood: **pulmonary vein** 1× 2024
- Heart rests **between two beats** 2× 2004, 2008
- Wrist pulse beats **at the heart's rate** 2× 2008, 2010
- Blood in a 70 kg adult: about **5-5.6 litres** (*) 1× 2017
- Blood: plasma and cells; **haemoglobin** makes it red (both) 2× 2025, one-liner
- Blood is a **connective tissue** 1× 2023
- Not correct of haemoglobin: "**slightly acidic**" (per key) 1× 2019
- Haemoglobin: **iron, red colour, oxygen carrier** (1, 2, 4) 4× 2001, 2004, 2005, 2016
- Contains iron: **myoglobin** 1× 2020
- Average BP: **120/80** mm Hg; **Hg = mercury** 3× 1991, 1993, 2015
- Universal donor: **O** 2× 2003, 2016
- Emergency transfusion: **O negative** 1× 2006
- Blood groups: **Karl Landsteiner** 1× one-liner
- Blood pH: **7.35-7.45** 1× 2007
- RBCs destroyed (and some WBCs formed) in the **spleen** 1× 2008
- Blood sugar in **mg per decilitre** 1× 2021
- **Thrombin**: **blood clotting** 1× 2011
- Alum stops bleeding: Al ions coagulate colloidal blood (A and R true, R explains A) 1× 2010
- "Plastic blood": developed in **Britain** 1× 2006

QUANTUM IAS & PCS

Blood group	Can give to	Can receive from
O negative	All (universal donor)	O negative only
AB positive	AB positive only	All (universal recipient)
A	A, AB	A, O
B	B, AB	B, O

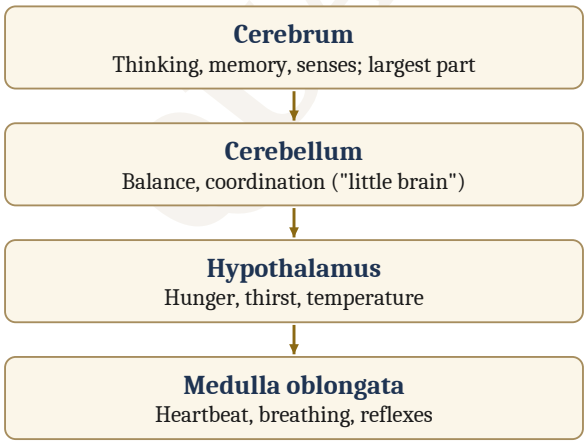
22.5 Excretion

- Detoxifies alcohol: **liver** 1× 2010
- Urine a day: about **1.5 litres** 1× 2004
- Correct: **all the blood** is filtered through the kidneys 1× 1994
- Urea removed from blood in the **kidney** 1× 2002
- **Dialyser**: does the **kidney's** work 1× 2002
- Not a kidney job: **blood sugar regulation** 1× 2011
- Kidney stones: **calcium oxalate** crystals 2× 2015, 2023

22.6 Nervous System

- Brain controls **all three** (heart rate, balance, thought) 2× 2016, one-liner
- "Little brain": **cerebellum** 1× 2023
- Feels no pain from a needle: **brain** tissue 1× 2017
- Body temperature **stays the same** in winter and summer 1× 2015
- Hunger and thirst centres: **hypothalamus** 1× 2010
- Normal temperature: **98.6 F (37 C)** 1× one-liner

PARTS OF THE BRAIN



22.7 Endocrine Glands and Hormones

- Urine volume: **ADH** (vasopressin) 1× 2010
- **Progesterone-ovaries, testosterone-testes, thyroxine-thyroid, insulin-pancreas** 1× 2010
- Both endocrine and exocrine: **pancreas** 1× 2019
- Second-largest gland, alkaline complete digestive juice: **pancreas** 1× 2008
- Stimulates pancreatic juice: **secretin** 1× 1996
- **Insulin**: a **protein**; from **beta cells** of the islets of Langerhans 3× 1993, 2015, one-liner
- Wrong pair: **gastrin-blood pressure** (gastrin makes stomach acid) 1× 2017
- Mother's hug and kiss release **oxytocin**; it also induces milk letdown in cattle 3× 2002, 2003, one-liner
- Not a protein (per key): **oxytocin** (a peptide) 1× 2008
- **Thyroxine** and **estrogen** are **hormones** 3× 2000, 2003, 2004
- **Fight or flight**: **adrenaline** 5× 2001, 2002, 2003, 2004, 2010
- Iodine deficiency: **goitre** 1× one-liner
- After menopause: **progesterone** stops; no option exact (*) 1× 2014
- Female sex hormone: **estrone** 1× 2017
- **Hormone-testosterone, enzyme-lipase, phospholipid-lecithin, polymer-polythene** 1× 1997
- Bright light is bad for sleep: light **lowers** melatonin (A true, R false) 1× 2014
- Smallest endocrine gland: **pituitary** per key (pineal is smaller) 1× 1996
- Sperm stay fertile for **one to two days** 1× 2010

TRAP ALERT

- **Pulmonary vein** carries oxygenated blood; pulmonary artery carries deoxygenated.
- Bile is made in the **liver**, stored in the gall bladder; it is not an enzyme.
- Melatonin rises in **darkness**, not light.
- **O** = universal donor; **AB** = universal recipient.

QUANTUM IAS & PCS

UP LINK

UP link: **SGPGIMS** and **KGMU** (Lucknow) run UP's major kidney transplant and dialysis units; the **PM National Dialysis Programme** gives free dialysis at district hospitals across UP.

LAST-MINUTE 10

1. Body water **65%**; oxygen = most abundant element
2. **206** bones; stapes smallest; jaw = strongest muscle
3. Enzymes = proteins; pepsin in the stomach
4. Bile from the liver; gallstones hit fat digestion
5. Alveoli = gas exchange; ATP = energy currency
6. Pulmonary vein = oxygenated; BP 120/80
7. O = universal donor; Landsteiner = blood groups; pH 7.35-7.45
8. Spleen = RBC graveyard; kidney stones = calcium oxalate
9. Cerebellum = balance; hypothalamus = hunger, thirst
10. Adrenaline = fight or flight; ADH = urine; oxytocin = milk letdown

Chapter 23

Vitamins and Nutrition

Built from 92 UPPSC questions (1990–2024). Focus: water- and fat-soluble vitamins, vitamin names and deficiency diseases, rich sources, proteins and amino acids, minerals, milk.

ASKED MOST · REVISE FIRST

- **Deficiency matching**: **A-night blindness, B1-beriberi, C-scurvy, D-rickets, B12-pernicious anaemia, folic acid-anaemia** **9×** 1999, 2002, 2010, 2015, 2016, 2017, 2023, 2023, 2024
- **Water-soluble**: **B and C**; fat-soluble: A, D, E, K **4×** 2002, 2019, 2021, 2023
- **Vitamin K** = **blood clotting** **4×** 1991, 2008, 2012, 2020
- **B12** has **cobalt** **6×** 2000, 2001, 2004, 2006, 2014, one-liner
- Highest protein: **soybean**; best all-amino-acid food: **egg** **4×** 2004, 2008, 2014, 2021

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

23.1 What Vitamins Are

- **Nutraceuticals**: nutrient plus medicinal effect **1×** 2009
- Vitamins named by **Casimir Funk** **1×** 2012
- "Accessory dietary factors": **vitamins** **1×** 2014
- Vitamins are needed in small amounts; they do **not kill bacteria** (A true, R false) **1×** 2002
- Not made in our body: **vitamin A** (made from carotene) **2×** 2011, 2013
- Made by gut bacteria: **vitamin K** and **B12** **1×** 2010
- Water-soluble: **B and C** **3×** 2002, 2021, 2023
- Not correct: **vitamin K is water-soluble** **1×** 2019
- B12 is stored in the liver for years; being water-soluble does not explain it (A and R true, R does not explain A) **1×** 2017
- Most vitamins are not made in the body (A true, R false) **1×** 2019
- The body does **not** make all vitamins; vitamins are essential (A false, R true) **2×** 2016, 2021

23.2 The B Vitamins

- **B1-thiamine, B2-riboflavin, B6-pyridoxine, B12-cobalamin** **2×** 2010, one-liner
- Wrong pair: **riboflavin-beriberi** (riboflavin lack causes **cheilosis**) **2×** 2002, one-liner
- Polished rice causes **beriberi** **2×** 2019, 2020
- Wrong pair: **B2-cod-liver oil** **1×** 2019

QUANTUM IAS & PCS

- Cobalt-containing vitamin: **B12** 4× 2001, 2004, 2006, one-liner
- **Thiamine** = **B1** 1× 1999
- Helps remember dreams: **B6** (per key) 2× 2010, 2013

23.3 Vitamin C

- Most vitamin C among vegetables: **chilli** 1× 2016
- Citrus vitamin for healthy skin: **C** 1× 1999
- Best source: **amla** 2× 1995, 2018
- Wound healing: **vitamin C** 1× one-liner
- Vitamin C helps absorb **iron** 1× 2010
- Not stored in the body: **vitamin C** 1× 2014

23.4 Vitamins A, D, E and K

- Vitamin D source: **sunlight**; made in the skin 2× 1993, one-liner
- Wrong pair: **vitamin D-antisterility** (that is E) 1× 2017
- Vitamin D: **calciferol** (cholecalciferol) 2× 1997, one-liner
- Night blindness: lack of **vitamin A** 1× 2006
- Wrong pair: **vitamin A-orange** (orange is vitamin C) 1× 2021
- Vitamin A stored in the **liver**; **retinol** for vision 2× 1997, one-liner
- Quitting greens alone need not cause night blindness (A false, R true) 1× 2011
- Rich in vitamin A: **carrot**; richest among options: **mango** 2× 2004, 2015
- Not a natural vitamin A source (per key): **milk** 1× 2009
- Vitamin E: **wheat germ oil** 1× 2016
- Clotting: **vitamin K** 4× 1991, 2008, 2012, 2020
- Wrong pair: **vitamin K-sterility** 1× 2008
- **K-alfalfa, D-cod-liver oil, E-wheat germ oil, C-lemon** 1× 2016

Vitamin	Name	Deficiency	Source
A	Retinol	Night blindness, xerophthalmia	Carrot, mango, liver
B1	Thiamine	Beriberi	Whole grains (not polished rice)
B2	Riboflavin	Cheilosis	Milk, eggs
B6	Pyridoxine	Anaemia, nerve problems	Grains, meat
B12	Cobalamin (cobalt)	Pernicious anaemia	Animal foods
C	Ascorbic acid	Scurvy	Amla, citrus, chilli
D	Calciferol	Rickets	Sunlight, cod-liver oil
E	Tocopherol	Infertility	Wheat germ oil
K	Phylloquinone	Poor clotting	Green leaves, alfalfa

23.5 Deficiency Diseases

- **A-blindness, B-beriberi, C-scurvy, D-rickets** 1× 2024
- **Night blindness-A, rickets-D, scurvy-C, beriberi-B** 3× 2010, 2015, 2023
- **C-scurvy, folic acid-anaemia, A-night blindness, B1-beriberi** 1× 2023
- Wrong pairs: **vitamin D-colour blindness, vitamin A-rheumatism** 2× 2016, 2017
- **Retinol-conjunctivitis, tocopherol-sterility, cyanocobalamin-pernicious anaemia, pyridoxine-mental illness** 2× 1999, 2002
- Not a vitamin: **thyroxine** 1× 2017

23.6 Energy Foods

- Most energy: **carbohydrate, fat, protein** together 1× 2021
- Most calories per gram: **fats** (9 kcal) 2× 1999, one-liner
- Apples help heart patients: rich in **potassium** 1× 2014

QUANTUM IAS & PCS

- Not a micronutrient: **sulphur** (a secondary macronutrient) 1× 2014
- Banana: about **116 kcal** per 100 g (per key) 1× 1998
- A glass of water: **zero** calories 1× 2015
- Essential diet constituent: **carbohydrate** 1× 2014
- No nitrogen in **carbohydrates** 1× one-liner
- Basmati grains lengthen on cooking: high **amylose** 1× 2015

23.7 Proteins and Amino Acids

- About 80% of living material (dry): **protein** 1× 1994
- Protein need: **higher in the old** (per key) 1× 2013
- Most protein: **soybean** 3× 2004, 2008, 2014
- Rich in protein and fat: **groundnut** 1× 2008
- Not correct of tyrosine: it guards against ageing free radicals 1× 2010
- Bread quality: **gluten** 1× 2010
- Tissues are built from **protein** 1× 1990
- **Alpha-keratin**: **skin**; hair and nails have **keratin** 2× 2003, one-liner
- Not a protein: **cotton** (cellulose) 1× 2019
- Does not denature protein: **infrared** rays 1× 2016
- Enzymes: **biocatalysts**, act where produced (both, per key) 1× 2019
- Not stored in the body: **amino acids** 1× 2015
- All essential amino acids: milk, egg, soybean (*); best: **egg** 2× 2016, 2021
- Sunflower oil is good; it is rich in **polyunsaturated fats** (A and R true, R explains A) 1× 2019

23.8 Minerals

- Acid-alkali balance: **phosphorus** (per key) 1× 2004
- Iron-rich fruit: **karonda**; iron-rich: **spinach** 2× 2006, 2008
- **B12-cobalt, haemoglobin-iron, chlorophyll-magnesium, brass-copper** 2× 2000, 2014
- Wrong pair: **washing soda-potassium** (it is sodium) 1× 2014

23.9 Milk

- Sweetness of milk: **lactose** 2× 2007, one-liner
- Cow milk is yellowish: **carotene** with casein; white colour: **casein** 3× 2014, one-liner, one-liner
- **Casein**: milk **protein** 1× 2008
- Wrong pair: **saccharin-carbohydrate** (it is an artificial sweetener) 1× 2017
- Lactating mother needs about **70 g** protein a day (per key) 1× 2008
- Mother's milk lacks **iron** 1× 2019

TRAP ALERT

- Water-soluble = **B and C** only; K is **fat**-soluble.
- B12 = **cobalt**; chlorophyll = **magnesium**; haemoglobin = **iron**.
- Vitamin E = fertility; vitamin K = clotting; vitamin D = rickets.
- Riboflavin (B2) lack = cheilosis, not beriberi.

UP LINK

UP link: UP's **Poshan Abhiyan** drives fight child **anaemia and stunting**; fortified rice (with iron, folic acid, B12) is now given through the PDS and mid-day meals in UP, and **Anganwadi** centres give take-home rations.

LAST-MINUTE 10

1. Funk named vitamins; vitamins = accessory factors
2. Water-soluble **B, C**; fat-soluble **A, D, E, K**
3. B1 = beriberi (polished rice); B2 = cheilosis
4. B12 = cobalt, pernicious anaemia
5. C = scurvy; amla best; helps iron absorption
6. D = sunlight, calciferol, rickets
7. A = retinol, night blindness, stored in liver
8. E = wheat germ oil, fertility; K = clotting
9. Fats = most calories; soybean = most protein; egg = all amino acids
10. Milk: lactose sweet, casein white, carotene yellow; lacks iron

Chapter 24

Plant Physiology

Built from 38 UPPSC questions (1996–2023). Focus: photosynthesis inputs and outputs, xylem and phloem, plant nutrients, plant hormones and their roles, crop practices and hydroponics.

ASKED MOST · REVISE FIRST

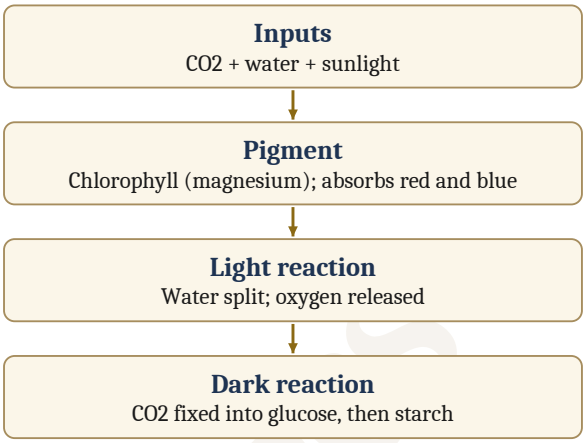
- ■ **Ethylene** ripens fruit (ethrel releases it) 6× 1996, 2007, 2008, 2017, 2023, one-liner
- ■ **Chlorophyll** has **magnesium**; photosynthesis needs **CO2**, releases O2 **from water** 5× 2011, 2012, 2013, 2014, one-liner
- ■ **Xylem** carries **water**; **phloem** carries **food** 4× 2012, 2015, 2015, 2023
- ■ **Plant hormones**: **auxin-apical dominance, gibberellin-elongation, cytokinin-cell division, abscisic acid-drought and dormancy**; insulin is not one 6× 2011, 2013, 2015, 2016, 2016, 2021
- • **Hydroponics** = growing plants **without soil** 2× 2009, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

24.1 Photosynthesis

- Element in chlorophyll: **magnesium** 2× 2012, 2013
- Plants make food by **photosynthesis** 1× 2013
- Sunlight becomes chemical energy; glucose turns into starch (A and R true, R explains A) 1× one-liner
- Essential gas: **CO2**; **oxygen** is not needed 2× 2011, one-liner
- Oxygen released comes from **water** 1× 2014
- Most of Earth's oxygen: **algae** (phytoplankton) 1× 2015
- Most effective light: **red** 1× 2014
- Artificial light **can** drive photosynthesis 1× 2008
- Best solar-to-chemical converter: **Chlorella** 1× 2014
- Does not add CO2 to the carbon cycle: **photosynthesis** 2× 2012, 2013

PHOTOSYNTHESIS IN ONE LINE



24.2 Transport in Plants

- **Phloem**: transports **food** (organic nutrients) 2× 2012, 2015
- **Xylem**: transports **water** 2× 2015, 2023

24.3 Mineral Nutrition

- Not a micronutrient: **magnesium** (a macronutrient) 1× 2019
- Plants take nitrogen as **nitrate** 2× 2016, 2021
- Yellowing and poor growth: lack of **nitrogen** 1× 2023
- More oil in mustard: **sulphur** 1× 2021
- Nodule formation in legumes: **boron** (per key) 1× 2019
- **Cuscuta** (amarbel) is a parasite; it feeds on the host's **stem** via haustoria, not leaves (A true, R false) 1× 2016
- **Crop logging**: plant analysis to judge **nutrient needs** 1× 2006

Nutrient	Role / deficiency
Nitrogen	Leaf growth; lack = yellowing (chlorosis)
Phosphorus	Roots, flowering
Potassium	Disease resistance, quality
Sulphur	Oil content (mustard)
Magnesium	Chlorophyll centre
Zinc	Khaira disease of rice
Boron	Flowering, nodules

24.4 Plant Hormones

QUANTUM IAS & PCS

- A plant hormone: **cytokinin** 3× 2015, 2016, 2021
- Not a plant hormone: **insulin** 1× 2011
- **Auxin-apical dominance, gibberellin-cell elongation, cytokinin-cell division, abscisic acid-growth inhibition** 1× 2016
- Drought tolerance: **abscisic acid** 1× 2013
- Correct (per key): weeping jelly from excess acid; **cytokinin** breaks dormancy (1 and 4) 1× 2013
- Ripening gas: **ethylene** (the key prints "ethane") 4× 1996, 2007, 2008, one-liner
- Hastens ripening: **ethrel** 1× 2023
- **Ripening-ethylene, baking soda-sodium bicarbonate, soft drink-CO2, quartz-sodium silicate** (per key) 1× 2017

24.5 Crop Practices

- **Gootee**: **vegetative propagation** (air layering) 1× 2021
- Seed changed every year: **hybrid seed** 1× 2010
- Crop rotation: a legume should be followed by a **non-legume** (only 2) 1× 2021
- Censer (pepper-pot) seed dispersal: **poppy** 1× 2010
- Clonal selection: **potato** 1× 2009
- Not correctly matched: key says **maize-tasseling** (*) 1× 2021
- **Hydroponics**: plants **without soil** 2× 2009, 2023

TRAP ALERT

- Oxygen from photosynthesis comes from **water**, not CO2.
- Magnesium is a **macro**nutrient; boron and zinc are micro.
- Abscisic acid = stress and dormancy; **gibberellin** is the main dormancy breaker.
- Fruit ripening gas is ethyl**ene**, not ethane.

UP LINK

UP link: **CIMAP** (Lucknow) and **NBRI** study medicinal and aromatic plant physiology; UP's **mustard** belt (Agra, Mathura, Etawah) is advised to use **sulphur** for higher oil content.

LAST-MINUTE 10

1. Chlorophyll = **magnesium**; red light most effective
2. O2 from **water**; algae make most O2
3. Chlorella = best solar converter
4. Xylem = water; phloem = food
5. Nitrate form of N; N lack = yellowing
6. Sulphur = mustard oil; boron = nodules
7. Cuscuta = parasite (amarbel)
8. Auxin = apical dominance; cytokinin = division; ABA = drought
9. Ethylene / ethrel = ripening
10. Hydroponics = no soil; gootee = air layering

Chapter 25

Economic Importance of Plants

Built from 35 UPPSC questions (2000–2023). Focus: improved crop varieties, biodiesel plants (Jatropha, Pongamia), caffeine and iodine, plant products and their parts, quinine and opium.

ASKED MOST · REVISE FIRST

- **Jatropha** = **biodiesel** plant; **Pongamia** (karanj) = leguminous petro-plant 4× 2004, 2010, 2021, 2021
- **Quinine** from **cinchona bark**; replaced by synthetic **chloroquine** 4× 2000, 2016, 2017, one-liner
- **Opium**: **latex of the unripe poppy fruit**; main constituent **morphine**; heroin and noscapine also from poppy 5× 2005, 2011, 2012, 2014, 2016
- **Seaweeds** are a major source of **iodine** 3× 2015, 2017, 2023
- **Caffeine** is in **tea and coffee** 2× 2016, 2017

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

25.1 Improved Varieties

- **Kanchan**: an improved variety of **amla** (Indian gooseberry) 1× 2015
- **Kashi Lalima**: red **okra** (IIVR, Varanasi) 1× 2021

- **Co 1148**: **sugarcane** 1× 2017
- First dwarf rice of India: **Jaya** 1× 2010
- Wrong pair: **linseed-Surya** 1× 2017

25.2 Oils, Biofuels and Industrial Products

- Green-synthesised nanoparticles for cancer research: **Parthenium** extract 1× 2004
- Peppermint oil is highest in the **leaf** 1× 2009
- **Litmus**: from **lichens** 1× 2010
- Biodiesel plant: **Jatropha** 3× 2004, 2021, 2021
- Leguminous petro-plant: **Pongam** (Pongamia) 1× 2010
- Petro-crops are rich in **hydrocarbons and lipids** (2 and 4) 1× 2009
- First biodiesel plant: **Andhra Pradesh** 1× 2004
- **Chilgoza**: seed of the **pine** 1× 2009
- Caffeine: **tea and coffee** both; tea's main stimulant 2× 2016, 2017
- Seaweeds: **iodine** 3× 2015, 2017, 2023

25.3 Spices and Plant Parts

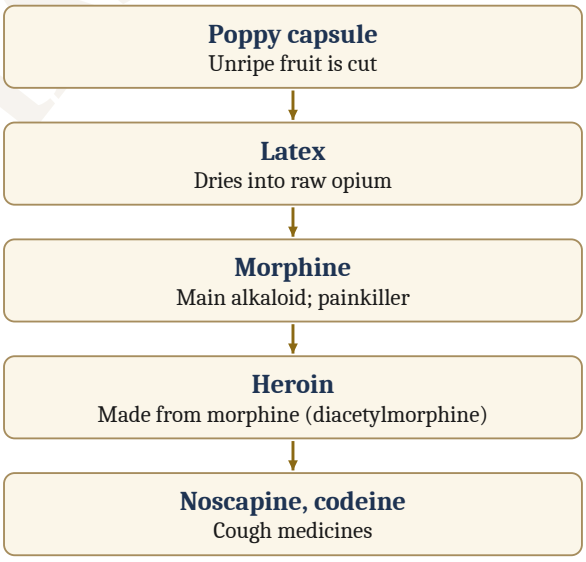
- **Opium-fruit, hing-root, rubber-stem, quinine-bark** 1× 2017
- **Asafoetida (hing)**: resin from the **stem base and roots** of Ferula 2× 2016, one-liner
- Main mosquito repellent plant: **neem** 1× 2004
- Tulsi's medicinal value: **phenols and flavonoids** 2× 2019, 2021
- Wrong pair: **rayon-modified starch** (rayon is regenerated **cellulose**) 2× 2019, 2022
- **Tapa cloth**: bark of plants of the **Moraceae** (mulberry) family 1× 2021
- **Beet-sugar, honey-glucose and fructose, cotton-cellulose, milk-lactose** (all four) 1× 2021

Product	Plant part
Opium	Latex of unripe poppy fruit
Hing (asafoetida)	Resin from root and stem base
Rubber	Latex from stem (bark)
Quinine	Cinchona bark
Clove	Flower bud
Saffron	Stigma
Cinnamon	Bark
Chilgoza	Pine seed

25.4 Medicinal and Drug Plants

- **Quinine**: **cinchona bark** 2× 2016, one-liner
- **Arteether** (antimalarial): from a **seed plant** (Artemisia annua) 1× 2005
- Synthetic drug that replaced quinine: **chloroquine** 1× 2000
- Hydroxychloroquine is fully **synthetic**; no plant fits (*) 1× 2016
- **Noscapine**: from **poppy** 1× 2012
- **Heroin**: from **opium poppy** 1× 2005
- Opium: **latex of the unripe fruit** 2× 2011, 2014
- Chief constituent of opium: **morphine** 1× 2016

FROM POPPY TO DRUGS



TRAP ALERT

- Opium comes from the **fruit**, not the seed or flower.
- Rayon is made from **cellulose**, not starch.
- Hing is a **resin** from roots and stem base, not a seed.
- Litmus = lichen; caffeine = tea **and** coffee.

QUANTUM IAS & PCS

UP LINK

UP link: **Kashi Lalima** (red okra) comes from **IIVR, Varanasi**; **Kanchan** amla was bred at **Narendra Deva Agricultural University** (Kumarganj, Ayodhya); **Barabanki** and **Ghazipur** are licensed **opium** poppy areas, and the **Government Opium Factory** is at **Ghazipur**.

LAST-MINUTE 10

1. Kanchan = amla; Kashi Lalima = okra; Co 1148 = sugarcane
2. Jaya = first dwarf rice
3. Jatropha = biodiesel; Pongamia = leguminous petro-plant
4. Litmus = lichen; chilgoza = pine seed
5. Caffeine = tea and coffee; seaweed = iodine
6. Hing = root/stem resin; neem = mosquito repellent
7. Rayon = cellulose; tapa cloth = Moraceae
8. Quinine = cinchona bark; chloroquine replaced it
9. Arteether = Artemisia; HCQ = synthetic
10. Opium = unripe fruit latex; morphine main; heroin from it

Chapter 26

Viral, Bacterial and Parasitic Diseases

Built from 70 UPPSC questions (1990–2024). Focus: viral diseases (dengue, polio, AIDS, hepatitis, Nipah, Ebola), bacterial diseases (TB, tetanus, cholera), vaccines, malaria and other parasites.

ASKED MOST · REVISE FIRST

- **Polio** vaccine: **Jonas Salk**; polio virus spreads through **polluted food and water** 6× 1991, 1993, 1995, one-liner, one-liner, one-liner
- **AIDS**: a virus (**retrovirus**, RNA + protein); tested by **ELISA**; India's strain **HIV-1C** 7× 2004, 2004, 2007, 2008, 2010, one-liner, one-liner
- **DPT** = **diphtheria, pertussis (whooping cough), tetanus**; not polio 4× 1994, 2004, 2013, one-liner
- **Tuberculosis** is **bacterial**: **BCG** vaccine, **Mantoux** test 7× 1990, 2010, 2015, 2016, 2017, one-liner, one-liner
- **Malaria** = **protozoan** (Plasmodium; **P. vivax** commonest in India) 5× 1998, 2008, 2008, 2012, 2023

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

26.1 Basic Terms and COVID-19

- Worldwide outbreak: **pandemic** 1× 2015
- Disease-causing microbe: **pathogen** 1× one-liner
- COVID-19 pause caused by **SARS-CoV-2** 1× 2024
- mRNA vaccine: **Moderna** 1× 2023
- **Sinovac**: **whole (inactivated) virus** vaccine 1× 2022

26.2 Mosquito-borne Viral Diseases

- Dengue lowers **platelets** 1× 2012
- Dengue mosquito: **Aedes** 1× 2022
- Not spread by tiger mosquitoes (Aedes): **Japanese encephalitis** (spread by **Culex**) 1× 2013
- Japanese encephalitis: a **virus** 1× one-liner

Mosquito	Disease
Aedes (tiger mosquito)	Dengue, chikungunya, Zika, yellow fever
Anopheles (female)	Malaria
Culex	Filaria, Japanese encephalitis

26.3 Other Viral Diseases

- **Hydrophobia** (rabies): **virus** 1× 2008
- Swelling of brain and spinal membranes: **meningitis** 1× 2008
- Infant diarrhoea deaths: **rotavirus** 1× 2010

QUANTUM IAS

- Polio virus enters through **polluted food and water** 2× 1993, one-liner
- Polio vaccine: **Jonas Salk** (injected); Sabin = oral 4× 1991, 1995, one-liner, one-liner
- Not viral: **tuberculosis** 1× 2017
- Common cold: **rhinovirus** 1× 1996
- **Hepatitis B**: a **DNA virus**; hits the **liver** 2× 2011, one-liner
- AIDS: caused by a **virus** (Acquired Immunodeficiency Syndrome) 2× 2010, one-liner
- HIV: **RNA + protein**; a **retrovirus** 2× 2008, one-liner
- Dominant strain in India: **HIV-1C** 1× 2004
- HTLV-III (HIV) discovered by **Robert Gallo** 1× 1999
- **ELISA** test: **AIDS** 2× 2004, 2007
- **Nipah** outbreak (2018): **Kerala** 1× 2018
- Smallpox is viral; isolation helps (A and R true, R explains A) 1× 1990
- Infectious cattle diseases: **FMD, anthrax, black quarter, cowpox** (all four) 1× 2003
- 2011 marked by the eradication of **rinderpest** 1× 2012
- Bird flu virus: **H5N1** 1× 2004
- **Ebola**: a deadly virus named after a **river** (Congo) 2× 2013, 2014

26.4 Bacterial Diseases

- Femur is the longest bone; cholera is bacterial (1 and 2); athlete's foot is **fungal** 1× 2008
- **Botulism** (food poisoning): **Clostridium** 2× 2013, one-liner
- **Haemophilus influenzae** (Hib): a **meningitis** in young children 1× 2022
- Bacterial disease: **tuberculosis** 2× 2010, one-liner
- Wrong pair: **genetic disease-tuberculosis** 1× 2016
- TB test: **Mantoux** test 1× 2015
- **Tetanus** = **lockjaw** 1× 2008
- Not from mosquitoes: **plague** (rat flea) 1× 1997
- Correct pair: **pneumonia-lungs** 1× 1996
- TB : lungs :: typhoid : **intestine** 1× 2023
- **Air-TB, water-cholera, contact-syphilis, wound-tetanus** 1× 1999
- Waterborne: **cholera** 1× 2015
- **BCG** vaccine: **TB**; give right **after birth** 2× 1990, one-liner
- Leprosy: **bacteria** 1× 2010
- Not viral: **typhoid** 1× 2008

Disease	Germ	Organ / note
Tuberculosis	Mycobacterium tuberculosis	Lungs; BCG, Mantoux
Typhoid	Salmonella typhi	Intestine; Widal test
Cholera	Vibrio cholerae	Intestine; waterborne
Tetanus	Clostridium tetani	Lockjaw; wounds
Diphtheria	Corynebacterium	Throat; toxoid vaccine
Leprosy	Mycobacterium leprae	Skin, nerves
Plague	Yersinia pestis	Rat flea

26.5 Vaccines and Antibiotics

- **DPT**: **diphtheria, whooping cough, tetanus** 3× 1994, 2004, 2013
- DPT does not protect against **polio** 1× one-liner
- Diphtheria vaccine: a **toxoid** 1× 2023
- **Sulpha drugs**: **antibacterial**; e.g. **Cibazole** 2× 2017, 2022
- Antibiotics are mostly from **bacteria** (actinomycetes) and fungi (per key) 1× 2021

26.6 Protozoan, Worm and Fungal Diseases

- Not common to malaria and dengue: **mosquito species** 1× 1998
- Protozoan disease: **malaria** 1× 2008
- About 65% of India's malaria: **P. vivax** (per key) 1× 2008
- **R21/Matrix-M**: malaria **vaccine** made by the **Serum Institute** 1× 2024
- Nematode disease: **filaria** 1× 1999
- Fungal (per key): **baldness** 1× 2016
- **Plague-bacteria, AIDS-virus, baldness-fungus, malaria-protozoa** 1× 2012
- **Peptic ulcer-bacteria, dengue-virus, elephant foot-worm, malaria-protozoa** 1× 2023

QUANTUM IAS & PCS

- **Sleeping sickness**: **Trypanosoma** (tsetse fly), not mosquitoes **2×** 2007, one-liner
- Leander Paes's brain problem (2003): a **cyst** (neurocysticercosis) **1×** 2005

TRAP ALERT

- **Tetanus** = lockjaw; **typhoid** = intestine; **TB** = lungs (all bacterial).
- **Japanese encephalitis** and filaria = Culex; dengue = **Aedes**; malaria = Anopheles.
- Plague spreads by **rat fleas**, not mosquitoes.
- Salk = injected polio vaccine; Sabin = oral.

UP LINK

UP link: **Japanese encephalitis/AES** hit **Gorakhpur** and eastern UP for decades; JE vaccination and the **"Dastak"** campaign (from 2017) cut deaths sharply; UP was also the last big polio reservoir before India was declared **polio-free (2014)**.

LAST-MINUTE 10

1. Pandemic = worldwide; SARS-CoV-2 = COVID-19; Moderna = mRNA
2. Dengue = Aedes, low platelets; JE = Culex (Gorakhpur)
3. Polio = Salk; spreads via food and water
4. Hepatitis B = DNA virus, liver; common cold = rhinovirus
5. AIDS = retrovirus; ELISA; HIV-1C in India; Gallo
6. H5N1 = bird flu; Ebola = river name; Nipah = Kerala 2018
7. TB = BCG, Mantoux; tetanus = lockjaw; typhoid = intestine
8. DPT = diphtheria, pertussis, tetanus; diphtheria = toxoid
9. Malaria = protozoa (P. vivax); R21 = Serum Institute vaccine
10. Filaria = nematode; sleeping sickness = Trypanosoma

Chapter 27

Genetic, Deficiency and Other Diseases; Plant Diseases

Built from 92 UPPSC questions (1991–2024). Focus: haemophilia and colour blindness, deficiency diseases, pollution diseases (Minamata, Itai-itai, blue baby), radioisotopes in medicine, eye and heart terms, crop diseases.

ASKED MOST · REVISE FIRST

- **Radioisotope matching**: **arsenic-74-tumour, cobalt-60-cancer (gamma rays), iodine-131-thyroid, sodium-24-blood flow, phosphorus-32-leukaemia** **9×** 2001, 2002, 2003, 2010, 2010, 2012, 2012, 2013, 2016
- **Pollution diseases**: **Minamata-mercury, Itai-itai-cadmium, blue baby-nitrate (methaemoglobin), knock-knee-fluoride** **7×** 2008, 2021, 2023, 2023, one-liner, one-liner, one-liner
- **Goitre** = **iodine** lack; **anaemia** = iron; **khaira** (rice) = zinc; **scurvy** = vitamin C **5×** 2005, 2007, 2007, 2011, 2015
- **Potassium** regulates the heartbeat; its lack gives **low blood pressure** (per key) **5×** 2001, 2002, 2003, 2004, 2008
- **Haemophilia** = sex-linked **hereditary** disease; night blindness is not genetic **4×** 2015, 2017, one-liner, one-liner

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

27.1 Genetic Disorders

- Not genetic: **night blindness** **1×** 2017
- Hereditary: **haemophilia** ("royal disease", sex-linked) **3×** 2015, one-liner, one-liner
- **Bubble baby disease** (SCID): baby kept in a **germ-free plastic bubble** **1×** 1997
- Wrong pair: **vitamin A-colour blindness** (colour blindness is genetic) **1×** 2016
- Colour-blind man and normal woman: the trait appears in the **daughters' sons** **1×** 2009

27.2 Deficiency and Lifestyle Diseases

- Arthritis (gout): **uric acid** deposits 2× 2013, one-liner
- Epidemic dropsy is from **argemone oil** in mustard oil, not rice husk 1× 1998
- Hysteria (per key): **young women** 1× 1991
- **Goitre**: lack of **iodine**; iodine is given for goitre 2× 2007, 2007
- Polished rice: **beriberi** 3× 2010, 2019, 2020
- **BMD** (Bone Mineral Density) test: **osteoporosis** 2× 2014, one-liner
- Not knowing when to stop eating: **bulimia** 1× 2016
- **Anosmia**: loss of **smell** 1× 2011
- **Sleep apnoea**: breathing stops briefly and repeatedly in sleep 1× 2014
- Heartbeat regulated by **potassium** 2× 2004, 2008
- Potassium deficiency (per key): **low blood pressure** 3× 2001, 2002, 2003
- Albumin in urine: **kidney** failure; prolonged fasting hurts the **kidneys** 2× 1998, one-liner
- Scurvy treated with **gooseberry** (amla) 1× 2011
- **Khaira-zinc, anaemia-iron, goitre-iodine, scurvy-vitamin C** 1× 2005
- **Marasmus-starvation, kwashiorkor-protein lack, TB-bacteria, hepatitis B-virus** 2× 2010, one-liner
- Iron deficiency: **anaemia** 1× 2015
- Heavy drinking: **liver cirrhosis** 1× 2008
- **Oral submucous fibrosis**: **gutka** 1× 2014
- **Bright's disease**: **kidney** 1× 1991
- Not controlled by vaccines: **diabetes** 1× 2005
- Diabetes: **pancreas**; tracked by **HbA1c** 3× 2019, 2024, one-liner
- Curry leaf claimed to help **diabetes** 1× 2004
- **Cerebral palsy**: **small children** 1× 2014
- Alzheimer's: **amyloid precursor protein** 1× 2013
- Mad cow disease: **prions** 1× 2009

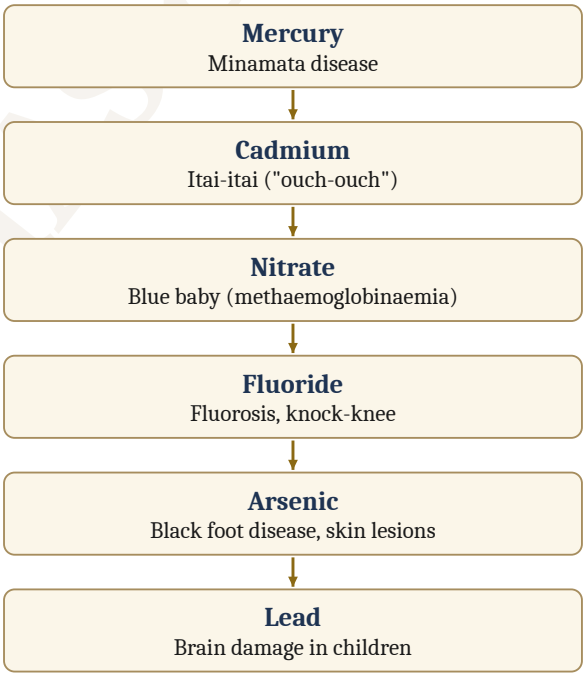
Disease	Cause
Goitre	Iodine deficiency
Anaemia	Iron deficiency
Kwashiorkor	Protein deficiency
Marasmus	Protein-energy starvation

Disease	Cause
Gout	Uric acid
Diabetes	Insulin lack (pancreas)
Cirrhosis	Alcohol
Mad cow disease	Prions

27.3 Pollution Diseases

- **Blue baby syndrome**: **methaemoglobin**, from **nitrate** in water 2× 2021, one-liner
- **Knock-knee**: **fluoride** pollution 1× 2023
- **Minamata**: **mercury** in fish and water (Japan, 1953) 3× 2023, one-liner, one-liner
- **Itai-itai**: **cadmium** 1× 2008

METAL AND CHEMICAL DISEASES



27.4 Radioisotopes and Cancer

- **As-74-tumour, Co-60-cancer, I-131-thyroid, Na-24-blood disorder** 3× 2001, 2002, 2003
- Wrong pair: **gold-198-retinal disorders** 1× 2010
- Speed of blood flow: **radio-sodium** 1× 2012
- Cobalt-60 therapy: it emits **gamma rays** 1× 2010
- Tumour detection: **arsenic-74** 2× 2010, 2013
- Leukaemia control: **phosphorus-32** 2× 2012, 2016
- **Leukaemia**: abnormal rise in **white blood cells** 1× 2016
- Latest cancer therapy: **IMRT** 1× 2017

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- **Interferon**: against **cancer** (and viruses) 1× 2010

27.5 Brain, Eyes and Medical Terms

- Least harmed by radiation: **brain** 2× 1999, 2011
- Brain diseases: **EEG** 2× 1993, one-liner
- **MRI** = Magnetic Resonance Imaging 2× 2008, 2012
- **Malaria-blood cells, filaria-lymph nodes, encephalitis-brain, leukaemia-bone marrow** 2× 1998, 2010
- **Astigmatism**: eye 1× 2004
- Myopic person sees **near** objects clearly 1× 2015
- Myopia corrected by a **concave lens** 1× 2003
- Long-sightedness: image forms **behind the retina** 1× 2017
- Blind spot and yellow spot: **retina** 1× 2008
- Light becomes nerve impulses at the **retina** 1× 1997
- Cataract: none of the options (it affects the **lens**) 1× 2017
- Pupil dilated with **atropine** 1× 2014
- Biological resurfacing: more than one joint fits (*) 1× 2014
- Cu-T common side effect: **bleeding** 1× 2014
- **Emphysema**: **lungs** 1× 2008
- Wrong pair: **aspirin-anaesthetic** (it is a painkiller) 1× 2017
- **Golden hour**: **heart attack** (and trauma) 1× 2014
- **Beta-blocker**: for **heart attack** 1× 2014
- Bypass graft vessel: **any of these** 1× 2014
- **Cath lab**: **cardiology** 1× 2014
- **CPR** = Cardiopulmonary Resuscitation 1× 2024
- Capsules: made of **gelatine** or **starch** 1× 2001

27.6 Plant Diseases

- Citrus canker = **bacteria**, red rot = **fungus**, potato black heart = **oxygen lack**; no code fits (*) 1× 2022
- **Rice-brown leaf spot, wheat-black rust, sugarcane-red rot, pigeon pea-wilt** 1× one-liner
- Wrong pair: **sugarcane-powdery mildew** 1× 2009
- Wrong pair: **rice-green hair** (per key) 1× 2015
- Maize chlorosis: **zinc** deficiency (white leaf tips) 2× 2006, one-liner
- **Karnal bunt**: **wheat** 1× 2016
- Rice **tungro** virus: **green leafhoppers** 1× 2009
- Not true: **false smut in barley** (it is a **rice** disease) 1× 2017
- Potato **black heart**: **oxygen deficiency** 1× 2008
- Okra **yellow vein mosaic**: spread by **whitefly** 2× 2002, one-liner
- Wrong pair: **paddy-wilt** 1× 2017
- **Spongy tissue**: **Alphonso** mango 1× 2002

Crop	Disease	Cause
Wheat	Black/brown rust, Karnal bunt	Fungus
Rice	Blast, brown spot, false smut	Fungus
Rice	Tungro	Virus (leafhopper)
Rice	Khaira	Zinc deficiency
Sugarcane	Red rot	Fungus
Potato	Late blight	Phytophthora
Potato	Black heart	Oxygen lack
Okra	Yellow vein mosaic	Virus (whitefly)
Citrus	Canker	Bacteria

TRAP ALERT

- **Minamata = mercury**; **Itai-itai = cadmium**; blue baby = **nitrate**.
- Cobalt-60 = **gamma** radiotherapy; I-131 = thyroid; P-32 = leukaemia.
- Cataract hits the **lens**; myopia sees **near** things; hypermetropia image is behind the retina.
- Khaira = **zinc**; maize white tips = zinc; potato black heart = **oxygen** lack.

UP LINK

UP link: **Ballia** and other eastern UP districts have **arsenic** in groundwater; **Unnao, Sonbhadra** and **Mathura-Agra** report **fluoride**; UP's sugarcane belt fights **red rot** (variety Co 0238 was hit hard), and **SGPGI Lucknow** runs nuclear medicine with I-131.

LAST-MINUTE 10

1. Haemophilia = sex-linked; colour blindness passes to daughters' sons
2. Gout = uric acid; goitre = iodine; khaira = zinc
3. Kwashiorkor = protein; marasmus = starvation; bulimia = overeating
4. Diabetes = pancreas, HbA1c; mad cow = prions
5. Minamata = Hg; Itai-itai = Cd; blue baby = nitrate; knock-knee = fluoride
6. Co-60 = gamma; I-131 = thyroid; P-32 = leukaemia; As-74 = tumour
7. EEG = brain; atropine dilates pupils; retina = blind and yellow spots
8. Golden hour = heart attack; cath lab = cardiology; CPR
9. Karnal bunt = wheat; tungro = leafhopper; red rot = sugarcane
10. Yellow vein mosaic = okra (whitefly); spongy tissue = Alphonso

Chapter 28

Genetic Engineering and Biotechnology

Built from 31 UPPSC questions (1999–2017). Focus: cloned animals, plasmids and phages, the Human Genome Project, stem cells, Bt cotton, golden rice and terminator seeds, DNA fingerprinting.

ASKED MOST · REVISE FIRST

- **Dolly** the **sheep** (1996) = first cloned mammal **2×** 2002, one-liner
- ■ **DNA fingerprinting**: paternity tests and crime; samples from blood, bone, hair, saliva **5×** 2002, 2004, 2006, 2008, 2010
- ■ **Golden rice** = transgenic rice rich in **beta-carotene** (vitamin A) **4×** 2000, 2012, one-liner, one-liner
- **Terminator technology**: seeds whose next generation is **sterile** **2×** 1999, one-liner
- **Bt cotton** carries a gene from a **bacterium** (Bacillus thuringiensis) **1×** 2010

■ asked 3 or more times • asked once or twice **4×** times asked, followed by the exam years

28.1 Cloning

- First cloned organism: **Dolly**, a **sheep** **2×** 2002, one-liner
- **Garima II**: a **cloned buffalo** (NDRI Karnal) **1×** 2009
- First all-green glowing transgenic pigs: **Taiwan** **1×** 2010
- **Injaz**: first cloned **camel** **1×** 2008
- Tried with some success: **disease resistance, growth promotion, animal cloning** (1, 2, 3); not human cloning **2×** 2001, 2003
- Genetic engineering vector: **plasmid** **1×** 2015
- **Bacteriophages** infect bacteria and are used in genetic engineering (both) **1×** 2017

Clone	Animal	Where / year
Dolly	Sheep	Scotland, 1996
Samrupa, Garima	Buffalo	NDRI Karnal, 2009-10
Injaz	Camel	Dubai, 2009
Noori	Pashmina goat	Kashmir, 2012

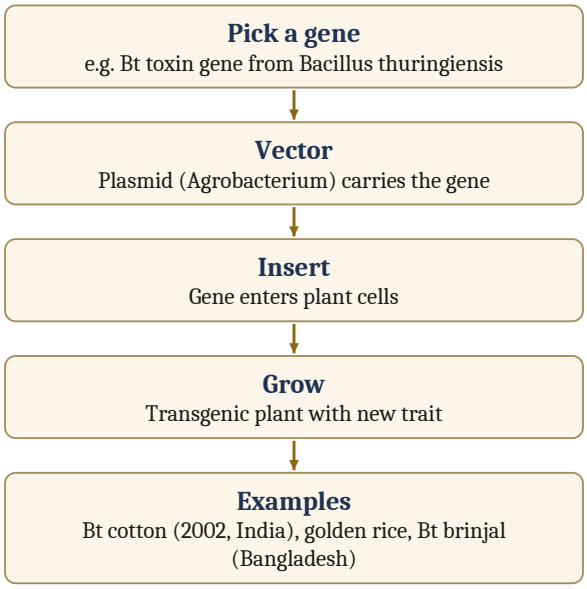
28.2 Genomes and Stem Cells

- **Human Genome Project**: identifying and **mapping human genes** and their sequence **1×** 2001
- Cells that become any cell type: **stem cells** **1×** 2006
- Non-controversial stem cells: **bone marrow** stem cells **1×** 2008
- **Plant Field Gene Bank, Banthara** (Lucknow): looks after **all the above** **2×** 2001, 2003

28.3 Transgenic Crops

- Bt cotton: gene from a **bacterium** **1×** 2010
- **Terminator technology**: seeds that give **sterile** next-generation seed **2×** 1999, one-liner
- **Orobanche** weed: **tobacco** fields **1×** 2014
- Transgenic plant: **golden rice** **1×** 2012
- ■ Golden rice helps consumers and farmers; its yellow comes from **beta-carotene** (A and R true, R explains A) **3×** 2000, one-liner, one-liner
- **Super rice**: **G.S. Khush** **1×** 2007

HOW A TRANSGENIC CROP IS MADE



28.4 Diagnosis and DNA Fingerprinting

- **Amniocentesis**: used to find **foetal sex** (now banned for that in India) 1× 2011
- **Genico** technology: advance **information on genetic diseases** 1× 1999
- DNA samples for crime: **blood, bone, hair, saliva** (all) 1× 2008
- **Biochip**: **RNA, DNA and protein** 1× 2008
- Paternity: **DNA fingerprinting** 2× 2002, 2010
- First crime solved by DNA (England): **1983** per key (the Pitchfork case, solved 1987-88) 1× 2006
- DNA fingerprinting used in **all** the areas 1× 2004
- Fingerprints on multi-coloured surfaces: **fluorescent powder** 1× 2013

28.5 Transgenics and Gene Therapy

- Not achieved by transgenics: **cloned animals** 1× 2000
- Gene therapy: several ways work (*) 1× 2004

TRAP ALERT

- Dolly = **sheep**; Injaz = camel; Garima = buffalo.
- Golden rice = **vitamin A** (beta-carotene), not iron.
- Bt gene comes from a **bacterium**, not a virus or fungus.
- Cloning is not a transgenic technique.

UP LINK

UP link: NBRI's **Plant Field Gene Bank** at **Banthara** (Lucknow) conserves plant genetic resources; the **National Bureau of Fish Genetic Resources** is also in **Lucknow**, and **IVRI Izatnagar** (Bareilly) works on animal biotechnology.

LAST-MINUTE 10

1. Dolly = first cloned mammal (sheep, 1996)
2. Garima II = buffalo; Injaz = camel; glowing pigs = Taiwan
3. Plasmid = vector; phages infect bacteria
4. Human Genome Project = gene mapping
5. Bone marrow = non-controversial stem cells
6. Bt cotton = bacterial gene
7. Terminator = sterile next-generation seed
8. Golden rice = beta-carotene; super rice = G.S. Khush
9. DNA fingerprinting = paternity, crime
10. Banthara = Plant Field Gene Bank (Lucknow)

Chapter 29

Biology Miscellaneous

Built from 133 UPPSC questions (1990–2024). Focus: health days and programmes, body chemistry, microbes in industry, medical tests and devices, animals and breeds, herbicides, and research institutes (many in UP).

ASKED MOST · REVISE FIRST

- **UP institutes: CDRI, CIMAP-Lucknow; IVRI-Bareilly; vegetable research-Varanasi; goat research-Makhdoom (Mathura); ALIMCO-Kanpur** 9× 1991, 2001, 2013, 2016, 2017, 2017, 2017, 2021, 2021
- **Health days: World AIDS Day 1 December, TB Day 24 March, Anti-Leprosy Day 30-31 January, Environment Day 5 June, Food Day 16 October** 5× 1992, 1999, 1999, 2002, 2009
- **Medical tests: EEG-brain, ECG-heart, EOG-eyes, EMG-muscle, audiogram-ear, mammogram-breast** 2× 2009, 2012
- **Endoscope** works by **total internal reflection** 5× 2008, 2009, 2010, 2014, 2014
- **Rayon** is regenerated **cellulose**, not modified starch 2× 2015, 2022
- **Three-chambered heart: frog** (amphibians) 2× 2013, 2016

■ asked 3 or more times • asked once or twice 4× times asked, followed by the exam years

29.1 Science News and Scientists

- Nobel Medicine 2024 (Ambros, Ruvkun): **microRNAs** 1× 2023
- **Birbal Sahni-palaeobotany, K.C. Mehta-plant pathology, P. Maheshwari-embryology, M.S. Swaminathan-Green Revolution** 1× 2021
- Science and Technology Policy 2003 aims: **all four** 1× 2008
- Not a science (Narlikar): **astrology** 1× 2018
- Science needs commitment to ideals and **honesty** (both) 1× 2024

29.2 Health Programmes and Days

- **Red Ribbon Express: AIDS awareness** 1× 2010
- **Mission Indradhanush**: immunising children against **seven** preventable diseases (now 12) 3× 2014, 2015, 2017
- **SANRAKSHA**: AIDS research and control project (Bengaluru) 1× 2014
- **NACO: AIDS** 1× 2006
- WHO Global Traditional Medicine Centre: **Jamnagar** 1× 2022
- Indian Biological Data Centre: **Haryana** (Faridabad) 1× 2023
- **1 December: World AIDS Day** 1× 2009
- Calendar order: **Anti-Leprosy (30 Jan), TB (24 Mar), Alzheimer's (21 Sep), AIDS (1 Dec)** 1× 2002
- **5 June-Environment, 1 December-AIDS, 5 September-Teachers, 16 October-Food** 1× 1992
- **TB-24 March, AIDS-1 December, Anti-Leprosy-30/31 January, Ozone-16 September** 1× 1999
- Minority Rights, AIDS, Animal, Food days: dates printed did not fit (*) 1× 1999
- Order: **Maritime (5 Apr), Health (7 Apr), Athletics (7 May), Red Cross (8 May)** 1× 2024
- Order: **Heritage (18 Apr), Light (16 May), Thyroid (25 May), No Tobacco (31 May)** 1× 2024

Day	Date
Anti-Leprosy Day	30 January
World TB Day	24 March
World Health Day	7 April
World Environment Day	5 June
Ozone Day	16 September
World Food Day	16 October
World AIDS Day	1 December

29.3 Body Chemistry

- **Cholesterol**: a **steroid** 2× 2005, 2021
- Trans fat lowers **HDL** (good cholesterol) 1× 2012
- Wrong pair: **mustard oil-lubricant** 1× 2023
- Yellow urine: **urochrome** 1× 2008
- **Antioxidants** protect cells from **free radicals** 1× 2012
- Silk, hair, nails, claws: **keratin** (per key; silk is fibroin) 1× 2009
- Lost in dehydration: **sodium chloride** 2× 1998, 2009
- Dehydrated person must not drink **sea water** 1× 2015
- Discomfort in humid heat: **sweat cannot evaporate** 1× 1995
- Most abundant organic compound: **cellulose** 2× 2012, 2014
- Fruit sweetness: **fructose** 1× 2017

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- Wrong pair: **pasteurisation-tea** 1× 2019
- **Acetic-vinegar, lactic-milk, butyric-butter, citric-lemon** 1× 2013
- **Bones-calcium phosphate, vinegar-acetic acid, souring milk-lactic acid, ripening-ethylene** 1× 2016
- **Vinegar-acetic, sour milk-lactic, bones-calcium phosphate, gastric juice-HCl** 1× 2016
- Wrong pair: **lactose-sour milk** (that is lactic acid) 1× 2015
- **Skin cancer-UV, noise-decibel, global warming-CO2, ozone hole-CFCs** 1× 2015
- Wrong pair: **antibiotic-quinine** (quinine is an antimalarial) 1× 2010

29.4 Microbes and Fermentation

- **Lactobacillus-curd, Rhizobium-nitrogen fixation, yeast-alcohol, Aspergillus-citric acid** 1× 2003
- Toddy fermentation: **Saccharomyces** (yeast) 3× 2010, one-liner, one-liner
- Microbe-made nanoparticle diode (India): **yeast** 1× 2002

29.5 Medical Devices and Medicine

- **EEG-brain, ECG-heart, EOG-eyes, EMG-muscle** 1× 2009
- **Audiogram-ear, ECG-heart, EEG-brain, mammogram-breast** 1× 2012
- Internal organs examined by **endoscope** 1× 2014
- Endoscope: **total internal reflection** 3× 2008, 2009, 2010
- What an endoscope is: **none of these** (per key) 1× 2014
- **Tetraplegia**: paralysis of **both arms and both legs** 1× 2015
- Electron beam therapy: **certain cancers** 1× 2008
- Dentist's mirror: **concave** 2× 2009, one-liner
- Foetal growth monitored by **ultrasonics** 1× 2006
- First heart transplant: **Christiaan Barnard** (1967) 2× 2004, 2021
- **Jarvik-7**: **artificial heart** 1× 2007
- **Pacemaker**: regulates heartbeat; natural one = **SA node** 2× 1999, 2007
- **Biojector**: painless **needleless injector** 1× 1997
- Wrong pairs: **antiseptic-aspirin, antiseptic-phenolphthalein** 2× 2003, 2006
- **Myoglobin-muscle, sarpagandha-tranquiliser, carcinoma-radiotherapy, haemoglobin-oxygen transport** 1× 2004
- **Bulimia-eating disorder, cholesterol-egg yolk, atropine-alkaloid, insulin-pancreas** 1× 2004
- Anaesthetics: **chloroform and nitrous oxide** (1 and 2) 2× 2002, one-liner
- **Cardiologist-heart, nephrologist-kidney, urologist-urinary tract, oculist-eye** 1× 2010
- Correct pair: **neurology-nervous system** 1× 2015
- Not banned for athletes: **colostrum** 1× 2003
- **Testosterone-hormone, codeine-sedative, caoutchouc-rubber, eugenol-clove oil** 1× 2008
- **Anthrax-biowarfare toxin, thalassaemia-defective gene, surrogacy-womb renting, transgenics-altering genomes** 1× 2003
- **Saheli**: CDRI's non-steroidal weekly **contraceptive pill** 1× 2017

29.6 Animals

- Dog that cannot bark: **dingo** (per key; basenji also) 1× 2010
- **Pheromones**: mostly **insects** 1× 2015
- Iodine from **Laminaria** (kelp) 2× 2004, one-liner
- **Cuckoo-no nest, lizard-sheds tail, blue whale-largest, chimpanzee-most intelligent** 1× 2004
- Stem fibre: **sunn hemp** 1× 2010
- Frog metamorphosis: **intestine shortens** 1× 2010
- Bull semen stored in **liquid nitrogen** 1× 2007
- High-yield cow: **Holstein** 1× 2006
- Wrong pairs: **sheep-Gir**, **sheep-Tharparkar** (both are cattle breeds) 2× 2021, 2021
- Three-chambered heart: **frog** (amphibians) 2× 2013, 2016
- Animals without a gall bladder: several fit (*) 1× 2004

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- First animal in space (key): a **dog** (Laika, 1957) 1× 2010
- **Angora wool**: **rabbit** 1× 2010
- Street-performing bear: **sloth bear** 1× 2008
- Largest brain for body size: **ant** 1× 2017
- Landmine detection: **bees** 1× 2002
- Cannot digest fibre well: **pig** 1× 1991
- Insect secretion: **lac** 1× 1996
- Light from living bodies: **bioluminescence** 1× 2014
- Mosquito-eating fish: **Gambusia** 1× 2011
- **Prallethrin**: mosquito repellent 1× 2017

29.7 Plants and Agriculture

- Herbicides spare crops as uptake rates differ (A and R true, R explains A) 1× 2002
- **2,4-D**: a **herbicide**; used in paddy 2× 1999, 2015
- Herbicide: **butachlor** 1× 2017
- Largest flower: **Rafflesia** 1× 1998
- Not needed for germination: **light** 1× 1990
- Mustard seed adulterant (per key): **poppy seeds**; argemone seeds cause dropsy 1× 2008
- Used in World War I: **mustard gas** 1× 2009
- Wax emulsion on fruit: **longer storage life** 1× 2010
- Clay holds most water, but its pores are **small** (A true, R false) 1× 2017
- **Eutrophication**: nutrient enrichment that **cuts dissolved oxygen** (both) 2× 2019, 2020
- Biodegradable: **wool** 1× 2022
- **Auxanometer**: rate of **growth** 1× 2010
- Tree age: **annual rings** 1× 2000
- Green Revolution seeds needed **more fertiliser and water** 1× 2001
- Paddy fields emit **methane** 1× 2006
- **Dapog** rice nursery: **Philippines** 1× 2010
- **Destructive Insects and Pests Act**: **1914** 1× 2023
- **Silicon carbide-artificial diamond, carbon fibre-aircraft, CO2-photosynthesis, CFC-refrigerator** 1× 2009
- Wrong pair: **rayon-modified starch** 2× 2015, 2022
- Wrong pair: **biological clock-biomass** 1× 2010

29.8 Research Institutes

- Indian Institute of Vegetable Research: **Varanasi**; not Gorakhpur 2× 2017, 2021
- **IVRI**: **Bareilly** (Izatnagar) 2× 2013, 2016
- **CDRI**: **Lucknow** 1× 1991
- **CIMAP**: **Lucknow** 1× 2021
- Wrong pair: **CECRI-Kolkata** (it is Karaikudi) 1× 2004
- High Security Animal Disease Laboratory: **Bhopal** 1× 2005
- Non-CSIR Lucknow institute: **BSIP** (DST) 1× 2005
- Institute for the mentally handicapped: **Hyderabad** (Secunderabad) 1× 2013
- **CDRI-Lucknow, Central Leprosy Institute-Agra, MNNIT-Allahabad, Sugar Technology-Kanpur** 1× 2001
- Wrong pair: **subtropical horticulture-Srinagar** (it is Lucknow) 1× 2017
- **ALIMCO** (artificial limbs): **Kanpur** 1× 2017
- **CDFD** (DNA fingerprinting): **Hyderabad** 1× 2013
- **FAO-Rome, Agmark lab-Nagpur, rice institute-Cuttack, potato institute-Shimla** 1× 2009
- International Potato Centre: **Lima, Peru** 1× 2021
- Mismatch: **NBRI-Kolkata** (it is Lucknow) 1× 2005
- **NBPGR**: **New Delhi** 1× 2013
- Goat research: **Makhdoom, Mathura** 1× 2017
- First dinosaurium: **Hyderabad** (per key) 1× 2004

Institute	Location (UP)
CDRI, CIMAP, NBRI, IITR, BSIP	Lucknow
IVRI	Izatnagar, Bareilly
Indian Institute of Vegetable Research	Varanasi
Central Institute for Research on Goats	Makhdoom, Mathura
ALIMCO	Kanpur
National Sugar Institute	Kanpur
Central JALMA (leprosy) Institute	Agra

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TRAP ALERT

- **Gir** and **Tharparkar** are **cattle** breeds, not sheep.
- CECRI = **Karaikudi**; NBRI = **Lucknow**; vegetable research = **Varanasi**, not Gorakhpur.
- Quinine is an antimalarial, not an antibiotic; aspirin is a painkiller, not an antiseptic.
- Frog = 3-chambered heart; fish = 2; birds and mammals = 4.

UP LINK

UP link: Lucknow alone hosts **CDRI** (Saheli pill), **CIMAP**, **NBRI**, **IITR**, **BSIP** and **NBFGR**; add **IVRI** (Bareilly), **IIVR** (Varanasi), **CIRG** (Makhdoom), **ALIMCO** and **NSI** (Kanpur), and **JALMA** (Agra).

LAST-MINUTE 10

1. AIDS Day **1 Dec**; TB Day **24 Mar**; Anti-Leprosy **30 Jan**
2. Mission Indradhanush = child immunisation
3. WHO traditional medicine centre = **Jamnagar**
4. Cholesterol = steroid; trans fat lowers HDL; urochrome = urine colour
5. EEG brain, ECG heart, EMG muscle, EOG eyes
6. Endoscope = TIR; Barnard = first heart transplant; Jarvik-7 = artificial heart
7. Saheli = CDRI pill; Gambusia eats larvae
8. Frog = 3-chambered heart; Angora = rabbit; Holstein = high-yield cow
9. 2,4-D, butachlor = herbicides; Rafflesia = largest flower
10. CDRI, CIMAP = Lucknow; IVRI = Bareilly; IIVR = Varanasi; ALIMCO = Kanpur

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